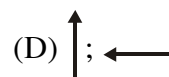
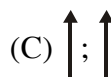
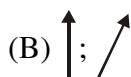
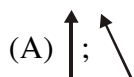
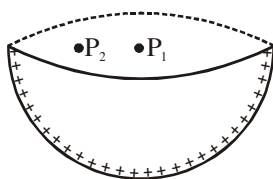
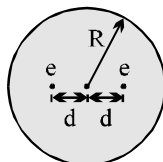


9. Consider a uniformly charged hemispherical shell shown below. Indicate the directions (not magnitude) of the electric field at the central point P_1 and an off-centre point P_2 on the drumhead of the shell.



ES0168

10. Using Thomson's model of the atom, consider an atom consisting of two electrons, each of charge $-e$, embedded in a sphere of charge $+2e$ and radius R . In equilibrium each electron is at distance d from the centre of the atom. What is equilibrium separation between electrons?



(A) R

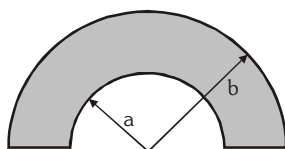
(B) $R/2$

(C) $R/3$

(D) $R/4$

ES0169

11. A non conducting semicircular disc (as shown in figure) has a uniform surface charge density σ . The electric potential at the centre of the disc :-



(A) $\frac{\sigma}{2\pi\epsilon_0} \frac{\ln(b/a)}{(b-a)}$

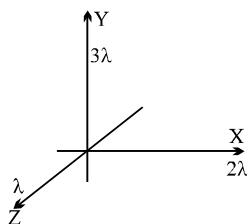
(B) $\frac{\sigma(b-a)}{2\epsilon_0}$

(C) $\frac{\sigma(b-a)}{4\epsilon_0}$

(D) $\frac{\sigma(b-a)}{4\pi\epsilon_0}$

ES0170

12. The diagram shows three infinitely long uniform line charges placed on the X, Y and Z axis. The work done in moving a unit positive charge from $(1, 1, 1)$ to $(0, 1, 1)$ is equal to



(A) $(\lambda \ln 2) / 2\pi\epsilon_0$

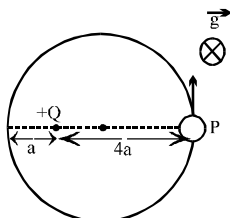
(B) $(\lambda \ln 2) / \pi\epsilon_0$

(C) $(3\lambda \ln 2) / 2\pi\epsilon_0$

(D) None

ES0171

13. The diagram shows a small bead of mass m carrying charge q . The bead can freely move on the smooth fixed ring placed on a smooth horizontal plane. In the same plane a charge $+Q$ has also been fixed as shown. The potential at the point P due to $+Q$ is V . The velocity with which the bead should be projected from the point P so that it can complete a circle should be greater than



- (A) $\sqrt{\frac{6qV}{m}}$ (B) $\sqrt{\frac{qV}{m}}$ (C) $\sqrt{\frac{3qV}{m}}$ (D) none

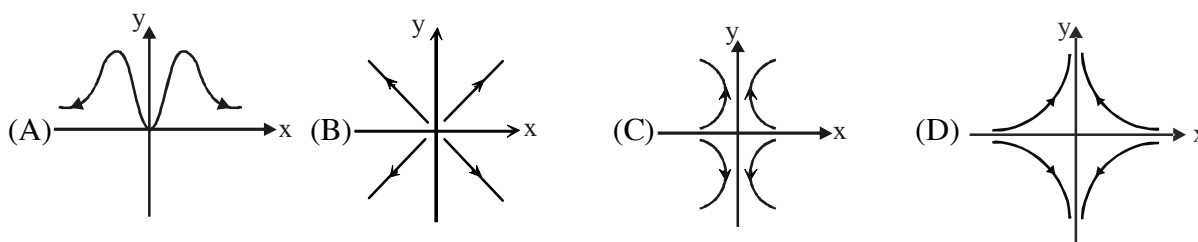
ES0172

14. A charged particle of charge Q is held fixed and another charged particle of mass m and charge q (of the same sign) is released from a distance r . The impulse of the force exerted by the external agent on the fixed charge by the time distance between Q and q becomes $2r$ is

- (A) $\sqrt{\frac{Qq}{4\pi\epsilon_0 mr}}$ (B) $\sqrt{\frac{Qqm}{4\pi\epsilon_0 r}}$ (C) $\sqrt{\frac{Qqm}{\pi\epsilon_0 r}}$ (D) $\sqrt{\frac{Qqm}{2\pi\epsilon_0 r}}$

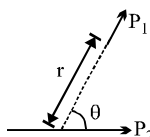
ES0173

15. In a certain region of space, the potential field depends on x and y coordinates as $V = (x^2 - y^2)$. The corresponding electric field lines in x - y plane are correctly represented by :



ES0174

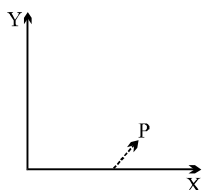
16. Two short electric dipoles are placed as shown. The energy of electric interaction between these dipoles will be :-



- (A) $\frac{2kP_1P_2 \cos\theta}{r^3}$ (B) $\frac{-2kP_1P_2 \cos\theta}{r^3}$ (C) $\frac{-2kP_1P_2 \sin\theta}{r^3}$ (D) $\frac{-4kP_1P_2 \cos\theta}{r^3}$

ES0175

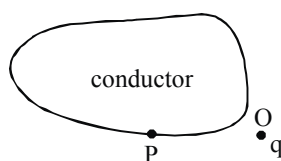
17. A small electric dipole $\vec{P}$ is placed on the X axis at the point (1, 0). The dipole vector forms an angle of 30° with the X axis. Consider a non uniform electric field to have been applied in the region given by the vector $\vec{E} = x^2\hat{i} + y^2\hat{j}$. What is the force acting on the dipole?



- (A) $2\vec{P} \cos 30^\circ (\hat{i} + 2\hat{j})$ (B) $2\vec{P} \cos 30^\circ (\hat{i})$ (C) $2\vec{P} \cos 30^\circ (2\hat{j})$ (D) None

ES0176

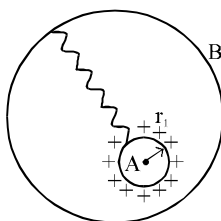
18. The density of charge at P on the conductor is σ . The resultant electric field near P will [$\hat{n}$ = unit normal vector at point P. $\vec{r}$ = vector along OP] where $K = 1/4 \pi \epsilon$



- (A) $\left(\frac{\sigma}{\epsilon_0}\right)\hat{n} + \left(\frac{Kq\vec{r}}{r^3}\right)$ (B) $\left(\frac{\sigma}{2\epsilon_0}\right)\hat{n} + \left(\frac{Kq\vec{r}}{r^3}\right)$ (C) $\left(\frac{\sigma}{\epsilon_0}\right)\hat{n}$ (D) $\left(\frac{\sigma}{2\epsilon_0}\right)\hat{n}$

ES0177

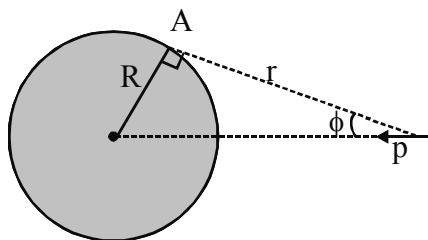
19. A metal sphere A of radius r_1 charged to a potential ϕ_1 is enveloped by a thin walled conducting spherical shell B of radius r_2 . Then ϕ_2 of the sphere A after it is connected by a thin wire to the shell B will be :-



- (A) $\phi_1 \frac{r_1}{r_2}$ (B) $\phi_1 \left(\frac{r_2}{r_1}\right)$ (C) $\phi_1 \left(1 - \frac{r_1}{r_2}\right)$ (D) $\phi_1 \left(\frac{r_1 r_2}{r_1 + r_2}\right)$

ES0178

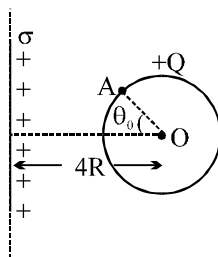
20. A dipole having dipole moment p is placed in front of a solid uncharged conducting sphere as shown in the diagram. The net potential at point A lying on the surface of the sphere is :-



- (A) $\frac{kpc \cos \phi}{r^2}$ (B) $\frac{kpc \cos^2 \phi}{r^2}$ (C) 0 (D) $\frac{2kpc \cos^2 \phi}{r^2}$

ES0179

21. A conducting sphere of radius R and charge Q is placed near a uniformly charged nonconducting infinitely large thin plate having surface charge density σ . Then find the potential at point A (on the surface of sphere) due to charge on sphere (here $K = \frac{1}{4\pi\epsilon_0}$, $\theta_0 = \frac{\pi}{3}$)



- (A) $K \frac{Q}{R} - \frac{\sigma}{4\epsilon_0} R$ (B) $K \frac{Q}{R} - \frac{\sigma R}{\epsilon_0}$ (C) $K \frac{Q}{R}$ (D) none of these

ES0180

22. The intensity of an electric field depends only on the coordinates x , y and z as follows :

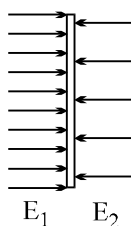
$$\vec{E} = a \frac{(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{3/2}} \text{ unit}$$

The electrostatic energy stored between two imaginary concentric spherical shells of radii R and $2R$ with centre at origin is :-

- (A) $\frac{4\pi\epsilon_0 a^2}{R}$ (B) $\frac{2\pi\epsilon_0 a^2}{R}$ (C) $\frac{\pi\epsilon_0 a^2}{R}$ (D) $\frac{\pi\epsilon_0 a^2}{2R}$

ES0181

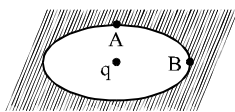
23. A charged large metal sheet is placed into uniform electric field, perpendicularly to the electric field lines. After placing the sheet into the field, the electric field on the left side of the sheet is $E_1 = 5 \times 10^5$ V/m and on the right it is $E_2 = 3 \times 10^5$ V/m. The sheet experiences a net electric force of 0.08 N. Find the area of one face of the sheet. Assume external field to remain constant after introducing the large sheet. Use $\left(\frac{1}{4\pi\epsilon_0}\right) = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$



- (A) $3.6 \pi \times 10^{-2} \text{ m}^2$ (B) $0.9 \pi \times 10^{-2} \text{ m}^2$ (C) $1.8 \pi \times 10^{-2} \text{ m}^2$ (D) none

ES0182

24. An ellipsoidal cavity is carved within a perfect conductor. A positive charge q is placed at the center of the cavity. The points A & B are on the cavity surface as shown in the figure. Then :

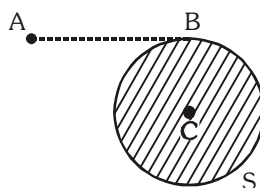


- (A) electric field near A in the cavity = electric field near B in the cavity
 (B) charge density at A = charge density at B
 (C) potential at A = potential at B
 (D) total electric field flux through the surface of the cavity is q/ϵ_0 .

ES0183

MULTIPLE CORRECT TYPE QUESTIONS

25. S is a solid neutral conducting sphere. A point charge q of $1 \times 10^{-6} \text{C}$ is placed at point A. C is the centre of sphere and AB is a tangent. $BC = 3\text{m}$ and $AB = 4\text{m}$.



- (A) The electric potential of the conductor is 1.8 kV.
 (B) The electric potential of the conductor is 2.25 kV.
 (C) The electric potential at B due to induced charges on the sphere is -0.45 kV .
 (D) The electric potential at B due to induced charges on the sphere is 0.45 kV .

ES0184

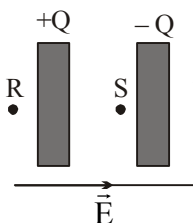
26. Four identical particles each having mass m and charge q are placed at the vertices of a square of side ℓ . All the particles are free to move without any friction and released simultaneously from rest. Then
 (A) At all instants, the particles remain at vertices of square whose edge length is changing
 (B) The configuration is changing (not remaining square) as the time passes

(C) The speed of the particles when one of the particles get displaced by $\frac{\ell}{\sqrt{2}}$ is $\sqrt{\frac{q^2}{8\pi\epsilon_0 m \ell} \left(2 + \frac{1}{\sqrt{2}} \right)}$

(D) Speed of the particles can not be found

ES0185

27. Two large thin conducting plates with small gap in between are placed in a uniform electric field E (perpendicular to the plates). Area of each plate is A and charges $+Q$ and $-Q$ are given to these plates as shown in the figure. If points R, S and T as shown in the figure are three points in space, then the



(A) field at point R is E

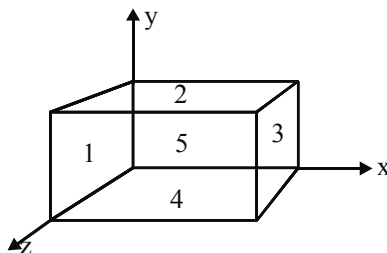
(B) field at point S is E

(C) field at point T is $\left(E + \frac{Q}{\epsilon_0 A} \right)$

(D) field at point S is $\left(E + \frac{Q}{A\epsilon_0} \right)$

ES0186

28. In a region of space, the electric field $\vec{E} = E_0 x \hat{i} + E_0 y \hat{j}$. Consider an imaginary cubical volume of edge 'a' with its edges parallel to the axes of coordinates. Now,



- (A) the total electric flux through the faces 1 and 3 is $E_0 a^3$
 (B) the charge inside the cubical volume is $2\epsilon_0 E_0 a^3$
 (C) the total electric flux through the faces 2 and 4 is $2E_0 a^3$
 (D) the charge inside the cubical volume is $\epsilon_0 E_0 a^3$

ES0187

29. Equipotential surfaces :-

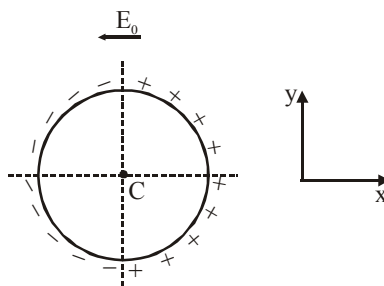
- (A) are closer in regions of large electric fields compared to regions of lower electric fields.
 (B) will be more crowded near sharp edges of a conductor.
 (C) will be more crowded near regions of large charge densities.
 (D) will always be equally spaced.

ES0188

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 30 and 31

A uniform ring of mass m and radius R can rotate freely about an axis passing through centre C and perpendicular to plane of paper. Half of ring is positively charge and other half is negatively charge. Uniform electric field E_0 is switched on along $-ve$ x -axis (axis are shown in figure) [Magnitude of charge density λ]



30. The dipole moment of ring is :-

- (A) $2 \lambda R^2$ (B) $4 \lambda R^2$ (C) $2 \pi \lambda R^2$ (D) $4 \pi \lambda R^2$

ES0189

31. If ring is slightly disturb from given position, find the angular speed of ring when it rotate by $\pi/2$.

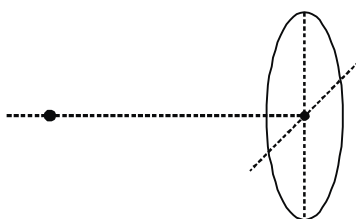
- (A) $2\sqrt{\frac{\lambda E_0}{m}}$ (B) $\sqrt{\frac{\lambda E_0}{m}}$ (C) $\sqrt{\frac{8\lambda E_0}{m}}$ (D) None

ES0189

**MATCHING LIST TYPE ($4 \times 4 \times 4$) SINGLE OPTION CORRECT
(THREE COLUMNS AND FOUR ROWS)**

Answer Q.32, 33 & 34 by appropriately matching the information given in the three columns of the following table.

Consider a non conducting ring of radius r and mass m and a particle of same mass, both at rest in free space. The particle is on the axis of the ring and far away from the ring. An amount Q of positive charge is uniformly distributed on the ring and the particle is given a positive charge q . The particle is imparted a velocity v towards the centre of the ring. Consider the consequences given in the columns and answer the following questions.



Column 1	Column 2	Column 3
(I) Maximum speed of the ring is v	(i) Minimum speed of the particle is $v/2$	(P) Final speed of the ring and particle is v and zero respectively.
(II) Maximum speed of the ring is $v/2$	(ii) Minimum speed of the particle is zero	(Q) Final speed of the ring and particle is zero and v respectively.
(III) Maximum speed of the ring is $\frac{v}{2} \left[1 + \sqrt{1 - \frac{Qq}{\pi\epsilon_0 mrv^2}} \right]$	(iii) Minimum speed of the particle is $\frac{v}{2} \left[1 + \sqrt{1 - \frac{Qq}{\pi\epsilon_0 mrv^2}} \right]$	(R) Final speed of the ring and particle is $v/4$ and $3v/4$ respectively.
(IV) Maximum speed of the ring is $\frac{v}{2} \left[1 - \sqrt{1 - \frac{Qq}{\pi\epsilon_0 mrv^2}} \right]$	(iv) Minimum speed of the particle is $\frac{v}{2} \left[1 - \sqrt{1 - \frac{Qq}{\pi\epsilon_0 mrv^2}} \right]$	(S) Final speed of the ring and particle is $v/2$ for both.

32. Which of following options is the correct representation if $v = \sqrt{\frac{Qq}{2\pi\epsilon_0 mr}}$

- (A) (II) (i) S (B) (I) (ii) P (C) (III) (iv) R (D) (IV) (iii) Q

ES0190

33. Which of following options is the correct representation if $v = \sqrt{\frac{Qq}{\pi\epsilon_0 mr}}$

- (A) (II) (i) S (B) (I) (ii) P (C) (III) (iv) R (D) (IV) (iii) Q

ES0190

34. Which of following options is the correct representation if $v = \sqrt{\frac{2Qq}{\pi\epsilon_0 mr}}$
- (A) (II) (i) S (B) (I) (ii) P (C) (III) (iv) Q (D) (IV) (iii) (Q)

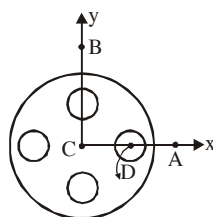
ES0190

MATRIX MATCH TYPE QUESTION

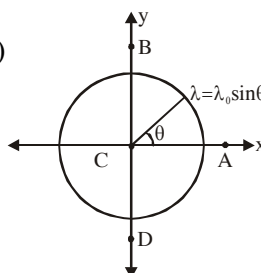
- 35.** Column-II shows some charge distributions and column-I has some statements about electric field at four points A, B, C, D. Match column-I with column-II.

Column-I

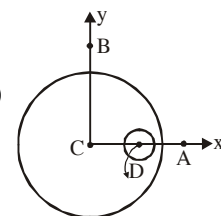
- (A) $\vec{E}_A$ has x component only (P)



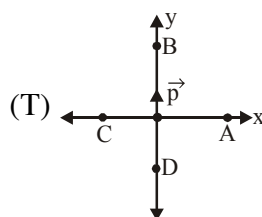
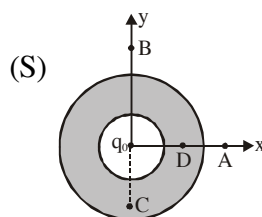
- (B) $\vec{E}_R$ has y component only (Q)



- (C) $\vec{E}_c$ has y component only (R)



- (D) $|\vec{E}_D|$ is zero



Column-II

A solid non conducting sphere of radius R of volumetric charge density ρ with four symmetrical spherical cavities. All the five sphere's centre lie in same plane.

A very small circular filament lying in xy-plane. All points lie in same plane. A, B and D are at large distance compared to radius of circle.

A charged spherical conductor
with an empty cavity in it.

A hollow thick spherical neutral conductor with a concentric cavity. Charge q_0 is placed inside at centre of cavity.

A small electric dipole $\vec{p}$ is placed at origin. A, B, C and D are four points at large distance from origin.

ES0191

EXERCISE-JM

1. Two identical charged spheres suspended from a common point by two massless string of length ℓ are initially a distance d ($d \ll \ell$) apart because of their mutual repulsion. The charge begins to leak from both the spheres at a constant rate. As a result the charges approach each other with a velocity v . Then as a function of distance x between them :- [AIEEE - 2011]

(1) $v \propto x^{1/2}$ (2) $v \propto x$ (3) $v \propto x^{-1/2}$ (4) $v \propto x^{-1}$

ES0192

2. The electrostatic potential inside a charged spherical ball is given by $\phi = ar^2 + b$ where r is the distance from the centre; a, b are constant. Then the charge density inside the ball is :-

[AIEEE - 2011]

(1) $-24\pi a \epsilon_0$ (2) $-6 a \epsilon_0$ (3) $-24\pi a \epsilon_0 r$ (4) $-6 a \epsilon_0 r$

ES0193

3. Two positive charges of magnitude ' q ' are placed at the ends of a side (side 1) of a square of side ' $2a$ '. Two negative charges of the same magnitude are kept at the other corners. Starting from rest, if a charge Q moves from the middle of side 1 to the centre of square, its kinetic energy at the centre of square is :- [AIEEE - 2011]

(1) $\frac{1}{4\pi \epsilon_0} \frac{2qQ}{a} \left(1 - \frac{1}{\sqrt{5}}\right)$

(2) zero

(3) $\frac{1}{4\pi \epsilon_0} \frac{2qQ}{a} \left(1 + \frac{1}{\sqrt{5}}\right)$

(4) $\frac{1}{4\pi \epsilon_0} \frac{2qQ}{a} \left(1 - \frac{2}{\sqrt{5}}\right)$

ES0194

4. This question has Statement-1 and Statement-2. Of the four choices given after the statements, choose the one that best describes the two statements. [AIEEE - 2012]

An insulating solid sphere of radius R has a uniformly positive charge density ρ . As a result of this uniform charge distribution there is a finite value of electric potential at the centre of the sphere, at the surface of the sphere and also at a point out side the sphere. The electric potential at infinity is zero.

Statement-1: When a charge ' q ' is taken from the centre to the surface of the sphere, its potential

energy changes by $\frac{q\rho}{3 \epsilon_0}$

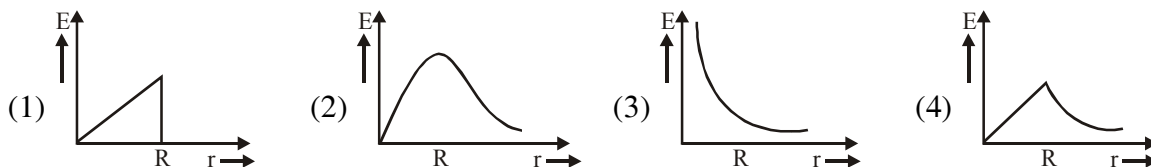
Statement-2 : The electric field at a distance r ($r < R$) from the centre of the sphere is $\frac{\rho r}{3 \epsilon_0}$

- (1) Statement-1 is true, Statement-2 is true and Statement-2 is the correct explanation of Statement-1.
 (2) Statement-1 is true, Statement-2 is true and Statement-2 is not the correct explanation of statement-1.
 (3) Statement-1 is true, Statement-2 is false
 (4) Statement-1 is false, Statement-2 is true

ES0195

5. In a uniformly charged sphere of total charge Q and radius R , the electric field E is plotted as a function of distance from the centre. The graph which would correspond to the above will be :-

[AIEEE - 2012]



ES0196

6. Let $[\epsilon_0]$ denote the dimensional formula of the permittivity of vacuum. If M = mass, L = Length, T = Time and A = electric current, then :

[JEE-Main-2013]

- (1) $[\epsilon_0] = [M^{-1} L^{-3} T^2 A]$ (2) $[\epsilon_0] = [M^{-1} L^{-3} T^4 A^2]$
 (3) $[\epsilon_0] = [M^{-1} L^2 T^{-1} A^{-2}]$ (4) $[\epsilon_0] = [M^{-1} L^2 T^{-1} A]$

ES0197

7. Two charges, each equal to q , are kept at $x = -a$ and $x = a$ on the x -axis. A particle of mass m and charge $q_0 = \frac{q}{2}$ is placed at the origin. If charge q_0 is given a small displacement ($y \ll a$) along the y -axis, the net force acting on the particle is proportional to

[JEE-Main-2013]

- (1) y (2) $-y$ (3) $\frac{1}{y}$ (4) $-\frac{1}{y}$

ES0198

8. A charge Q is uniformly distributed over a long rod AB of length L as shown in the figure. The electric potential at the point O lying at a distance L from the end A is :-

[JEE-Main-2013]

- (1) $\frac{Q}{8\pi \epsilon_0 L}$ (2) $\frac{3Q}{4\pi \epsilon_0 L}$ (3) $\frac{Q}{4\pi \epsilon_0 L \ln 2}$ (4) $\frac{Q \ln 2}{4\pi \epsilon_0 L}$

ES0199

9. Assume that an electric field $\vec{E} = 30x^2 \hat{i}$ exists in space. Then the potential difference $V_A - V_O$, where V_O is the potential at the origin and V_A the potential at $x = 2$ m is :-

[JEE-Main-2014]

- (1) -80 J (2) 80 J (3) 120 J (4) -120 J

ES0200

10. A uniformly charged solid sphere of radius R has potential V_0 (measured with respect to ∞) on its surface. For this sphere the equipotential surfaces with potentials $\frac{3V_0}{2}$, $\frac{5V_0}{4}$, $\frac{3V_0}{4}$ and $\frac{V_0}{4}$ have radius

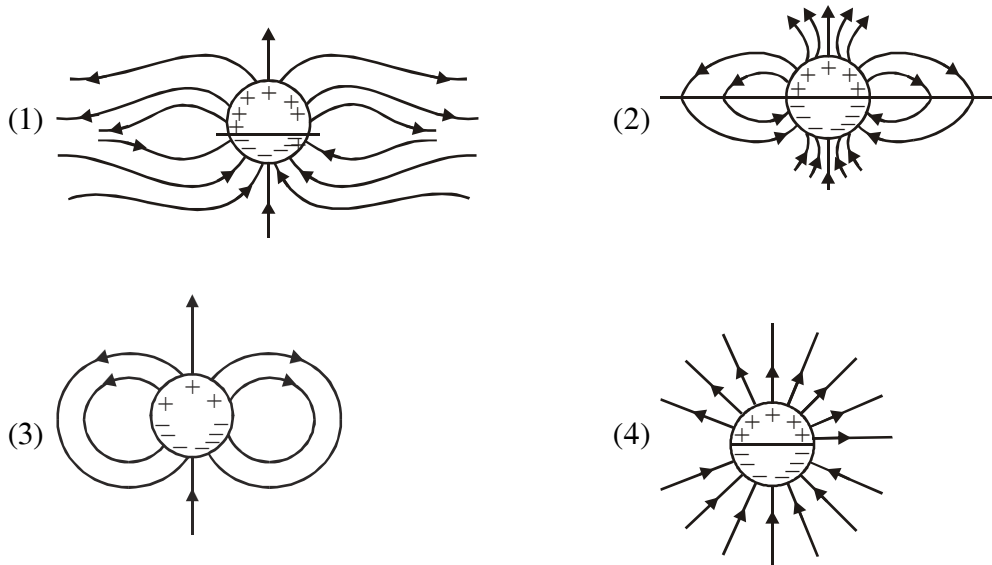
R_1 , R_2 , R_3 and R_4 respectively. Then

[JEE-Main-2015]

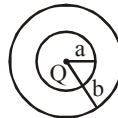
- (1) $R_1 = 0$ and $R_2 < (R_4 - R_3)$ (2) $2R < R_4$
 (3) $R_1 = 0$ and $R_2 > (R_4 - R_3)$ (4) $R_1 \neq 0$ and $(R_2 - R_1) > (R_4 - R_3)$

ES0201

11. A long cylindrical shell carries positive surface charge σ in the upper half and negative surface charge $-\sigma$ in the lower half. The electric field lines around the cylinder will look like figure given in: (figures are schematic and not drawn to scale)
- [JEE-Main-2015]**

**ES0202**

12. The region between two concentric spheres of radii 'a' and 'b', respectively (see figure), has volume charge density $\rho = \frac{A}{r}$, where A is a constant and r is the distance from the centre. At the centre of the spheres is a point charge Q. The value of A such that the electric field in the region between the spheres will be constant, is :- **[JEE-Main-2016]**



- $$(1) \frac{2Q}{\pi a^2} \qquad (2) \frac{Q}{2\pi a^2} \qquad (3) \frac{Q}{2\pi(b^2 - a^2)} \qquad (4) \frac{2Q}{\pi(a^2 - b^2)}$$

ES0203

13. An electric dipole has a fixed dipole moment $\vec{p}$, which makes angle θ with respect to x-axis. When subjected to an electric field $\vec{E}_1 = E\hat{i}$, it experiences a torque $\vec{T}_1 = \tau\hat{k}$. When subjected to another electric field $\vec{E}_2 = \sqrt{3}E\hat{j}$ it experiences torque $\vec{T}_2 = -\vec{T}_1$. The angle θ is : **[JEE-Main-2017]**
- (1) 60° (2) 90° (3) 30° (4) 45°

ES0204

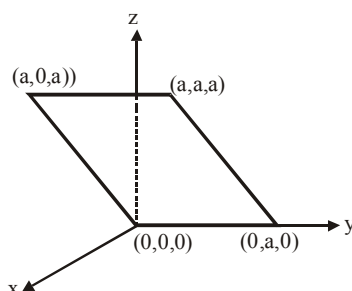
14. Three concentric metal shells A, B and C of respective radii a , b and c ($a < b < c$) have surface charge densities $+\sigma$, $-\sigma$ and $+\sigma$ respectively. The potential of shell B is :- **[JEE-Main-2018]**

- $$(1) \frac{\sigma}{\varepsilon_0} \left[\frac{a^2 - b^2}{b} + c \right] \quad (2) \frac{\sigma}{\varepsilon_0} \left[\frac{b^2 - c^2}{b} + a \right] \quad (3) \frac{\sigma}{\varepsilon_0} \left[\frac{b^2 - c^2}{c} + a \right] \quad (4) \frac{\sigma}{\varepsilon_0} \left[\frac{a^2 - b^2}{a} + c \right]$$

ES0205

EXERCISE-(JA)

1. Consider an electric field $\vec{E} = E_0 \hat{x}$, where E_0 is a constant. The flux through the shaded area (as shown in the figure) due to this field is [IIT-JEE 2011]



- (A) $2E_0a^2$ (B) $\sqrt{2}E_0a^2$ (C) E_0a^2 (D) $\frac{E_0a^2}{\sqrt{2}}$

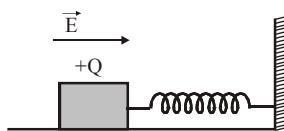
ES0206

2. A spherical metal shell A of radius R_A and a solid metal sphere B of radius $R_B (< R_A)$ are kept far apart and each is given charge $+Q$. Now they are connected by a thin metal wire. Then [IIT-JEE 2011]

- (A) $E_A^{inside} = 0$ (B) $Q_A > Q_B$ (C) $\frac{\sigma_A}{\sigma_B} = \frac{R_B}{R_A}$ (D) $E_A^{on\ surface} < E_B^{on\ surface}$

ES0207

3. A wooden block performs SHM on a frictionless surface with frequency, ν_0 . The block carries a charge $+Q$ on its surface. If now a uniform electric field $\vec{E}$ is switched-on as shown, then the SHM of the block will be :- [IIT-JEE 2011]



- (A) of the same frequency and with shifted mean position
 (B) of the same frequency and with the same mean position
 (C) of changed frequency and with shifted mean position
 (D) of changed frequency and with the same mean position

ES0208

4. Which of the following statement(s) is/are correct? [IIT-JEE 2011]
- (A) If the electric field due to a point charge varies as $r^{-2.5}$ instead of r^{-2} , then the Gauss law will still be valid.
- (B) The Gauss law can be used to calculate the field distribution around an electric dipole.
- (C) If the electric field between two point charges is zero somewhere, then the sign of the two charges is the same.
- (D) The work done by the external force in moving a unit positive charge from point A at potential V_A to point B at potential V_B is $(V_B - V_A)$

ES0209

5. Four point charges, each of $+q$, are rigidly fixed at the four corners of a square planar soap film of side ' α '. The surface tension of the soap film is γ . The system of charges and planar film are in equilibrium,

and $a = k \left[\frac{q^2}{\gamma} \right]^{1/N}$, where ' k ' is a constant. Then N is [IIT-JEE 2011]

ES0210

Paragraph for Question Nos. 6 and 7

A dense collection of equal number of electrons and positive ions is called neutral plasma. Certain solids containing fixed positive ions surrounded by free electrons can be treated as neutral plasma. Let ' N ' be the number density of free electrons, each of mass ' m '. When the electrons are subjected to an electric field, they are displaced relatively away from the heavy positive ions. If the electric field becomes zero, the electrons begins to oscillate about the positive ions with a natural angular frequency ' ω_p ', which is called the plasma frequency. To sustain the oscillations, a time varying electric field needs to be applied that has an angular frequency ω , where a part of the energy is absorbed and a part of it is reflected. As ω approaches ω_p , all the free electrons are set to resonance together and all the energy is reflected. This is the explanation of high reflectivity of metals.

[IIT-JEE 2011]

6. Taking the electronic charge as ' e ' and the permittivity as ' ϵ_0 ', use dimensional analysis to determine the correct expression for ω_p .

(A) $\sqrt{\frac{Ne}{m\epsilon_0}}$ (B) $\sqrt{\frac{m\epsilon_0}{Ne}}$ (C) $\sqrt{\frac{Ne^2}{m\epsilon_0}}$ (D) $\sqrt{\frac{m\epsilon_0}{Ne^2}}$

ES0211

7. Estimate the wavelength at which plasma reflection will occur for a metal having the density of electrons $N \approx 4 \times 10^{27} \text{ m}^{-3}$. Take $\epsilon_0 \approx 10^{-11}$ and $m \approx 10^{-30}$, where these quantities are in proper SI units

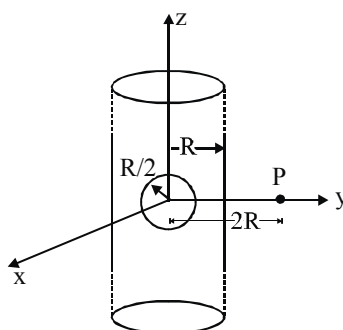
(A) 800 nm (B) 600 nm (C) 300 nm (D) 200 nm

ES0211

8. An infinitely long solid cylinder of radius R has a uniform volume charge density ρ . It has a spherical cavity of radius $R/2$ with its centre on the axis of the cylinder, as shown in the figure. The magnitude of the electric field at the point P, which is at a distance $2R$ from the axis of the cylinder, is given by the

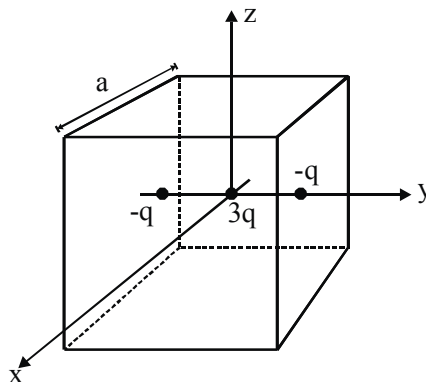
expression $\frac{23\rho R}{16k\epsilon_0}$. The value of k is

[IIT-JEE 2012]



ES0212

9. A cubical region of side a has its centre at the origin. It encloses three fixed point charges, $-q$ at $(0, -a/4, 0)$, $+3q$ at $(0, 0, 0)$ and $-q$ at $(0, +a/4, 0)$. Choose the correct option(s). [IIT-JEE 2012]



- (A) The net electric flux crossing the plane $x = +a/2$ is equal to the net electric flux crossing the plane $x = -a/2$
- (B) The net electric flux crossing the plane $y = +a/2$ is more than the net electric flux crossing the plane $y = -a/2$.
- (C) The net electric flux crossing the entire region is $\frac{q}{\epsilon_0}$
- (D) The net electric flux crossing the plane $z = +a/2$ is equal to the net electric flux crossing the plane $z = -a/2$.

ES0213

10. Two large vertical and parallel metal plates having a separation of 1 cm are connected to a DC voltage source of potential difference X . A proton is released at rest midway between the two plates. It is found to move at 45° to the vertical JUST after release. Then X is nearly

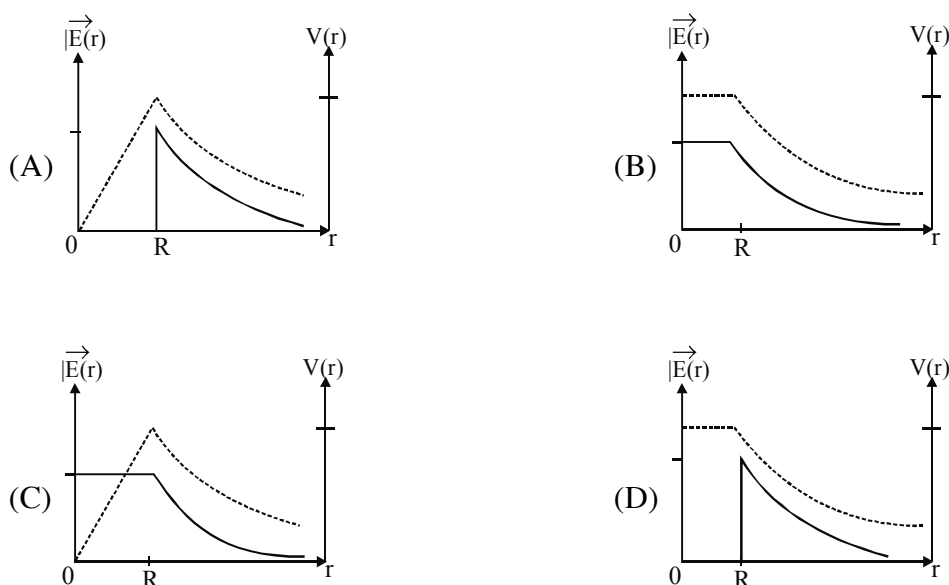
[IIT-JEE 2012]

- (A) 1×10^{-5} V (B) 1×10^{-7} V
(C) 1×10^{-9} V (D) 1×10^{-10} V

ES0214

11. Consider a thin spherical shell of radius R with its centre at the origin, carrying uniform positive surface charge density. The variation of the magnitude of the electric field $|\vec{E}(r)|$ and the electric potential $V(r)$ with the distance r from the centre, is best represented by which graph?

[IIT-JEE 2012]



ES0215

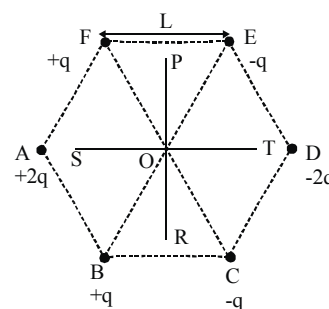
12. Six point charges are kept at the vertices of a regular hexagon of side L and centre O , as shown in figure. Given

that $K = \frac{1}{4\pi\epsilon_0} \frac{q}{L^2}$, which of the following statement(s) is(are)

correct?

[IIT-JEE 2012]

- (A) The electric field at O is $6K$ along OD .
(B) The potential at O is zero.
(C) The potential at all points on the line PR is same.
(D) The potential at all points on the line ST is same.



ES0216

13. Two non-conducting solid spheres of radii R and $2R$, having uniform volume charge densities ρ_1 and ρ_2 respectively, touch each other. The net electric field at a distance $2R$ from the centre of the smaller sphere, along the line joining the centres of the spheres, is zero. The ratio $\frac{\rho_1}{\rho_2}$ can be

[JEE-Advance-2013]

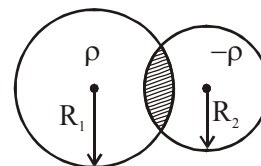
- (A) -4 (B) $-\frac{32}{25}$ (C) $\frac{32}{25}$ (D) 4

ES0217

14. Two non-conducting spheres of radii R_1 and R_2 and carrying uniform volume charge densities $+\rho$ and $-\rho$, respectively, are placed such that they partially overlap, as shown in the figure. At all points in the overlapping region :-

[JEE-Advance-2013]

- (A) the electrostatic field is zero
(B) the electrostatic potential is constant
(C) the electrostatic field is constant in magnitude
(D) the electrostatic field has same direction



ES0218

15. Let $E_1(r)$, $E_2(r)$ and $E_3(r)$ be the respective electric fields at a distance r from a point charge Q , an infinitely long wire with constant linear charge density λ , and an infinite plane with uniform surface charge density σ . If $E_1(r_0) = E_2(r_0) = E_3(r_0)$ at a given distance r_0 , then :-

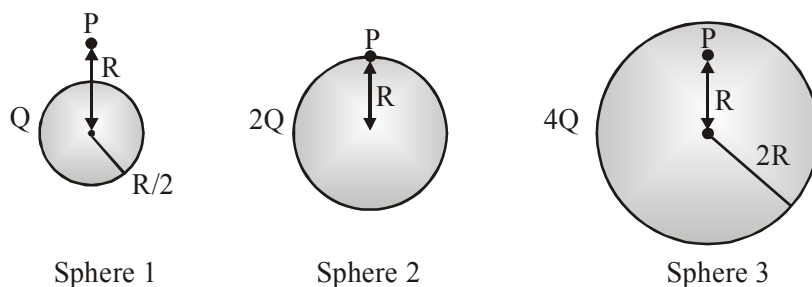
[JEE-Advance-2014]

- (A) $Q = 4\sigma\pi r_0^2$ (B) $r_0 = \frac{\lambda}{2\pi\sigma}$
(C) $E_1(r_0/2) = 2E_2(r_0/2)$ (D) $E_2(r_0/2) = 4E_3(r_0/2)$

ES0219

16. Charges Q , $2Q$ and $4Q$ are uniformly distributed in three dielectric solid spheres 1, 2 and 3 of radii $R/2$, R and $2R$ respectively, as shown in figure. If magnitudes of the electric fields at point P at a distance R from the centre of spheres 1, 2 and 3 are E_1 , E_2 and E_3 respectively, then

[JEE-Advance-2014]



- (A) $E_1 > E_2 > E_3$ (B) $E_3 > E_1 > E_2$ (C) $E_2 > E_1 > E_3$ (D) $E_3 > E_2 > E_1$

ES0220

17. Four charges Q_1, Q_2, Q_3 and Q_4 of same magnitude are fixed along the x axis at $x = -2a, -a, +a$ and $+2a$, respectively. A positive charge q is placed on the positive y axis at a distance $b > 0$. Four options of the signs of these charges are given in List I. The direction of the forces on the charge q is given in List II. Match List I with List II and select the correct answer using the code given below the lists.

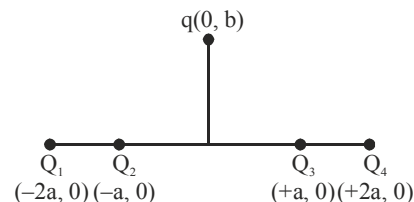
[JEE-Advance-2014]

List-I

- (P) Q_1, Q_2, Q_3, Q_4 all positive
 (Q) Q_1, Q_2 positive ; Q_3, Q_4 negative
 (R) Q_1, Q_4 positive ; Q_2, Q_3 negative
 (S) Q_1, Q_3 positive ; Q_2, Q_4 negative

List-II

- (1) $+x$
 (2) $-x$
 (3) $+y$
 (4) $-y$



Code :

(A) P-3, Q-1, R-4, S-2

(B) P-4, Q-2, R-3, S-1

(C) P-3, Q-1, R-2, S-4

(D) P-4, Q-2, R-1, S-3

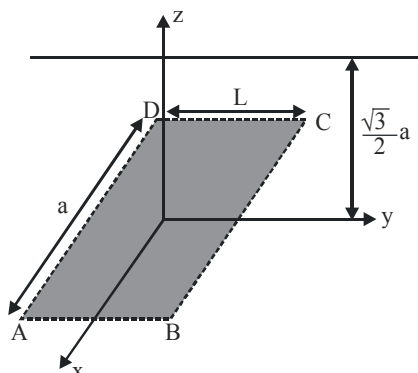
ES0221

18. An infinitely long uniform line charge distribution of charge per unit length λ lies parallel to the y-axis in the y-z plane at $z = \frac{\sqrt{3}}{2}a$ (see figure). If the magnitude of the flux of the electric field through the

rectangular surface ABCD lying in the x-y plane with its centre at the origin is $\frac{\lambda L}{n\epsilon_0}$ (ϵ_0 = permittivity

of free space) then the value of n is.

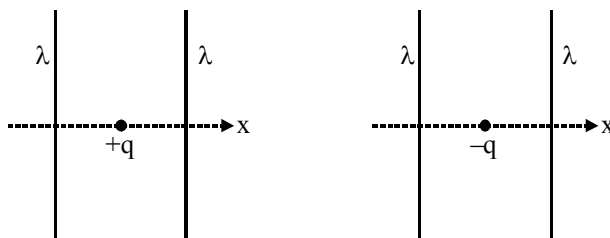
[JEE-Advance-2015]



ES0222

19. The figures below depict two situations in which two infinitely long static line charges of constant positive line charge density λ are kept parallel to each other. In their resulting electric field, point charges q and $-q$ are kept in equilibrium between them. The point charges are confined to move in the x direction only. If they are given a small displacement about their equilibrium positions, then the correct statement(s) is (are) :

[JEE-Advance-2015]

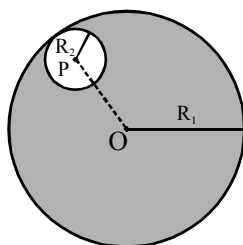


- (A) Both charges execute simple harmonic motion
 (B) Both charges will continue moving in the direction of their displacement
 (C) Charge $+q$ executes simple harmonic motion while charge $-q$ continues moving in the direction of its displacement.
 (D) Charge $-q$ executes simple harmonic motion while charge $+q$ continues moving in the direction of its displacement.

ES0223

20. Consider a uniform spherical distribution of radius R_1 centred at the origin O . In this distribution, a spherical cavity of radius R_2 , centred at P with distance $OP = a = R_1 - R_2$ (see figure) is made. If the electric field inside the cavity at position $\vec{r}$ is $\vec{E}(\vec{r})$, then the correct statement(s) is(are) :

[JEE-Advance-2015]



- (A) $\vec{E}$ is uniform, its magnitude is independent of R_2 but its direction depends on $\vec{r}$
 (B) $\vec{E}$ is uniform, its magnitude depends on R_2 and its direction depends on $\vec{r}$
 (C) $\vec{E}$ is uniform, its magnitude is independent of a but its direction depends on $\vec{a}$
 (D) $\vec{E}$ is uniform and both its magnitude and direction depend on $\vec{a}$

ES0224

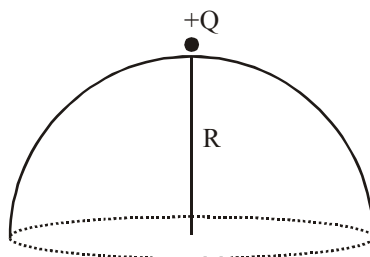
21. A length-scale (ℓ) depends on the permittivity (ϵ) of a dielectric material, Boltzmann constant k_B , the absolute temperature T , the number per unit volume (n) of certain charged particles, and the charge (q) carried by each of the particles, Which of the following expressions(s) for ℓ is(are) dimensionally correct?

[JEE-Advance-2016]

- (A) $\ell = \sqrt{\left(\frac{nq^2}{\epsilon k_B T}\right)}$ (B) $\ell = \sqrt{\left(\frac{\epsilon k_B T}{nq^2}\right)}$ (C) $\ell = \sqrt{\left(\frac{q^2}{\epsilon n^{2/3} k_B T}\right)}$ (D) $\ell = \sqrt{\left(\frac{q^2}{\epsilon n^{1/3} k_B T}\right)}$

ES0225

22. A point charge $+Q$ is placed just outside an imaginary hemispherical surface of radius R as shown in the figure. Which of the following statements is/are correct ? [JEE-Advance-2017]

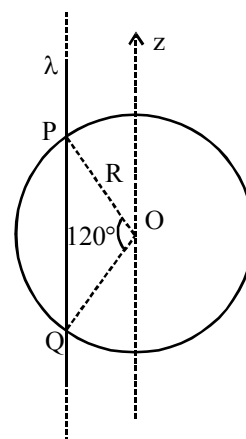


- (A) The circumference of the flat surface is an equipotential
- (B) The electric flux passing through the curved surface of the hemisphere is $-\frac{Q}{2\epsilon_0} \left(1 - \frac{1}{\sqrt{2}}\right)$
- (C) Total flux through the curved and the flat surfaces is $\frac{Q}{\epsilon_0}$
- (D) The component of the electric field normal to the flat surface is constant over the surface.

ES0226

23. An infinitely long thin non-conducting wire is parallel to the z -axis and carries a uniform line charge density λ . It pierces a thin non-conducting spherical shell of radius R in such a way that the arc PQ subtends an angle 120° at the centre O of the spherical shell, as shown in the figure. The permittivity of free space is ϵ_0 . Which of the following statements is (are) true ?

[JEE-Advance-2018]



- (A) The electric flux through the shell is $\sqrt{3} R \lambda / \epsilon_0$
- (B) The z -component of the electric field is zero at all the points on the surface of the shell
- (C) The electric flux through the shell is $\sqrt{2} R \lambda / \epsilon_0$
- (D) The electric field is normal to the surface of the shell at all points

ES0227

24. A particle, of mass 10^{-3} kg and charge 1.0 C, is initially at rest. At time $t = 0$, the particle comes under the influence of an electric field $\vec{E}(t) = E_0 \sin \omega t \hat{i}$ where $E_0 = 1.0 \text{ N C}^{-1}$ and $\omega = 10^3 \text{ rad s}^{-1}$. Consider the effect of only the electrical force on the particle. Then the maximum speed, in ms^{-1} , attained by the particle at subsequent times is..... [JEE-Advance-2018]

ES0228

25. The electric field E is measured at a point $P(0,0,d)$ generated due to various charge distributions and the dependence of E on d is found to be different for different charge distributions. List-I contains different relations between E and d . List-II describes different electric charge distributions, along with their locations. Match the functions in List-I with the related charge distributions in List-II.

[JEE-Advance-2018]

- | List-I | List-II |
|------------------------------|---|
| P. E is independent of d | 1. A point charge Q at the origin |
| Q. $E \propto \frac{1}{d}$ | 2. A small dipole with point charges Q at $(0,0,\ell)$ and $-Q$ at $(0,0,-\ell)$.
Take $2\ell \ll d$ |
| R. $E \propto \frac{1}{d^2}$ | 3. An infinite line charge coincident with the x -axis, with uniform linear charge density λ . |
| S. $E \propto \frac{1}{d^3}$ | 4. Two infinite wires carrying uniform linear charge density parallel to the x -axis. The one along $(y=0, z=\ell)$ has a charge density $+\lambda$ and the one along $(y=0, z=-\ell)$ has a charge density $-\lambda$. Take $2\ell \ll d$ |
| | 5. Infinite plane charge coincident with the xy -plane with uniform surface charge density |
- (A) $P \rightarrow 5$; $Q \rightarrow 3, 4$; $R \rightarrow 1$; $S \rightarrow 2$
 (B) $P \rightarrow 5$; $Q \rightarrow 3, ;$ $R \rightarrow 1,4$; $S \rightarrow 2$
 (C) $P \rightarrow 5$; $Q \rightarrow 3, ;$ $R \rightarrow 1,2$; $S \rightarrow 4$
 (D) $P \rightarrow 4$; $Q \rightarrow 2, 3$; $R \rightarrow 1$; $S \rightarrow 5$

ES0229

26. A thin spherical insulating shell of radius R carries a uniformly distributed charge such that the potential at its surface is V_0 . A hole with a small area $\alpha 4\pi R^2$ ($\alpha \ll 1$) is made on the shell without affecting the rest of the shell. Which one of the following statements is correct ?

[JEE-Advance-2019]

- (1) The ratio of the potential at the center of the shell to that of the point at $\frac{1}{2}R$ from center towards the hole will be $\frac{1-\alpha}{1-2\alpha}$
- (2) The magnitude of electric field at the center of the shell is reduced by $\frac{\alpha V_0}{2R}$
- (3) The magnitude of electric field at a point, located on a line passing through the hole and shell's center on a distance $2R$ from the center of the spherical shell will be reduced by $\frac{\alpha V_0}{2R}$
- (4) The potential at the center of the shell is reduced by $2\alpha V_0$

ES0230

27. A charged shell of radius R carries a total charge Q . Given Φ as the flux of electric field through a closed cylindrical surface of height h , radius r and with its center same as that of the shell. Here, center of the cylinder is a point on the axis of the cylinder which is equidistant from its top and bottom surfaces. Which of the following option(s) is/are correct ? [ϵ_0 is the permittivity of free space]

[JEE-Advance-2019]

- (1) If $h > 2R$ and $r > R$ then $\Phi = \frac{Q}{\epsilon_0}$ (2) If $h < \frac{8R}{5}$ and $r = \frac{3R}{5}$ then $\Phi = 0$
- (3) If $h > 2R$ and $r = \frac{4R}{5}$ then $\Phi = \frac{Q}{5\epsilon_0}$ (4) If $h > 2R$ and $r = \frac{3R}{5}$ then $\Phi = \frac{Q}{5\epsilon_0}$

ES0231

28. An electric dipole with dipole moment $\frac{p_0}{\sqrt{2}}(\hat{i} + \hat{j})$ is held fixed at the origin O in the presence of an uniform electric field of magnitude E_0 . If the potential is constant on a circle of radius R centered at the origin as shown in figure, then the correct statement(s) is/are:

(ϵ_0 is permittivity of free space, $R \gg$ dipole size)

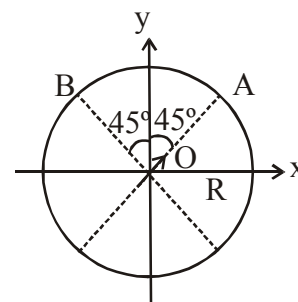
[JEE-Advance-2019]

- (1) $R = \left(\frac{p_0}{4\pi\epsilon_0 E_0} \right)^{1/3}$
- (2) The magnitude of total electric field on any two points

of the circle will be same

- (3) Total electric field at point A is $\vec{E}_A = \sqrt{2}E_0(\hat{i} + \hat{j})$

- (4) Total electric field at point B is $\vec{E}_B = 0$



ES0232

ELECTROSTATICS

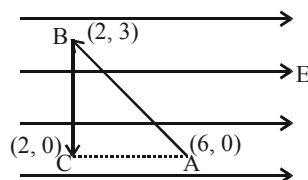
(CBSE Previous Year's Questions)

1. An electrostatic field line cannot be discontinuous. Why ? [1; CBSE-2005]
2. Define electric field intensity. Write its S.I unit. Write the magnitude and direction of electric field intensity due to an electric dipole of length $2a$ at the mid- point of the line joining the two charges. [2; CBSE-2005]
3. State Gauss' theorem. Apply this theorem to obtain the expression for the electric field intensity at a point due to an infinitely long, thin, uniformly charged straight wire. [3; CBSE-2005]
4. Define the term electric dipole moment. Is it a scalar or a vector quantity? [1; CBSE-2006]
5. A point charge ' q ' is placed at O as shown in the figure. [2; CBSE-2006]
Is $V_p - V_Q$ positive or negative when (i) $q > 0$, (ii) $q < 0$? Justify your answer.



6. Using Gauss's theorem, show mathematically that for any point outside the shell, the field due to a uniformly charged thin spherical shell is the same as if the entire charge of the shell is concentrated at the centre. Why do you expect the electric field inside the shell to be zero according to this theorem? [3; CBSE-2006]
7. Two point charges $4\mu\text{C}$ and $-2\mu\text{C}$ are separated by a distance of 1 m in air. Calculate at what point on the line joining the two charges is the electric potential zero. [1; CBSE-2007]
8. State Gauss's theorem in electrostatics. Apply this theorem to derive an expression for electric field intensity at a point near an infinitely long straight charged wire. [2; CBSE-2007]
9. A $500\mu\text{C}$ Charge is at the centre of a square of side 10cm. Find the work done in moving a charge of $10\mu\text{C}$ between two diagonally opposite points on the square. [1; CBSE-2008]
10. Derive the expression for the electric potential at any point along the axial line of an electric dipole. [1; CBSE-2008]
11. (i) Can two equi- potential surfaces intersect each other? Give reasons.
(ii) Two charges $-q$ and $+q$ are located at points A (0,0, $-a$) and B (0,0, $+a$) respectively. How much work is done in moving a test charge from point P (7,0,0) to Q (-3,0,0)? [2; CBSE-2009]
12. State Gauss's law in electrostatics. Using this law derive an expression for the electric field due to a uniformly charged infinite plane sheet. [3; CBSE-2009]
13. Name the physical quantity whose S.I. unit is JC^{-1} . Is it a scalar or a vector quantity ? [1; CBSE-2010]
14. Define electric dipole moment. Write its S.I. unit. [1; CBSE-2011]
15. A hollow metal sphere of radius 5 cm is charged such that the potential on its surface is 10 V, What is the potential at the centre of the sphere ? [1; CBSE-2011]

16. A thin straight infinitely long conducting wire having charge density λ is enclosed by a cylindrical surface of radius r and length l , its axis coinciding with the length of the wire. Find the expression for the electric flux through the surface of the cylinder. [2; CBSE-2011]
17. Plot a graph showing the variation of coulomb force (F) versus $\left(\frac{1}{r^2}\right)$, where r is the distance between the two charges of each pair of charges: $(1\mu\text{C}, 2\mu\text{C})$ and $(2\mu\text{C}, -3\mu\text{C})$, interpret the graphs obtained. [2; CBSE-2011]
18. Two wires of equal length, one of copper and the other of manganin have the same resistance. Which wire is thicker? [1; CBSE-2012]
19. A charge ' q ' is moved without acceleration from A to C along the path from A to B and then from B to C in electric field E as shown in the figure, (i) Calculate the potential difference between A and C. (ii) At which point (of the two) is the electric potential more and why?



20. An electric dipole is held in a uniform electric field.
- (i) Show that the net force acting on it is zero
- (ii) The dipole is aligned parallel to the field. Find the work done in rotating it through the angle of 180° . [2; CBSE-2012]
21. Two charges of magnitudes $-2Q$ and $+Q$ are located at points $(a, 0)$ and $(4a, 0)$ respectively. What is the electric flux due to these charges through a sphere of radius ' $3a$ ' with its centre at the origin? [CBSE-2013]
22. (a) Define electric dipole moment. Is it a scalar or a vector? Derive the expression for the electric field of a dipole at a point on the equatorial plane of the dipole.
- (b) Draw the equipotential surfaces due to an electric dipole. Locate the points where the potential due to the dipole is zero.

OR

Using Gauss' law deduce the expression for the electric field due to a uniformly charged spherical conducting shell of radius R at a point (i) outside and (ii) inside the shell.

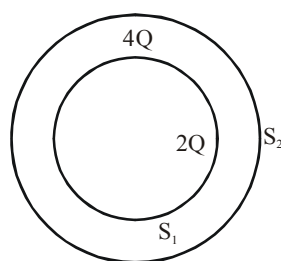
Plot a graph showing variation of electric field as a function of $r > R$ and $r < R$. (r being the distance from the centre of the shell) [CBSE-2013]

23. Why do the electrostatic field lines not form closed loops? [CBSE-2014]

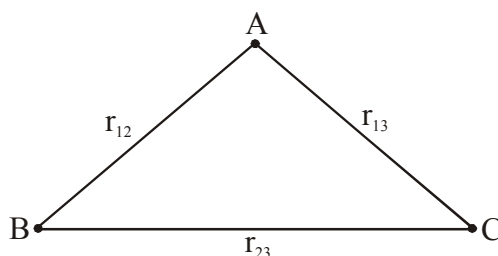
24. Draw a labelled diagram of Van de Graaff generator. State its working principle to show how by introducing a small charged sphere, into a larger sphere, a large amount of charge can be transferred to the outer sphere state the use of this machine and also point out its limitations.

OR

- (a) Deduce the expression for the torque acting on a dipole of dipole moment $\vec{p}$ in the presence of a uniform electric field $\vec{E}$.
- (b) Consider two hollow concentric spheres, S_1 and S_2 , enclosing charges $2Q$ and $4Q$ respectively as shown in the figure, (i) Find out the ratio of the electric flux through them, (ii) How will the electric flux through the sphere S_1 change if a medium of dielectric constant ' ϵ_r ' is introduced in the space inside S_1 in place of air ? Deduce the necessary expression. [CBSE-2014]



25. Determine the distance of closest approach when an alpha particle of kinetic energy 4.5 MeV strikes a nucleus of $Z = 80$, stops and reverses its direction. [2; CBSE-2015]
26. (a) State Gauss's law in electrostatics. Show, with the help of a suitable example along with the figure, that the outward flux due to a point charge ' q ', in vacuum within a closed surface, is independent of its size or shape and is given by q/ϵ_0 . [5; CBSE-2015]

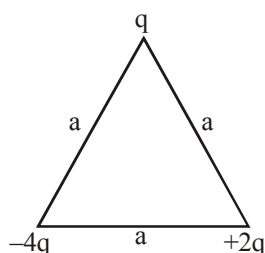


- (b) Two parallel uniformly charged infinite plane sheets, '1' and '2', have charge densities $+\sigma$ and $-\sigma$ respectively. Give the magnitude and direction of the net electric field at a point
- in between the two sheets and
 - outside near the sheet '1'.
27. What is the amount of work done in moving a point charge Q around a circular arc of radius ' r ' at the centre of which another point charge ' q ' is located? [1; CBSE-2016]
28. Find the electric field intensity due to a uniformly charged spherical shell at a point (i) outside the shell and (ii) inside the shell. Plot the graph of electric field with distance from the centre of the shell. [3; CBSE-2016]

29. (a) Explain why, for any charge configuration, the equipotential surface through a point is normal to the electric field at the point. [5; CBSE-2016]

Draw a sketch of equipotential surfaces due to a single charge ($-q$), depicting the electric field lines due to the charge.

- (b) Obtain an expression for the work done to dissociate the system of three charges placed at the vertices of an equilateral triangle of side 'a' as shown below.



30. (a) Derive an expression for the electric field E due to a dipole of length ' $2a$ ' at a point distant r from the centre of the dipole on the axial line. [CBSE-2017]
- (b) Draw a graph of E versus r for $r \gg a$.
- (c) If this dipole were kept in a uniform external electric field E_0 , diagrammatically represent the position of the dipole in stable and unstable equilibrium and write the expressions for the torque acting on the dipole in both the cases.

OR

- (a) Use Gauss's theorem to find the electric field due to a uniformly charged infinitely large plane thin sheet with surface charge density σ .
- (b) An infinitely large thin plane sheet has a uniform surface charge density $+\sigma$. Obtain the expression for the amount of work done in bringing a point charge q from infinity to a point, distant r , in front of the charged plane sheet.

ANSWER KEY

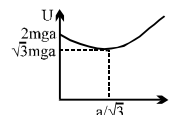
EXERCISE (S-1)

1. Ans. 80 2. Ans. 003 3. Ans. $(1 + 2\sqrt{2})/4$ Q 4. Ans. 9
 5. Ans. 3 6. Ans. 4 7. Ans. 400 8. Ans. 9.30
 9. Ans. 3 10. Ans. 8 11. Ans. 2 12. Ans. 1
 13. Ans. 2 14. Ans. 4 15. Ans. 576 16. Ans. 3
 17. Ans. 4 18. Ans. 4 19. Ans. 6 20. Ans. 0
 21. Ans. $-\frac{kq^2}{a}(3 - \sqrt{2})$ 22. Ans. 2 23. Ans. 90 24. Ans. 2
 25. Ans. 2 26. Ans. $2 \tan^{-1}\left(\frac{\sigma q}{2\epsilon_0 mg}\right)$ 27. Ans. 20
 28. Ans. $\left[\frac{2KQq}{mR}\left(\frac{r-R}{r} + \frac{3}{8}\right)\right]^{1/2}$ 29. Ans. 2 30. Ans. 4
 31. Ans. $\frac{kP}{\sqrt{2}y^3}(-\hat{i} - 2\hat{j})$ 32. Ans. 028
 33. Ans. (a) $K.E = \frac{P}{4\pi\epsilon_0} \frac{Q}{d^2}$, (b) $\frac{QP}{2\pi\epsilon_0 d^3}$ along positive x-axis
 34. Ans. (a) (i) $V = \frac{k(q_a + q_b)}{r}$, $E = \frac{k(q_a + q_b)}{r^2}$; (ii) $\frac{k(q_a + q_b)}{R} + \frac{kq_b}{r} - \frac{kq_b}{b}$; $\frac{kq_b}{r^2}$
 (b) $\sigma_R = \left(\frac{q_a + q_b}{4\pi R^2}\right)$, $\sigma_a = -\frac{q_a}{4\pi a^2}$; $\sigma_b = -\frac{q_b}{4\pi b^2}$; (c) $f = 0$
 35. Ans. $-Q/3$ 36. Ans. 3 37. Ans. $V' = \left(\frac{a}{3t}\right)^{1/3} V$
 38. Ans. $-\epsilon E, \epsilon E$ and so on

EXERCISE (S-2)

1. Ans. $\frac{\sigma}{\pi \epsilon_0}$ 2. Ans. a 3. Ans. $\pi K d t \rho R^2$ 4. Ans. 3
 5. Ans. 3 6. Ans. 8
 7. Ans. $v_0 = 3 \text{ m/s}$; K.E. at the origin $= (27 - 10\sqrt{6}) \times 10^{-4} \text{ J}$ approx. $2.5 \times 10^{-4} \text{ J}$
 8. Ans. 6 9. Ans. $\lambda R E_0 \hat{i}$ 10. Ans. $\sqrt{4\pi\epsilon_0 Ka}$
 11. Ans. $W_{\text{first step}} = \left(\frac{8}{3} - \frac{4}{\sqrt{5}}\right) \frac{Kq^2}{r}$, $W_{\text{second step}} = 0$, $W_{\text{total}} = 0$

12. Ans. (a) $H = \frac{4a}{3}$, (b) $U = mg \left[2\sqrt{h^2 + a^2} - h \right]$ equilibrium at $h = \frac{a}{\sqrt{3}}$,



13. Ans. (i) $-Q$, $+Q$ (ii) $E_A = 0$, can't be found (iii) can't be found (iv) can't be found (v) 0 (vi) No, induced charge on outer surface]

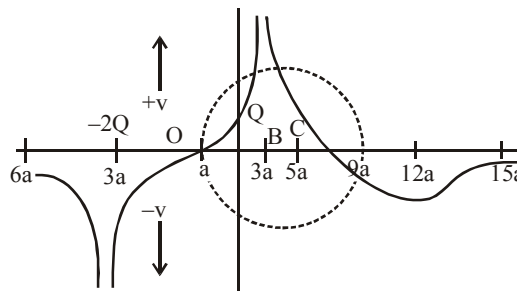
14. Ans. 180

15. Ans. $F = \frac{Q^2}{32\pi\epsilon_0 R^4} (R^2 - h^2)$

16. Ans. (a) radius = $4a$ & center $(5a, 0)$

(b) $V = \frac{1}{4\pi\epsilon_0} \left[\frac{-2Q}{|(x+3a)|} + \frac{Q}{|(x-3a)|} \right]$

(c) speed $v = \sqrt{\frac{1}{4\pi\epsilon_0} \times \frac{qQ}{2ma}}$



17. Ans. 7

EXERCISE (O-1)

- | | | | | | |
|--|-------------------------|--|------------------|------------------|--------------|
| 1. Ans. (C) | 2. Ans. (B) | 3. Ans. (C) | 4. Ans. (C) | 5. Ans. (B) | 6. Ans. (A) |
| 7. Ans. (D) | 8. Ans. (B) | 9. Ans. (A) | 10. Ans. (A) | 11. Ans. (A) | 12. Ans. (C) |
| 13. Ans. (A) | 14. Ans. (C) | 15. Ans. (A) | 16. Ans. (D) | 17. Ans. (B) | 18. Ans. (C) |
| 19. Ans. (D) | 20. Ans. (B) | 21. Ans. (D) | 22. Ans. (A) | 23. Ans. (D) | 24. Ans. (A) |
| 25. Ans. (D) | 26. Ans. (D) | 27. Ans. (D) | 28. Ans. (B) | 29. Ans. (C) | 30. Ans. (C) |
| 31. Ans. (D) | 32. Ans. (C) | 33. Ans. (D) | 34. Ans. (D) | 35. Ans. (A) | 36. Ans. (B) |
| 37. Ans. (A) | 38. Ans. (B) | 39. Ans. (D) | 40. Ans. (B) | 41. Ans. (A) | 42. Ans. (C) |
| 43. Ans. (C) | 44. Ans. (C) | 45. Ans. (C) | 46. Ans. (A) | 47. Ans. (C) | 48. Ans. (B) |
| 49. Ans. (B) | 50. Ans. (A) | 51. Ans. (B) | 52. Ans. (C) | 53. Ans. (A) | 54. Ans. (A) |
| 55. Ans. (D) | 56. Ans. (B) | 57. Ans. (A) | 58. Ans. (A) | 59. Ans. (C) | 60. Ans. (C) |
| 61. Ans. (A) | 62. Ans. a. (B); b. (D) | 63. Ans. (C) | 64. Ans. (B) | 65. Ans. (D) | |
| 66. Ans. (A) | 67. Ans. (D) | 68. Ans. (D) | 69. Ans. (A) | 70. Ans. (C) | 71. Ans. (D) |
| 72. Ans. (B) | 73. Ans. (B) | 74. Ans. (A) | 75. Ans. (A) | 76. Ans. (C) | 77. Ans. (A) |
| 78. Ans. (C) | 79. Ans. (C) | 80. Ans. (A,B) | 81. Ans. (A,B,C) | 82. Ans. (A,B,D) | |
| 83. Ans. (A,B,C) | 84. Ans. (A,B,C) | 85. Ans. (AC) | 86. Ans. (A,C,D) | | |
| 87. Ans. (A,B,C) | 88. Ans. (A,B,C) | 89. Ans. (B,D) | 90. Ans. (A,C) | | |
| 91. Ans. (A, D) | 92. Ans. (B,C,D) | 93. Ans. (A, B, D) | 94. Ans. (A, B) | | |
| 95. Ans. (A,B,C) | 96. Ans. (B) | 97. Ans. (D) | 98. Ans. (C) | 99. Ans. (A) | |
| 100. Ans. (C) | 101. Ans. (A) | 102. Ans. (A) | 103. Ans. (A) | 104. Ans. (B) | |
| 105. Ans. (C) | 106. Ans. (B) | 107. Ans. (A)→(Q); (B)→(S); (C)→(R); (D)→(P) | | | |
| 108. Ans. (A) → (Q, R) ; (B) → (Q,T) ; (C) → (S,T) (D) → (P) | | | | | |
| 109. Ans. (A) – S; (B) – P; (C) – R; (D) – Q | | | | | |
| 110. Ans. A → Q; B → P, S; C → P, Q, R; D → Q, R | | | | | |
| 111. Ans. (A)-Q,S, (B)-P,R (C)-P,R (D)-Q,R | | | | | |

EXERCISE (O-2)

- | | | | | | |
|-----------------|-------------------------------------|----------------|----------------|----------------|------------------|
| 1. Ans. (A) | 2. Ans. (B) | 3. Ans. (B) | 4. Ans. (C) | 5. Ans. (C) | 6. Ans. (A) |
| 7. Ans. (A) | 8. Ans. (B) | 9. Ans. (C) | 10. Ans. (A) | 11. Ans. (C) | 12. Ans. (B) |
| 13. Ans. (A) | 14. Ans. (B) | 15. Ans. (D) | 16. Ans. (B) | 17. Ans. (B) | 18. Ans. (C) |
| 19. Ans. (A) | 20. Ans. (B) | 21. Ans. (A) | 22. Ans. (C) | 23. Ans. (A) | 24. Ans. (C) |
| 25. Ans. (A, C) | | 26. Ans. (A,C) | 27. Ans. (A,D) | 28. Ans. (A,B) | 29. Ans. (A,B,C) |
| 30. Ans. (B) | 31. Ans. (C) | 32. Ans. (B) | 33. Ans. (A) | 34. Ans. (D) | |
| 35. Ans. (A) | P,R,S (B) P,Q,R,S,T (C) Q,T (D) R,S | | | | |

EXERCISE-JM

- | | | | | | |
|---------------------|---------------------|--------------------|--------------------------|---------------------|---------------------|
| 1. Ans. (3) | 2. Ans. (2) | 3. Ans. (1) | 4. Ans. (4) | 5. Ans. (4) | 6. Ans. (2) |
| 7. Ans. (1) | 8. Ans. (4) | 9. Ans. (1) | 10. Ans. (1 or 2) | 11. Ans. (3) | 12. Ans. (2) |
| 13. Ans. (1) | 14. Ans. (1) | | | | |

EXERCISE-(JA)

- | | | | | |
|-----------------------|---------------------------|-------------------------|--------------------------------|---------------------|
| 1. Ans. (C) | 2. Ans. (A,B,C,D) | 3. Ans. (A) | 4. Ans. (C,D) | 5. Ans. 3 |
| 6. Ans. (C) | 7. Ans. (B) | 8. Ans. 6 | 9. Ans. (A,C,D) | 10. Ans. (C) |
| 11. Ans. (D) | 12. Ans. (A, B, C) | 13. Ans. (B, D) | 14. Ans. (C, D) | 15. Ans. (C) |
| 16. Ans. (C) | 17. Ans. (A) | 18. Ans. 6 | 19. Ans. (C) | 20. Ans. (D) |
| 21. Ans. (B,D) | 22. Ans. (A) (B) | 23. Ans. (A,B) | 24. Ans. 2 [1.99, 2.01] | |
| 25. Ans. (B) | 26. Ans. (1) | 27. Ans. (1,2,4) | 28. Ans. (1,4) | |

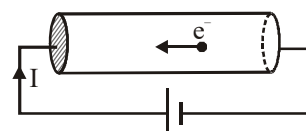
CURRENT ELECTRICITY

KEY CONCEPT

In previous chapters we deal largely with electrostatics that is, with charges at rest. With this chapter we begin to focus on electric currents, that is, charges in motion.

ELECTRIC CURRENT

Electric charges in motion constitute an electric current. Any medium having practically free electric charges, free to migrate is a conductor of electricity. The electric charge flows from higher potential energy state to lower potential energy state.



Positive charge flows from higher to lower potential and negative charge flows from lower to higher. Metals such as gold, silver, copper, aluminium etc. are good conductors.

When charge flows in a conductor from one place to the other, then the rate of flow of charge is called electric current (I). When there is a transfer of charge from one point to other point in a conductor, we say that there is an electric current through the area. If the moving charges are positive, the current is in the direction of motion of charge. If they are negative the current is opposite to the direction of motion. If a charge ΔQ crosses an area in time Δt then the average electric current through the area, during this time as

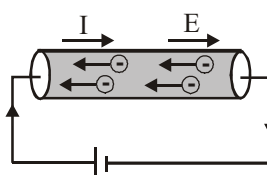
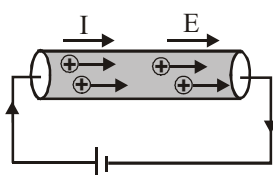
• Average current $I_{av} = \frac{\Delta Q}{\Delta t}$

• Instantaneous current $I = \lim_{\Delta t \rightarrow 0} \frac{\Delta Q}{\Delta t} = \frac{dQ}{dt}$

- Current is a fundamental quantity with dimension $[M^0 L^0 T^0 A^1]$
- Current is a scalar quantity with its SI unit **ampere**.

Ampere : The current through a conductor is said to be one ampere if one coulomb of charge is flowing per second through a cross-section of wire.

- The conventional direction of current is the direction of flow of positive charge or applied field. It is opposite to direction of flow of negatively charged electrons.



- The conductor remains uncharged when current flows through it because the charge entering at one end per second is equal to charge leaving the other end per second.
- For a given conductor current does not change with change in its cross-section because current is simply rate of flow of charge.
- If there are n particles per unit volume each having a charge q and moving with velocity v then

current through cross-sectional area A is $I = \frac{\Delta q}{\Delta t} = nqvA$

- If a charge q is moving in a circle of radius r with speed v then its time period is $T = 2\pi r/v$. The equivalent current $I = \frac{q}{T} = \frac{qv}{2\pi r}$.

Behavior of conductor in absence of applied potential difference :

In absence of applied potential difference electrons have random motion. The average displacement and average velocity is zero. There is no flow of current due to thermal motion of free electrons in a conductor.

The free electrons present in a conductor gain energy from temperature of surrounding and move randomly in the conductor.

The speed gained by virtue of temperature is called as thermal speed of an electron $\frac{1}{2}mv_{\text{rms}}^2 = \frac{3}{2}kt$

So thermal speed $v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$ where m is mass of electron

At room temperature $T = 300 \text{ K}$, $v_{\text{rms}} = 10^5 \text{ m/s}$

- **Mean free path λ :** ($\lambda \sim 10 \text{ \AA}$) $= \lambda = \frac{\text{total distance travelled}}{\text{number of collisions}}$
- **Relaxation time :** The time taken by an electron between two successive collisions is called as relaxation time τ : ($\tau \sim 10^{-14} \text{ s}$), Relaxation time : $\tau = \frac{\text{total time taken}}{\text{number of collisions}}$

Behavior of conductor in presence of applied potential difference :

When two ends of a conductors are joined to a battery then one end is at higher potential and another at lower potential. This produces an electric field inside the conductor from point of higher to lower potential

$E = \frac{V}{L}$ where V = emf of the battery, L = length of the conductor.

The field exerts an electric force on free electrons causing acceleration of each electron.

Acceleration of electron $\vec{a} = \frac{\vec{F}}{m} = \frac{-e\vec{E}}{m}$

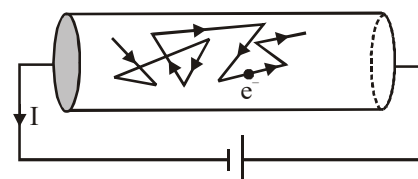
DRIFT VELOCITY

Drift velocity is defined as the velocity with which the free electrons get drifted towards the positive terminal under the effect of the applied external electric field. In addition to its thermal velocity, due to acceleration given by applied electric field, the electron acquires a velocity component in a direction opposite to the direction of the electric field. The gain in velocity due to the applied field is very small and is lost in the next collision.

At any given time, an electron has a velocity $\vec{v}_1 = \vec{u}_1 + \vec{a}\tau_1$, where $\vec{u}_1$ = the thermal velocity and $\vec{a}\tau_1$ = the velocity acquired by the electron under the influence of the applied electric field.

τ_1 = the time that has elapsed since the last collision. Similarly, the velocities of the other electrons are $\vec{v}_2 = \vec{u}_2 + \vec{a}\tau_2$, $\vec{v}_3 = \vec{u}_3 + \vec{a}\tau_3$, ..., $\vec{v}_N = \vec{u}_N + \vec{a}\tau_N$.

The average velocity of all the free electrons in the conductor is equal to the drift velocity $\vec{v}_d$ of the free electrons



Under the action of electric field :
Random motion of an electron
with superimposed drift

$$\vec{v}_d = \frac{\vec{v}_1 + \vec{v}_2 + \vec{v}_3 + \dots + \vec{v}_N}{N} = \frac{(u_1 + \vec{a}\tau_1) + (\vec{u}_2 + \vec{a}\tau_2) + \dots + (\vec{u}_N + \vec{a}\tau_N)}{N} = \frac{(\vec{u}_1 + \vec{u}_2 + \dots + \vec{u}_N)}{N} + \vec{a} \left(\frac{\tau_1 + \tau_2 + \dots + \tau_N}{N} \right)$$

$$\therefore \frac{\vec{u}_1 + \vec{u}_2 + \dots + \vec{u}_N}{N} = 0 \quad \therefore \vec{v}_d = \vec{a} \left(\frac{\tau_1 + \tau_2 + \dots + \tau_N}{N} \right) \Rightarrow \vec{v}_d = \vec{a}\tau = -\frac{e\vec{E}}{m}\tau$$

Note : Order of drift velocity is 10^{-4} m/s.

Relation between current and drift velocity :

Let n = number density of free electrons and A = area of cross-section of conductor.

Number of free electrons in conductor of length $L = nAL$, Total charge on these free electrons

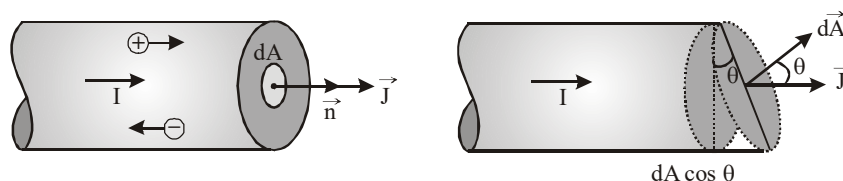
$$\Delta q = neAL$$

Time taken by drifting electrons to cross conductor $\Delta t = \frac{L}{v_d}$ $\therefore$ current $I = \frac{\Delta q}{\Delta t} = neAL \left(\frac{v_d}{L} \right) = neAv_d$

CURRENT DENSITY (J)

Current is a macroscopic quantity and deals with the overall rate of flow of charge through a section. To specify the current with direction in the microscopic level at a point, the term current density is introduced. Current density at any point inside a conductor is defined as a vector having magnitude equal to current per unit area surrounding that point. Remember area is normal to the direction of charge flow (or current passes) through that point.

- Current density at point P is given by $\vec{J} = \frac{dI}{dA} \vec{n}$



- If the cross-sectional area is not normal to the current, but makes an angle θ with the direction of current then $J = \frac{dI}{dA \cos \theta} \Rightarrow dI = J dA \cos \theta = \vec{J} \cdot d\vec{A} \Rightarrow I = \int \vec{J} \cdot d\vec{A}$
- Current density $\vec{J}$ is a vector quantity. It's direction is same as that of $\vec{E}$. It's S.I. unit is ampere/m² and dimension $[L^{-2}A]$.

Ex. The current density at a point is $\vec{J} = (2 \times 10^4 \hat{j}) \text{ Jm}^{-2}$.

Find the rate of charge flow through a cross sectional area $\vec{S} = (2\hat{i} + 3\hat{j}) \text{ cm}^2$

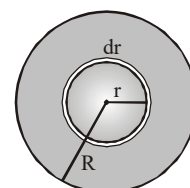
Sol. The rate of flow of charge = current = $I = \int \vec{J} \cdot d\vec{S} \Rightarrow I = \vec{J} \cdot \vec{S} = (2 \times 10^4) [\hat{j} \cdot (2\hat{i} + 3\hat{j})] \times 10^{-4} \text{ A} = 6 \text{ A}$

Ex. A potential difference applied to the ends of a wire made up of an alloy drives a current through it. The current density varies as $J = 3 + 2r$, where r is the distance of the point from the axis. If R be the radius of the wire, then the total current through any cross section of the wire.

Sol. Consider a circular strip of radius r and thickness dr

$$dI = \vec{J} \cdot d\vec{S} = (3 + 2r)(2\pi r dr) \cos 0^\circ = 2\pi(3r + 2r^2) dr$$

$$I = \int_0^R 2\pi(3r + 2r^2) dr = 2\pi \left(\frac{3r^2}{2} + \frac{2}{3}r^3 \right)_0^R = 2\pi \left(\frac{3R^2}{2} + \frac{2R^3}{3} \right) \text{ units}$$



RELATION BETWEEN CURRENT DENSITY, CONDUCTIVITY AND ELECTRIC FIELD

Let the number of free electrons per unit volume in a conductor = n

Total number of electrons in dx distance = $n(Adx)$

Total charge $dQ = n(Adx)e$

$$\text{Current } I = \frac{dQ}{dt} = nAe \frac{dx}{dt} = neAv_d, \text{ Current density } J = \frac{I}{A} = nev_d$$

$$= ne \left(\frac{eE}{m} \right) \tau \because v_d = \left(\frac{eE}{m} \right) \tau \Rightarrow J = \left(\frac{ne^2 \tau}{m} \right) E \Rightarrow J = \sigma E, \text{ where conductivity } \sigma = \frac{ne^2 \tau}{m}$$

σ depends only on the material of the conductor and its temperature.

In vector form $\vec{J} = \sigma \vec{E}$ Ohm's law (at microscopic level)

RELATION BETWEEN POTENTIAL DIFFERENCE AND CURRENT (Ohm's Law)

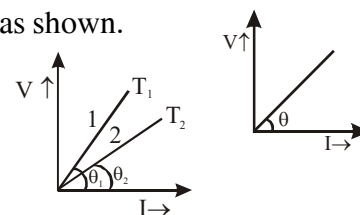
If the physical conditions of the conductor (length, temperature, mechanical strain etc.) remains same, then the current flowing through the conductor is directly proportional to the potential difference across its two ends i.e. $I \propto V \Rightarrow V = IR$ where R is a proportionality constant, known as electric resistance. Ohm's law (at macroscopic level)

- Ohm's law is not a universal law. The substances, which obey ohm's law are known as ohmic.
- Graph between V and I for a metallic conductor is a straight line as shown.

$$\text{Slope of the line} = \tan \theta = \frac{V}{I} = R$$

At different temperatures V - I curves are different.

Here $\tan \theta_1 > \tan \theta_2$ So $R_1 > R_2$ i.e. $T_1 > T_2$



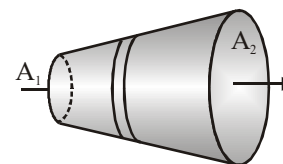
- 1 ampere of current means the flow of 6.25×10^{18} electrons per second through any cross section of conductor.
- Current is a scalar quantity but current density is a vector quantity.
- Order of free electron density in conductors = 10^{28} electrons/ m^3

Terms	Thermal speed v_T	Mean free path λ	Relaxation time τ	Drift speed v_d
Order	10^5 m/s	$10 \text{ \AA}$	10^{-14} s	10^{-4} m/s

- If a steady current flows in a metallic conductor of non uniform cross section.

Current density and drift velocity depends on area

$$I_1 = I_2, A_1 < A_2 \Rightarrow J_1 > J_2, E_1 > E_2, v_{d1} > v_{d2}$$



- If the temperature of the conductor increases, the amplitude of the vibrations of the positive ions in the conductor also increase. Due to this, the free electrons collide more frequently with the vibrating ions and as a result, the average relaxation time decreases.

Ex. What will be the number of electron passing through a heater wire in one minute, if it carries a current of 8 A.

Sol. $I = \frac{Ne}{t} \Rightarrow N = \frac{It}{e} = \frac{8 \times 60}{1.6 \times 10^{-19}} = 3 \times 10^{21}$ electrons

Ex. A current of 1.34 A exists in a copper wire of cross-section 1.0 mm^2 . Assuming each copper atom contributes one free electron. Calculate the drift speed of the free electrons in the wire. The density of copper is 8990 kg/m^3 and atomic mass = 63.50.

Sol. Mass of 1 m^3 volume of the copper is = $8990 \text{ kg} = 8990 \times 10^3 \text{ g}$

$$\text{Number of moles in } 1 \text{ m}^3 = \frac{8990 \times 10^3}{63.5} = 1.4 \times 10^5$$

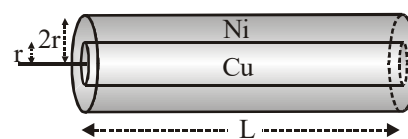
Since each mole contains 6×10^{23} atoms therefore number of atoms in 1 m^3

$$n = (1.4 \times 10^5) \times (6 \times 10^{23}) = 8.4 \times 10^{28}$$

$$\therefore I = neAv_d \therefore v_d = \frac{I}{neA} = \frac{1.34}{8.4 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-6}} = 10^{-4} \text{ m/s} = 0.1 \text{ mm/s} (\because 1 \text{ mm}^2 = 10^{-6} \text{ m}^2)$$

Ex. A copper wire of length ' ℓ ' and radius ' r ' is nickel plated till its final radius is $2r$. If the resistivity of the copper and nickel are ρ_{Cu} and ρ_{Ni} , then find the equivalent resistance of wire?

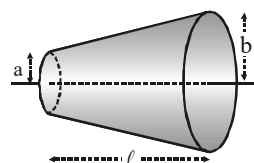
Sol. $R = \rho \frac{\ell}{A}$; Resistance of copper wire $R_{\text{Cu}} = \rho_{\text{Cu}} \frac{\ell}{\pi r^2}$
($\because A = \pi r^2$)



$$\therefore A_{\text{Ni}} = \pi(2r)^2 - \pi r^2 = 3\pi r^2 \Rightarrow \text{Resistance of Nickel wire } R_{\text{Ni}} = \rho_{\text{Ni}} \frac{\ell}{3\pi r^2}$$

$$\text{Both wire are connected in parallel. So equivalent resistance } R = \frac{R_{\text{Cu}} R_{\text{Ni}}}{R_{\text{Cu}} + R_{\text{Ni}}} = \left(\frac{\rho_{\text{Cu}} \rho_{\text{Ni}}}{3\rho_{\text{Cu}} + \rho_{\text{Ni}}} \right) \frac{\ell}{\pi r^2}$$

Ex. Figure shows a conductor of length ℓ carrying current I and having a circular cross-section. The radius of cross section varies linearly from a to b . Assuming that $(b - a) \ll \ell$. Calculate current density at distance x from left end.



Sol. Since radius at left end is a and that of right end is b ,
Therefore increase in radius over length ℓ is $(b - a)$.

$$\text{Hence rate of increase of radius per unit length} = \left(\frac{b - a}{\ell} \right) \text{ Increase in radius over length } x = \left(\frac{b - a}{\ell} \right) x$$

$$\text{Since radius at left end is } a \text{ so radius at distance } x, r = a + \left(\frac{b - a}{\ell} \right) x$$

$$\text{Area at this particular section } A = \pi r^2 = \pi \left[a + \left(\frac{b - a}{\ell} \right) x \right]^2$$

$$\text{Hence current density } J = \frac{I}{A} = \frac{I}{\pi r^2} = \frac{I}{\pi \left[a + \frac{x(b - a)}{\ell} \right]^2}$$

RESISTANCE

The resistance of a conductor is the opposition which the conductor offers to the flow of charge. When a potential difference is applied across a conductor, free electrons get accelerated and collide with positive ions and their motion is thus opposed. This opposition offered by the ions is called resistance of the conductor.

Resistance is the property of a conductor by virtue of which it opposes the flow of current in it.

Unit : ohm, volt/ampere,

Dimension = $M L^2 T^{-3} A^{-2}$

Resistance depends on :

- Length of the conductor ($R \propto \ell$)

- Area of cross-section of the conductor $R \propto \frac{1}{A}$

- Nature of material of the conductor $R = \frac{\rho \ell}{A}$

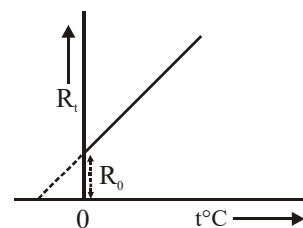
- Temperature $R_t = R_0 (1 + \alpha \Delta t)$

Where

R_t = Resistance at $t^\circ C$, R_0 = Resistance at $0^\circ C$

Δt = Change in temperature, α = Temperature coefficient of resistance

[For metals : α positive for semiconductors : α negative]

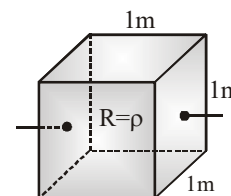


RESISTIVITY

Resistivity : $\rho = RA/\ell$ if $\ell = 1m$, $A = 1m^2$ then $\rho = R$

The specific resistance of a material is equal to the resistance of the wire of that material with unit cross – section area and unit length.

Resistivity depends on (i) Nature of material (ii) Temperature of material ρ does not depend on the size and shape of the material because it is the characteristic property of the conductor material.



Specific use of conducting materials :

- The **heating element** of devices like heater, geyser, press etc are made of **nichrome** because it has high resistivity and high melting point. It does not react with air and acquires steady state when red hot at $800^\circ C$.
- **Fuse wire** is made of **tin lead alloy** because it has low melting point and low resistivity. The fuse is used in series, and melts to produce open circuit when current exceeds the safety limit.
- **Resistances** of resistance box are made of **manganin** or **constantan** because they have moderate resistivity and very small temperature coefficient of resistance. The resistivity is nearly independent of temperature.
- The **filament of bulb** is made up of **tungsten** because it has low resistivity, high melting point of $3300 K$ and gives light at $2400 K$. The bulb is filled with inert gas because at high temperature it reacts with air forming oxide.
- The **connection wires** are made of **copper** because it has low resistance and resistivity.

KIRCHHOFF'S LAW

There are two laws given by Kirchhoff for determination of potential difference and current in different branches of any complicated network.

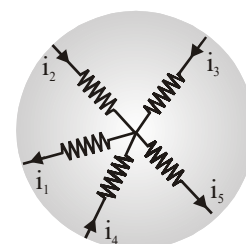
Law of conservation of charge is a consequence of continuity equation

- **First law (Junction Law or Current Law)**

In an electric circuit, the algebraic sum of the current meeting at any junction in the circuit is zero or Sum of the currents entering the Junction is equal to sum of the current leaving the Junction. $\Sigma i = 0$

$$i_1 - i_2 - i_3 - i_4 + i_5 = 0 \Rightarrow i_1 + i_5 = i_2 + i_3 + i_4$$

This is based on law of conservation of charge.



• **Second law (loop rule or potential law)**

In any closed circuit the algebraic sum of all potential differences and e.m.f. is zero. $\Sigma E - \Sigma IR = 0$ while moving from negative to positive terminal inside the cell, e.m.f. is taken as positive while moving in the direction of current in a circuit the potential drop (i.e. IR) across resistance is taken as positive.

This law is based on law of conservation of energy.

COMBINATION OF RESISTANCE

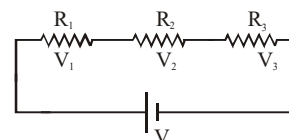
Series Combination

- Same current passes through each resistance
- Voltage across each resistance is directly proportional to its value

$$V_1 = IR_1, V_2 = IR_2, V_3 = IR_3$$

- Sum of the voltage across resistance is equal to the voltage applied across the circuit.

$$V = V_1 + V_2 + V_3 \Rightarrow IR = IR_1 + IR_2 + IR_3 \Rightarrow R = R_1 + R_2 + R_3 \text{ Where } R = \text{equivalent resistance}$$

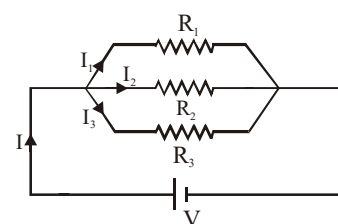


Parallel Combination

- There is same drop of potential across each resistance.
- Current in each resistance is inversely proportional to the

$$\text{value of resistance. } I_1 = \frac{V}{R_1}, I_2 = \frac{V}{R_2}, I_3 = \frac{V}{R_3}$$

- Current flowing in the circuit is sum of the currents in individual resistance.



$$I = I_1 + I_2 + I_3 \Rightarrow \frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} \Rightarrow \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

- To get maximum resistance, resistance must be connected in series and in series the resultant is greater than largest individual.
- To get minimum resistance, resistance must be connected in parallel and the equivalent resistance of parallel combination is lower than the value of lowest resistance in the combination.

• **In general :**

- Resistivity of alloys is greater than their metals.
- Temperature coefficient of alloys is lower than pure metals.
- The resistivity of an insulator (e.g. amber) is greater than the metal by a factor of 10^{22}

Ex. The resistance $4R, 16R, 64R \dots \infty$ are connected in series. Find their equivalent resistance.

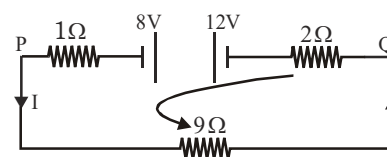
Sol. Resultant of the given combination $R_{eq} = 4R + 16R + 64R + \dots \infty = \infty$

Ex. Resistance $R, 2R, 4R, 8R \dots \infty$ are connected in parallel. What is their resultant resistance ?

Sol.
$$\frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{2R} + \frac{1}{4R} + \frac{1}{8R} \dots \infty = \frac{1}{R} \left[1 + \frac{1}{2} + \frac{1}{4} + \dots \infty \right] = \frac{1}{R} \left[\frac{1}{1 - \frac{1}{2}} \right] = \frac{2}{R} \Rightarrow R_{eq} = \frac{R}{2}$$

Ex. In the given circuit calculate potential difference between the points P and Q.

Sol. Applying Kirchhoff's voltage law (KVL)



$$12 - 8 = (1)I + (9)I + (2)I \Rightarrow I = \frac{1}{3} \text{ A}$$

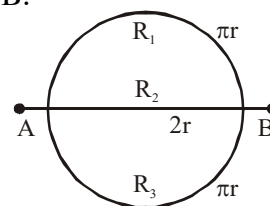
Potential difference between the points P and Q, $V_P - V_Q = 9 \times \frac{1}{3} = 3 \text{ volt}$

Ex. A wire of $\rho_L = 10^{-6} \Omega/\text{m}$ is turned in the form of a circle of diameter 2 m. A piece of same material is connected in diameter AB. Then find resistance between A and B.

Sol. $\therefore R = \rho_L \times \text{length}$

$$\therefore R_1 = \pi \times 10^{-6} \Omega, R_2 = 2 \times 10^{-6} \Omega, R_3 = \pi \times 10^{-6} \Omega$$

$$\frac{1}{R_{AB}} = \frac{1}{\pi \times 10^{-6}} + \frac{1}{2 \times 10^{-6}} + \frac{1}{\pi \times 10^{-6}}; R_{AB} = 0.88 \times 10^{-6} \text{ ohm.}$$



CELL

Cell convert chemical energy into electrical energy.

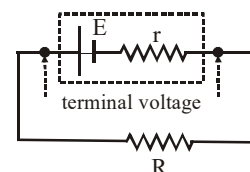
ELECTRO MOTIVE FORCE (E. M. F.)

The potential difference across the terminals of a cell when it is not giving any current is called emf of the cell. The energy given by the cell in the flow of unit charge in the whole circuit (including the cell) is called the emf of the cell.

- emf depends on : (i) nature of electrolyte (ii) metal of electrodes
- emf does not depend on : (i) area of plates (ii) distance between the electrodes (iii) quantity of electrolyte (iv) size of cell

TERMINAL VOLTAGE (V)

- When current is drawn through the cell or current is supplied to cell then, the potential difference across its terminals called terminal voltage.
- When I current is drawn from cell, then terminal voltage is less than it's e.m.f. $V = E - Ir$

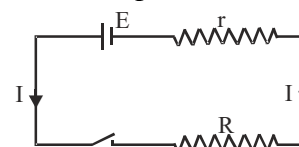


- **Terminal Potential Difference :** The potential difference between the two electrodes of a cell in a closed circuit i.e. when current is being drawn from the cell is called terminal potential difference.

(a) When cell is discharging :

Current inside the cell is from cathode to anode.

$$\text{Current } I = \frac{E}{r + R} \Rightarrow E = IR + Ir = V + Ir \Rightarrow V = E - Ir$$

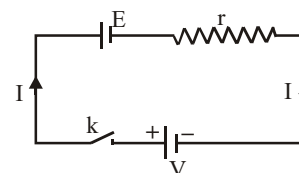


When current is drawn from the cell potential difference is less than emf of cell. Greater is the current drawn from the cell smaller is the terminal voltage. When a large current is drawn from a cell its terminal voltage is reduced.

(b) When cell is charging :

Current inside the cell is from anode to cathode.

$$\text{Current } I = \frac{V - E}{r} \Rightarrow V = E + Ir$$



During charging terminal potential difference is greater than emf of cell.

(c) When cell is in open circuit :

$$\text{In open circuit } R = \infty \therefore I = \frac{E}{R + r} = 0 \Rightarrow V = E$$

In open circuit terminal potential difference is equal to emf and is the maximum potential difference which a cell can provide.

(d) **When cell is short circuited :**

In short circuit $R = 0 \Rightarrow I = \frac{E}{R + r} = \frac{E}{r}$ and $V = IR = 0$

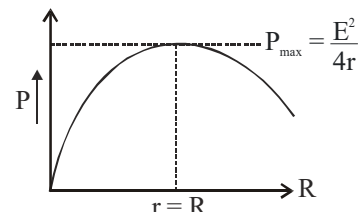
In short circuit current from cell is maximum and terminal potential difference is zero.

(e) **Power transferred to load by cell :**

$$P = I^2 R = \frac{E^2 R}{(r + R)^2} \Rightarrow P = P_{\max} \text{ if } \frac{dP}{dR} = 0 \Rightarrow r = R$$

Power transferred by cell to load is maximum when

$$r = R \text{ and } P_{\max} = \frac{E^2}{4r} = \frac{E^2}{4R}$$



COMBINATION OF CELLS

• **Series combination**

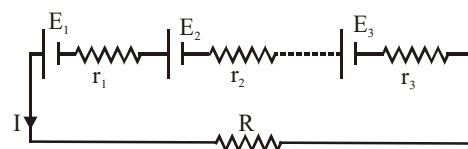
When the cells are connected in series the total e.m.f. of the series combination is equal to the sum of the e.m.f.'s of the individual cells and internal resistance of the cells also come in series.

Equivalent internal resistance $r = r_1 + r_2 + r_3 + \dots$

Equivalent emf $= E = E_1 + E_2 + E_3 + \dots$

$$\text{Current } I = \frac{E_{\text{net}}}{r_{\text{net}} + R}, \text{ If all } n \text{ cell are identical then } I = \frac{nE}{nr + R}$$

• If $nr \gg R$, $I = \frac{E}{r} \approx$ current from one cell • If $nr \ll R$, $I = \frac{nE}{R} \approx n \times$ current from one cell



• **Parallel combination**

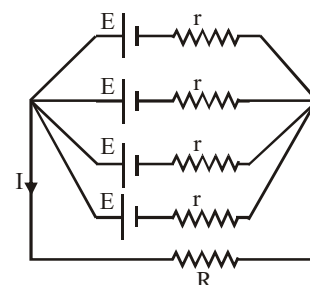
When the cells are connected in parallel, the total e.m.f. of the parallel combination remains equal to the e.m.f. of a single cell and internal resistance of the cell also come in parallel. If m identical cell

connected in parallel then total internal resistance of this combination $r_{\text{net}} = \frac{r}{m}$. Total e.m.f. of this combination $= E$

$$\text{Current in the circuit } I = \frac{E}{R + \frac{r}{m}} = \frac{mE}{mR + r}$$

If $r \ll mR$ $I = E/R =$ Current from one cell

If $r \gg mR$ $I = \frac{mE}{r} = m \times$ current from one cell



• **Mixed combination**

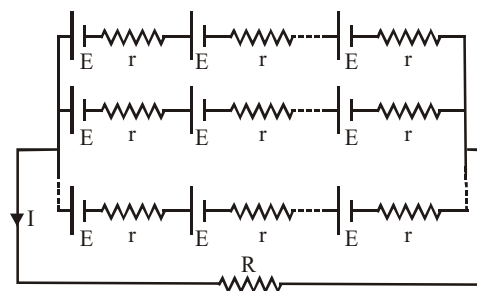
If n cells connected in series and there are m such branches in the circuit then total number of identical cell in this circuit is nm . The internal resistance of the cells connected in a row $= nr$. Since there are such m rows,

Total internal resistance of the circuit $r_{\text{net}} = \frac{nr}{m}$

Total e.m.f. of the circuit = total e.m.f. of the cells connected in a row

$$E_{\text{net}} = nE$$

$$\text{Current in the circuit } I = \frac{E_{\text{net}}}{R + r_{\text{net}}} = \frac{nE}{R + \frac{nr}{m}}$$



Current in the circuit is maximum when external resistance in the circuit is equal to the total internal

resistance of the cells $R = \frac{nr}{m}$

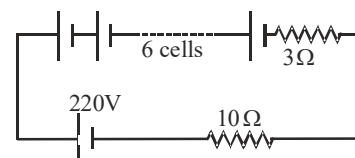
- At the time of charging a cell when current is supplied to the cell, the terminal voltage is greater than the e.m.f. E , $V = E + Ir$
- Series combination is useful when internal resistance is less than external resistance of the cell.
- Parallel combination is useful when internal resistance is greater than external resistance of the cell.
- Power in R (given resistance) is maximum, if its value is equal to net resistance of remaining circuit.
- Internal resistance of ideal cell = 0
- if external resistance is zero then current given by circuit is maximum.

Ex. A battery of six cells each of e.m.f. 2 V and internal resistance 0.5Ω is being charged by D. C. mains of e.m.f. 220 V by using an external resistance of 10Ω . What will be the charging current.

Sol. Net e.m.f. of the battery = 12 V and total internal resistance = 3Ω

Total resistance of the circuit = $3 + 10 = 13 \Omega$

$$\text{Charging current } I = \frac{\text{Net e.m.f.}}{\text{total resistance}} = \frac{220 - 12}{13} = 16 \text{ A}$$



Ex. A battery of six cells each of e.m.f. 2 V and internal resistance 0.5Ω is being charged by D. C. mains of e.m.f. 220 V by using an external resistance of 10Ω . What is the potential difference across the battery ?

Sol. In case of charging of battery, terminal potential $V = E + Ir = 12 + 16 \times 3 = 60$ volt.

Ex. Four identical cells each of e.m.f. 2 V are joined in parallel providing supply of current to external circuit consisting of two 15Ω resistors joined in parallel. The terminal voltage of the equivalent cell as read by an ideal voltmeter is 1.6 V calculate the internal resistance of each cell.

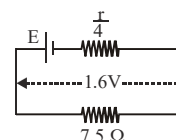
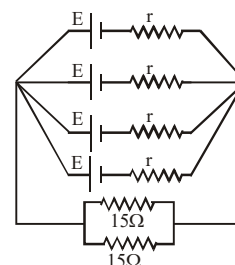
Sol. Total internal resistance of the combination $r_{\text{eq}} = \frac{r}{4}$

Total e.m.f. $E_{\text{eq}} = 2 \text{ V}$

$$\text{Total external resistance } R = \frac{15 \times 15}{15 + 15} = \frac{15}{2} = 7.5 \Omega$$

$$\text{Current drawn from equivalent cell } I = \frac{\text{terminal potential}}{\text{external resistance}} = \frac{1.6}{7.5} \text{ A}$$

$$\therefore E - I \left(\frac{r}{4} \right) = 1.6 \quad \therefore E - I \left(\frac{r}{4} \right) = 1.6 \Rightarrow r = 7.5 \Omega$$



Ex. The e.m.f. of a primary cell is 2 V, when it is shorted then it gives a current of 4 A. Calculate internal resistance of primary cell.

Sol. $I = \frac{E}{r + R}$, If cell is shorted then $R = 0$, $I = \frac{E}{r} \therefore r = \frac{E}{I} = \frac{2}{4} = 0.5 \Omega$

Ex. n rows each containing m cells in series, are joined in parallel. Maximum current is taken from this combination in a 3Ω resistance. If the total number of cells used is 24 and internal resistance of each cell is 0.5Ω , find the value of m and n.

Sol. Total number of cell $mn = 24$, For maximum current $\frac{mr}{n} = R \Rightarrow 0.5 m = 3 n$, $m = \frac{3n}{0.5} = 6n$

$$\therefore 6n \times n = 24 \Rightarrow n = 2 \text{ and } m \times 2 = 24 \Rightarrow m = 12$$

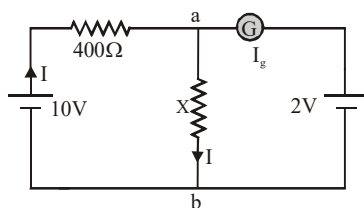
Ex. In the given circuit calculate potential difference between A and B.

Sol. First applying KVL on left mesh $2 - 3 I_1 - 2 I_1 = 0 \Rightarrow I_1 = 0.4 \text{ amp}$.
Now applying KVL on right mesh. $4 - 5 I_2 - 3 I_2 = 0 \Rightarrow I_2 = 0.5 \text{ amp}$.

Potential difference between points A and B

$$V_A - V_B = -3 \times 0.4 - 4 + 3 \times 0.5 = -3.7 \text{ volt.}$$

Ex. In the following circuit diagram, the galvanometer reading is zero. If the internal resistance of cells are negligible then what is the value of X ?

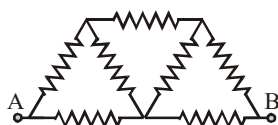


Sol. $\therefore I_g = 0 \quad \therefore I = \frac{10}{400 + X}$

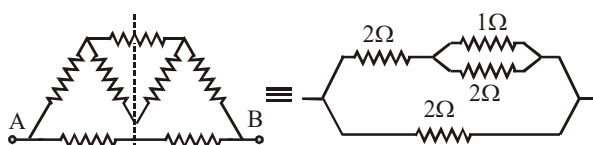
also potential difference across X is $2V \Rightarrow I X = 2$

$$\frac{10X}{400 + X} = 2 \left(\because I = \frac{10}{400 + X} \right) \Rightarrow X = 100 \Omega$$

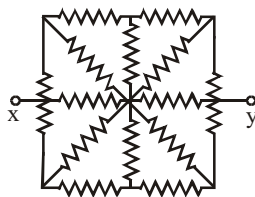
Ex. Each resistance is of 1Ω in the circuit diagram shown in figure. Find out equivalent resistance between A and B



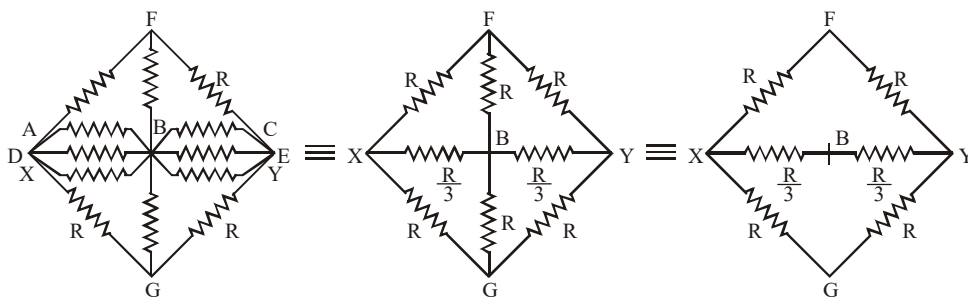
Sol. By symmetric line method $R_{AB} = (2 + 1 \parallel 2) \parallel 2 = \frac{8}{7} \Omega$



Ex. Identical resistance of resistance R are connected as in figure then find out net resistance between x and y .



Sol. Given circuit can be modified according to following figures



$$\frac{1}{R_{xy}} = \frac{1}{2R} + \frac{3}{2R} + \frac{1}{2R} = \frac{5}{2R} \Rightarrow R_{xy} = \frac{2R}{5}$$

HEATING EFFECT OF CURRENT

CAUSE OF HEATING

The potential difference applied across the two ends of conductor sets up electric field. Under the effect of electric field, electrons accelerate and as they move, they collide against the ions and atoms in the conductor, the energy of electrons transferred to the atoms and ions appears as heat.

- Joules's Law of Heating**

When a current I is made to flow through a passive or ohmic resistance R for time t , heat Q is produced such that

$$Q = I^2 R t = P \times t = V I t = \frac{V^2}{R} t$$

Heat produced in conductor does not depend upon the direction of current.

- SI unit** : joule ;

Practical Units : 1 kilowatt hour (kWh)

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ joule} = 1 \text{ unit}$$

$$1 \text{ BTU (British Thermal Unit)} = 1055 \text{ J}$$

- Power** : $P = V I = \frac{V^2}{R} = I^2 R$

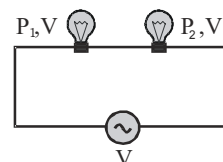
- SI unit** : Watt

The watt-hour meter placed on the premises of every consumer records the electrical energy consumed.

- Series combination of resistors (bulbs)**

$$\text{Total power consumed } P_{\text{total}} = \frac{P_1 P_2}{P_1 + P_2}$$

$$\text{If } n \text{ bulbs are identical } P_{\text{total}} = \frac{P}{n}$$



In series combination of bulbs : Brightness $\propto$ Power consumed by bulb $\propto V \propto R \propto \frac{1}{P_{\text{rated}}}$

Bulb of lesser wattage will shine more. For same current $P = I^2 R$ $P \propto R$ $R \uparrow \Rightarrow P \uparrow$

• **Parallel combination of resistors (bulbs)**

Total power consumed $P_{\text{total}} = P_1 + P_2$

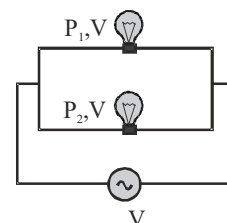
If n bulbs are identical $P_{\text{total}} = nP$

In parallel combination of bulbs

Brightness $\propto$ Power consumed by bulb $\propto I \propto \frac{1}{R}$

Bulb of greater wattage will shine more.

For same V more power will be consumed in smaller resistance $P \propto \frac{1}{R}$



- Two identical heater coils gives total heat H_s when connected in series and H_p when connected in parallel than $\frac{H_p}{H_s} = 4$ [In this, it is assumed that supply voltage is same]

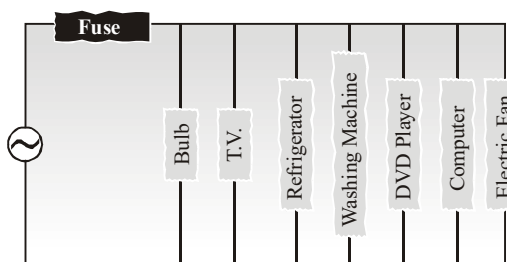
- If a heater boils m kg water in time T_1 and another heater boils the same water in T_2 , then both connected in series will boil the same water in time $T_s = T_1 + T_2$ and in parallel $T_p = \frac{T_1 T_2}{T_1 + T_2}$ [Use

time taken $\propto$ Resistance]

- Instruments based on heating effect of current, works on both A.C. and D.C. Equal value of A.C. (RMS) and D.C. produces, equal heating effect. That why brightness of bulb is same whether it is operated by A.C. or same value D.C.

FUSE WIRE

The fuse wire for an electric circuit is chosen keeping in view the value of safe current through the circuit.



- The fuse wire should have high resistance per unit length and low melting point.
- However the melting point of the material of fuse wire should be above the temperature that will be reached on the passage of the current through the circuit
- A fuse wire is made of alloys of lead (Pb) and tin (Sn).
- Length of fuse wire is immaterial.
- The material of the filament of a heater should have high resistivity and high melting point.
- The temperature of the wire increases to such a value at which, the heat produced per second equals

heat lost per second due to radiation from the surface of wire $I^2 \left(\frac{\rho \ell}{\pi r^2} \right) = H \times 2\pi r \ell$ $I^2 \propto r^3$

H = heat lost per second per unit area due to radiation.

Ex. An electric heater and an electric bulb are rated 500 W, 220 V and 100 W, 220 V respectively. Both are connected in series to a 220 V a.c. mains. Calculate power consumed by (i) heater (ii) bulb.

Sol. $P = \frac{V^2}{R}$ or $R = \frac{V^2}{P}$, For heater. Resistance $R_h = \frac{(220)^2}{500} = 96.8 \Omega$,

For bulb resistance $R_L = \frac{(220)^2}{100} = 484 \Omega$

Current in the circuit when both are connected in series $I = \frac{V}{R_L + R_h} = \frac{220}{484 + 96.8} = 0.38 \text{ A}$

(i) Power consumed by heater $= I^2 R_h = (0.38)^2 \times 96.8 = 13.98 \text{ W}$

(ii) Power consumed by bulb $= I^2 R_L = (0.38)^2 \times 484 = 69.89 \text{ W}$

Ex. A heater coil is rated 100 W, 200 V. It is cut into two identical parts. Both parts are connected together in parallel, to the same source of 200 V. Calculate the energy liberated per second in the new combination.

Sol. $\therefore P = \frac{V^2}{R}$ $\therefore R = \frac{V^2}{P} = \frac{(200)^2}{100} = 400 \Omega$

Resistance of half piece $= \frac{400}{2} = 200 \Omega$

Resistance of pieces connected in parallel $= \frac{400}{2} = 100 \Omega$

Energy liberated/second $P = \frac{V^2}{R} = \frac{200 \times 200}{100} = 400 \text{ W}$

Ex. The power of a heater is 500W at 800°C. What will be its power at 200°C. If $\alpha = 4 \times 10^{-4}$ per °C?

Sol. $P = \frac{V^2}{R} \therefore \frac{P_{200}}{P_{800}} = \frac{R_{800}}{R_{200}} = \frac{R_0(1 + 4 \times 10^{-4} \times 800)}{R_0(1 + 4 \times 10^{-4} \times 200)} \Rightarrow P_{200} = \frac{500 \times 1.32}{1.08} = 611 \text{ W}$

Ex. When a battery sends current through a resistance R_1 for time t , the heat produced in the resistor is Q . When the same battery sends current through another resistance R_2 for time t , the heat produced in R_2 is again Q . Determine the internal resistance of battery.

Sol. $\left[\frac{E}{R_1 + r} \right]^2 R_1 = \left[\frac{E}{R_2 + r} \right]^2 R_2 \Rightarrow r = \sqrt{R_1 R_2}$

Ex. How much time heater will take to increase the temperature of 100 g water by 50°C if resistance of heating coil is 484Ω and supply voltage is 220V a.c.

Sol. Heat given by heater = heat taken by water $\Rightarrow \frac{V^2}{R} t = ms \Delta \theta \Rightarrow \frac{220 \times 220}{484}$

$t = (100 \times 10^{-3}) (4.2 \times 10^3) (50) \Rightarrow t = 210 \text{ s}$

GALVANOMETER

The instrument used to measure strength of current, by measuring the deflection of the coil due to torque produced by a magnetic field, is known as galvanometer.

SHUNT

The small resistance connected in parallel to galvanometer coil, in order to control current flowing through the galvanometer, is known as shunt.

• Merits of shunt

- (i) To protect the galvanometer coil from burning.
- (ii) Any galvanometer can be converted into ammeter of desired range with the help of shunt.
- (iii) The range an ammeter can be changed by using shunt resistance of different values.

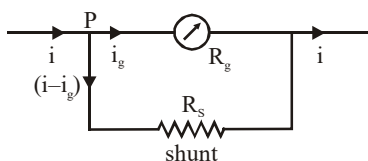
• Demerits of shunt

Shunt resistance decreases the sensitivity of galvanometer.

CONVERSION OF GALVANOMETER INTO AMMETER

A galvanometer can be converted into an ammeter by connecting low resistance in parallel to its coil.

- The value of shunt resistance to be connected in parallel to galvanometer coil is given by : $R_s = \frac{R_g i_g}{i - i_g}$



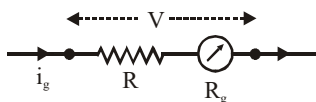
Where i = Range of ammeter

i_g = Current required for full scale deflection of galvanometer.

R_g = Resistance of galvanometer coil.

CONVERSION OF GALVANOMETER INTO VOLTMETER

- The galvanometer can be converted into voltmeter by connecting high resistance in series with its coil.
- The high resistance to be connected in series with galvanometer coil is given by $R = \frac{V}{i_g} - R_g$



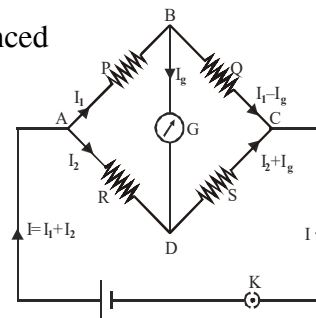
- The rate of variation of deflection depends upon the magnitude of deflection itself and so the accuracy of the instrument.
- A suspended coil galvanometer can measure currents of the order of 10^{-9} ampere.
- I_g is the current for full scale deflection. If the current for a deflection, of one division on the galvanometer scale is k and N is the total number of divisions on one side of the zero of galvanometer scale, then $I_g = k \times N$.
- A ballistic galvanometer is a specially designed moving coil galvanometer, used to measure charge flowing through the circuit for small time intervals.

WHEAT STONE BRIDGE

- The configuration in the adjacent figure is called Wheat Stone Bridge.
- If current in galvanometer is zero ($I_g = 0$) then bridge is said to be balanced

$$V_D = V_B \Rightarrow I_1 P = I_2 R \text{ \& } I_1 Q = I_2 S \Rightarrow \frac{P}{Q} = \frac{R}{S}$$

- If $\frac{P}{Q} < \frac{R}{S}$ then $V_B > V_D$ and current will flow from B to D.
- If $\frac{P}{Q} > \frac{R}{S}$ then $V_B < V_D$ and current will flow from D to B.



METRE BRIDGE

It is based on principle of whetstone bridge. It is used to find out unknown resistance of wire. AC is 1 m long uniform wire. R.B. is known resistance and S is unknown resistance. A cell is connected across 1 m long wire and Galvanometer is connected between Jockey and midpoint D. To find out unknown resistance we touch jockey from A to C and find balance condition. Let balance is at B point on wire.

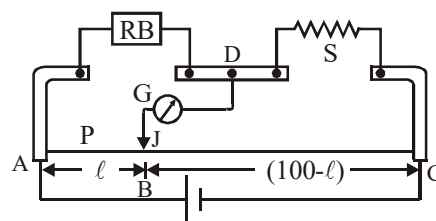
$$AB = \ell \text{ cm}$$

$$BC = (100 - \ell) \text{ cm}$$

$$P = r \ell$$

$$Q = r(100 - \ell)$$

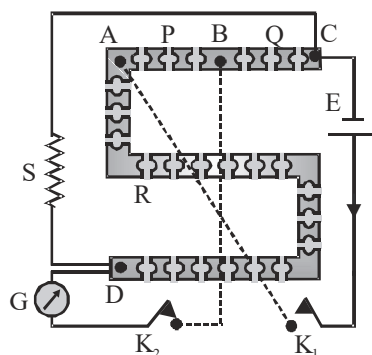
where r = resistance per unit length on wire.



$$\text{At balance condition : } \frac{P}{Q} = \frac{R}{S} \Rightarrow \frac{r\ell}{r(100-\ell)} = \frac{R}{S} \Rightarrow S = \frac{(100-\ell)}{\ell} R$$

POST OFFICE BOX

It is also based on wheat stone bridge. The resistance of 10Ω , 100Ω , and 1000Ω are often connected between AB and BC. These are known as ratio arms. Resistance from 1Ω to 5000Ω are connected between A and D, this is known arm. Unknown resistance is connected between C and D.



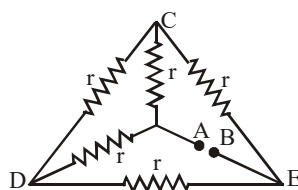
A cell is connected between A and C with key K_1 and Galvanometer is connected between B and D with key K_2 .

First we select ratio of resistance Q and P. For given value of S we will take value of resistance from known arm in such a way that Galvanometer show null deflection $S = \frac{Q}{P} R$. On decreasing the value of

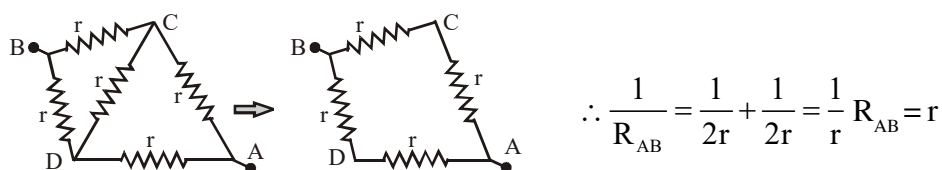
$\frac{Q}{P}$ the sensitivity of the box increases. It is used to find out the breakage in telegraph line in post and telegraph offices.

- To increase the range of an ammeter a shunt is connected in parallel with the galvanometer.
- To convert an ammeter of range I ampere and resistance $R_g \Omega$ into an ammeter of range nI ampere, the value of resistance to be connected in parallel will be $R_g(n-1)$
- To increase the range of a voltmeter a high resistance is connected in series with it.
- To convert a voltmeter of resistance $R_g \Omega$ and range V volt into a voltmeter of range nV volt, the value of resistance to be connected in series will be $(n-1)R_g$.
- Resistance of ideal ammeter is zero & resistance of ideal voltmeter is infinite.
- The bridge is most sensitive when the resistance in all the four branches of the bridge is of same order.

Ex. In the adjoining network of resistors each is of resistance $r \Omega$. Find the equivalent resistance between point A and B



Sol. Given circuit is balanced Wheat stone Bridge



Ex. A 100 volt voltmeter whose resistance is $20 \text{ k}\Omega$ is connected in series to a very high resistance R. When it is joined in a line of 110 volt, it reads 5 volt. What is the magnitude of resistance R ?

Sol. When voltmeter connected in 110 volt line, Current through the voltmeter $I = \frac{110}{(20 \times 10^3 + R)}$

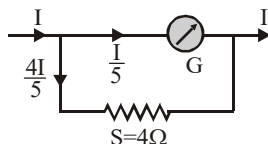
The potential difference across the voltmeter $V = IR_v$

$$\Rightarrow 5 = \frac{110 \times 20 \times 10^3}{(20 \times 10^3 + R)}$$

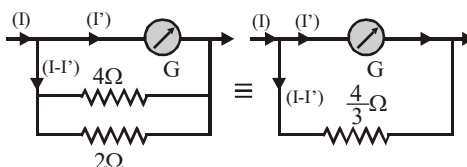
$$\Rightarrow 20 \times 10^3 + R = 440 \times 10^3 \Rightarrow R = 420 \times 10^3 \Omega$$

Ex. When a shunt of 4Ω is attached to a galvanometer, the deflection reduces to $1/5^{\text{th}}$. If an additional shunt of 2Ω is attached what will be the deflection ?

Sol. Initial condition : When shunt of 4Ω used $\frac{I}{5} \times G = \frac{4}{5} I \times 4 \Rightarrow G = 16\Omega$



When additional shunt of 2Ω used $I' \times 16 = (I - I') \frac{4}{3} \Rightarrow I' = \frac{I}{13}$



$\therefore$ it will reduce to $\frac{I}{13}$ of the initial deflection

Ex. A galvanometer having 30 divisions has current sensitivity of $20\mu\text{A}/\text{division}$. It has a resistance of 25Ω .

- How will you convert it into an ammeter measuring upto 1 ampere.
- How will you convert this ammeter into a voltmeter upto 1 volt.

Sol The current required for full scale deflection $I_g = 20\mu\text{A} \times 30 = 600\mu\text{A} = 6 \times 10^{-4}\text{A}$

- To convert it into ammeter, a shunt is required in parallel with it

$$\text{shunt resistance } R'_s = \frac{I_g R_g}{(I - I_g)} = \left(\frac{6 \times 10^{-4}}{1 - 6 \times 10^{-4}} \right) 25 = 0.015\Omega$$

- To convert galvanometer into voltmeter, a high resistance in series with it is required

$$\text{series resistance } R = \frac{V}{i_g} - R_g = \frac{1}{6 \times 10^{-4}} - 25 = 1666.67 - 25 = 1641.67\Omega$$

POTENTIOMETER

• Necessity of potentiometer

Practically voltmeter has a finite resistance. (ideally it should be ∞) in other words it draws some current from the circuit. To overcome this problem potentiometer is used because at the instant of measurement, it draws no current from the circuit. It means its effective resistance is infinite.

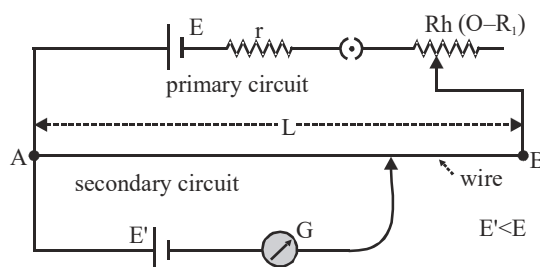
• Working principle of potentiometer

Any unknown potential difference is balanced on a known potential difference which is uniformly distributed over entire length of potentiometer wire. This process is named as zero deflection or null deflection method.

• Potentiometer wire

Made up of alloys of magnin, constantan, Eureka. Specific properties of these alloys are high specific resistance, negligible temperature co-efficient of resistance (α). Invariability of resistance of potentiometer wire over a long period.

CIRCUITS OF POTENTIOMETER



- **Primary circuit** contains constant source of voltage rheostat or Resistance Box
- **Secondary, Unknown or galvanometer circuit**
Let ρ = Resistance per unit length of potentiometer wire
- **Potential gradient (x) (V/m)**
The fall of potential per unit length of potentiometer wire is called potential gradient.

$$x = \frac{V}{L} = \frac{\text{current} \times \text{resistance of potentiometer wire}}{\text{length of potentiometer wire}} = I \left(\frac{R}{L} \right)$$

The potential gradient depends only on primary circuit and is independent of secondary circuit.

- **Applications of potentiometer**

• To measure potential difference across a resistance. • To find out emf of a cell • Comparison of two emfs E_1/E_2 • To find out internal resistance of a primary cell • Comparison of two resistance. • To find out an unknown resistance which is connected in series with the given resistance.	• To find out current in a given circuit • Calibration of an ammeter or to have a check on reading of (A) • Calibration of a voltmeter or to have a check on reading of (V) • To find out thermocouple emf (e_t) (mV or mV)
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Different between potentiometer and voltmeter

Potentiometer	Voltmeter
• It measures the unknown emf very accurately • While measuring emf it does not draw any current from the driving source of know emf. • While measuring unknown potential difference the resistance of potentiometer becomes infinite. • It is based on zero deflection method. • It has a high sensitivity. • it is used for various applications like measurement of internal resistance of cell, calibration of ammeter and voltmeter, measurement of thermo emf, comparison of emf's etc.	• It measures the unknown emf approximately. • While measuring emf it draws some current from the source of emf. • While measuring unknown potential difference the resistance of voltmeter is high but finite. • It is based on deflection method. • Its sensitivity is low. • It is only used to measured emf or unknown potential difference.

Ex. There is a definite potential difference between the two ends of a potentiometer. Two cells are connected in such a way that first time help each other, and second time they oppose each other. They are balanced on the potentiometer wire at 120 cm and 60 cm length respectively. Compare the electromotive force of the cells.

Sol. Suppose the potential gradient along the potentiometer wire = x and the emf's of the two cells are E_1 and E_2 .

When the cells help each other, the resultant emf = $(E_1 + E_2)$

$$E_1 + E_2 = x \times 120 \text{ cm} \dots(i)$$

When the cells oppose each other, the resultant emf = $(E_1 - E_2)$

$$E_1 - E_2 = x \times 60 \text{ cm} \dots(ii)$$

$$\text{From equation (i) and (ii)} \quad \frac{E_1 + E_2}{E_1 - E_2} = \frac{120 \text{ cm}}{60 \text{ cm}} = \frac{2}{1} \Rightarrow E_1 + E_2 = 2(E_1 - E_2) \Rightarrow 3E_2 = E_1 \Rightarrow \frac{E_1}{E_2} = \frac{3}{1}$$

COLOUR CODE FOR CARBON RESISTORS

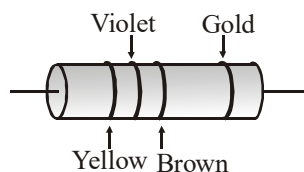
Colour	Strip A	Strip B	Strip C	Strip D (Tolerance)
Black	0	0	10^0	
Brown	1	1	10^1	
Red	2	2	10^2	
Orange	3	3	10^3	
Yellow	4	4	10^4	
Green	5	5	10^5	
Blue	6	6	10^6	
Violet	7	7	10^7	
Grey	8	8	10^8	
White	9	9	10^9	
Gold	-	-	10^{-1}	$\pm 5 \%$
Silver	-	-	10^{-2}	$\pm 10 \%$
No colour	-	-	-	$\pm 20 \%$

Aid to memory BBROY
Great Britain
Very Good watch of gold & silver

Ex. Draw a colour code for $42 \text{ k } \Omega \pm 10\%$ carbon resistance.

Sol. According to colour code colour for digit 4 is yellow, for digit 2 it is red, for 3 colour is orange and 10% tolerance is represented by silver colour. So colour code should be yellow, red, orange and silver.

Ex. What is resistance of following resistor.



Sol. Number for yellow is 4, Number of violet is 7

Brown colour gives multiplier 10^1 , Gold gives a tolerance of $\pm 5\%$

So resistance of resistor is $47 \times 10^1 \Omega \pm 5\% = 470 \pm 5\% \Omega$.

EXERCISE # S-1

Microscopic analysis

1. A copper wire of length L , and cross section area A carries a current I . If the specific resistance of copper is ρ , the electric field in the wire is _____.

CE0001

2. A copper wire carries a current density j (= current per unit area). Assuming that n = No. of free electrons per unit volume, e = electronic charge, $\langle v \rangle$ = average speed due to thermal agitation. The distance which will be covered by an electron during its displacement l along the wire _____.

CE0002

3. The total momentum of electrons in a straight wire of length ℓ carrying a current I is _____ (mass of electron = m_e , charge of electron = e)

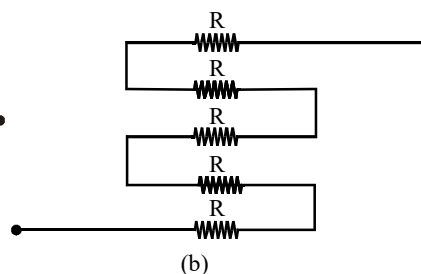
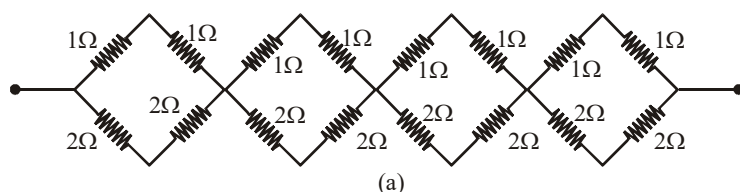
CE0003

4. Two conductors are made of the same material and have the same length. Conductor A is a solid wire of diameter 1mm. Conductor B is a hollow tube of outer diameter 2mm and inner diameter 1mm. Find the ratio of resistance R_A to R_B .

CE0004

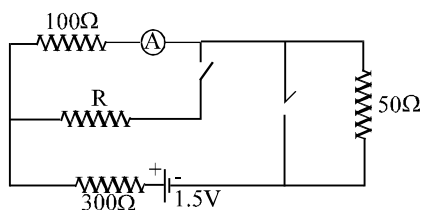
Ohm's law and circuit analysis

5. (a) Given n resistors each of resistance R , how will you combine them to get the (i) maximum (ii) minimum effective resistance? What is the ratio of the maximum to minimum resistance?
(b) Given the resistances of 1Ω , 2Ω , 3Ω , how will be combine them to get an equivalent resistance of
(i) $(11/3)\Omega$ (ii) $(11/5)\Omega$ (iii) 6Ω (iv) $(6/11)\Omega$?
(c) Determine the equivalent resistance of networks shown in figures



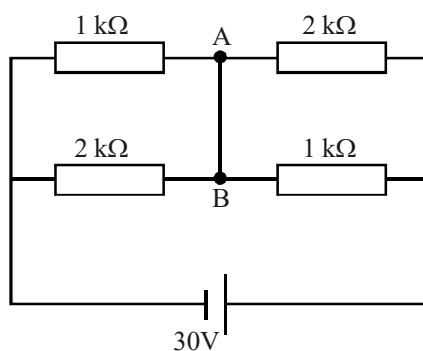
CE0005

6. In the circuit shown in figure the reading of ammeter is the same with both switches open as with both closed. Then find the resistance R . (ammeter is ideal)



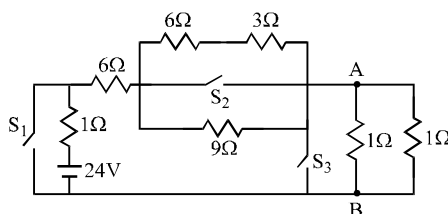
CE0006

7. Find the current (in mA) in the wire between points A and B.



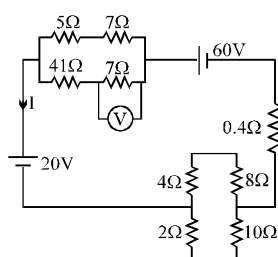
CE0007

8. If the switches S_1 , S_2 and S_3 in the figure are arranged such that current through the battery is minimum, find the voltage across points A and B.



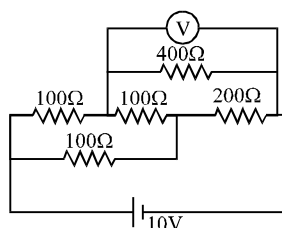
CE0008

9. Find the current I & voltage V in the circuit shown.



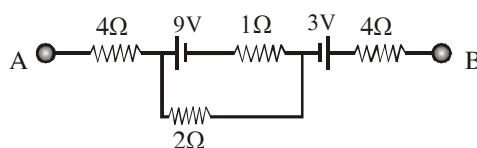
CE0009

10. An electrical circuit is shown in the figure. Calculate the potential difference across the resistance of $400\ \Omega$, as will be measured by the voltmeter V of resistance $400\ \Omega$, either by applying Kirchhoff's rules or otherwise.



CE0010

11. In the circuit shown in figure potential difference between point A and B is 16V . Find the current passing through $2\ \Omega$ resistance.

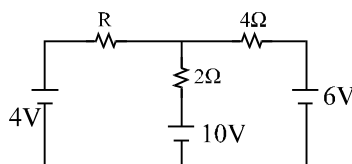


CE0011

12. An enquiring physics student connects a cell to a circuit and measures the current drawn from the cell to I_1 . When he joins a second identical cell in series with the first, the current becomes I_2 . When the cells are connected in parallel, the current through the circuit is I_3 . Show that relation between the current is $3 I_3 I_2 = 2 I_1 (I_2 + I_3)$

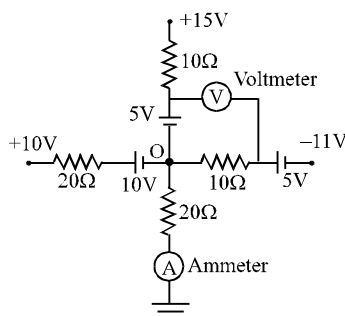
CE0012

13. For what value of R in circuit, current through $4\ \Omega$ resistance is zero.



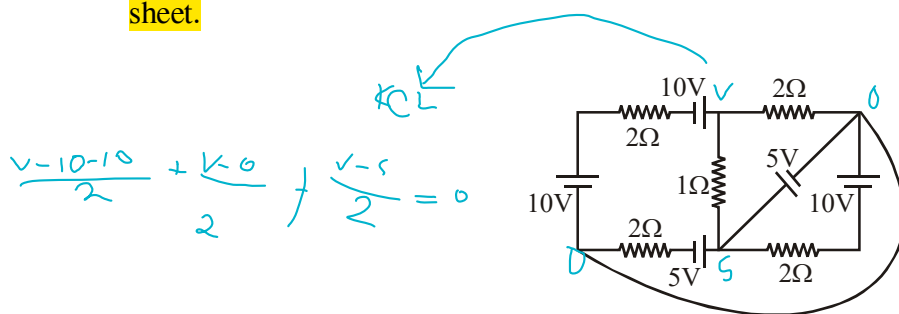
CE0013

14. The potential of certain points in the circuit are maintained at the values indicated. The Voltmeter and Ammeter are ideal. Find the potential of the cross junction point in the circuit (at center O) and the readings of Voltmeter and Ammeter. All cells are ideal.



CE0014

15. In the given circuit diagram, the current through the 1Ω resistor is given by I amp. Fill 2 I in OMR sheet.



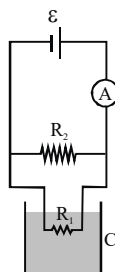
CE0015

Joule heating

16. If a cell of constant E.M.F. produces the same amount of the heat during the same time in two independent resistors R_1 and R_2 , when they are separately connected across the terminals of the cell, one after the another, find the internal resistance of the cell.

CE0016

17. The coil of a calorimeter C has a resistance of $R_1 = 60\Omega$. The coil R_1 is connected to the circuit as shown in figure. What is the rise in temperature ($^{\circ}\text{C}$) of 240 grams of water poured into the calorimeter when it is heated for 7 minutes during which a current flows through the coil and the ammeter shows 3A? The resistance $R_2 = 30\Omega$. [Disregard the resistances of the battery and the ammeter, and the heat losses and heat capacity of the calorimeter and the resistor and specific heat of water = $4200 \text{ J/kg}^{\circ}\text{C}$]

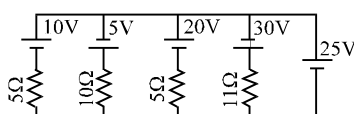


CE0017

18. An electric kettle has two windings. When one of them is switched on, the water in the kettle begins to boil in 15 minutes, and when the other is switched on it takes 30 minutes for water to boil. If the two windings are joined in series and switched on, water in the kettle begin to boil in $\frac{\alpha}{4}$ hr. Assuming no heat loss to the surrounding fill the value of α in OMR sheet.

CE0018

19. Find the current through 25V cell & power supplied by 20V cell in the figure shown.



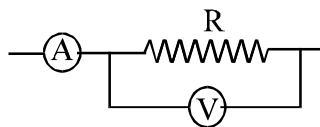
CE0019

20. A person decides to use his bath tub water to generate electric power to run a 40 watt bulb. The bath tub is located at a height of 10 m from the ground & it holds 200 litres of water. If we install a water driven wheel generator on the ground, at what rate should the water drain from the bath tub to light bulb? How long can we keep the bulb on, if the bath tub was full initially. The efficiency of generator is 90%. ($g = 10\text{m/s}^2$)

CE0020

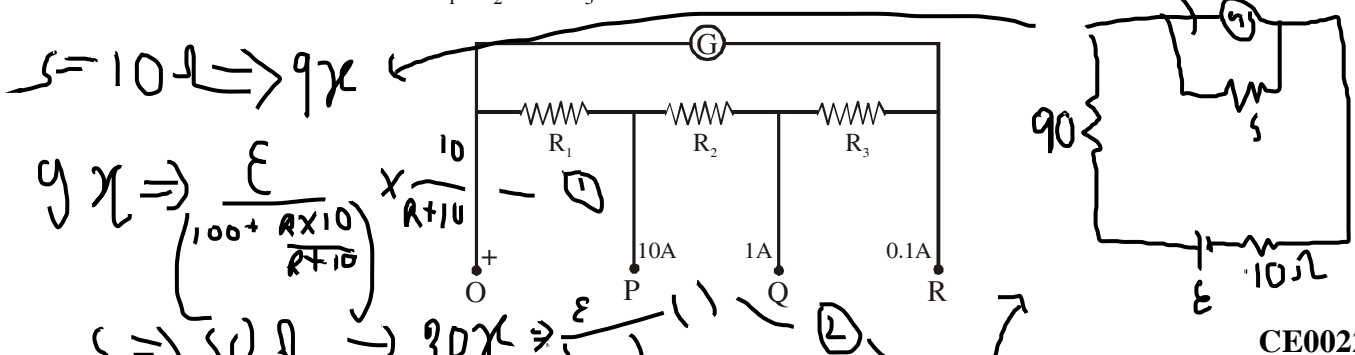
Instruments

21. A part of a circuit is shown in figure. Here reading of ammeter is 5 ampere and voltmeter is 96V & voltmeter resistance is 480 ohm. Then find the resistance R



CE0021

22. The resistance of the galvanometer G in the circuit is 25Ω . The meter deflects full scale for a current of 10 mA. The meter behaves as an ammeter of three different ranges. The range is 0–10A, if the terminals O and P are taken; range is 0–1 A between O and Q; range is 0–0.1 A between O and R. Calculate the resistance R_1 , R_2 and R_3 .



CE0022

23. A galvanometer having 50 divisions provided with a variable shunt is used to measure the current as an ammeter when connected in series with a resistance of 90Ω and a battery of internal resistance 10Ω . It is observed that when the shunt resistance are 10Ω , 50Ω , respectively the deflection are respectively 9 & 30 divisions. What is the resistance of the galvanometer? Further if the full scale deflection of the galvanometer movement is 300 mA, find the emf of the cell.

CE0023

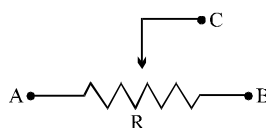
24. Draw the circuit for experimental verification of Ohm's law using a source of variable D.C. voltage, a main resistance of 100Ω , two galvanometers and two resistances of values $10^6\Omega$ and $10^{-3}\Omega$ respectively. Clearly show the positions of the voltmeter and the ammeter. [IIT-JEE' 2004]

CE0024

25. How a battery is to be connected so that shown rheostat will behave like a potential divider? Also indicate the points about which output can be taken. [IIT-JEE' 2003]

WV Sir

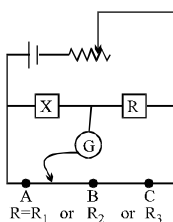
Notes



CE0025

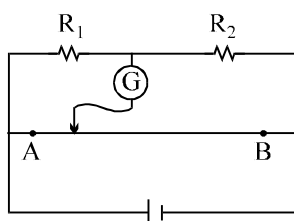
26. An unknown resistance X is to be determined using resistances R_1 , R_2 or R_3 . Their corresponding null points are A, B and C. Find which of the above will give the most accurate reading and why?

[IIT-JEE 2005]



CE0026

27. In the figure shown for which values of R_1 and R_2 the balance point for Jockey is at 40 cm from A. When R_2 is shunted by a resistance of 10Ω , balance shifts to 50 cm. Find R_1 and R_2 . ($AB = 1$ m):



CE0027

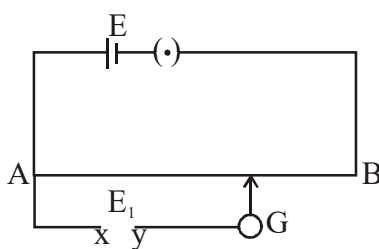
28. While doing an experiment with potentiometer it was found that the deflection is one sided and two cases are possible

- (i) the deflection decreased while moving from one end A of the wire to the end B;
(ii) the deflection increased while the jockey was moved towards the end B. Then

- (a) Which terminal + or -ve of the cell E_1 is connected at X in case (i) and how is E_1 related to E ?
(b) Which terminal of the cell E_1 is connected at X in case (ii)?

NV S10 Notes

Potentiometer



CE0028

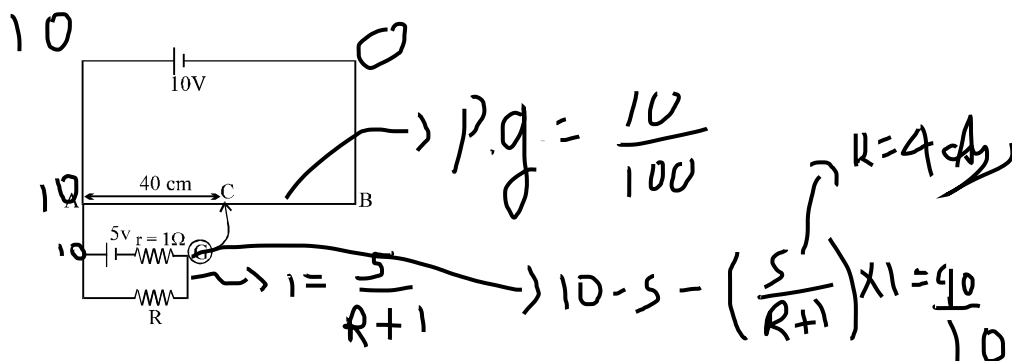
29. In a potentiometer arrangement, a cell of emf 1.25 V gives a balance point at 35.0 cm length of the wire. If the cell is replaced by another cell and the balance point shifts to 63.0 cm, what is the emf of the second cell?

$E \propto l$ → Balance Length

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

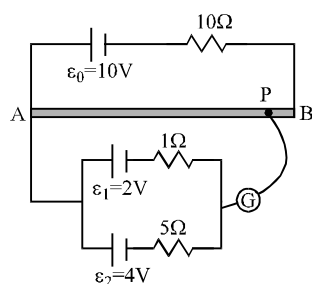
CE0029

30. A potentiometer wire AB is 100 cm long and has a total resistance of 10 ohm. If the galvanometer shows zero deflection at the position C, then find the value of unknown resistance R.



CE0030

31. A battery of emf $\varepsilon_0 = 10$ V is connected across a 1 m long uniform wire having resistance $10\Omega/m$. Two cells of emf $\varepsilon_1 = 2$ V and $\varepsilon_2 = 4$ V having internal resistances 1Ω and 5Ω respectively are connected as shown in the figure. If a galvanometer shows no deflection at the point P, find the distance of point P from the point A.



CE0031

EXERCISE # S-2

1. A long cylinder with uniformly charged surface and cross sectional radius $a = 1.0$ cm moves with a constant velocity $v = 10$ m/s, along its axis. An electric field strength at the surface of the cylinder is equal to $E = 0.9$ KV/cm. Find the resulting convection current, that is, the current caused by mechanical transfer of charge.

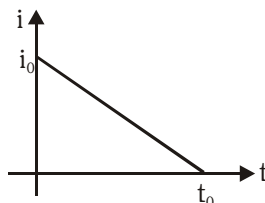
CE0032

2. A resistance R of thermal coefficient of resistivity $= \alpha$ is connected in parallel with a resistance $= 3R$, having thermal coefficient of resistivity $= 2\alpha$. Find the value of α_{eff} .

CE0033

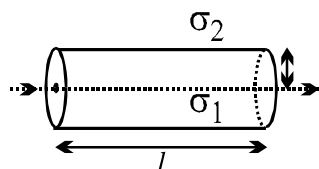
3. Relation between current in conductor and time is shown in figure then determine.

- (i) Total charge flow through the conductor
(ii) Write expression of current in terms of time
(iii) If resistance of conductor is R then total heat dissipated across resistance R is



CE0034

4. A long conductor of circular cross-section has radius r and length l as shown in the figure. The conductivity of the material near the axis is σ_1 and increases linearly with the distance from axis and becomes σ_2 near the surface. Find the resistance of the conductor if the current enters from the one end and leaves from the other end.



CE0035

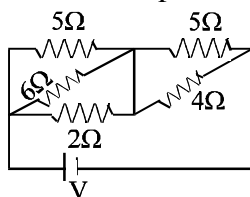
5. (a) The current density across a cylindrical conductor of radius R varies according to the equation

$J = J_0 \left(1 - \frac{r}{R} \right)$, where r is the distance from the axis. Thus the current density is a maximum J_0 at the axis $r = 0$ and decreases linearly to zero at the surface $r = R$. Calculate the current in terms of J_0 and the conductor's cross sectional area is $A = \pi R^2$.

(b) Suppose that instead the current density is a maximum J_0 at the surface and decreases linearly to zero at the axis so that $J = J_0 \frac{r}{R}$. Calculate the current.

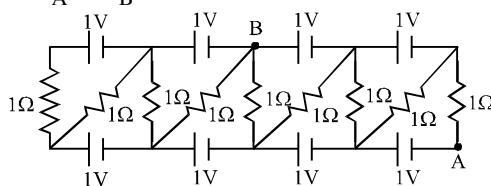
CE0036

6. Find the resistor in which maximum heat will be produced.



CE0037

7. Find the potential difference $V_A - V_B$ for the circuit shown in the figure.



CE0038

8. A galvanometer (coil resistance 99Ω) is converted into an ammeter using a shunt of 1Ω and connected as shown in the figure (i). The ammeter reads $3A$. The same galvanometer is converted into a voltmeter by connecting a resistance of 101Ω in series. This voltmeter is connected as shown in figure (ii). Its reading is found to be $4/5$ of the full scale reading. Find
- internal resistance r of the cell
 - range of the ammeter and voltmeter
 - full scale deflection current of the galvanometer

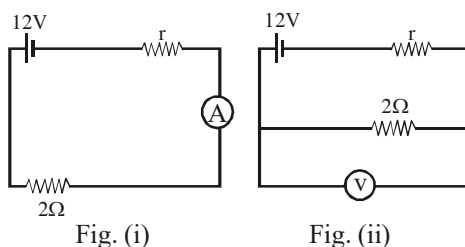


Fig. (i)

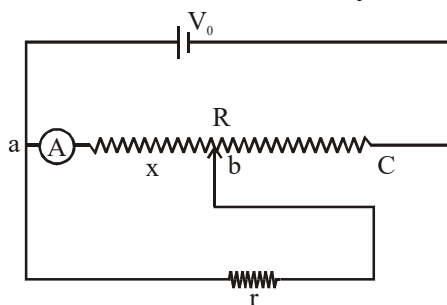
Fig. (ii)

CE0039

9. An accumulator of emf $2V$ and negligible internal resistance is connected across a uniform wire of length $10m$ and resistance 30Ω . The appropriate terminals of a cell of emf $1.5V$ and internal resistance 1Ω is connected to one end of the wire, and the other terminal of the cell is connected through a sensitive galvanometer to a slider on the wire. What length of the wire will be required to produce zero deflection of the galvanometer? How will the balancing change (a) when a coil of resistance 5Ω is placed in series with the accumulator, (b) the cell of $1.5V$ is shunted with 5Ω resistor?

CE0040

10. A constant voltage $V_0 (= 12V)$ is applied to a potential divider of resistance $R (= 4\Omega)$, connected to an ideal ammeter. A constant resistor $r (= 2\Omega)$ is connected to the sliding contact of the potential divider (as shown). Find the minimum current (in A) measured by ammeter.



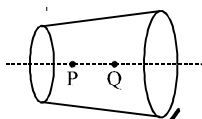
CE0041

EXERCISE # O-1

SINGLE CORRECT TYPE QUESTIONS

Microscopic analysis

1. A wire has a non-uniform cross-section as shown in figure. A steady current flows through it. The drift speed of electrons at points P and Q is v_P and v_Q .



- (A) $v_P = v_Q$ (B) $v_P < v_Q$ (C) $v_P > v_Q$ (D) Data insufficient

CE0042

2. Two wires each of radius of cross section r but of different materials are connected together end to end (in series). If the densities of charge carriers in the two wires are in the ratio 1 : 4, the drift velocity of electrons in the two wires will be in the ratio:

- (A) 1 : 2 (B) 2 : 1 (C) 4 : 1 (D) 1 : 4

CE0043

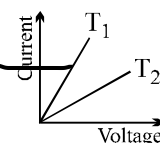
3. An insulating pipe of cross-section area 'A' contains an electrolyte which has two types of ions → their charges being $-e$ and $+2e$. A potential difference applied between the ends of the pipe result in the drifting of the two types of ions, having drift speed = v ($-ve$ ion) and $v/4$ ($+ve$ ion). Both ions have the same number per unit volume = n . The current flowing through the pipe is

- (A) $nev A/2$ (B) $nev A/4$ (C) $5nev A/2$ (D) $3nev A/2$

CE0044

4. The current in a metallic conductor is plotted against voltage at two different temperatures T_1 and T_2 . Which is correct

- (A) $T_1 > T_2$ (B) $T_1 < T_2$ (C) $T_1 = T_2$ (D) none



CE0045

5. A metal rod of length 10 cm and a rectangular cross-section of $1 \text{ cm} \times \frac{1}{2} \text{ cm}$ is connected to a battery across opposite faces. The resistance will be

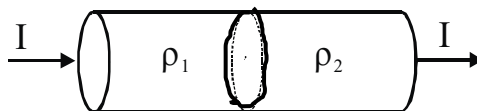
- (A) maximum when the battery is connected across $1 \text{ cm} \times \frac{1}{2} \text{ cm}$ faces. (B) maximum when the battery is connected across $10 \text{ cm} \times 1 \text{ cm}$ faces. (C) maximum when the battery is connected across $10 \text{ cm} \times \frac{1}{2} \text{ cm}$ faces. (D) same irrespective of the three faces.

CE0046

6. Consider a current carrying wire (current I) in the shape of a circle. Note that as the current progresses along the wire, the direction of $\mathbf{j}$ (current density) changes in an exact manner, while the current I remain unaffected. The agent that is *essentially* responsible for is
- (A) source of emf.
 (B) electric field produced by charges accumulated on the surface of wire.
 (C) the charges just behind a given segment of wire which push them just the right way by repulsion.
 (D) the charges ahead.

CE0047

7. Two long straight cylindrical conductors with resistivities ρ_1 and ρ_2 respectively are joined together as shown in figure. If current I flows through the conductors, the magnitude of the total free charge at the interface of the two conductors is :-



- (A) zero
 (B) $\frac{(\rho_1 - \rho_2) I \epsilon_0}{2}$
 (C) $\epsilon_0 I |\rho_1 - \rho_2|$
 (D) $\epsilon_0 I |\rho_1 + \rho_2|$

CE0048

8. **Statement 1 :** The drift speed of electrons in metals is small (in the order of a few mm/s) and the charge of an electron is also very small ($= 1.6 \times 10^{-19} \text{C}$), yet we can obtain a large current in a metal. and

Statement 2 : At room temperature, the thermal speed of electron is very high (about 10^7 times the drift speed).

- (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True ; Statement-2 is not a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.

CE0049

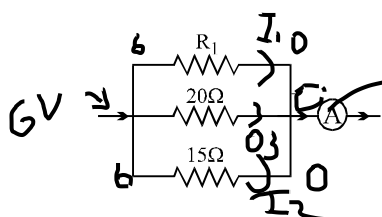
Ohm's law & circuit analysis

9. A storage battery is connected to a charger for charging with a voltage of 12.5 Volts. The internal resistance of the storage battery is 1Ω . When the charging current is 0.5 A, the emf of the storage battery is:

- (A) 13 Volts
 (B) 12.5 Volts
 (C) 12 Volts
 (D) 11.5 Volts

CE0050

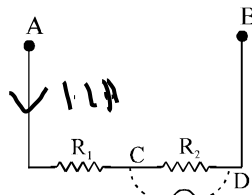
10. In the given circuit the current flowing through the resistance 20 ohms is 0.3 ampere while the ammetre reads 0.8 ampere. What is the value of R_1 ?



- (A) 30 ohms
 (B) 40 ohms
 (C) 50 ohms
 (D) 60 ohms

CE0051

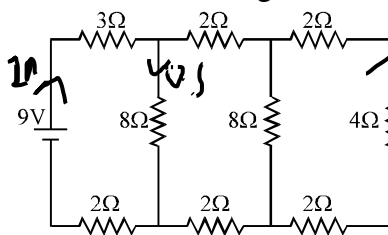
11. Resistances R_1 and R_2 each 60Ω are connected in series as shown in figure. The Potential difference between A and B is kept 120 volt. Then what will be the reading of voltmeter connected between the point C & D if resistance of voltmeter is 120Ω .



- (A) 48 V (B) 24 V (C) 40V (D) None

CE0052

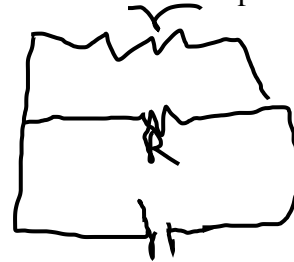
12. In the circuit shown in the figure, the current through :



- (A) the 3Ω resistor is 0.50 A (B) the 3Ω resistor is 0.25 A
(C) 4Ω resistor is 0.50 A (D) the 4Ω resistor is 0.25 A

CE0053

13. A simple circuit contains an ideal battery and a resistance R . If a second resistor is placed in parallel with the first,
(A) the potential across R will decrease
(B) the current through R will decreased
(C) the current delivered by the battery will increase
(D) the power dissipated by R will increased.



CE0054

14. The equivalent resistance of a group of resistances is R . If another resistance is connected in parallel to the group, its new equivalent becomes R_1 & if it is connected in series to the group, its new equivalent becomes R_2 we have :

- (A) $R_1 > R$ (B) $R_1 < R$ (C) $R_2 > R$ (D) $R_2 < R$

CE0055

15. An energy source will supply a constant current into the load, if its internal resistance is-

[AIEEE - 2005]

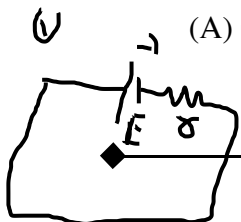
- (A) equal to the resistance of the load (B) very large as compared to the load resistance
(C) zero (D) non-zero but less than the resistance of the load

CE0056

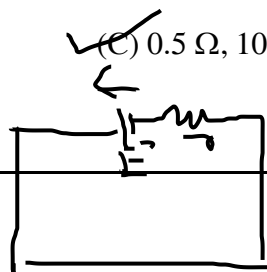
16. When a current of 4 A flows within a battery from its positive to negative terminal, the potential difference across the battery is 12 volts. The potential difference across the battery is 9 volts when a current of 2 A flows within it from its negative to its positive terminal. The internal resistance and the e.m.f. of the battery are :-

- (A) 0.1Ω , 4V (B) 0.2Ω , 5V (C) 0.5Ω , 10V (D) 0.7Ω , 10V

CE0057



$$12 = E + 4r \quad \text{--- (1)}$$



$$9 = E - 2r \quad \text{--- (2)}$$

$$\text{①} - \text{②} \Rightarrow 3 = 6r \Rightarrow r = 0.5\Omega$$

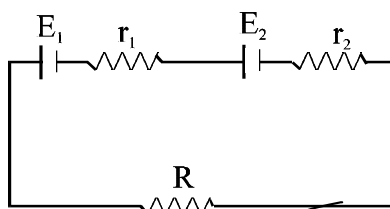
$$E = 10$$

17. Two resistances of equal magnitude R and having temperature coefficient α_1 and α_2 respectively are connected in parallel. The temperature coefficient of the parallel combination is, approximately

(A) $2(\alpha_1 + \alpha_2)$ (B) $\frac{\alpha_1 \alpha_2}{\alpha_1 + \alpha_2}$ (C) $\frac{\alpha_1 - \alpha_2}{2}$ (D) $\frac{\alpha_1 + \alpha_2}{2}$

CE0058

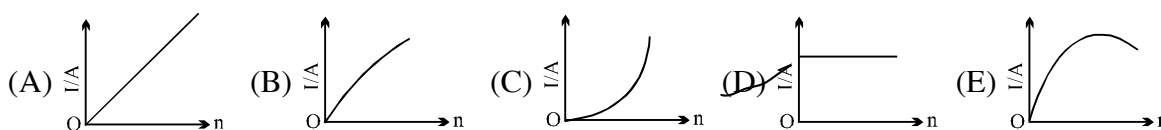
18. Under what condition current passing through the resistance R can be increased by short circuiting the battery of emf E_2 . The internal resistances of the two batteries are r_1 and r_2 respectively.



- (A) $E_2 r_1 > E_1 (R + r_2)$ (B) $E_1 r_2 > E_2 (R + r_1)$
(C) $E_2 r_2 > E_1 (R + r_2)$ (D) $E_1 r_1 > E_2 (R + r_1)$

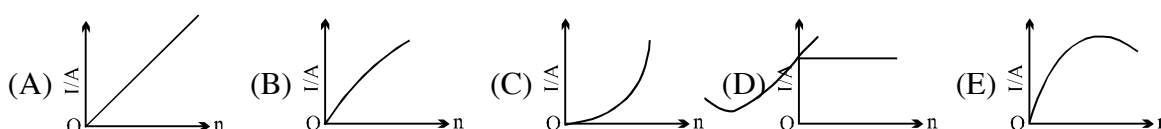
CE0059

19. A battery consists of a variable number n of identical cells having internal resistance connected in series. The terminals of the battery are short circuited and the current I measured. Which one of the graph below shows the relationship between I and n ?



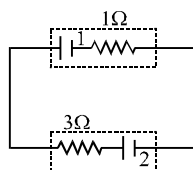
CE0060

20. In **previous problem**, if the cell had been connected in parallel (instead of in series) which of the above graphs would have shown the relationship between total current I and n ?



CE0061

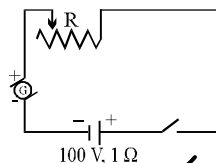
21. In the figure shown, battery 1 has emf = 6 V and internal resistance = 1Ω . Battery 2 has emf = 2 V and internal resistance = 3Ω . The wires have negligible resistance. What is the potential difference across the terminals of battery 2?



- (A) 4 V (B) 1.5 V (C) 5 V (D) 0.5 V

CE0062

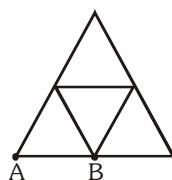
22. The battery in the diagram is to be charged by the generator G. The generator has a terminal voltage of 120 volts when the charging current is 10 amperes. The battery has an emf of 100 volts and an internal resistance of 1 ohm. In order to charge the battery at 10 amperes charging current, the resistance R should be set at :-



- (A) 0.1 Ω (B) 0.5 Ω (C) 1.0 Ω (D) 5.0 Ω

CE0063

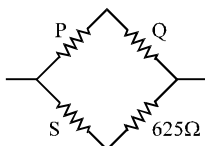
23. In the diagram resistance between any two junctions is R. Equivalent resistance across terminals A and B is :-



- (A) $\frac{11R}{7}$ (B) $\frac{18R}{11}$ (C) $\frac{7R}{11}$ (D) $\frac{11R}{18}$

CE0064

24. A Wheatstone's bridge is balanced with a resistance of 625 Ω in the third arm, where P, Q and S are in the 1st, 2nd and 4th arm respectively. If P and Q are interchanged, the resistance in the third arm has to be increased by 51 Ω to secure balance. The unknown resistance in the fourth arm is :-



- (A) 625 Ω (B) 650 Ω (C) 676 Ω (D) 600 Ω

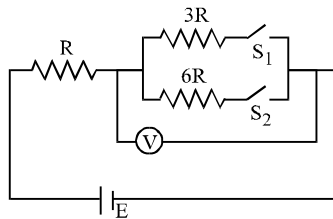
CE0065

25. One end of a Nichrome wire of length 2L and cross-sectional area A is attached to an end of another Nichrome wire of length L and cross-sectional area 2A. If the free end of the longer wire is at an electric potential of 8.0 volts, and the free end of the shorter wire is at an electric potential of 1.0 volt, the potential at the junction of the two wires is equal to :-

- (A) 2.4 V (B) 3.2 V (C) 4.5 V (D) 5.6 V

CE0066

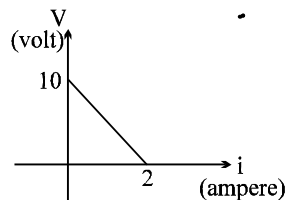
26. In the circuit shown in figure reading of voltmeter is V_1 when only S_1 is closed, reading of voltmeter is V_2 when only S_2 is closed. The reading of voltmeter is V_3 when both S_1 and S_2 are closed then



- (A) $V_2 > V_1 > V_3$ (B) $V_3 > V_2 > V_1$ (C) $V_3 > V_1 > V_2$ (D) $V_1 > V_2 > V_3$

CE0067

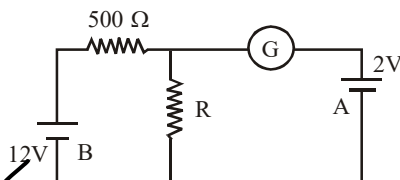
27. A battery of emf E and internal resistance r is connected across a resistance R . Resistance R can be adjusted to any value greater than or equal to zero. A graph is plotted between the current (i) passing through the resistance and potential difference (V) across it. Select the correct alternative.



- (A) internal resistance of battery is 5Ω
 (B) emf of the battery is 20V
 (C) maximum current which can be taken from the battery is 4A
 (D) $V-i$ graph can never be a straight line as shown in figure.

CE0068

28. In the circuit, the galvanometer G shows zero deflection. If the batteries A and B have negligible internal resistance, the value of the resistor R will be- [AIEEE - 2005]

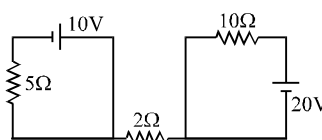


- (A) 200Ω (B) 100Ω (C) 500Ω (D) 1000Ω

CE0069

29. In the figure shown the current through 2Ω resistor is

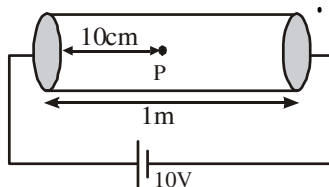
[IIT-JEE 2005 (Scr)]



- (A) 2 A (B) 0 A (C) 4 A (D) 6 A

CE0070

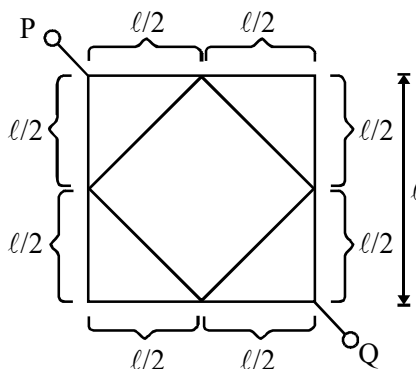
30. A cylindrical solid of length 1m and radius 1 m is connected across a source of emf 10V and negligible internal resistance shown in figure. The resistivity of the rod as a function of x (x measured from left end) is given by $\rho = bx$ [where b is a positive constant]. Find the electric field (in SI unit) at point P at a distance 10 cm from left end.



- (A) 1 (B) 2 (C) 3 (D) 4

CE0071

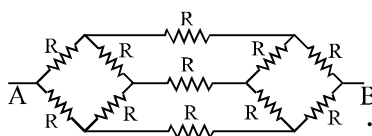
31. If the wire has resistivity ρ and cross sectional area A , the equivalent resistance between P and Q is :-



- (A) $\frac{\rho l}{\sqrt{2}A}$ (B) $\frac{\sqrt{2}\rho l}{A}$ (C) $\frac{2\rho l}{A}$ (D) $\frac{\rho l}{A}$

CE0072

32. The equivalent resistance between the terminal points A and B in the network shown in figure is :-



- (A) $\frac{7R}{5}$ (B) $\frac{5R}{6}$ (C) $\frac{7R}{12}$ (D) $\frac{5R}{12}$

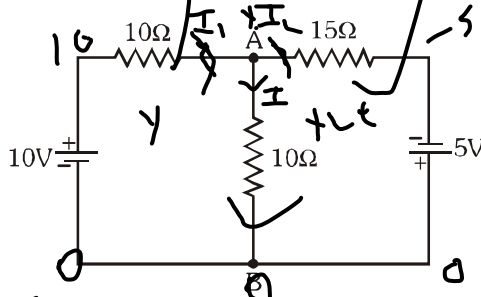
CE0073

33. An electric bell has a resistance of 5Ω and requires a current of 0.25 A to work it. Assuming that the resistance of the bell wire is 1Ω per 15m and that the bell push is 90m distance from the bell. How many cells each of emf 1.4V and internal resistance 2Ω , will be required to work the circuit-

- (A) 3 (B) 4 (C) 5 (D) Can't be determined

CE0074

34. A circuit is arranged as shown. Then, the current from A to B is



- (A) +500 mA (B) +250 mA (C) -250 mA (D) -500 mA

Joule heating

35. Power generated across a uniform wire connected across a supply is H . If the wire is cut into n equal parts and all the parts are connected in parallel across the same supply, the total power generated in the wire is

- (A) $\frac{H}{n^2}$ (B) $n^2 H$ (C) nH (D) $\frac{H}{n}$

CE0076

36. When electric bulbs of same power, but different marked voltage are connected in series across the power line, their brightness will be :

- (A) proportional to their marked voltage
(B) inversely proportional to their marked voltage
(C) proportional to the square of their marked voltage
(D) inversely proportional to the square of their marked voltage

CE0077

37. Two bulbs rated (25 W – 220V) and (100W – 220V) are connected in series to a 440 V line. Which one is likely to fuse?

- (A) 25 W bulb (B) 100 W bulb (C) both bulbs (D) none

CE0078

38. Rate of dissipation of Joule's heat in resistance per unit volume is (symbols have usual meaning)

- (A) σE (B) σI (C) $J E$ (D) None

CE0079

39. If the length of the filament of a heater is reduced by 10%, the power of the heater will

- (A) increase by about 9% (B) increase by about 11%
(C) increase by about 19% (D) decrease by about 10%

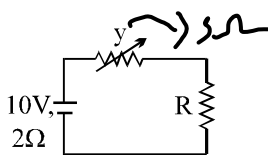
CE0080

40. Two bulbs one of 200 volts, 60 watts & the other of 200 volts, 100 watts are connected in series to a 200 volt supply. The power consumed will be

- (A) 37.5 watt (B) 160 watt (C) 62.5 watt (D) 110 watt

CE0081

41. In the figure shown the power generated in y is maximum when $y = 5\Omega$. Then R is :-

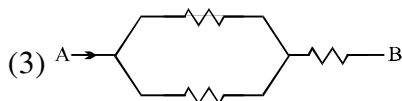
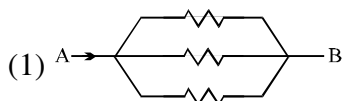


$$I = \frac{10}{R + 7}$$

- (A) 2Ω (B) 6Ω (C) 5Ω (D) 3Ω

CE0082

42. Arrange the order of power dissipated in the given circuits, if the same current is passing through all circuits and each resistor is ' r ' [IIT-JEE' 2003 (Scr)]



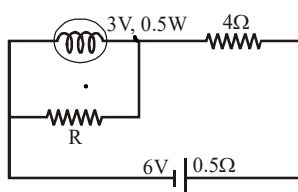
- (A) $P_2 > P_3 > P_4 > P_1$ (B) $P_3 > P_2 > P_4 > P_1$ (C) $P_4 > P_3 > P_2 > P_1$ (D) $P_1 > P_2 > P_3 > P_4$

CE0083

43. A rigid container with thermally insulated walls contains a coil of resistance 100Ω , carrying current 1 A. Change in internal energy after 5 min will be [IIT-JEE 2005]
(A) zero (B) 10 kJ (C) 20 kJ (D) 30 kJ

CE0084

44. The value of the resistance R in the circuit shown below so that electric bulb consumes the rated power is :-

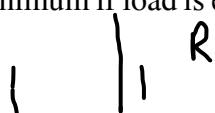


- (A) 4 ohm (B) 6 ohm (C) 8 ohm (D) 10 ohm

CE0085

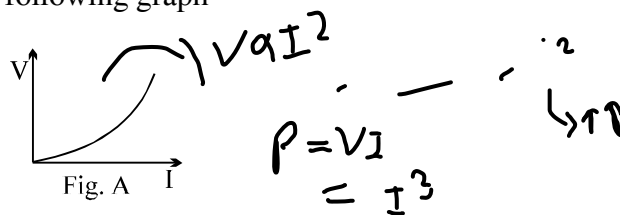
45. A variable load R is connected to a voltage source of internal resistance r . Then choose the **INCORRECT** statement out of the following :-

- (A) If $R = r$, maximum power is transferred to the load!
(B) If current is maximum, power transfer to load is also maximum
(C) If $R \ll r$, the voltage source supplied a fixed current to the load
(D) Power supplied to load is minimum if load is either too low or too high



CE0086

46. The variation of current (I) and voltage (V) is as shown in figure A. The variation of power with current I is best shown by which of the following graph



CE0087

Instruments

47. A galvanometer has a resistance of 20Ω and reads full-scale when 0.2 V is applied across it. To convert it into a 10 A ammeter, the galvanometer coil should have a

(A) 0.01Ω resistor connected across it (B) 0.02Ω resistor connected across it
(C) 200Ω resistor connected in series with it (D) 2000Ω resistor connected in series with it

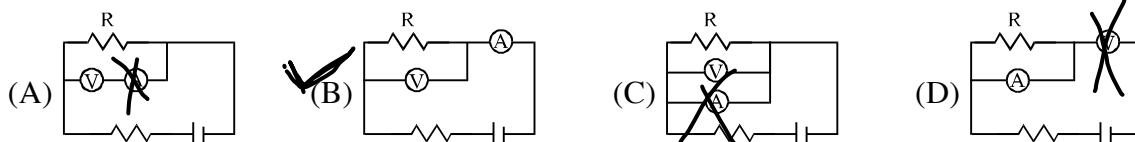
CE0088

48. A galvanometer coil has a resistance 90Ω and full scale deflection current 10 mA . A 910Ω resistance is connected in series with the galvanometer to make a voltmeter. If the least count of the voltmeter is 0.1 V , the number of divisions on its scale is

(A) 90 (B) 91 (C) 100 (D) none

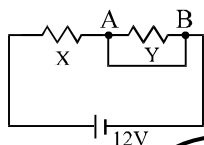
CE0089

49. Which of the following wiring diagrams could be used to experimentally determine R using ohm's law? Assume an ideal voltmeter and an ideal ammeter.



CE0090

50. When an ammeter of negligible internal resistance is inserted in series with circuit it reads 1 A . When the voltmeter of very large resistance is connected across X it reads 1 V . When the point A and B are shorted by a conducting wire, the voltmeter measures 10 V across the battery. The internal resistance of the battery is equal to :-

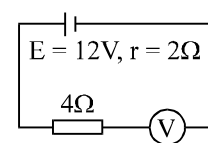


(A) zero (B) 0.5Ω (C) 0.2Ω (D) 0.1Ω

CE0091

51. By error, a student places moving-coil voltmeter V (nearly ideal) in series with the resistance in a circuit in order to read the current, as shown. The voltmeter reading will be

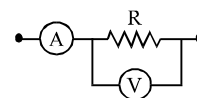
(A) 0 (B) 4 V
(C) 6 V (D) 12 V



CE0092

52. In the circuit shown the readings of ammeter and voltmeter are 4A and 20V respectively. The meters are non ideal, then R is :

(A) 5Ω (B) less than 5Ω
(C) greater than 5Ω (D) between 4Ω & 5Ω



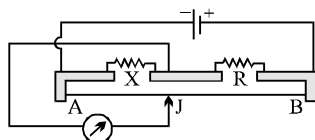
CE0093

53. In a balanced wheat stone bridge, current in the galvanometer is zero. It remains zero when:

[1] battery emf is increased
[2] all resistances are increased by 10 ohms
[3] all resistances are made five times
[4] the battery and the galvanometer are interchanged
(A) only [1] is correct (B) [1], [2] and [3] are correct
(C) [1], [3] and [4] are correct (D) [1] and [3] are correct

CE0094

54. The figure shows a metre-bridge circuit, with $AB = 100$ cm, $X = 12\Omega$ and $R = 18\Omega$, and the jockey J in the position of balance. If R is now made 8Ω , through what distance will J have to be moved to obtain balance?



(A) 10 cm (B) 20 cm (C) 30 cm (D) 40 cm

CE0095

55. In a metre bridge experiment, null point is obtained at 20 cm from one end of the wire when resistance X is balanced against another resistance Y . If $X < Y$, then where will be the new position of the null point from the same end, if one decides to balance a resistance of $4X$ against Y ? [AIEEE - 2004]

(A) 50 cm (B) 80 cm (C) 40 cm (D) 70 cm

CE0096

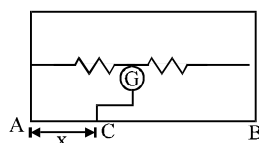
56. A resistance R is to be measured using a meter bridge. Student chooses the standard resistance S to be 100Ω . He finds the null point at $l_1 = 2.9$ cm. He is told to attempt to improve the accuracy.

Which of the following is a useful way?

(A) He should measure l_1 more accurately.
(B) He should change S to 1000Ω and repeat the experiment.
(C) He should change S to 3Ω and repeat the experiment.
(D) He should give up hope of a more accurate measurement with a meter bridge.

CE0097

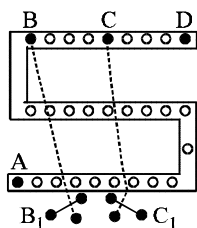
57. In the given circuit, no current is passing through the galvanometer. If the cross-sectional diameter of AB is doubled then for null point of galvanometer the value of AC would [IIT-JEE' 2003 (Scr)]



(A) x (B) $x/2$ (C) $2x$ (D) None

CE0098

58. For the post office box arrangement to determine the value of unknown resistance, the unknown resistance should be connected between [IIT-JEE' 2004 (Scr)]



- (A) B and C (B) C and D (C) A and D (D) B₁ and C₁

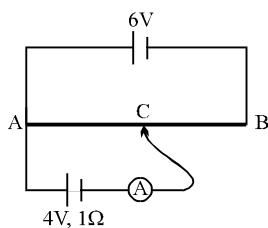
CE0099

59. A potentiometer wire has length 10 m and resistance 10Ω. It is connected to a battery of EMF 11 volt and internal resistance 1 Ω, then the potential gradient in the wire is :-

- (A) 10 V/m (B) 1 V/m (C) 0.1 V/m (D) none

CE0100

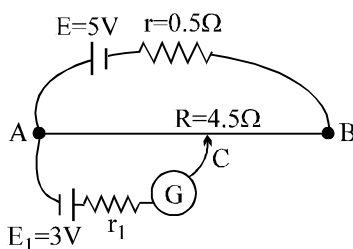
60. A 6 V battery of negligible internal resistance is connected across a uniform wire of length 1 m. The positive terminal of another battery of emf 4V and internal resistance 1 Ω is joined to the point A as shown in figure. The ammeter shows zero deflection when the jockey touches the wire at the point C. The AC is equal to :-



- (A) 2/3 m (B) 1/3 m (C) 3/5 m (D) 1/2 m

CE0101

61. In the given potentiometer circuit length of the wire AB is 3 m and resistance is $R = 4.5 \Omega$. The length AC for no deflection in galvanometer is :-



- (A) 2 m (B) 1.8 m (C) dependent on r_1 (D) none of these

CE0102

62. The length of a potentiometer wire is ℓ . A cell of emf E is balanced at a length $\ell/3$ from the positive end of the wire. If the length of the wire is increased by $\ell/2$. At what distance will the same cell give a balance point.

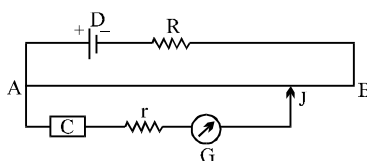
- (A) $\frac{2\ell}{3}$ (B) $\frac{\ell}{2}$ (C) $\frac{\ell}{6}$ (D) $\frac{4\ell}{3}$

CE0103

63. Two cells of emf's approximately 5V and 10V are to be accurately compared using a potentiometer of length 400cm.
- (A) The battery that runs the potentiometer should have voltage of 8V.
- (B) The battery of potentiometer can have a voltage of 15V and R adjusted so that the potential drop across the wire slightly exceeds 10V.
- (C) The first portion of 50 cm of wire itself should have a potential drop of 10V.
- (D) Potentiometer is usually used for comparing resistances and not voltages.

CE0104

64. In the given potentiometer circuit, the resistance of the potentiometer wire AB is R_0 . C is a cell of internal resistance r . The galvanometer G does not give zero deflection for any position of the jockey J. Which of the following cannot be a reason for this ?



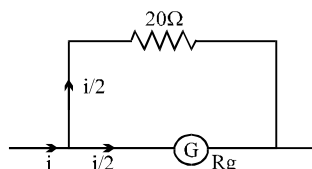
- (A) $r > R_0$ (B) $R \gg R_0$
- (C) emf of C > emf of D (D) The negative terminal of C is connected to A.

CE0105

65. A resistor has a color code of green, blue; brown and silver. What is its resistance?
- (A) $56 \Omega \pm 5\%$ (B) $560 \Omega \pm 10\%$ (C) $560 \Omega \pm 5\%$ (D) $5600 \Omega \pm 10\%$

CE0106

66. In a galvanometer, the deflection becomes one half when the galvanometer is shunted by a 20Ω resistor. The galvanometer resistance is

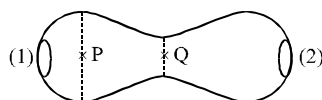


- (A) 5Ω (B) 10Ω (C) 40Ω (D) 20Ω

CE0107

MULTIPLE CORRECT TYPE QUESTIONS

67. A metallic conductor of irregular cross-section is as shown in the figure. A constant potential difference is applied across the ends (1) and (2). Then :



- (A) the current at the cross-section P equals the current at the cross-section Q
- (B) the electric field intensity at P is less than that at Q.
- (C) the rate of heat generated per unit time at Q is greater than that at P
- (D) the number of electrons crossing per unit area of cross-section at P is less than that at Q.

CE0108

68. A current passes through an ohmic conductor of nonuniform cross section. Which of the following quantities are independent of the cross-section?
- (A) the charge crossing in a given time interval.
 - (B) drift speed
 - (C) current density
 - (D) free-electron density

CE0109

69. A battery is of emf E is being charged from a charger such that positive terminal of the battery is connected to terminal A of charger and negative terminal of the battery is connected to terminal B of charger. The internal resistance of the battery is r .
- (A) Potential difference across points A and B must be more than E .
 - (B) A must be at higher potential than B
 - (C) In battery, current flows from positive terminal to the negative terminal
 - (D) No current flows through battery

CE0110

70. Two identical fuses are rated at 10A. If they are joined
- (A) in parallel, the combination acts as a fuse of rating 20A
 - (B) in parallel, the combination acts as a fuse of rating 5A
 - (C) in series, the combination acts as a fuse of rating 10A.
 - (D) in series, the combination acts as a fuse of rating 20A.

CE0111

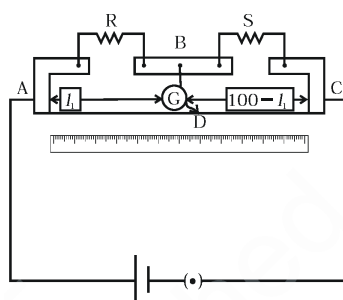
71. A micrometer has a resistance of 100Ω and a full scale range of $50\mu\text{A}$. It can be used as a voltmeter or a higher range ammeter provided a resistance is added to it. Pick the correct range and resistance combination(s).
- (A) 50 V range with $10\text{ k}\Omega$ resistance in series.
 - (B) 10 V range with $200\text{ k}\Omega$ resistance in series.
 - (C) 5 mA range with 1Ω resistance in parallel.
 - (D) 10 mA range with $1\text{ k}\Omega$ resistance in parallel.

CE0112

72. Mark out the correct options.
- (A) An ammeter should have small resistance.
 - (B) An ammeter should have large resistance.
 - (C) A voltmeter should have small resistance.
 - (D) A voltmeter should have large resistance.

CE0113

73. In a meter bridge the point D is a neutral point as shown in figure.

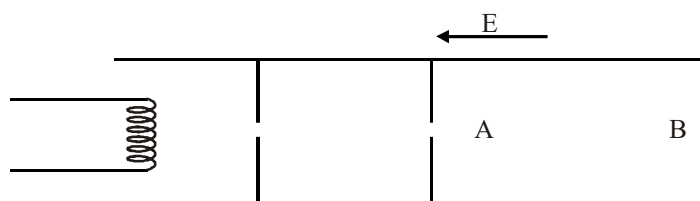


- (A) The meter bridge can have no other neutral point for this set of resistances.
 (B) When the jockey contacts a point on meter wire left of D, current flows to B from the wire through galvanometer.
 (C) When the jockey contacts a point on the meter wire to the right of D, current flows from B to the wire through galvanometer.
 (D) When R is increased, the neutral point shifts to left.

CE0114

MATRIX MATCH TYPE QUESTION

74. Electrons are emitted by a hot filament and are accelerated by an electric field as shown in figure. The two stops at the left ensure that the electron beam has a uniform cross-section. Match the entries of column-I with column-II as electron move from A to B :



Column-I

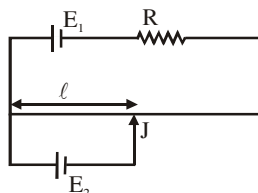
- (A) Speed of an electron
 (B) Number of free electrons per unit volume
 (C) Current density
 (D) Electric potential

Column-II

- (P) Increases
 (Q) Decreases
 (R) Remains same
 (S) any of the above is possible

CE0115

75. In the potentiometer arrangement shown in figure, null point is obtained at length ℓ .



Column-I

- (A) If E_1 is increased
 (B) If R is increased
 (C) If E_2 is increased

Column-II

- (P) ℓ should increase
 (Q) ℓ should decrease
 (R) ℓ should remain the same to again get the null point

CE0116

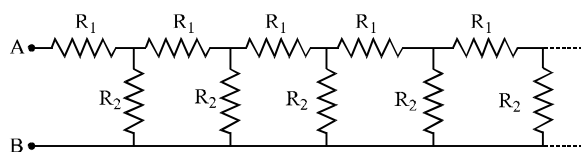
EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

- Which of the following quantities do not change when an ohmic resistor connected to a battery is heated due to the current?
 (A) drift speed (B) resistivity
 (C) resistance (D) number of free electrons

CE0117

- Consider an infinite ladder network shown in figure. A voltage V is applied between the points A and B. This applied value of voltage is halved after each section.



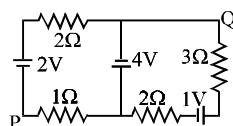
- (A) $R_1/R_2 = 1$ (B) $R_1/R_2 = 1/2$ (C) $R_1/R_2 = 2$ (D) $R_1/R_2 = 3$

CE0118

- A wire of length L and 3 identical cells of negligible internal resistances are connected in series. Due to the current, the temperature of the wire is raised by ΔT in time t . N number of similar cells is now connected in series with a wire of the same material and cross section but of length $2L$. The temperature of the wire is raised by the same amount ΔT in the same time t . The value of N is :
 (A) 4 (B) 6 (C) 8 (D) 9

CE0119

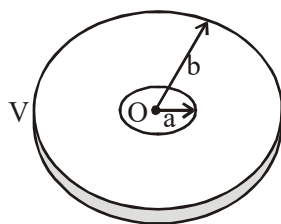
- In the circuit shown, what is the potential difference V_{PQ} ?



- (A) +3V (B) +2V (C) -2V (D) none

CE0120

- A circular portion is cut of a disc of thickness t , its resistivity is ρ and radii of disc are a and b ($b > a$). A potential difference is maintained between outer and inner cylindrical surfaces of the disc. What is resistance of the disc ?



- (A) $\frac{\rho}{2\pi t} \ln\left(\frac{b}{a}\right)$ (B) $\rho\left(\frac{1}{a} - \frac{1}{b}\right)$ (C) $2\pi\rho t\left(\frac{1}{a^2} - \frac{1}{b^2}\right)$ (D) $\frac{\rho}{2\pi t}\left(\frac{b^2 - a^2}{ab}\right)$

CE0121

6. The wire shown in figure has a uniform cross-section A.

$$\overline{\quad\quad\quad} \quad \begin{matrix} x=0 & & x=L \end{matrix}$$

Resistivity of the material of wire is given by $\rho = \rho_0 \left(\frac{L}{L+x} \right)$. A potential difference V is applied across the wire :-

(A) Resistance of wire is $\frac{\rho_0 L}{A} \cdot \ln(2)$

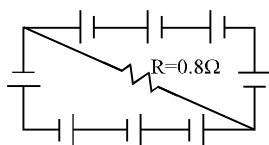
(B) Current density is variable inside the wire

(C) Electric field at $x=0$ is $\frac{2V}{(\ln 2) \cdot L}$

(D) Electric field at $x=L$ is $\frac{V}{(\ln 2)L}$

CE0122

7. A circuit is comprised of eight identical batteries and a resistor $R = 0.8\Omega$. Each battery has an emf of 1.0 V and internal resistance of 0.2Ω . The voltage difference across any of the battery is



(A) 0.5V

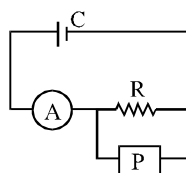
(B) 1.0V

(C) 0 V

(D) 2 V

CE0123

8. An ammeter A of finite resistance, and a resistor R are joined in series to an ideal cell C. A potentiometer P is joined in parallel to R. The ammeter reading is I_0 and the potentiometer reading is V_0 . P is now replaced by a voltmeter of finite resistance. The ammeter reading now is I and the voltmeter reading is V.



(A) $I > I_0, V < V_0$

(B) $I > I_0, V = V_0$

(C) $I = I_0, V < V_0$

(D) $I < I_0, V = V_0$

CE0124

9. The sensitivity of post-office box for determination of resistance of 5Ω is maximum when, P and Q both are

(A) 1 Ω

(B) 10 Ω

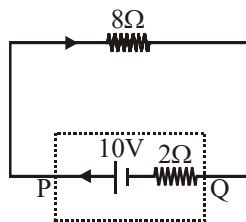
(C) 100 Ω

(D) 1000 Ω

CE0125

MULTIPLE CORRECT TYPE QUESTIONS

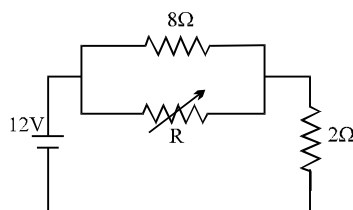
10. A battery of emf 10 volt and internal resistance 2Ω is connected to an external resistance 8Ω as shown in the figure :-



- (A) Work done due to conservative electric field while a unit positive charge passes through battery from Q to P (along the arrow) is 8 Joule.
 (B) Work done due to conservative electric field while a unit positive charge passes through battery from Q to P (along the arrow) is -8 Joule.
 (C) Work done due to conservative electric field while a unit positive charge passes through 8Ω along the arrow is -8 Joule.
 (D) Work done due to non conservative electric field while a unit positive charge moves from Q to P (along the arrow) is 10 Joule.

CE0126

11. The value of the resistance R in figure is adjusted such that power dissipated in the 2Ω resistor is maximum. Under this condition

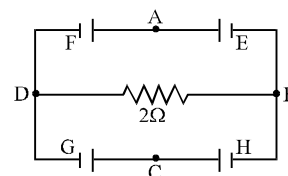


- (A) $R = 0$
 (B) $R = 8\Omega$
 (C) power dissipated in the 2Ω resistor is 72 W.
 (D) power dissipated in the 2Ω resistor is 8 W.

CE0127

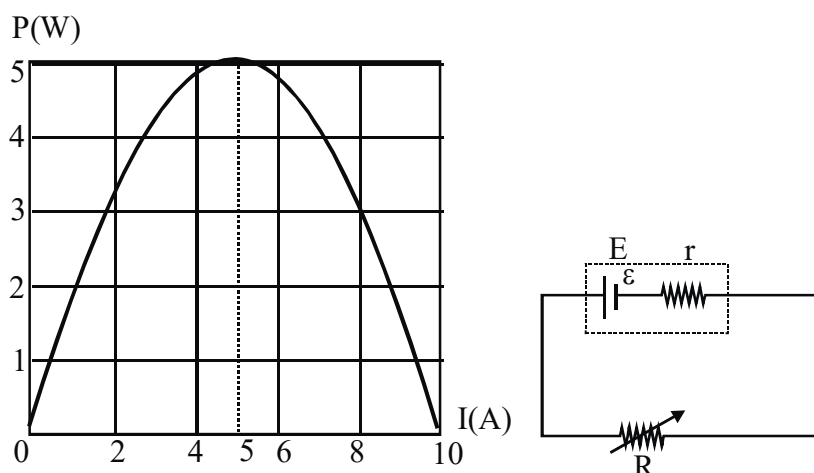
12. In the circuit shown E, F, G and H are cells of e.m.f. 2V, 1V, 3V and 1V respectively and their internal resistances are 2Ω , 1Ω , 3Ω and 1Ω respectively.

- (A) $V_D - V_B = -2/13$ V
 (B) $V_D - V_B = 2/13$ V
 (C) $V_G = 21/13$ V = potential difference across G.
 (D) $V_H = 19/13$ V = potential difference across H.



CE0128

13. Figure shows the net power dissipated in R versus the current in a simple circuit shown.



- (A) The internal resistance of battery is 0.2Ω (B) The emf of battery is $2V$
 (C) R at which power is $5W$ is 2.5Ω (D) At $i = 2A$, power is $3.2W$

CE0129

14. In a potentiometer wire experiment the emf of a battery in the primary circuit is $20V$ and its internal resistance is 5Ω . There is a resistance box in series with the battery and the potentiometer wire, whose resistance can be varied from 120Ω to 170Ω . Resistance of the potentiometer wire is 75Ω . The following potential differences can be measured using this potentiometer.

- (A) $5V$ (B) $6V$ (C) $7V$ (D) $8V$

CE0130

EXERCISE-JM

1. Two conductors have the same resistance at 0°C but their temperature coefficients of resistance are α_1 and α_2 . The respective temperature coefficients of their series and parallel combinations are nearly :
[AIEEE-2010]

(1) $\frac{\alpha_1 + \alpha_2}{2}, \frac{\alpha_1 + \alpha_2}{2}$ (2) $\frac{\alpha_1 + \alpha_2}{2}, \alpha_1 + \alpha_2$ (3) $\alpha_1 + \alpha_2, \frac{\alpha_1 + \alpha_2}{2}$ (4) $\alpha_1 + \alpha_2, \frac{\alpha_1 \alpha_2}{\alpha_1 + \alpha_2}$

CE0131

2. If a wire is stretched to make it 0.1 % longer its resistance will :-
(1) decrease by 0.2% (2) decrease by 0.05% (3) increase by 0.05% (4) increase by 0.2%
[AIEEE-2011]

CE0132

3. If $400\ \Omega$ of resistance is made by adding four $100\ \Omega$ resistance of tolerance 5%, then the tolerance of the combination is :
(1) 20% (2) 5% (3) 10% (4) 15%
[AIEEE - 2011]

CE0133

4. The current in the primary circuit of a potentiometer is 0.2 A. The specific resistance and cross-section of the potentiometer wire are 4×10^{-7} ohm metre and 8×10^{-7} m² respectively. The potential gradient will be equal to :-
(1) 0.2 V/m (2) 1 V/m (3) 0.5 V/m (4) 0.1 V/m
[AIEEE - 2011]

CE0134

5. Two electric bulbs marked 25W-220 V and 100 W-220 V are connected in series to a 440 V supply. Which of the bulbs will fuse ?
(1) Neither (2) Both (3) 100 W (4) 25 W
[AIEEE - 2012]

CE0135

6. The supply voltage to a room is 120V. The resistance of the lead wires is $6\ \Omega$. A 60 W bulb is already switched on. What is the decrease of voltage across the bulb, when a 240 W heater is switched on in parallel to the bulb?
(1) zero Volt (2) 2.9 Volt (3) 13.3 Volt (4) 10.04 Volt
[JEE-Main2013]

CE0136

7. This question has Statement I and Statement II. Of the four choice given after the Statements, choose the one that best describes the two Statements.
[JEE-Main 2013]

Statement-I : Higher the range, greater is the resistance of ammeter.

Statement-II : To increase the range of ammeter, additional shunt needs to be used across it.

- (1) Statement-I is true, Statement-II is true, Statement-II is the **correct** explanation of Statement-I
(2) Statement-I is true, Statement-II is true, Statement-II is **not** the correct explanation of Statement-I.
(3) Statement-I is **true**, Statement-II is false.
(4) Statement-I is **false**, Statement-II is true.

CE0137

8. In a large building, there are 15 bulbs of 40 W, 5 bulbs of 100 W, 5 fans of 80 W and 1 heater of 1kW. The voltage of the electric mains is 220 V. The minimum capacity of the main fuse of the building will be : [JEE-Main 2014]

(1) 12 A (2) 14 A (3) 8 A (4) 10 A

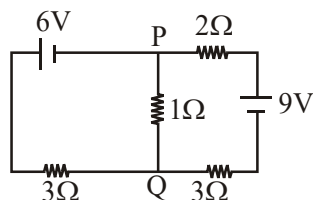
CE0138

9. When 5V potential difference is applied across a wire of length 0.1 m, the drift speed of electrons is $2.5 \times 10^{-4} \text{ ms}^{-1}$. If the electron density in the wire is $8 \times 10^{28} \text{ m}^{-3}$, the resistivity of the material is close to :- [JEE-Main 2015]

(1) $1.6 \times 10^{-6} \Omega\text{m}$ (2) $1.6 \times 10^{-5} \Omega\text{m}$ (3) $1.6 \times 10^{-8} \Omega\text{m}$ (4) $1.6 \times 10^{-7} \Omega\text{m}$

CE0139

10. In the circuit shown, the current in the 1Ω resistor is :- [JEE-Main 2015]



(1) 0.13 A, from Q to P (2) 0.13 A, from P to Q
(3) 1.3 A, from P to Q (4) 0A

CE0140

11. A galvanometer having a coil resistance of 100Ω gives a full scale deflection, when a current of 1 mA is passed through it. The value of the resistance, which can convert this galvanometer into ammeter giving a full scale deflection for a current of 10A, is :- [JEE-Main 2016]

(1) 3Ω (2) 0.01Ω (3) 2Ω (4) 0.1Ω

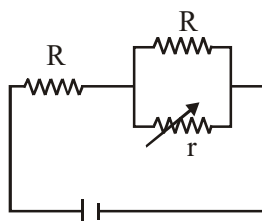
CE0141

12. A 50Ω resistance is connected to a battery of 5V. A galvanometer of resistance 100Ω is to be used as an ammeter to measure current through the resistance, for this a resistance r_s is connected to the galvanometer. Which of the following connections should be employed if the measured current is within 1% of the current without the ammeter in the circuit ? [JEE-Mains (Online) 2016]

(1) $r_s = 1\Omega$ in series with galvanometer (2) $r_s = 0.5\Omega$ in parallel with the galvanometer
(3) $r_s = 0.5\Omega$ in series with the galvanometer (4) $r_s = 1\Omega$ in parallel with galvanometer

CE0142

13. In the circuit shown, the resistance r is a variable resistance. If for $r = fR$, the heat generation in r is maximum then the value of f is : [JEE-Mains (Online) 2016]



(1) 1 (2) $\frac{1}{4}$ (3) $\frac{1}{2}$ (4) $\frac{3}{4}$

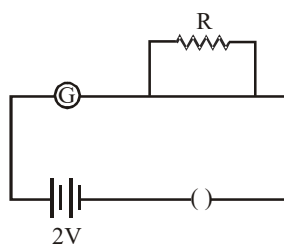
CE0143

14. The resistance of an electrical toaster has a temperature dependence given by $R(T) = R_0 [1 + \alpha(T - T_0)]$ in its range of operation. At $T_0 = 300$ K, $R = 100 \Omega$ and at $T = 500$ K, $R = 120 \Omega$. The toaster is connected to a voltage source at 200 V and its temperature is raised at a constant rate from 300 to 500 K in 30 s. The total work done in raising the temperature is :- [JEE-Mains (Online) 2016]

- (1) $400 \ln \frac{1.5}{1.3}$ J (2) 300 J (3) $400 \ln \frac{5}{6}$ J (4) $200 \ln \frac{2}{3}$ J

CE0144

15. A galvanometer has a 50 division scale. Battery has no internal resistance. It is found that there is deflection of 40 divisions when $R = 2400 \Omega$. Deflection becomes 20 divisions when resistance taken from resistance box is 4900 Ω . Then we can conclude : [JEE-Mains (Online) 2016]



- (1) Resistance of galvanometer is 200 Ω .
 (2) Resistance required on R.B. for a deflection of 10 divisions is 9800 Ω .
 (3) Full scale deflection current is 2 mA.
 (4) Current sensitivity of galvanometer is 20 μ A/division.

CE0145

16. Which of the following statements is false ? [JEE-Main 2017]

- (1) A rheostat can be used as a potential divider
 (2) Kirchhoff's second law represents energy conservation
 (3) Wheatstone bridge is the most sensitive when all the four resistances are of the same order of magnitude.
 (4) In a balanced wheatstone bridge if the cell and the galvanometer are exchanged, the null point is disturbed.

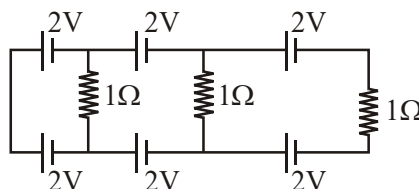
CE0146

17. When a current of 5 mA is passed through a galvanometer having a coil of resistance 15 Ω , it shows full scale deflection. The value of the resistance to be put in series with the galvanometer to convert it into a voltmeter of range 0 – 10 V is :- [JEE-Main 2017]

- (1) $2.535 \times 10^3 \Omega$ (2) $4.005 \times 10^3 \Omega$ (3) $1.985 \times 10^3 \Omega$ (4) $2.045 \times 10^3 \Omega$

CE0147

18. In the above circuit the current in each resistance is :- [JEE-Main 2017]



- (1) 0.5 A (2) 0 A (3) 1 A (4) 0.25 A

CE0148

19. In a potentiometer experiment, it is found that no current passes through the galvanometer when the terminals of the cell are connected across 52 cm of the potentiometer wire. If the cell is shunted by a resistance of $5\ \Omega$, a balance is found when the cell is connected across 40 cm of the wire. Find the internal resistance of the cell. **[JEE-Main 2018]**

(1) $1.5\ \Omega$ (2) $2\ \Omega$ (3) $2.5\ \Omega$ (4) $1\ \Omega$

CE0149

20. On interchanging the resistances, the balance point of a meter bridge shifts to the left by 10 cm. The resistance of their series combination is $1\ \text{k}\Omega$. How much was the resistance on the left slot before interchanging the resistances ? **[JEE-Main 2018]**

(1) $505\ \text{k}\Omega$ (2) $550\ \text{k}\Omega$ (3) $910\ \text{k}\Omega$ (4) $990\ \text{k}\Omega$

CE0150

21. Two batteries with e.m.f $12\ \text{V}$ and $13\ \text{V}$ are connected in parallel across a load resistor of $10\ \Omega$. The internal resistances of the two batteries are $1\ \Omega$ and $2\ \Omega$ respectively. The voltage across the load lies between. **[JEE-Main 2018]**

(1) $11.5\ \text{V}$ and $11.6\ \text{V}$ (2) $11.4\ \text{V}$ and $11.5\ \text{V}$
(3) $11.7\ \text{V}$ and $11.8\ \text{V}$ (4) $11.6\ \text{V}$ and $11.7\ \text{V}$

CE0151

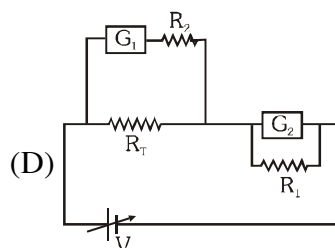
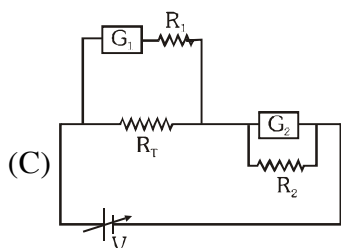
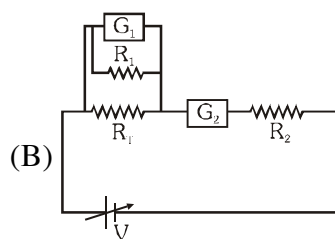
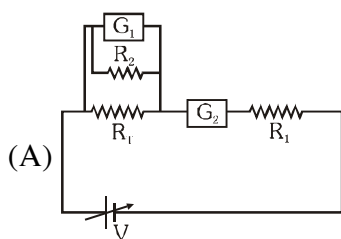
EXERCISE - JA

1. Incandescent bulbs are designed by keeping in mind that the resistance of their filament increases with the increase in temperature. If at room temperature, 100 W, 60 W and 40 W bulbs have filament resistance R_{100} , R_{60} and R_{40} , respectively, the relation between these resistances is [IIT-JEE 2010]

(A) $\frac{1}{R_{100}} = \frac{1}{R_{40}} + \frac{1}{R_{60}}$ (B) $R_{100} = R_{40} + R_{60}$ (C) $R_{100} > R_{60} > R_{40}$ (D) $\frac{1}{R_{100}} > \frac{1}{R_{60}} > \frac{1}{R_{40}}$

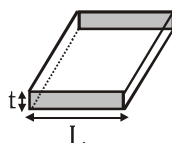
CE0152

2. To verify Ohm's law, a student is provided with a test resistor R_T , a high resistance R_1 , a small resistance R_2 , two identical galvanometers G_1 and G_2 , and a variable voltage source V . The correct circuit to carry out the experiment is :- [IIT-JEE 2010]



CE0153

3. Consider a thin square sheet of side L and thickness t , made of a material of resistivity ρ . The resistance between two opposite faces, shown by the shaded areas in the figure is [IIT-JEE 2010]



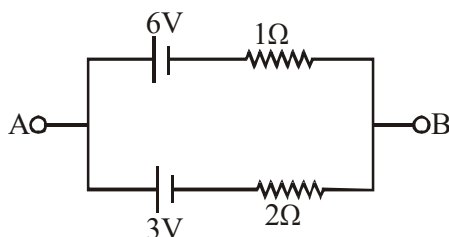
- (A) directly proportional to L (B) directly proportional to t
(C) independent of L (D) independent of t

CE0154

4. When two identical batteries of internal resistance 1Ω each are connected in series across a resistor R , the rate of heat produced in R is J_1 . When the same batteries are connected in parallel across R , the rate is J_2 . If $J_1 = 2.25 J_2$ then the value of R in Ω is [IIT-JEE 2010]

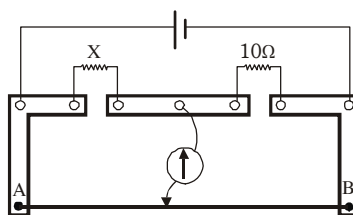
CE0155

5. Two batteries of different emfs and different internal resistances are connected as shown. The voltage across AB in volts is [IIT-JEE 2011]



CE0156

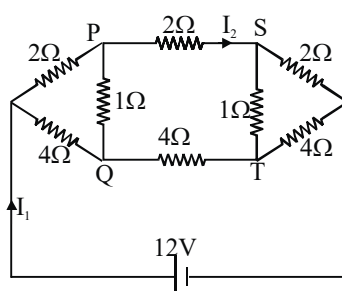
6. A meter bridge is set-up as shown, to determine an unknown resistance 'X' using a standard 10Ω resistor. The galvanometer shows null point when tapping-key is at 52 cm mark. The end-corrections are 1 cm and 2 cm respectively for the ends A and B . The determined value of 'X' is [IIT-JEE 2011]



- (A) 10.2 ohm (B) 10.6 ohm (C) 10.8 ohm (D) 11.1 ohm

CE0157

7. For the resistance network shown in the figure, choose the correct option(s). [IIT-JEE 2012]



- (A) the current through PQ is zero (B) $I_1 = 3\text{ A}$
(C) The potential at S is less than that at Q (D) $I_2 = 2\text{ A}$

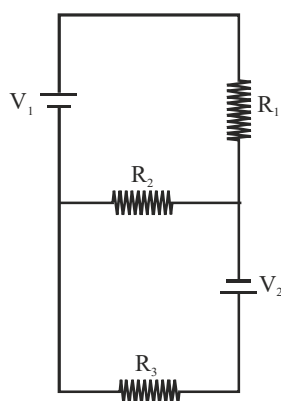
CE0158

8. Heater of an electric kettle is made of a wire of length L and diameter d . It takes 4 minutes to raise the temperature of 0.5 kg water by 40 K. This heater is replaced by a new heater having two wires of the same material, each of length L and diameter $2d$. The way these wires are connected is given in the options. How much time in minutes will it take to raise the temperature of the same amount of water by 40 K ? [JEE Advanced 2014]

- (A) 4 if wires are in parallel (B) 2 if wires are in series
(C) 1 if wires are in series (D) 0.5 if wires are in parallel

CE0159

9. Two ideal batteries of emf V_1 and V_2 and three resistances R_1 , R_2 and R_3 are connected as shown in the figure. The current in resistance R_2 would be zero if :- [JEE Advanced 2014]



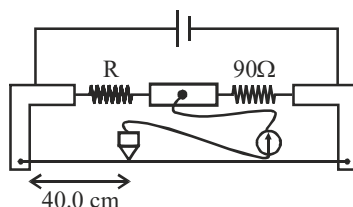
- (A) $V_1 = V_2$ and $R_1 = R_2 = R_3$ (B) $V_1 = V_2$ and $R_1 = 2R_2 = R_3$
(C) $V_1 = 2V_2$ and $2R_1 = 2R_2 = R_3$ (D) $2V_1 = V_2$ and $2R_1 = R_2 = R_3$

CE0160

10. A galvanometer gives full scale deflection with 0.006 A current. By connecting it to a $4990\ \Omega$ resistance, it can be converted into a voltmeter of range 0 - 30 V. If connected to a $\frac{2n}{249}\ \Omega$ resistance, it becomes an ammeter of range 0 - 1.5 A. The value of n is :- [JEE Advanced 2014]

CE0161

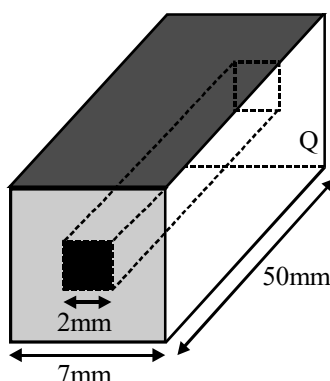
11. During an experiment with a metre bridge, the galvanometer shows a null point when the jockey is pressed at 40.0 cm using a standard resistance of $90\ \Omega$, as shown in the figure. The least count of the scale used in the metre bridge is 1 mm. The unknown resistance is :- [JEE Advanced 2014]



- (A) $60 \pm 0.15\ \Omega$ (B) $135 \pm 0.56\ \Omega$ (C) $60 \pm 0.25\ \Omega$ (D) $135 \pm 0.23\ \Omega$

CE0162

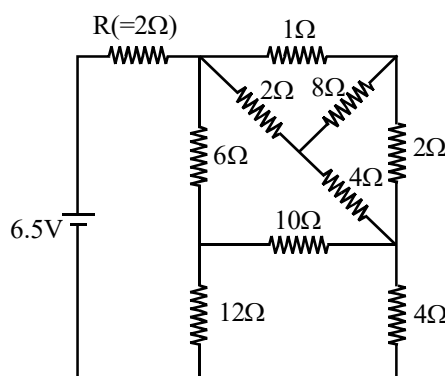
12. In an aluminum (Al) bar of square cross section, a square hole is drilled and is filled with iron (Fe) as shown in the figure. The electrical resistivities of Al and Fe are $2.7 \times 10^{-8} \Omega \text{ m}$ and $1.0 \times 10^{-7} \Omega \text{ m}$, respectively. The electrical resistance between the two faces P and Q of the composite bar is : [JEE Advanced-2015]



- (A) $\frac{2475}{64} \mu\Omega$ (B) $\frac{1875}{64} \mu\Omega$ (C) $\frac{1875}{49} \mu\Omega$ (D) $\frac{2475}{132} \mu\Omega$

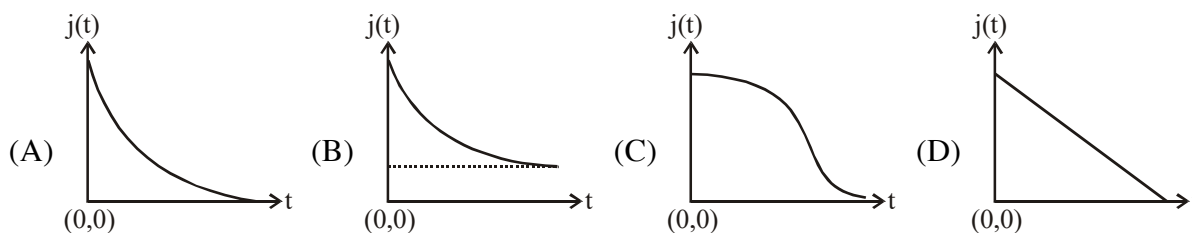
CE0163

13. In the following circuit, the current through the resistor $R (= 2\Omega)$ is I Amperes. The value of I is [JEE Advanced-2015]



CE0164

14. An infinite line charge of uniform electric charge density λ lies along the axis of an electrically conducting infinite cylindrical shell of radius R . At time $t = 0$, the space inside the cylinder is filled with a material of permittivity ϵ and electrical conductivity σ . The electrical conduction in the material follows Ohm's law. Which one of the following graphs best describes the subsequent variation of the magnitude of current density $j(t)$ at any point in the material? [JEE Advanced-2016]



CE0165

15. An incandescent bulb has a thin filament of tungsten that is heated to high temperature by passing an electric current. The hot filament emits black-body radiation. The filament is observed to break up at random locations after a sufficiently long time of operation due to non-uniform evaporation of tungsten from the filament. If the bulb is powered at constant voltage, which of the following statement(s) is(are) true? [JEE Advanced-2016]

- (A) The temperature distribution over the filament is uniform
- (B) The resistance over small sections of the filament decreases with time
- (C) The filament emits more light at higher band of frequencies before it breaks up
- (D) The filament consumes less electrical power towards the end of the life of the bulb

CE0166

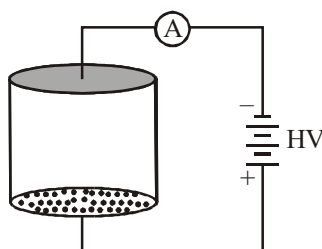
16. Consider two identical galvanometers and two identical resistors with resistance R . If the internal resistance of the galvanometers $R_G < R/2$, which of the following statement(s) about any one of the galvanometers is(are) true? [JEE Advanced-2016]

- (A) The maximum voltage range is obtained when all the components are connected in series
- (B) The maximum voltage range is obtained when the two resistors and one galvanometer are connected in series, and the second galvanometer is connected in parallel to the first galvanometer
- (C) The maximum current range is obtained when all the components are connected in parallel
- (D) The maximum current range is obtained when the two galvanometers are connected in series and the combination is connected in parallel with both the resistors.

CE0167

Paragraph for Questions No. 17 and 18

Consider an evacuated cylindrical chamber of height h having rigid conducting plates at the ends and an insulating curved surface as shown in the figure. A number of spherical balls made of a light weight and soft material and coated with a conducting material are placed on the bottom plate. The balls have a radius $r \ll h$. Now a high voltage source (HV) is connected across the conducting plates such that the bottom plate is at $+V_0$ and the top plate at $-V_0$. Due to their conducting surface, the balls will get charged, will become equipotential with the plate and are repelled by it. The balls will eventually collide with the top plate, where the coefficient of restitution can be taken to be zero due to the soft nature of the material of the balls. The electric field in the chamber can be considered to be that of a parallel plate capacitor. Assume that there are no collision between the balls and the interaction between them is negligible. (Ignore gravity) [JEE Advanced-2016]



17. Which of the following statements is correct ?
- (A) The balls will bounce back to the bottom plate carrying the opposite charge they went up with
(B) the balls will execute simple harmonic motion between the two plates
(C) The balls will bounce back to the bottom plate carrying the same charge they went up with
(D) The balls will stick to the top plate and remain there

CE0168

18. The average current in the steady state registered by the ammeter in the circuit will be :

- (A) Proportional to $V_0^{1/2}$ (B) Proportional to V_0^2
(C) Proportional to the potential V_0 (D) Zero

CE0168

19. Two identical moving coil galvanometer have $10\ \Omega$ resistance and full scale deflection at $2\ \mu\text{A}$ current. One of them is converted into a voltmeter of $100\ \text{mV}$ full scale reading and the other into an Ammeter of $1\ \text{mA}$ full scale current using appropriate resistors. These are then used to measure the voltage and current in the Ohm's law experiment with $R = 1000\ \Omega$ resistor by using an ideal cell. Which of the following statement(s) is/are correct ?

[JEE Advanced-2019]

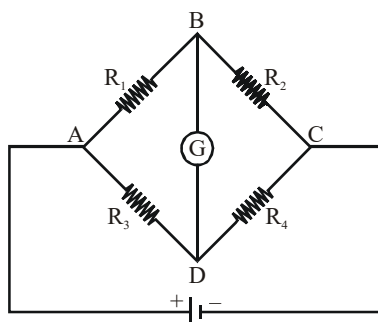
- (1) The measured value of R will be $978\ \Omega < R < 982\ \Omega$.
(2) The resistance of the Voltmeter will be $100\ \text{k}\Omega$.
(3) The resistance of the Ammeter will be $0.02\ \Omega$ (round off to 2^{nd} decimal place)
(4) If the ideal cell is replaced by a cell having internal resistance of $5\ \Omega$ then the measured value of R will be more than $1000\ \Omega$.

CE0169

CURRENT ELECTRICITY

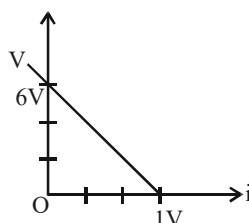
(CBSE Previous Year's Questions)

1. Explain how does the resistivity of a conductor depend upon (i) number density 'n' of free electrons, and (ii) relaxation time ' λ '. [2; CBSE-2004]
2. Two cells of emf 1.5 V and 2 V and internal resistance 1 ohm and 2 ohm respectively are connected in parallel to pass a current in the same direction through an external resistance of 5 ohm.
(i) Draw the circuit diagram.
(ii) Using Kirchhoff's laws, calculate the current through each branch of the circuit and potential difference across the 5 ohm resistor. [3; CBSE-2005]
3. You are given 'n' resistors, each of resistance 'r'. These are first connected to get minimum possible resistance. In the second case, these are again connected differently to get maximum possible resistance. Compute the ratio between the minimum and maximum values of resistance so obtained. [2; CBSE-2006]
4. Draw a circuit diagram using a metre bridge and write the necessary mathematical relation used to determine the value of an unknown resistance. Why cannot such an arrangement be used for measuring very low resistances? [2; CBSE-2006]
5. Which one of the two, an ammeter or a millianuneter, has a higher resistance and why? [2; CBSE-2006]
6. The given figure shows a network of resistances R_1 , R_2 , R_3 and R_4 .



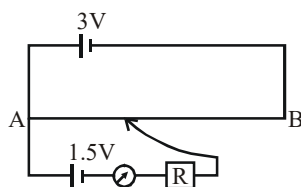
Using Kirchhoff's laws, establish the balance condition for the network. [2; CBSE-2007]

7. The plot of the variation of potential difference across a combination of three identical cells in series, versus current is as shown below. What is the emf of each, cell [1; CBSE-2008]

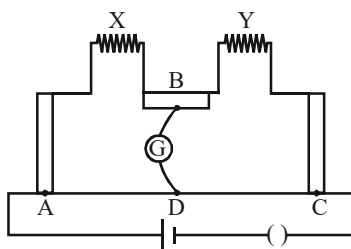


8. Prove that the current density of a metallic conductor is directly proportional to the drift speed of electrons. [3; CBSE-2008]

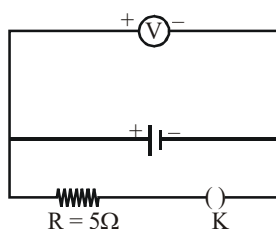
9. A number of identical cells, n each of emf E , internal resistance r connected in series are charged by a d.c source of emf E' , using a resistor R . (i) Draw the circuit arrangement. (ii) Deduce the expressions for (a) the charging current and (b) the potential difference across the combination of the cells. [3; CBSE-2008]
10. A potentiometer wire of length 1m is connected to a drive cell of emf 3V as shown in the figure. When a cell of 1.5 V emf is used in the secondary circuit, the balance point is found to be 60cm. On replacing this cell and using a cell of unknown emf, the balance point shifts to 80cm



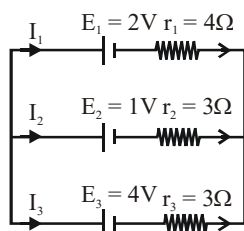
- (i) Calculate unknown emf of the cell. [3; CBSE-2008]
 (ii) Explain with reason, whether the circuit works, if the drive cell is replaced with a cell emf 1V.
 (iii) Does the high resistance R , used in the secondary circuit affect the balance point? Justify your answer.
11. A cell of emf ' E ' and internal resistance ' r ' is connected across a variable resistor ' R '. Plot a graph showing the variation of terminal potential ' V ' with Resistance R . [2; CBSE-2009]
12. Derive an expression for drift velocity of the free electrons in a conductor in terms of relaxation time. [2; CBSE-2009]
13. The figure shows experimental set up of a bridge. When the two unknown resistances X and Y are inserted, the null point D is obtained 40 cm from the end A . when a resistance of 10 is connected in series with X , the null point shifts by 10 cm. find the position of the null point when the 10 resistance is instead connected in series with resistance ' Y '. Determine the values of the resistances X and Y . [3; CBSE-2009]



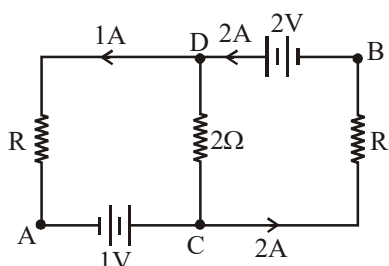
14. Write any two factors on which internal resistance of a cell depends. The reading on a high resistance voltmeter, when a cell is connected across it, is 2.2 V. When the terminals of the cell are also connected to a resistance of 5Ω as shown in the figure, the voltmeter reading drops to 1.8 V. Find the internal resistance of the cell. [3; CBSE-2010]



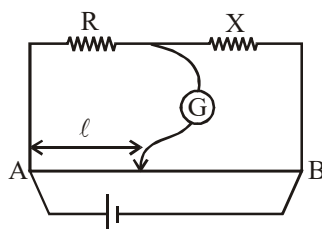
15. State Kirchhoff's rules. Use these rules to write the expressions for the currents I_1 , I_2 and I_3 in the circuit diagram shown. [3; CBSE-2010]



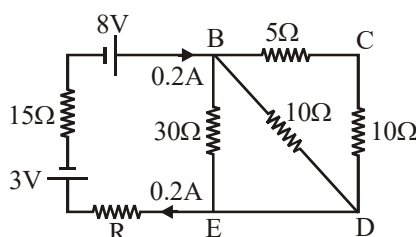
16. In the given circuit, assuming point A to be at zero potential, use Kirchhoff's rules to determine the potential at point B. [2; CBSE-2011]



17. In the meter bridge experiment, balance point was observed at J with $AJ = l$.
 (i) The values of R and X were doubled and then interchanged. What would be the new position of balance point ?
 (ii) If the galvanometer and battery are interchanged at the balance position, how will balance point get affected ? [2; CBSE-2011]



18. Two heating elements of resistance R_1 and R_2 when operated at a constant supply of voltage, V , consume powers P_1 and P_2 respectively. Deduce the expressions for the power of their combination when they are, in turn, connected in (i) series and (ii) parallel across the same voltage supply. [3; CBSE-2011]
19. Calculate the value of the resistance R in the circuit shown in the figure so that the current in the circuit is $0.2A$. What would be the potential difference between points B and E? [3; CBSE-2012]

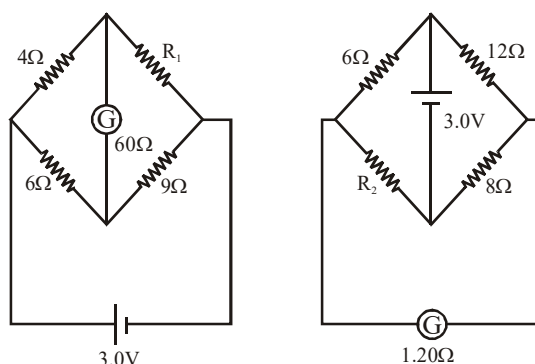


20. Define relaxation time of the free electrons drifting in a conductor. How is it related to the drift velocity of free electrons? Use this relation to deduce the expression for the electrical resistivity of the material. [3; .CBSE-2012]
21. Two identical cells, each of emf E , having negligible internal resistance, are connected in parallel with each other across an external resistance R . What is the current through this resistance? [CBSE-2013]
22. Explain the term 'drift velocity' of electrons in a conductor. Hence obtain the expression for the current through a conductor in terms of 'drift velocity'. [CBSE-2013]

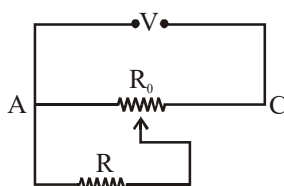
OR

Describe briefly, with the help of a circuit diagram, how a potentiometer is used to determine the internal resistance of a cell.

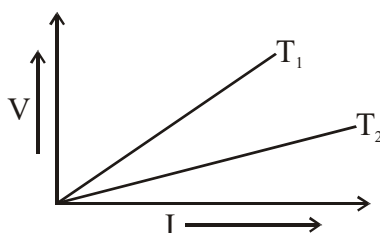
23. Define the current sensitivity of a galvanometer. Write its S. I. unit. Figure shows two circuits each having a galvanometer and a battery of 3V. When the galvanometers in each arrangement do not show any deflection, obtain the ratio R_1/R_2 . [CBSE-2013]



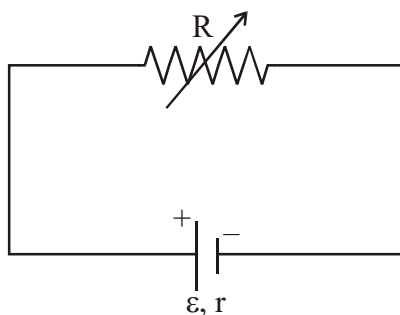
24. Estimate the average drift speed of conduction electrons in a copper wire of cross-sectional area $2.5 \times 10^{-7} \text{ m}^2$ carrying a current of 1.8 A. Assume the density of conduction electrons to be $9 \times 10^{28} \text{ m}^{-3}$. [CBSE-2014]
25. A cell of emf ' E ' and internal resistance ' r ' is connected across a variable resistor ' R '. Plot a graph showing variation of terminal voltage ' V ' of the cell versus the current ' I '. Using the plot, show how the emf of the cell and its internal resistance can be determined. [CBSE-2014]
26. A resistance of $R\Omega$ draws current from a potentiometer as shown in the figure. The potentiometer has a total resistance $R_0\Omega$. A voltage V is supplied to the potentiometer. Derive an expression for the voltage across R when the sliding contact is in the middle of the potentiometer. [CBSE-2014]



27. $V - I$ graph for a metallic wire at two different temperatures T_1 and T_2 is as shown in the figure. Which of the two temperatures is higher and why ? [1; CBSE-2015]

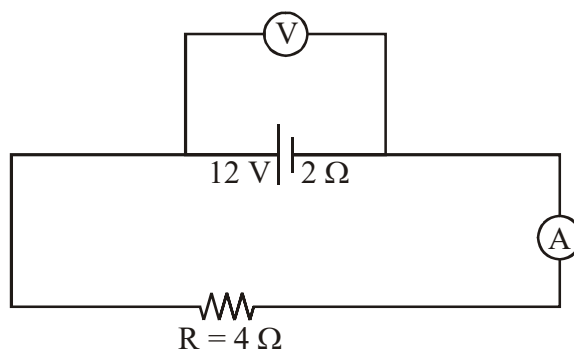


28. A variable resistor R is connected across a cell of emf ε and internal resistance r as shown in the figure. Draw a plot showing the variation of (i) terminal voltage V and (ii) the current I , as a function of R . [2; CBSE-2015]

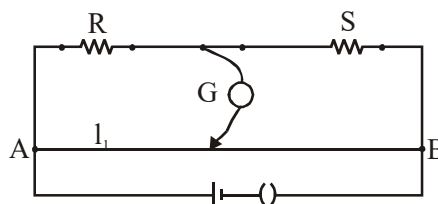


29. A potential difference V is applied across a conductor of length L and diameter D . How is the drift velocity, v_d , of charge carriers in the conductor affected when (i) V is halved, (ii) L is doubled and (iii) D is halved ? Justify your answer in each case. [3; CBSE-2015]
30. Define mobility of a charge carrier. What is its relation with relaxation time ? [1; CBSE-2016]
31. When 5V potential difference is applied across a wire of length 0.1m, the drift speed of electrons is 2.5×10^{-4} m/s. If the electron density in the wire is $8 \times 10^{28} \text{ m}^{-3}$, calculate the resistivity of the material of wire. [2; CBSE-2016]
32. Two identical cells of emf 1.5 V each joined in parallel supply energy to an external circuit consisting of two resistances of 7Ω each joined in parallel. A very high resistance voltmeter reads the terminal voltage of cells to be 1.4 V. Calculate the internal resistance of each cell. [3; CBSE-2016]
33. Nichrome and copper wires of same length and same radius are connected in series. Current I is passed through them. Which wire gets heated up more ? Justify your answer. [CBSE-2017]

34. (a) The potential difference applied across a given resistor is altered so that the heat produced per second increases by a factor of 9. By what factor does the applied potential difference change ?
 (b) In the figure shown, an ammeter A and a resistor of $4\ \Omega$ are connected to the terminals of the source. The emf of the source is 12 V having an internal resistance of $2\ \Omega$. Calculate the voltmeter and ammeter readings. [CBSE-2017]

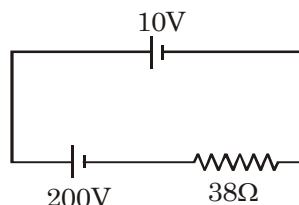


35. (a) Write the principle of working of a metre bridge.
 (b) In a metre bridge, the balance point is found at a distance l_1 with resistance R and S as shown in the figure. [CBSE-2017]



An unknown resistance X is now connected in parallel to the resistance S and the balance point is found at a distance l_2 . Obtain a formula for X in terms of l_1 , l_2 and S.

36. Two electric bulbs P and Q have their resistance in the ratio of 1 : 2. They are connected in series across a battery. Find the ratio of the power dissipation in these bulbs. [CBSE-2018]
 37. A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance $38\ \Omega$ as shown in the figure. Find the value of current in the circuit. [CBSE-2018]



In a potentiometer arrangement for determining the emf of a cell, the balance point of the cell in open circuit is 350 cm. When a resistance of $9\ \Omega$ is used in the external circuit of the cell, the balance point shifts to 300 cm. Determine the internal resistance of the cell.

38. (a) Define the term 'conductivity' of a metallic wire. Write its SI unit. [CBSE-2018]
 (b) Using the concept of free electrons in a conductor, derive the expression for the conductivity of a wire in terms of number density and relaxation time. Hence obtain the relation between current density and the applied electric field E.

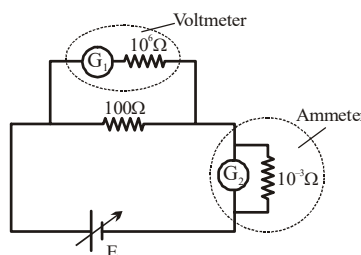
ANSWER KEY

EXERCISE - S-1

1. Ans. $I\rho/A$ 2. Ans. $S = enl\langle v \rangle / j$ 3. Ans. $p = I m_e l/e$ 4. Ans. 3 : 1
 5. Ans. (a) (i) in series, (ii) all in parallel: n^2 . (b)
 (i) Join 1Ω , 2Ω in parallel and the combination in series with 3Ω ,
 (ii) parallel combination of 2Ω and 3Ω in series with 1Ω , (iii) all in series, (iv) all in parallel.
 (c) (i) $(16/3)\Omega$, (ii) $5R$.
 6. Ans. 600Ω 7. Ans. 7.5 mA 8. Ans. 1 V 9. Ans. $I = 2.5\text{ A}$, $V = 3.5\text{ Volts}$
 10. Ans. $20/3\text{ V}$ 11. Ans. 3.5 A 13. Ans. 1Ω 14. Ans. $x = \frac{4}{3}\text{ V}$, $12\frac{1}{3}\text{ V}$, $\frac{1}{15}\text{ A}$
 15. Ans. 5 16. Ans. $\sqrt{R_1 R_2}$ 17. Ans. 25 18. Ans. 3 19. Ans. 12 A , -20 W
 20. Ans. $4/9\text{ kg/sec.}$, 450 sec 21. Ans. 20 ohm 22. Ans. $R_1 = 0.0278\Omega$, $R_2 = 0.25\Omega$, $R_3 = 2.5\Omega$

23. Ans. 233.3Ω , 144 V

24. Ans.



25. Ans. Battery should be connected across A and B. Out put can be taken across the terminals A and C or B and C
 26. Ans. This is true for $r_1 = r_2$; So R_2 given most accurate value 27. Ans. $\frac{10}{3}\Omega$, 5Ω
 28. Ans. (a) +ve, $E_t > E$ (b) -ve 29. Ans. 2.25 V 30. Ans. 4 ohm 31. Ans. 46.67 cm

EXERCISE - S-2

1. Ans. $5 \times 10^{-7}\text{ A}$ 2. Ans. $\alpha_{\text{eff}} = \frac{5}{4}\alpha$ 3. Ans. (i) $\frac{1}{2}i_0 t_0$ (ii) $i = i_0 \left(1 - \frac{t}{t_0}\right)$ (iii) $\frac{R t_0 i_0^2}{3}$
 4. Ans. $\frac{1}{R} = \frac{\pi r^2}{3l} (2\sigma_2 + \sigma_1)$ 5. Ans. (a) $J_0 A/3$; (b) $2J_0 A/3$ 6. Ans. 4Ω
 7. Ans. $-\frac{22}{9}\text{ V}$ 8. Ans. (i) 1.01Ω (ii) $0-5\text{ A}$, $0-10\text{ V}$, (ii) 0.05 A
 9. Ans. 7.5 m , 8.75 m , 6.25 m 10. Ans. 2

EXERCISE - O-1**SINGLE CORRECT TYPE QUESTIONS**

- | | | | |
|--------------|----------------|--------------|--------------|
| 1. Ans. (C) | 2. Ans. (C) | 3. Ans. (D) | 4. Ans. (B) |
| 5. Ans. (A) | 6. Ans. (B) | 7. Ans. (C) | 8. Ans. (B) |
| 9. Ans. (C) | 10. Ans. (D) | 11. Ans. (A) | 12. Ans. (D) |
| 13. Ans. (C) | 14. Ans. (B,C) | 15. Ans. (B) | 16. Ans. (C) |
| 17. Ans. (D) | 18. Ans. (B) | 19. Ans. (D) | 20. Ans. (A) |
| 21. Ans. (C) | 22. Ans. (C) | 23. Ans. (D) | 24. Ans. (B) |
| 25. Ans. (A) | 26. Ans. (A) | 27. Ans. (A) | 28. Ans. (B) |
| 29. Ans. (B) | 30. Ans. (B) | 31. Ans. (A) | 32. Ans. (A) |
| 33. Ans. (C) | 34. Ans. (B) | 35. Ans. (B) | 36. Ans. (C) |
| 37. Ans. (A) | 38. Ans. (C) | 39. Ans. (B) | 40. Ans. (A) |
| 41. Ans. (D) | 42. Ans. (A) | 43. Ans. (D) | 44. Ans. (B) |
| 45. Ans. (B) | 46. Ans. (B) | 47. Ans. (B) | 48. Ans. (C) |
| 49. Ans. (B) | 50. Ans. (C) | 51. Ans. (D) | 52. Ans. (C) |
| 53. Ans. (C) | 54. Ans. (B) | 55. Ans. (A) | 56. Ans. (C) |
| 57. Ans. (A) | 58. Ans. (C) | 59. Ans. (B) | 60. Ans. (A) |
| 61. Ans. (D) | 62. Ans. (B) | 63. Ans. (B) | 64. Ans. (A) |
| 65. Ans. (B) | 66. Ans. (D) | | |

MULTIPLE CORRECT TYPE QUESTIONS

- | | | | |
|--------------------|----------------|-------------------|----------------|
| 67. Ans. (A,B,C,D) | 68. Ans. (A,D) | 69. Ans. (A,B,C) | 70. Ans. (A,C) |
| 71. Ans. (B,C) | 72. Ans. (A,D) | 73. Ans. (A,B, C) | |

MATRIX MATCH TYPE QUESTION

- | | |
|-----------------------------|------------------------|
| 74. Ans. A-P; B-Q; C-R; D-P | 75. Ans. A-Q; B-P; C-P |
|-----------------------------|------------------------|

EXERCISE - O-2**SINGLE CORRECT TYPE QUESTIONS**

- | | | | | |
|-------------|-------------|-------------|-------------|-------------|
| 1. Ans. (D) | 2. Ans. (B) | 3. Ans. (B) | 4. Ans. (B) | 5. Ans. (A) |
| 6. Ans. (A) | 7. Ans. (C) | 8. Ans. (A) | 9. Ans. (A) | |

MULTIPLE CORRECT TYPE QUESTIONS

- | | | | | |
|-----------------|----------------|------------------|-------------------|------------------|
| 10. Ans. (B, D) | 11. Ans. (A,C) | 12. Ans. (A,C,D) | 13. Ans. (A, B,D) | 14. Ans. (A,B,C) |
|-----------------|----------------|------------------|-------------------|------------------|

EXERCISE - JM

- | | | | | |
|--------------|--------------|--------------|--------------|--------------|
| 1. Ans. (1) | 2. Ans. (4) | 3. Ans. (2) | 4. Ans. (4) | 5. Ans. (4) |
| 6. Ans. (4) | 7. Ans. (4) | 8. Ans. (1) | 9. Ans. (2) | 10. Ans. (2) |
| 12. Ans. (2) | 13. Ans. (3) | 14. Ans. (1) | 15. Ans. (4) | 16. Ans. (4) |
| 17. Ans. (3) | 18. Ans. (2) | 19. Ans. (1) | 20. Ans. (2) | 21. Ans. (1) |

EXERCISE - JA

- | | | | | |
|----------------|-------------------|---------------|-----------------|----------------|
| 1. Ans. (D) | 2. Ans. (C) | 3. Ans. (C) | 4. Ans. 4 | 5. Ans. 5 |
| 6. Ans. (B) | 7. Ans. (A,B,C,D) | 8. Ans. (B,D) | 9. Ans. (A,B,D) | 10. Ans. 5 |
| 11. Ans. (C) | 12. Ans. (B) | 13. Ans. 1 | 14. Ans. (A) | 15. Ans. (C,D) |
| 16. Ans. (B,C) | 17. Ans. (A) | 18. Ans. (B) | 19. Ans. (A,C) | |

Important Notes

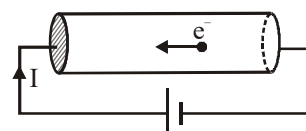
CURRENT ELECTRICITY

KEY CONCEPT

In previous chapters we deal largely with electrostatics that is, with charges at rest. With this chapter we begin to focus on electric currents, that is, charges in motion.

ELECTRIC CURRENT

Electric charges in motion constitute an electric current. Any medium having practically free electric charges, free to migrate is a conductor of electricity. The electric charge flows from higher potential energy state to lower potential energy state.



Positive charge flows from higher to lower potential and negative charge flows from lower to higher. Metals such as gold, silver, copper, aluminium etc. are good conductors.

When charge flows in a conductor from one place to the other, then the rate of flow of charge is called electric current (I). When there is a transfer of charge from one point to other point in a conductor, we say that there is an electric current through the area. If the moving charges are positive, the current is in the direction of motion of charge. If they are negative the current is opposite to the direction of motion. If a charge ΔQ crosses an area in time Δt then the average electric current through the area, during this time as

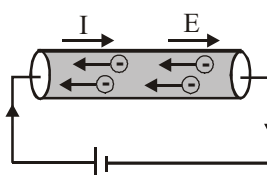
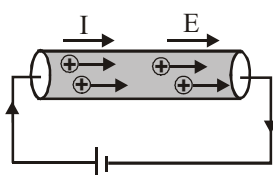
• Average current $I_{av} = \frac{\Delta Q}{\Delta t}$

• Instantaneous current $I = \lim_{\Delta t \rightarrow 0} \frac{\Delta Q}{\Delta t} = \frac{dQ}{dt}$

- Current is a fundamental quantity with dimension $[M^0 L^0 T^0 A^1]$
- Current is a scalar quantity with its SI unit **ampere**.

Ampere : The current through a conductor is said to be one ampere if one coulomb of charge is flowing per second through a cross-section of wire.

- The conventional direction of current is the direction of flow of positive charge or applied field. It is opposite to direction of flow of negatively charged electrons.



- The conductor remains uncharged when current flows through it because the charge entering at one end per second is equal to charge leaving the other end per second.
- For a given conductor current does not change with change in its cross-section because current is simply rate of flow of charge.
- If there are n particles per unit volume each having a charge q and moving with velocity v then

current through cross-sectional area A is $I = \frac{\Delta q}{\Delta t} = nqvA$

- If a charge q is moving in a circle of radius r with speed v then its time period is $T = 2\pi r/v$. The equivalent current $I = \frac{q}{T} = \frac{qv}{2\pi r}$.

Behavior of conductor in absence of applied potential difference :

In absence of applied potential difference electrons have random motion. The average displacement and average velocity is zero. There is no flow of current due to thermal motion of free electrons in a conductor.

The free electrons present in a conductor gain energy from temperature of surrounding and move randomly in the conductor.

The speed gained by virtue of temperature is called as thermal speed of an electron $\frac{1}{2}mv_{\text{rms}}^2 = \frac{3}{2}kt$

So thermal speed $v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$ where m is mass of electron

At room temperature $T = 300 \text{ K}$, $v_{\text{rms}} = 10^5 \text{ m/s}$

- **Mean free path λ :** ($\lambda \sim 10 \text{ \AA}$) $= \lambda = \frac{\text{total distance travelled}}{\text{number of collisions}}$
- **Relaxation time :** The time taken by an electron between two successive collisions is called as relaxation time τ : ($\tau \sim 10^{-14} \text{ s}$), Relaxation time : $\tau = \frac{\text{total time taken}}{\text{number of collisions}}$

Behavior of conductor in presence of applied potential difference :

When two ends of a conductors are joined to a battery then one end is at higher potential and another at lower potential. This produces an electric field inside the conductor from point of higher to lower potential

$E = \frac{V}{L}$ where V = emf of the battery, L = length of the conductor.

The field exerts an electric force on free electrons causing acceleration of each electron.

Acceleration of electron $\vec{a} = \frac{\vec{F}}{m} = \frac{-e\vec{E}}{m}$

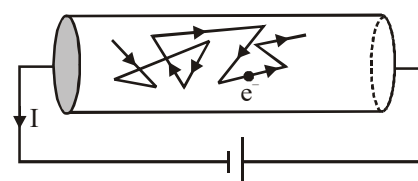
DRIFT VELOCITY

Drift velocity is defined as the velocity with which the free electrons get drifted towards the positive terminal under the effect of the applied external electric field. In addition to its thermal velocity, due to acceleration given by applied electric field, the electron acquires a velocity component in a direction opposite to the direction of the electric field. The gain in velocity due to the applied field is very small and is lost in the next collision.

At any given time, an electron has a velocity $\vec{v}_1 = \vec{u}_1 + \vec{a}\tau_1$, where $\vec{u}_1$ = the thermal velocity and $\vec{a}\tau_1$ = the velocity acquired by the electron under the influence of the applied electric field.

τ_1 = the time that has elapsed since the last collision. Similarly, the velocities of the other electrons are $\vec{v}_2 = \vec{u}_2 + \vec{a}\tau_2$, $\vec{v}_3 = \vec{u}_3 + \vec{a}\tau_3$, ..., $\vec{v}_N = \vec{u}_N + \vec{a}\tau_N$.

The average velocity of all the free electrons in the conductor is equal to the drift velocity $\vec{v}_d$ of the free electrons



Under the action of electric field :
Random motion of an electron
with superimposed drift

$$\vec{v}_d = \frac{\vec{v}_1 + \vec{v}_2 + \vec{v}_3 + \dots + \vec{v}_N}{N} = \frac{(u_1 + \vec{a}\tau_1) + (\vec{u}_2 + \vec{a}\tau_2) + \dots + (\vec{u}_N + \vec{a}\tau_N)}{N} = \frac{(\vec{u}_1 + \vec{u}_2 + \dots + \vec{u}_N)}{N} + \vec{a} \left(\frac{\tau_1 + \tau_2 + \dots + \tau_N}{N} \right)$$

$$\therefore \frac{\vec{u}_1 + \vec{u}_2 + \dots + \vec{u}_N}{N} = 0 \quad \therefore \vec{v}_d = \vec{a} \left(\frac{\tau_1 + \tau_2 + \dots + \tau_N}{N} \right) \Rightarrow \vec{v}_d = \vec{a}\tau = -\frac{e\vec{E}}{m}\tau$$

Note : Order of drift velocity is 10^{-4} m/s.

Relation between current and drift velocity :

Let n = number density of free electrons and A = area of cross-section of conductor.

Number of free electrons in conductor of length $L = nAL$, Total charge on these free electrons

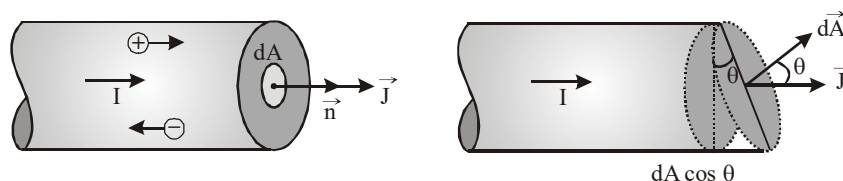
$$\Delta q = neAL$$

Time taken by drifting electrons to cross conductor $\Delta t = \frac{L}{v_d}$ $\therefore$ current $I = \frac{\Delta q}{\Delta t} = neAL \left(\frac{v_d}{L} \right) = neAv_d$

CURRENT DENSITY (J)

Current is a macroscopic quantity and deals with the overall rate of flow of charge through a section. To specify the current with direction in the microscopic level at a point, the term current density is introduced. Current density at any point inside a conductor is defined as a vector having magnitude equal to current per unit area surrounding that point. Remember area is normal to the direction of charge flow (or current passes) through that point.

- Current density at point P is given by $\vec{J} = \frac{dI}{dA} \vec{n}$



- If the cross-sectional area is not normal to the current, but makes an angle θ with the direction of current then $J = \frac{dI}{dA \cos \theta} \Rightarrow dI = J dA \cos \theta = \vec{J} \cdot d\vec{A} \Rightarrow I = \int \vec{J} \cdot d\vec{A}$
- Current density $\vec{J}$ is a vector quantity. It's direction is same as that of $\vec{E}$. It's S.I. unit is ampere/m² and dimension $[L^{-2}A]$.

Ex. The current density at a point is $\vec{J} = (2 \times 10^4 \hat{j}) \text{ Jm}^{-2}$.

Find the rate of charge flow through a cross sectional area $\vec{S} = (2\hat{i} + 3\hat{j}) \text{ cm}^2$

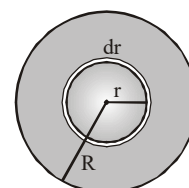
Sol. The rate of flow of charge = current = $I = \int \vec{J} \cdot d\vec{S} \Rightarrow I = \vec{J} \cdot \vec{S} = (2 \times 10^4) [\hat{j} \cdot (2\hat{i} + 3\hat{j})] \times 10^{-4} \text{ A} = 6 \text{ A}$

Ex. A potential difference applied to the ends of a wire made up of an alloy drives a current through it. The current density varies as $J = 3 + 2r$, where r is the distance of the point from the axis. If R be the radius of the wire, then the total current through any cross section of the wire.

Sol. Consider a circular strip of radius r and thickness dr

$$dI = \vec{J} \cdot d\vec{S} = (3 + 2r)(2\pi r dr) \cos 0^\circ = 2\pi(3r + 2r^2) dr$$

$$I = \int_0^R 2\pi(3r + 2r^2) dr = 2\pi \left(\frac{3r^2}{2} + \frac{2}{3}r^3 \right)_0^R = 2\pi \left(\frac{3R^2}{2} + \frac{2R^3}{3} \right) \text{ units}$$



RELATION BETWEEN CURRENT DENSITY, CONDUCTIVITY AND ELECTRIC FIELD

Let the number of free electrons per unit volume in a conductor = n

Total number of electrons in dx distance = $n(Adx)$

Total charge $dQ = n(Adx)e$

$$\text{Current } I = \frac{dQ}{dt} = nAe \frac{dx}{dt} = neAv_d, \text{ Current density } J = \frac{I}{A} = nev_d$$

$$= ne \left(\frac{eE}{m} \right) \tau \because v_d = \left(\frac{eE}{m} \right) \tau \Rightarrow J = \left(\frac{ne^2 \tau}{m} \right) E \Rightarrow J = \sigma E, \text{ where conductivity } \sigma = \frac{ne^2 \tau}{m}$$

σ depends only on the material of the conductor and its temperature.

In vector form $\vec{J} = \sigma \vec{E}$ Ohm's law (at microscopic level)

RELATION BETWEEN POTENTIAL DIFFERENCE AND CURRENT (Ohm's Law)

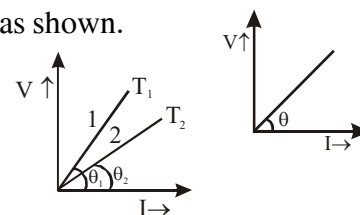
If the physical conditions of the conductor (length, temperature, mechanical strain etc.) remains same, then the current flowing through the conductor is directly proportional to the potential difference across its two ends i.e. $I \propto V \Rightarrow V = IR$ where R is a proportionality constant, known as electric resistance. Ohm's law (at macroscopic level)

- Ohm's law is not a universal law. The substances, which obey ohm's law are known as ohmic.
- Graph between V and I for a metallic conductor is a straight line as shown.

$$\text{Slope of the line} = \tan \theta = \frac{V}{I} = R$$

At different temperatures V - I curves are different.

Here $\tan \theta_1 > \tan \theta_2$ So $R_1 > R_2$ i.e. $T_1 > T_2$



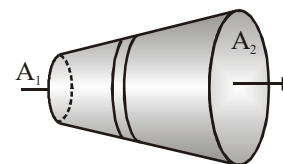
- 1 ampere of current means the flow of 6.25×10^{18} electrons per second through any cross section of conductor.
- Current is a scalar quantity but current density is a vector quantity.
- Order of free electron density in conductors = 10^{28} electrons/ m^3

Terms	Thermal speed v_T	Mean free path λ	Relaxation time τ	Drift speed v_d
Order	10^5 m/s	$10 \text{ \AA}$	10^{-14} s	10^{-4} m/s

- If a steady current flows in a metallic conductor of non uniform cross section.

Current density and drift velocity depends on area

$$I_1 = I_2, A_1 < A_2 \Rightarrow J_1 > J_2, E_1 > E_2, v_{d1} > v_{d2}$$



- If the temperature of the conductor increases, the amplitude of the vibrations of the positive ions in the conductor also increase. Due to this, the free electrons collide more frequently with the vibrating ions and as a result, the average relaxation time decreases.

Ex. What will be the number of electron passing through a heater wire in one minute, if it carries a current of 8 A.

Sol. $I = \frac{Ne}{t} \Rightarrow N = \frac{It}{e} = \frac{8 \times 60}{1.6 \times 10^{-19}} = 3 \times 10^{21}$ electrons

Ex. A current of 1.34 A exists in a copper wire of cross-section 1.0 mm^2 . Assuming each copper atom contributes one free electron. Calculate the drift speed of the free electrons in the wire. The density of copper is 8990 kg/m^3 and atomic mass = 63.50.

Sol. Mass of 1 m^3 volume of the copper is = $8990 \text{ kg} = 8990 \times 10^3 \text{ g}$

$$\text{Number of moles in } 1 \text{ m}^3 = \frac{8990 \times 10^3}{63.5} = 1.4 \times 10^5$$

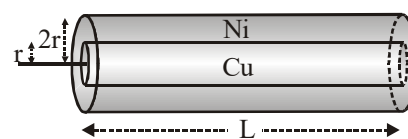
Since each mole contains 6×10^{23} atoms therefore number of atoms in 1 m^3

$$n = (1.4 \times 10^5) \times (6 \times 10^{23}) = 8.4 \times 10^{28}$$

$$\therefore I = neAv_d \therefore v_d = \frac{I}{neA} = \frac{1.34}{8.4 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-6}} = 10^{-4} \text{ m/s} = 0.1 \text{ mm/s} (\because 1 \text{ mm}^2 = 10^{-6} \text{ m}^2)$$

Ex. A copper wire of length ' ℓ ' and radius ' r ' is nickel plated till its final radius is $2r$. If the resistivity of the copper and nickel are ρ_{Cu} and ρ_{Ni} , then find the equivalent resistance of wire?

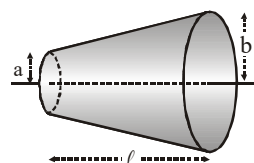
Sol. $R = \rho \frac{\ell}{A}$; Resistance of copper wire $R_{\text{Cu}} = \rho_{\text{Cu}} \frac{\ell}{\pi r^2}$
($\because A = \pi r^2$)



$$\therefore A_{\text{Ni}} = \pi(2r)^2 - \pi r^2 = 3\pi r^2 \Rightarrow \text{Resistance of Nickel wire } R_{\text{Ni}} = \rho_{\text{Ni}} \frac{\ell}{3\pi r^2}$$

$$\text{Both wire are connected in parallel. So equivalent resistance } R = \frac{R_{\text{Cu}} R_{\text{Ni}}}{R_{\text{Cu}} + R_{\text{Ni}}} = \left(\frac{\rho_{\text{Cu}} \rho_{\text{Ni}}}{3\rho_{\text{Cu}} + \rho_{\text{Ni}}} \right) \frac{\ell}{\pi r^2}$$

Ex. Figure shows a conductor of length ℓ carrying current I and having a circular cross-section. The radius of cross section varies linearly from a to b . Assuming that $(b - a) \ll \ell$. Calculate current density at distance x from left end.



Sol. Since radius at left end is a and that of right end is b ,
Therefore increase in radius over length ℓ is $(b - a)$.

$$\text{Hence rate of increase of radius per unit length} = \left(\frac{b-a}{\ell} \right) \text{ Increase in radius over length } x = \left(\frac{b-a}{\ell} \right) x$$

$$\text{Since radius at left end is } a \text{ so radius at distance } x, r = a + \left(\frac{b-a}{\ell} \right) x$$

$$\text{Area at this particular section } A = \pi r^2 = \pi \left[a + \left(\frac{b-a}{\ell} \right) x \right]^2$$

$$\text{Hence current density } J = \frac{I}{A} = \frac{I}{\pi r^2} = \frac{I}{\pi \left[a + \frac{x(b-a)}{\ell} \right]^2}$$

RESISTANCE

The resistance of a conductor is the opposition which the conductor offers to the flow of charge. When a potential difference is applied across a conductor, free electrons get accelerated and collide with positive ions and their motion is thus opposed. This opposition offered by the ions is called resistance of the conductor.

Resistance is the property of a conductor by virtue of which it opposes the flow of current in it.

Unit : ohm, volt/ampere,

Dimension = $M L^2 T^{-3} A^{-2}$

Resistance depends on :

- Length of the conductor ($R \propto \ell$)

- Area of cross-section of the conductor $R \propto \frac{1}{A}$

- Nature of material of the conductor $R = \frac{\rho \ell}{A}$

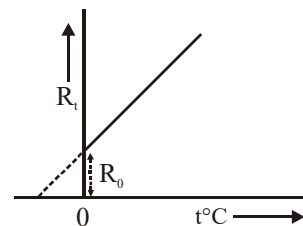
- Temperature $R_t = R_0 (1 + \alpha \Delta t)$

Where

R_t = Resistance at $t^\circ C$, R_0 = Resistance at $0^\circ C$

Δt = Change in temperature, α = Temperature coefficient of resistance

[For metals : α positive for semiconductors : α negative]

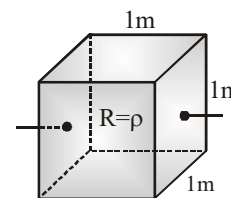


RESISTIVITY

Resistivity : $\rho = RA/\ell$ if $\ell = 1m$, $A = 1m^2$ then $\rho = R$

The specific resistance of a material is equal to the resistance of the wire of that material with unit cross – section area and unit length.

Resistivity depends on (i) Nature of material (ii) Temperature of material ρ does not depend on the size and shape of the material because it is the characteristic property of the conductor material.



Specific use of conducting materials :

- The **heating element** of devices like heater, geyser, press etc are made of **nichrome** because it has high resistivity and high melting point. It does not react with air and acquires steady state when red hot at $800^\circ C$.
- **Fuse wire** is made of **tin lead alloy** because it has low melting point and low resistivity. The fuse is used in series, and melts to produce open circuit when current exceeds the safety limit.
- **Resistances** of resistance box are made of **manganin** or **constantan** because they have moderate resistivity and very small temperature coefficient of resistance. The resistivity is nearly independent of temperature.
- The **filament of bulb** is made up of **tungsten** because it has low resistivity, high melting point of $3300 K$ and gives light at $2400 K$. The bulb is filled with inert gas because at high temperature it reacts with air forming oxide.
- The **connection wires** are made of **copper** because it has low resistance and resistivity.

KIRCHHOFF'S LAW

There are two laws given by Kirchhoff for determination of potential difference and current in different branches of any complicated network.

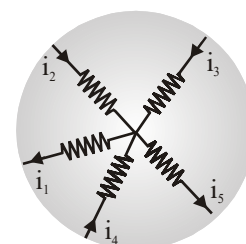
Law of conservation of charge is a consequence of continuity equation

- **First law (Junction Law or Current Law)**

In an electric circuit, the algebraic sum of the current meeting at any junction in the circuit is zero or Sum of the currents entering the Junction is equal to sum of the current leaving the Junction. $\Sigma i = 0$

$$i_1 - i_2 - i_3 - i_4 + i_5 = 0 \Rightarrow i_1 + i_5 = i_2 + i_3 + i_4$$

This is based on law of conservation of charge.



• **Second law (loop rule or potential law)**

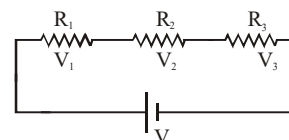
In any closed circuit the algebraic sum of all potential differences and e.m.f. is zero. $\Sigma E - \Sigma IR = 0$ while moving from negative to positive terminal inside the cell, e.m.f. is taken as positive while moving in the direction of current in a circuit the potential drop (i.e. IR) across resistance is taken as positive.

This law is based on law of conservation of energy.

COMBINATION OF RESISTANCE

Series Combination

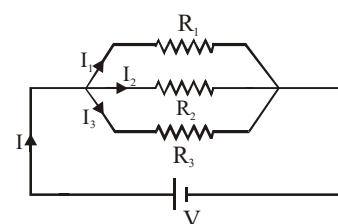
- Same current passes through each resistance
- Voltage across each resistance is directly proportional to its value
 $V_1 = IR_1, V_2 = IR_2, V_3 = IR_3$
- Sum of the voltage across resistance is equal to the voltage applied across the circuit.
 $V = V_1 + V_2 + V_3 \Rightarrow IR = IR_1 + IR_2 + IR_3 \Rightarrow R = R_1 + R_2 + R_3$ Where R = equivalent resistance



Parallel Combination

- There is same drop of potential across each resistance.
- Current in each resistance is inversely proportional to the

value of resistance. $I_1 = \frac{V}{R_1}, I_2 = \frac{V}{R_2}, I_3 = \frac{V}{R_3}$



- Current flowing in the circuit is sum of the currents in individual resistance.

$$I = I_1 + I_2 + I_3 \Rightarrow \frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} \Rightarrow \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

- To get maximum resistance, resistance must be connected in series and in series the resultant is greater than largest individual.
- To get minimum resistance, resistance must be connected in parallel and the equivalent resistance of parallel combination is lower than the value of lowest resistance in the combination.

• **In general :**

- Resistivity of alloys is greater than their metals.
- Temperature coefficient of alloys is lower than pure metals.
- The resistivity of an insulator (e.g. amber) is greater than the metal by a factor of 10^{22}

Ex. The resistance $4R, 16R, 64R \dots \infty$ are connected in series. Find their equivalent resistance.

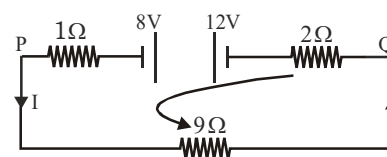
Sol. Resultant of the given combination $R_{eq} = 4R + 16R + 64R + \dots \infty = \infty$

Ex. Resistance $R, 2R, 4R, 8R \dots \infty$ are connected in parallel. What is their resultant resistance ?

Sol. $\frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{2R} + \frac{1}{4R} + \frac{1}{8R} \dots \infty = \frac{1}{R} \left[1 + \frac{1}{2} + \frac{1}{4} + \dots \infty \right] = \frac{1}{R} \left[\frac{1}{1 - \frac{1}{2}} \right] = \frac{2}{R} \Rightarrow R_{eq} = \frac{R}{2}$

Ex. In the given circuit calculate potential difference between the points P and Q.

Sol. Applying Kirchhoff's voltage law (KVL)



$$12 - 8 = (1)I + (9)I + (2)I \Rightarrow I = \frac{1}{3} \text{ A}$$

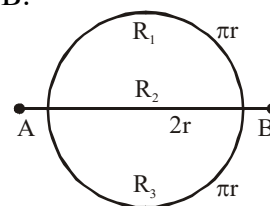
Potential difference between the points P and Q, $V_P - V_Q = 9 \times \frac{1}{3} = 3 \text{ volt}$

Ex. A wire of $\rho_L = 10^{-6} \Omega/\text{m}$ is turned in the form of a circle of diameter 2 m. A piece of same material is connected in diameter AB. Then find resistance between A and B.

Sol. $\therefore R = \rho_L \times \text{length}$

$$\therefore R_1 = \pi \times 10^{-6} \Omega, R_2 = 2 \times 10^{-6} \Omega, R_3 = \pi \times 10^{-6} \Omega$$

$$\frac{1}{R_{AB}} = \frac{1}{\pi \times 10^{-6}} + \frac{1}{2 \times 10^{-6}} + \frac{1}{\pi \times 10^{-6}}; R_{AB} = 0.88 \times 10^{-6} \text{ ohm.}$$



CELL

Cell convert chemical energy into electrical energy.

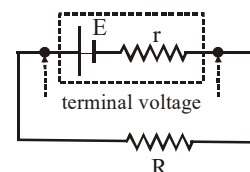
ELECTRO MOTIVE FORCE (E. M. F.)

The potential difference across the terminals of a cell when it is not giving any current is called emf of the cell. The energy given by the cell in the flow of unit charge in the whole circuit (including the cell) is called the emf of the cell.

- emf depends on : (i) nature of electrolyte (ii) metal of electrodes
- emf does not depend on : (i) area of plates (ii) distance between the electrodes (iii) quantity of electrolyte (iv) size of cell

TERMINAL VOLTAGE (V)

- When current is drawn through the cell or current is supplied to cell then, the potential difference across its terminals called terminal voltage.
- When I current is drawn from cell, then terminal voltage is less than it's e.m.f. $V = E - Ir$

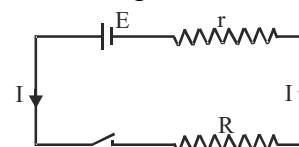


- **Terminal Potential Difference :** The potential difference between the two electrodes of a cell in a closed circuit i.e. when current is being drawn from the cell is called terminal potential difference.

(a) When cell is discharging :

Current inside the cell is from cathode to anode.

$$\text{Current } I = \frac{E}{r + R} \Rightarrow E = IR + Ir = V + Ir \Rightarrow V = E - Ir$$

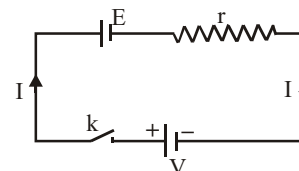


When current is drawn from the cell potential difference is less than emf of cell. Greater is the current drawn from the cell smaller is the terminal voltage. When a large current is drawn from a cell its terminal voltage is reduced.

(b) When cell is charging :

Current inside the cell is from anode to cathode.

$$\text{Current } I = \frac{V - E}{r} \Rightarrow V = E + Ir$$



During charging terminal potential difference is greater than emf of cell.

(c) When cell is in open circuit :

$$\text{In open circuit } R = \infty \therefore I = \frac{E}{R + r} = 0 \Rightarrow V = E$$

In open circuit terminal potential difference is equal to emf and is the maximum potential difference which a cell can provide.

(d) **When cell is short circuited :**

In short circuit $R = 0 \Rightarrow I = \frac{E}{R + r} = \frac{E}{r}$ and $V = IR = 0$

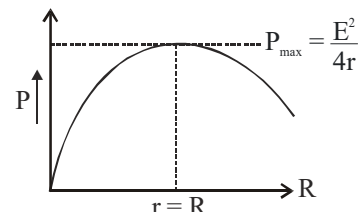
In short circuit current from cell is maximum and terminal potential difference is zero.

(e) **Power transferred to load by cell :**

$$P = I^2 R = \frac{E^2 R}{(r + R)^2} \Rightarrow P = P_{\max} \text{ if } \frac{dP}{dR} = 0 \Rightarrow r = R$$

Power transferred by cell to load is maximum when

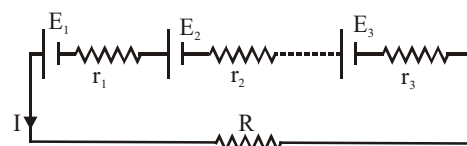
$$r = R \text{ and } P_{\max} = \frac{E^2}{4r} = \frac{E^2}{4R}$$



COMBINATION OF CELLS

• **Series combination**

When the cells are connected in series the total e.m.f. of the series combination is equal to the sum of the e.m.f.'s of the individual cells and internal resistance of the cells also come in series.



Equivalent internal resistance $r = r_1 + r_2 + r_3 + \dots$

Equivalent emf $= E = E_1 + E_2 + E_3 + \dots$

$$\text{Current } I = \frac{E_{\text{net}}}{r_{\text{net}} + R}, \text{ If all } n \text{ cell are identical then } I = \frac{nE}{nr + R}$$

• If $nr \gg R$, $I = \frac{E}{r} \approx$ current from one cell • If $nr \ll R$, $I = \frac{nE}{R} \approx n \times$ current from one cell

• **Parallel combination**

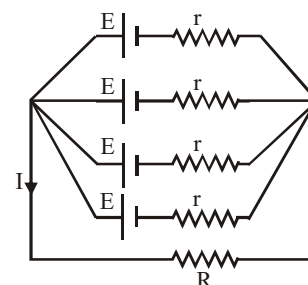
When the cells are connected in parallel, the total e.m.f. of the parallel combination remains equal to the e.m.f. of a single cell and internal resistance of the cell also come in parallel. If m identical cell

connected in parallel then total internal resistance of this combination $r_{\text{net}} = \frac{r}{m}$. Total e.m.f. of this combination $= E$

$$\text{Current in the circuit } I = \frac{E}{R + \frac{r}{m}} = \frac{mE}{mR + r}$$

If $r \ll mR$ $I = E/R =$ Current from one cell

If $r \gg mR$ $I = \frac{mE}{r} = m \times$ current from one cell



• **Mixed combination**

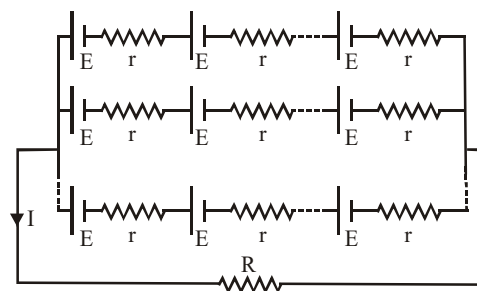
If n cells connected in series and there are m such branches in the circuit then total number of identical cell in this circuit is nm . The internal resistance of the cells connected in a row $= nr$. Since there are such m rows,

Total internal resistance of the circuit $r_{\text{net}} = \frac{nr}{m}$

Total e.m.f. of the circuit = total e.m.f. of the cells connected in a row

$$E_{\text{net}} = nE$$

$$\text{Current in the circuit } I = \frac{E_{\text{net}}}{R + r_{\text{net}}} = \frac{nE}{R + \frac{nr}{m}}$$



Current in the circuit is maximum when external resistance in the circuit is equal to the total internal

resistance of the cells $R = \frac{nr}{m}$

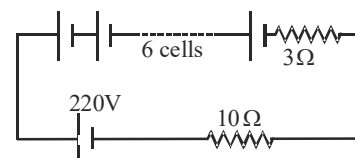
- At the time of charging a cell when current is supplied to the cell, the terminal voltage is greater than the e.m.f. E , $V = E + Ir$
- Series combination is useful when internal resistance is less than external resistance of the cell.
- Parallel combination is useful when internal resistance is greater than external resistance of the cell.
- Power in R (given resistance) is maximum, if its value is equal to net resistance of remaining circuit.
- Internal resistance of ideal cell = 0
- if external resistance is zero then current given by circuit is maximum.

Ex. A battery of six cells each of e.m.f. 2 V and internal resistance 0.5Ω is being charged by D. C. mains of e.m.f. 220 V by using an external resistance of 10Ω . What will be the charging current.

Sol. Net e.m.f. of the battery = 12 V and total internal resistance = 3Ω

Total resistance of the circuit = $3 + 10 = 13 \Omega$

$$\text{Charging current } I = \frac{\text{Net e.m.f.}}{\text{total resistance}} = \frac{220 - 12}{13} = 16 \text{ A}$$



Ex. A battery of six cells each of e.m.f. 2 V and internal resistance 0.5Ω is being charged by D. C. mains of e.m.f. 220 V by using an external resistance of 10Ω . What is the potential difference across the battery ?

Sol. In case of charging of battery, terminal potential $V = E + Ir = 12 + 16 \times 3 = 60 \text{ volt}$.

Ex. Four identical cells each of e.m.f. 2 V are joined in parallel providing supply of current to external circuit consisting of two 15Ω resistors joined in parallel. The terminal voltage of the equivalent cell as read by an ideal voltmeter is 1.6 V calculate the internal resistance of each cell.

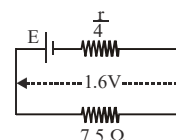
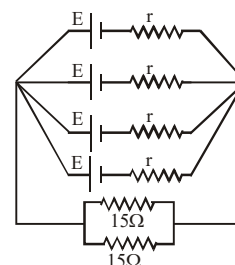
Sol. Total internal resistance of the combination $r_{\text{eq}} = \frac{r}{4}$

Total e.m.f. $E_{\text{eq}} = 2 \text{ V}$

$$\text{Total external resistance } R = \frac{15 \times 15}{15 + 15} = \frac{15}{2} = 7.5 \Omega$$

$$\text{Current drawn from equivalent cell } I = \frac{\text{terminal potential}}{\text{external resistance}} = \frac{1.6}{7.5} \text{ A}$$

$$\therefore E - I \left(\frac{r}{4} \right) = 1.6 \quad \therefore E - I \left(\frac{r}{4} \right) = 1.6 \Rightarrow r = 7.5 \Omega$$



Ex. The e.m.f. of a primary cell is 2 V, when it is shorted then it gives a current of 4 A. Calculate internal resistance of primary cell.

Sol. $I = \frac{E}{r + R}$, If cell is shorted then $R = 0$, $I = \frac{E}{r} \therefore r = \frac{E}{I} = \frac{2}{4} = 0.5 \Omega$

Ex. n rows each containing m cells in series, are joined in parallel. Maximum current is taken from this combination in a 3Ω resistance. If the total number of cells used is 24 and internal resistance of each cell is 0.5Ω , find the value of m and n .

Sol. Total number of cell $mn = 24$, For maximum current $\frac{mr}{n} = R \Rightarrow 0.5 m = 3 n$, $m = \frac{3n}{0.5} = 6n$

$$\therefore 6n \times n = 24 \Rightarrow n = 2 \text{ and } m \times 2 = 24 \Rightarrow m = 12$$

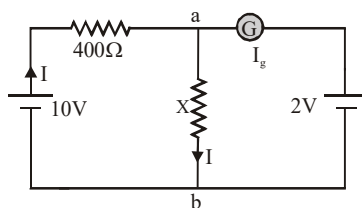
Ex. In the given circuit calculate potential difference between A and B.

Sol. First applying KVL on left mesh $2 - 3 I_1 - 2 I_1 = 0 \Rightarrow I_1 = 0.4 \text{ amp}$.
Now applying KVL on right mesh. $4 - 5 I_2 - 3 I_2 = 0 \Rightarrow I_2 = 0.5 \text{ amp}$.

Potential difference between points A and B

$$V_A - V_B = -3 \times 0.4 - 4 + 3 \times 0.5 = -3.7 \text{ volt.}$$

Ex. In the following circuit diagram, the galvanometer reading is zero. If the internal resistance of cells are negligible then what is the value of X ?

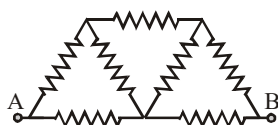


Sol. $\therefore I_g = 0 \quad \therefore I = \frac{10}{400 + X}$

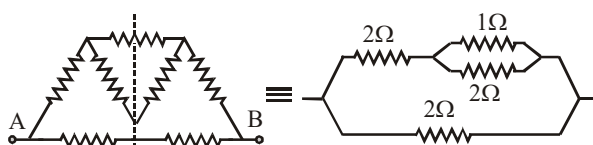
also potential difference across X is $2V \Rightarrow I X = 2$

$$\frac{10X}{400 + X} = 2 \left(\because I = \frac{10}{400 + X} \right) \Rightarrow X = 100 \Omega$$

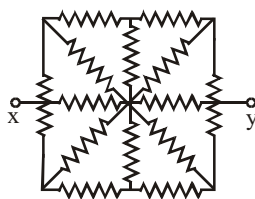
Ex. Each resistance is of 1Ω in the circuit diagram shown in figure. Find out equivalent resistance between A and B



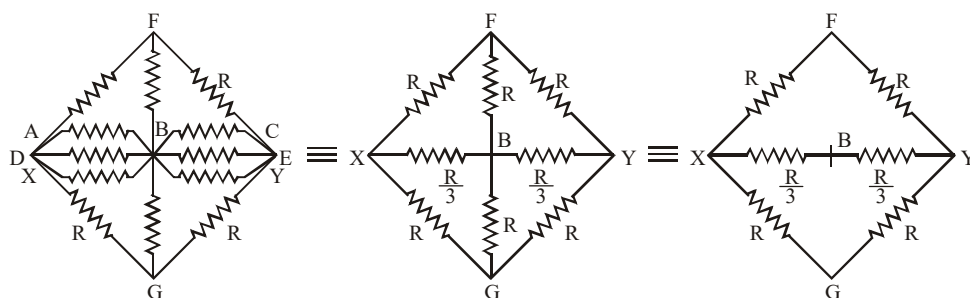
Sol. By symmetric line method $R_{AB} = (2 + 1 \parallel 2) \parallel 2 = \frac{8}{7} \Omega$



Ex. Identical resistance of resistance R are connected as in figure then find out net resistance between x and y .



Sol. Given circuit can be modified according to following figures



$$\frac{1}{R_{xy}} = \frac{1}{2R} + \frac{3}{2R} + \frac{1}{2R} = \frac{5}{2R} \Rightarrow R_{xy} = \frac{2R}{5}$$

HEATING EFFECT OF CURRENT

CAUSE OF HEATING

The potential difference applied across the two ends of conductor sets up electric field. Under the effect of electric field, electrons accelerate and as they move, they collide against the ions and atoms in the conductor, the energy of electrons transferred to the atoms and ions appears as heat.

- Joules's Law of Heating**

When a current I is made to flow through a passive or ohmic resistance R for time t , heat Q is produced such that

$$Q = I^2 R t = P \times t = V I t = \frac{V^2}{R} t$$

Heat produced in conductor does not depend upon the direction of current.

- SI unit** : joule ;

Practical Units : 1 kilowatt hour (kWh)

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ joule} = 1 \text{ unit}$$

$$1 \text{ BTU (British Thermal Unit)} = 1055 \text{ J}$$

- Power** : $P = V I = \frac{V^2}{R} = I^2 R$

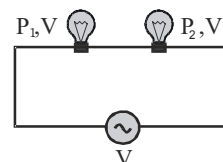
- SI unit** : Watt

The watt-hour meter placed on the premises of every consumer records the electrical energy consumed.

- Series combination of resistors (bulbs)**

$$\text{Total power consumed } P_{\text{total}} = \frac{P_1 P_2}{P_1 + P_2}$$

$$\text{If } n \text{ bulbs are identical } P_{\text{total}} = \frac{P}{n}$$



In series combination of bulbs : Brightness $\propto$ Power consumed by bulb $\propto V \propto R \propto \frac{1}{P_{\text{rated}}}$

Bulb of lesser wattage will shine more. For same current $P = I^2 R$ $P \propto R$ $R \uparrow \Rightarrow P \uparrow$

• **Parallel combination of resistors (bulbs)**

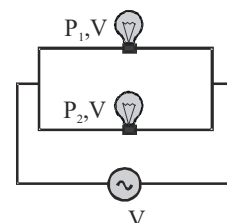
Total power consumed $P_{\text{total}} = P_1 + P_2$

If n bulbs are identical $P_{\text{total}} = nP$

In parallel combination of bulbs

Brightness $\propto$ Power consumed by bulb $\propto I \propto \frac{1}{R}$

Bulb of greater wattage will shine more.



For same V more power will be consumed in smaller resistance $P \propto \frac{1}{R}$

- Two identical heater coils gives total heat H_s when connected in series and H_p when connected in parallel than $\frac{H_p}{H_s} = 4$ [In this, it is assumed that supply voltage is same]

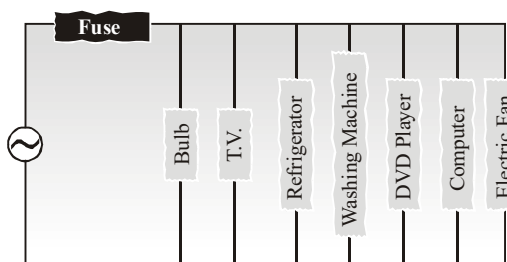
- If a heater boils m kg water in time T_1 and another heater boils the same water in T_2 , then both connected in series will boil the same water in time $T_s = T_1 + T_2$ and in parallel $T_p = \frac{T_1 T_2}{T_1 + T_2}$ [Use

time taken $\propto$ Resistance]

- Instruments based on heating effect of current, works on both A.C. and D.C. Equal value of A.C. (RMS) and D.C. produces, equal heating effect. That why brightness of bulb is same whether it is operated by A.C. or same value D.C.

FUSE WIRE

The fuse wire for an electric circuit is chosen keeping in view the value of safe current through the circuit.



- The fuse wire should have high resistance per unit length and low melting point.
- However the melting point of the material of fuse wire should be above the temperature that will be reached on the passage of the current through the circuit
- A fuse wire is made of alloys of lead (Pb) and tin (Sn).
- Length of fuse wire is immaterial.
- The material of the filament of a heater should have high resistivity and high melting point.
- The temperature of the wire increases to such a value at which, the heat produced per second equals

heat lost per second due to radiation from the surface of wire $I^2 \left(\frac{\rho \ell}{\pi r^2} \right) = H \times 2\pi r \ell$ $I^2 \propto r^3$

H = heat lost per second per unit area due to radiation.

Ex. An electric heater and an electric bulb are rated 500 W, 220 V and 100 W, 220 V respectively. Both are connected in series to a 220 V a.c. mains. Calculate power consumed by (i) heater (ii) bulb.

Sol. $P = \frac{V^2}{R}$ or $R = \frac{V^2}{P}$, For heater. Resistance $R_h = \frac{(220)^2}{500} = 96.8 \Omega$,

For bulb resistance $R_L = \frac{(220)^2}{100} = 484 \Omega$

Current in the circuit when both are connected in series $I = \frac{V}{R_L + R_h} = \frac{220}{484 + 96.8} = 0.38 \text{ A}$

(i) Power consumed by heater $= I^2 R_h = (0.38)^2 \times 96.8 = 13.98 \text{ W}$

(ii) Power consumed by bulb $= I^2 R_L = (0.38)^2 \times 484 = 69.89 \text{ W}$

Ex. A heater coil is rated 100 W, 200 V. It is cut into two identical parts. Both parts are connected together in parallel, to the same source of 200 V. Calculate the energy liberated per second in the new combination.

Sol. $\therefore P = \frac{V^2}{R}$ $\therefore R = \frac{V^2}{P} = \frac{(200)^2}{100} = 400 \Omega$

Resistance of half piece $= \frac{400}{2} = 200 \Omega$

Resistance of pieces connected in parallel $= \frac{400}{2} = 100 \Omega$

Energy liberated/second $P = \frac{V^2}{R} = \frac{200 \times 200}{100} = 400 \text{ W}$

Ex. The power of a heater is 500W at 800°C. What will be its power at 200°C. If $\alpha = 4 \times 10^{-4}$ per °C?

Sol. $P = \frac{V^2}{R} \therefore \frac{P_{200}}{P_{800}} = \frac{R_{800}}{R_{200}} = \frac{R_0(1 + 4 \times 10^{-4} \times 800)}{R_0(1 + 4 \times 10^{-4} \times 200)} \Rightarrow P_{200} = \frac{500 \times 1.32}{1.08} = 611 \text{ W}$

Ex. When a battery sends current through a resistance R_1 for time t , the heat produced in the resistor is Q . When the same battery sends current through another resistance R_2 for time t , the heat produced in R_2 is again Q . Determine the internal resistance of battery.

Sol. $\left[\frac{E}{R_1 + r} \right]^2 R_1 = \left[\frac{E}{R_2 + r} \right]^2 R_2 \Rightarrow r = \sqrt{R_1 R_2}$

Ex. How much time heater will take to increase the temperature of 100 g water by 50°C if resistance of heating coil is 484Ω and supply voltage is 220V a.c.

Sol. Heat given by heater = heat taken by water $\Rightarrow \frac{V^2}{R} t = ms \Delta \theta \Rightarrow \frac{220 \times 220}{484}$

$t = (100 \times 10^{-3}) (4.2 \times 10^3) (50) \Rightarrow t = 210 \text{ s}$

GALVANOMETER

The instrument used to measure strength of current, by measuring the deflection of the coil due to torque produced by a magnetic field, is known as galvanometer.

SHUNT

The small resistance connected in parallel to galvanometer coil, in order to control current flowing through the galvanometer, is known as shunt.

• Merits of shunt

- (i) To protect the galvanometer coil from burning.
- (ii) Any galvanometer can be converted into ammeter of desired range with the help of shunt.
- (iii) The range an ammeter can be changed by using shunt resistance of different values.

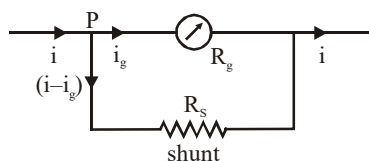
• Demerits of shunt

Shunt resistance decreases the sensitivity of galvanometer.

CONVERSION OF GALVANOMETER INTO AMMETER

A galvanometer can be converted into an ammeter by connecting low resistance in parallel to its coil.

- The value of shunt resistance to be connected in parallel to galvanometer coil is given by : $R_s = \frac{R_g i_g}{i - i_g}$



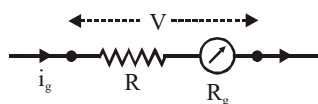
Where i = Range of ammeter

i_g = Current required for full scale deflection of galvanometer.

R_g = Resistance of galvanometer coil.

CONVERSION OF GALVANOMETER INTO VOLTMETER

- The galvanometer can be converted into voltmeter by connecting high resistance in series with its coil.
- The high resistance to be connected in series with galvanometer coil is given by $R = \frac{V}{i_g} - R_g$



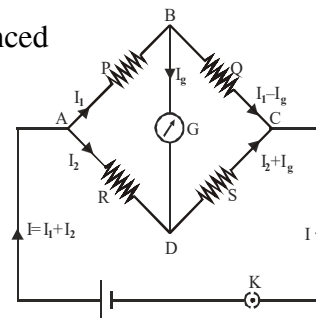
- The rate of variation of deflection depends upon the magnitude of deflection itself and so the accuracy of the instrument.
- A suspended coil galvanometer can measure currents of the order of 10^{-9} ampere.
- I_g is the current for full scale deflection. If the current for a deflection, of one division on the galvanometer scale is k and N is the total number of divisions on one side of the zero of galvanometer scale, then $I_g = k \times N$.
- A ballistic galvanometer is a specially designed moving coil galvanometer, used to measure charge flowing through the circuit for small time intervals.

WHEAT STONE BRIDGE

- The configuration in the adjacent figure is called Wheat Stone Bridge.
- If current in galvanometer is zero ($I_g = 0$) then bridge is said to be balanced

$$V_D = V_B \Rightarrow I_1 P = I_2 R \text{ \& } I_1 Q = I_2 S \Rightarrow \frac{P}{Q} = \frac{R}{S}$$

- If $\frac{P}{Q} < \frac{R}{S}$ then $V_B > V_D$ and current will flow from B to D.
- If $\frac{P}{Q} > \frac{R}{S}$ then $V_B < V_D$ and current will flow from D to B.



METRE BRIDGE

It is based on principle of whetstone bridge. It is used to find out unknown resistance of wire. AC is 1 m long uniform wire. R.B. is known resistance and S is unknown resistance. A cell is connected across 1 m long wire and Galvanometer is connected between Jockey and midpoint D. To find out unknown resistance we touch jockey from A to C and find balance condition. Let balance is at B point on wire.

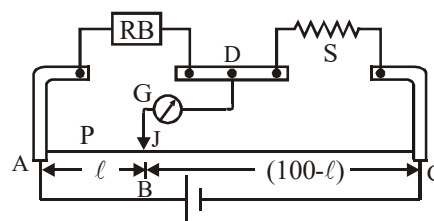
$$AB = \ell \text{ cm}$$

$$BC = (100 - \ell) \text{ cm}$$

$$P = r \ell$$

$$Q = r(100 - \ell)$$

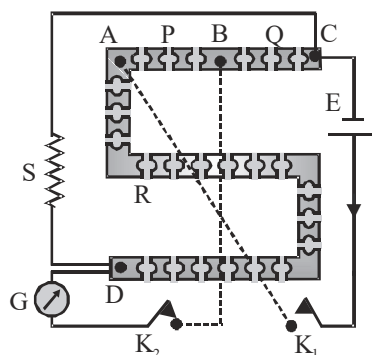
where r = resistance per unit length on wire.



$$\text{At balance condition : } \frac{P}{Q} = \frac{R}{S} \Rightarrow \frac{r\ell}{r(100-\ell)} = \frac{R}{S} \Rightarrow S = \frac{(100-\ell)}{\ell} R$$

POST OFFICE BOX

It is also based on wheat stone bridge. The resistance of 10Ω , 100Ω , and 1000Ω are often connected between AB and BC. These are known as ratio arms. Resistance from 1Ω to 5000Ω are connected between A and D, this is known arm. Unknown resistance is connected between C and D.



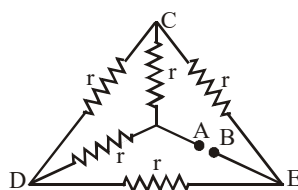
A cell is connected between A and C with key K_1 and Galvanometer is connected between B and D with key K_2 .

First we select ratio of resistance Q and P. For given value of S we will take value of resistance from known arm in such a way that Galvanometer show null deflection $S = \frac{Q}{P} R$. On decreasing the value of

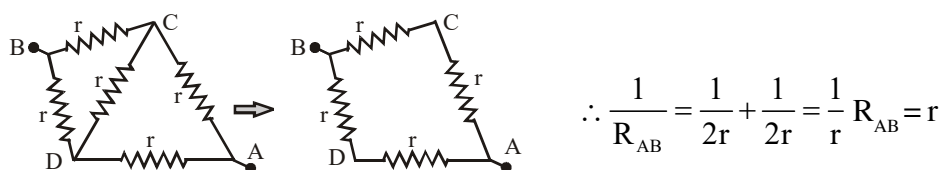
$\frac{Q}{P}$ the sensitivity of the box increases. It is used to find out the breakage in telegraph line in post and telegraph offices.

- To increase the range of an ammeter a shunt is connected in parallel with the galvanometer.
- To convert an ammeter of range I ampere and resistance $R_g \Omega$ into an ammeter of range nI ampere, the value of resistance to be connected in parallel will be $R_g(n-1)$
- To increase the range of a voltmeter a high resistance is connected in series with it.
- To convert a voltmeter of resistance $R_g \Omega$ and range V volt into a voltmeter of range nV volt, the value of resistance to be connected in series will be $(n-1)R_g$.
- Resistance of ideal ammeter is zero & resistance of ideal voltmeter is infinite.
- The bridge is most sensitive when the resistance in all the four branches of the bridge is of same order.

Ex. In the adjoining network of resistors each is of resistance $r \Omega$. Find the equivalent resistance between point A and B



Sol. Given circuit is balanced Wheat stone Bridge



Ex. A 100 volt voltmeter whose resistance is $20 \text{ k}\Omega$ is connected in series to a very high resistance R. When it is joined in a line of 110 volt, it reads 5 volt. What is the magnitude of resistance R ?

Sol. When voltmeter connected in 110 volt line, Current through the voltmeter $I = \frac{110}{(20 \times 10^3 + R)}$

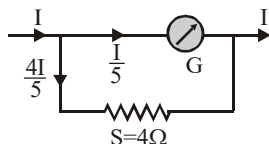
The potential difference across the voltmeter $V = IR_v$

$$\Rightarrow 5 = \frac{110 \times 20 \times 10^3}{(20 \times 10^3 + R)}$$

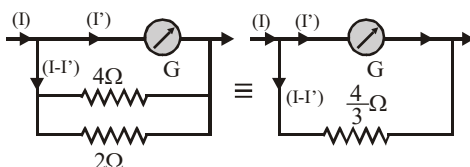
$$\Rightarrow 20 \times 10^3 + R = 440 \times 10^3 \Rightarrow R = 420 \times 10^3 \Omega$$

Ex. When a shunt of 4Ω is attached to a galvanometer, the deflection reduces to $1/5^{\text{th}}$. If an additional shunt of 2Ω is attached what will be the deflection ?

Sol. Initial condition : When shunt of 4Ω used $\frac{I}{5} \times G = \frac{4}{5} I \times 4 \Rightarrow G = 16\Omega$



When additional shunt of 2Ω used $I' \times 16 = (I - I') \frac{4}{3} \Rightarrow I' = \frac{I}{13}$



$\therefore$ it will reduce to $\frac{I}{13}$ of the initial deflection

Ex. A galvanometer having 30 divisions has current sensitivity of $20\mu\text{A}/\text{division}$. It has a resistance of 25Ω .

- How will you convert it into an ammeter measuring upto 1 ampere.
- How will you convert this ammeter into a voltmeter upto 1 volt.

Sol The current required for full scale deflection $I_g = 20\mu\text{A} \times 30 = 600\mu\text{A} = 6 \times 10^{-4}\text{A}$

- To convert it into ammeter, a shunt is required in parallel with it

$$\text{shunt resistance } R'_s = \frac{I_g R_g}{(I - I_g)} = \left(\frac{6 \times 10^{-4}}{1 - 6 \times 10^{-4}} \right) 25 = 0.015\Omega$$

- To convert galvanometer into voltmeter, a high resistance in series with it is required

$$\text{series resistance } R = \frac{V}{i_g} - R_g = \frac{1}{6 \times 10^{-4}} - 25 = 1666.67 - 25 = 1641.67\Omega$$

POTENTIOMETER

• Necessity of potentiometer

Practically voltmeter has a finite resistance. (ideally it should be ∞) in other words it draws some current from the circuit. To overcome this problem potentiometer is used because at the instant of measurement, it draws no current from the circuit. It means its effective resistance is infinite.

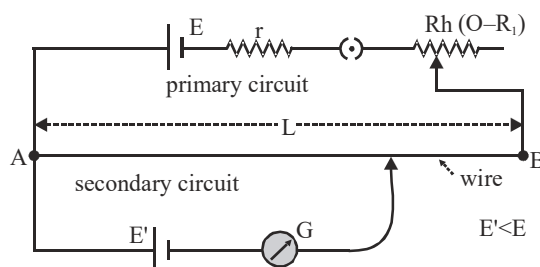
• Working principle of potentiometer

Any unknown potential difference is balanced on a known potential difference which is uniformly distributed over entire length of potentiometer wire. This process is named as zero deflection or null deflection method.

• Potentiometer wire

Made up of alloys of magnin, constantan, Eureka. Specific properties of these alloys are high specific resistance, negligible temperature co-efficient of resistance (α). Invariability of resistance of potentiometer wire over a long period.

CIRCUITS OF POTENTIOMETER



- **Primary circuit** contains constant source of voltage rheostat or Resistance Box
- **Secondary, Unknown or galvanometer circuit**
Let ρ = Resistance per unit length of potentiometer wire
- **Potential gradient (x) (V/m)**
The fall of potential per unit length of potentiometer wire is called potential gradient.

$$x = \frac{V}{L} = \frac{\text{current} \times \text{resistance of potentiometer wire}}{\text{length of potentiometer wire}} = I \left(\frac{R}{L} \right)$$

The potential gradient depends only on primary circuit and is independent of secondary circuit.

- **Applications of potentiometer**

• To measure potential difference across a resistance. • To find out emf of a cell • Comparison of two emfs E_1/E_2 • To find out internal resistance of a primary cell • Comparison of two resistance. • To find out an unknown resistance which is connected in series with the given resistance.	• To find out current in a given circuit • Calibration of an ammeter or to have a check on reading of (A) • Calibration of a voltmeter or to have a check on reading of (V) • To find out thermocouple emf (e_t) (mV or mV)
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Different between potentiometer and voltmeter

Potentiometer	Voltmeter
• It measures the unknown emf very accurately • While measuring emf it does not draw any current from the driving source of know emf. • While measuring unknown potential difference the resistance of potentiometer becomes infinite. • It is based on zero deflection method. • It has a high sensitivity. • it is used for various applications like measurement of internal resistance of cell, calibration of ammeter and voltmeter, measurement of thermo emf, comparison of emf's etc.	• It measures the unknown emf approximately. • While measuring emf it draws some current from the source of emf. • While measuring unknown potential difference the resistance of voltmeter is high but finite. • It is based on deflection method. • Its sensitivity is low. • It is only used to measured emf or unknown potential difference.

Ex. There is a definite potential difference between the two ends of a potentiometer. Two cells are connected in such a way that first time help each other, and second time they oppose each other. They are balanced on the potentiometer wire at 120 cm and 60 cm length respectively. Compare the electromotive force of the cells.

Sol. Suppose the potential gradient along the potentiometer wire = x and the emf's of the two cells are E_1 and E_2 .

When the cells help each other, the resultant emf = $(E_1 + E_2)$

$$E_1 + E_2 = x \times 120 \text{ cm} \dots(i)$$

When the cells oppose each other, the resultant emf = $(E_1 - E_2)$

$$E_1 - E_2 = x \times 60 \text{ cm} \dots(ii)$$

$$\text{From equation (i) and (ii)} \quad \frac{E_1 + E_2}{E_1 - E_2} = \frac{120 \text{ cm}}{60 \text{ cm}} = \frac{2}{1} \Rightarrow E_1 + E_2 = 2(E_1 - E_2) \Rightarrow 3E_2 = E_1 \Rightarrow \frac{E_1}{E_2} = \frac{3}{1}$$

COLOUR CODE FOR CARBON RESISTORS

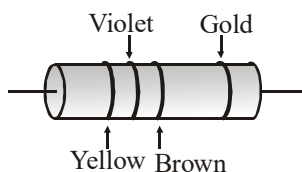
Colour	Strip A	Strip B	Strip C	Strip D (Tolerance)
Black	0	0	10^0	
Brown	1	1	10^1	
Red	2	2	10^2	
Orange	3	3	10^3	
Yellow	4	4	10^4	
Green	5	5	10^5	
Blue	6	6	10^6	
Violet	7	7	10^7	
Grey	8	8	10^8	
White	9	9	10^9	
Gold	-	-	10^{-1}	$\pm 5 \%$
Silver	-	-	10^{-2}	$\pm 10 \%$
No colour	-	-	-	$\pm 20 \%$

Aid to memory BBROY
Great Britain
Very Good watch of gold & silver

Ex. Draw a colour code for $42 \text{ k } \Omega \pm 10\%$ carbon resistance.

Sol. According to colour code colour for digit 4 is yellow, for digit 2 it is red, for 3 colour is orange and 10% tolerance is represented by silver colour. So colour code should be yellow, red, orange and silver.

Ex. What is resistance of following resistor.



Sol. Number for yellow is 4, Number of violet is 7

Brown colour gives multiplier 10^1 , Gold gives a tolerance of $\pm 5\%$

So resistance of resistor is $47 \times 10^1 \Omega \pm 5\% = 470 \pm 5\% \Omega$.

EXERCISE # S-1

Microscopic analysis

1. A copper wire of length L , and cross section area A carries a current I . If the specific resistance of copper is ρ , the electric field in the wire is _____.

CE0001

2. A copper wire carries a current density j (= current per unit area). Assuming that n = No. of free electrons per unit volume, e = electronic charge, $\langle v \rangle$ = average speed due to thermal agitation. The distance which will be covered by an electron during its displacement l along the wire _____.

CE0002

3. The total momentum of electrons in a straight wire of length ℓ carrying a current I is _____ (mass of electron = m_e , charge of electron = e)

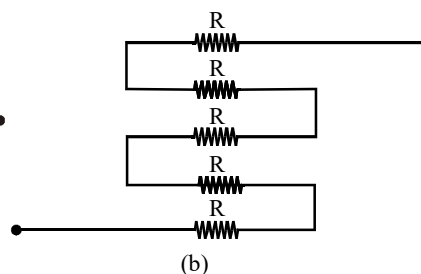
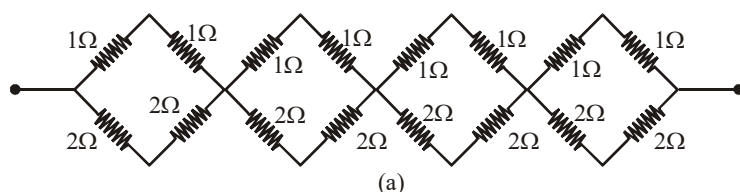
CE0003

4. Two conductors are made of the same material and have the same length. Conductor A is a solid wire of diameter 1mm. Conductor B is a hollow tube of outer diameter 2mm and inner diameter 1mm. Find the ratio of resistance R_A to R_B .

CE0004

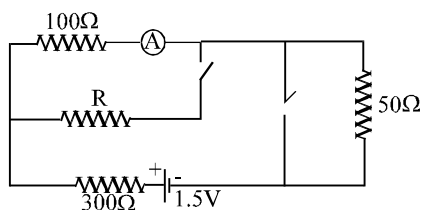
Ohm's law and circuit analysis

5. (a) Given n resistors each of resistance R , how will you combine them to get the (i) maximum (ii) minimum effective resistance? What is the ratio of the maximum to minimum resistance?
(b) Given the resistances of 1Ω , 2Ω , 3Ω , how will be combine them to get an equivalent resistance of
(i) $(11/3)\Omega$ (ii) $(11/5)\Omega$ (iii) 6Ω (iv) $(6/11)\Omega$?
(c) Determine the equivalent resistance of networks shown in figures



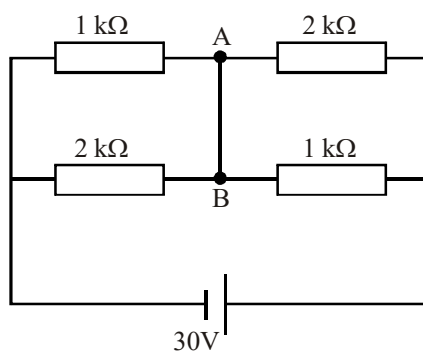
CE0005

6. In the circuit shown in figure the reading of ammeter is the same with both switches open as with both closed. Then find the resistance R . (ammeter is ideal)



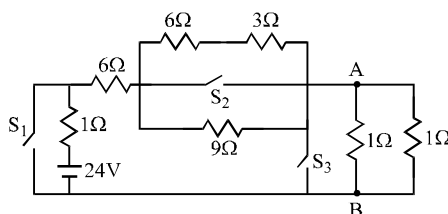
CE0006

7. Find the current (in mA) in the wire between points A and B.



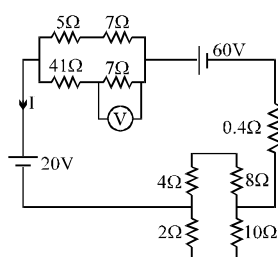
CE0007

8. If the switches S_1 , S_2 and S_3 in the figure are arranged such that current through the battery is minimum, find the voltage across points A and B.



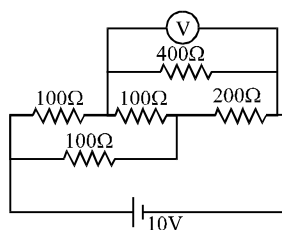
CE0008

9. Find the current I & voltage V in the circuit shown.



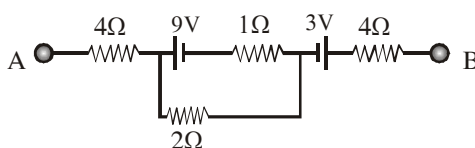
CE0009

10. An electrical circuit is shown in the figure. Calculate the potential difference across the resistance of $400\ \Omega$, as will be measured by the voltmeter V of resistance $400\ \Omega$, either by applying Kirchhoff's rules or otherwise.



CE0010

11. In the circuit shown in figure potential difference between point A and B is 16V . Find the current passing through $2\ \Omega$ resistance.

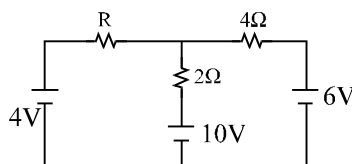


CE0011

12. An enquiring physics student connects a cell to a circuit and measures the current drawn from the cell to I_1 . When he joins a second identical cell in series with the first, the current becomes I_2 . When the cells are connected in parallel, the current through the circuit is I_3 . Show that relation between the current is $3 I_3 I_2 = 2 I_1 (I_2 + I_3)$

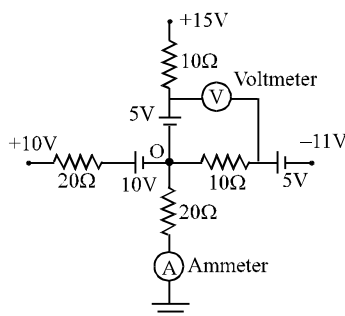
CE0012

13. For what value of R in circuit, current through $4\ \Omega$ resistance is zero.



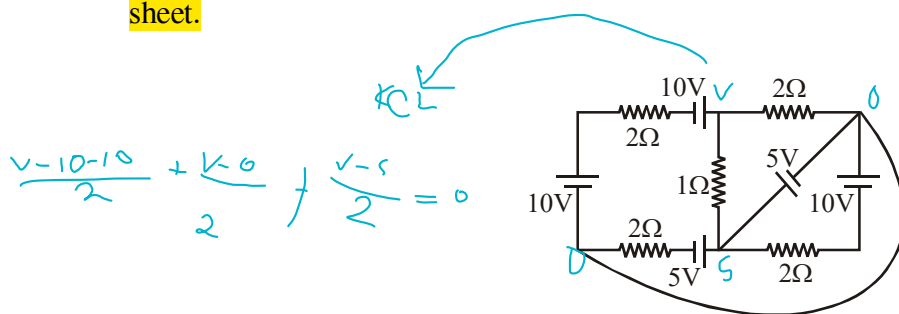
CE0013

14. The potential of certain points in the circuit are maintained at the values indicated. The Voltmeter and Ammeter are ideal. Find the potential of the cross junction point in the circuit (at center O) and the readings of Voltmeter and Ammeter. All cells are ideal.



CE0014

15. In the given circuit diagram, the current through the 1Ω resistor is given by I amp. Fill 2 I in OMR sheet.



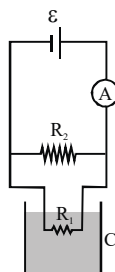
CE0015

Joule heating

16. If a cell of constant E.M.F. produces the same amount of the heat during the same time in two independent resistors R_1 and R_2 , when they are separately connected across the terminals of the cell, one after the another, find the internal resistance of the cell.

CE0016

17. The coil of a calorimeter C has a resistance of $R_1 = 60\Omega$. The coil R_1 is connected to the circuit as shown in figure. What is the rise in temperature ($^{\circ}\text{C}$) of 240 grams of water poured into the calorimeter when it is heated for 7 minutes during which a current flows through the coil and the ammeter shows 3A? The resistance $R_2 = 30\Omega$. [Disregard the resistances of the battery and the ammeter, and the heat losses and heat capacity of the calorimeter and the resistor and specific heat of water = $4200 \text{ J/kg}^{\circ}\text{C}$]

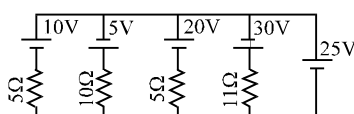


CE0017

18. An electric kettle has two windings. When one of them is switched on, the water in the kettle begins to boil in 15 minutes, and when the other is switched on it takes 30 minutes for water to boil. If the two windings are joined in series and switched on, water in the kettle begin to boil in $\frac{\alpha}{4}$ hr. Assuming no heat loss to the surrounding fill the value of α in OMR sheet.

CE0018

19. Find the current through 25V cell & power supplied by 20V cell in the figure shown.



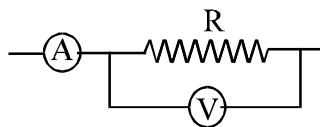
CE0019

20. A person decides to use his bath tub water to generate electric power to run a 40 watt bulb. The bath tub is located at a height of 10 m from the ground & it holds 200 litres of water. If we install a water driven wheel generator on the ground, at what rate should the water drain from the bath tub to light bulb? How long can we keep the bulb on, if the bath tub was full initially. The efficiency of generator is 90%. ($g = 10\text{m/s}^2$)

CE0020

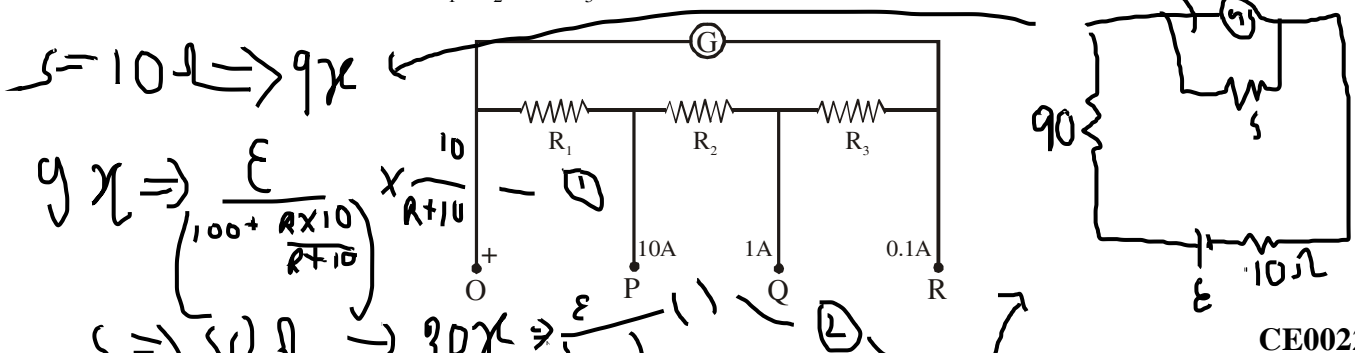
Instruments

21. A part of a circuit is shown in figure. Here reading of ammeter is 5 ampere and voltmeter is 96V & voltmeter resistance is 480 ohm. Then find the resistance R



CE0021

22. The resistance of the galvanometer G in the circuit is 25Ω . The meter deflects full scale for a current of 10 mA. The meter behaves as an ammeter of three different ranges. The range is 0–10A, if the terminals O and P are taken; range is 0–1 A between O and Q; range is 0–0.1 A between O and R. Calculate the resistance R_1 , R_2 and R_3 .



CE0022

23. A galvanometer having 50 divisions provided with a variable shunt is used to measure the current as an ammeter when connected in series with a resistance of 90Ω and a battery of internal resistance 10Ω . It is observed that when the shunt resistance are 10Ω , 50Ω , respectively the deflection are respectively 9 & 30 divisions. What is the resistance of the galvanometer? Further if the full scale deflection of the galvanometer movement is 300 mA, find the emf of the cell.

CE0023

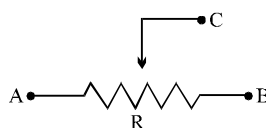
24. Draw the circuit for experimental verification of Ohm's law using a source of variable D.C. voltage, a main resistance of 100Ω , two galvanometers and two resistances of values $10^6\Omega$ and $10^{-3}\Omega$ respectively. Clearly show the positions of the voltmeter and the ammeter. [IIT-JEE' 2004]

CE0024

25. How a battery is to be connected so that shown rheostat will behave like a potential divider? Also indicate the points about which output can be taken. [IIT-JEE' 2003]

WV Sir

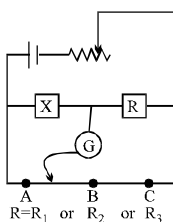
Notes



CE0025

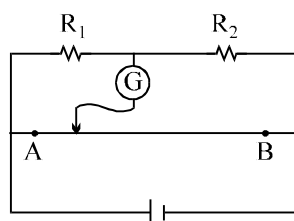
26. An unknown resistance X is to be determined using resistances R_1 , R_2 or R_3 . Their corresponding null points are A, B and C. Find which of the above will give the most accurate reading and why?

[IIT-JEE 2005]



CE0026

27. In the figure shown for which values of R_1 and R_2 the balance point for Jockey is at 40 cm from A. When R_2 is shunted by a resistance of $10\ \Omega$, balance shifts to 50 cm. Find R_1 and R_2 . ($AB = 1\text{ m}$):



CE0027

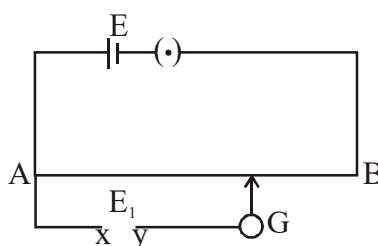
28. While doing an experiment with potentiometer it was found that the deflection is one sided and two cases are possible

- (i) the deflection decreased while moving from one end A of the wire to the end B;
(ii) the deflection increased while the jockey was moved towards the end B. Then

- (a) Which terminal + or -ve of the cell E_1 is connected at X in case (i) and how is E_1 related to E ?
(b) Which terminal of the cell E_1 is connected at X in case (ii)?

NV S10 Notes

Potentiometer



CE0028

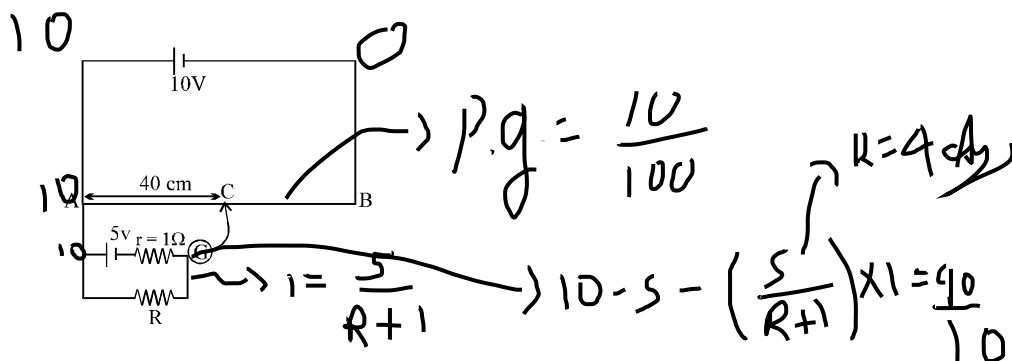
29. In a potentiometer arrangement, a cell of emf 1.25 V gives a balance point at 35.0 cm length of the wire. If the cell is replaced by another cell and the balance point shifts to 63.0 cm, what is the emf of the second cell?

$E \propto l$ — Balance Length

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

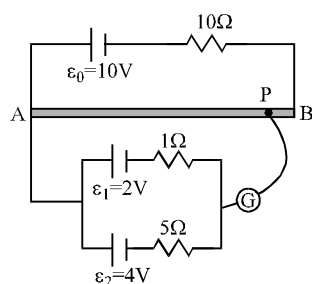
CE0029

30. A potentiometer wire AB is 100 cm long and has a total resistance of 10 ohm. If the galvanometer shows zero deflection at the position C, then find the value of unknown resistance R.



CE0030

31. A battery of emf $\varepsilon_0 = 10$ V is connected across a 1 m long uniform wire having resistance $10\Omega/\text{m}$. Two cells of emf $\varepsilon_1 = 2$ V and $\varepsilon_2 = 4$ V having internal resistances 1Ω and 5Ω respectively are connected as shown in the figure. If a galvanometer shows no deflection at the point P, find the distance of point P from the point A.



CE0031

EXERCISE # S-2

1. A long cylinder with uniformly charged surface and cross sectional radius $a = 1.0$ cm moves with a constant velocity $v = 10$ m/s, along its axis. An electric field strength at the surface of the cylinder is equal to $E = 0.9$ KV/cm. Find the resulting convection current, that is, the current caused by mechanical transfer of charge.

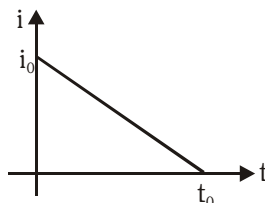
CE0032

2. A resistance R of thermal coefficient of resistivity $= \alpha$ is connected in parallel with a resistance $= 3R$, having thermal coefficient of resistivity $= 2\alpha$. Find the value of α_{eff} .

CE0033

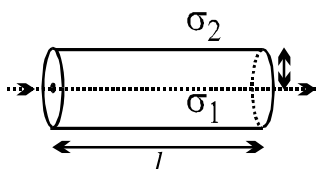
3. Relation between current in conductor and time is shown in figure then determine.

- (i) Total charge flow through the conductor
 (ii) Write expression of current in terms of time
 (iii) If resistance of conductor is R then total heat dissipated across resistance R is



CE0034

4. A long conductor of circular cross-section has radius r and length l as shown in the figure. The conductivity of the material near the axis is σ_1 and increases linearly with the distance from axis and becomes σ_2 near the surface. Find the resistance of the conductor if the current enters from the one end and leaves from the other end.



CE0035

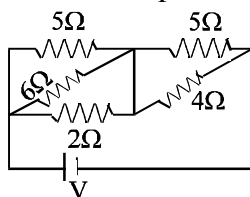
5. (a) The current density across a cylindrical conductor of radius R varies according to the equation

$J = J_0 \left(1 - \frac{r}{R} \right)$, where r is the distance from the axis. Thus the current density is a maximum J_0 at the axis $r = 0$ and decreases linearly to zero at the surface $r = R$. Calculate the current in terms of J_0 and the conductor's cross sectional area is $A = \pi R^2$.

(b) Suppose that instead the current density is a maximum J_0 at the surface and decreases linearly to zero at the axis so that $J = J_0 \frac{r}{R}$. Calculate the current.

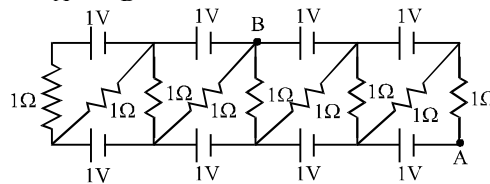
CE0036

6. Find the resistor in which maximum heat will be produced.



CE0037

7. Find the potential difference $V_A - V_B$ for the circuit shown in the figure.



CE0038

8. A galvanometer (coil resistance 99Ω) is converted into an ammeter using a shunt of 1Ω and connected as shown in the figure (i). The ammeter reads $3A$. The same galvanometer is converted into a voltmeter by connecting a resistance of 101Ω in series. This voltmeter is connected as shown in figure (ii). Its reading is found to be $4/5$ of the full scale reading. Find
- internal resistance r of the cell
 - range of the ammeter and voltmeter
 - full scale deflection current of the galvanometer

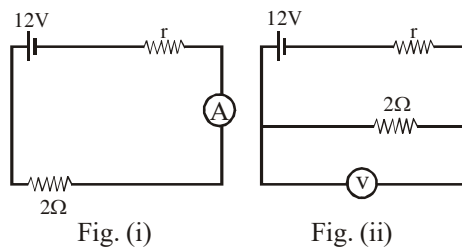


Fig. (i)

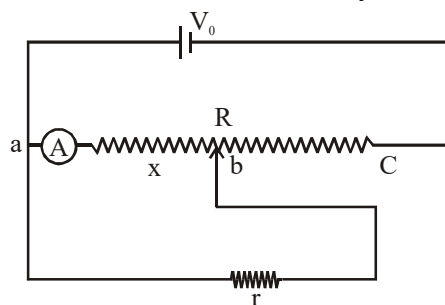
Fig. (ii)

CE0039

9. An accumulator of emf $2V$ and negligible internal resistance is connected across a uniform wire of length $10m$ and resistance 30Ω . The appropriate terminals of a cell of emf $1.5V$ and internal resistance 1Ω is connected to one end of the wire, and the other terminal of the cell is connected through a sensitive galvanometer to a slider on the wire. What length of the wire will be required to produce zero deflection of the galvanometer? How will the balancing change (a) when a coil of resistance 5Ω is placed in series with the accumulator, (b) the cell of $1.5V$ is shunted with 5Ω resistor?

CE0040

10. A constant voltage $V_0 (= 12V)$ is applied to a potential divider of resistance $R (= 4\Omega)$, connected to an ideal ammeter. A constant resistor $r (= 2\Omega)$ is connected to the sliding contact of the potential divider (as shown). Find the minimum current (in A) measured by ammeter.



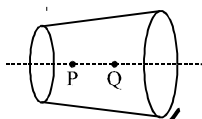
CE0041

EXERCISE # O-1

SINGLE CORRECT TYPE QUESTIONS

Microscopic analysis

1. A wire has a non-uniform cross-section as shown in figure. A steady current flows through it. The drift speed of electrons at points P and Q is v_P and v_Q .



- (A) $v_P = v_Q$ (B) $v_P < v_Q$ (C) $v_P > v_Q$ (D) Data insufficient

CE0042

2. Two wires each of radius of cross section r but of different materials are connected together end to end (in series). If the densities of charge carriers in the two wires are in the ratio 1 : 4, the drift velocity of electrons in the two wires will be in the ratio:

- (A) 1 : 2 (B) 2 : 1 (C) 4 : 1 (D) 1 : 4

CE0043

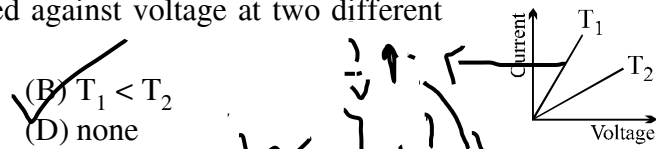
3. An insulating pipe of cross-section area 'A' contains an electrolyte which has two types of ions → their charges being $-e$ and $+2e$. A potential difference applied between the ends of the pipe result in the drifting of the two types of ions, having drift speed = v ($-ve$ ion) and $v/4$ ($+ve$ ion). Both ions have the same number per unit volume = n . The current flowing through the pipe is

- (A) $nev A/2$ (B) $nev A/4$ (C) $5nev A/2$ (D) $3nev A/2$

CE0044

4. The current in a metallic conductor is plotted against voltage at two different temperatures T_1 and T_2 . Which is correct

- (A) $T_1 > T_2$ (B) $T_1 < T_2$ (C) $T_1 = T_2$ (D) none



CE0045

5. A metal rod of length 10 cm and a rectangular cross-section of $1 \text{ cm} \times \frac{1}{2} \text{ cm}$ is connected to a battery across opposite faces. The resistance will be

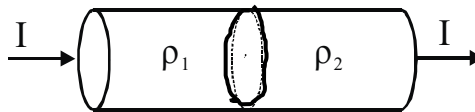
- (A) maximum when the battery is connected across $1 \text{ cm} \times \frac{1}{2} \text{ cm}$ faces. (B) maximum when the battery is connected across $10 \text{ cm} \times 1 \text{ cm}$ faces. (C) maximum when the battery is connected across $10 \text{ cm} \times \frac{1}{2} \text{ cm}$ faces. (D) same irrespective of the three faces.

CE0046

6. Consider a current carrying wire (current I) in the shape of a circle. Note that as the current progresses along the wire, the direction of $\mathbf{j}$ (current density) changes in an exact manner, while the current I remain unaffected. The agent that is *essentially* responsible for is
- (A) source of emf.
 (B) electric field produced by charges accumulated on the surface of wire.
 (C) the charges just behind a given segment of wire which push them just the right way by repulsion.
 (D) the charges ahead.

CE0047

7. Two long straight cylindrical conductors with resistivities ρ_1 and ρ_2 respectively are joined together as shown in figure. If current I flows through the conductors, the magnitude of the total free charge at the interface of the two conductors is :-



- (A) zero
 (B) $\frac{(\rho_1 - \rho_2) I \epsilon_0}{2}$
 (C) $\epsilon_0 I |\rho_1 - \rho_2|$
 (D) $\epsilon_0 I |\rho_1 + \rho_2|$

CE0048

8. **Statement 1 :** The drift speed of electrons in metals is small (in the order of a few mm/s) and the charge of an electron is also very small ($= 1.6 \times 10^{-19} \text{C}$), yet we can obtain a large current in a metal. and

- Statement 2 :** At room temperature, the thermal speed of electron is very high (about 10^7 times the drift speed).

- (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True ; Statement-2 is not a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.

CE0049

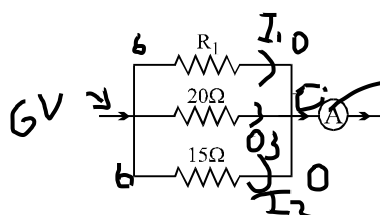
Ohm's law & circuit analysis

9. A storage battery is connected to a charger for charging with a voltage of 12.5 Volts. The internal resistance of the storage battery is 1Ω . When the charging current is 0.5 A, the emf of the storage battery is:

- (A) 13 Volts
 (B) 12.5 Volts
 (C) 12 Volts
 (D) 11.5 Volts

CE0050

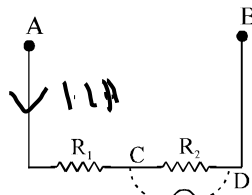
10. In the given circuit the current flowing through the resistance 20 ohms is 0.3 ampere while the ammetre reads 0.8 ampere. What is the value of R_1 ?



- (A) 30 ohms
 (B) 40 ohms
 (C) 50 ohms
 (D) 60 ohms

CE0051 $\frac{5-4}{60}$
 $R_1 = 10$

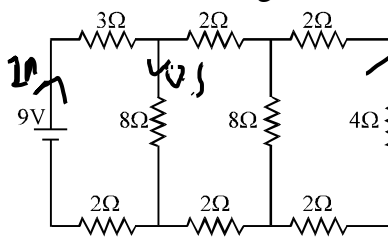
11. Resistances R_1 and R_2 each 60Ω are connected in series as shown in figure. The Potential difference between A and B is kept 120 volt. Then what will be the reading of voltmeter connected between the point C & D if resistance of voltmeter is 120Ω .



- (A) 48 V (B) 24 V (C) 40V (D) None

CE0052

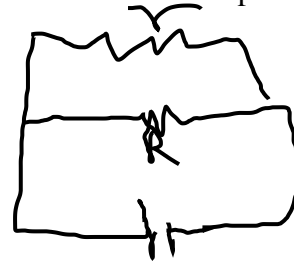
12. In the circuit shown in the figure, the current through :



- (A) the 3Ω resistor is 0.50 A (B) the 3Ω resistor is 0.25 A
(C) 4Ω resistor is 0.50 A (D) the 4Ω resistor is 0.25 A

CE0053

13. A simple circuit contains an ideal battery and a resistance R . If a second resistor is placed in parallel with the first,
(A) the potential across R will decrease
(B) the current through R will decreased
(C) the current delivered by the battery will increase
(D) the power dissipated by R will increased.



CE0054

14. The equivalent resistance of a group of resistances is R . If another resistance is connected in parallel to the group, its new equivalent becomes R_1 & if it is connected in series to the group, its new equivalent becomes R_2 we have :

- (A) $R_1 > R$ (B) $R_1 < R$ (C) $R_2 > R$ (D) $R_2 < R$

CE0055

15. An energy source will supply a constant current into the load, if its internal resistance is-

[AIEEE - 2005]

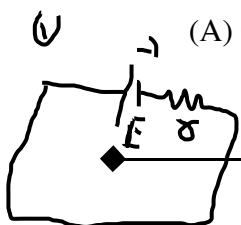
- (A) equal to the resistance of the load (B) very large as compared to the load resistance
(C) zero (D) non-zero but less than the resistance of the load

CE0056

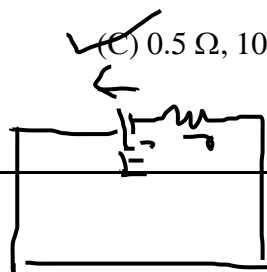
16. When a current of 4 A flows within a battery from its positive to negative terminal, the potential difference across the battery is 12 volts. The potential difference across the battery is 9 volts when a current of 2 A flows within it from its negative to its positive terminal. The internal resistance and the e.m.f. of the battery are :-

- (A) 0.1Ω , 4V (B) 0.2Ω , 5V (C) 0.5Ω , 10V (D) 0.7Ω , 10V

CE0057



$$12 = E + 4r \quad (1)$$



$$9 = E - 2r \quad (2)$$

$$3 = 6r \quad r = 0.5\Omega$$

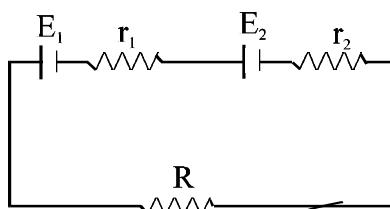
$$E = 10$$

17. Two resistances of equal magnitude R and having temperature coefficient α_1 and α_2 respectively are connected in parallel. The temperature coefficient of the parallel combination is, approximately

(A) $2(\alpha_1 + \alpha_2)$ (B) $\frac{\alpha_1 \alpha_2}{\alpha_1 + \alpha_2}$ (C) $\frac{\alpha_1 - \alpha_2}{2}$ (D) $\frac{\alpha_1 + \alpha_2}{2}$

CE0058

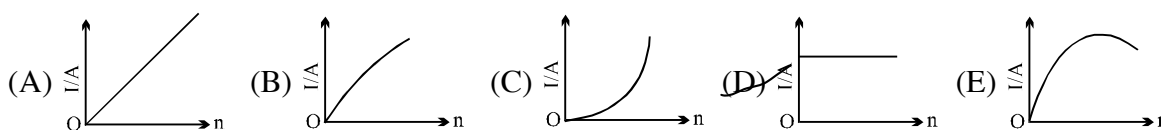
18. Under what condition current passing through the resistance R can be increased by short circuiting the battery of emf E_2 . The internal resistances of the two batteries are r_1 and r_2 respectively.



- (A) $E_2 r_1 > E_1 (R + r_2)$ (B) $E_1 r_2 > E_2 (R + r_1)$
(C) $E_2 r_2 > E_1 (R + r_2)$ (D) $E_1 r_1 > E_2 (R + r_1)$

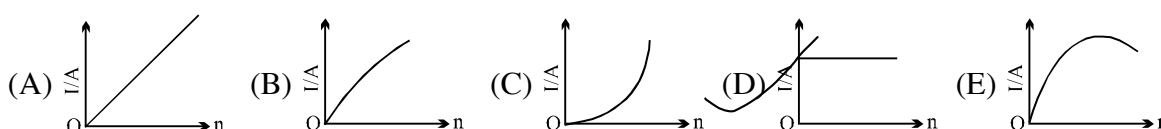
CE0059

19. A battery consists of a variable number n of identical cells having internal resistance connected in series. The terminals of the battery are short circuited and the current I measured. Which one of the graph below shows the relationship between I and n ?



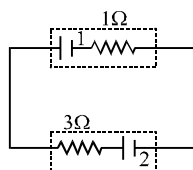
CE0060

20. In **previous problem**, if the cell had been connected in parallel (instead of in series) which of the above graphs would have shown the relationship between total current I and n ?



CE0061

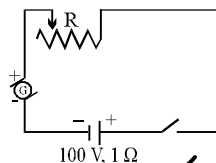
21. In the figure shown, battery 1 has emf = 6 V and internal resistance = 1Ω . Battery 2 has emf = 2 V and internal resistance = 3Ω . The wires have negligible resistance. What is the potential difference across the terminals of battery 2?



- (A) 4 V (B) 1.5 V (C) 5 V (D) 0.5 V

CE0062

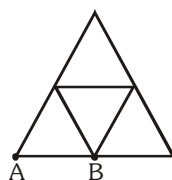
22. The battery in the diagram is to be charged by the generator G. The generator has a terminal voltage of 120 volts when the charging current is 10 amperes. The battery has an emf of 100 volts and an internal resistance of 1 ohm. In order to charge the battery at 10 amperes charging current, the resistance R should be set at :-



- (A) 0.1 Ω (B) 0.5 Ω (C) 1.0 Ω (D) 5.0 Ω

CE0063

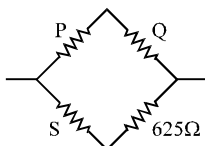
23. In the diagram resistance between any two junctions is R. Equivalent resistance across terminals A and B is :-



- (A) $\frac{11R}{7}$ (B) $\frac{18R}{11}$ (C) $\frac{7R}{11}$ (D) $\frac{11R}{18}$

CE0064

24. A Wheatstone's bridge is balanced with a resistance of 625 Ω in the third arm, where P, Q and S are in the 1st, 2nd and 4th arm respectively. If P and Q are interchanged, the resistance in the third arm has to be increased by 51 Ω to secure balance. The unknown resistance in the fourth arm is :-



- (A) 625 Ω (B) 650 Ω (C) 676 Ω (D) 600 Ω

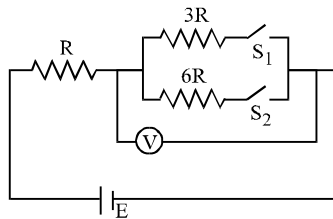
CE0065

25. One end of a Nichrome wire of length 2L and cross-sectional area A is attached to an end of another Nichrome wire of length L and cross-sectional area 2A. If the free end of the longer wire is at an electric potential of 8.0 volts, and the free end of the shorter wire is at an electric potential of 1.0 volt, the potential at the junction of the two wires is equal to :-

- (A) 2.4 V (B) 3.2 V (C) 4.5 V (D) 5.6 V

CE0066

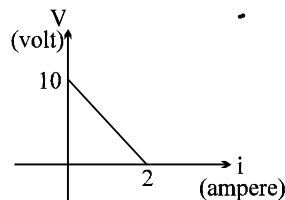
26. In the circuit shown in figure reading of voltmeter is V_1 when only S_1 is closed, reading of voltmeter is V_2 when only S_2 is closed. The reading of voltmeter is V_3 when both S_1 and S_2 are closed then



- (A) $V_2 > V_1 > V_3$ (B) $V_3 > V_2 > V_1$ (C) $V_3 > V_1 > V_2$ (D) $V_1 > V_2 > V_3$

CE0067

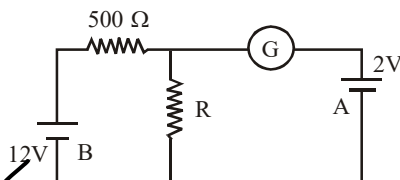
27. A battery of emf E and internal resistance r is connected across a resistance R . Resistance R can be adjusted to any value greater than or equal to zero. A graph is plotted between the current (i) passing through the resistance and potential difference (V) across it. Select the correct alternative.



- (A) internal resistance of battery is 5Ω
 (B) emf of the battery is 20V
 (C) maximum current which can be taken from the battery is 4A
 (D) $V-i$ graph can never be a straight line as shown in figure.

CE0068

28. In the circuit, the galvanometer G shows zero deflection. If the batteries A and B have negligible internal resistance, the value of the resistor R will be- [AIEEE - 2005]

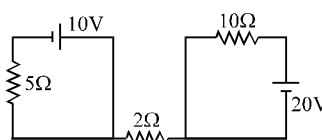


- (A) 200Ω (B) 100Ω (C) 500Ω (D) 1000Ω

CE0069

29. In the figure shown the current through 2Ω resistor is

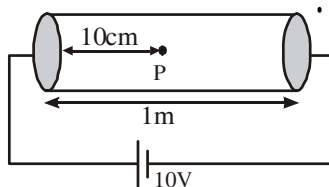
[IIT-JEE 2005 (Scr)]



- (A) 2 A (B) 0 A (C) 4 A (D) 6 A

CE0070

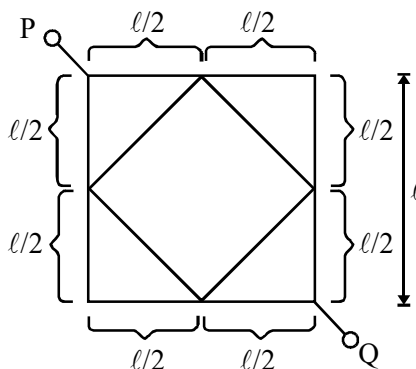
30. A cylindrical solid of length 1m and radius 1 m is connected across a source of emf 10V and negligible internal resistance shown in figure. The resistivity of the rod as a function of x (x measured from left end) is given by $\rho = bx$ [where b is a positive constant]. Find the electric field (in SI unit) at point P at a distance 10 cm from left end.



- (A) 1 (B) 2 (C) 3 (D) 4

CE0071

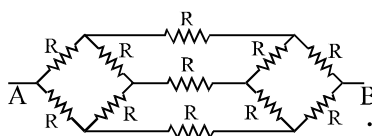
31. If the wire has resistivity ρ and cross sectional area A , the equivalent resistance between P and Q is :-



- (A) $\frac{\rho l}{\sqrt{2}A}$ (B) $\frac{\sqrt{2}\rho l}{A}$ (C) $\frac{2\rho l}{A}$ (D) $\frac{\rho l}{A}$

CE0072

32. The equivalent resistance between the terminal points A and B in the network shown in figure is :-



- (A) $\frac{7R}{5}$ (B) $\frac{5R}{6}$ (C) $\frac{7R}{12}$ (D) $\frac{5R}{12}$

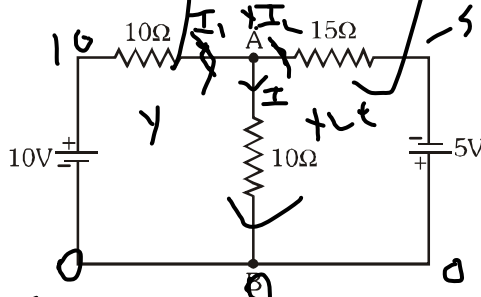
CE0073

33. An electric bell has a resistance of 5Ω and requires a current of 0.25 A to work it. Assuming that the resistance of the bell wire is 1Ω per 15m and that the bell push is 90m distance from the bell. How many cells each of emf 1.4V and internal resistance 2Ω , will be required to work the circuit-

- (A) 3 (B) 4 (C) 5 (D) Can't be determined

CE0074

34. A circuit is arranged as shown. Then, the current from A to B is



- (A) +500 mA (B) +250 mA (C) -250 mA (D) -500 mA

Joule heating

35. Power generated across a uniform wire connected across a supply is H . If the wire is cut into n equal parts and all the parts are connected in parallel across the same supply, the total power generated in the wire is

- (A) $\frac{H}{n^2}$ (B) $n^2 H$ (C) nH (D) $\frac{H}{n}$

CE0076

36. When electric bulbs of same power, but different marked voltage are connected in series across the power line, their brightness will be :

- (A) proportional to their marked voltage
(B) inversely proportional to their marked voltage
(C) proportional to the square of their marked voltage
(D) inversely proportional to the square of their marked voltage

CE0077

37. Two bulbs rated (25 W – 220V) and (100W – 220V) are connected in series to a 440 V line. Which one is likely to fuse?

- (A) 25 W bulb (B) 100 W bulb (C) both bulbs (D) none

CE0078

38. Rate of dissipation of Joule's heat in resistance per unit volume is (symbols have usual meaning)

- (A) σE (B) σI (C) $J E$ (D) None

CE0079

39. If the length of the filament of a heater is reduced by 10%, the power of the heater will

- (A) increase by about 9% (B) increase by about 11%
(C) increase by about 19% (D) decrease by about 10%

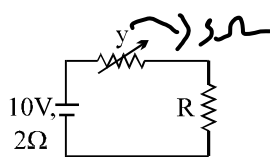
CE0080

40. Two bulbs one of 200 volts, 60 watts & the other of 200 volts, 100 watts are connected in series to a 200 volt supply. The power consumed will be

- (A) 37.5 watt (B) 160 watt (C) 62.5 watt (D) 110 watt

CE0081

41. In the figure shown the power generated in y is maximum when $y = 5\Omega$. Then R is :-

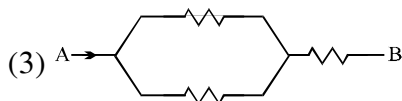
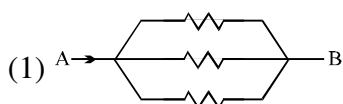


$$I = \frac{10}{R + 7}$$

- (A) 2Ω (B) 6Ω (C) 5Ω (D) 3Ω

CE0082

42. Arrange the order of power dissipated in the given circuits, if the same current is passing through all circuits and each resistor is ' r ' [IIT-JEE' 2003 (Scr)]



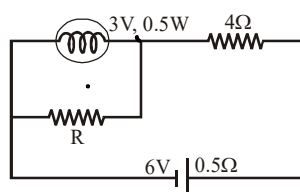
- (A) $P_2 > P_3 > P_4 > P_1$ (B) $P_3 > P_2 > P_4 > P_1$ (C) $P_4 > P_3 > P_2 > P_1$ (D) $P_1 > P_2 > P_3 > P_4$

CE0083

43. A rigid container with thermally insulated walls contains a coil of resistance 100Ω , carrying current 1 A. Change in internal energy after 5 min will be [IIT-JEE 2005]
(A) zero (B) 10 kJ (C) 20 kJ (D) 30 kJ

CE0084

44. The value of the resistance R in the circuit shown below so that electric bulb consumes the rated power is :-

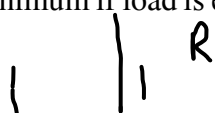


- (A) 4 ohm (B) 6 ohm (C) 8 ohm (D) 10 ohm

CE0085

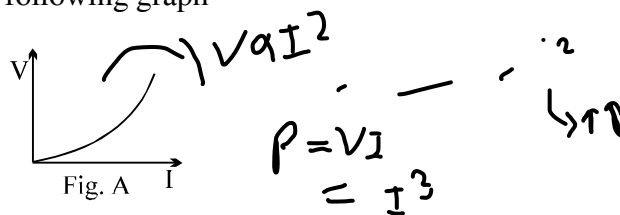
45. A variable load R is connected to a voltage source of internal resistance r . Then choose the **INCORRECT** statement out of the following :-

- (A) If $R = r$, maximum power is transferred to the load!
(B) If current is maximum, power transfer to load is also maximum
(C) If $R \ll r$, the voltage source supplied a fixed current to the load
(D) Power supplied to load is minimum if load is either too low or too high



CE0086

46. The variation of current (I) and voltage (V) is as shown in figure A. The variation of power with current I is best shown by which of the following graph



CE0087

Instruments

47. A galvanometer has a resistance of 20Ω and reads full-scale when 0.2 V is applied across it. To convert it into a 10 A ammeter, the galvanometer coil should have a

(A) 0.01Ω resistor connected across it (B) 0.02Ω resistor connected across it
(C) 200Ω resistor connected in series with it (D) 2000Ω resistor connected in series with it

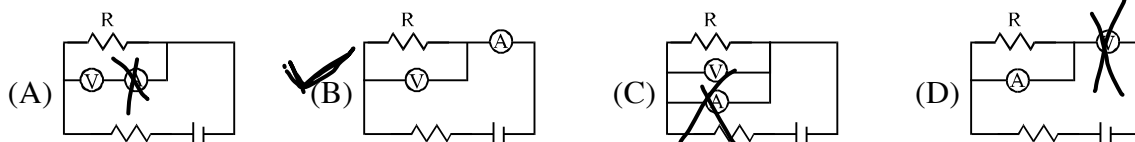
CE0088

48. A galvanometer coil has a resistance 90Ω and full scale deflection current 10 mA . A 910Ω resistance is connected in series with the galvanometer to make a voltmeter. If the least count of the voltmeter is 0.1 V , the number of divisions on its scale is

(A) 90 (B) 91 (C) 100 (D) none

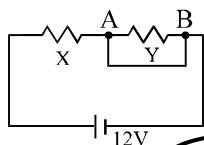
CE0089

49. Which of the following wiring diagrams could be used to experimentally determine R using ohm's law? Assume an ideal voltmeter and an ideal ammeter.



CE0090

50. When an ammeter of negligible internal resistance is inserted in series with circuit it reads 1 A . When the voltmeter of very large resistance is connected across X it reads 1 V . When the point A and B are shorted by a conducting wire, the voltmeter measures 10 V across the battery. The internal resistance of the battery is equal to :-

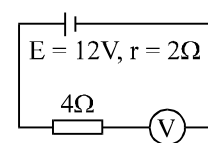


(A) zero (B) 0.5Ω (C) 0.2Ω (D) 0.1Ω

CE0091

51. By error, a student places moving-coil voltmeter V (nearly ideal) in series with the resistance in a circuit in order to read the current, as shown. The voltmeter reading will be

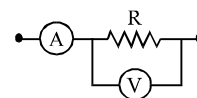
(A) 0 (B) 4 V
(C) 6 V (D) 12 V



CE0092

52. In the circuit shown the readings of ammeter and voltmeter are 4A and 20V respectively. The meters are non ideal, then R is :

(A) 5Ω (B) less than 5Ω
(C) greater than 5Ω (D) between 4Ω & 5Ω



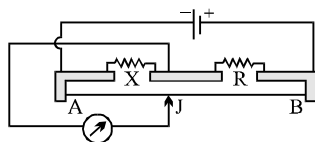
CE0093

53. In a balanced wheat stone bridge, current in the galvanometer is zero. It remains zero when:

[1] battery emf is increased
[2] all resistances are increased by 10 ohms
[3] all resistances are made five times
[4] the battery and the galvanometer are interchanged
(A) only [1] is correct (B) [1], [2] and [3] are correct
(C) [1], [3] and [4] are correct (D) [1] and [3] are correct

CE0094

54. The figure shows a metre-bridge circuit, with $AB = 100$ cm, $X = 12\Omega$ and $R = 18\Omega$, and the jockey J in the position of balance. If R is now made 8Ω , through what distance will J have to be moved to obtain balance?



(A) 10 cm (B) 20 cm (C) 30 cm (D) 40 cm

CE0095

55. In a metre bridge experiment, null point is obtained at 20 cm from one end of the wire when resistance X is balanced against another resistance Y . If $X < Y$, then where will be the new position of the null point from the same end, if one decides to balance a resistance of $4X$ against Y ? [AIEEE - 2004]

(A) 50 cm (B) 80 cm (C) 40 cm (D) 70 cm

CE0096

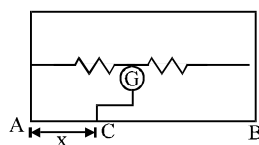
56. A resistance R is to be measured using a meter bridge. Student chooses the standard resistance S to be 100Ω . He finds the null point at $l_1 = 2.9$ cm. He is told to attempt to improve the accuracy.

Which of the following is a useful way?

(A) He should measure l_1 more accurately.
(B) He should change S to 1000Ω and repeat the experiment.
(C) He should change S to 3Ω and repeat the experiment.
(D) He should give up hope of a more accurate measurement with a meter bridge.

CE0097

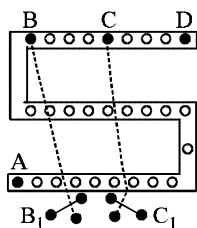
57. In the given circuit, no current is passing through the galvanometer. If the cross-sectional diameter of AB is doubled then for null point of galvanometer the value of AC would [IIT-JEE' 2003 (Scr)]



(A) x (B) $x/2$ (C) $2x$ (D) None

CE0098

58. For the post office box arrangement to determine the value of unknown resistance, the unknown resistance should be connected between [IIT-JEE' 2004 (Scr)]



- (A) B and C (B) C and D (C) A and D (D) B₁ and C₁

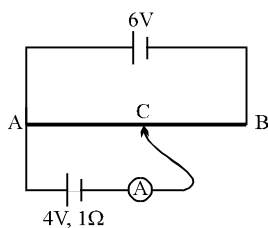
CE0099

59. A potentiometer wire has length 10 m and resistance 10Ω. It is connected to a battery of EMF 11 volt and internal resistance 1 Ω, then the potential gradient in the wire is :-

- (A) 10 V/m (B) 1 V/m (C) 0.1 V/m (D) none

CE0100

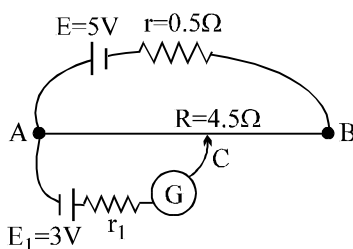
60. A 6 V battery of negligible internal resistance is connected across a uniform wire of length 1 m. The positive terminal of another battery of emf 4V and internal resistance 1 Ω is joined to the point A as shown in figure. The ammeter shows zero deflection when the jockey touches the wire at the point C. The AC is equal to :-



- (A) 2/3 m (B) 1/3 m (C) 3/5 m (D) 1/2 m

CE0101

61. In the given potentiometer circuit length of the wire AB is 3 m and resistance is $R = 4.5 \Omega$. The length AC for no deflection in galvanometer is :-



- (A) 2 m (B) 1.8 m (C) dependent on r_1 (D) none of these

CE0102

62. The length of a potentiometer wire is ℓ . A cell of emf E is balanced at a length $\ell/3$ from the positive end of the wire. If the length of the wire is increased by $\ell/2$. At what distance will the same cell give a balance point.

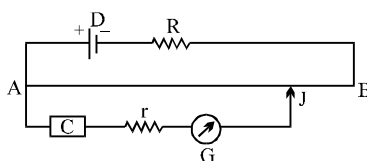
- (A) $\frac{2\ell}{3}$ (B) $\frac{\ell}{2}$ (C) $\frac{\ell}{6}$ (D) $\frac{4\ell}{3}$

CE0103

63. Two cells of emf's approximately 5V and 10V are to be accurately compared using a potentiometer of length 400cm.
- (A) The battery that runs the potentiometer should have voltage of 8V.
- (B) The battery of potentiometer can have a voltage of 15V and R adjusted so that the potential drop across the wire slightly exceeds 10V.
- (C) The first portion of 50 cm of wire itself should have a potential drop of 10V.
- (D) Potentiometer is usually used for comparing resistances and not voltages.

CE0104

64. In the given potentiometer circuit, the resistance of the potentiometer wire AB is R_0 . C is a cell of internal resistance r . The galvanometer G does not give zero deflection for any position of the jockey J. Which of the following cannot be a reason for this ?



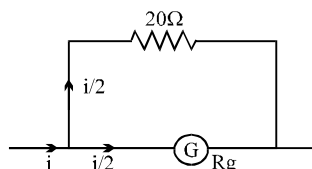
- (A) $r > R_0$ (B) $R \gg R_0$
 (C) emf of C > emf of D (D) The negative terminal of C is connected to A.

CE0105

65. A resistor has a color code of green, blue; brown and silver. What is its resistance?
- (A) $56 \Omega \pm 5\%$ (B) $560 \Omega \pm 10\%$ (C) $560 \Omega \pm 5\%$ (D) $5600 \Omega \pm 10\%$

CE0106

66. In a galvanometer, the deflection becomes one half when the galvanometer is shunted by a 20Ω resistor. The galvanometer resistance is

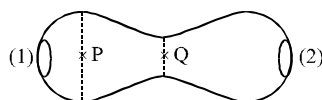


- (A) 5Ω (B) 10Ω (C) 40Ω (D) 20Ω

CE0107

MULTIPLE CORRECT TYPE QUESTIONS

67. A metallic conductor of irregular cross-section is as shown in the figure. A constant potential difference is applied across the ends (1) and (2). Then :



- (A) the current at the cross-section P equals the current at the cross-section Q
- (B) the electric field intensity at P is less than that at Q.
- (C) the rate of heat generated per unit time at Q is greater than that at P
- (D) the number of electrons crossing per unit area of cross-section at P is less than that at Q.

CE0108

68. A current passes through an ohmic conductor of nonuniform cross section. Which of the following quantities are independent of the cross-section?
- (A) the charge crossing in a given time interval.
 - (B) drift speed
 - (C) current density
 - (D) free-electron density

CE0109

69. A battery is of emf E is being charged from a charger such that positive terminal of the battery is connected to terminal A of charger and negative terminal of the battery is connected to terminal B of charger. The internal resistance of the battery is r .
- (A) Potential difference across points A and B must be more than E .
 - (B) A must be at higher potential than B
 - (C) In battery, current flows from positive terminal to the negative terminal
 - (D) No current flows through battery

CE0110

70. Two identical fuses are rated at 10A. If they are joined
- (A) in parallel, the combination acts as a fuse of rating 20A
 - (B) in parallel, the combination acts as a fuse of rating 5A
 - (C) in series, the combination acts as a fuse of rating 10A.
 - (D) in series, the combination acts as a fuse of rating 20A.

CE0111

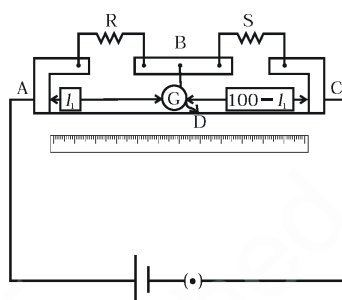
71. A micrometer has a resistance of 100Ω and a full scale range of $50\mu\text{A}$. It can be used as a voltmeter or a higher range ammeter provided a resistance is added to it. Pick the correct range and resistance combination(s).
- (A) 50 V range with $10\text{ k}\Omega$ resistance in series.
 - (B) 10 V range with $200\text{ k}\Omega$ resistance in series.
 - (C) 5 mA range with 1Ω resistance in parallel.
 - (D) 10 mA range with $1\text{ k}\Omega$ resistance in parallel.

CE0112

72. Mark out the correct options.
- (A) An ammeter should have small resistance.
 - (B) An ammeter should have large resistance.
 - (C) A voltmeter should have small resistance.
 - (D) A voltmeter should have large resistance.

CE0113

73. In a meter bridge the point D is a neutral point as shown in figure.

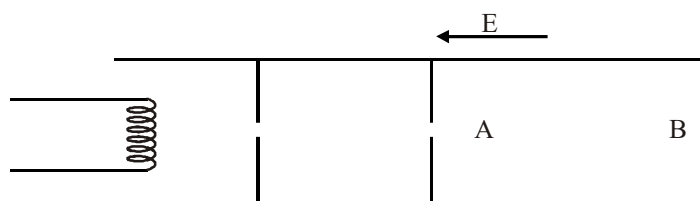


- (A) The meter bridge can have no other neutral point for this set of resistances.
 (B) When the jockey contacts a point on meter wire left of D, current flows to B from the wire through galvanometer.
 (C) When the jockey contacts a point on the meter wire to the right of D, current flows from B to the wire through galvanometer.
 (D) When R is increased, the neutral point shifts to left.

CE0114

MATRIX MATCH TYPE QUESTION

74. Electrons are emitted by a hot filament and are accelerated by an electric field as shown in figure. The two stops at the left ensure that the electron beam has a uniform cross-section. Match the entries of column-I with column-II as electron move from A to B :



Column-I

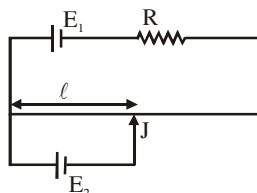
- (A) Speed of an electron
 (B) Number of free electrons per unit volume
 (C) Current density
 (D) Electric potential

Column-II

- (P) Increases
 (Q) Decreases
 (R) Remains same
 (S) any of the above is possible

CE0115

75. In the potentiometer arrangement shown in figure, null point is obtained at length ℓ .



Column-I

- (A) If E_1 is increased
 (B) If R is increased
 (C) If E_2 is increased

Column-II

- (P) ℓ should increase
 (Q) ℓ should decrease
 (R) ℓ should remain the same to again get the null point

CE0116

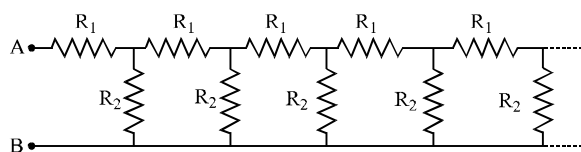
EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

- Which of the following quantities do not change when an ohmic resistor connected to a battery is heated due to the current?
 (A) drift speed (B) resistivity
 (C) resistance (D) number of free electrons

CE0117

- Consider an infinite ladder network shown in figure. A voltage V is applied between the points A and B. This applied value of voltage is halved after each section.



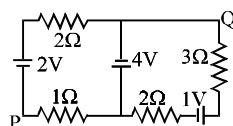
- (A) $R_1/R_2 = 1$ (B) $R_1/R_2 = 1/2$ (C) $R_1/R_2 = 2$ (D) $R_1/R_2 = 3$

CE0118

- A wire of length L and 3 identical cells of negligible internal resistances are connected in series. Due to the current, the temperature of the wire is raised by ΔT in time t . N number of similar cells is now connected in series with a wire of the same material and cross section but of length $2L$. The temperature of the wire is raised by the same amount ΔT in the same time t . The value of N is :
 (A) 4 (B) 6 (C) 8 (D) 9

CE0119

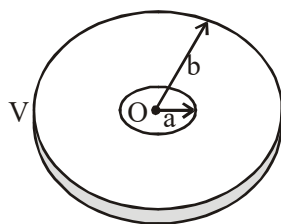
- In the circuit shown, what is the potential difference V_{PQ} ?



- (A) +3V (B) +2V (C) -2V (D) none

CE0120

- A circular portion is cut of a disc of thickness t , its resistivity is ρ and radii of disc are a and b ($b > a$). A potential difference is maintained between outer and inner cylindrical surfaces of the disc. What is resistance of the disc ?



- (A) $\frac{\rho}{2\pi t} \ln\left(\frac{b}{a}\right)$ (B) $\rho\left(\frac{1}{a} - \frac{1}{b}\right)$ (C) $2\pi\rho t\left(\frac{1}{a^2} - \frac{1}{b^2}\right)$ (D) $\frac{\rho}{2\pi t}\left(\frac{b^2 - a^2}{ab}\right)$

CE0121

6. The wire shown in figure has a uniform cross-section A.

$$\overline{\quad\quad\quad} \quad \begin{matrix} x=0 & & x=L \end{matrix}$$

Resistivity of the material of wire is given by $\rho = \rho_0 \left(\frac{L}{L+x} \right)$. A potential difference V is applied across the wire :-

(A) Resistance of wire is $\frac{\rho_0 L}{A} \cdot \ln(2)$

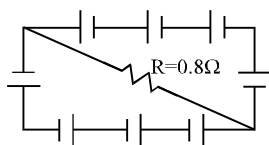
(B) Current density is variable inside the wire

(C) Electric field at $x=0$ is $\frac{2V}{(\ln 2) \cdot L}$

(D) Electric field at $x=L$ is $\frac{V}{(\ln 2)L}$

CE0122

7. A circuit is comprised of eight identical batteries and a resistor $R = 0.8\Omega$. Each battery has an emf of 1.0 V and internal resistance of 0.2Ω . The voltage difference across any of the battery is



(A) 0.5V

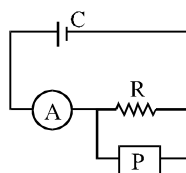
(B) 1.0V

(C) 0 V

(D) 2 V

CE0123

8. An ammeter A of finite resistance, and a resistor R are joined in series to an ideal cell C. A potentiometer P is joined in parallel to R. The ammeter reading is I_0 and the potentiometer reading is V_0 . P is now replaced by a voltmeter of finite resistance. The ammeter reading now is I and the voltmeter reading is V.



(A) $I > I_0, V < V_0$

(B) $I > I_0, V = V_0$

(C) $I = I_0, V < V_0$

(D) $I < I_0, V = V_0$

CE0124

9. The sensitivity of post-office box for determination of resistance of 5Ω is maximum when, P and Q both are

(A) 1 Ω

(B) 10 Ω

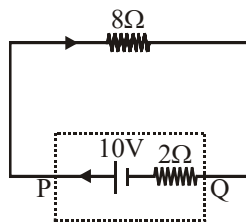
(C) 100 Ω

(D) 1000 Ω

CE0125

MULTIPLE CORRECT TYPE QUESTIONS

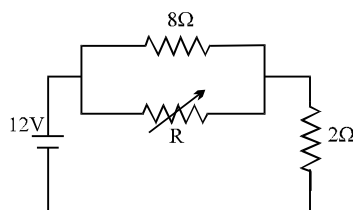
10. A battery of emf 10 volt and internal resistance 2Ω is connected to an external resistance 8Ω as shown in the figure :-



- (A) Work done due to conservative electric field while a unit positive charge passes through battery from Q to P (along the arrow) is 8 Joule.
 (B) Work done due to conservative electric field while a unit positive charge passes through battery from Q to P (along the arrow) is -8 Joule.
 (C) Work done due to conservative electric field while a unit positive charge passes through 8Ω along the arrow is -8 Joule.
 (D) Work done due to non conservative electric field while a unit positive charge moves from Q to P (along the arrow) is 10 Joule.

CE0126

11. The value of the resistance R in figure is adjusted such that power dissipated in the 2Ω resistor is maximum. Under this condition

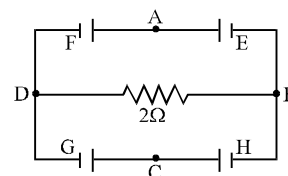


- (A) $R = 0$
 (B) $R = 8\Omega$
 (C) power dissipated in the 2Ω resistor is 72 W.
 (D) power dissipated in the 2Ω resistor is 8 W.

CE0127

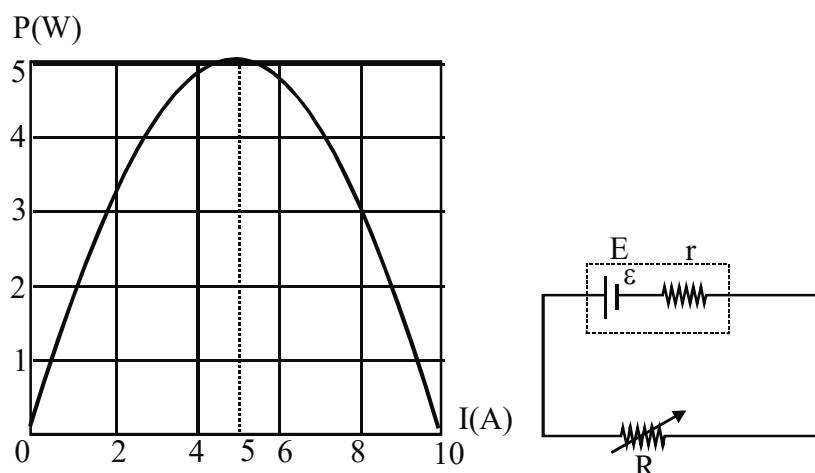
12. In the circuit shown E, F, G and H are cells of e.m.f. 2V, 1V, 3V and 1V respectively and their internal resistances are 2Ω , 1Ω , 3Ω and 1Ω respectively.

- (A) $V_D - V_B = -2/13$ V
 (B) $V_D - V_B = 2/13$ V
 (C) $V_G = 21/13$ V = potential difference across G.
 (D) $V_H = 19/13$ V = potential difference across H.



CE0128

13. Figure shows the net power dissipated in R versus the current in a simple circuit shown.



- (A) The internal resistance of battery is 0.2Ω (B) The emf of battery is 2V
(C) R at which power is 5W is 2.5Ω (D) At $i = 2$ A, power is 3.2 W

CE0129

14. In a potentiometer wire experiment the emf of a battery in the primary circuit is 20V and its internal resistance is 5Ω . There is a resistance box in series with the battery and the potentiometer wire, whose resistance can be varied from 120Ω to 170Ω . Resistance of the potentiometer wire is 75Ω . The following potential differences can be measured using this potentiometer.

- (A) 5V (B) 6V (C) 7V (D) 8V

CE0130

EXERCISE-JM

1. Two conductors have the same resistance at 0°C but their temperature coefficients of resistance are α_1 and α_2 . The respective temperature coefficients of their series and parallel combinations are nearly :
[AIEEE-2010]

(1) $\frac{\alpha_1 + \alpha_2}{2}, \frac{\alpha_1 + \alpha_2}{2}$ (2) $\frac{\alpha_1 + \alpha_2}{2}, \alpha_1 + \alpha_2$ (3) $\alpha_1 + \alpha_2, \frac{\alpha_1 + \alpha_2}{2}$ (4) $\alpha_1 + \alpha_2, \frac{\alpha_1 \alpha_2}{\alpha_1 + \alpha_2}$

CE0131

2. If a wire is stretched to make it 0.1 % longer its resistance will :-
(1) decrease by 0.2% (2) decrease by 0.05% (3) increase by 0.05% (4) increase by 0.2%
[AIEEE-2011]

CE0132

3. If $400\ \Omega$ of resistance is made by adding four $100\ \Omega$ resistance of tolerance 5%, then the tolerance of the combination is :
(1) 20% (2) 5% (3) 10% (4) 15%
[AIEEE - 2011]

CE0133

4. The current in the primary circuit of a potentiometer is 0.2 A. The specific resistance and cross-section of the potentiometer wire are 4×10^{-7} ohm metre and 8×10^{-7} m² respectively. The potential gradient will be equal to :-
(1) 0.2 V/m (2) 1 V/m (3) 0.5 V/m (4) 0.1 V/m
[AIEEE - 2011]

CE0134

5. Two electric bulbs marked 25W-220 V and 100 W-220 V are connected in series to a 440 V supply. Which of the bulbs will fuse ?
(1) Neither (2) Both (3) 100 W (4) 25 W
[AIEEE - 2012]

CE0135

6. The supply voltage to a room is 120V. The resistance of the lead wires is $6\ \Omega$. A 60 W bulb is already switched on. What is the decrease of voltage across the bulb, when a 240 W heater is switched on in parallel to the bulb?
(1) zero Volt (2) 2.9 Volt (3) 13.3 Volt (4) 10.04 Volt
[JEE-Main2013]

CE0136

7. This question has Statement I and Statement II. Of the four choice given after the Statements, choose the one that best describes the two Statements.
[JEE-Main 2013]

Statement-I : Higher the range, greater is the resistance of ammeter.

Statement-II : To increase the range of ammeter, additional shunt needs to be used across it.

- (1) Statement-I is true, Statement-II is true, Statement-II is the **correct** explanation of Statement-I
(2) Statement-I is true, Statement-II is true, Statement-II is **not** the correct explanation of Statement-I.
(3) Statement-I is **true**, Statement-II is false.
(4) Statement-I is **false**, Statement-II is true.

CE0137

8. In a large building, there are 15 bulbs of 40 W, 5 bulbs of 100 W, 5 fans of 80 W and 1 heater of 1kW. The voltage of the electric mains is 220 V. The minimum capacity of the main fuse of the building will be : [JEE-Main 2014]

(1) 12 A (2) 14 A (3) 8 A (4) 10 A

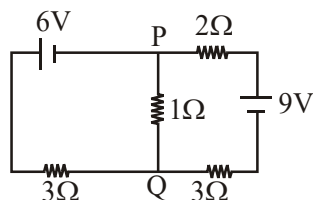
CE0138

9. When 5V potential difference is applied across a wire of length 0.1 m, the drift speed of electrons is $2.5 \times 10^{-4} \text{ ms}^{-1}$. If the electron density in the wire is $8 \times 10^{28} \text{ m}^{-3}$, the resistivity of the material is close to :- [JEE-Main 2015]

(1) $1.6 \times 10^{-6} \Omega \text{m}$ (2) $1.6 \times 10^{-5} \Omega \text{m}$ (3) $1.6 \times 10^{-8} \Omega \text{m}$ (4) $1.6 \times 10^{-7} \Omega \text{m}$

CE0139

10. In the circuit shown, the current in the 1Ω resistor is :- [JEE-Main 2015]



- (1) 0.13 A, from Q to P (2) 0.13 A, from P to Q
(3) 1.3 A, from P to Q (4) 0A

CE0140

11. A galvanometer having a coil resistance of 100Ω gives a full scale deflection, when a current of 1 mA is passed through it. The value of the resistance, which can convert this galvanometer into ammeter giving a full scale deflection for a current of 10A, is :- [JEE-Main 2016]

(1) 3Ω (2) 0.01Ω (3) 2Ω (4) 0.1Ω

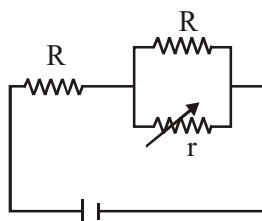
CE0141

12. A 50Ω resistance is connected to a battery of 5V. A galvanometer of resistance 100Ω is to be used as an ammeter to measure current through the resistance, for this a resistance r_s is connected to the galvanometer. Which of the following connections should be employed if the measured current is within 1% of the current without the ammeter in the circuit ? [JEE-Mains (Online) 2016]

- (1) $r_s = 1\Omega$ in series with galvanometer (2) $r_s = 0.5\Omega$ in parallel with the galvanometer
(3) $r_s = 0.5\Omega$ in series with the galvanometer (4) $r_s = 1\Omega$ in parallel with galvanometer

CE0142

13. In the circuit shown, the resistance r is a variable resistance. If for $r = fR$, the heat generation in r is maximum then the value of f is : [JEE-Mains (Online) 2016]



- (1) 1 (2) $\frac{1}{4}$ (3) $\frac{1}{2}$ (4) $\frac{3}{4}$

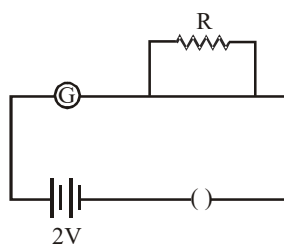
CE0143

14. The resistance of an electrical toaster has a temperature dependence given by $R(T) = R_0 [1 + \alpha(T - T_0)]$ in its range of operation. At $T_0 = 300$ K, $R = 100 \Omega$ and at $T = 500$ K, $R = 120 \Omega$. The toaster is connected to a voltage source at 200 V and its temperature is raised at a constant rate from 300 to 500 K in 30 s. The total work done in raising the temperature is :- [JEE-Mains (Online) 2016]

- (1) $400 \ln \frac{1.5}{1.3}$ J (2) 300 J (3) $400 \ln \frac{5}{6}$ J (4) $200 \ln \frac{2}{3}$ J

CE0144

15. A galvanometer has a 50 division scale. Battery has no internal resistance. It is found that there is deflection of 40 divisions when $R = 2400 \Omega$. Deflection becomes 20 divisions when resistance taken from resistance box is 4900 Ω . Then we can conclude : [JEE-Mains (Online) 2016]



- (1) Resistance of galvanometer is 200 Ω .
 (2) Resistance required on R.B. for a deflection of 10 divisions is 9800 Ω .
 (3) Full scale deflection current is 2 mA.
 (4) Current sensitivity of galvanometer is 20 μ A/division.

CE0145

16. Which of the following statements is false ? [JEE-Main 2017]

- (1) A rheostat can be used as a potential divider
 (2) Kirchhoff's second law represents energy conservation
 (3) Wheatstone bridge is the most sensitive when all the four resistances are of the same order of magnitude.
 (4) In a balanced wheatstone bridge if the cell and the galvanometer are exchanged, the null point is disturbed.

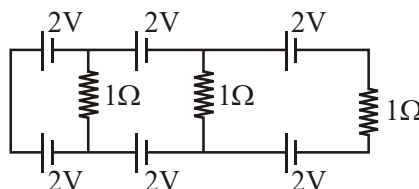
CE0146

17. When a current of 5 mA is passed through a galvanometer having a coil of resistance 15 Ω , it shows full scale deflection. The value of the resistance to be put in series with the galvanometer to convert it into a voltmeter of range 0 – 10 V is :- [JEE-Main 2017]

- (1) $2.535 \times 10^3 \Omega$ (2) $4.005 \times 10^3 \Omega$ (3) $1.985 \times 10^3 \Omega$ (4) $2.045 \times 10^3 \Omega$

CE0147

18. In the above circuit the current in each resistance is :- [JEE-Main 2017]



- (1) 0.5 A (2) 0 A (3) 1 A (4) 0.25 A

CE0148

19. In a potentiometer experiment, it is found that no current passes through the galvanometer when the terminals of the cell are connected across 52 cm of the potentiometer wire. If the cell is shunted by a resistance of $5\ \Omega$, a balance is found when the cell is connected across 40 cm of the wire. Find the internal resistance of the cell. **[JEE-Main 2018]**

(1) $1.5\ \Omega$ (2) $2\ \Omega$ (3) $2.5\ \Omega$ (4) $1\ \Omega$

CE0149

20. On interchanging the resistances, the balance point of a meter bridge shifts to the left by 10 cm. The resistance of their series combination is $1\ \text{k}\Omega$. How much was the resistance on the left slot before interchanging the resistances ? **[JEE-Main 2018]**

(1) $505\ \text{k}\Omega$ (2) $550\ \text{k}\Omega$ (3) $910\ \text{k}\Omega$ (4) $990\ \text{k}\Omega$

CE0150

21. Two batteries with e.m.f $12\ \text{V}$ and $13\ \text{V}$ are connected in parallel across a load resistor of $10\ \Omega$. The internal resistances of the two batteries are $1\ \Omega$ and $2\ \Omega$ respectively. The voltage across the load lies between. **[JEE-Main 2018]**

(1) $11.5\ \text{V}$ and $11.6\ \text{V}$ (2) $11.4\ \text{V}$ and $11.5\ \text{V}$
(3) $11.7\ \text{V}$ and $11.8\ \text{V}$ (4) $11.6\ \text{V}$ and $11.7\ \text{V}$

CE0151

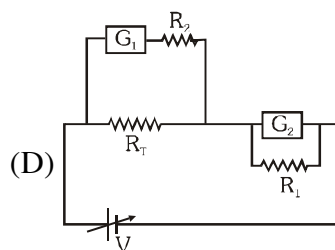
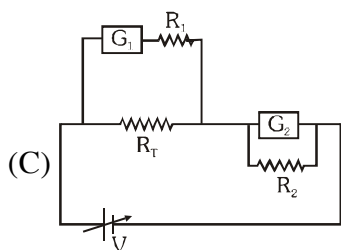
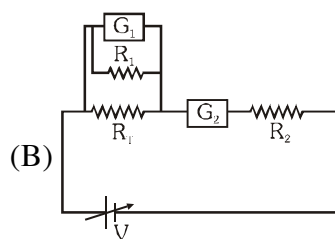
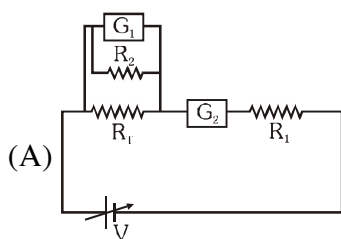
EXERCISE - JA

1. Incandescent bulbs are designed by keeping in mind that the resistance of their filament increases with the increase in temperature. If at room temperature, 100 W, 60 W and 40 W bulbs have filament resistance R_{100} , R_{60} and R_{40} , respectively, the relation between these resistances is [IIT-JEE 2010]

(A) $\frac{1}{R_{100}} = \frac{1}{R_{40}} + \frac{1}{R_{60}}$ (B) $R_{100} = R_{40} + R_{60}$ (C) $R_{100} > R_{60} > R_{40}$ (D) $\frac{1}{R_{100}} > \frac{1}{R_{60}} > \frac{1}{R_{40}}$

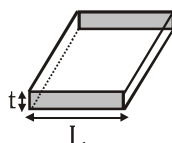
CE0152

2. To verify Ohm's law, a student is provided with a test resistor R_T , a high resistance R_1 , a small resistance R_2 , two identical galvanometers G_1 and G_2 , and a variable voltage source V . The correct circuit to carry out the experiment is :- [IIT-JEE 2010]



CE0153

3. Consider a thin square sheet of side L and thickness t , made of a material of resistivity ρ . The resistance between two opposite faces, shown by the shaded areas in the figure is [IIT-JEE 2010]



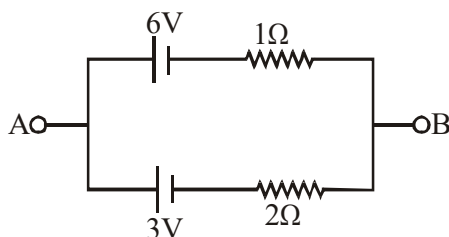
- (A) directly proportional to L (B) directly proportional to t
(C) independent of L (D) independent of t

CE0154

4. When two identical batteries of internal resistance 1Ω each are connected in series across a resistor R , the rate of heat produced in R is J_1 . When the same batteries are connected in parallel across R , the rate is J_2 . If $J_1 = 2.25 J_2$ then the value of R in Ω is [IIT-JEE 2010]

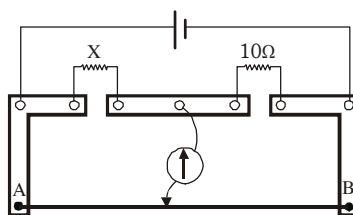
CE0155

5. Two batteries of different emfs and different internal resistances are connected as shown. The voltage across AB in volts is [IIT-JEE 2011]



CE0156

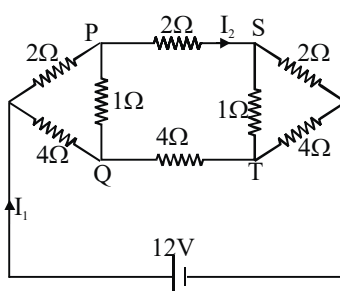
6. A meter bridge is set-up as shown, to determine an unknown resistance 'X' using a standard 10Ω resistor. The galvanometer shows null point when tapping-key is at 52 cm mark. The end-corrections are 1 cm and 2 cm respectively for the ends A and B . The determined value of 'X' is [IIT-JEE 2011]



- (A) 10.2 ohm (B) 10.6 ohm (C) 10.8 ohm (D) 11.1 ohm

CE0157

7. For the resistance network shown in the figure, choose the correct option(s). [IIT-JEE 2012]



- (A) the current through PQ is zero (B) $I_1 = 3\text{ A}$
(C) The potential at S is less than that at Q (D) $I_2 = 2\text{ A}$

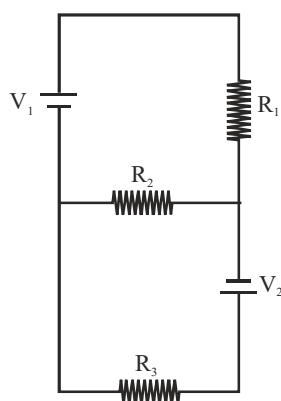
CE0158

8. Heater of an electric kettle is made of a wire of length L and diameter d . It takes 4 minutes to raise the temperature of 0.5 kg water by 40 K. This heater is replaced by a new heater having two wires of the same material, each of length L and diameter $2d$. The way these wires are connected is given in the options. How much time in minutes will it take to raise the temperature of the same amount of water by 40 K ? [JEE Advanced 2014]

- (A) 4 if wires are in parallel (B) 2 if wires are in series
(C) 1 if wires are in series (D) 0.5 if wires are in parallel

CE0159

9. Two ideal batteries of emf V_1 and V_2 and three resistances R_1 , R_2 and R_3 are connected as shown in the figure. The current in resistance R_2 would be zero if :- [JEE Advanced 2014]



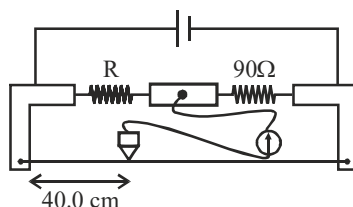
- (A) $V_1 = V_2$ and $R_1 = R_2 = R_3$ (B) $V_1 = V_2$ and $R_1 = 2R_2 = R_3$
(C) $V_1 = 2V_2$ and $2R_1 = 2R_2 = R_3$ (D) $2V_1 = V_2$ and $2R_1 = R_2 = R_3$

CE0160

10. A galvanometer gives full scale deflection with 0.006 A current. By connecting it to a $4990\ \Omega$ resistance, it can be converted into a voltmeter of range 0 - 30 V. If connected to a $\frac{2n}{249}\ \Omega$ resistance, it becomes an ammeter of range 0 - 1.5 A. The value of n is :- [JEE Advanced 2014]

CE0161

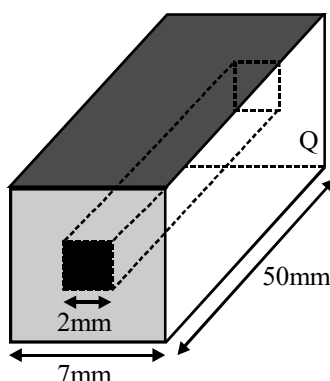
11. During an experiment with a metre bridge, the galvanometer shows a null point when the jockey is pressed at 40.0 cm using a standard resistance of $90\ \Omega$, as shown in the figure. The least count of the scale used in the metre bridge is 1 mm. The unknown resistance is :- [JEE Advanced 2014]



- (A) $60 \pm 0.15\ \Omega$ (B) $135 \pm 0.56\ \Omega$ (C) $60 \pm 0.25\ \Omega$ (D) $135 \pm 0.23\ \Omega$

CE0162

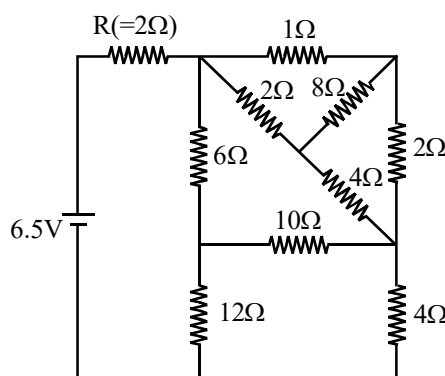
12. In an aluminum (Al) bar of square cross section, a square hole is drilled and is filled with iron (Fe) as shown in the figure. The electrical resistivities of Al and Fe are $2.7 \times 10^{-8} \Omega \text{ m}$ and $1.0 \times 10^{-7} \Omega \text{ m}$, respectively. The electrical resistance between the two faces P and Q of the composite bar is : [JEE Advanced-2015]



- (A) $\frac{2475}{64} \mu\Omega$ (B) $\frac{1875}{64} \mu\Omega$ (C) $\frac{1875}{49} \mu\Omega$ (D) $\frac{2475}{132} \mu\Omega$

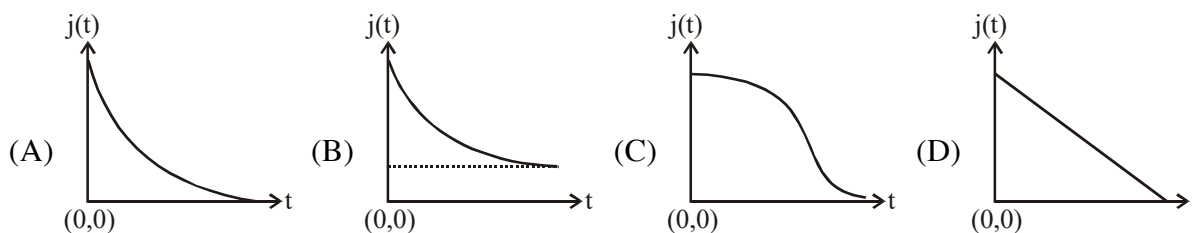
CE0163

13. In the following circuit, the current through the resistor $R (= 2\Omega)$ is I Amperes. The value of I is [JEE Advanced-2015]



CE0164

14. An infinite line charge of uniform electric charge density λ lies along the axis of an electrically conducting infinite cylindrical shell of radius R . At time $t = 0$, the space inside the cylinder is filled with a material of permittivity ϵ and electrical conductivity σ . The electrical conduction in the material follows Ohm's law. Which one of the following graphs best describes the subsequent variation of the magnitude of current density $j(t)$ at any point in the material? [JEE Advanced-2016]



CE0165

15. An incandescent bulb has a thin filament of tungsten that is heated to high temperature by passing an electric current. The hot filament emits black-body radiation. The filament is observed to break up at random locations after a sufficiently long time of operation due to non-uniform evaporation of tungsten from the filament. If the bulb is powered at constant voltage, which of the following statement(s) is(are) true? [JEE Advanced-2016]

- (A) The temperature distribution over the filament is uniform
- (B) The resistance over small sections of the filament decreases with time
- (C) The filament emits more light at higher band of frequencies before it breaks up
- (D) The filament consumes less electrical power towards the end of the life of the bulb

CE0166

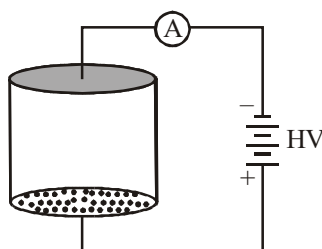
16. Consider two identical galvanometers and two identical resistors with resistance R . If the internal resistance of the galvanometers $R_G < R/2$, which of the following statement(s) about any one of the galvanometers is(are) true? [JEE Advanced-2016]

- (A) The maximum voltage range is obtained when all the components are connected in series
- (B) The maximum voltage range is obtained when the two resistors and one galvanometer are connected in series, and the second galvanometer is connected in parallel to the first galvanometer
- (C) The maximum current range is obtained when all the components are connected in parallel
- (D) The maximum current range is obtained when the two galvanometers are connected in series and the combination is connected in parallel with both the resistors.

CE0167

Paragraph for Questions No. 17 and 18

Consider an evacuated cylindrical chamber of height h having rigid conducting plates at the ends and an insulating curved surface as shown in the figure. A number of spherical balls made of a light weight and soft material and coated with a conducting material are placed on the bottom plate. The balls have a radius $r \ll h$. Now a high voltage source (HV) is connected across the conducting plates such that the bottom plate is at $+V_0$ and the top plate at $-V_0$. Due to their conducting surface, the balls will get charged, will become equipotential with the plate and are repelled by it. The balls will eventually collide with the top plate, where the coefficient of restitution can be taken to be zero due to the soft nature of the material of the balls. The electric field in the chamber can be considered to be that of a parallel plate capacitor. Assume that there are no collision between the balls and the interaction between them is negligible. (Ignore gravity) [JEE Advanced-2016]



17. Which of the following statements is correct ?
- (A) The balls will bounce back to the bottom plate carrying the opposite charge they went up with
(B) the balls will execute simple harmonic motion between the two plates
(C) The balls will bounce back to the bottom plate carrying the same charge they went up with
(D) The balls will stick to the top plate and remain there

CE0168

18. The average current in the steady state registered by the ammeter in the circuit will be :

- (A) Proportional to $V_0^{1/2}$ (B) Proportional to V_0^2
(C) Proportional to the potential V_0 (D) Zero

CE0168

19. Two identical moving coil galvanometer have $10\ \Omega$ resistance and full scale deflection at $2\ \mu\text{A}$ current. One of them is converted into a voltmeter of $100\ \text{mV}$ full scale reading and the other into an Ammeter of $1\ \text{mA}$ full scale current using appropriate resistors. These are then used to measure the voltage and current in the Ohm's law experiment with $R = 1000\ \Omega$ resistor by using an ideal cell. Which of the following statement(s) is/are correct ?

[JEE Advanced-2019]

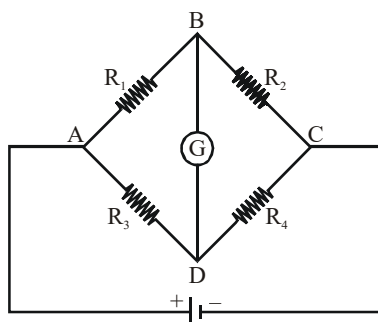
- (1) The measured value of R will be $978\ \Omega < R < 982\ \Omega$.
(2) The resistance of the Voltmeter will be $100\ \text{k}\Omega$.
(3) The resistance of the Ammeter will be $0.02\ \Omega$ (round off to 2^{nd} decimal place)
(4) If the ideal cell is replaced by a cell having internal resistance of $5\ \Omega$ then the measured value of R will be more than $1000\ \Omega$.

CE0169

CURRENT ELECTRICITY

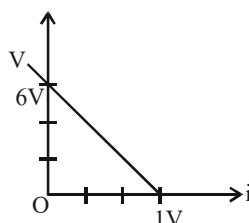
(CBSE Previous Year's Questions)

1. Explain how does the resistivity of a conductor depend upon (i) number density ' n ' of free electrons, and (ii) relaxation time ' λ '. [2; CBSE-2004]
2. Two cells of emf 1.5 V and 2 V and internal resistance 1 ohm and 2 ohm respectively are connected in parallel to pass a current in the same direction through an external resistance of 5 ohm.
(i) Draw the circuit diagram.
(ii) Using Kirchhoff's laws, calculate the current through each branch of the circuit and potential difference across the 5 ohm resistor. [3; CBSE-2005]
3. You are given ' n ' resistors, each of resistance ' r '. These are first connected to get minimum possible resistance. In the second case, these are again connected differently to get maximum possible resistance. Compute the ratio between the minimum and maximum values of resistance so obtained. [2; CBSE-2006]
4. Draw a circuit diagram using a metre bridge and write the necessary mathematical relation used to determine the value of an unknown resistance. Why cannot such an arrangement be used for measuring very low resistances? [2; CBSE-2006]
5. Which one of the two, an ammeter or a millianuneter, has a higher resistance and why? [2; CBSE-2006]
6. The given figure shows a network of resistances R_1 , R_2 , R_3 and R_4 .



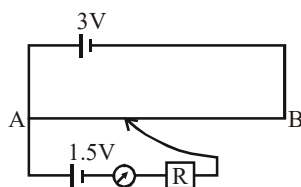
Using Kirchhoff's laws, establish the balance condition for the network. [2; CBSE-2007]

7. The plot of the variation of potential difference across a combination of three identical cells in series, versus current is as shown below. What is the emf of each, cell [1; CBSE-2008]

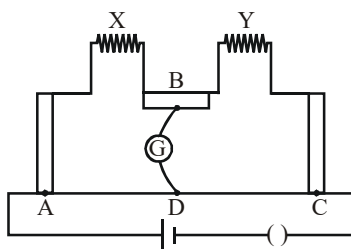


8. Prove that the current density of a metallic conductor is directly proportional to the drift speed of electrons. [3; CBSE-2008]

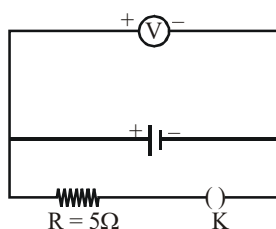
9. A number of identical cells, n each of emf E , internal resistance r connected in series are charged by a d.c source of emf E' , using a resistor R . (i) Draw the circuit arrangement. (ii) Deduce the expressions for (a) the charging current and (b) the potential difference across the combination of the cells. [3; CBSE-2008]
10. A potentiometer wire of length 1m is connected to a drive cell of emf 3V as shown in the figure. When a cell of 1.5 V emf is used in the secondary circuit, the balance point is found to be 60cm. On replacing this cell and using a cell of unknown emf, the balance point shifts to 80cm



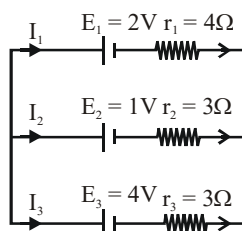
- (i) Calculate unknown emf of the cell. [3; CBSE-2008]
 (ii) Explain with reason, whether the circuit works, if the drive cell is replaced with a cell emf 1V.
 (iii) Does the high resistance R , used in the secondary circuit affect the balance point? Justify your answer.
11. A cell of emf ' E ' and internal resistance ' r ' is connected across a variable resistor ' R '. Plot a graph showing the variation of terminal potential ' V ' with Resistance R . [2; CBSE-2009]
12. Derive an expression for drift velocity of the free electrons in a conductor in terms of relaxation time. [2; CBSE-2009]
13. The figure shows experimental set up of a bridge. When the two unknown resistances X and Y are inserted, the null point D is obtained 40 cm from the end A . when a resistance of 10 is connected in series with X , the null point shifts by 10 cm. find the position of the null point when the 10 resistance is instead connected in series with resistance ' Y '. Determine the values of the resistances X and Y . [3; CBSE-2009]



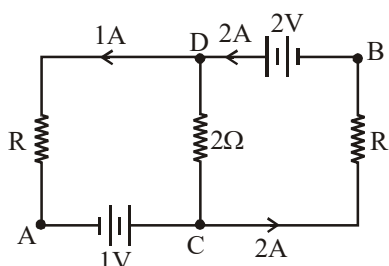
14. Write any two factors on which internal resistance of a cell depends. The reading on a high resistance voltmeter, when a cell is connected across it, is 2.2 V. When the terminals of the cell are also connected to a resistance of 5Ω as shown in the figure, the voltmeter reading drops to 1.8 V. Find the internal resistance of the cell. [3; CBSE-2010]



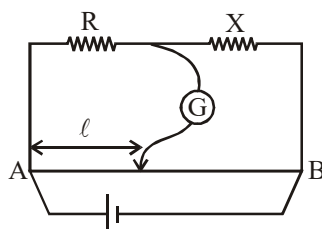
15. State Kirchhoff's rules. Use these rules to write the expressions for the currents I_1 , I_2 and I_3 in the circuit diagram shown. [3; CBSE-2010]



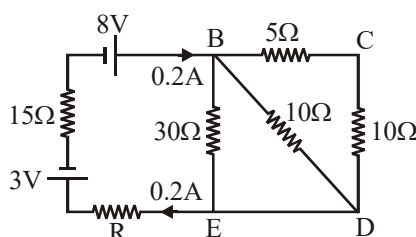
16. In the given circuit, assuming point A to be at zero potential, use Kirchhoff's rules to determine the potential at point B. [2; CBSE-2011]



17. In the meter bridge experiment, balance point was observed at J with $AJ = l$.
 (i) The values of R and X were doubled and then interchanged. What would be the new position of balance point ?
 (ii) If the galvanometer and battery are interchanged at the balance position, how will balance point get affected ? [2; CBSE-2011]



18. Two heating elements of resistance R_1 and R_2 when operated at a constant supply of voltage, V , consume powers P_1 and P_2 respectively. Deduce the expressions for the power of their combination when they are, in turn, connected in (i) series and (ii) parallel across the same voltage supply. [3; CBSE-2011]
19. Calculate the value of the resistance R in the circuit shown in the figure so that the current in the circuit is $0.2A$. What would be the potential difference between points B and E? [3; CBSE-2012]

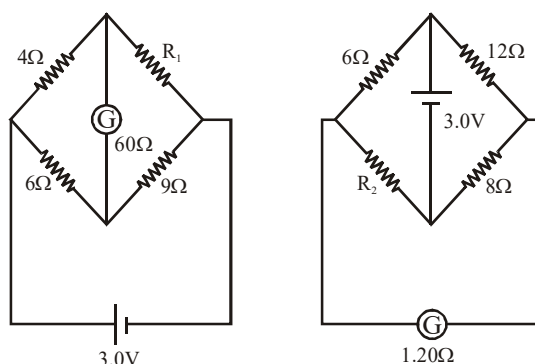


20. Define relaxation time of the free electrons drifting in a conductor. How is it related to the drift velocity of free electrons? Use this relation to deduce the expression for the electrical resistivity of the material. [3; .CBSE-2012]
21. Two identical cells, each of emf E , having negligible internal resistance, are connected in parallel with each other across an external resistance R . What is the current through this resistance? [CBSE-2013]
22. Explain the term 'drift velocity' of electrons in a conductor. Hence obtain the expression for the current through a conductor in terms of 'drift velocity'. [CBSE-2013]

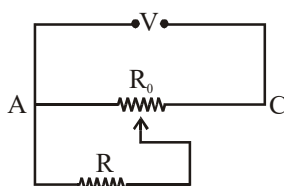
OR

Describe briefly, with the help of a circuit diagram, how a potentiometer is used to determine the internal resistance of a cell.

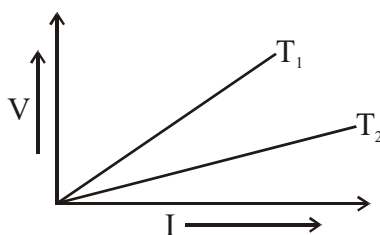
23. Define the current sensitivity of a galvanometer. Write its S. I. unit. Figure shows two circuits each having a galvanometer and a battery of 3V. When the galvanometers in each arrangement do not show any deflection, obtain the ratio R_1/R_2 . [CBSE-2013]



24. Estimate the average drift speed of conduction electrons in a copper wire of cross-sectional area $2.5 \times 10^{-7} \text{ m}^2$ carrying a current of 1.8 A. Assume the density of conduction electrons to be $9 \times 10^{28} \text{ m}^{-3}$. [CBSE-2014]
25. A cell of emf ' E ' and internal resistance ' r ' is connected across a variable resistor ' R '. Plot a graph showing variation of terminal voltage ' V ' of the cell versus the current ' I '. Using the plot, show how the emf of the cell and its internal resistance can be determined. [CBSE-2014]
26. A resistance of $R\Omega$ draws current from a potentiometer as shown in the figure. The potentiometer has a total resistance $R_0\Omega$. A voltage V is supplied to the potentiometer. Derive an expression for the voltage across R when the sliding contact is in the middle of the potentiometer. [CBSE-2014]

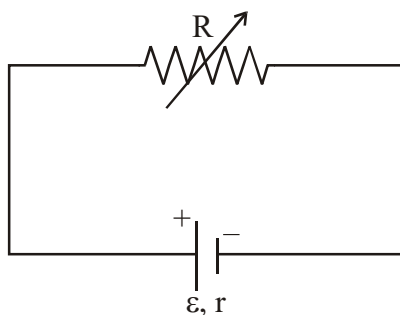


27. $V - I$ graph for a metallic wire at two different temperatures T_1 and T_2 is as shown in the figure. Which of the two temperatures is higher and why ? [1; CBSE-2015]



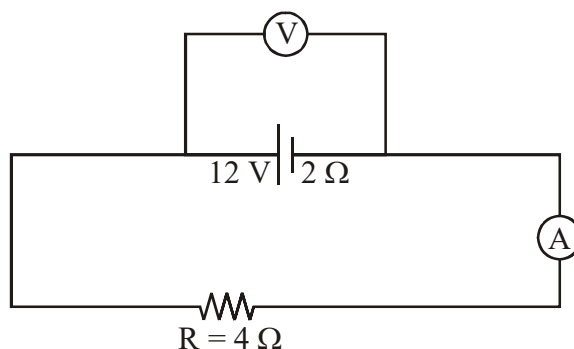
28. A variable resistor R is connected across a cell of emf ε and internal resistance r as shown in the figure. Draw a plot showing the variation of (i) terminal voltage V and (ii) the current I , as a function of R .

[2; CBSE-2015]

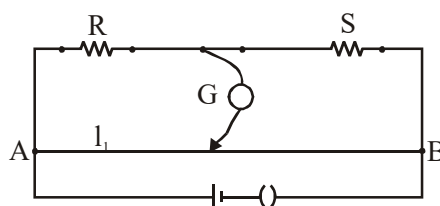


29. A potential difference V is applied across a conductor of length L and diameter D . How is the drift velocity, v_d , of charge carriers in the conductor affected when (i) V is halved, (ii) L is doubled and (iii) D is halved ? Justify your answer in each case. [3; CBSE-2015]
30. Define mobility of a charge carrier. What is its relation with relaxation time ? [1; CBSE-2016]
31. When 5V potential difference is applied across a wire of length 0.1m, the drift speed of electrons is 2.5×10^{-4} m/s. If the electron density in the wire is $8 \times 10^{28} \text{ m}^{-3}$, calculate the resistivity of the material of wire. [2; CBSE-2016]
32. Two identical cells of emf 1.5 V each joined in parallel supply energy to an external circuit consisting of two resistances of 7Ω each joined in parallel. A very high resistance voltmeter reads the terminal voltage of cells to be 1.4 V. Calculate the internal resistance of each cell. [3; CBSE-2016]
33. Nichrome and copper wires of same length and same radius are connected in series. Current I is passed through them. Which wire gets heated up more ? Justify your answer. [CBSE-2017]

34. (a) The potential difference applied across a given resistor is altered so that the heat produced per second increases by a factor of 9. By what factor does the applied potential difference change ?
 (b) In the figure shown, an ammeter A and a resistor of $4\ \Omega$ are connected to the terminals of the source. The emf of the source is 12 V having an internal resistance of $2\ \Omega$. Calculate the voltmeter and ammeter readings. [CBSE-2017]

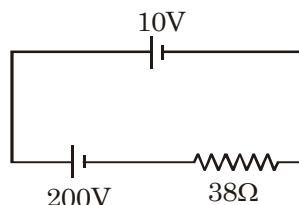


35. (a) Write the principle of working of a metre bridge.
 (b) In a metre bridge, the balance point is found at a distance l_1 with resistance R and S as shown in the figure. [CBSE-2017]



An unknown resistance X is now connected in parallel to the resistance S and the balance point is found at a distance l_2 . Obtain a formula for X in terms of l_1 , l_2 and S.

36. Two electric bulbs P and Q have their resistance in the ratio of 1 : 2. They are connected in series across a battery. Find the ratio of the power dissipation in these bulbs. [CBSE-2018]
 37. A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance $38\ \Omega$ as shown in the figure. Find the value of current in the circuit. [CBSE-2018]



In a potentiometer arrangement for determining the emf of a cell, the balance point of the cell in open circuit is 350 cm. When a resistance of $9\ \Omega$ is used in the external circuit of the cell, the balance point shifts to 300 cm. Determine the internal resistance of the cell.

38. (a) Define the term 'conductivity' of a metallic wire. Write its SI unit. [CBSE-2018]
 (b) Using the concept of free electrons in a conductor, derive the expression for the conductivity of a wire in terms of number density and relaxation time. Hence obtain the relation between current density and the applied electric field E.

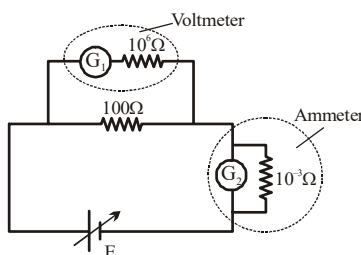
ANSWER KEY

EXERCISE - S-1

1. Ans. $I\rho/A$ 2. Ans. $S = e n l \langle v \rangle / j$ 3. Ans. $p = I m_e l/e$ 4. Ans. 3 : 1
 5. Ans. (a) (i) in series, (ii) all in parallel: n^2 . (b)
 (i) Join 1Ω , 2Ω in parallel and the combination in series with 3Ω ,
 (ii) parallel combination of 2Ω and 3Ω in series with 1Ω , (iii) all in series, (iv) all in parallel.
 (c) (i) $(16/3)\Omega$, (ii) $5R$.
 6. Ans. 600Ω 7. Ans. 7.5 mA 8. Ans. 1 V 9. Ans. $I = 2.5\text{ A}$, $V = 3.5\text{ Volts}$
 10. Ans. $20/3\text{ V}$ 11. Ans. 3.5 A 13. Ans. 1Ω 14. Ans. $x = \frac{4}{3}\text{ V}$, $12\frac{1}{3}\text{ V}$, $\frac{1}{15}\text{ A}$
 15. Ans. 5 16. Ans. $\sqrt{R_1 R_2}$ 17. Ans. 25 18. Ans. 3 19. Ans. 12 A , -20 W
 20. Ans. $4/9\text{ kg/sec.}$, 450 sec 21. Ans. 20 ohm 22. Ans. $R_1 = 0.0278\Omega$, $R_2 = 0.25\Omega$, $R_3 = 2.5\Omega$

23. Ans. 233.3Ω , 144 V

24. Ans.



25. Ans. Battery should be connected across A and B. Out put can be taken across the terminals A and C or B and C
 26. Ans. This is true for $r_1 = r_2$; So R_2 given most accurate value 27. Ans. $\frac{10}{3}\Omega$, 5Ω
 28. Ans. (a) +ve, $E_t > E$ (b) -ve 29. Ans. 2.25 V 30. Ans. 4 ohm 31. Ans. 46.67 cm

EXERCISE - S-2

1. Ans. $5 \times 10^{-7}\text{ A}$ 2. Ans. $\alpha_{\text{eff}} = \frac{5}{4}\alpha$ 3. Ans. (i) $\frac{1}{2}i_0 t_0$ (ii) $i = i_0 \left(1 - \frac{t}{t_0}\right)$ (iii) $\frac{R t_0 i_0^2}{3}$
 4. Ans. $\frac{1}{R} = \frac{\pi r^2}{3l} (2\sigma_2 + \sigma_1)$ 5. Ans. (a) $J_0 A/3$; (b) $2J_0 A/3$ 6. Ans. 4Ω
 7. Ans. $-\frac{22}{9}\text{ V}$ 8. Ans. (i) 1.01Ω (ii) $0-5\text{ A}$, $0-10\text{ V}$, (ii) 0.05 A
 9. Ans. 7.5 m , 8.75 m , 6.25 m 10. Ans. 2

EXERCISE - O-1**SINGLE CORRECT TYPE QUESTIONS**

- | | | | |
|--------------|----------------|--------------|--------------|
| 1. Ans. (C) | 2. Ans. (C) | 3. Ans. (D) | 4. Ans. (B) |
| 5. Ans. (A) | 6. Ans. (B) | 7. Ans. (C) | 8. Ans. (B) |
| 9. Ans. (C) | 10. Ans. (D) | 11. Ans. (A) | 12. Ans. (D) |
| 13. Ans. (C) | 14. Ans. (B,C) | 15. Ans. (B) | 16. Ans. (C) |
| 17. Ans. (D) | 18. Ans. (B) | 19. Ans. (D) | 20. Ans. (A) |
| 21. Ans. (C) | 22. Ans. (C) | 23. Ans. (D) | 24. Ans. (B) |
| 25. Ans. (A) | 26. Ans. (A) | 27. Ans. (A) | 28. Ans. (B) |
| 29. Ans. (B) | 30. Ans. (B) | 31. Ans. (A) | 32. Ans. (A) |
| 33. Ans. (C) | 34. Ans. (B) | 35. Ans. (B) | 36. Ans. (C) |
| 37. Ans. (A) | 38. Ans. (C) | 39. Ans. (B) | 40. Ans. (A) |
| 41. Ans. (D) | 42. Ans. (A) | 43. Ans. (D) | 44. Ans. (B) |
| 45. Ans. (B) | 46. Ans. (B) | 47. Ans. (B) | 48. Ans. (C) |
| 49. Ans. (B) | 50. Ans. (C) | 51. Ans. (D) | 52. Ans. (C) |
| 53. Ans. (C) | 54. Ans. (B) | 55. Ans. (A) | 56. Ans. (C) |
| 57. Ans. (A) | 58. Ans. (C) | 59. Ans. (B) | 60. Ans. (A) |
| 61. Ans. (D) | 62. Ans. (B) | 63. Ans. (B) | 64. Ans. (A) |
| 65. Ans. (B) | 66. Ans. (D) | | |

MULTIPLE CORRECT TYPE QUESTIONS

- | | | | |
|--------------------|----------------|-------------------|----------------|
| 67. Ans. (A,B,C,D) | 68. Ans. (A,D) | 69. Ans. (A,B,C) | 70. Ans. (A,C) |
| 71. Ans. (B,C) | 72. Ans. (A,D) | 73. Ans. (A,B, C) | |

MATRIX MATCH TYPE QUESTION

- | | |
|-----------------------------|------------------------|
| 74. Ans. A-P; B-Q; C-R; D-P | 75. Ans. A-Q; B-P; C-P |
|-----------------------------|------------------------|

EXERCISE - O-2**SINGLE CORRECT TYPE QUESTIONS**

- | | | | | |
|-------------|-------------|-------------|-------------|-------------|
| 1. Ans. (D) | 2. Ans. (B) | 3. Ans. (B) | 4. Ans. (B) | 5. Ans. (A) |
| 6. Ans. (A) | 7. Ans. (C) | 8. Ans. (A) | 9. Ans. (A) | |

MULTIPLE CORRECT TYPE QUESTIONS

- | | | | | |
|-----------------|----------------|------------------|-------------------|------------------|
| 10. Ans. (B, D) | 11. Ans. (A,C) | 12. Ans. (A,C,D) | 13. Ans. (A, B,D) | 14. Ans. (A,B,C) |
|-----------------|----------------|------------------|-------------------|------------------|

EXERCISE - JM

- | | | | | |
|--------------|--------------|--------------|--------------|--------------|
| 1. Ans. (1) | 2. Ans. (4) | 3. Ans. (2) | 4. Ans. (4) | 5. Ans. (4) |
| 6. Ans. (4) | 7. Ans. (4) | 8. Ans. (1) | 9. Ans. (2) | 10. Ans. (2) |
| 12. Ans. (2) | 13. Ans. (3) | 14. Ans. (1) | 15. Ans. (4) | 16. Ans. (4) |
| 17. Ans. (3) | 18. Ans. (2) | 19. Ans. (1) | 20. Ans. (2) | 21. Ans. (1) |

EXERCISE - JA

- | | | | | |
|----------------|-------------------|---------------|-----------------|----------------|
| 1. Ans. (D) | 2. Ans. (C) | 3. Ans. (C) | 4. Ans. 4 | 5. Ans. 5 |
| 6. Ans. (B) | 7. Ans. (A,B,C,D) | 8. Ans. (B,D) | 9. Ans. (A,B,D) | 10. Ans. 5 |
| 11. Ans. (C) | 12. Ans. (B) | 13. Ans. 1 | 14. Ans. (A) | 15. Ans. (C,D) |
| 16. Ans. (B,C) | 17. Ans. (A) | 18. Ans. (B) | 19. Ans. (A,C) | |

Important Notes

CAPACITANCE

KEY CONCEPTS

CONCEPT OF CAPACITANCE

When a conductor is charged then its potential rises. The increase in potential is directly proportional to the charge given to the conductor. $Q \propto V \Rightarrow Q = CV$

The constant C is known as the capacity of the conductor.

Capacitance is a scalar quantity with dimension $C = \frac{Q}{V} = \frac{Q^2}{W} = \frac{A^2 T^2}{M^1 L^{-2} T^{-2}} = M^{-1} L^{-2} T^4 A^2$

Unit :- farad, coulomb/volt

The capacity of a conductor is independent of the charge given or its potential raised. It is also independent of nature of material and thickness of the conductor. Theoretically infinite amount of charge can be given to a conductor. But practically the electric field becomes so large that it causes ionisation of medium surrounding it. The charge on conductor leaks reducing its potential.

THE CAPACITANCE OF A SPHERICAL CONDUCTOR

When a charge Q is given to a isolated spherical conductor then its potential rises.

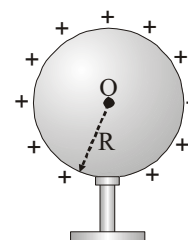
$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{R} \Rightarrow C = \frac{Q}{V} = 4\pi\epsilon_0 R$$

If conductor is placed in a medium then

$$C_{\text{medium}} = 4\pi\epsilon R = 4\pi\epsilon_0 \epsilon_r R$$

Capacitance depends upon :

- (a) Size and Shape of Conductor
- (b) Surrounding medium
- (c) Presence of other conductors nearby



CONDENSER/CAPACITOR

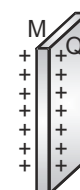
The pair of conductor of opposite charges on which sufficient quantity of charge may be accommodated is defined as condenser.

• Principle of a Condenser

It is based on the fact that capacitance can be increased by reducing potential keeping the charge constant.

Consider a conducting plate M which is given a charge Q such that its potential rises to V then

$$C = \frac{Q}{V}$$



Let us place another identical conducting plate N parallel to it such that charge is induced on plate N (as shown in figure). If V_- is the potential at M due to induced negative charge on N and V_+ is the potential at M due to induced positive charge on N, then

$$C' = \frac{Q}{V'} = \frac{Q}{V + V_+ - V_-}$$

Since $V' < V$ (as the induced negative charge lies closer to the plate M in comparison to induced positive charge). $\Rightarrow C' > C$ Further, if N is earthed from the outer side (see figure) then $V'' = V_+ - V_-$ ($\because$ the entire positive charge flows to the earth)

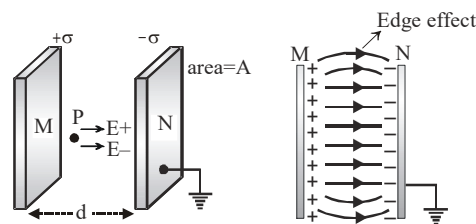
$$C'' = \frac{Q}{V''} = \frac{Q}{V - V_-} \Rightarrow C'' \gg C$$

If an identical earthed conductor is placed in the vicinity of a charged conductor then the capacitance of the charged conductor increases appreciable. This is the principle of a parallel plate capacitor.

• Parallel Plate Capacitor

(i) Capacitance

It consists of two metallic plates M and N each of area A at separation d. Plate M is positively charged and plate N is earthed. If ϵ_r is the dielectric constant of the material medium and E is the field at a point P that exists between the two plates, then



I step : Finding electric field $E = E_+ + E_- = \frac{\sigma}{2\epsilon} + \frac{\sigma}{2\epsilon} = \frac{\sigma}{\epsilon} = \frac{\sigma}{\epsilon_0 \epsilon_r}$ [$\epsilon = \epsilon_0 \epsilon_r$]

II step : Finding potential difference $V = Ed = \frac{\sigma}{\epsilon_0 \epsilon_r} d = \frac{qd}{A\epsilon_0 \epsilon_r}$ ($\because E = \frac{V}{d}$ and $\sigma = \frac{q}{A}$)

III step : Finding capacitance $C = \frac{q}{V} = \frac{\epsilon_r \epsilon_0 A}{d}$

(ii) Force between the plates

The two plates of capacitor attract each other because they are oppositely charged.

Electric field due to positive plate $E = \frac{\sigma}{2\epsilon_0} = \frac{Q}{2\epsilon_0 A}$

Force on negative charge $-Q$ is $F = -QE = -\frac{Q^2}{2\epsilon_0 A}$

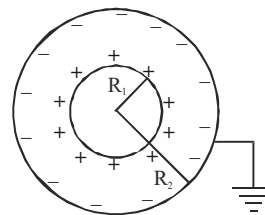
Magnitude of force $F = \frac{Q^2}{2\epsilon_0 A} = \frac{1}{2} \epsilon_0 A E^2$

Force per unit area or energy density or electrostatic pressure $= \frac{F}{A} = u = p = \frac{1}{2} \epsilon_0 E^2$

• **Spherical Capacitor**
Outer sphere is earthed

When a charge Q is given to inner sphere it is uniformly distributed on its surface. A charge $-Q$ is induced on inner surface of outer sphere. The charge $+Q$ induced on outer surface of outer sphere flows to earth as it is grounded.

$E = 0$ for $r < R_1$ and $E = 0$ for $r > R_2$



$$\text{Potential of inner sphere } V_1 = \frac{Q}{4\pi\epsilon_0 R_1} + \frac{-Q}{4\pi\epsilon_0 R_2} \Rightarrow \frac{Q}{4\pi\epsilon_0} \left(\frac{R_2 - R_1}{R_1 R_2} \right)$$

As outer surface is earthed so potential $V_2 = 0$

$$\text{Potential difference between plates } V = V_1 - V_2 = \frac{Q}{4\pi\epsilon_0} \frac{(R_2 - R_1)}{R_1 R_2}$$

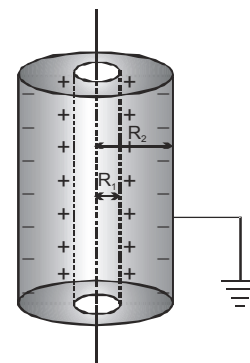
$$\text{So } C = \frac{Q}{V} = 4\pi\epsilon_0 \frac{R_1 R_2}{R_2 - R_1} \text{ (in air or vacuum)}$$

• **Cylindrical Capacitor**

When a charge Q is given to inner cylinder it is uniformly distributed on its surface. A charge $-Q$ is induced on inner surface of outer cylinder. The charge $+Q$ induced on outer surface of outer cylinder flows to earth as it is grounded.

$$\text{Electrical field between cylinders } E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{Q/\ell}{2\pi\epsilon_0 r}$$

$$\text{Potential difference between plates } V = \int_{R_1}^{R_2} \frac{Q}{2\pi\epsilon_0 r \ell} dr = \frac{Q}{2\pi\epsilon_0 \ell} \ln \left(\frac{R_2}{R_1} \right)$$



$$\text{Capacitance } C = \frac{Q}{V} = \frac{2\pi\epsilon_0 \ell}{\log_e (R_2 / R_1)}$$

Ex. The stratosphere acts as a conducting layer for the earth. If the stratosphere extends beyond 50 km from the surface of earth, then calculate the capacitance of the spherical capacitor formed between stratosphere and earth's surface. Take radius of earth as 6400 km.

Sol. The capacitance of a spherical capacitor is $C = 4\pi\epsilon_0 \left(\frac{ab}{b-a} \right)$

$b =$ radius of the top of stratosphere layer $= 6400 \text{ km} + 50 \text{ km} = 6450 \text{ km} = 6.45 \times 10^6 \text{ m}$

$a =$ radius of earth $= 6400 \text{ km} = 6.4 \times 10^6 \text{ m}$

$$\therefore C = \frac{1}{9 \times 10^9} \times \frac{6.45 \times 10^6 \times 6.4 \times 10^6}{6.45 \times 10^6 - 6.4 \times 10^6} = 0.092 \text{ F}$$

Ex. A cylindrical capacitor has two co-axial cylinders of length 15 cm and radii 1.5 cm and 1.4 cm. The outer cylinder is earthed and the inner cylinder is given a charge of $3.5 \mu\text{C}$. Determine the capacitance of the system and the potential of the inner cylinder.

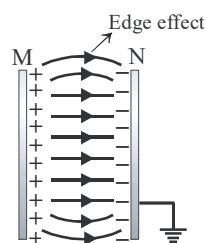
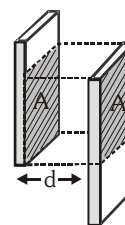
Sol. $\ell = 15 \text{ cm} = 15 \times 10^{-2} \text{ m}$; $a = 1.4 \text{ cm} = 1.4 \times 10^{-2} \text{ m}$;
 $b = 1.5 \text{ cm} = 1.5 \times 10^{-2} \text{ m}$; $q = 3.5 \mu\text{C} = 3.5 \times 10^{-6} \text{ C}$

$$\text{Capacitance } C = \frac{2\pi\epsilon_0\ell}{2.303 \log_{10} \left(\frac{b}{a} \right)} = \frac{2\pi \times 8.854 \times 10^{-12} \times 15 \times 10^{-2}}{2.303 \log_{10} \frac{1.5 \times 10^{-2}}{1.4 \times 10^{-2}}} = 1.21 \times 10^{-8} \text{ F}$$

Since the outer cylinder is earthed, the potential of the inner cylinder will be equal to the potential difference between them. Potential of inner cylinder, is

$$V = \frac{q}{C} = \frac{3.5 \times 10^{-6}}{1.2 \times 10^{-10}} = 2.89 \times 10^4 \text{ V}$$

- If one of the plates of parallel plate capacitor slides relatively than C decrease (As overlapping area decreases).
- If both the plates of parallel plate capacitor are touched each other resultant charge and potential became zero.
- Electric field between the plates of a capacitor is shown in figure. Non-uniformity of electric field at the boundaries of the plates is negligible if the distance between the plates is very small as compared to the length of the plates.



$\vec{E}$ = uniform in the centre
 $\vec{E}$ = non-uniform at the edges

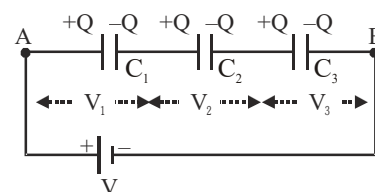
COMBINATION OF CAPACITOR

Capacitor in series:

In this arrangement of capacitors the charge has no alternative path(s) to flow.

- The charges on each capacitor are equal
i.e. $Q = C_1 V_1 = C_2 V_2 = C_3 V_3$
- The total potential difference across AB is shared by the capacitors in the inverse ratio of the capacitances $V = V_1 + V_2 + V_3$
If C_s is the net capacitance of the series combination, then

$$\frac{Q}{C_s} = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3} \Rightarrow \frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$



• **Capacitors in parallel**

In such arrangement of capacitors the charge has an alternative path(s) to flow.

- (i) The potential difference across each capacitor is same and equal the

$$\text{total potential applied. i.e. } V = V_1 = V_2 = V_3 \Rightarrow V = \frac{Q_1}{C_1} = \frac{Q_2}{C_2} = \frac{Q_3}{C_3}$$

- (ii) The total charge Q is shared by each capacitor in the direct ratio of the capacitances. $Q = Q_1 + Q_2 + Q_3$

If C_p is the net capacitance for the parallel combination of capacitors :

$$C_p V = C_1 V + C_2 V + C_3 V \Rightarrow C_p = C_1 + C_2 + C_3$$

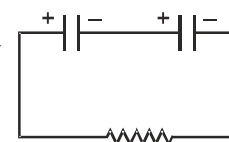
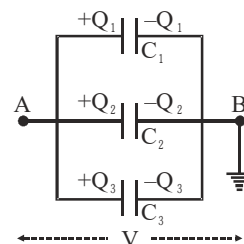
- For a given voltage to store maximum energy capacitors should be connected in parallel.

- If N identical capacitors each having breakdown voltage V are joined in

- (i) series then the break down voltage of the combination is equal to NV

- (ii) parallel then the breakdown voltage of the combination is equal to V .

- Two capacitors are connected in series with a battery. Now battery is removed and loose wires connected together then final charge on each capacitor is zero.



- If N identical capacitors are connected then $C_{\text{series}} = \frac{C}{N}$, $C_{\text{parallel}} = NC$

- In DC capacitor's offers infinite resistance in steady state, so there will be no current flows through capacitor branch.

Ex. Capacitor $C, 2C, 4C, \dots \infty$ are connected in parallel, then what will be their effective capacitance ?

Sol. Let the resultant capacitance be $C_{\text{resultant}} = C + 2C + 4C + \dots \infty = C[1 + 2 + 4 + \dots \infty] = C \times \infty = \infty$

Ex. An infinite number of capacitors of capacitance $C, 4C, 16C \dots \infty$ are connected in series then what will be their resultant capacitance ?

Sol. Let the equivalent capacitance of the combination $= C_{\text{eq}}$

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C} + \frac{1}{4C} + \frac{1}{16C} + \dots \infty = \left[1 + \frac{1}{4} + \frac{1}{16} + \dots \infty \right] \frac{1}{C} \text{ (this is G. P. series)}$$

$$\Rightarrow S_{\infty} = \frac{a}{1-r} \text{ first term } a = 1, \text{ common ratio } r = \frac{1}{4}$$

$$\Rightarrow \frac{1}{C_{\text{eq}}} = \frac{1}{1 - \frac{1}{4}} \times \frac{1}{C} \Rightarrow C_{\text{eq}} = \frac{3}{4} C$$

ENERGY STORED IN A CHARGED CONDUCTOR/CAPACITOR

Let C is capacitance of a conductor. On being connected to a battery. It charges to a potential V from zero potential. If q is charge on the conductor at that time then $q = CV$. Let battery supplies small amount of charge dq to the conductor at constant potential V . Then small amount of work done by the battery against the force exerted by existing charge is

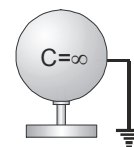
$$dW = Vdq = \frac{q}{C} dq \Rightarrow W = \int_0^Q \frac{q}{C} dq = \frac{1}{C} \left[\frac{q^2}{2} \right]_0^Q \Rightarrow W = \frac{Q^2}{2C}$$

where Q is the final charge acquired by the conductor. This work done is stored as potential energy, so

$$U = \frac{Q^2}{2C} = \frac{1}{2} \frac{(CV)^2}{C} = \frac{1}{2} CV^2 = \frac{1}{2} \left(\frac{Q}{V} \right) V^2 = \frac{1}{2} QV \therefore U = \frac{Q^2}{2C} = \frac{1}{2} CV^2 = \frac{1}{2} QV$$

- As the potential of the Earth is assumed to be zero, capacity of earth or a conductor

connected to earth will be infinite $C = \frac{q}{V} = \frac{q}{0} = \infty$



- Actual capacity of the Earth $C = 4\pi\epsilon_0 R = \frac{1}{9 \times 10^9} \times 64 \times 10^5 = 711 \mu\text{F}$
- Work done by battery $W_b = (\text{charge given by battery}) \times (\text{emf}) = QV$ but

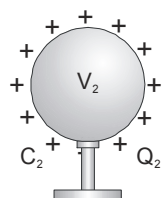
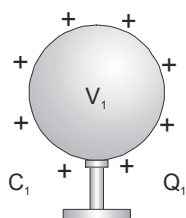
$$\text{Energy stored in conductor} = \frac{1}{2} QV$$

so 50% energy supplied by the battery is lost in form of heat.

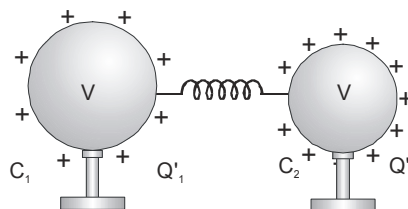
REDISTRIBUTION OF CHARGES AND LOSS OF ENERGY

When two charged conductors are connected by a conducting wire then charge flows from a conductor at higher potential to that at lower potential. This flow of charge stops when the potential of two conductors became equal.

Let the amounts of charges after the conductors are connected are Q_1' and Q_2' respectively and potential is V then



(Before connection)



(After connection)

- Common potential**

According to law of Conservation of charge

$$\Rightarrow C_1 V_1 + C_2 V_2 = C_1 V + C_2 V$$

$$Q_{\text{before connection}} = Q_{\text{after connection}}$$

Common potential after connection

$$V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

Charges after connection

$$Q_1' = C_1 V = C_1 \left(\frac{Q_1 + Q_2}{C_1 + C_2} \right) = \left(\frac{C_1}{C_1 + C_2} \right) Q \quad (Q : \text{Total charge on system})$$

$$Q_2' = C_2 V = C_2 \left(\frac{Q_1 + Q_2}{C_1 + C_2} \right) = \left(\frac{C_2}{C_1 + C_2} \right) Q$$

$$\text{Ratio of the charges after redistribution} \quad \frac{Q_1'}{Q_2'} = \frac{C_1 V}{C_2 V} = \frac{R_1}{R_2} \quad (\text{in case of spherical conductors})$$

Loss of energy in redistribution

When charge flows through the conducting wire then **energy is lost mainly on account of Joule effect**, electrical energy is converted into heat energy, so change in energy of this system,

$$\Delta U = U_f - U_i \Rightarrow \left(\frac{1}{2} C_1 V^2 + \frac{1}{2} C_2 V^2 \right) - \left(\frac{1}{2} C_1 V_1^2 + \frac{1}{2} C_2 V_2^2 \right) \Rightarrow \Delta U = -\frac{1}{2} \left(\frac{C_1 C_2}{C_1 + C_2} \right) (V_1 - V_2)^2$$

Here negative sign indicates that energy of the system decreases in the process.

Ex. The plates of a capacitor are charged to a potential difference of 100 V and then connected across a resistor. The potential difference across the capacitor decays exponentially with respect to time. After one second the potential difference between the plates of the capacitor is 80 V. What is the fraction of the stored energy which has been dissipated ?

Sol. Energy losses $\Delta U = \frac{1}{2} C V_0^2 - \frac{1}{2} C V^2$

$$\text{Fractional energy loss} \quad \frac{\Delta U}{U_0} = \frac{\frac{1}{2} C V_0^2 - \frac{1}{2} C V^2}{\frac{1}{2} C V_0^2} = \frac{V_0^2 - V^2}{V_0^2} = \frac{(100)^2 - (80)^2}{(100)^2} = \frac{20 \times 180}{(100)^2} = \frac{9}{25}$$

Ex. Two uniformly charged spherical drops at potential V coalesce to form a larger drop. If capacity of each smaller drop is C then find capacity and potential of larger drop.

Sol. When drops coalesce to form a larger drop then total charge and volume remains conserved. If r is radius and q is charge on smaller drop then $C = 4 \pi \epsilon_0 r$ and $q = CV$

$$\text{Equating volume we get} \quad \frac{4}{3} \pi R^3 = 2 \times \frac{4}{3} \pi r^3 \Rightarrow R = 2^{1/3} r$$

$$\text{Capacitance of larger drop} \quad C' = 4 \pi \epsilon_0 R = 2^{1/3} C$$

$$\text{Charge on larger drop} \quad Q = 2q = 2CV$$

$$\text{Potential of larger drop} \quad V' = \frac{Q}{C'} = \frac{2CV}{2^{1/3} C} = 2^{2/3} V$$

EFFECT OF DIELECTRIC

- The insulators in which microscopic local displacement of charges takes place in presence of electric field are known as **dielectrics**.
- Dielectrics are non conductors upto certain value of field depending on its nature. If the field exceeds this limiting value called **dielectric strength** they lose their insulating property and begin to conduct.

- Dielectric strength** is defined as the maximum value of electric field that a dielectric can tolerate without breakdown. Unit is volt/metre. Dimensions $M^1 L^1 T^{-3} A^{-1}$

Polar dielectrics

- In absence of external field the centres of positive and negative charge do not coincide-due to asymmetric shape of molecules.
- Each molecule has permanent dipole moment.
- The dipoles are randomly oriented so average dipole moment per unit volume of polar dielectric in absence of external field is nearly zero.
- In presence of external field dipoles tend to align in direction of field.

Ex. Water, Alcohol, CO_2 , HCl , NH_3

Non polar dielectrics

- In absence of external field the centre of positive and negative charge coincides in these atoms or molecules because they are symmetric.
- The dipole moment is zero in normal state.
- In presence of external field they acquire induced dipole moment.

Ex. Nitrogen, Oxygen, Benzene, Methane

Polarisation :

The alignment of dipole moments of permanent or induced dipoles in the direction applied electric field is called polarisation.

- Polarisation vector $\vec{P}$**

This is a vector quantity which describes the extent to which molecules of dielectric become polarized by an electric field or oriented in direction of field.

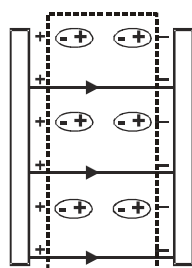
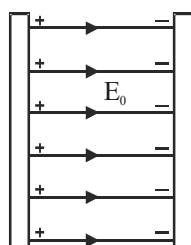
$\vec{P}$ = the dipole moment per unit volume of dielectric = $n\vec{p}$

where n is number of atoms per unit volume of dielectric and $\vec{p}$ is dipole moment of an atom or molecule.

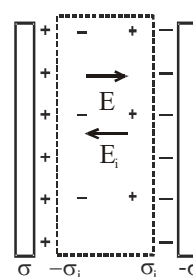
$$\vec{P} = n\vec{p} = \frac{q_i d}{A d} = \left(\frac{q_i}{A} \right) = \sigma_i = \text{induced surface charge density.}$$

Unit of $\vec{P}$ is C/m^2

Dimension is $L^{-2} T^1 A^1$



Dielectric slab



Let E_0 , V_0 , C_0 be electric field, potential difference and capacitance in absence of dielectric. Let E , V , C are electric field, potential difference and capacitance in presence of dielectric respectively.

$$\text{Electric field in absence of dielectric } E_0 = \frac{V_0}{d} = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}$$

Electric field in presence of dielectric $E = E_0 - E_i = \frac{\sigma - \sigma_i}{\epsilon_0} = \frac{Q - Q_i}{\epsilon_0} = \frac{V}{d}$

Capacitance in absence of dielectric $C_0 = \frac{Q}{V_0}$

Capacitance in presence of dielectric $C = \frac{Q - Q_i}{V}$

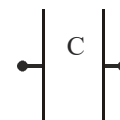
The dielectric constant or relative permittivity K

or $\epsilon_r = \frac{E_0}{E} = \frac{V_0}{V} = \frac{C}{C_0} = \frac{Q}{Q - Q_i} = \frac{\sigma}{\sigma - \sigma_i} = \frac{\epsilon}{\epsilon_0}$

From $K = \frac{Q}{Q - Q_i} \Rightarrow Q_i = Q \left(1 - \frac{1}{K}\right)$ and $K = \frac{\sigma}{\sigma - \sigma_i} \Rightarrow \sigma_i = \sigma \left(1 - \frac{1}{K}\right)$

CAPACITY OF DIFFERENT CONFIGURATION

In case of parallel plate capacitor $C = \frac{\epsilon_0 A}{d}$



If capacitor is partially filled with dielectric

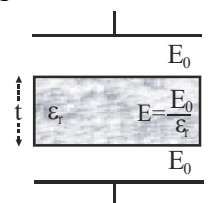
When the dielectric is filled partially between plates, the thickness of dielectric slab is t ($t < d$).

If no slab is introduced between the plates of the capacitor, then a field E_0 given by $E_0 = \frac{\sigma}{\epsilon_0}$, exists in a space d .

On inserting the slab of thickness t , a field $E = \frac{E_0}{\epsilon_r}$ exists inside the slab of

thickness t and a field E_0 exists in remaining space $(d - t)$. If V is total potential then $V = E_0(d - t) + E t$

$\Rightarrow V = E_0 \left[d - t + \left(\frac{E}{E_0} \right) t \right] \therefore \frac{E_0}{E} = \epsilon_r = \text{Dielectric constant}$

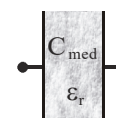


$\Rightarrow V = \frac{\sigma}{\epsilon_0} \left[d - t + \frac{t}{\epsilon_r} \right] = \frac{q}{A \epsilon_0} \left[d - t + \frac{t}{\epsilon_r} \right] \Rightarrow C = \frac{q}{V} = \frac{\epsilon_0 A}{d - t \left(1 - \frac{1}{\epsilon_r} \right)} = \frac{\epsilon_0 A}{d - t \left(1 - \frac{1}{\epsilon_r} \right)} \dots (i)$

If medium is fully present between the space.

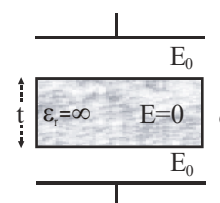
$\therefore t = d$

Now from equation (i) $C_{\text{medium}} = \frac{\epsilon_0 \epsilon_r A}{d}$



If capacitor is partially filled by a conducting slab of thickness $(t < d)$.

$\therefore \epsilon_r = \infty$ for conductor $C = \frac{\epsilon_0 A}{d - t \left(1 - \frac{1}{\infty} \right)} = \frac{\epsilon_0 A}{(d - t)}$



DISTANCE AND AREA DIVISION BY DIELECTRIC

Distance Division

- (i) Distance is divided and area remains same.
- (ii) Capacitors are in series.

(iii) Individual capacitances are $C_1 = \frac{\epsilon_0 \epsilon_{r_1} A}{d_1}$, $C_2 = \frac{\epsilon_0 \epsilon_{r_2} A}{d_2}$

These two are in series $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} \Rightarrow \frac{1}{C} = \frac{d_1}{\epsilon_0 \epsilon_{r_1} A} + \frac{d_2}{\epsilon_0 \epsilon_{r_2} A}$

$$\Rightarrow \frac{1}{C} = \frac{1}{\epsilon_0 A} \left[\frac{d_1 \epsilon_{r_2} + d_2 \epsilon_{r_1}}{\epsilon_{r_1} \epsilon_{r_2}} \right] \Rightarrow C = \epsilon_0 A \left[\frac{\epsilon_{r_1} \epsilon_{r_2}}{d_1 \epsilon_{r_2} + d_2 \epsilon_{r_1}} \right]$$

Special case : If $d_1 = d_2 = \frac{d}{2} \Rightarrow C = \frac{\epsilon_0 A}{d} \left[\frac{2 \epsilon_{r_1} \epsilon_{r_2}}{\epsilon_{r_1} + \epsilon_{r_2}} \right]$

Area Division

- (i) Area is divided and distance remains same.
- (ii) Capacitors are in parallel.

(iii) Individual capacitances are $C_1 = \frac{\epsilon_0 \epsilon_{r_1} A_1}{d}$, $C_2 = \frac{\epsilon_0 \epsilon_{r_2} A_2}{d}$

These two are in parallel so $C = C_1 + C_2 = \frac{\epsilon_0 \epsilon_{r_1} A_1}{d} + \frac{\epsilon_0 \epsilon_{r_2} A_2}{d} = \frac{\epsilon_0}{d} (\epsilon_{r_1} A_1 + \epsilon_{r_2} A_2)$

Special case : If $A_1 = A_2 = \frac{A}{2}$ Then $C = \frac{\epsilon_0 A}{d} \left(\frac{\epsilon_{r_1} + \epsilon_{r_2}}{2} \right)$

Variable Dielectric Constant :

If the dielectric constant is variable, then equivalent capacitance can be obtained by selecting an element as per the given condition and then integrating.

- (i) If different elements are in parallel, then $C = \int dC$, where dC = capacitance of selected differential element.
- (ii) If different element are in series, then $\frac{1}{C} = \int d\left(\frac{1}{C}\right)$ is solved to get equivalent capacitance C .

FORCE ON A DIELECTRIC IN A CAPACITOR

Consider a differential displacement dx of the dielectric as shown in figure always keeping the net force on it zero so that the dielectric moves slowly without acceleration. Then,

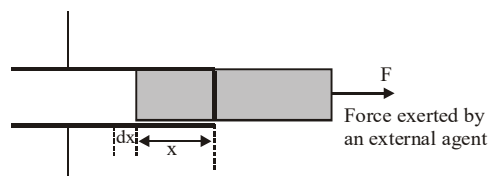
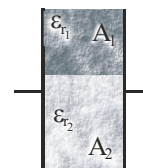
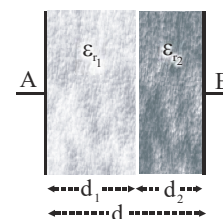
$W_{\text{Electrostatic}} + W_F = 0$, where W_F denotes the work done by external agent in displacement dx

$$W_F = -W_{\text{Electrostatic}} \quad W_F = \Delta U$$

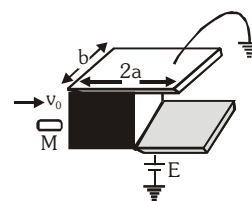
$$\Rightarrow -F \cdot dx = \frac{Q^2}{2} d\left[\frac{1}{C}\right] \left[U = \frac{Q^2}{2C} \right] \Rightarrow -F \cdot dx = \frac{-Q^2}{2C^2} dC \Rightarrow F = \frac{Q^2}{2C^2} \left(\frac{dC}{dx} \right)$$

This is also true for the force between the plates of the capacitor. If the capacitor has battery connected

to it, then as the p.d. across the plates is maintained constant. $V = \frac{Q}{C} \Rightarrow F = \frac{1}{2} V^2 \frac{dC}{dx}$.



Ex. A parallel plate capacitor is half filled with a dielectric (K) of mass M. Capacitor is attached with a cell of emf E. Plates are held fixed on smooth insulating horizontal surface. A bullet of mass M hits the dielectric elastically and it is found that dielectric just leaves out the capacitor. Find speed of bullet and the current as a function of time.



Sol. Since collision is elastic $\therefore$ Velocity of dielectric after collision is v_0 . Dielectric will move and when it is coming out of capacitor a force is applied on

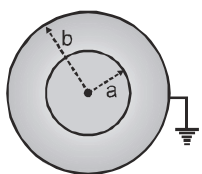
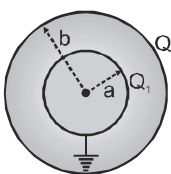
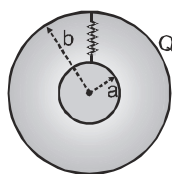
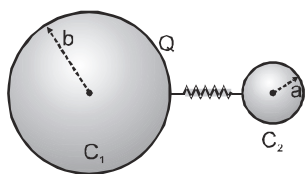
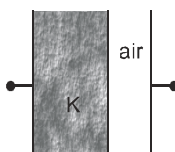
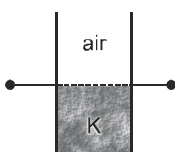
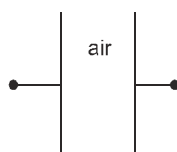
it by the capacitor
$$F = \frac{-dU}{dx} = \frac{-E^2 \epsilon_0 b (K-1)}{2d}$$

Which decreases its speed to zero, till it comes out it travels a distance a.

$$\frac{1}{2} M v_0^2 = \frac{E^2 \epsilon_0 b (K-1) a}{2d} \Rightarrow v_0 = E \left[\frac{\epsilon_0 a b (K-1)}{M d} \right]^{1/2}$$

$$i = v E \frac{dC}{dx} \left[\text{Since, } i = \frac{dq}{dt} = \frac{d}{dt} (EC) = E \frac{dC}{dt} = \frac{EdC}{dx} \frac{dx}{dt} \right]$$

$$i = \left(v_0 - \frac{F}{M} t \right) E \epsilon_0 b \frac{(K-1)}{d} \text{ for } t_0 < t < \left(t_0 + \frac{v_0 M}{F} \right) \text{ (where, } t_0 = \frac{a}{v_0} \text{)}$$

Spherical capacitor outer is earthed	Inner is earthed and outer is given a charge	Connected and outer is given a charge	Connected spheres
			
$C = \frac{4\pi\epsilon_0 ab}{b-a}$ ($b > a$)	$C = \frac{4\pi\epsilon_0 b^2}{b-a}$ ($b > a$)	$C = 4\pi\epsilon_0 b$	$C = C_1 + C_2$ $C = 4\pi\epsilon_0 (a+b)$
			
$C_1 = \left[\frac{2K}{K+1} \right] C$	$C_2 = \left[\frac{K+1}{2} \right] C$	$C_3 = C$ when no dielectric is used	
$C_2 > C_1 > C_3$			

Ex. A capacitor has two circular plates whose radius are 8cm and distance between them is 1mm. When mica (dielectric constant = 6) is placed between the plates, calculate the capacitance of this capacitor and the energy stored when it is given potential of 150 volt.

Sol. Area of plate $\pi r^2 = \pi \times (8 \times 10^{-2})^2 = 0.0201 \text{ m}^2$ and $d = 1\text{mm} = 1 \times 10^{-3} \text{ m}$

$$\text{Capacity of capacitor } C = \frac{\epsilon_0 \epsilon_r A}{d} = \frac{8.85 \times 10^{-12} \times 6 \times 0.0201}{1 \times 10^{-3}} = 1.068 \times 10^{-9} \text{ F}$$

Potential difference $V = 150$ volt

Energy stored $U = \frac{1}{2} CV^2 = \frac{1}{2} \times (1.068 \times 10^{-9}) \times (150)^2 = 1.2 \times 10^{-5} \text{ J}$

Ex. A parallel-plate capacitor is formed by two plates, each of area 100 cm^2 , separated by a distance of 1 mm . A dielectric of dielectric constant 5.0 and dielectric strength $1.9 \times 10^7 \text{ V/m}$ is filled between the plates. Find the maximum charge that can be stored on the capacitor without causing any dielectric breakdown.

Sol. If the charge on the capacitor = Q

the surface charge density $\sigma = \frac{Q}{A}$ and the electric field = $\frac{Q}{KA\epsilon_0}$.

This electric field should not exceed the dielectric strength $1.9 \times 10^7 \text{ V/m}$.

$\therefore$ if the maximum charge which can be given is Q

then $\frac{Q}{KA\epsilon_0} = 1.9 \times 10^7 \text{ V/m}$, $\therefore A = 100 \text{ cm}^2 = 10^{-2} \text{ m}^2$

$\Rightarrow Q = (5.0) \times (10^{-2}) \times (8.85 \times 10^{-12}) \times (1.9 \times 10^7) = 8.4 \times 10^{-6} \text{ C}$.

Ex. The distance between the plates of a parallel-plate capacitor is 0.05 m . A field of $3 \times 10^4 \text{ V/m}$ is established between the plates of capacitor by connecting with battery. Now capacitor is disconnected from the battery and an uncharged metal plate of thickness 0.01 m is inserted between the plates of capacitor. Calculate new potential difference between the plates of capacitor. What would be the potential difference if a plate of same thickness and dielectric constant $K = 2$ is introduced in place of metal plate?

Sol. (i) In case of a capacitor as $E = (V/d)$, the potential difference between the plates before the introduction of metal plate

$$V = E \times d = 3 \times 10^4 \times 0.05 = 1.5 \text{ kV}$$

(ii) Now as after charging battery is removed, capacitor is isolated so $q = \text{constant}$. If C' and V' are the

capacity and potential after the introduction of plate $q = CV = C'V'$ i.e., $V' = \frac{C}{C'} V$

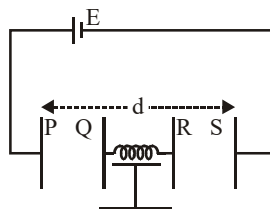
And as $C = \frac{\epsilon_0 A}{d}$ and $C' = \frac{\epsilon_0 A}{(d-t) + (t/K)}$, $V' = \frac{(d-t) + (t/K)}{d} \times V$

So in case of metal plate as $K = \infty$, $V_M = \left[\frac{d-t}{d} \right] \times V = \left[\frac{0.05-0.01}{0.05} \right] \times 1.5 = 1.2 \text{ kV}$

And if instead of metal plate, dielectric with $K = 2$ is introduced

$$V_D = \left[\frac{(0.05-0.01) + (0.01/2)}{0.05} \right] \times 1.5 = 1.35 \text{ kV}$$

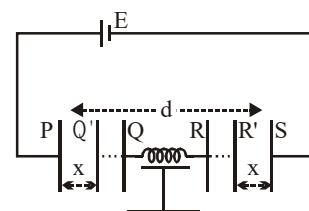
Ex. Two parallel plate capacitors with area A are connected through a conducting spring of natural length ℓ in series as shown. Plates P and S have fixed positions at separation d . Now the plates are connected by a battery of emf E as shown. If the extension in the spring in equilibrium is equal to the separation between the plates, find the spring constant k .



Sol. At any time distance between plates P and Q, R and S is same because force acting on them is same. Let charge on capacitors be q and separation between plates P and Q, R and S be x

Capacitance of capacitor PQ, $C_1 = \frac{\epsilon_0 A}{x}$

Capacitance of capacitor RS, $C_2 = \frac{\epsilon_0 A}{x}$ From KVL $\frac{q}{C_1} + \frac{q}{C_2} = E \Rightarrow q = \frac{\epsilon_0 AE}{2x}$



At this moment extension in spring, $y = d - 2x - \ell$.

Force on plate Q towards P, $F_1 = \frac{q^2}{2A\epsilon_0} = \frac{\epsilon_0^2 A^2 E^2}{8Ax^2\epsilon_0} = \frac{A\epsilon_0 E^2}{8x^2}$

Spring force on plate Q due to extension in spring, $F_2 = ky$

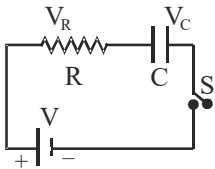
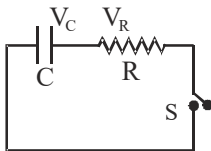
At equilibrium, separation between plates = extension in spring

Thus $x = y = d - 2x - \ell \Rightarrow x = \frac{d - \ell}{3}$... (i) and $F_1 = F_2$... (ii)

From eq. (i) and (ii), $\frac{A\epsilon_0 E^2}{8x^2} = ky = kx \Rightarrow x = \left(\frac{A\epsilon_0 E^2}{8k} \right)^{1/3}$... (iii)

From eq. (i) and (iii), $\left(\frac{d - \ell}{3} \right) = \frac{A\epsilon_0 E^2}{8k} \Rightarrow k = \frac{A\epsilon_0 E^2 27}{8(d - \ell)^3}$

CHARGING & DISCHARGING OF A CAPACITOR

Charging	Discharging
- When a capacitor, resistance, battery, and key is connected in series and key is closed, then  Charge at any instant $V = V_C + V_R = \frac{Q}{C} + IR = \frac{Q}{C} + \frac{dQ}{dt} R$ $Q = CV \left[1 - e^{-t/RC} \right] = Q_0 \left[1 - e^{-t/RC} \right]$ At $t = \tau = RC = \text{time constant}$ $Q = Q_0 [1 - e^{-1}] = 0.632 Q_0$ So, in charging, charge increases to 63.2% of charge in the time equal to τ. Current at any instant $i = dQ/dt = i_0 e^{-t/RC} \quad \{i_0 = Q_0/RC\}$ Potential at any instant $V = V_0 (1 - e^{-t/RC})$	- When a charged capacitor, resistance and keys is connected in series and key is closed. Then energy stored in capacitor is used to circulate current in the circuit.  Charge at any instant $V_C + V_R = 0$ $Q = Q_0 e^{-t/RC}$ At $t = \tau = RC = \text{time constant}$ $Q = Q_0 e^{-1} = 0.368 Q_0$ So, in discharging, charge decreases to 36.8% of the initial charge in the time equal to τ. Current at any instant $i = dQ/dt = -i_0 e^{-t/RC} \quad \{i_0 = Q_0/RC\}$ Potential at any instant $V = V_0 e^{-t/RC}$

Ex. Find the time constant for given circuit if

$$R_1 = 4\Omega, R_2 = 12\Omega, C_1 = 3\mu\text{F} \text{ and } C_2 = 6\mu\text{F}.$$

Sol. Given circuit can be reduced to :

$$C = \frac{C_1 C_2}{C_1 + C_2} = \frac{3 \times 6}{3 + 6} = 2\mu\text{F}, \quad R = \frac{R_1 R_2}{R_1 + R_2} = \frac{4 \times 12}{4 + 12} = 3\Omega$$

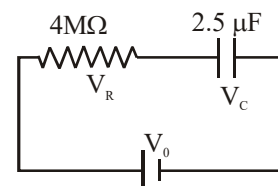
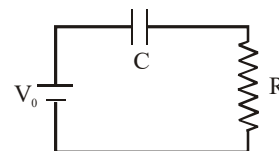
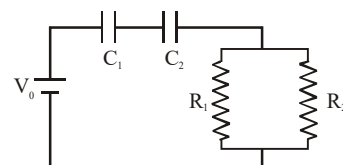
$$\text{Time constant} = RC = (3)(2 \times 10^{-6}) = 6\mu\text{s}$$

Ex. A capacitor of $2.5 \mu\text{F}$ is charged through a series resistor of $4\text{M}\Omega$. In what time the potential drop across the the capacitor will become 3 times that of the resistor. (Given : $\ln 2 = 0.693$)

Sol. $V_C = V_0(1 - e^{-t/RC}) \therefore V_C = 3V_R \therefore V_0 = V_C + \frac{V_C}{3} \Rightarrow V_C = \frac{3}{4} V_0$

$$\Rightarrow \frac{3}{4} V_0 = V_0(1 - e^{-t/RC}) \Rightarrow \frac{3}{4} = 1 - e^{-t/RC} \Rightarrow \frac{1}{4} = e^{-t/RC} \Rightarrow 4 = e^{t/RC}$$

$$\Rightarrow \frac{t}{RC} = \ln 4 \Rightarrow t = RC \ln 4 = 2RC \ln 2 = 2 \times 4 \times 10^6 \times 2.5 \times 10^{-6} \times 0.693 = 13.86 \text{ s}$$



EXERCISE (S-1)

Capacitance

- Two large parallel conducting plates are 34 mm apart and carry equal but opposite charges on their facing surfaces. An electron placed midway between the plates experiences a force of 3.2×10^{-16} N. What is the potential difference (in volts) between the plates?

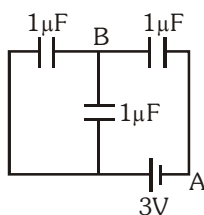
CP0001

- 2 conducting objects one with charge of $+Q$ and another with $-Q$ are kept on x-axis at $x = 0$ and $x = 1$ respectively. The electric field on the x-axis is given by $3Q \left(x^2 + \frac{4}{3} \right)$. If the capacitance of this configuration of objects is C . Then fill $\frac{1}{C}$ (in F^{-1}) in OMR sheet.

CP0002

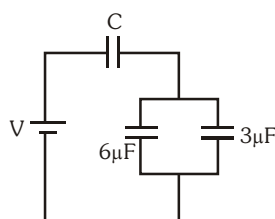
Capacitor Circuits

- If potential of A is 5V, then potential of B in volt is



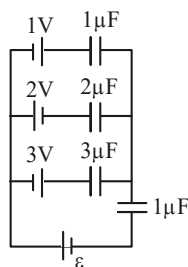
CP0003

- If charge on $3\mu F$ capacitor is $3\mu C$. Find the charge on capacitor of capacitance C in μC .



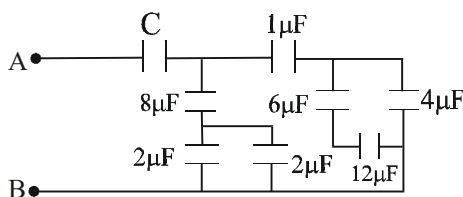
CP0004

- In the figure shown, find the e.m.f. ε for which charge on $2\mu F$ capacitor is $4\mu C$.



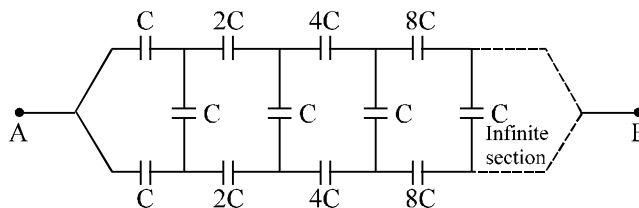
CP0005

6. In the following circuit, the resultant capacitance between A and B is $1 \mu\text{F}$. Find the value of C.



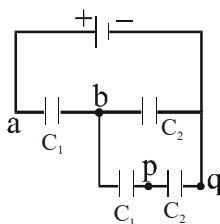
CP0006

7. Find the equivalent capacitance of the circuit between point A and B.



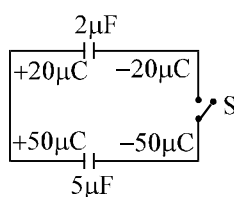
CP0007

8. In the given network if potential difference between p and q is 2V and $C_2 = 3C_1$. Then find the potential difference between a & b.



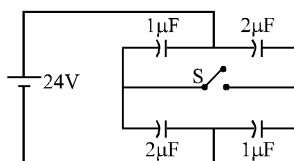
CP0008

9. Find heat produced in the circuit shown in figure on closing the switch S.



CP0009

10. The connections shown in figure are established with the switch S open. How much charge will flow through the switch if it is closed ?



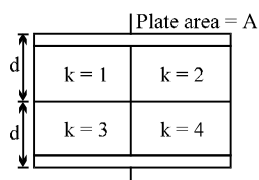
CP0010

- 11.** The plates of a parallel plate capacitor are given charges $+4Q$ and $-2Q$. The capacitor is then connected across an uncharged capacitor of same capacitance as first one ($= C$). Find the final potential difference between the plates of the first capacitor.

CP0011

Dielectrics

- 12.** Find the capacitance of the system shown in figure.

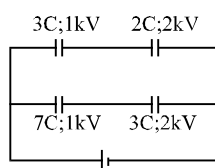


CP0012

- 13.** The plates of a parallel plate capacitor are charged upto 100 volt. A 2 mm thick plate is inserted between the plates, then to maintain the same potential difference, the distance between the capacitor plates is increased by 1.6 mm. Find the dielectric constant of the plate.

CP0013

- 14.** The diagram shows four capacitors with capacitances and break down voltages as mentioned. What should be the maximum value of the external emf source such that no capacitor breaks down?



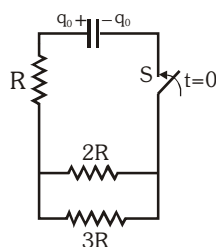
CP0014

- 15.** Two square metallic plates of 1 m side are kept 0.01 m apart, like a parallel plate capacitor, in air in such a way that one of their edges is perpendicular, to an oil surface in a tank filled with an insulating oil. The plates are connected to a battery of e.m.f. 500 volt . The plates are then lowered vertically into the oil at a speed of 0.001 m/s. Calculate the current drawn from the battery during the process.
[di-electric constant of oil = 11, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N}^2 \text{ m}^2$]

CP0015

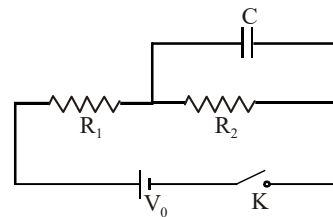
R-C Circuits

- 16.** At $t = 0$ charge on capacitor is q_0 . Now switch S is closed. Heat loss in $3R$ is $x \times 10^{-6}$ J. Then find the value of x . [Given $q_0 = 15 \mu\text{C}$, $C = 6/55 \mu\text{F}$]



CP0016

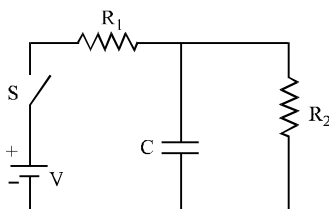
17. In the connection shown in the figure the switch K is open and the capacitor is uncharged. Then we close the switch and let the capacitor charge up to the maximum and open the switch again. Then (Use the following data : $V_0 = 30$ V, $R_1 = 10$ k Ω , $R_2 = 5$ k Ω .)
- the current through R_1 be I_1 immediately after closing the switch
 - the current through R_2 be I_2 a long time after the switch was closed
 - the current through R_2 be I_3 immediately after reopening the switch



Find the value of $\frac{I_1}{I_2 I_3}$ (in ampere⁻¹).

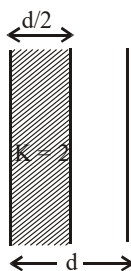
CP0017

18. In the given circuit, the switch S is closed at time $t = 0$. The charge Q on the capacitor at any instant t is given by $Q(t) = Q_0 (1 - e^{-\alpha t})$. Find the value of Q_0 and α in terms of given parameters shown in the circuit. [IIT-JEE 2005]



CP0018

19. A certain series RC circuit is formed using a resistance R , a capacitor without dielectric having a capacitance $C = 2$ F and a battery of emf $E = 3$ V. The circuit is completed and it is allowed to attain the steady state. After this, at $t = 0$, half the thickness of the capacitor is filled with a dielectric of constant $K = 2$ as shown in the figure. The system is again allowed to attain a steady state. What will be the heat generated (in joule) in the circuit between $t = 0$ and $t = \infty$?



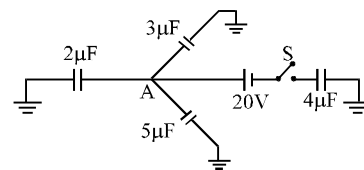
CP0019

20. A capacitor filled with dielectric of permittivity $\epsilon = 2.1$ loses half the charge acquired during a time interval $\tau = 3.0$ min. Assuming the charge to leak only through the dielectric filler, calculate its resistivity.

CP0020

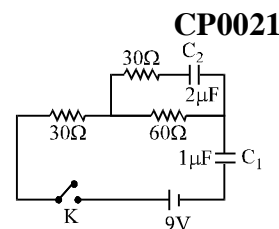
EXERCISE (S-2)

1. Three capacitors of $2\mu\text{F}$, $3\mu\text{F}$ and $5\mu\text{F}$ are independently charged with batteries of emf's 5V, 20V and 10V respectively. After disconnecting from the voltage sources. These capacitors are connected as shown in figure with their positive polarity plates are connected to A and negative polarity is earthed. Now a battery of 20V and an uncharged capacitor of $4\mu\text{F}$ capacitance are connected to the junction A as shown with a switch S. When switch is closed, find :



- (a) the potential of the junction A.
(b) final charges on all four capacitors.

2. For the arrangement shown in the figure, the key is closed at $t = 0$. C_2 is initially uncharged while C_1 has a charge of $2\mu\text{C}$.



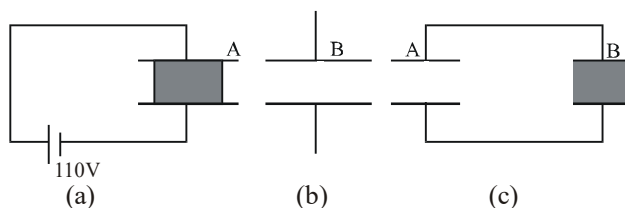
- (a) Find the current coming out of the battery just after switch is closed.
(b) Find the charge on the capacitors in the steady state condition.

3. A potential difference of 300 V is applied between the plates of a parallel plate capacitor spaced 1 cm apart. A plane parallel glass plate with a thickness of 0.5 cm and a plane parallel paraffin plate with a thickness of 0.5 cm are placed in the space between the capacitor plates find :

- (i) Intensity of electric field in each layer.
(ii) The drop of potential in each layer.
(iii) Surface charge density on the capacitor. Given that : $k_{\text{glass}} = 6$, $k_{\text{paraffin}} = 2$

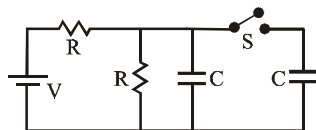
4. Two parallel plate capacitors A & B have the same separation $d = 8.85 \times 10^{-4} \text{ m}$ between the plates. The plate areas of A & B are 0.04 m^2 & 0.02 m^2 respectively. A slab of di-electric constant (relative permittivity) $K=9$ has dimensions such that it can exactly fill the space between the plates of capacitor B.

- (i) The di-electric slab is placed inside A as shown in the figure (a) A is then charged to a potential difference of 110 volt. Calculate the capacitance of A and the energy stored in it.
(ii) The battery is disconnected & then the di-electric slab is removed from A . Find the work done by the external agency in removing the slab from A .
(iii) The same di-electric slab is now placed inside B, filling it completely. The two capacitors A & B are then connected as shown in figure (c). Calculate the energy stored in the system.



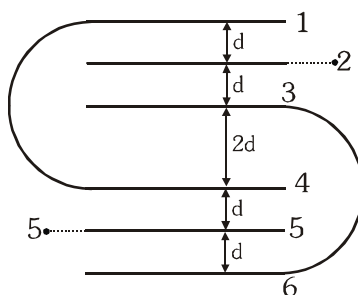
CP0024

5. In the figure shown initially switch is open for a long time. Now the switch is closed at $t = 0$. Find the charge on the rightmost capacitor as a function of time given that it was initially uncharged.



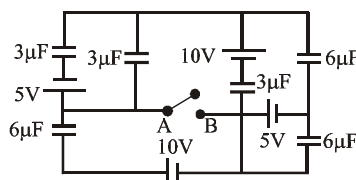
CP0025

6. There are six plates of equal area A and separation between the plates is d ($d \ll A$) are arranged as shown in figure. The equivalent capacitance between points 2 and 5, is $\alpha \frac{\epsilon_0 A}{d}$. Then find the value of α .



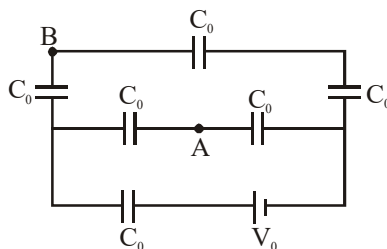
CP0026

7. Find the charge flow through the switch from A to B when it is closed.



CP0027

8. In the arrangement shown in figure, find the potential difference $V_B - V_A$ (in Volt). (Take $V_0 = 55$ V)



CP0028

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Capacitance

1. A parallel plate capacitor has $d = 1\text{mm}$ and $C = 1\text{ F}$ with no medium inside will have :-

(A) Area = $36\pi \times 10^6$ (B) Area = $4\pi \times 10^6$
(C) Area = $6\pi \times 10^6$ (D) Area = $\pi \times 10^6$

CP0029

2. A capacitor connected with battery is kept in a box so :-

(A) Net flux will come out from the box (B) Net flux will get in the box
(C) Net flux is zero (D) Net flux depends on polarity of battery

CP0030

3. Choose the CORRECT statement :-

(A) C will increase on increasing Q (B) C will increase on decreasing Q
(C) C will increase on decreasing V (D) C doesnot depend on Q & V

CP0031

4. The plate areas and plate separations of parallel plate capacitors are

Capacitor 1 : area A_0 , separation d_0

Capacitor 2 : area $2A_0$, separation $2d_0$

Capacitor 3 : area $2A_0$, separation $d_0/2$

Capacitor 4 : area $A_0/2$, separation $2d_0$

Capacitor 5 : area A_0 , separation $d_0/2$

Rank these according to their capacitances, least to greatest

(A) 1, 2, 3, 4, 5 (B) 5, 4, 3, 2, 1
(C) 5, 3 and 4 tie ; then 1, 2 (D) 4, 1 and 2 tie ; then 5, 3

CP0032

Capacitor circuits

5. A $2\text{-}\mu\text{F}$ and a $1\text{-}\mu\text{F}$ capacitor are connected in parallel and a potential difference is applied across the combination. The $2\text{-}\mu\text{F}$ capacitor has :

(A) Twice the charge of the $1\text{-}\mu\text{F}$ capacitor
(B) Half the charge of the $1\text{-}\mu\text{F}$ capacitor
(C) Twice the potential difference of the $1\text{-}\mu\text{F}$
(D) Half the potential difference of the $1\text{-}\mu\text{F}$ capacitor

CP0033

6. A $2\text{-}\mu\text{F}$ and a $1\text{-}\mu\text{F}$ capacitor are connected in series and a potential difference is applied across the combination. The $2\text{-}\mu\text{F}$ capacitor has :
- (A) Twice the charge of the $1\text{-}\mu\text{F}$ capacitor
 (B) Half the charge of the $1\text{-}\mu\text{F}$ capacitor
 (C) Twice the potential difference of the $1\text{-}\mu\text{F}$
 (D) Half the potential difference of the $1\text{-}\mu\text{F}$ capacitor

CP0034

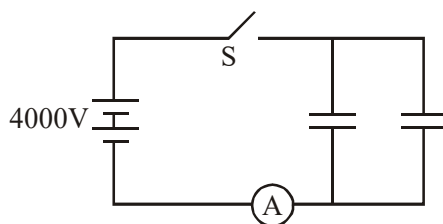
7. Capacitor C_1 and C_2 are connected in parallel and a potential difference is applied to the combination. If the capacitor that is equivalent to the combination has the same potential difference, then the charge on the equivalent capacitor is the same as :
- (A) The charge on C_1
 (B) The sum of the charges on C_1 and C_2
 (C) The difference of the charges on C_1 and C_2
 (D) The product of the charges on C_1 and C_2

CP0035

8. A $2\text{-}\mu\text{F}$ and a $1\text{-}\mu\text{F}$ capacitor are connected in series and charged from a battery. They store charges P and Q , respectively. When disconnected and charged separately using the same battery, they have charges R and S , respectively. Then :
- (A) $R > S > Q = P$ (B) $P > Q > R = S$ (C) $R > P = Q > S$ (D) $R = P > S = Q$

CP0036

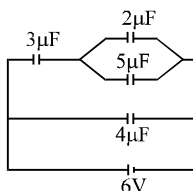
9. Each of the two $25\text{-}\mu\text{F}$ capacitors shown is initially uncharged. How many coulombs of charge pass through the ammeter A after the switch S is closed ?



- (A) 0.10 (B) 0.20 (C) 10 (D) 0.05

CP0037

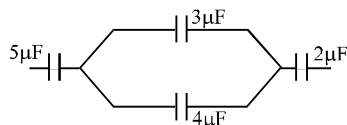
10. In the circuit shown in figure, the ratio of charges on $5\text{-}\mu\text{F}$ and $4\text{-}\mu\text{F}$ capacitor is :-



- (A) $4/5$ (B) $3/5$ (C) $3/8$ (D) $1/2$

CP0038

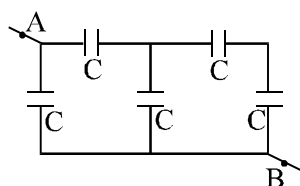
11. If charge on left plate of the $5\mu\text{F}$ capacitor in the circuit segment shown in the figure is $-20\mu\text{C}$, the charge on the right plate of $3\mu\text{F}$ capacitor is :-



- (A) $+8.57\mu\text{C}$ (B) $-8.57\mu\text{C}$ (C) $+11.42\mu\text{C}$ (D) $-11.42\mu\text{C}$

CP0039

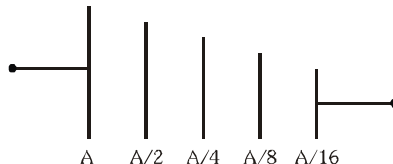
12. What is the equivalent capacitance of the system of capacitors between A & B as shown in the figure.



- (A) $\frac{7}{6}C$ (B) $1.6C$ (C) C (D) None

CP0040

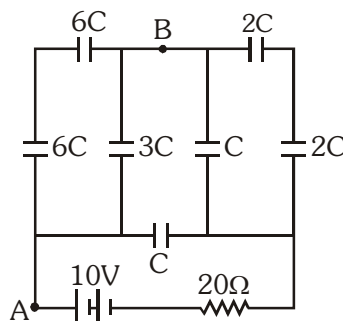
13. 5 Conducting plates each are placed face to face & equi-spaced at distance d . Area of each plate is half the previous plate. If area of first plate is A . Then the equivalent capacitance of the system shown is :-



- (A) $\frac{\epsilon_0 A}{d}$ (B) $\frac{\epsilon_0 A}{10d}$ (C) $\frac{\epsilon_0 A}{20d}$ (D) $\frac{\epsilon_0 A}{30d}$

CP0041

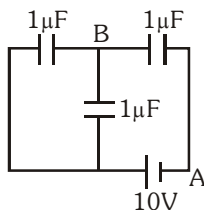
14. For the circuit shown here, the potential difference between points A and B is :-



- (A) 2.5 V (B) 7.5 V (C) 10 V (D) Zero

CP0042

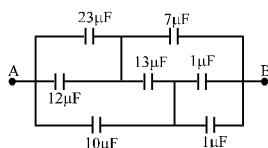
15. If potential of A is 10V, then potential of B is :-



- (A) $25/3$ V (B) $50/3$ V (C) $100/3$ V (D) 50 V

CP0043

16. Find the equivalent capacitance across A & B :-



- (A) $\frac{28}{3} \mu\text{f}$ (B) $\frac{15}{2} \mu\text{F}$ (C) $15 \mu\text{F}$ (D) none

CP0044

17. A capacitor of capacitance C is charged to a potential difference V from a cell and then disconnected from it. A charge $+Q$ is now given to its positive plate. The potential difference across the capacitor is now :-

- (A) V (B) $V + \frac{Q}{C}$ (C) $V + \frac{Q}{2C}$ (D) $V - \frac{Q}{C}$, if $V < CV$

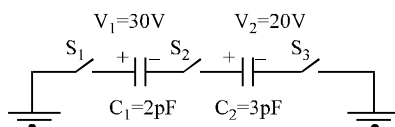
CP0045

18. A parallel plate capacitor is made by stacking n equally spaced plates connected alternatively. If the capacitance between any two adjacent plates is C , then the resultant capacitance is :- [AIEEE-2005]
- (A) $(n-1)C$ (B) $(n+1)C$ (C) C (D) nC

CP0046

19. For the circuit shown, which of the following statements is true ?

[IIT-JEE 1999]



- (A) with S_1 closed, $V_1 = 15$ V, $V_2 = 20$ V
 (B) with S_3 closed, $V_1 = V_2 = 25$ V
 (C) with S_1 & S_2 closed, $V_1 = V_2 = 0$
 (D) with S_1 & S_2 closed, $V_1 = 30$ V, $V_2 = 20$ V

CP0047

Force & energy

20. A $2\text{-}\mu\text{F}$ and a $1\text{-}\mu\text{F}$ capacitor are connected in series and charged by a battery. They store energies P and Q, respectively. When disconnected and charged separately using the same battery, they store energies R and S, respectively. Then

(A) $R > P > S > Q$ (B) $P > Q > R > S$ (C) $R > P > Q > S$ (D) $R > S > Q > P$

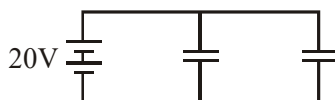
CP0048

21. Capacitors A and B are identical. Capacitor A is charged so it stores 4J of energy and capacitor B is uncharged. The capacitor are then connected in parallel. The total stored energy in the capacitors is now:

(A) 16 J (B) 8 J (C) 4 J (D) 2 J

CP0049

22. To store a total of 4×10^{-4} J of energy in the two identical capacitors shown, each should have a capacitance of :



(A) $0.10\text{ }\mu\text{F}$ (B) $0.50\text{ }\mu\text{F}$, $0.10\text{ }\mu\text{F}$ (C) $1.0\text{ }\mu\text{F}$ (D) $1.5\text{ }\mu\text{F}$

CP0050

23. A battery is used to charged a parallel-plate capacitor, after which it is disconnected. Then the plates are pulled apart to twice their original separation. This process will doubled the :

(A) capacitance (B) surface charge density on each plate
(C) stored energy (D) electric field between the two places3

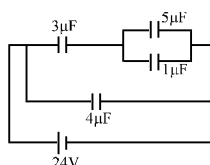
CP0051

24. A parallel-plate capacitor has a plate area of 0.3 m^2 and a plate separation of 0.1 mm . If the charge on each plate has a magnitude of $5 \times 10^{-6}\text{ C}$ then the force exerted by one plate on the other has a magnitude of about :

(A) 0 (B) 5 N (C) $1 \times 10^4\text{ N}$ (D) $9 \times 10^5\text{ N}$

CP0052

25. In the circuit shown, the energy stored in $1\text{ }\mu\text{F}$ capacitor is :-



(A) $40\text{ }\mu\text{J}$ (B) $64\text{ }\mu\text{J}$ (C) $32\text{ }\mu\text{J}$ (D) none

CP0053

26. A parallel plate capacitor has an electric field of 10^5 V/m between the plates. If the charge on the capacitor plate is $1\text{ }\mu\text{C}$, then the force on each capacitor plate is :-

(A) 0.1 N (B) 0.05 N (C) 0.02 N (D) 0.01 N

CP0054

27. Consider a capacitor connected with a battery, capacitor is in steady state. Now plates of capacitor are drawn apart so as to double the separation in two cases.

Case :

- (i) Battery remains connected
- (ii) Battery is disconnected

Mark the **CORRECT** statement.

- (A) In case (i) energy of capacitor increases
- (B) In case (i) work done by battery is positive
- (C) In case (ii) energy of capacitor increases
- (D) In case (ii) potential difference across capacitor decreases

CP0055

28. Two identical capacitors, have the same capacitance C . One of them is charged to potential V_1 and the other to V_2 . The negative ends of the capacitors are connected together. When the positive ends are also connected, the decrease in energy of the combined system is :- [IIT-JEE 2002 (Scr)]

- (A) $\frac{1}{4}C(V_1^2 - V_2^2)$ (B) $\frac{1}{4}C(V_1^2 + V_2^2)$ (C) $\frac{1}{4}C(V_1 - V_2)^2$ (D) $\frac{1}{4}C(V_1 + V_2)^2$

CP0056

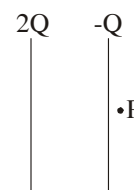
29. In the figure shown the plates of a parallel plate capacitor have unequal charges. Its capacitance is ' C '. P is a point outside the capacitor and close to the plate of charge $-Q$. The distance between the plates is ' d ' then which statement is wrong :-

- (A) A point charge at point ' P ' will experience electric force due to capacitor

- (B) The potential difference between the plates will be $\frac{3Q}{2C}$

- (C) The energy stored in the electric field in the region between the plates is $\frac{9Q^2}{8C}$

- (D) The force on one plate due to the other plate is $\frac{Q^2}{2\pi\epsilon_0 d^2}$



CP0057

Dielectrics

30. A dielectric slab is slowly inserted between the plates of a parallel plate capacitor, while the potential difference between the plates is held constant by a battery. As it is being inserted :

- (A) the capacitance, the potential difference between the plates, and the charge on the positive plate all increase
- (B) the capacitance, the potential difference between the plates, and the charge on the positive plate all decrease
- (C) the potential difference between the plates increases, the charge on the positive plate decreases, and the capacitance remains the same
- (D) the capacitance and the charge on the positive plate increase but the potential difference between the plates remains the same

CP0058

31. An air-filled parallel-plate capacitor has a capacitance of 1 pF. The plate separation is then doubled and a wax dielectric is inserted, completely filling the space between the plates. As a result, the capacitance becomes 2 pF. The dielectric of the wax is
 (A) 0.25 (B) 0.5 (C) 2.0 (D) 4.0

CP0059

32. One of materials listed below is to be placed between two identical metal sheets, with no, air gap, to form a parallel-plate capacitor. Which produces the greatest capacitance ?
 (A) material of thickness 0.1 mm and dielectric constant 2
 (B) material of thickness 0.2 mm and dielectric constant 3
 (C) material of thickness 0.3 mm and dielectric constant 2
 (D) material of thickness 0.4 mm and dielectric constant 8

CP0060

33. Two parallel-plate capacitors with the same plate area but different capacitance are connected in parallel to a battery. Both capacitors are filled with air. The quantity that is the same for both capacitors when they are fully charged is :
 (A) potential difference (B) energy density
 (C) electric field between the plates (D) charge on the positive plate

CP0061

34. A capacitor stores $60\mu\text{C}$ charge when connected across a battery. When the gap between the plates is filled with a dielectric, a charge of $120\mu\text{C}$ flows through the battery. The dielectric constant of the material inserted is :
 (A) 1 (B) 2 (C) 3 (D) none

CP0062

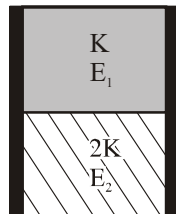
35. Condenser A has a capacity of $15\mu\text{F}$ when it is filled with a medium of dielectric constant 15. Another condenser B has a capacity $1\mu\text{F}$ with air between the plates. Both are charged separately by a battery of 100V. After charging, both are connected in parallel without the battery and the dielectric material being removed. The common potential now is :-
 (A) 400V (B) 800V (C) 1200V (D) 1600V

CP0063

36. Three capacitors $2\mu\text{F}$, $3\mu\text{F}$ and $5\mu\text{F}$ can withstand voltages to 3V, 2V and 1V respectively. Their series combination can withstand a maximum voltage equal to :-
 (A) 5 Volts (B) $(31/6)$ Volts (C) $(26/5)$ Volts (D) None

CP0064

37. A parallel plate capacitor is connected from a cell and then isolated from it. Two dielectric slabs of dielectric constant K and $2K$ are now introduced in the region between upper half and lower half of the plate (as shown in figure). The electric field intensity in upper half of dielectric is E_1 and lower half is E_2 then



- (A) $E_1 = 2E_2$
 (B) Electrostatic potential energy of upper half is less than that of lower half
 (C) Induced charges on both slabs are same
 (D) Charge distribution on the plates remains same after insertion of dielectric

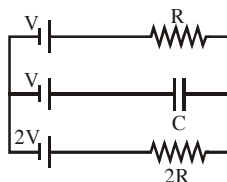
CP0065

38. Two point charges exert a force F_0 on each other when placed in vacuum. Now the charges are increased to four times, separation between them is doubled and the system is placed in an insulating medium. Now they experience the same force. What should be the dielectric constant of the medium?
 (A) 3 (B) 4 (C) 2 (D) 5

CP0066

R-C Circuits

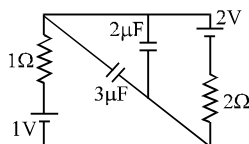
39. In the given circuit, with steady current the potential drop across the capacitor must be :-



- (A) V (B) $\frac{V}{2}$ (C) $\frac{V}{3}$ (D) $\frac{2V}{3}$

CP0067

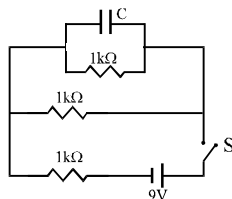
40. In the circuit shown, the charge on the $3\mu\text{F}$ capacitor at steady state will be



- (A) $6\mu\text{C}$ (B) $4\mu\text{C}$ (C) $\frac{2}{3}\mu\text{C}$ (D) $3\mu\text{C}$

CP0068

41. A capacitor $C = 100 \mu\text{F}$ is connected to three resistor each of resistance $1 \text{ k}\Omega$ and a battery of emf 9 V . The switch S has been closed for long time so as to charge the capacitor. When switch S is opened, the capacitor discharges with time constant :-



- (A) 33 ms (B) 5 ms (C) 3.3 ms (D) 50 ms

CP0069

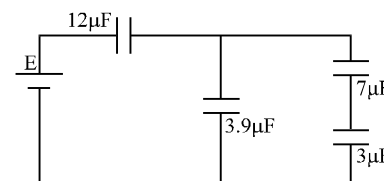
42. An uncharged capacitor of capacitance $4 \mu\text{F}$, a battery of emf 12 volt and a resistor of $2.5 \text{ M}\Omega$ are connected in series. The time after which $v_c = 3v_R$ is (take $\ln 2 = 0.693$) [IIT-JEE' 2005 (Scr)]
(A) 6.93 sec. (B) 13.86 sec. (C) 20.52 sec. (D) none of these

CP0070

MULTIPLE CORRECT TYPE QUESTIONS

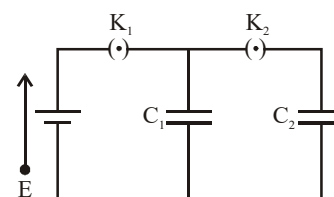
Capacitor Circuits

43. Four capacitors and a battery are connected as shown. The potential drop across the $7 \mu\text{F}$ capacitor is 6 V . Then the :
(A) potential difference across the $3 \mu\text{F}$ capacitor is 10 V
(B) charge on the $3 \mu\text{F}$ capacitor is $42 \mu\text{C}$
(C) e.m.f. of the battery is 30 V
(D) potential difference across the $12 \mu\text{F}$ capacitor is 10 V .



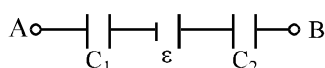
CP0071

44. In the circuit shown in figure initially key K_1 is closed and key K_2 is open. Then K_1 is opened and K_2 is closed (order is important). [Take Q'_1 and Q'_2 as charges on C_1 and C_2 and V_1 and V_2 as voltage respectively. Then
(A) charge on C_1 gets redistributed such that $V_1 = V_2$
(B) charge on C_1 gets redistributed such that $Q'_1 = Q'_2$
(C) charge on C_1 gets redistributed such that $C_1 V_1 + C_2 V_2 = C_1 E$
(D) charge on C_1 gets redistributed such that $Q'_1 + Q'_2 = Q$



CP0072

45. A circuit shown in the figure consists of a battery of emf 10 V and two capacitance C_1 and C_2 of capacitances $1.0 \mu\text{F}$ and $2.0 \mu\text{F}$ respectively. The potential difference $V_A - V_B$ is 5 V



- (A) charge on capacitor C_1 is equal to charge on capacitor C_2
(B) Voltage across capacitor C_1 is 5 V .
(C) Voltage across capacitor C_2 is 10 V
(D) Energy stored in capacitor C_1 is two times the energy stored in capacitor C_2 .

CP0073

Dielectrics

- 46.** A parallel plate capacitor has a parallel slab of copper inserted between and parallel to the two plates, without touching the plates. The capacity of the capacitor after the introduction of the copper sheet is:
- (A) minimum when the copper slab touches one of the plates.
(B) maximum when the copper slab touches one of the plates.
(C) invariant for all positions of the slab between the plates.
(D) greater than that before introducing the slab.

CP0074

47. A parallel plate air-core capacitor is connected across a source of constant potential difference. When a dielectric plate is introduced between the two plates then :
- (A) some charge from the capacitor will flow back into the source.
 - (B) some extra charge from the source will flow back into the capacitor.
 - (C) the electric field intensity between the two plate does not change.
 - (D) the electric field intensity between the two plates will decrease.

CP0075

- 48.** A parallel plate capacitor of plate area A and plate separation d is charged to potential difference V and then the battery is disconnected. A slab of dielectric constant K is then inserted between the plates of the capacitor so as to fill the space between the plates. If Q , E and W denote, the magnitude of charge on each plate, the electric field between the plates (after the slab is inserted) and the work done on the system respectively in question, then in the process of inserting the slab

$$(A) \quad Q = \frac{\epsilon_0 A V}{d}$$

$$(B) \quad Q = \frac{\epsilon_0 K A V}{d}$$

$$(C) E = \frac{V}{K_d}$$

$$(D) \ W = -\frac{\epsilon_0 A V^2}{2d} \left(1 - \frac{1}{K} \right)$$

CP0076

49. The capacitance of a parallel plate capacitor is C when the region between the plate has air. This region is now filled with a dielectric slab of dielectric constant k . The capacitor is connected to a cell of emf E , and the slab is taken out
- (A) charge $CE(k - 1)$ flows through the cell
- (B) energy $E^2C(k - 1)$ is absorbed by the cell.
- (C) the energy stored in the capacitor is reduced by $E^2C(k - 1)$
- (D) the external agent has to do $\frac{1}{2}E^2C(k - 1)$ amount of work to take the slab out.

CP0077

50. A capacitor of capacity C_0 is connected to a battery of emf V_0 . When steady state is attained a dielectric slab of dielectric constant K is slowly introduced in the capacitor. Mark the Correct statement(s), in final steady state :-

- (A) Magnitude of induced charge on the each surface of slab is $C_0 V_0 (K - 1)$
 (B) Net electric force due to induced charges on the plate is zero.

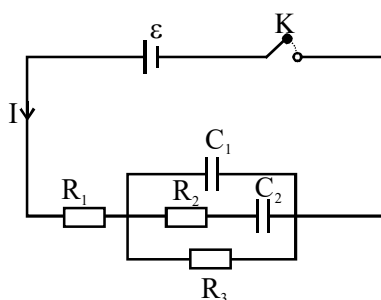
(C) Force of attraction between plates of capacitor is $\frac{K(C_0 V_0)^2}{2 \epsilon_0 A}$

(D) Net field due to induced charges in dielectric slab is $\frac{8V_0 (k-1)^2}{K \epsilon_0 A}$

CP0078

R-C Circuits

51. Mark the **CORRECT** statement(s) regarding the current I through the battery in the circuit shown in figure.



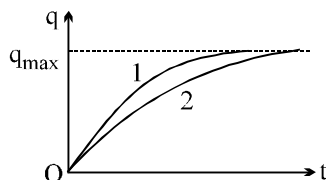
- (A) Immediately after the key K is closed, $I = \frac{\epsilon}{R_1}$
 (B) Immediately after the key K is closed, $I = \frac{\epsilon}{R_1 + R_3}$
 (C) Long time after key K is closed, $I = \frac{\epsilon}{R_1 + R_3}$
 (D) Long time after key K is closed, $I = \frac{\epsilon}{R_1 + R_2}$

CP0079

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 52 & 53

The charge across the capacitor in two different RC circuits 1 and 2 are plotted as shown in figure.



52. Choose the correct statement(s) related to the two circuits.

- (A) Both the capacitors are charged to the same charge.
- (B) The emf's of cells in both the circuit are equal.
- (C) The emf's of the cells may be different.
- (D) The emf E_1 is more than E_2

CP0080

53. Identify the correct statement(s) related to the R_1 , R_2 , C_1 and C_2 of the two RC circuits.

- (A) $R_1 > R_2$ if $E_1 = E_2$
- (B) $C_1 < C_2$ if $E_1 = E_2$
- (C) $R_1 C_1 > R_2 C_2$
- (D) $\frac{R_1}{R_2} < \frac{C_2}{C_1}$

CP0081

MATRIX MATCH TYPE QUESTION

54.

Column-I

Column-II

- | | |
|--|---|
| (A) Plates of an isolated, charged, parallel plate, air core capacitor are slowly pulled apart. | (P) Electric energy stored inside capacitor increases in the process. |
| (B) A dielectric is slowly inserted inside an isolated and charged parallel plate air cored capacitor to completely fill the space between plates. | (Q) Force between the two plates of the capacitor remain unchanged. |
| (C) Plates of a parallel plate capacitor connected across a battery are slowly pulled apart. | (R) Electric field in the region between plates remain unchanged. |
| (D) A dielectric slab is slowly inserted inside a parallel plate capacitor connected across a battery to completely fill the space between plates. | (S) Total electric energy stored inside capacitor decreases in the process. |
| | (T) Electric field in the region decreases. |

CP0082

EXERCISE (O-2)

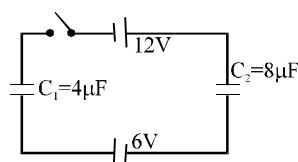
SINGLE CORRECT TYPE QUESTIONS

1. Three long concentric conducting cylindrical shells have radii R , $2R$ and $2\sqrt{2}R$. Inner and outer shells are connected to each other. The capacitance across middle and inner shells per unit length is:

- (A) $\frac{1}{3} \frac{\epsilon_0}{\ln 2}$ (B) $\frac{6\pi \epsilon_0}{\ln 2}$ (C) $\frac{\pi \epsilon_0}{2 \ln 2}$ (D) None

CP0083

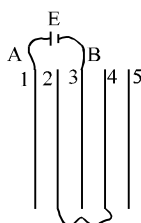
2. In the circuit shown initially C_1 & C_2 are uncharged. After closing the switch



- (A) The charge on C_2 is greater than that on C_1
 (B) The charge on C_1 and C_2 are the same
 (C) The potential drops across C_1 and C_2 are the same
 (D) The potential drops across C_2 is greater than that across C_1

CP0084

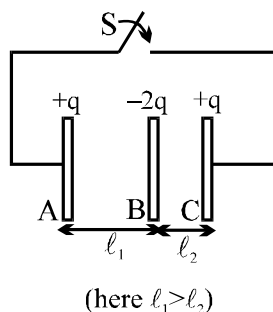
3. Five conducting parallel plates having area A and separation between them d , are placed as shown in the figure. Plate number 2 and 4 are connected wire and between point A and B, a cell of emf E is connected. The charge flown through the cell is :-



- (A) $\frac{3}{4} \frac{\epsilon_0 A E}{d}$ (B) $\frac{2}{3} \frac{\epsilon_0 A E}{d}$ (C) $\frac{4\epsilon_0 A E}{d}$ (D) $\frac{\epsilon_0 A E}{2d}$

CP0085

4. Three long conducting plate A, B & C having charges $+q$, $-2q$ & $+q$ as shown in figure. Here plate A and C are fixed. If the switch S is closed. The middle plate (B) will start moving in



- (A) Leftward direction
(B) Rightward direction
(C) will not move
(D) First move leftward & then rightward

CP0086

5. A capacitor of capacitance C is initially charged to a potential difference of V volt. Now it is connected to a battery of $2V$ Volt with opposite polarity. The ratio of heat generated to the final energy stored in the capacitor will be :-

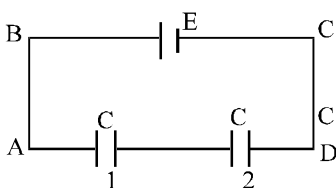
- (A) 1.75 (B) 2.25 (C) 2.5 (D) 1/2

CP0087

6. A conducting body 1 has some initial charge Q , and its capacitance is C . There are two other conducting bodies, 2 and 3, having capacitances : $C_2 = 2C$ and $C_3 \rightarrow \infty$. Bodies 2 and 3 are initially uncharged. "Body 2 is touched with body 1. Then, body 2 is removed from body 1 and touched with body 3, and then removed." This process is repeated N times. Then, the charge on body 1 at the end must be
- (A) $Q/3^N$ (B) $Q/3^{N-1}$ (C) Q/N^3 (D) None

CP0088

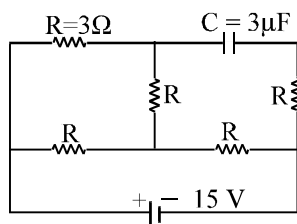
7. In the adjoining figure, capacitor (1) and (2) have a capacitance ' C ' each. When the dielectric of dielectric constant K is inserted between the plates of one of the capacitor, the total charge flowing through battery is :-



- (A) $\frac{KCE}{K+1}$ from B to C
(B) $\frac{KCE}{K+1}$ from C to B
(C) $\frac{(K-1)CE}{2(K+1)}$ from B to C
(D) $\frac{(K-1)CE}{2(K+1)}$ from C to B

CP0089

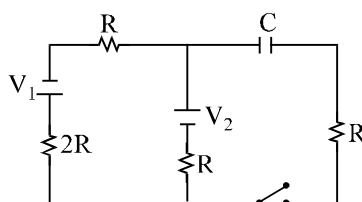
8. In the circuit shown, the cell is ideal, with $\text{emf} = 15 \text{ V}$. Each resistance is of 3Ω . The potential difference across the capacitor is :-



- (A) zero (B) 9 V (C) 12 V (D) 15 V

CP0090

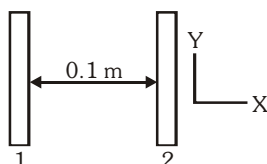
9. In the transient circuit shown the time constant of the circuit is :



- (A) $\frac{5}{3}RC$ (B) $\frac{5}{2}RC$ (C) $\frac{7}{4}RC$ (D) $\frac{7}{3}RC$

CP0091

10. Two insulating plates are both uniformly charged in such a way that the potential difference between them is $V_2 - V_1 = 20 \text{ V}$. (i.e., plate 2 is at a higher potential). The plates are separated by $d = 0.1 \text{ m}$ and can be treated as infinitely large. An electron is released from rest on the inner surface of plate 1. What is its speed when it hits plate 2 ? ($e = 1.6 \times 10^{-19} \text{ C}$, $m_0 = 9.11 \times 10^{-31} \text{ kg}$) [AIEEE-2006]



- (A) $2.65 \times 10^6 \text{ m/s}$ (B) $7.02 \times 10^{12} \text{ m/s}$ (C) $1.87 \times 10^6 \text{ m/s}$ (D) $32 \times 10^{-19} \text{ m/s}$

CP0092

11. A battery is used to charge a parallel plate capacitor till the potential difference between the plates becomes equal to the electromotive force of the battery. The ratio of the energy stored in the capacitor and the work done by the battery will be- [AIEEE-2007]

- (A) 1 (B) 2 (C) 1/4 (D) 1/2

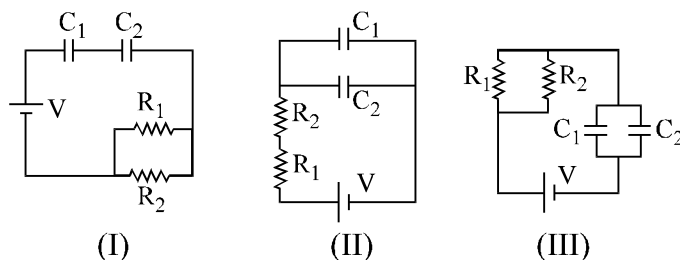
CP0093

12. A parallel plate condenser with a dielectric of dielectric constant K between the plates has a capacity C and is charged to a potential V volts. The dielectric slab is slowly removed from between the plates and then re-inserted. The net work done by the system in this process is- [AIEEE-2007]

- (A) $\frac{1}{2}(K-1)CV^2$ (B) $CV^2(K-1)/K$ (C) $(K-1)CV^2$ (D) zero

CP0094

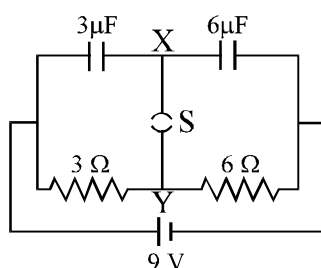
13. Given : $R_1 = 1\Omega$, $R_2 = 2\Omega$, $C_1 = 2\mu\text{F}$, $C_2 = 4\mu\text{F}$. The time constants (in μs) for the circuits I, II, III are respectively [IIT-JEE 2006]



- (A) 18, 8/9, 4 (B) 18, 4, 8/9 (C) 4, 8/9, 18 (D) 8/9, 18, 4

CP0095

14. A circuit is connected as shown in the figure with the switch S open. When the switch is closed, the total amount of charge that flows from Y to X is :- [IIT-JEE 2007]

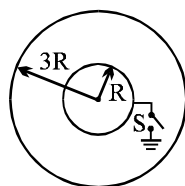


- (A) 0 (B) $54\mu\text{C}$ (C) $27\mu\text{C}$ (D) $81\mu\text{C}$

CP0096

MULTIPLE CORRECT TYPE QUESTIONS

15. A parallel-plate capacitor is connected to a cell. Its positive plate A and its negative plate B have charges $+Q$ and $-Q$ respectively. A third plate C, identical to A and B, with charge $+Q$, is now introduced midway between A and B, parallel to them. Which of the following are correct?
- (A) The charge on the inner face of B is now $-\frac{3Q}{2}$
- (B) There is no change in the potential difference between A and B.
- (C) The potential difference between A and C is one-third of the potential difference between B and C.
- (D) The charge on the inner face of A is now $Q/2$.
16. Two thin conducting shells of radii R and $3R$ are shown in the figure. The outer shell carries a charge $+Q$ and the inner shell is neutral. The inner shell is earthed with the help of a switch S. [IIT-JEE 2007]



- (A) With the switch S open, the potential of the inner sphere is equal to that of the outer.
- (B) When the switch S is closed, the potential of the inner sphere becomes zero.
- (C) With the switch S closed, the charge attained by the inner sphere is $-Q/3$.
- (D) By closing the switch the capacitance of the system increases.

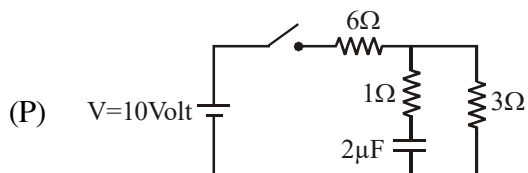
CP0098

MATRIX MATCH TYPE QUESTION

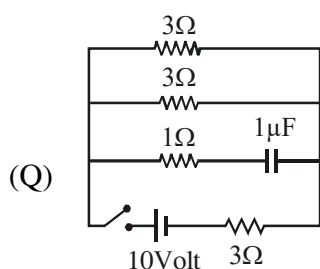
17. Match the following. In each of the cases shown below, find the time constant of the circuit (in μs) after switch is closed.

List-I

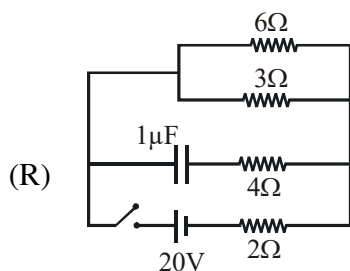
List-II



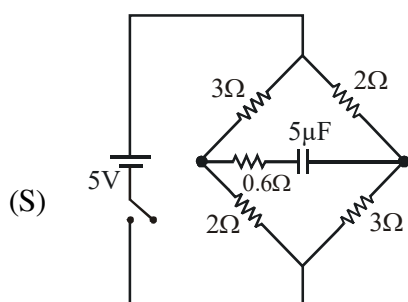
(1) 6



(2) 2



(3) 5



(4) 15

Codes :

	P	Q	R	S
(A)	1	2	1	4
(B)	3	1	4	2
(C)	1	2	3	4
(D)	2	4	1	3

CP0099

EXERCISE (J-M)

1. Let C be the capacitance of a capacitor discharging through a resistor R . Suppose t_1 is the time taken for the energy stored in the capacitor to reduce to half its initial value and t_2 is the time taken for the charge to reduce to one-fourth its initial value. Then the ratio t_1/t_2 will be : [AIEEE-2010]
 (1) 2 (2) 1 (3) $1/2$ (4) $1/4$

CP0100

2. Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle of 30° with each other. When suspended in a liquid of density 0.8 g cm^{-3} , the angle remains the same. If density of the material of the sphere is 1.6 g cm^{-3} , the dielectric constant of the liquid is : [AIEEE - 2010]
 (1) 1 (2) 4 (3) 3 (4) 2

CP0101

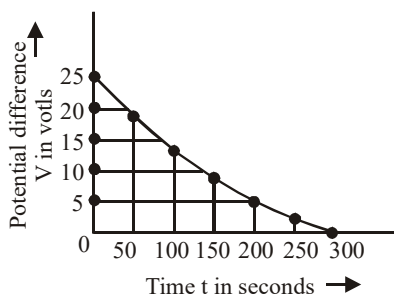
3. A resistor ' R ' and $2\mu\text{F}$ capacitor in series is connected through a switch to 200 V direct supply. Across the capacitor is a neon bulb that lights up at 120 V . Calculate the value of R to make the bulb light up 5s after the switch has been closed. ($\log_{10} 2.5 = 0.4$) [AIEEE-2011]
 (1) $2.7 \times 10^6 \Omega$ (2) $3.3 \times 10^7 \Omega$ (3) $1.3 \times 10^4 \Omega$ (4) $1.7 \times 10^5 \Omega$

CP0102

4. Combination of two identical capacitors, a resistor R and a dc voltage source of voltage 6V is used in an experiment on (C-R) circuit. It is found that for a parallel combination of the capacitor the time in which the voltage of the fully charged combination reduces to half its original voltage is 10 second . For series combination the time needed for reducing the voltage of the fully charged series combination by half is :- [AIEEE-2011]
 (1) 20 second (2) 10 second (3) 5 second (4) 2.5 second

CP0103

5. The figure shows an experimental plot for discharging of a capacitor in an R-C circuit. The time constant τ of this circuit lies between:- [AIEEE 2012]



- (1) 100 sec and 150 sec (2) 150 sec and 200 sec
 (3) 0 and 50 sec (4) 50 sec and 100 sec

CP0104

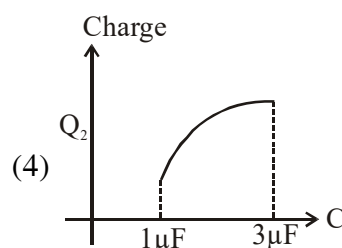
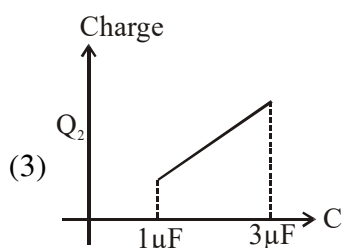
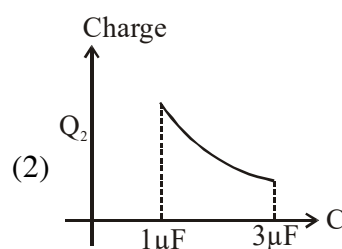
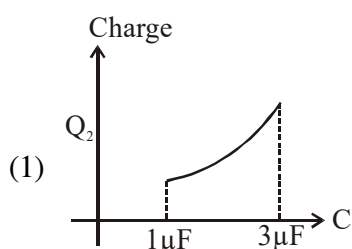
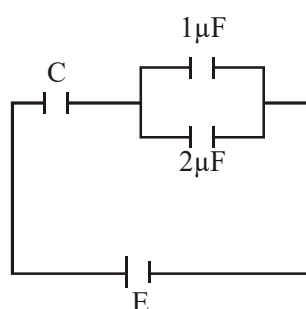
6. Two capacitors C_1 and C_2 are charged to 120V and 200V respectively. It is found that by connecting them together the potential on each one can be made zero. Then : [JEE-Main-2013]
 (1) $5C_1 = 3C_2$ (2) $3C_1 = 5C_2$ (3) $3C_1 + 5C_2 = 0$ (4) $9C_1 = 4C_2$

CP0105

7. A parallel plate capacitor is made of two circular plates separated by a distance of 5 mm and with a dielectric of dielectric constant 2.2 between them. When the electric field in the dielectric is 3×10^4 V/m, the charge density of the positive plate will be close to : [JEE-Main-2014]
 (1) 3×10^4 C/m² (2) 6×10^4 C/m² (3) 6×10^{-7} C/m² (4) 3×10^{-7} C/m²

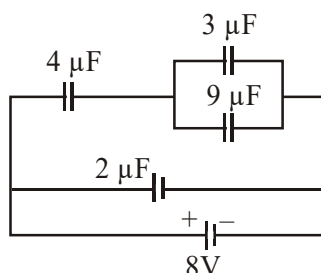
CP0106

8. In the given circuit, charge Q_2 on the $2\mu\text{F}$ capacitor changes as C is varied from $1\mu\text{F}$ to $3\mu\text{F}$. Q_2 as a function of ' C ' is given properly by : (figures are drawn schematically and are not to scale) :- [JEE-Main-2015]



CP0107

9. A combination of capacitors is set up as shown in the figure. The magnitude of the electric field, due to a point charge Q (having a charge equal to the sum of the charges on the $4\mu\text{F}$ and $9\mu\text{F}$ capacitors), at a point 30 m from it, would equal: [JEE-Main-2016]



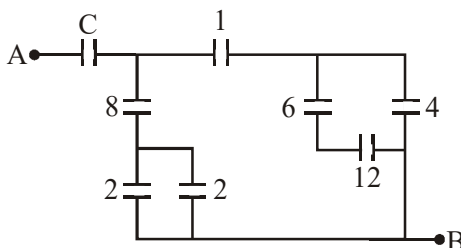
- (1) 480 N/C (2) 240 N/C (3) 360 N/C (4) 420 N/C

CP0108

10. Three capacitors each of $4 \mu\text{F}$ are to be connected in such a way that the effective capacitance is $6 \mu\text{F}$. This can be done by connecting them : [JEE-Main online-2016]
- (1) two in parallel and one in series (2) all in parallel
(3) two in series and one in parallel (4) all in series

CP0109

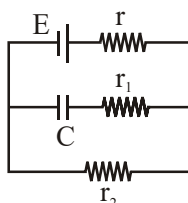
11. Figure shows a network of capacitors where the numbers indicates capacitances in micro Farad. The value of capacitance C if the equivalent capacitance between point A and B is to be $1 \mu\text{F}$ is :- [JEE-Main online-2016]



- (1) $\frac{33}{23} \mu\text{F}$ (2) $\frac{34}{23} \mu\text{F}$ (3) $\frac{31}{23} \mu\text{F}$ (4) $\frac{32}{23} \mu\text{F}$

CP0110

12. In the given circuit diagram when the current reaches steady state in the circuit, the charge on the capacitor of capacitance C will be : [JEE-Main-2017]



- (1) $CE \frac{r_2}{(r + r_2)}$ (2) $CE \frac{r_1}{(r_1 + r)}$ (3) CE (4) $CE \frac{r_1}{(r_2 + r)}$

CP0111

13. A capacitance of $2 \mu\text{F}$ is required in an electrical circuit across a potential difference of 1.0 kV . A large number of $1 \mu\text{F}$ capacitors are available which can withstand a potential difference of not more than 300 V . The minimum number of capacitors required to achieve this is : [JEE-Main-2017]
- (1) 24 (2) 32 (3) 2 (4) 16

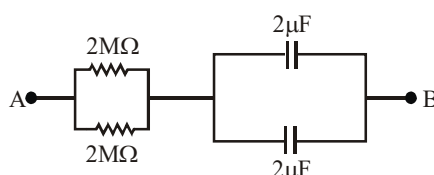
CP0112

14. A parallel plate capacitor of capacitance 90 pF is connected to a battery of emf 20 V . If a dielectric material of dielectric constant $K = \frac{5}{3}$ is inserted between the plates, the magnitude of the induced charge will be :- [JEE-Main-2018]
- (1) 0.3 nC (2) 2.4 nC (3) 0.9 nC (4) 1.2 nC

CP0113

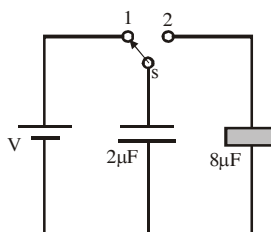
EXERCISE (J-A)

1. At time $t = 0$, a battery of 10V is connected across points A and B in the given circuit. If the capacitors have no charge initially, at what time (in seconds) does the voltage across them become 4V?
[Take : $\ln 5 = 1.6$, $\ln 3 = 1.1$] [IIT-JEE 2010]



CP0114

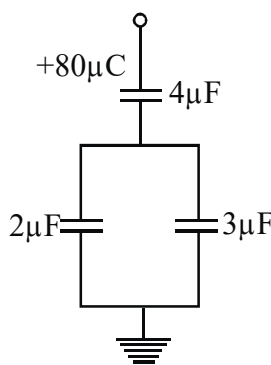
2. A $2\mu\text{F}$ capacitor is charged as shown in figure. The percentage of its stored energy dissipated after the switch S is turned to position 2 is :- [IIT-JEE 2011]



- (A) 0% (B) 20% (C) 75% (D) 80%

CP0115

3. In the given circuit, a charge of $+80\mu\text{C}$ is given to the upper plate of the $4\mu\text{F}$ capacitor. Then in the steady state, the charge on the upper plate of the $3\mu\text{F}$ capacitor is :- [IIT-JEE 2012]

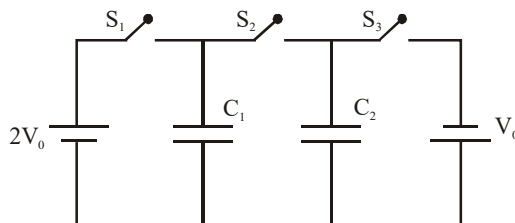


- (A) $+32\mu\text{C}$ (B) $+40\mu\text{C}$ (C) $+48\mu\text{C}$ (D) $+80\mu\text{C}$

CP0116

4. In the circuit shown in the figure, there are two parallel plate capacitors each of the capacitance C . The switch S_1 is pressed first to fully charge the capacitor C_1 and then released. The switch S_2 is then pressed to charge the capacitor C_2 . After some time, S_2 is released and then S_3 is pressed. After some time,

[IIT-JEE 2013]

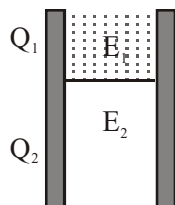


- (A) the charge on the upper plate of C_1 is $2CV_0$
 (B) the charge on the upper plate of C_1 is CV_0
 (C) the charge on the upper plate of C_2 is 0.
 (D) the charge on the upper plate of C_2 is $-CV_0$

CP0117

5. A parallel plate capacitor has a dielectric slab of dielectric constant K between its plates that covers $1/3$ of the area of its plates, as shown in the figure. The total capacitance of the capacitor is C while that of the portion with dielectric in between is C_1 . When the capacitor is charged, the plate area covered by the dielectric gets charge Q_1 and the rest of the area gets charge Q_2 . The electric field in the dielectric is E_1 and that in the other portion is E_2 . Choose the correct option/options, ignoring edge effects.

[JEE-Advance 2014]



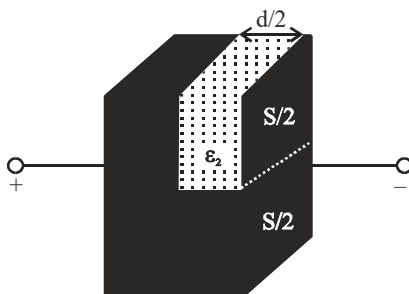
- (A) $\frac{E_1}{E_2} = 1$ (B) $\frac{E_1}{E_2} = \frac{1}{K}$ (C) $\frac{Q_1}{Q_2} = \frac{3}{K}$ (D) $\frac{C}{C_1} = \frac{2+K}{K}$

CP0118

6. A parallel plate capacitor having plates of area S and plate separation d , has capacitance C_1 in air. When two dielectrics of different relative permittivities ($\epsilon_1 = 2$ and $\epsilon_2 = 4$) are introduced between the

two plates as shown in the figure, the capacitance becomes C_2 . The ratio $\frac{C_2}{C_1}$ is :-

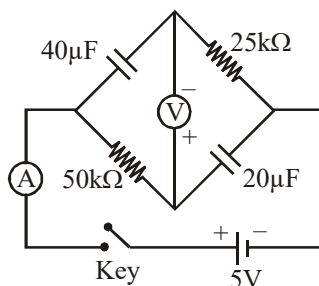
[JEE-Advance 2015]



- (A) $\frac{6}{5}$ (B) $\frac{5}{3}$ (C) $\frac{7}{5}$ (D) $\frac{7}{3}$

CP0119

7. In the circuit shown below, the key is pressed at time $t = 0$. Which of the following statement(s) is(are) true? [JEE-Advance 2016]



- (A) The voltmeter displays $-5V$ as soon as the key is pressed, and displays $+5V$ after a long time
 (B) The voltmeter will display $0V$ at time $t = \ln 2$ seconds
 (C) The current in the ammeter becomes $1/e$ of the initial value after 1 second
 (D) The current in the ammeter becomes zero after a long time

PARAGRAPH-1

Consider a simple RC circuit as shown in figure 1.

Process 1 : In the circuit the switch S is closed at $t = 0$ and the capacitor is fully charged to voltage V_0 (i.e., charging continues for time $T \gg RC$). In the process some dissipation (E_D) occurs across the resistance R . The amount of energy finally stored in the fully charged capacitor is E_C .

Process 2 : In a different process the voltage is first set to $\frac{V_0}{3}$ and maintained for a charging time

$T \gg RC$. Then the voltage is raised to $\frac{2V_0}{3}$ without discharging the capacitor and again maintained for a time $T \gg RC$. The process is repeated one more time by raising the voltage to V_0 and the capacitor is charged to the same final voltage V_0 as in Process 1.

These two processes are depicted in Figure 2.

[JEE-Advance 2017]

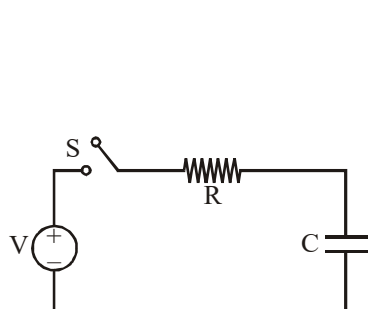


Figure 1

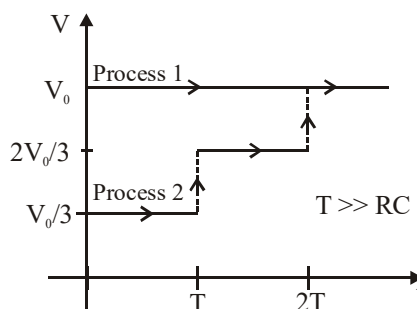


Figure 2

8. In Process 1, the energy stored in the capacitor E_C and heat dissipated across resistance E_D are related by :-

- (A) $E_C = E_D$ (B) $E_C = 2E_D$ (C) $E_C = \frac{1}{2} E_D$ (D) $E_C = E_D \ln 2$

CP0120

CP0121

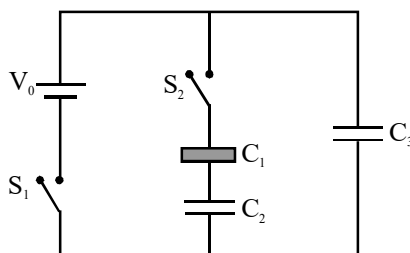
9. In Process 2, total energy dissipated across the resistance E_D is :-

(A) $E_D = \frac{1}{3} \left(\frac{1}{2} CV_0^2 \right)$ (B) $E_D = 3 \left(\frac{1}{2} CV_0^2 \right)$ (C) $E_D = \frac{1}{2} CV_0^2$ (D) $E_D = 3 CV_0^2$

CP0122

10. Three identical capacitors C_1 , C_2 and C_3 have a capacitance of $1.0 \mu\text{F}$ each and they are uncharged initially. They are connected in a circuit as shown in the figure and C_1 is then filled completely with a dielectric material of relative permittivity ϵ_r . The cell electromotive force (emf) $V_0 = 8\text{V}$. First the switch S_1 is closed while the switch S_2 is kept open. When the capacitor C_3 is fully charged, S_1 is opened and S_2 is closed simultaneously. When all the capacitors reach equilibrium, the charge on C_3 is found to be $5\mu\text{C}$. The value of ϵ_r .

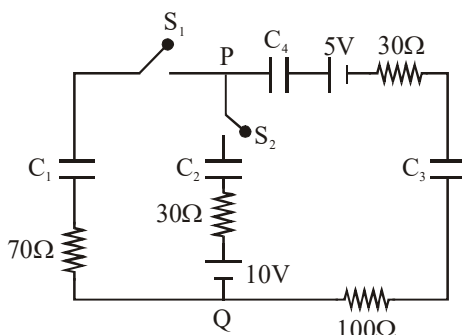
[JEE-Advance 2018]



CP0123

11. In the circuit shown, initially there is no charge on capacitors and keys S_1 and S_2 are open. The values of the capacitors are $C_1 = 10 \mu\text{F}$, $C_2 = 30 \mu\text{F}$ and $C_3 = C_4 = 80 \mu\text{F}$.

[JEE-Advance 2019]



Which of the statement(s) is/are correct ?

- (1) The keys S_1 is kept closed for long time such that capacitors are fully charged. Now key S_2 is closed, at this time, the instantaneous current across 30Ω resistor (between points P and Q) will be 0.2 A (round off to 1^{st} decimal place).
- (2) If key S_1 is kept closed for long time such that capacitors are fully charged, the voltage difference between points P and Q will be 10 V .
- (3) At time $t = 0$, the key S_1 is closed, the instantaneous current in the closed circuit will be 25 mA .
- (4) If key S_1 is kept closed for long time such that capacitors are fully charged, the voltage across the capacitors C_1 will be 4 V .

CP0124

12. A parallel plate capacitor of capacitance C has spacing d between two plates having area A . The region between the plates is filled with N dielectric layers, parallel to its plates, each with thickness

$\delta = \frac{d}{N}$. The dielectric constant of the m^{th} layer is $K_m = K \left(1 + \frac{m}{N} \right)$. For a very large N ($> 10^3$), the

capacitance C is $\propto \left(\frac{K \epsilon_0 A}{d \ln 2} \right)$. The value of α will be _____.

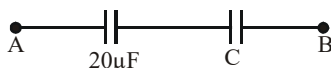
$[\epsilon_0$ is the permittivity of free space]

[JEE-Advance 2019]
CP0125

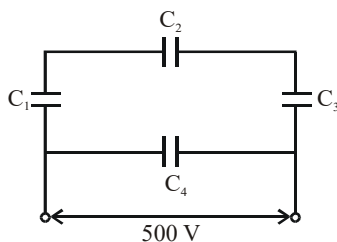
CAPACITANCE

(CBSE Previous Year's Questions)

1. A parallel capacitor is to be designed with a voltage rating 1 kV using a material of dielectric constant 3 and dielectric strength about 10^7 Vm^{-1} . For safety we would like the field 'never to exceed say, 10% of the dipole strength. What minimum area of the plates is required to have a capacitance of 50 PF?
[2; CBSE-2005]
2. A $4\mu\text{F}$ capacitor is charged by a 200 V supply. The supply is then disconnected and the charged capacitor is connected to another uncharged $2\mu\text{F}$ capacitor. How much electrostatic energy of the first capacitor is lost in the process of attaining the steady situation?
[2; CBSE-2005]
3. Two capacitors of capacitance of $6\mu\text{F}$ and $12\mu\text{F}$ are connected in series with a battery. The voltage across the $6\mu\text{F}$ capacitor is 2 V. Compute the total battery voltage.
[2; CBSE-2006]
4. A parallel plate capacitor with air between the plates has a capacitance of 8 pF. The separation between the plates is now reduced by half and the space between them is filled with a medium of dielectric constant 5. Calculate the value of capacitance of the capacitor in the second case.
[2; CBSE-2006]
5. The equivalent capacitance of the combination between A and B in the given figure is $4\mu\text{F}$.

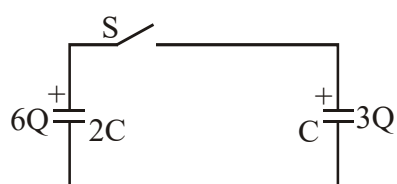


- (i) Calculate capacitance of the capacitor C.
 - (ii) Calculate charge on each capacitor if a 12 V battery is connected across terminals A and B.
 - (iii) What will be the potential drop across each capacitor?
[3; CBSE-2009]
6. A network of four capacitors each of $12\mu\text{F}$ capacitance is connected to a 500 V supply as shown in the figure. Determine (a) equivalent capacitance of the network and (b) charge on each capacitor.
[3; CBSE-2010]

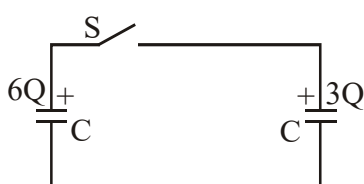


7. A parallel plate capacitor is being charged by a time varying current. Explain briefly how Ampere's circuital law is generalized to incorporate the effect due to the displacement current.
[2; CBSE-2011]
8. Net capacitance of three identical capacitor in series is $1\mu\text{F}$. What will be their net capacitance if connected in parallel? Find the ratio of energy stored in the two configurations if they are both connected to the same source.
[2; CBSE-2011]
9. A capacitor of capacitance 'C' is being charged by connecting it across a dc source along with an ammeter. Will the ammeter show a momentary deflection during the process of charging? If so, how would you explain this momentary deflection and the resulting continuity of current in the circuit? Write the expression for the current inside the capacitor.
[2; CBSE-2012]

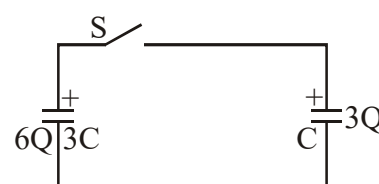
10. Deduce the expression for the electrostatic energy stored in a capacitor of capacitance 'C' and having charge 'Q'.
How will the (i) energy stored and (ii) the electric field inside the capacitor be affected when it is completely filled with a dielectric material of dielectric constant K? [3; CBSE-2012]
11. A slab of material of dielectric constant K has the same area as that of the plates of a parallel plate capacitor but has the thickness $d/2$, where d is the separation between the plates. Find out the expression for its capacitance when the slab is inserted between the plates of the capacitor. [CBSE-2013]
12. A parallel plate capacitor of capacitance C is charged to a potential V. It is then connected to another uncharged capacitor having the same capacitance. Find out the ratio of the energy stored in the combined system to that stored initially in the single capacitor. [CBSE-2014]
13. Define dielectric constant of a medium. What is its S.I. unit? [1; CBSE-2015]
14. Three circuits, each consisting of a switch 'S' and two capacitors, are initially charged, as shown in the figure. After the switch has been closed, in which circuit will the charge on the left-hand capacitor (i) increase, (ii) decrease and (iii) remain same? Give reasons. [3; CBSE-2015]



(a)

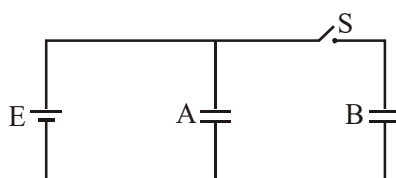


(b)



(c)

15. (a) Distinguish, with the help of a suitable diagram, the difference in the behaviour of a conductor and a dielectric placed in an external electric field. How does polarised dielectric modify the original external field? [5; CBSE-2016]
- (b) A capacitor of capacitance C is charged fully by connecting it to a battery of emf E. It is then disconnected from the battery. If the separation between the plates of the capacitor is now doubled, how will the following change?
- charge stored by the capacitor.
 - field strength between the plates.
 - energy stored by the capacitor.
- Justify your answer in each case.
16. Two identical parallel plate capacitors A and B are connected to a battery of V volts with the switch S closed. The switch is now opened and the free space between the plates of the capacitors is filled with a dielectric of dielectric constant K. Find the ratio of the total electrostatic energy stored in both capacitors before and after the introduction of the dielectric. [3; CBSE-2017]



ANSWER KEY

EXERCISE (S-1)

1. Ans. 68 2. Ans. 5 3. Ans. 7 4. Ans. 9
5. Ans. 6V, 34V 6. Ans. $\frac{32}{23} \mu\text{F}$ 7. Ans. (C) 8. Ans. 30 V
9. Ans. 0 10. Ans. $12\mu\text{C}$ 11. Ans. $3Q/2C$ 12. Ans. $\frac{25}{24} \frac{\epsilon_0 A}{d}$
13. Ans. 5 14. Ans. 2.5 kV 15. Ans. 4.425×10^{-9} Ampere
16. Ans. 225 17. Ans. 750 18. Ans. $Q_0 = \frac{CVR_2}{R_1 + R_2}$ and $\alpha = \frac{R_1 + R_2}{CR_1R_2}$
19. Ans. $\frac{3}{4} \text{ J}$ 20. Ans. $\rho = \tau/\epsilon_0 \epsilon \ln 2 = 1.4 \times 10^{13} \Omega\cdot\text{m}$.

EXERCISE (S-2)

1. Ans. (a) $\frac{100}{7}$ volts; (b) 28.56 μC , 42.84 μC , 71.4 μC , 22.88 μC
2. Ans. (a) $\frac{7}{50} \text{ A}$ or $\frac{11}{50} \text{ A}$, (b) $Q_1 = 9\mu\text{C}$, $Q_2 = 0$
3. Ans. (i) $1.5 \times 10^4 \text{ V/m}$, $4.5 \times 10^4 \text{ V/m}$, (ii) 75 V, 225 V, (iii) $8 \times 10^{-7} \text{ C/m}^2$
4. Ans. (i) $0.2 \times 10^{-8} \text{ F}$, $1.2 \times 10^{-5} \text{ J}$; (ii) $4.84 \times 10^{-5} \text{ J}$; (iii) $1.1 \times 10^{-5} \text{ J}$
5. Ans. $q = \frac{CV}{2} \left(1 - \frac{1}{2} e^{-t/RC} \right)$ 6. Ans. 1 7. Ans. 69 mC
8. Ans. 5

EXERCISE (O-1)

1. Ans. (A) 2. Ans. (C) 3. Ans. (D) 4. Ans. (D) 5. Ans. (A)
6. Ans. (D) 7. Ans. (B) 8. Ans. (A) 9. Ans. (B) 10. Ans. (C)
11. Ans. (A) 12. Ans. (B) 13. Ans. (D) 14. Ans. (A) 15. Ans. (B)
16. Ans. (B) 17. Ans. (C) 18. Ans. (A) 19. Ans. (D) 20. Ans. (D)
21. Ans. (D) 22. Ans. (C) 23. Ans. (C) 24. Ans. (B) 25. Ans. (C)
26. Ans. (B) 27. Ans. (C) 28. Ans. (C) 29. Ans. (D) 30. Ans. (D)
31. Ans. (D) 32. Ans. (A,D) 33. Ans. (A) 34. Ans. (C) 35. Ans. (B)
36. Ans. (B) 37. Ans. (B) 38. Ans. (B) 39. Ans. (C) 40. Ans. (B)
41. Ans. (D) 42. Ans. (B) 43. Ans. (B, C, D) 44. Ans. (A,C,D) 45. Ans. (A, D)
46. Ans. (C, D) 47. Ans. (B, C) 48. Ans. (A, C, D) 49. Ans. (A, B, D) 50. Ans. (A, B)
51. Ans. (A, C) 52. Ans. (A, C) 53. Ans. (D)
54. Ans. (A) $\rightarrow$ (PQR); (B) $\rightarrow$ (QST); (C) $\rightarrow$ (ST); (D) $\rightarrow$ (PR)

EXERCISE (O-2)

- | | | | | |
|--------------------|--------------------|--------------|--------------|--------------|
| 1. Ans. (B) | 2. Ans. (B) | 3. Ans. (B) | 4. Ans. (B) | 5. Ans. (B) |
| 6. Ans. (A) | 7. Ans. (D) | 8. Ans. (C) | 9. Ans. (C) | 10. Ans. (A) |
| 11. Ans. (D) | 12. Ans. (D) | 13. Ans. (D) | 14. Ans. (C) | |
| 15. Ans. (A,B,C,D) | 16. Ans. (A,B,C,D) | 17. Ans. (C) | | |

EXERCISE (J-M)

- | | | | | |
|--------------|--------------|--------------|--------------|--------------|
| 1. Ans. (3) | 2. Ans. (4) | 3. Ans. (2) | 4. Ans. (4) | 5. Ans. (1) |
| 6. Ans. (2) | 7. Ans. (3) | 8. Ans. (4) | 9. Ans. (4) | 10. Ans. (3) |
| 11. Ans. (4) | 12. Ans. (1) | 13. Ans. (2) | 14. Ans. (4) | |

EXERCISE (J-A)

- | | | | | |
|----------------|-------------------|-------------|---------------|---------------|
| 1. Ans. 2 | 2. Ans. (D) | 3. Ans. (C) | 4. Ans. (B,D) | 5. Ans. (A,D) |
| 6. Ans. (D) | 7. Ans. (A,B,C,D) | 8. Ans. (A) | 9. Ans. (A) | 10. Ans. 1.50 |
| 11. Ans. (3,4) | 12. Ans. (1.00) | | | |

MAGNETIC EFFECT OF CURRENT

KEY CONCEPT

The branch of physics which deals with the magnetism due to electric current or moving charge (i.e. electric current is equivalent to the charges or electrons in motion) is called electromagnetism.

Biot-Savart Law

Currents which arise due to the motion of charges are the source of magnetic fields. When charges move in a conducting wire and produce a current I , the magnetic field at any point P due to the current can be calculated by adding up the magnetic field contributions, $d\vec{B}$, from small segments of the wire $d\vec{s}$ (Figure).

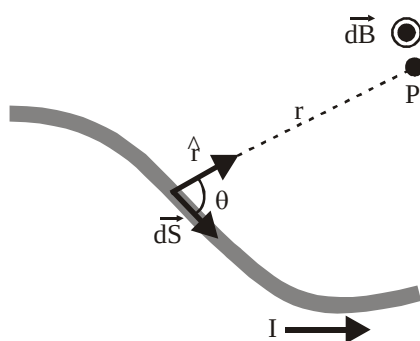


Figure : Magnetic field $d\vec{B}$ at point P due to a current-carrying element $Id\vec{s}$

These segments can be thought of as a vector quantity having a magnitude of the length of the segment and pointing in the direction of the current flow. The infinitesimal current source can then be written as $Id\vec{s}$.

Let r denote as the distance from the current source to the field point P and $\hat{r}$ the corresponding unit vector. The Biot-Savart law gives an expression for the magnetic field contribution, $d\vec{B}$, from the current source, $Id\vec{s}$,

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{Id\vec{s} \times \hat{r}}{r^2}$$

where μ_0 is a constant called the permeability of free space:

$$\mu_0 = 4\pi \times 10^{-7} \text{ T.m/A here Tesla (T) is SI unit of } \vec{B}$$

Adding up these contributions to find the magnetic field at the point P requires integrating over the current source,

$$\vec{B} = \int d\vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{s} \times \hat{r}}{r^2}$$

- According to $d\vec{B} = \frac{\mu_0}{4\pi} \frac{Id\vec{\ell} \times \vec{r}}{r^3}$, direction of magnetic field vector $d\vec{B}$ is always perpendicular to the plane of vectors $(Id\vec{\ell})$ and $(\vec{r})$, where plane of $(Id\vec{\ell})$ and $(\vec{r})$ is the plane of wire.
- Magnetic field on the axis of current carrying conductor is always zero ($\theta = 0^\circ$ or $\theta = 180^\circ$)

RIGHT HAND THUMB RULE

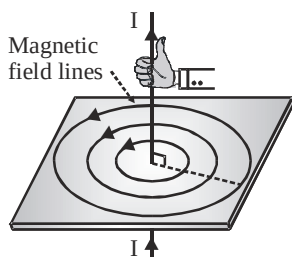
This rule gives the pattern of magnetic field lines due to current carrying wire.

(i) Straight current

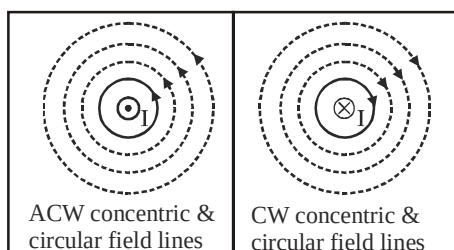
Thumb $\rightarrow$ In the direction of current

Curling fingers $\rightarrow$ Gives field line pattern

Case I : wire in the plane of the paper



Case II : Wire is $\perp$ to the plane of the paper.

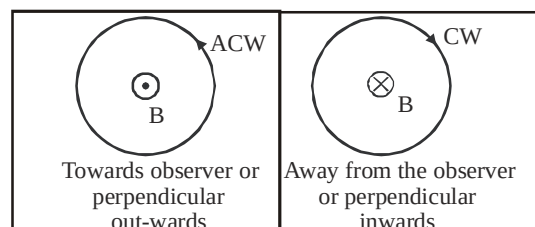


(ii) Circular current

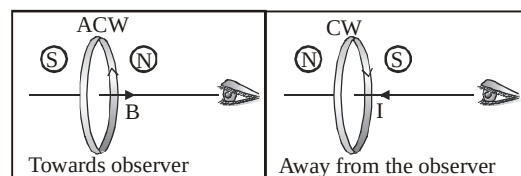
Curling fingers $\rightarrow$ In the direction of current,

Thumb $\rightarrow$ Gives field line pattern

Case I : wire in the plane of the paper



Case II : Wire is $\perp$ to the plane of the paper

**APPLICATION OF BIOT-SAVART LAW :**

- Magnetic field surrounding a thin straight current carrying conductor**

AB is a straight conductor carrying current i from B to A.

At a point P, whose perpendicular distance from AB is $OP = a$, the direction of field is perpendicular to the plane of paper, inwards (represented by a cross)

$\ell = a \tan \theta \Rightarrow d\ell = a \sec^2 \theta d\theta \dots (i)$

$\alpha = 90^\circ - \theta$ & $r = a \sec \theta$

- By Biot-Savart's law**

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{id\ell \sin \alpha}{r^2} \otimes \text{ (due to a current element } id\ell \text{ at point P)}$$

$$\Rightarrow B = \int dB = \int \frac{\mu_0}{4\pi} \frac{id\ell \sin \alpha}{r^2} \text{ (due to wire AB)} \therefore B = \frac{\mu_0 i}{4\pi} \int \cos \theta d\theta$$

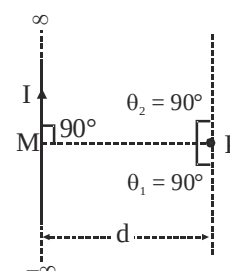
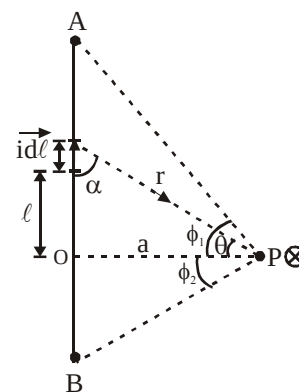
Taking limits of integration as $-\phi_2$ to ϕ_1

$$B = \frac{\mu_0 i}{4\pi a} \int_{-\phi_2}^{\phi_1} \cos \theta d\theta = \frac{\mu_0 i}{4\pi a} [\sin \theta]_{-\phi_2}^{\phi_1} = \frac{\mu_0 i}{4\pi a} [\sin \phi_1 + \sin \phi_2] \text{ (inwards)}$$

Ex. Magnetic field due to infinite length wire at point 'P'

Sol. $B_p = \frac{\mu_0 I}{4\pi d} [\sin 90^\circ + \sin 90^\circ]$

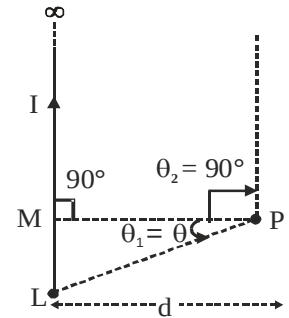
$$B_p = \frac{\mu_0 I}{2\pi d}$$



Ex. Magnetic field due to semi infinite length wire at point 'P'

Sol. $B_p = \frac{\mu_0 I}{4\pi d} [\sin\theta + \sin 90^\circ]$

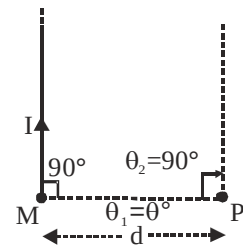
$$B_p = \frac{\mu_0 I}{4\pi d} [\sin\theta + 1]$$



Ex. Magnetic field due to special semi infinite length wire at point 'P'

Sol. $B_p = \frac{\mu_0 I}{4\pi d} [\sin 0^\circ + \sin 90^\circ]$

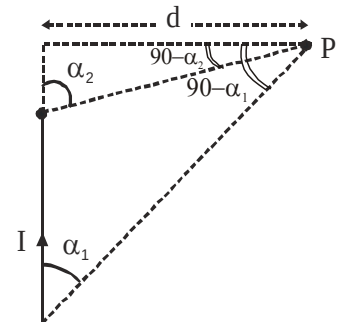
$$B_p = \frac{\mu_0 I}{4\pi d}$$



Ex. If point 'P' lies out side the line of wire then magnetic field at point 'P' :

Sol. $B_p = \frac{\mu_0 I}{4\pi d} [\sin(90 - \alpha_1) - \sin(90 - \alpha_2)]$

$$= \frac{\mu_0 I}{4\pi d} (\cos \alpha_1 - \cos \alpha_2)$$



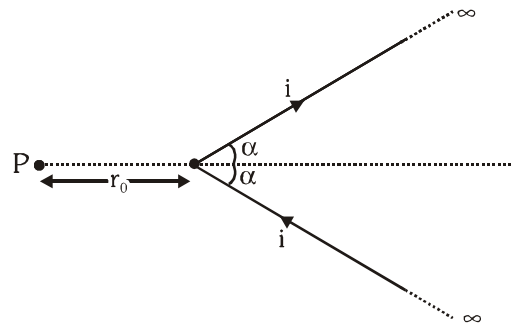
Ex. A current carrying wire in the form of 'V' alphabet is kept as shown in the figure. Magnetic field intensity at point P which lies on the angular bisector of V is

(A) $\frac{\mu_0 i}{4\pi r_0} [1 - \cos \alpha]$

(B) $\frac{\mu_0 i}{2\pi r_0} [1 - \cos \alpha]$

(C) $\frac{\mu_0 i}{4\pi r_0} \frac{[1 - \cos \alpha]}{\sin \alpha}$

(D) $\frac{\mu_0 i}{2\pi r_0} \frac{[1 - \cos \alpha]}{\sin \alpha}$



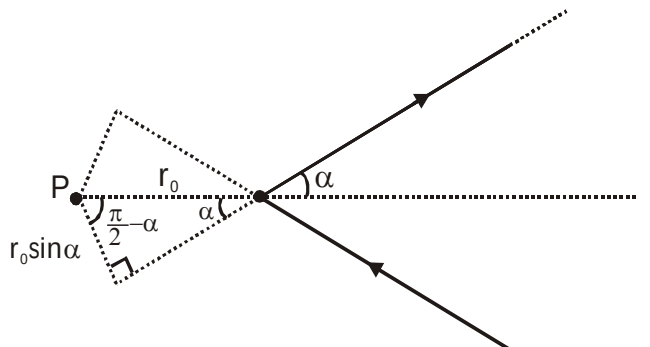
Ans. (D)

Sol. By using formula for stragith wire i.e.

$$B = \frac{\mu_0 I}{4\pi d} (\sin \theta_1 + \sin \theta_2)$$

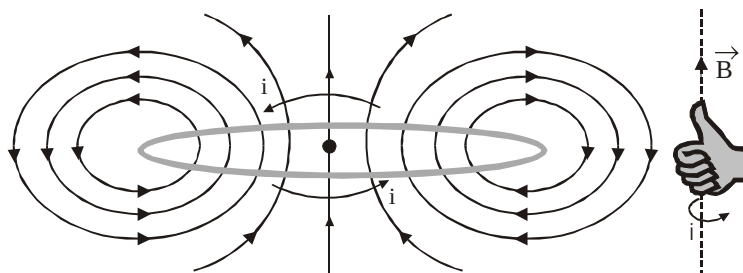
$$B = 2 \left[\frac{\mu_0 i}{4\pi r_0 \sin \alpha} (\sin 90^\circ - \sin(90^\circ - \alpha)) \right]$$

$$= \frac{\mu_0 i}{2\pi r_0 \sin \alpha} [1 - \cos \alpha]$$



- Magnetic field due to a loop of current**

Magnetic field lines due to a loop of wire are shown in the figure

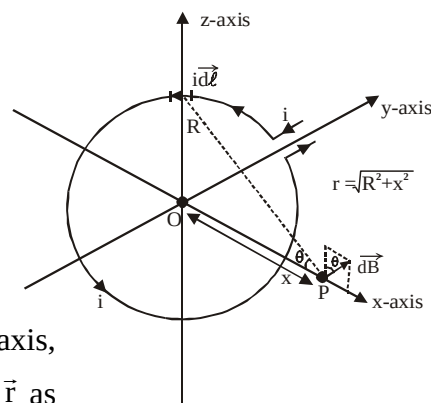


The direction of magnetic field on the axis of current loop can be determined by right hand thumb rule. If fingers of right hand are curled in the direction of current, the stretched thumb is in the direction of magnetic field.

- Calculation of magnetic field**

Consider a current loop placed in y-z plane carrying current i in anticlockwise sense as seen from positive x-axis. Due to a small current element $id\vec{\ell}$ shown in the figure, the magnetic field at P

is given by
$$dB = \frac{\mu_0}{4\pi} \frac{id\ell \sin 90^\circ}{r^2}.$$



The angle between $id\vec{\ell}$ and $\vec{r}$ is 90° because $id\vec{\ell}$ is along y-axis, while $\vec{r}$ lies in x-z plane. The direction of $\vec{dB}$ is perpendicular to $\vec{r}$ as shown. The vector $\vec{dB}$ can be resolved into two components, $dB \cos \theta$ along z-axis and $dB \sin \theta$ along x-axis.

For any two diametrically opposite current elements, the components along x-axis add up, while the other two components cancel out. Therefore, the field at P is due to x-component of field only. Hence, we have

$$\therefore B = \frac{\mu_0}{4\pi} \frac{i \times 2\pi R^2}{(R^2 + x^2)^{3/2}} \quad \left(\because r = \sqrt{R^2 + x^2} \right)$$

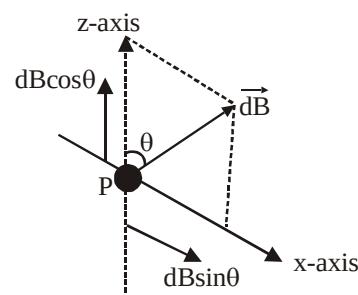
(a) At the centre, $x = 0$, $B_{\text{centre}} = \frac{\mu_0 i}{2R}$

(b) At points very close to centre, $x \ll R \Rightarrow B = \frac{\mu_0 i}{2} \left(1 + \frac{x^2}{R^2} \right)^{-3/2} = \frac{\mu_0 i}{2} \left(1 - \frac{3x^2}{2R^2} \right)$

(c) At points far off from the centre, $x \gg R \Rightarrow B = \frac{\mu_0 \ell}{4\pi} \frac{2\pi R^2}{x^3}$

(d) The result in point (c) is also expressed as $B = \frac{\mu_0}{4\pi} \frac{2M}{x^3}$

where $M = \ell \times \pi R^2$, is called magnetic dipole moment.



Ex. Find the magnetic field at the centre of a current carrying conductor bent in the form of an arc subtending angle θ at its centre. Radius of the arc is R .

Sol. Let the arc lie in x-y plane with its centre at the origin.

Consider a small current element $id\vec{\ell}$ as shown.

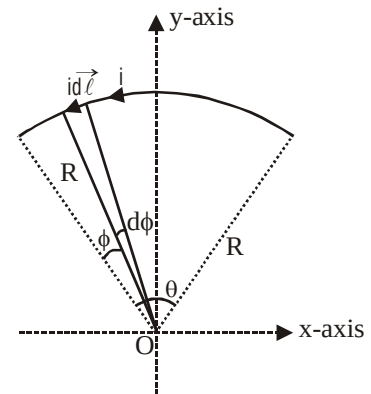
The field due to this element at the centre is

$$dB = \frac{\mu_0}{4\pi} \frac{id\ell \sin 90^\circ}{R^2} \quad (\because id\vec{\ell} \text{ and } R \text{ are perpendicular})$$

$$\text{Now } d\ell = R d\phi \therefore dB = \frac{\mu_0}{4\pi} \frac{iR d\phi}{R^2} \Rightarrow dB = \frac{\mu_0}{4\pi} \frac{i}{R} d\phi$$

The direction of field is outward perpendicular to plane of paper

$$\text{Total magnetic field } B = \int dB \therefore B = \frac{\mu_0 i}{4\pi R} \int_0^\theta d\phi = \frac{\mu_0 i}{4\pi R} [\phi]_0^\theta \therefore B = \frac{\mu_0 i}{4\pi R} \theta$$



Ex. Find the magnetic field at the centre of a current carrying conductor bent in the form of an arc subtending angle α_1 and α_2 at the centre.

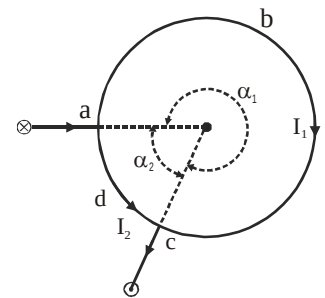
Sol. Magnetic field at the centre of arc abc and adc wire of circuit loop

$$B_{abc} = \frac{\mu_0 I_1 \alpha_1}{4\pi r} \text{ and } B_{adc} = \frac{\mu_0 I_2 \alpha_2}{4\pi r} \Rightarrow \frac{B_{abc}}{B_{adc}} = \frac{I_1 \alpha_1}{I_2 \alpha_2}$$

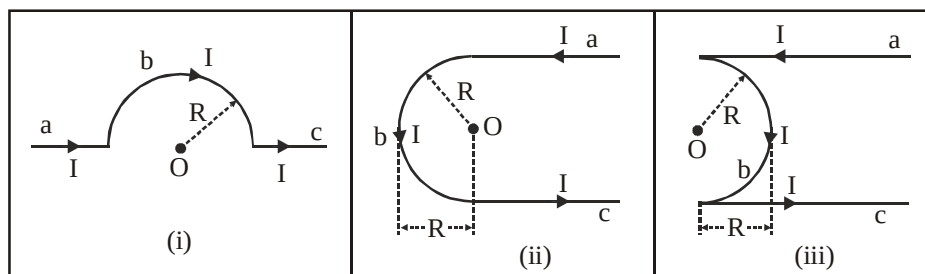
$$\therefore \text{angle} = \frac{\text{arc length}}{\text{radius}} \Rightarrow \frac{\alpha_1}{\alpha_2} = \frac{\ell_1}{\ell_2}$$

$$\therefore V = I_1 R_1 = I_2 R_2 \Rightarrow \frac{I_1}{I_2} = \frac{R_2}{R_1} \Rightarrow \frac{I_1}{I_2} = \frac{\ell_2}{\ell_1} (\because R = \frac{\rho \ell}{A} \Rightarrow R \propto \ell)$$

$$\therefore \frac{B_{abc}}{B_{adc}} = \left(\frac{\ell_2}{\ell_1} \right) \left(\frac{\ell_1}{\ell_2} \right) \Rightarrow \frac{B_{\alpha_1}}{B_{\alpha_2}} = \frac{1}{1}$$



Ex. Calculate the field at the centre of a semi-circular wire of radius R in situations depicted in figure (i), (ii) and (iii) if the straight wire is of infinite length.



Sol. The magnetic field due to a straight current carrying wire of infinite length, for a point at a distance R

from one of its ends is zero if the point is along its length and $\frac{\mu_0 I}{4\pi R}$ if the point is on a line perpendicular

to its length while at the centre of a semicircular coil is $\frac{\mu_0 I}{4R}$ so net magnetic field at the centre of

semicircular wire is $\vec{B}_R = \vec{B}_a + \vec{B}_b + \vec{B}_c$

$$(i) \quad \vec{B}_R = 0 + \frac{\mu_0}{4} \frac{I}{R} \otimes + 0 = \frac{\mu_0 I}{4R} \otimes (\text{into the page})$$

$$(ii) \quad \vec{B}_R = \frac{\mu_0}{4\pi} \frac{I}{R} \odot + \frac{\mu_0}{4} \frac{I}{R} \odot + \frac{\mu_0}{4\pi} \frac{I}{R} \odot = \frac{\mu_0}{4\pi} \frac{I}{R} [\pi + 2] \odot (\text{out of the page})$$

$$(iii) \quad \vec{B}_R = \frac{\mu_0}{4\pi} \frac{I}{R} \odot + \frac{\mu_0}{4} \frac{I}{R} \otimes + \frac{\mu_0}{4\pi} \frac{I}{R} \odot = \frac{\mu_0}{4\pi} \frac{I}{R} [\pi - 2] \otimes (\text{into the page})$$

Ex. A long wire bent as shown in the figure carries current I . If the radius of the semi-circular portion is " a " then find the magnetic induction at the centre C .

Sol. Due to semi circular part

$$\vec{B}_1 = \frac{\mu_0 I}{4a} (-\hat{i})$$

due to parallel parts of currents

$$\vec{B}_2 = 2 \times \frac{\mu_0 I}{4\pi a} (-\hat{k}), \quad B_{\text{net}} = B_C = \vec{B}_1 + \vec{B}_2 = \frac{\mu_0 I}{4a} (-\hat{i}) + \frac{\mu_0 I}{2\pi a} (-\hat{k})$$

$$\text{magnitude of resultant field } B = \sqrt{B_1^2 + B_2^2} = \frac{\mu_0 I}{4\pi a} \sqrt{\pi^2 + 4}$$

Ex. A thin insulated wire forms a plane spiral of $N = 100$ tight turns carrying a current $I = 8$ mA. The radii of inside and outside turns (Fig.) are equal to $a = 50$ mm and $b = 100$ mm. Find the magnetic induction at the centre of the spiral;

$$\text{Ans. } B = \frac{\mu_0 IN \ln(b/a)}{2(b-a)} = 7 \mu\text{T};$$

Sol. From Biot-Savart's law, the magnetic induction due to a circular current carrying wire loop at its centre is given by,

$$B_r = \frac{\mu_0}{2r} i$$

The plane spiral is made up of concentric circular loops, having different radii, varying from a to b . Therefore, the total magnetic induction at the centre,

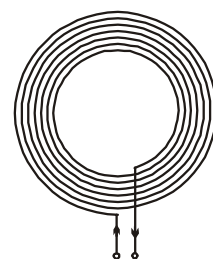
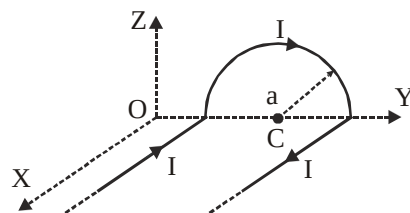
$$B_0 = \int \frac{\mu_0}{2r} dN$$

where $\frac{\mu_0}{2r} i$ is the contribution of one turn of radius r and dN is the number of turns in the interval $(r, r + dr)$.

$$\text{i.e. } dN = \frac{N}{b-a} dr$$

Substituting in equation (1) and integrating the result over r between a and b , we obtain,

$$B_0 = \int_a^b \frac{\mu_0 i}{2r} \frac{N}{(b-a)} dr = \frac{\mu_0 i N}{2(b-a)} \ln \frac{b}{a}$$



AMPERE'S CIRCUITAL LAW

Ampere's circuital law states that the line integral of the magnetic field around any closed path in free space or vacuum is equal to μ_0 times of net current or total current which crosses through the

area bounded by the closed path. Mathematically $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 \Sigma I$

This law is independent of size and shape of the closed path.

Any current outside the closed path is not included in writing the right hand side of law

Note :

- This law is suitable for infinite long and symmetrical current distribution.
- Radius of cross section of thick cylindrical conductor and current density must be given to apply this law.

APPLICATION OF AMPERE'S CIRCUITAL LAW

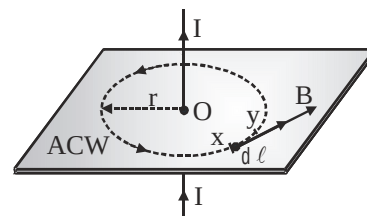
Magnetic field due to infinite long thin current carrying straight conductor

Consider a circle of radius 'r'. Let XY be the small element of length $d\ell$. $\vec{B}$ and $d\vec{\ell}$ are in the same direction because direction is along the tangent of the circle. By A.C.L.

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 \Sigma I, \oint B d\ell \cos \theta = \mu_0 I \quad (\text{where } \theta = 0^\circ)$$

$$\oint B d\ell \cos 0^\circ = \mu_0 I \Rightarrow B \oint d\ell = \mu_0 I \quad (\text{where } \oint d\ell = 2\pi r)$$

$$B (2\pi r) = \mu_0 I \Rightarrow B = \frac{\mu_0 I}{2\pi r}$$



Magnetic field due to infinite long solid cylindrical conductor

- For a point inside the cylinder $r < R$, Current from area πR^2 is = I

$$\text{so current from area } \pi r^2 \text{ is } = \frac{I}{\pi R^2} (\pi r^2) = \frac{I r^2}{R^2}$$

By Ampere circuital law for circular path 1 of radius r

$$B_{in} (2\pi r) = \mu_0 I' = \mu_0 \frac{I r^2}{R^2} \Rightarrow B_{in} = \frac{\mu_0 I r}{2\pi R^2} \Rightarrow B_{in} \propto r$$

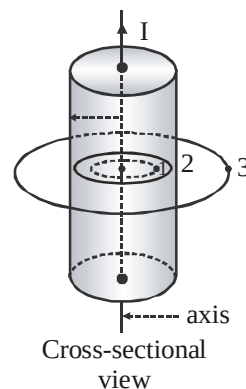
- For a point on the axis of the cylinder ($r = 0$); $B_{axis} = 0$
- For a point on the surface of cylinder ($r = R$)

By Ampere circuital law for circular path 2 of radius R

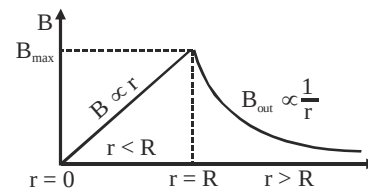
$$B_s (2\pi R) = \mu_0 I \Rightarrow B_s = \frac{\mu_0 I}{2\pi R} \quad (\text{it is maximum})$$

- For a point outside the cylinder ($r > R$) :-
By Ampere circuital law for circular path 3 of radius r

$$B_{out} (2\pi r) = \mu_0 I \Rightarrow B_{out} = \frac{\mu_0 I}{2\pi r} \Rightarrow B_{out} \propto \frac{1}{r}$$



Cross-sectional view



Magnetic field outside the cylindrical conductor does not depend upon nature (thick/thin or solid/hollow) of the conductor as well as its radius of cross section.

• **Magnetic field due to infinite long hollow cylindrical conductor**

• For a point at a distance r such that $r < a < b$ $B_1 = 0$

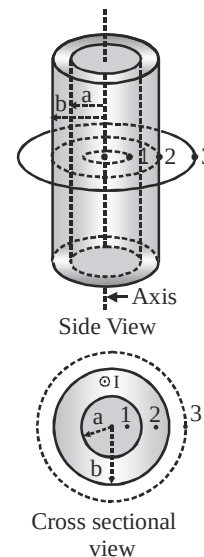
• For a point at a distance r such that $a < r < b$

$$B_2(2\pi r) = \mu_0 I' \Rightarrow B_2(2\pi r) = \mu_0 I \left(\frac{r^2 - a^2}{b^2 - a^2} \right)$$

$$B_2 = \frac{\mu_0 I}{2\pi r} \left(\frac{r^2 - a^2}{b^2 - a^2} \right) \begin{cases} r = a \text{ (inner surface)} \Rightarrow B_{is} = 0 \\ r = b \text{ (outer surface)} \Rightarrow B_{os} = \frac{\mu_0 I}{2\pi b} \text{ (maximum)} \end{cases}$$

• For a point at a distance r such that $r > b > a$, $B_3(2\pi r) = \mu_0 I \Rightarrow B_3 = \frac{\mu_0 I}{2\pi r}$

• For a point at the axis of cylinder $r = 0$ $B_{axis} = 0$

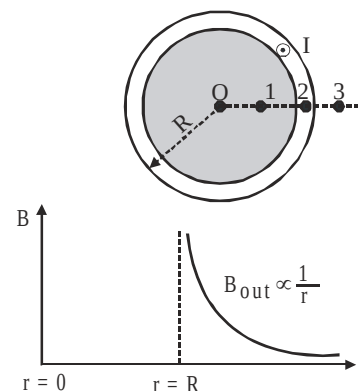


Magnetic field at specific positions for thin hollow cylindrical conductor

At point 1 $B_1 = 0$

At point 2 $B_2 = \frac{\mu_0 I}{2\pi R}$ (maximum) [outer surface] and
 $B_2 = 0$ (minimum) [inner surface]

At point 3 $B_3 = \frac{\mu_0 I}{2\pi r}$ (for the point on axis $B_{axis} = 0$)



Ex. Non-Uniform Current Density : Consider an infinitely long, cylindrical conductor of radius R carrying a current I with a non-uniform current density $J = \alpha r$ where α is a constant. Find the magnetic field everywhere.

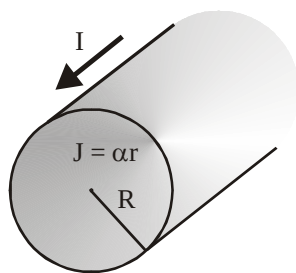


Figure : Non-uniform current density

Solution :

The problem can be solved by using the Ampere's law:

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_{enc}$$

Where the enclosed current I_{enc} is given by

$$I_{enc} = \int \vec{J} \cdot d\vec{A} = \int (\alpha r')(2\pi r' dr')$$

- (a) For $r < R$, the enclosed current is $I_{\text{enc}} = \int_0^r 2\pi\alpha r'^2 dr' = \frac{2\pi\alpha r^3}{3}$

Applying Ampere's law, the magnetic field at P_1 is given by

$$B_1(2\pi r) = \frac{2\mu_0\pi\alpha r^3}{3} \text{ or } B_1 = \frac{\alpha\mu_0}{3} r^2$$

The direction of the magnetic field $\vec{B}_1$ is tangential to the Amperian loop which encloses the current.

- (b) For $r > R$, the enclosed current is : $I_{\text{enc}} = \int_0^R 2\pi\alpha r'^2 dr' = \frac{2\pi\alpha R^3}{3}$

which yields $B_2(2\pi r) = \frac{2\mu_0\pi\alpha R^3}{3}$

Thus, the magnetic field at a point P_2 outside the conductor is ;

$$B_2 = \frac{\alpha\mu_0 R^3}{3r}$$

A plot of B as a function of r is shown in figure.

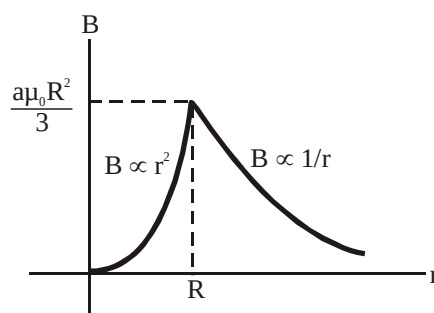
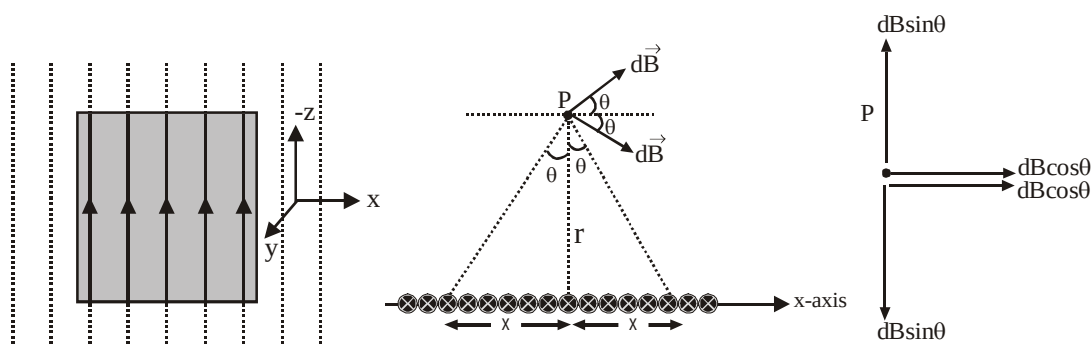


Figure : The magnetic field as a function of distance away from the conductor

Magnetic field due to an infinite plane sheet of current



An infinite sheet of current lies in x - z plane, carrying current along z -axis. The field at any point P on y is along a line parallel to x - z plane. We can take a rectangular amperian loop as shown. If you traverse the loop in clockwise direction, inward current will be positive.

By Ampere circuital law, $\oint_{PQRS} \vec{B} \cdot d\vec{\ell} = \mu_0 \ell_{\text{enclosed}} \dots (i)$

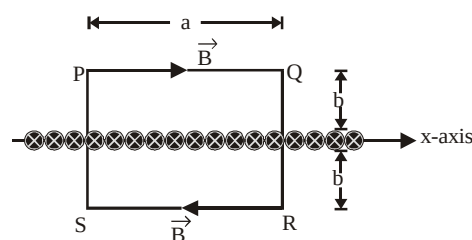
Let λ represents current per unit length.

The current enclosed is given by $\ell_{\text{enclosed}} = \lambda a$

$$\text{Now, } \oint_{PQRS} \vec{B} \cdot d\vec{\ell} = \int_{PQ} \vec{B} \cdot d\vec{\ell} + \int_{QR} \vec{B} \cdot d\vec{\ell} + \int_{RS} \vec{B} \cdot d\vec{\ell} + \int_{SP} \vec{B} \cdot d\vec{\ell}$$

$$\text{Now, } \int_{QR} \vec{B} \cdot d\vec{\ell} = \int_{SP} \vec{B} \cdot d\vec{\ell} = 0 \text{ as } \vec{B} \perp d\vec{\ell}$$

$$\text{Also, } \int_{PQ} \vec{B} \cdot d\vec{\ell} + \int_{RS} \vec{B} \cdot d\vec{\ell} = 2 \times B \times a \left(\text{as } \vec{B} \parallel d\vec{\ell} \right) \therefore 2B \times a = \mu_0 \lambda a \Rightarrow B = \frac{\mu_0 \lambda}{2}$$



MAGNETIC FIELD DUE TO SOLENOID

It is a coil which has length and used to produce uniform magnetic field of long range. It consists a conducting wire which is tightly wound over a cylindrical frame in the form of helix. All the adjacent turns are electrically insulated to each other. The magnetic field at a point on the axis of a solenoid can be obtained by superposition of field due to large number of identical circular turns having their centres on the axis of solenoid.

Magnetic field due to a long solenoid

A solenoid is a tightly wound helical coil of wire. If length of solenoid is large, as compared to its radius, then in the central region of the solenoid, a reasonably uniform magnetic field is present. Figure shows a part of long solenoid with number of turns/length n . We can find the field by using Ampere circuital law.

Consider a rectangular loop ABCD. For this loop $\oint_{ABCD} \vec{B} \cdot d\vec{\ell} = \mu_0 i_{\text{enc}}$

Now

$$\oint_{ABCD} \vec{B} \cdot d\vec{\ell} = \oint_{AB} \vec{B} \cdot d\vec{\ell} + \oint_{BC} \vec{B} \cdot d\vec{\ell} + \oint_{CD} \vec{B} \cdot d\vec{\ell} + \oint_{DA} \vec{B} \cdot d\vec{\ell} = B \times a$$

This is because $\oint_{AB} \vec{B} \cdot d\vec{\ell} = \oint_{CD} \vec{B} \cdot d\vec{\ell} = 0$, $\vec{B} \perp d\vec{\ell}$.

$$\text{And, } \oint_{DA} \vec{B} \cdot d\vec{\ell} = 0$$

($\therefore \vec{B}$ outside the solenoid is negligible)

$$\text{Now, } i_{\text{enc}} = (n \times a) \times i \Rightarrow B \times a = \mu_0 (n \times a \times i)$$

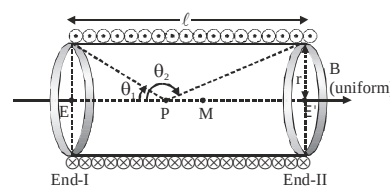
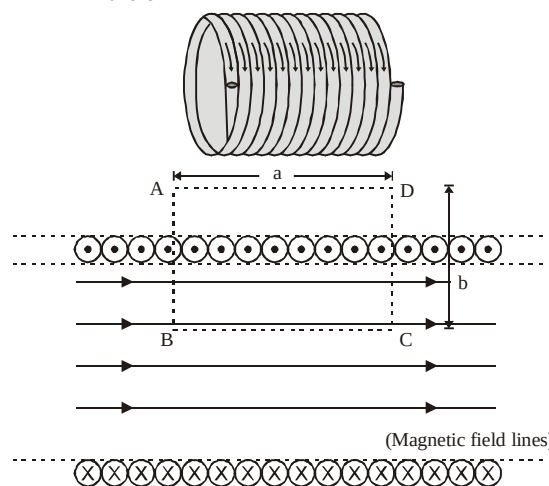
$$\Rightarrow B = \mu_0 n i$$

Finite length solenoid :

Its length and diameter are comparable. By the concept of BSL magnetic field at the axial point 'P' obtained as :

$$B_p = \frac{\mu_0 n I}{2} (\cos \theta_1 - \cos \theta_2)$$

Angle θ_1 and θ_2 both measured in same sense from the axis of the solenoid to end vectors.



Infinite length solenoid :

Its length very large as compared to its diameter i.e. ends of solenoid tends to infinity.

(a) Magnetic field at axial point which is well inside the solenoid

$$\theta_1 \approx 0^\circ \text{ and } \theta_2 \approx 180^\circ \Rightarrow B \approx \frac{\mu_0 n I}{2} [\cos 0^\circ - \cos 180^\circ] \approx \frac{\mu_0 n I}{2} [(1) - (-1)] \approx \mu_0 n I$$

(b) Magnetic field at both axial end points of solenoid

$$\theta_1 = 90^\circ \text{ and } \theta_2 \approx 180^\circ \Rightarrow B \approx \frac{\mu_0 n I}{2} [\cos 90^\circ - \cos 180^\circ] \approx \frac{\mu_0 n I}{2} [(0) - (-1)] \approx \frac{\mu_0 n I}{2}$$

Ex. The length of solenoid is 0.1m. and its diameter is very small. A wire is wound over it in two layers. The number of turns in inner layer is 50 and that of outer layer is 40. The strength of current flowing in two layers in opposite direction is 3A. Then find magnetic induction at the middle of the solenoid.

Sol. Direction of magnetic field due to both layers is opposite, as direction of current is opposite so

$$\begin{aligned} B_{\text{net}} &= B_1 - B_2 = \mu_0 n_1 I_1 - \mu_0 n_2 I_2 = \mu_0 \frac{N_1}{\ell} I - \mu_0 \frac{N_2}{\ell} I \quad (\because I_1 = I_2 = I) \\ &= \frac{\mu_0 I}{\ell} (N_1 - N_2) = \frac{4\pi \times 10^{-7} \times 3}{0.1} (50 - 40) = 12\pi \times 10^{-5} \text{ T} \end{aligned}$$

Ex. Find out magnetic field at axial point 'P' of solenoid shown in figure (where turn density 'n' and current through it is I)

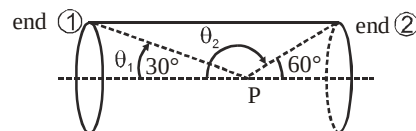
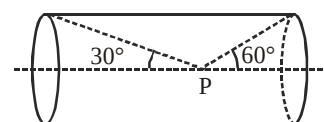
Sol. Magnetic field at point 'P' due to finite length solenoid

$$B_P = \frac{\mu_0 n I}{2} [\cos \theta_1 - \cos \theta_2],$$

where $\theta_1 = 30^\circ$ (CW),

$$\theta_2 = (180^\circ - 60^\circ) = 120^\circ \text{ (CW)} = \frac{\mu_0 n I}{2} [\cos 30^\circ - \cos 120^\circ]$$

$$= \frac{\mu_0 n I}{2} \left[\frac{\sqrt{3}}{2} - \left(-\frac{1}{2} \right) \right] = \frac{\mu_0 n I}{4} (\sqrt{3} + 1)$$

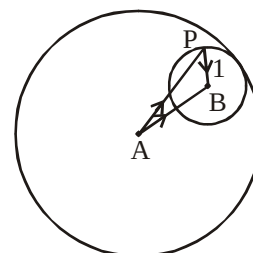


Ex. Inside a long straight uniform wire of round cross-section there is a long round cylindrical cavity whose axis is parallel to the axis of the wire and displaced from the latter by a distance I. A direct current of density j flows along the wire. Find the magnetic induction inside the cavity. Consider, in particular, the case $I = 0$.

Sol. We can think of the given current which will be assumed uniform, as arising due to a negative current, flowing in the cavity, superimposed on the true current, everywhere including the cavity. Then from the previous problem, by superposition

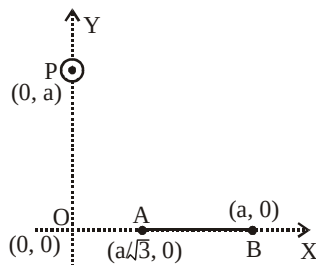
$$\vec{B} = \frac{1}{2} \mu_0 \vec{j} \times (\vec{AP} - \vec{BP}) = \frac{1}{2} \mu_0 \vec{j} \times \vec{\ell}$$

If $\vec{\ell}$ vanishes so that the cavity is concentric with the conductor, there is no magnetic field in the cavity.



Ex. An infinite current carrying conductor, parallel to z-axis is situated at point P as shown in the figure. Value

of $\int_A \vec{B} \cdot d\vec{\ell}$ is given by $\alpha \frac{\mu_0 i}{96}$, then fill the value of α in OMR sheet ?

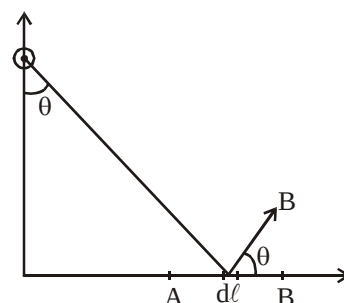


Ans. 4

Sol. $d\ell = d(a \tan \theta) = a \sec^2 \theta d\theta$.

$$B = \frac{\mu_0 i}{2\pi a} \cos \theta$$

$$\therefore \int \vec{B} \cdot d\vec{\ell} = \int_{\pi/6}^{\pi/4} \frac{\mu_0 i}{2\pi a} \cos \theta \cdot a \sec^2 \theta d\theta \cdot \cos \theta = \frac{\mu_0 i}{2\pi} \int_{\pi/6}^{\pi/4} d\theta = \frac{\mu_0 i}{2\pi} \cdot \frac{\pi}{12} = \frac{\mu_0 i}{24}$$



MOTION OF A CHARGED PARTICLE IN A MAGNETIC FIELD

Motion of a charged particle when it is moving collinear with the field magnetic field is not affected by the field (i.e. if motion is just along or opposite to magnetic field) ($\because F = 0$). The following two cases are possible :

• Case I :

When the charged particle is moving perpendicular to the field.

The angle between $\vec{B}$ and $\vec{v}$ is $\theta = 90^\circ$. So the force will be maximum ($= qvB$) and always perpendicular to motion (and also field); Hence the charged particle will move along a circular path (with its plane

perpendicular to the field). Centripetal force is provided by the force qvB , So $\frac{mv^2}{r} = qvB \Rightarrow r = \frac{mv}{qB}$

Angular frequency of circular motion, called cyclotron or gyro-frequency. $\omega = \frac{v}{r} = \frac{qB}{m}$

and the time period, $T = \frac{2\pi}{\omega} = 2\pi \frac{m}{qB}$ i.e., time period (or frequency) is independent of speed of particle and radius of the orbit. Time period depends only on the field B and the nature of the particle, i.e., specific charge (q/m) of the particle.

This principle has been used in a large number of devices such as cyclotron (a particle accelerator), bubble-chamber (a particle detector) or mass-spectrometer etc.

Note that $\vec{F}_B$ is always perpendicular to $\vec{v}$ and $\vec{B}$, and cannot change the particle's speed v (and thus the kinetic energy). In other words, magnetic force cannot speed up or slow down a charged particle.

Consequently, $\vec{F}_B$ can do no work on the particle :

$$dW = \vec{F}_B \cdot d\vec{s} = q(\vec{v} \times \vec{B}) \cdot \vec{v} dt = q(\vec{v} \times \vec{v}) \cdot \vec{B} dt = 0$$

Case II :

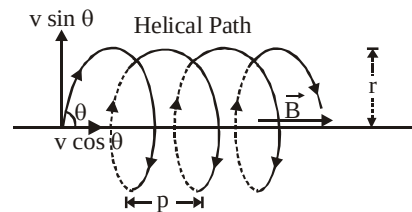
The charged particle is moving at an angle θ to the field :

($\theta \neq 0^\circ, 90^\circ$ or 180°)

Resolving the velocity of the particle along and perpendicular to the field. The particle moves with constant velocity $v \cos \theta$ along the field

($\because$ no force acts on a charged particle when it moves parallel to the field).

And at the same time it is also moving with velocity $v \sin \theta$ perpendicular to the field due to which it will describe a circle (in a plane perpendicular to the field)



Radius of the circular path $r = \frac{m(v \sin \theta)}{qB}$ and Time period $T = \frac{2\pi r}{v \sin \theta} = \frac{2\pi m}{qB}$

So the resultant path will be a helix with its axis parallel to the field $\vec{B}$ as shown in fig.

The pitch p of the helix = linear distance travelled in one rotation $p = T(v \cos \theta) = \frac{2\pi m}{qB}(v \cos \theta)$

Ex. An electron emitted by a heated cathode and accelerated through a potential difference of 2.0 kV enters a region with uniform magnetic field of 0.15 T. Determine the radius of the trajectory of the electron if the field is –

(a) Transverse to its initial velocity (b) Makes an angle of 30° with the initial velocity

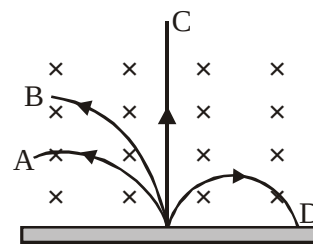
[Given : $m_e = 9 \times 10^{-31}$ kg]

Sol. $\frac{1}{2}mv^2 = eV \Rightarrow v = \sqrt{\frac{2eV}{m}} = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 2 \times 10^3}{9 \times 10^{-31}}} = \frac{8}{3} \times 10^7$ m/s

(a) Radius $r_1 = \frac{mv}{qB} = \frac{9 \times 10^{-31} \times (8/3) \times 10^7}{1.6 \times 10^{-19} \times 0.15} = 10^{-3}$ m = 1 mm

(b) Radius $r_2 = \frac{mv \sin \theta}{qB} = r_1 \sin \theta = 1 \times \sin 30^\circ = 1 \times \frac{1}{2} = 0.5$ mm

Ex. A neutron, a proton, an electron an α -particle enter a region of constant magnetic field with equal velocities. The magnetic field is along the inwards normal to the plane of the paper. The tracks of the particles are shown in fig. Relate the tracks to the particles.



Sol. Force on a charged particle in magnetic field $\vec{F} = q(\vec{v} \times \vec{B})$

For neutron $q = 0$, $F = 0$ hence it will pass undeflected.

i.e., tracks C corresponds to neutron.

If the particle is negatively charged, i.e. electron. $\vec{F} = -e(\vec{v} \times \vec{B})$

It will experience a force to the right; so track D corresponds to electron.

If the charge on particle is positive. It will experience a force to the left; so both tracks A and B corresponds to positively charged particles (i.e., protons and α -particles). When motion of charged particle perpendicular to the magnetic field the path is a circle with radius

$r = \frac{mv}{qB}$ i.e. $r \propto \frac{m}{q}$ and as $\left(\frac{m}{q}\right)_\alpha = \left(\frac{4m}{2e}\right)$ while $\left(\frac{m}{q}\right)_p = \frac{m}{e} \Rightarrow \left(\frac{m}{q}\right)_\alpha > \left(\frac{m}{q}\right)_p$

So $r_\alpha > r_p \Rightarrow$ track B to α -particle and A corresponds to proton.

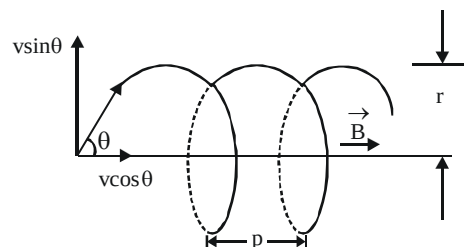
Ex. A beam of protons with velocity 4×10^5 m/s enters a uniform magnetic field of 0.3 tesla at an angle of 60° to the magnetic field. Find the radius of the helical path taken by the proton beam. Also find the pitch of helix. Mass of proton = 1.67×10^{-27} kg.

Sol. Radius of helix $r = \frac{mv \sin \theta}{qB}$ ($\because$ component of velocity $\perp$ to field is $v \sin \theta$)

$$= \frac{(1.67 \times 10^{-27})(4 \times 10^5) \sqrt{\frac{3}{2}}}{(1.6 \times 10^{-19})0.3} = \frac{2}{\sqrt{3}} \times 10^{-2} \text{ m} = 1.2 \text{ cm}$$

Again, pitch $p = v \cos \theta \times T$ (where $T = \frac{2\pi r}{v \sin \theta}$)

$$\therefore p = \frac{v \cos \theta \times 2\pi r}{v \sin \theta} = \frac{\cos 60^\circ \times 2\pi \times (1.2 \times 10^{-2})}{\sin 60^\circ} = 4.35 \times 10^{-2} \text{ m} = 4.35 \text{ cm}$$



Ex. The region between $x = 0$ and $x = L$ is filled with uniform, steady magnetic field $B_0 \hat{k}$. A particle of mass m , positive charge q and velocity $v_0 \hat{i}$ travels along X-axis and enters the region of magnetic field. Neglect the gravity throughout the question.

- Find the value of L if the particle emerges from the region of magnetic field with its final velocity at an angle 30° to its initial velocity.
- Find the final velocity of the particle and the time spent by it in the magnetic field, if the magnetic field now extends upto $2.1 L$.

Sol. (a) The particle is moving with velocity $v_0 \hat{i}$, perpendicular to magnetic field $B_0 \hat{k}$. Hence the particle will move along

a circular arc OA of radius $r = \frac{mv_0}{qB_0}$

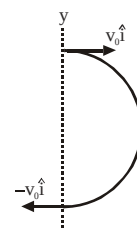
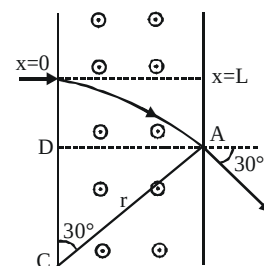
Let the particle leave the magnetic field at A.

$$\text{From } \triangle CDA, \sin 60^\circ = \frac{AD}{CA} = \frac{L}{r} \Rightarrow L = r \sin 30^\circ = \frac{r}{2} \therefore L = \frac{mv_0}{2qB_0}$$

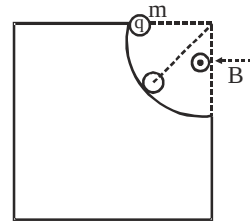
- As the magnetic field extends upto $2.1 L$ i.e., $L > 2r$, so the particle completes half cycle before leaving the magnetic field, as shown in figure.

The magnetic field is always perpendicular to velocity vector, therefore the magnitude of velocity will remain the same.

$$\therefore \text{Final velocity} = v_0(-\hat{i}) = -v_0 \hat{i} \quad \text{Time spent in magnetic field} = \frac{\pi r}{v_0} = \frac{\pi m}{qB_0}$$



Ex. A charged sphere of mass m and charge q starts sliding from rest on a vertical fixed circular track of radius R from the position as shown in figure. There exists a uniform and constant horizontal magnetic field of induction B . Find the maximum force exerted by the track on the sphere.

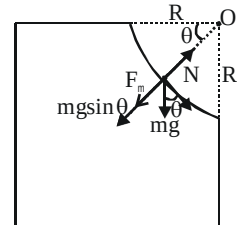


Sol. Magnetic force on sphere $F_m = qvB$ (directed radially outward)

$$\therefore N - mg \sin \theta - qvB = \frac{mv^2}{R}$$

$$\Rightarrow N = \frac{mv^2}{R} + mg \sin \theta + qvB$$

Hence, at $\theta = \pi/2$ we get $N_{\max} = \frac{2mgR}{R} + mg + qB\sqrt{2gR} = 3mg + qB\sqrt{2gR}$



Ex. A particle of charge q and mass m starts moving from the origin under the action of an electric field $\vec{E} = E_0 \hat{i}$ and magnetic field $\vec{B} = B_0 \hat{i}$ with velocity $\vec{v} = v_0 \hat{j}$. The speed of the particle will become $2v_0$ after a time :-

(A) $t = \frac{2mv_0}{qE}$ (B) $t = \frac{2Bq}{mv_0}$ (C) $t = \frac{\sqrt{3} Bq}{mv_0}$ (D) $t = \frac{\sqrt{3} mv_0}{qE}$

Ans. (D)

Sol. Charged particle will move in a Helical path.

$$\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}$$

$$\vec{a} = \frac{qE\hat{i}}{m} \text{ this part will increase x component of velocity}$$

+

$$\frac{qv_0 \times B}{m} \text{ (in y-z plane) this term will provide centripetal acceleration.}$$

$$v_x = \frac{qE}{m} \cdot t$$

$$v = \sqrt{v_x^2 + v_0^2}$$

$$2v_0 = \sqrt{\left(\frac{qE}{m}t\right)^2 + v_0^2}$$

$$\sqrt{3}v_0 = \frac{qE}{m}t$$

$$t = \frac{\sqrt{3}mv_0}{qE}$$

(1) Velocity Selector :

In the presence of both electric field $\vec{E}$ and magnetic field $\vec{B}$, the total force on a charged particle is

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

This is known as the Lorentz force. By combining the two fields, particles which move with a certain velocity can be selected. This was the principle used by J. J. Thomson to measure the charge-to-mass ratio of the electrons. In Figure the schematic diagram of Thomson's apparatus is depicted.

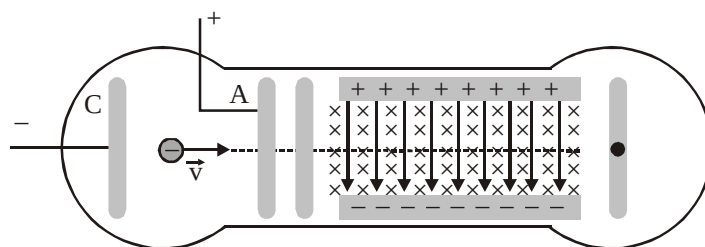


Figure : Thomson's apparatus

The electrons with charge $q = -e$ and mass m are emitted from the cathode C and then accelerated toward slit A. Let the potential difference between A and C be $V_A - V_C = \Delta V$. The change in potential energy is equal to the external work done in accelerating the electrons: $\Delta U = W_{\text{ext}} = q\Delta V = -e\Delta V$. By energy conservation, the kinetic energy gained is $\Delta K = -\Delta U = mv^2/2$. Thus, the speed of the electrons is given by

$$v = \sqrt{\frac{2e\Delta V}{m}}$$

If the electrons further pass through a region where there exists a downward uniform electric field, the electrons, being negatively charged, will be deflected upward. However, if in addition to the electric field, a magnetic field directed into the page is also applied, then the electrons will experience an additional downward magnetic force $-e\vec{v} \times \vec{B}$. When the two forces exactly cancel, the electrons will move in a straight path. From Eq., we see that when the condition for the cancellation of the two forces is given by $eE = evB$, which implies

$$v = \frac{E}{B}$$

In other words, only those particles with speed $v = E/B$ will be able to move in a straight line. Combining the two equations, we obtain

$$\frac{e}{m} = \frac{E^2}{2(\Delta V)B^2}$$

By measuring E , ΔV and B , the charge-to-mass ratio can be readily determined. The most precise measurement to date is $e/m = 1.758820174(71) \times 10^{11} \text{ C/kg}$.

(2) Mass Spectrometer :

Various methods can be used to measure the mass of an atom. One possibility is through the use of a mass spectrometer. The basic feature of a *Bainbridge* mass spectrometer is illustrated in Figure. A particle carrying a charge $+q$ is first sent through a velocity selector.

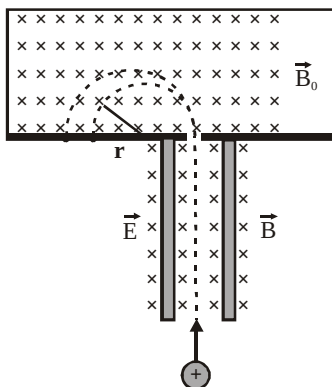


Figure : A Bainbridge mass spectrometer

The applied electric and magnetic fields satisfy the relation $E = vB$ so that the trajectory of the particle is a straight line. Upon entering a region where a second magnetic field $\vec{B}_0$ pointing into the page has been applied, the particle will move in a circular path with radius r and eventually strike the photographic plate. Using Eq., we have

$$r = \frac{mv}{qB_0}$$

Since $v = E/B$, the mass of the particle can be written as

$$m = \frac{qB_0 r}{v} = \frac{qB_0 Br}{E}$$

Ex. Particle A with charge q and mass m_A and particle B with charge $2q$ and mass m_B , are accelerated from rest by a potential difference ΔV , and subsequently deflected by a uniform magnetic field into semicircular paths. The radii of the trajectories by particle A and B are R and $2R$, respectively. The direction of the magnetic field is perpendicular to the velocity of the particle. What is their mass ratio?

Sol. The kinetic energy gained by the charges is equal to

$$\frac{1}{2}mv^2 = q\Delta V$$

which yields $v = \sqrt{\frac{2q\Delta V}{m}}$

The charges move in semicircles, since the magnetic force points radially inward and provides the source of the centripetal force :

$$\frac{mv^2}{r} = qvB$$

The radius of the circle can be readily obtained as :

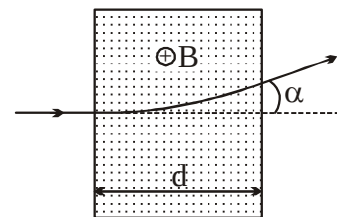
$$r = \frac{mv}{qB} = \frac{m}{qB} \sqrt{\frac{2q\Delta V}{m}} = \frac{1}{B} \sqrt{\frac{2m\Delta V}{q}}$$

which shows that r is proportional to $(m/q)^{1/2}$. The mass ratio can then be obtained from

$$\frac{r_A}{r_B} = \frac{(m_A/q_A)^{1/2}}{(m_B/q_B)^{1/2}} \Rightarrow \frac{R}{2R} = \frac{(m_A/q)^{1/2}}{(m_B/2q)^{1/2}}$$

which gives $\frac{m_A}{m_B} = \frac{1}{8}$

Ex. A proton accelerated by a potential difference $V = 500$ kV flies through a uniform transverse magnetic field with induction $B = 0.51$ T. The field occupies a region of space $d = 10$ cm in thickness (Fig.). Find the angle α through which the proton deviates from the initial direction of its motion.



Ans. $\alpha = \arcsin \left(dB \sqrt{\frac{q}{2mV}} \right) = 30^\circ$

Sol. From the figure,

$$\sin \alpha = \frac{d}{R} = \frac{dqB}{mv}$$

As radius of the arc $R = \frac{mv}{qB}$, where v is the velocity of the particle, when it enters into the field.

From initial condition of the problem,

$$qV = \frac{1}{2}mv^2 \quad \text{or, } v = \sqrt{\frac{2qV}{m}}$$

$$\text{Hence, } \sin \alpha = \frac{dqB}{m\sqrt{2qV/m}} = dB \sqrt{\frac{q}{2mV}}$$

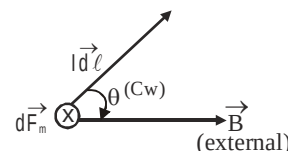
$$\text{and } \alpha = \sin^{-1} \left(dB \sqrt{\frac{q}{2mV}} \right) = 30^\circ, \text{ on putting the values.}$$

CURRENT CARRYING CONDUCTOR IN MAGNETIC FIELD

When a current carrying conductor placed in magnetic field, a magnetic force exerts on each free electron which are present inside the conductor. The resultant of these forces on all the free electrons is called magnetic force on conductor.

• Magnetic force on current element

Through experiments Ampere established that when current element $I d\vec{\ell}$ is placed in magnetic field $\vec{B}$, it experiences a



magnetic force $d\vec{F}_m = I(d\vec{\ell} \times \vec{B})$

- Current element in a magnetic field does not experience any force if the current in it is parallel or anti-parallel with the field $\theta = 0^\circ$ or 180°

$$dF_m = 0 \text{ (min.)}$$

- Current element in a magnetic field experiences maximum force if the current in it is perpendicular with the field $\theta = 90^\circ$

$$dF_m = BId\ell \text{ (max.)}$$

- Magnetic force on current element is always perpendicular to the current element vector and magnetic field vector. $d\vec{F}_m \perp I d\vec{\ell}$ and $d\vec{F}_m \perp \vec{B}$ (always)

- Total magnetic force on straight current carrying conductor in uniform magnetic field given as

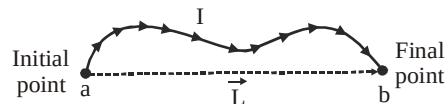
$$\vec{F}_m = \int_i^f d\vec{F}_m \left[\int_i^f d\vec{\ell} \right] = I \times \vec{B}, \quad \vec{F}_m = I(\vec{L} \times \vec{B})$$



Where $\vec{L} = \int_i^f d\vec{\ell}$, vector sum of all length elements from initial to final point, which is in accordance with the law of vector addition and $|\vec{L}| = \text{length of the conductor}$.

- Total magnetic force on arbitrary shape current carrying conductor in uniform magnetic field $\vec{B}$ is

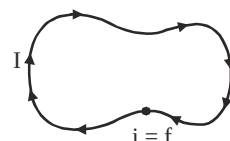
$$\int_i^f d\vec{F}_m = I \left[\int_i^f d\vec{\ell} \right] \times \vec{B}, \quad \vec{F}_m = I(\vec{L} \times \vec{B}) \quad (L = ab)$$



Where $\vec{L} = \int_i^f d\vec{\ell}$, vector sum of all length elements from initial to final point or displacement between free ends of an arbitrary conductor from initial to final point.

- A current carrying closed loop (or coil) of any shape placed in uniform magnetic field then no net magnetic force act on it (Torque may or may not be zero)

$$\vec{L} = \int_i^f d\vec{\ell} = 0 \quad \text{or} \quad \oint d\vec{\ell} = 0$$



So net magnetic force acting on a current carrying closed loop $\vec{F}_m = 0$ (always)

- When a current carrying closed loop (or coil) of any shape placed in non uniform magnetic field then net magnetic force is always acts on it (Torque may or may not be zero)

Ex. : Magnetic Force on a Semi-Circular Loop

Consider a closed semi-circular loop lying in the xy plane carrying a current I in the counterclockwise direction, as shown in Figure.

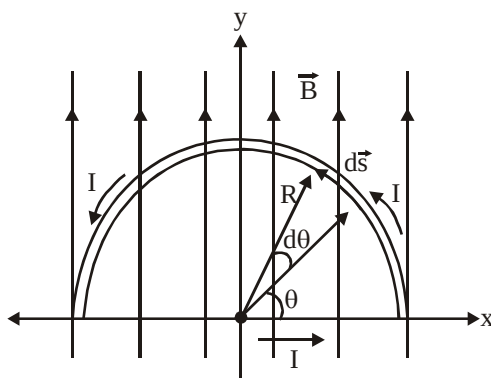


Figure 8.3.6 Semi-circular loop carrying a current I

A uniform magnetic field pointing in the $+y$ direction is applied. Find the magnetic force acting on the straight segment and the semicircular arc.

- Sol.** Let $\vec{B} = B\hat{j}$ and $\vec{F}_1$ and $\vec{F}_2$ the forces acting on the straight segment and the semicircular parts, respectively. Using Eq. and noting that the length of the straight segment is $2R$, the magnetic force is

$$\vec{F}_1 = I(2R\hat{i}) \times (B\hat{j}) = 2IRB\hat{k}$$

where $\hat{k}$ is directed out of the page.

To evaluate $\vec{F}_2$, we first note that the differential length element $d\vec{s}$ on the semicircle can be written as $d\vec{s} = ds\hat{\theta} = R d\theta(-\sin\theta\hat{i} + \cos\theta\hat{j})$. The force acting on the length element is $d\vec{F}$ is :

$$d\vec{s} = I d\vec{s} \times \vec{B} = IR d\theta (-\sin\theta \hat{i} + \cos\theta \hat{j}) \times (B\hat{j}) = -IBR \sin\theta d\theta \hat{k}$$

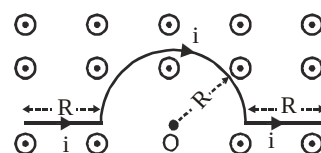
Here we see that $d\vec{F}_2$ points into the page. Integrating over the entire semi-circular arc, we have

Thus, the net force acting on the semi-circular wire is

$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 = \vec{0}$$

This is consistent from our previous claim that the net magnetic force acting on a closed current-carrying loop must be zero.

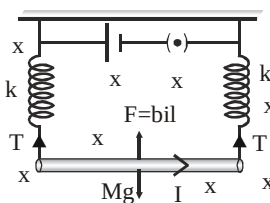
Ex. A wire bent as shown in fig carries a current i and is placed in a uniform field of magnetic induction $\vec{B}$ that emerges from the plane of the figure. Calculate the force acting on the wire.



Sol. The total force on the whole wire is

$$F_m = I|\vec{L}|B = I(R + 2R + R)B = 4RIB$$

Ex. A metal rod of mass 10 gm and length 25 cm is suspended on two springs as shown in figure. The springs are extended by 4 cm. When a 20 ampere current passes through the rod it rises by 1 cm. Determine the magnetic field assuming acceleration due to gravity to be 10 m/s^2 .



Sol. Let tension in each spring is $= T_0$

Initially the rod will be in equilibrium if $2T_0 = Mg$

$$\text{then } T_0 = kx_0 \quad \dots(i)$$

Now when the current I is passed through the rod it will experience a force $F = BIL$ vertically up; so in this situation for its equilibrium,

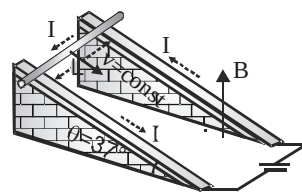
$$2T + BIL = Mg \quad \text{with } T = kx \quad \dots(ii) \quad (x = 4 - 1 = 3\text{cm})$$

$$\text{So from eq. (i) and eq.(ii) } \frac{T}{T_0} = \frac{Mg - BIL}{Mg}$$

$$\Rightarrow \frac{x}{x_0} = 1 - \frac{BIL}{Mg}$$

$$\Rightarrow B = \frac{Mg(x_0 - x)}{ILx_0} = \frac{10 \times 10^{-3} \times 10 \times 3 \times 10^{-2}}{20 \times 25 \times 10^{-2} \times 4 \times 10^{-2}} = 1.5 \times 10^{-2} \text{ T}$$

Ex. Two conducting rails are connected to a source of e.m.f. and form an incline as shown in fig. A bar of mass 50 g slides without friction down the incline through a vertical magnetic field B . If the length of the bar is 50 cm and a current of 2.5 A is provided by the battery, for what value of B will the bar slide at a constant velocity? [$g = 10 \text{ m/s}^2$]

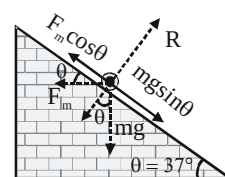


Sol. Force on current carrying wire $F = BIL$

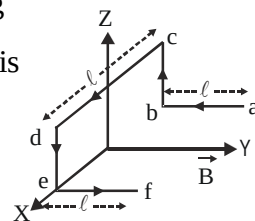
The rod will move down the plane with constant velocity only if

$$F \cos \theta = mg \sin \theta \Rightarrow BIL \cos \theta = mg \sin \theta$$

$$\text{or, } B = \frac{mg}{IL} \tan \theta = \frac{50 \times 10^{-3} \times 10}{2.5 \times 50 \times 10^{-2}} \times \frac{3}{4} = 0.3 \text{ T}$$



Ex. A wire abcdef with each side of length ' ℓ ' bent as shown in figure and carrying a current I is placed in a uniform magnetic field B parallel to $+y$ direction. What is the force experienced by the wire.

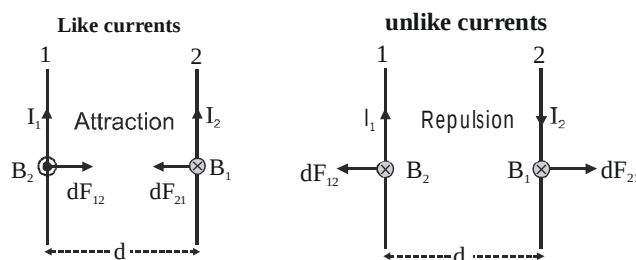


Sol. Magnetic force on wire abcdef in uniform magnetic field is $\vec{F}_m = I(\vec{L} \times \vec{B})$,

$\vec{L}$ is displacement between free ends of the conductor from initial to final point. $\vec{L} = (\ell) \hat{i}$ and $\vec{B} = (B) \hat{j}$

$$F_m = I(\vec{L} \times \vec{B}) = BIL (\hat{i} \times \hat{j}) = BI\ell (\hat{k}) = BI\ell, \text{ along } +z \text{ direction}$$

MAGNETIC FORCE BETWEEN TWO PARALLEL CURRENT CARRYING CONDUCTORS



The net magnetic force acts on a current carrying conductor due to its own field is zero. So consider two infinite long parallel conductors separated by distance ' d ' carrying currents I_1 and I_2 .

Magnetic field at each point on conductor (ii) due to current I_1 is $B_1 = \frac{\mu_0 I_1}{2\pi d}$

[uniform field for conductor (2)]

Magnetic field at each point on conductor (i) due to current I_2 is $B_2 = \frac{\mu_0 I_2}{2\pi d}$

[Uniform field for conductor (1)]

consider a small element of length ' $d\ell$ ' on each conductor. These elements are right angle to the external magnetic field, so magnetic force experienced by elements of each conductor given as

$$dF_{12} = B_2 I_1 d\ell = \left(\frac{\mu_0 I_2}{2\pi d} \right) I_1 d\ell \quad \dots(i) \quad (\text{Where } I_1 d\ell \perp B_2)$$

$$dF_{21} = B_1 I_2 d\ell = \left(\frac{\mu_0 I_1}{2\pi d} \right) I_2 d\ell \quad \dots(ii) \quad (\text{Where } I_2 d\ell \perp B_1)$$

Where dF_{12} is magnetic force on element of conductor (i), due field of conductor (i) and dF_{21} is magnetic force on element of conductor (ii), due to field of conductor (i).

Magnetic force per unit length of each conductor is $\frac{dF_{12}}{d\ell} = \frac{dF_{21}}{d\ell} = \frac{\mu_0 I_1 I_2}{2\pi d}$

$$f = \frac{\mu_0 I_1 I_2}{2\pi d} \text{ N/m (in S.I.)}$$

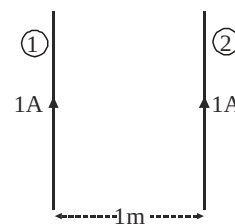
$$f = \frac{2 I_1 I_2}{d} \text{ dyne/cm (In C.G.S.)}$$

Definition of ampere :

Magnetic force/unit length for both infinite length conductor gives as

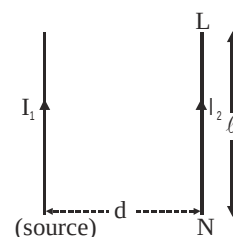
$$f = \frac{\mu_0 I_1 I_2}{2\pi d} = \frac{(4\pi \times 10^{-7})(1)(1)}{2\pi(1)} = 2 \times 10^{-7} \text{ N/m}$$

'Ampere' is the current which, when passed through each of two parallel infinite long straight conductors placed in free space at a distance of 1 m from each other, produces between them a force of $2 \times 10^{-7} \text{ N/m}$



- Force scale $f = \frac{\mu_0 I_1 I_2}{2\pi d}$ is applicable when at least one conductor must be of infinite length so it behaves like source of uniform magnetic field for other conductor.

Magnetic force on conductor 'LN' is $F_{LN} = f \times \ell \Rightarrow F_{LN} = \left(\frac{\mu_0 I_1 I_2}{2\pi d} \right) \ell$

**Torque on a Current Loop :**

What happens when we place a rectangular loop carrying a current I in the xy plane and switch on a uniform magnetic field $\vec{B} = B\hat{i}$ which runs parallel to the plane of the loop, as shown in Figure (a)?

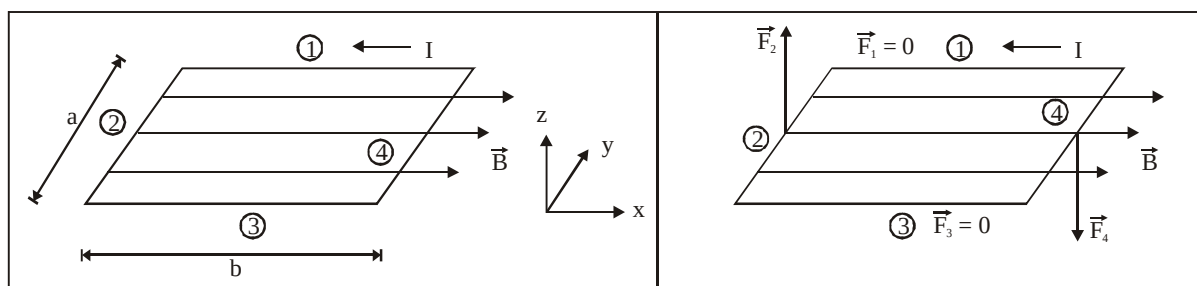


Figure : (a) A rectangular current loop placed in a uniform magnetic field.
(b) The magnetic forces acting on sides 2 and 4.

From Eq., we see the magnetic forces acting on sides 1 and 3 vanish because the length vectors $\vec{\ell}_1 = -b\hat{i}$ and $\vec{\ell}_3 = b\hat{i}$ are parallel and anti-parallel to $\vec{B}$ and their cross products vanish. On the other hand, the magnetic forces acting on segments 2 and 4 are non-vanishing:

$$\begin{cases} \vec{F}_2 = I(-a\hat{j}) \times (B\hat{i}) = IaB\hat{k} \\ \vec{F}_4 = I(a\hat{j}) \times (B\hat{i}) = -IaB\hat{k} \end{cases}$$

with $\vec{F}_2$ pointing out of the page and $\vec{F}_4$ into the page. Thus, the net force on the rectangular loop is

$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 = \vec{0}$$

as expected. Even though the net force on the loop vanishes, the forces $\vec{F}_2$ and $\vec{F}_4$ will produce a torque which causes the loop to rotate about the y -axis (Figure). The torque with respect to the center of the loop is

$$\begin{aligned}\vec{\tau} &= \left(-\frac{b}{2}\hat{i}\right) \times \vec{F}_2 + \left(\frac{b}{2}\hat{i}\right) \times \left(IaB\hat{k} + \left(\frac{b}{2}\hat{i}\right) \times (-IaB\hat{k})\right) \\ &= \left(\frac{IabB}{2} + \frac{IabB}{2}\right)\hat{j} = IabB\hat{j} = IAB\hat{j}\end{aligned}$$

where $A = ab$ represents the area of the loop and the positive sign indicates that the rotation is clockwise about the y -axis. It is convenient to introduce the area vector $\vec{A} = A\hat{n}$ where $\hat{n}$ is a unit vector in the direction normal to the plane of the loop. The direction of the positive sense of $\hat{n}$ is set by the conventional right-hand rule. In our case, we have $\hat{n} = +\hat{k}$. The above expression for torque can then be rewritten as

$$\vec{\tau} = I\vec{A} \times \vec{B}$$

Notice that the magnitude of the torque is at a maximum when $\vec{B}$ is parallel to the plane of the loop (or perpendicular to).

Consider now the more general situation where the loop (or the area vector $\vec{A}$) makes an angle θ with respect to the magnetic field.

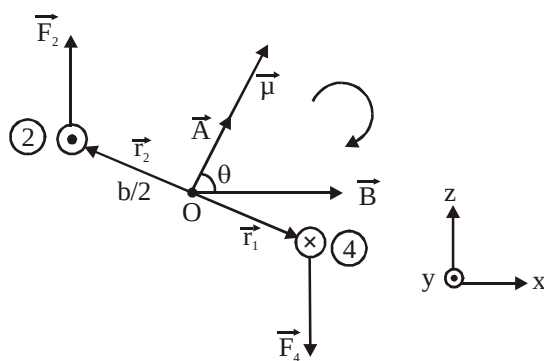


Figure : Rotation of a rectangular current loop

From Figure, the lever arms and can be expressed as:

$$\vec{r}_2 = \frac{b}{2}(-\sin\theta\hat{i} + \cos\theta\hat{k}) = -\vec{r}_4$$

and the net torque becomes

$$\vec{\tau} = \vec{r}_2 \times \vec{F}_2 + \vec{r}_4 \times \vec{F}_4 = 2\vec{r}_2 \times \vec{F}_2 = 2 \cdot \frac{b}{2}(-\sin\theta\hat{i} + \cos\theta\hat{k}) \times (IaB\hat{k})$$

$$jIabB \sin\theta\hat{j} = I\vec{A} \times \vec{B}$$

For a loop consisting of N turns, the magnitude of the torque is

$$\tau = NIAB \sin\theta$$

The quantity $NI\vec{A}$ is called the magnetic dipole moment $\vec{\mu}$

$$\vec{\mu} = NI\vec{A}$$

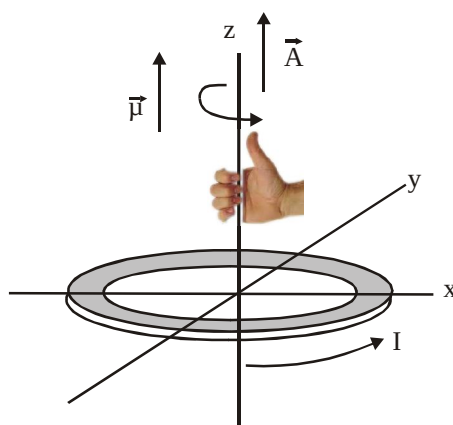


Figure : Right-hand rule for determining the direction of $\vec{\mu}$

The direction of $\vec{\mu}$ is the same as the area vector $\vec{A}$ (perpendicular to the plane of the loop) and is determined by the right-hand rule (Figure). The SI unit for the magnetic dipole moment is ampere-meter² ($\text{A}\cdot\text{m}^2$). Using the expression for $\vec{\mu}$, the torque exerted on a current-carrying loop can be rewritten as

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

The above equation is analogous to $\vec{\tau} = \vec{p} \times \vec{E}$ in Eq., the torque exerted on an electric dipole moment $\vec{p}$ in the presence of an electric field $\vec{E}$.

Configuratin energy of current loop in uniform magnetic field.

Recalling that the potential energy for an electric dipole is $U = -\vec{p} \cdot \vec{E}$ [see Eq.], a similar form is expected for the magnetic case. The work done by an external agent to rotate the magnetic dipole from an angle θ_0 to θ is given by

$$W_{\text{ext}} = \int_{\theta_0}^{\theta} (\mu B \sin \theta') d\theta' = \mu B (\cos \theta_0 - \cos \theta)$$

$$= \Delta U = U - U_0$$

Once again, $W_{\text{ext}} = -W$, where W is the work done by the magnetic field. Choosing $U_0 = 0$ at $\theta_0 = \pi/2$, the dipole in the presence of an external field then has a potential energy as

$$U = -\mu B \cos \theta = -\vec{\mu} \cdot \vec{B}$$

The configuration is at a stable equilibrium when $\vec{\mu}$ is aligned parallel to $\vec{B}$, making U a minimum with $U_{\text{min}} = -\mu B$. On the other hand, when $\vec{\mu}$ and $\vec{B}$ are anti-parallel, $U_{\text{max}} = +\mu B$ is a maximum and the system is unstable.

Ex. A non-conducting thin disc of radius R charged uniformly over one side with surface density σ rotates about its axis with an angular velocity ω . Find:

- the magnetic induction at the centre of the disc;
- the magnetic moment of the disc.

Ans. (a) $B = \frac{1}{2} \mu_0 \sigma \omega R$; (b) $p_m = \frac{1}{4} \pi \sigma \omega R^4$

Sol. (a) Let us take a ring element of radius r and thickness dr , then charge on the ring element.,
 $dq = \sigma 2\pi r dr$

and current, due to this element, $di = \frac{(\sigma 2\pi r dr)\omega}{2\pi} = \sigma\omega r dr$

So, magnetic induction at the centre, due to this element : $dB = \frac{\mu_0}{2} \frac{di}{r}$

and hence, from symmetry : $B = \int dB = \int_0^R \frac{\mu_0 \sigma \omega r dr}{r} = \frac{\mu_0}{2} \sigma \omega R$

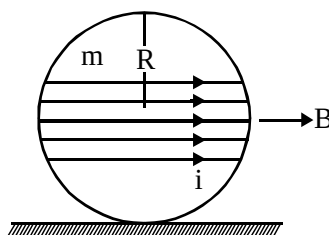
(b) Magnetic moment of the element, considered,

$$dp_m = (di) \pi r^2 = \sigma\omega r dr \pi r^2 = \sigma\pi\omega r^3 dr$$

Hence, the sought magnetic moment,

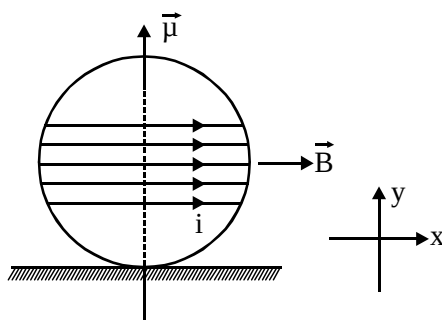
$$p_m = \int dp_m = \int_0^R \sigma\pi\omega r^3 dr = \sigma\pi\omega \frac{R^4}{4}$$

Ex. A wire is wrapped $N = 10$ times over a solid sphere of mass $m = 5\text{kg}$, current $I = 2\text{A}$, which is placed on a smooth horizontal surface. A horizontal magnetic field of induction $|\vec{B}| = 10\text{T}$ is present. Find the angular acceleration experienced by the sphere. Assume that the mass of the wire is negligible compared to the mass of the sphere. If answer is $20n\pi$. Write value of n .



Ans. 5

Sol. (a) The net torque acting on the sphere is



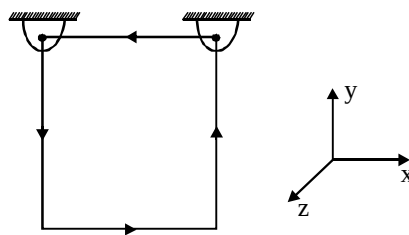
$$\vec{\tau} = \vec{\mu} \times \vec{B} = (NiA\hat{j}) \times (B\hat{i}) = -NiAB\hat{k}, \text{ where } A = \pi R^2$$

$$\text{or } \vec{\tau} = -N\pi R^2 i B \hat{k}$$

(b) $\alpha = \frac{\tau}{I_C}$ ($\because$ the sphere is free to rotate, it must rotate about the centroidal axis)

$$= -\frac{N\pi R^2 i B}{\frac{2}{5}mR^2} \hat{k} \quad \left(\because I_C = \frac{2}{5}mR^2 \right) = \frac{5N\pi i B}{2m} \hat{k}$$

Ex. A current carrying uniform square frame is suspended from hinged supports as shown in the figure such that it can freely rotate about its upper side. The length and mass of each side of the frame is 2m and 4kg respectively. A uniform magnetic field $\vec{B} = (3\hat{i} + 4\hat{j})$ is applied. When the wire frame is rotated to 45° from vertical and released it remains in equilibrium. If the magnitude of current (in A) in the wire frame is I then find $\left(\frac{3}{5}\right)I$.



Ans. 6

Sol. $\vec{\mu}$ (Magnetic moment of loop) when it is lifted by $45^\circ = i\ell^2 \left(\frac{\hat{j} + \hat{k}}{\sqrt{2}} \right)$

$$\therefore \vec{\tau} \text{ due to magnetic field} = \vec{\mu} \times \vec{B} = \frac{i\ell^2}{\sqrt{2}} \left[(\hat{j} + \hat{k}) \times (3\hat{i} + 4\hat{j}) \right]$$

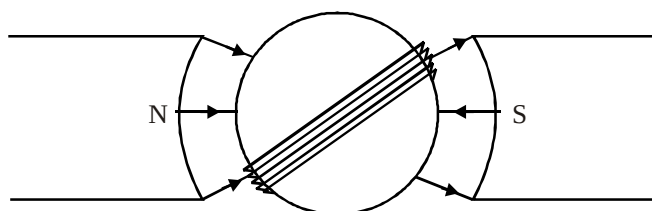
$$\vec{\tau} \text{ due to mg (about top edge)} = 4mg \frac{\ell}{2} \cos 45^\circ \hat{i}$$

$\therefore$ For equilibrium net torque along X-axis = 0

$$\therefore \frac{4mg\ell}{2\sqrt{2}} = \frac{4i\ell^2}{\sqrt{2}} \Rightarrow i = \frac{mg}{2\ell} = 10\text{A}$$

MOVING COIL GALVANOMETER :

The main parts of a moving-coil galvanometer are shown in figure.



The current to be measured is passed through the galvanometer. As the coil is in the magnetic field $\vec{B}$ of the permanent magnet, a torque $\vec{\Gamma} = n i \vec{A} \times \vec{B}$ acts on the coil. Here n = number of turns, i = current in the coil, $\vec{A}$ = area-vector of the coil and $\vec{B}$ = magnetic field at the site of the coil. This torque deflects the coil from its equilibrium position.

The pole pieces are made cylindrical. As a result, the magnetic field at the arms of the coil remains parallel to the plane of the coil everywhere even as the coil rotates. The deflecting torque is then $\Gamma = niAB$. As the upper end of the suspension strip W is fixed, the strip gets twisted when the coil rotates. This produces a restoring torque acting on the coil. If the deflection of the coil is θ and the torsional constant of the suspension strip is k , the restoring torque is $k\theta$. The coil will stay at a deflection θ where

$$niAB = k\theta$$

$$\text{or, } i = \frac{k}{nAB} \theta$$

Hence, the current is proportional to the deflection. The constant $\frac{k}{nAB}$ is called the galvanometer constant.

We define the **current sensitivity** of the galvanometer as the deflection per unit current. From Eq. this current sensitivity is.

$$\frac{\phi}{I} = \frac{NAB}{k}$$

A convenient way for the manufacturer to increase the sensitivity is to increase the number of turns N . We choose galvanometers having sensitivities of value, required by our experiment.

We define the **voltage sensitivity** as the deflection per unit volt of applied potential difference

$$\frac{\phi}{I} = \left(\frac{NAB}{k} \right) \frac{I}{V} = \left(\frac{NAB}{k} \right) \frac{1}{R}$$

An interesting point to note is that increasing the current sensitivity may not necessarily increase the voltage sensitivity. If $N \rightarrow 2N$, i.e., we double the number of turns, then

$$\frac{\phi}{I} \rightarrow 2 \frac{\phi}{I}$$

Thus, the current sensitivity doubles. However, the resistance of the galvanometer is also likely to double, since it is proportional to the length of the wire. In eq. $N \rightarrow 2N$, and $R \rightarrow 2R$, thus the voltage sensitivity,

$$\frac{\phi}{V} \rightarrow \frac{\phi}{V}$$

remains unchanged.

ATOMIC MAGNETISM

An atomic orbital electron, which doing bounded uniform circular motion around nucleus. A current constitutes with this orbital motion and hence orbit behaves like current carrying loop. Due to this magnetism produces at nucleus position. This phenomenon called as 'atomic magnetism'.

Bohr's postulates :

$$(i) \quad \frac{mv^2}{r} = \frac{kze^2}{r^2} \quad (ii) \quad L = mvr = n \left(\frac{h}{2\pi} \right), \text{ where } n = 1, 2, 3, \dots$$

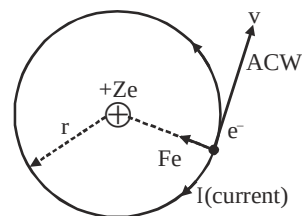
Basic elements of atomic magnetism :

$$(a) \quad \text{Orbital current :- } I = ef = \frac{e}{T} = \frac{ev}{2\pi r} = \frac{e\omega}{2\pi}$$

(b) **Magnetic induction at nucleus position :-** As circular orbit behaves like current carrying loop, so

$$\text{magnetic induction at nucleus position } B_N = \frac{\mu_0 I}{2r}$$

$$B_N = \frac{\mu_0 ef}{2r} = \frac{\mu_0 e}{2Tr} = \frac{\mu_0 ev}{4\pi r^2} = \frac{\mu_0 e\omega}{4\pi r}$$



(c) **Magnetic moment of circular orbit :-** Magnetic dipole moment of circular orbit

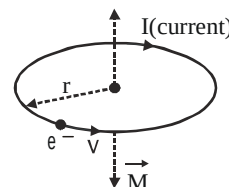
$$M = IA \text{ where } A \text{ is area of circular orbit. } M = ef(\pi r^2) = \frac{\pi e r^2}{T} = \frac{evr}{2} = \frac{e\omega r^2}{2}$$

• **Relation between magnetic moment and angular momentum of orbital electron**

$$\text{Magnetic moment } M = \frac{evr}{2} \times \frac{m}{m} = \frac{eL}{2m} \quad (\because \text{angular momentum } L = mvr)$$

$$\text{Vector form} \quad \vec{M} = \frac{-e\vec{L}}{2m}$$

For orbital electron its $\vec{M}$ and $\vec{L}$ both are antiparallel axial vectors.



A NONCONDUCTING CHARGED BODY IS ROTATED WITH SOME ANGULAR SPEED.

In this case the ratio of magnetic moment and angular momentum is constant which is equal to $\frac{q}{2m}$

here q = charge and m = the mass of the body.

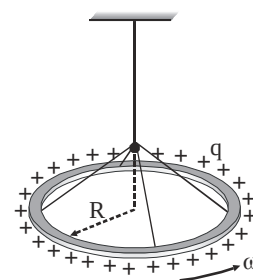
Ex. :- In case of a ring, of mass m , radius R and charge q distributed on its circumference.

$$\text{Angular momentum } L = I\omega = (mR^2)(\omega) \quad \dots (i)$$

$$\text{Magnetic moment } M = iA = (qf)(\pi R^2)$$

$$M = (q) \left(\frac{\omega}{2\pi} \right) (\pi R^2) = q \frac{\omega R^2}{2} \quad \dots (ii)$$

$$\therefore f = \frac{\omega}{2\pi} \text{ From Eqs. (i) and (ii) } \frac{M}{L} = \frac{q}{2m}$$



Although this expression is derived for simple case of a ring, it holds good for other bodies also. For

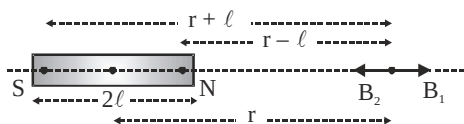
example, for a disc or a sphere. $M = \frac{qL}{2m} \Rightarrow M = \frac{q(I\omega)}{2m}$, where $L = I\omega$

Rigid body	Ring	Disc	Solid sphere	Spherical shell
Moment of inertia (I)	mR^2	$\frac{mR^2}{2}$	$\frac{2}{5} mR^2$	$\frac{2}{3} mR^2$
Magnetic moment = $\frac{qI\omega}{2m}$	$\frac{q\omega R^2}{2}$	$\frac{q\omega R^2}{4}$	$\frac{q\omega R^2}{5}$	$\frac{q\omega R^2}{3}$

SUPPLEMENT FOR JEE-MAINS

MISCELLANEOUS

Magnetic field of long Bar magnet



(i) At Axial position :-

Magnetic field at point 'P' due to north pole $B_1 = \frac{\mu_0}{4\pi} \frac{m}{(r - \ell)^2}$ (away from north pole)

Magnetic field at point 'P' due to south pole $B_2 = \frac{\mu_0}{4\pi} \frac{m}{(r + \ell)^2}$ (towards north pole)

Net magnetic field at point 'P'

$$B_{\text{axis}} = B_1 - B_2, (\because B_1 > B_2) = \frac{\mu_0 m}{4\pi} \left[\frac{1}{(r - \ell)^2} - \frac{1}{(r + \ell)^2} \right] = \frac{\mu_0 m}{4\pi} \left[\frac{4r\ell}{(r^2 - \ell^2)^2} \right]$$

$$B_{\text{axis}} = \frac{\mu_0}{4\pi} \frac{2Mr}{(r^2 - \ell^2)^2}, \text{ where } M = m(2\ell)$$

If magnet is short $r \gg \ell$, then $B_{\text{axis}} \simeq \frac{\mu_0}{4\pi} \frac{2M}{r^3}$

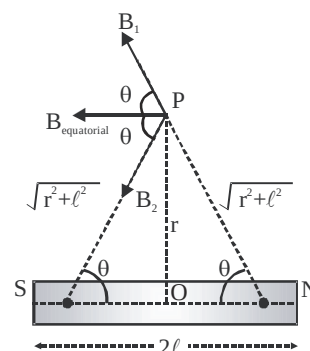
(ii) At equatorial position :-

Magnetic field at point 'P' due to north pole :-

$$B_1 = \frac{\mu_0}{4\pi} \frac{m}{(\sqrt{r^2 + \ell^2})^2} \dots (1) \quad (\text{along NP line})$$

Magnetic field at point 'P' due to south pole :-

$$B_2 = \frac{\mu_0}{4\pi} \frac{m}{(\sqrt{r^2 + \ell^2})^2} \dots (2) \quad (\text{along PS line})$$



From equation (1) & (2) $B_1 = B_2 = \frac{\mu_0}{4\pi} \cdot \frac{m}{r^2 + \ell^2} = B$ (Let)

Net magnetic field at point 'P'

$$B_{\text{eq}} = 2 B \cos\theta = 2 \cdot \frac{\mu_0}{4\pi} \frac{m}{(r^2 + \ell^2)} \cos\theta, \quad \left[\text{where } \cos\theta = \frac{\ell}{\sqrt{(r^2 + \ell^2)}} \right]$$

$$= 2 \cdot \frac{\mu_0}{4\pi} \frac{m}{(r^2 + \ell^2)} \frac{\ell}{\sqrt{(r^2 + \ell^2)}}$$

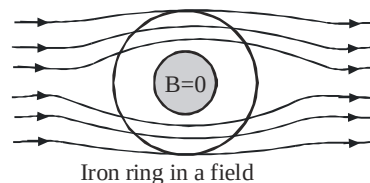
$$B_{eq.} = \frac{\mu_0}{4\pi} \cdot \frac{M}{(r^2 + \ell^2)^{3/2}},$$

where $M = m(2\ell)$

If magnet is short $r \gg \ell$, then $B_{eq.} \approx \frac{\mu_0}{4\pi} \cdot \frac{M}{r^3}$

Magnetic shielding

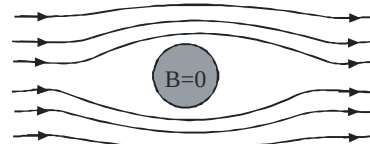
If a soft iron ring is placed in magnetic field, most of the lines are found to pass through the ring and no lines pass through the space inside the ring. The inside of the ring is thus protected against any external magnetic effect. This phenomenon is called magnetic screening or shielding and is used to protect costly wrist-watches and other instruments from



Iron ring in a field

external magnetic fields by enclosing them in a soft-iron case or box.

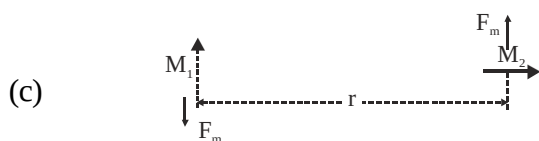
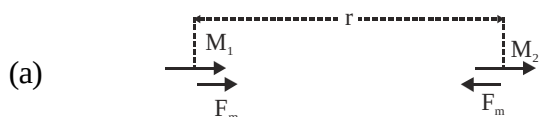
- (i) Super conductors also provides perfect magnetic screening due to exclusion of lines of force. This effect is called '**Meissner effect**'
- (ii) Relative magnetic permeability of super conductor is zero. So we can say that super conductors behaves like perfect diamagnetic.



Super conductor in a field

Dipole - Dipole Interactions :

S.No. Relative position of dipoles



Magnetic force (F_m)

$$\frac{\mu_0}{4\pi} \cdot \frac{6M_1M_2}{r^4} \text{ (along } r\text{)}$$

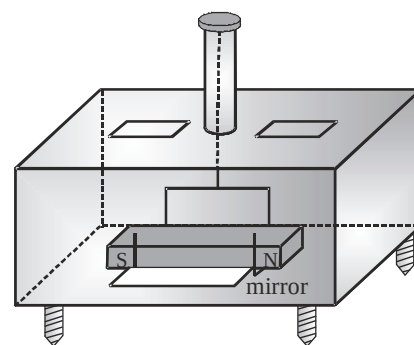
$$\frac{\mu_0}{4\pi} \cdot \frac{3M_1M_2}{r^4} \text{ (along } r\text{)}$$

$$\frac{\mu_0}{4\pi} \cdot \frac{3M_1M_2}{r^4} \text{ (perpendicular to } r\text{)}$$

VIBRATION MAGNETOMETER :

It is an instrument used to compare the horizontal components of magnetic field of earth of two different places, to compare magnetic fields and magnetic moments of two bar magnets. It is also called oscillation magnetometer.

Principle : This device works on the principle, that whenever a freely suspended bar magnet horizontal component in earth magnetic field (B_H) is slightly disturbed from its equilibrium position then, it will experience a torque and executes angular S.H.M. *Rotation is possible only in horizontal plane.



Angular S.H.M of magnetic dipole :- When a dipole is suspended in a uniform magnetic field it will align itself parallel to field. Now if it is given a small angular displacement θ about its equilibrium position. The restoring torque acts on it :

$$\tau = -M B_H \sin \theta \Rightarrow I \alpha = -M B_H \sin \theta = -M B_H \theta, (\because \sin \theta \simeq \theta)$$

$$\Rightarrow \alpha = \frac{M B_H}{I} (-\theta) \Rightarrow \alpha = \omega^2 (-\theta) \Rightarrow \omega^2 = \frac{M B_H}{I}$$

$$\text{The time period of angular S.H.M.} \Rightarrow T = 2\pi \sqrt{\frac{I}{M B_H}}$$

M = magnetic moment of bar magnet

I = moment of inertia of bar magnet about its geometric axis

Comparison of magnetic moments of magnets of the same size

Let the two magnets of same size have moment of inertia I and magnetic moments M_1 and M_2 . Suspend the two given magnets turn by turn in the metal stirrup of the vibration magnetometer and note the time period in each case.

$$\text{Then } T_1 = 2\pi \sqrt{\frac{I}{M_1 B}} \text{ and } T_2 = 2\pi \sqrt{\frac{I}{M_2 B}}$$

$$\text{Dividing, } \frac{T_1}{T_2} = \sqrt{\frac{M_2}{M_1}} \text{ or } \frac{M_1}{M_2} = \frac{T_2^2}{T_1^2}$$

Since T_1 and T_2 are known therefore the ratio $\frac{M_2}{M_1}$ can be determined.

Comparison of magnetic moments of magnets of different sizes

Let the two magnets have moments of inertia I_1 and I_2 and magnetic moments M_1 and M_2 respectively. Place the two given magnets one upon the other as shown in Fig. (a). This combination is called 'sum combination'. It has moment of inertia $(I_1 + I_2)$ and magnetic moment $(M_1 + M_2)$. Put this combination in the magnetometer and set it into oscillations. The time period T_1 is determined.

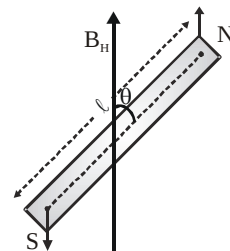
$$T_1 = 2\pi \sqrt{\frac{I_1 + I_2}{(M_1 + M_2)B}} \quad \dots\dots\dots(1)$$

Now, the two magnets are placed as shown in Fig. (b). This combination is called 'difference combination'. It has moment of inertia $(I_1 + I_2)$ and magnetic moment $(M_1 - M_2)$. This combination is put in the magnetometer and its time period T_2 is determined.

$$T_2 = 2\pi \sqrt{\frac{I_1 + I_2}{(M_1 - M_2)B}} \quad \dots\dots\dots(2)$$

$$\text{Dividing, } \frac{T_1}{T_2} = \sqrt{\frac{M_1 - M_2}{M_1 + M_2}} \quad [\text{from equation (1) and (2)}]$$

knowing T_1 and T_2 , we can determine $\frac{M_1}{M_2}$.



(a) sum combination



(b) difference combination

☉ **Comparison of earth's magnetic field at two different places**

Let the vibrating magnet have moment of inertia I and magnetic moment M . Let it be vibrated in places where earth's magnetic field is B_{H_1} and B_{H_2} .

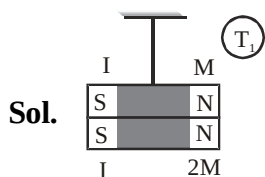
$$\text{Then, } T_1 = 2\pi \sqrt{\frac{I}{MB_{H_1}}} \text{ and } T_2 = 2\pi \sqrt{\frac{I}{MB_{H_2}}}$$

T_1 and T_2 are determined by placing magnetometer at two different places, turn by turn.

$$\text{Dividing, } \frac{T_1}{T_2} = \sqrt{\frac{B_{H_2}}{B_{H_1}}} \text{ or } \frac{T_1^2}{T_2^2} = \frac{B_{H_2}}{B_{H_1}} = \frac{B_2 \cos \theta_2}{B_1 \cos \theta_1} \Rightarrow \frac{B_1}{B_2} = \frac{T_2^2 \cos \theta_2}{T_1^2 \cos \theta_1}$$

Knowing T_1 , T_2 and θ_1 , θ_2 the ratio $\frac{B_1}{B_2}$ can be determined.

Ex. Magnetic moments of two identical magnets are M and $2M$ respectively. Both are combined in such a way that their similar poles are same side. The time period in this is case ' T_1 '. If polarity of one of the magnets is reversed its period becomes ' T_2 ' then find out ratio of their time periods respectively.

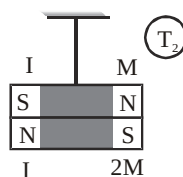


$$M_{\text{system}} = 2M + M = 3M$$

$$I_{\text{system}} = 2I$$

$$T = 2\pi \sqrt{\frac{I}{MB_H}} \quad (I_{\text{system}} \rightarrow \text{same}, B_H \rightarrow \text{same}) \quad \boxed{T \propto \frac{1}{\sqrt{M}}};$$

$$\frac{T_1}{T_2} = \sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{M}{3M}} = \frac{1}{\sqrt{3}}$$



$$M_{\text{system}} = 2M - M = M$$

$$I_{\text{system}} = 2I$$

Ex. A magnet is suspended in such a way when it oscillates in the horizontal plane. It makes 20 oscillations per minute at a place where dip angle is 30° and 15 oscillations per min at a place where dip angle is 60° . Find the ratio of the total earth's magnetic field at the two places.

Sol. $f = \frac{1}{2\pi} \sqrt{\frac{MB_H}{I}}$

$$\Rightarrow f^2 = \frac{1}{4\pi^2} \cdot \frac{MB \cos \theta}{I} \quad \text{I and M are same in given cases}$$

$$\frac{B_1}{B_2} = \frac{f_1^2}{f_2^2} \times \frac{\cos \theta_2}{\cos \theta_1} = \frac{20 \times 20}{15 \times 15} \times \frac{\cos 60^\circ}{\cos 30^\circ} = \frac{16}{9\sqrt{3}}$$

NEUTRAL POINT

It is a point where net magnetic field is zero.

At this point magnetic field of bar magnet or current carrying coil or current carrying wire is just neutralised by magnetic field of earth. (B_H)

A compass needle placed at this neutral point can set itself in any direction.

Location of Neutral Points :

- (a) **When N-pole of magnet directed towards North :-** Two neutral points symmetrically located on equatorial line of magnet. Let distance of each neutral point from centre of magnet is 'y' then

$$B_{eq} = B_H$$

$$B_H = \frac{\mu_0}{4\pi} \cdot \frac{M}{(y^2 + \ell^2)^{3/2}}$$

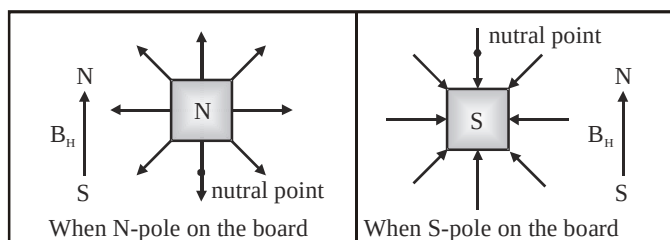
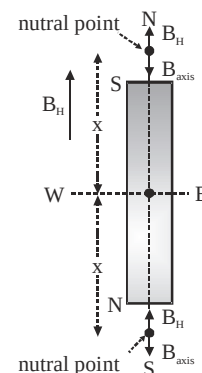
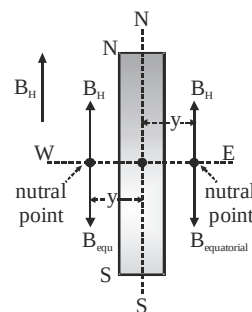
$$\frac{\mu_0}{4\pi} \cdot \frac{M}{y^3} = B_H \quad (\text{If } y \gg \ell)$$

- (b) **When S-pole of magnet directed towards North :-** Two neutral points symmetrically located on the axial line of magnet. Let distance of each neutral point from centre of the magnet is x, then

$$B_{axis} = B_H \Rightarrow B_H = \frac{\mu_0}{4\pi} \cdot \frac{2Mx}{(x^2 - \ell^2)^2}$$

$$\frac{\mu_0}{4\pi} \cdot \frac{2M}{x^3} = B_H \quad (\text{If } x \gg \ell)$$

- (c) If magnet is held vertically on the board, then only one neutral point is obtained on the horizontal board.



Ex. The magnetic field at a point x on the axis of a small bar magnet is equal to the field at a point y on the equator of the same magnet. Find the ratio of the distances of x and y from the centre of the magnet.

Sol. $B_{axis} = B_{equatorial} \Rightarrow \frac{\mu_0}{4\pi} \frac{2M}{x^3} = \frac{\mu_0}{4\pi} \frac{M}{y^3}$

$$\Rightarrow \frac{2}{x^3} = \frac{1}{y^3} \Rightarrow \frac{x^3}{y^3} = \frac{2}{1}$$

$$\Rightarrow \frac{x}{y} = 2^{1/3}$$

Ex. A coil of 0.1 m radius and 100 turns placed perpendicular magnetic meridian. When current of 2 ampere is flow through the coil then the neutral point is obtained at the centre. Find out magnetising field of earth.

Sol. Magnetic field at centre of coil $B = \frac{\mu_0 NI}{2R} = \mu_0 H_c$ ($\because H_c = H_e$)

$$\text{Magnetising field of earth } H_e = \frac{NI}{2R} = \frac{100 \times 2}{2 \times 0.1} = 1000 \text{ A/m}$$

Ex. A short magnet of moment 6.75 A-m^2 produces a neutral point on its axis. If the horizontal component of earth's magnetic field $5 \times 10^{-5} \text{ Wb/m}^2$, Calculate the distance of the neutral point from the centre.

Sol. $B_H = \frac{2KM}{d^3} \Rightarrow d = \left(\frac{2KM}{B_H} \right)^{1/3} = \left(\frac{2 \times 10^{-7} \times 6.75}{5 \times 10^{-5}} \right)^{1/3} = 0.3 \text{ m} = 30 \text{ cm}$

GEO-MAGNETISM

Important definitions :

- (a) **Geographic axis** : It is a straight line passing through the geographical poles of the earth. It is also called axis of rotation or polar axis of the earth.
- (b) **Geographic Meridian (GM)** : It is a vertical plane at any place which passing through geographical axis of the earth.

- (c) **Geographic equator** : It is a great circle on the surface of the earth, in a plane perpendicular to the geographic axis. All the points on the geographic equator are at equal distance from the geographic poles.

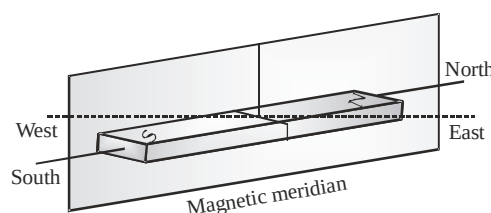
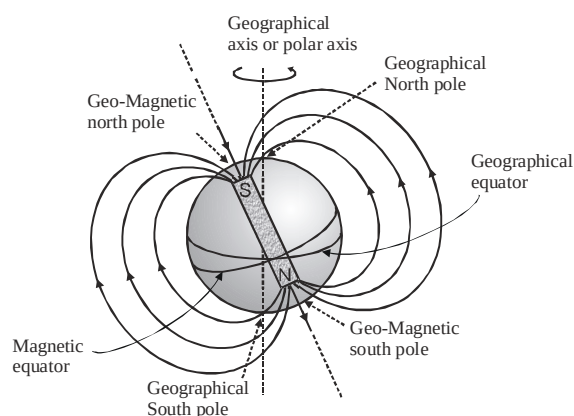
A great plane which passes through geographic equator and perpendicular to the geographic axis called **geographic equatorial plane**. This plane cuts the earth in two equal parts, a part has geographic north called **northern hemisphere (NHS)** and another part has geographic south called **southern hemisphere (SHS)**.

- (d) **Magnetic axis** : It is a straight line passing through magnetic poles of the earth. It is inclined to the geographic axis at nearly 17° .

- (e) **Magnetic Meridian (MM)** : (i) It is a vertical plane at any place which passing through magnetic axis of the earth. (ii) It is a vertical plane at any place which passing through axis of free suspended bar magnet or magnetic needle.

(iii) It is a vertical plane at any place which contains all the magnetic field lines of earth of that place.

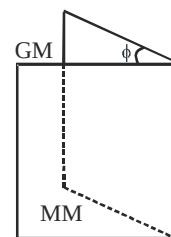
- (f) **Magnetic equator** : It is a great circle on the surface of the earth, in a plane perpendicular to the magnetic axis. All the points on the magnetic equator are at equal distance from the magnetic poles.



MAIN ELEMENTS OF EARTH'S MAGNETIC FIELD

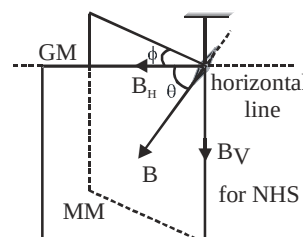
Angle of declination (ϕ)

At a given place the acute angle between geographic meridian and the magnetic meridian is called angle of declination, i.e. at a given place it is the angle between the geographical north south direction and the direction indicated by a magnetic compass needle in its equilibrium.



Angle of dip (θ)

- (i) It is an angle which the direction of resultant magnetic field of the earth subtends with the horizontal line in magnetic meridian at the given place.
- (ii) It is an angle which the axis of freely suspended magnetic needle (up or down) subtends with the horizontal line in magnetic meridian at a given place.

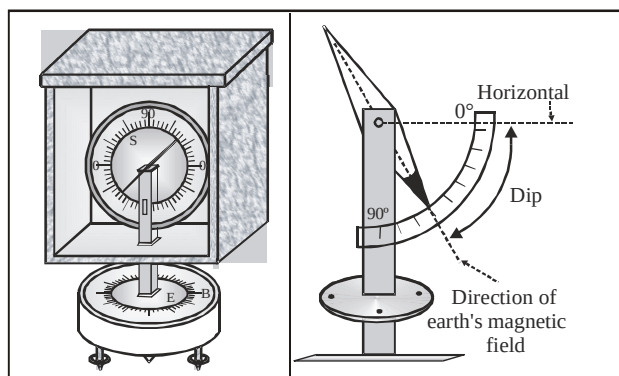


In northern hemisphere, north pole of freely suspended magnetic needle will dip downwards i.e. towards the earth surface. In southern hemisphere, south pole of freely suspended magnetic needle will dip downwards i.e. towards the earth surface.

Dip circle : Angle of dip at a place is measured by the instrument called 'Dip-circle' in which a magnetic needle is free to rotate in vertical plane. About its horizontal axis. The ends of the needle move over a vertical scale graduated in degree.

Horizontal component of earth magnetic field (B_H)

Horizontal component of earth magnetic field at a given place is the component of resultant magnetic field of the earth along the horizontal line in magnetic meridian.



$$B_H = B \cos \theta \quad \text{and} \quad B_V = B \sin \theta \quad \dots\dots(1)$$

$$\text{so that } \tan \theta = \frac{B_V}{B_H} \quad \text{and} \quad B = \sqrt{B_H^2 + B_V^2} \quad \dots\dots(2)$$

At magnetic poles $\theta = 90^\circ$

$B_H = 0$ and only B_V exist

❖ At magnetic equator $\theta = 0^\circ$

$B_V = 0$ and only B_H exist

ϕ decides the plane in which magnetic field lies at any place, (ϕ) and (θ) decides the direction of magnetic field and (θ) and (B_H) decides the magnitude of the field.

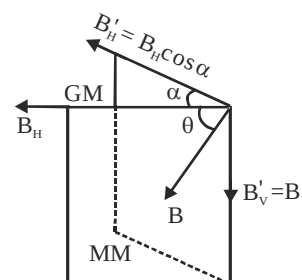
Apparent angle of dip (θ') :

When the plane of vertical scale of dip circle is in the magnetic meridian, the needle rest in the direction of earth's magnetic field. The angle made by the needle with the horizontal is called true dip or actual dip. If the plane of vertical scale of dip circle not kept in magnetic meridian, then the needle will not indicate the correct direction of earth magnetic field.

In this situation the angle made by the needle with the horizontal is called the apparent angle of dip. Suppose the dip circle is set at an angle α to the magnetic meridian. Effective horizontal component in this plane will be $B_H \cos \alpha$ and no effect on vertical component B_V

$$\text{Apparent angle of dip } \tan \theta' = \frac{B'_V}{B'_H}$$

$$\Rightarrow \tan \theta' = \frac{B_V}{B_H \cos \alpha} \Rightarrow \tan \theta' = \frac{\tan \theta}{\cos \alpha}$$



❖ For a vertical plane other than magnetic meridian

$$\alpha > 0 \Rightarrow \cos \alpha < 1 \Rightarrow \tan \theta' > \tan \theta \Rightarrow \theta' > \theta,$$

so apparent angle of dip is always more than actual angle of dip at any place.

❖ For a vertical plane perpendicular to magnetic meridian

$$\alpha = 90^\circ \Rightarrow \tan \theta' = \frac{\tan \theta}{\cos 90^\circ} = \infty$$

$\theta' = 90^\circ$, so in a plane perpendicular to magnetic meridian dip needle becomes just verticle.

Ex. At a certain place, the horizontal component of earth's magnetic field is $\sqrt{3}$ times of the vertical component. What the angle of dip at that place.

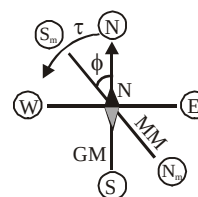
$$\text{Sol. } B_H = \sqrt{3} B_V, \tan \theta = \frac{B_V}{B_H} = \frac{B_V}{\sqrt{3} B_V} = \frac{1}{\sqrt{3}} = \tan 30^\circ \Rightarrow \theta = 30^\circ$$

Ex. A compass needle of magnetic moment is 60 A-m^2 pointing towards geographical north at a certain place where the horizontal component of earth's magnetic field is $40 \mu\text{T}$, experiences a torque $1.2 \times 10^{-3} \text{ N-m}$. What is the declination of that place.

$$\text{Sol. } \tau = MB \sin \phi$$

$$\Rightarrow \sin \phi = \frac{\tau}{MB} = \frac{1.2 \times 10^{-3}}{24 \times 10^{-4}} = \frac{1}{2}$$

$$\Rightarrow \phi = 30^\circ$$



Ex. If the dip circle is set at 45° to the magnetic meridian, then the apparent dip is 30° . Calculate the true dip.

Sol. $\tan \theta' = \frac{\tan \theta}{\cos \alpha}$

$$\tan \theta = \tan \theta' \cos \alpha = \tan 30^\circ \cos 45^\circ = \frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{2}}$$

$$\tan \theta = \frac{1}{\sqrt{6}} \Rightarrow \theta = \tan^{-1} \left(\frac{1}{\sqrt{6}} \right)$$

Ex. A magnetic needle is free to rotate in a vertical plane and that plane makes an angle of 60° with magnetic meridian. If the needle stays in a direction making an angle of $\tan^{-1} \left(\frac{2}{\sqrt{3}} \right)$ with the horizontal direction, what would be the actual dip at that place ?

Sol. $\tan \theta = \tan \theta' \cos \alpha \quad (\because \theta' = \tan^{-1} \left(\frac{2}{\sqrt{3}} \right), \alpha = 60^\circ)$

$$\therefore \tan \theta = \tan \left(\tan^{-1} \frac{2}{\sqrt{3}} \right) \cos 60^\circ$$

$$\tan \theta = \frac{2}{\sqrt{3}} \times \frac{1}{2} = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$$

Ex. A 1-meter long narrow solenoid having 1000 turns is placed in magnetic meridian. Find the current in the solenoid which neutralises the earth's horizontal field of 0.36 oersted at the centre of the solenoid.

Sol. The magnetic field intensity at the centre of solenoid is $H = ni = 1000 i \text{ A/m} = 4\pi i \text{ oersted}$
($\because 1 \text{ amp/meter} = 4\pi \times 10^{-3} \text{ oersted}$)

Since it neutralises the earth's field of 0.36 oersted, it is equal and opposite to the earth's field.

$$\therefore 4\pi i = 0.36$$

$$\Rightarrow i = \frac{0.36}{4 \times 3.14} = 0.0286 \text{ ampere} = 28.6 \text{ milli-ampere or } 28.6 \text{ mA}$$

Ex. If θ_1 and θ_2 are angles of dip in two vertical planes at right angle to each other and θ is true dip then prove $\cot^2 \theta = \cot^2 \theta_1 + \cot^2 \theta_2$.

Sol. If the vertical plane in which dip is θ_1 subtends an angle α with meridian than other vertical plane in which dip is θ_2 and is perpendicular to first will make an angle of $90 - \alpha$ with magnetic meridian. If θ_1 and θ_2 are apparent dips then

$$\tan \theta_1 = \frac{B_V}{B_H \cos \alpha}$$

$$\tan \theta_2 = \frac{B_V}{B_H \cos(90 - \alpha)} = \frac{B_V}{B_H \sin \alpha}$$

$$\cot^2 \theta_1 + \cot^2 \theta_2 = \frac{1}{(\tan \theta_1)^2} + \frac{1}{(\tan \theta_2)^2} = \frac{B_H^2 \cos^2 \alpha + B_H^2 \sin^2 \alpha}{B_V^2} = \frac{B_H^2}{B_V^2} = \left(\frac{B \cos \theta}{B \sin \theta} \right)^2 = \cot^2 \theta$$

$$\text{So } \cot^2 \theta_1 + \cot^2 \theta_2 = \cot^2 \theta$$

Ex. Considering earth as a short bar magnet show that the angle of dip θ is related to magnetic latitude λ as $\tan\theta = 2\tan\lambda$

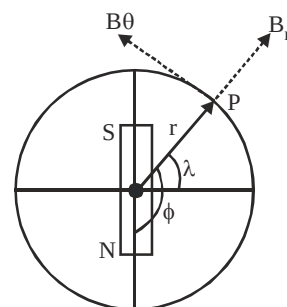
Sol. For a magnetic dipole the field components at point P (r, ϕ) are given as

$$B_r = \frac{\mu_0}{4\pi} \frac{2M \cos \phi}{r^3} \quad B_\theta = \frac{\mu_0}{4\pi} \frac{M \sin \phi}{r^3}$$

$$\tan \theta = \frac{B_v}{B_H} = -\frac{B_r}{B_\theta} = -\frac{\mu_0}{4\pi} \frac{2M \cos \phi}{r^3} \frac{4\pi r^3}{\mu_0 M \sin \phi}$$

$$\text{or } \tan \theta = -2 \cot \phi$$

$$\text{From figure } \phi = \frac{\pi}{2} + \lambda \quad \text{So } \tan \theta = -2 \cot \left(\frac{\pi}{2} + \lambda \right) \text{ or } \tan \theta = 2 \tan \lambda$$



Ex. At a certain location in Africa, a compass points 12° west of the geographic north. The north tip of the magnetic needle of a dip circle placed in the plane of the magnetic meridian points 60° above the horizontal. The horizontal component of the earth's field is measured to be 0.16 G. Specify the direction and magnitude of the earth's field at the location.

Sol. From formula, $B_H = B \cos \theta$

$$B = \frac{B_H}{\cos \theta} = B_H \sec \theta = 0.16 \times 2 = 0.32 \text{ G.}$$

The earth's field is 0.32 G, in direction 60° upwards from horizontal, in a plane (magnetic meridian) 12° West of geographical meridian.

Ex. A dip circle shows an apparent dip of 60° at a place where the true dip is 45° . If the dip circle is rotated through 90° what apparent dip will it show?

Sol. Let θ_1 and θ_2 be apparent dip shown by dip circle in two perpendicular positions then true dip θ is given by

$$\cot^2 \theta = \cot^2 \theta_1 + \cot^2 \theta_2 \text{ or } \cot^2 45^\circ = \cot^2 60^\circ + \cot^2 \theta_2$$

$$\text{or } \cot^2 \theta_2 = \frac{2}{3} \text{ or } \cot \theta_2 = 0.816 \text{ giving } \theta_2 = 51^\circ$$

IMPORTANT DEFINITIONS AND RELATIONS

Magnetising field or Magnetic intensity ($\vec{H}$)

Field in which a material is placed for magnetisation, called as magnetising field.

$$\text{Magnetising field } \vec{H} = \frac{\vec{B}_0}{\mu_0} = \frac{\text{magnetic field}}{\text{permeability of free space}}$$

SI Unit $\vec{H}$: ampere/meter

Intensity of magnetisation ($\vec{I}$)

When a magnetic material is placed in magnetising field then induced dipole moment per unit volume of

that material is known as intensity of magnetisation $\vec{I} = \frac{\vec{M}}{V}$

SI Unit : ampere/meter $[\because \frac{\vec{M}}{V} = \frac{IA}{V} = \frac{\text{ampere} \times \text{meter}^2}{\text{meter}^3}]$

Magnetic susceptibility (χ_m)

$$\chi_m = \frac{I}{H} \quad [\text{It is a scalar with no units and dimensions}]$$

Physically it represent the ease with which a magnetic material can be magnetised

A material with more χ_m , can be change into magnet easily.

Magnetic permeability μ

$$\mu = \frac{B_m}{H} = \frac{\text{total magnetic field inside the material}}{\text{magnetising field}}$$

It measures the degree to which a magnetic material can be penetrated (or permeated) by the magnetic field lines

$$\text{SI Unit of } \mu : \mu = \frac{B_m}{H} = \frac{\text{Wb/m}^2}{\text{A/m}} = \frac{\text{Wb}}{\text{A-m}} = \frac{\text{H-A}}{\text{A-m}} = \frac{\text{H}}{\text{m}}$$

$$[\because \phi = LI \therefore \text{weber} = \text{henry} - \text{ampere}]$$

$$\text{Relative permeability } \mu_r = \frac{\mu}{\mu_0}$$

It has no units and dimensions.

Relation between permeability and susceptibility

When a magnetic material is placed in magnetic field $\vec{B}_0$ for magnetisation then total magnetic field

in material $\vec{B}_m = \vec{B}_0 + \vec{B}_i$, where $\vec{B}_i$ = induced field.

$$\therefore \vec{B}_0 = \mu_0 \vec{H}; \vec{B}_i = \mu_0 \vec{I}$$

$$\therefore \vec{B}_m = \mu_0 \vec{H} + \mu_0 \vec{I} \Rightarrow \vec{B}_m = \mu_0 (\vec{H} + \vec{I}) = \mu_0 \vec{H} \left(1 + \frac{I}{H} \right)$$

$$\Rightarrow \frac{B}{H} = \mu_0 \left(1 + \frac{I}{H} \right) \Rightarrow \mu = \mu_0 (1 + \chi_m) \Rightarrow \mu_r = 1 + \chi_m$$

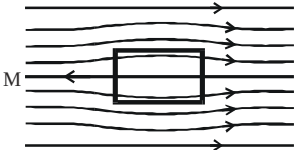
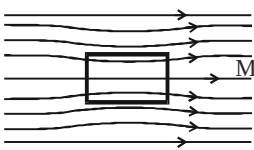
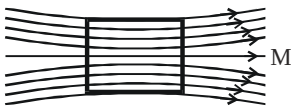
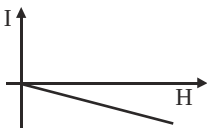
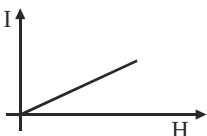
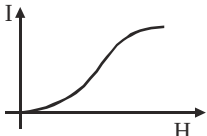
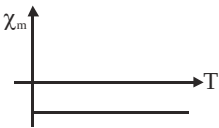
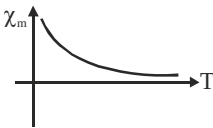
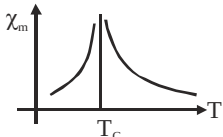
$$\text{for vacuum } \chi_m = 0, (\because \mu_r = 1)$$

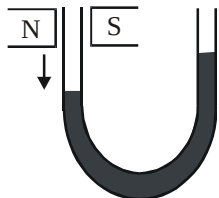
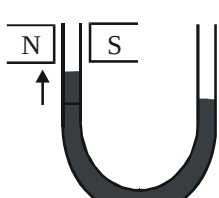
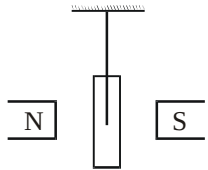
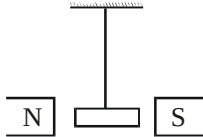
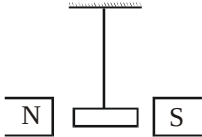
$$\text{at STP for air } \chi_m = 0.04$$

$$(\because \text{at S.T.P. for air } \mu_r = 1.04)$$

CLASSIFICATION OF MAGNETIC MATERIALS

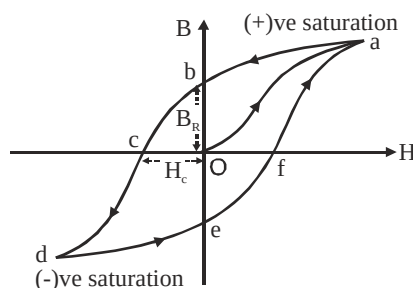
On the basis of magnetic properties of the materials [as magnetisation intensity (I), Susceptibility (χ) and relative permeability (μ_r)] Faraday divide these materials in three classes –

PROPERTIES	DIAMAGNETIC	PARAMAGNETIC	FERROMAGNETIC
Cause of magnetism	Orbital motion of electrons	Spin motion of electrons	Formation of domains
Substance placed in uniform magnetic field.	Poor magnetisation in opposite direction. Here $B_m < B_0$ 	Poor magnetisation in same direction. Here $B_m > B_0$ 	Strong magnetisation in same direction. Here $B_m \gg B_0$ 
$I - H$ curve	$I \rightarrow$ Small, negative, varies linearly with field 	$I \rightarrow$ Small, positive, varies linearly with field 	$I \rightarrow$ very large, positive & varies non-linearly with field 
$\chi_m - T$ curve	$\chi_m \rightarrow$ small, negative & temperature independent $\chi_m \propto T^0$ 	$\chi_m \rightarrow$ small, positive & varies inversely with temp. $\chi_m \propto \frac{1}{T}$ (Curie law) 	$\chi_m \rightarrow$ very large, positive & temp. dependent $\chi_m \propto \frac{1}{T - T_C}$ (Curie Weiss law) (for $T > T_C$) (T_C = Curie temperature)  $T_C(\text{Iron}) = 770^\circ\text{C}$ or 1043K
μ_r	$(\mu < \mu_0) \quad 1 > \mu_r > 0$	$2 > \mu_r > 1 \quad (\mu > \mu_0)$	$\mu_r \gg 1 \quad (\mu \gg \mu_0)$
Magnetic moment of single atom	Atoms do not have any permanent magnetic moment	Atoms have permanent magnetic moment which are randomly oriented. (i.e. in absence of external magnetic field the magnetic moment of whole material is zero)	Atoms have permanent magnetic moment which are organised in domains.

PROPERTIES	DIAMAGNETIC	PARAMAGNETIC	FERROMAGNETIC
Behaviour of substance in Nonuniform magnetic field	It moves from stronger to weaker magnetic field.  Weak Field  Strong Field  Level depressed in that limb	It moves with weak force from weaker magnetic field to stronger magnetic field.  Weak Field  Strong Field  Level slightly rises	Strongly attract from weaker magnetic field to stronger magnetic field.  Weak Field  Strong Field
When rod of material is suspended between poles of magnet.	It becomes perpendicular to the direction of external magnetic field. 	If there is strong magnetic field in between the poles then rod becomes parallel to the magnetic field. 	Weak magnetic field between magnetic poles can make rod parallel to field direction. 
Magnetic moment of substance in presence of external magnetic field	Value $\vec{M}$ is very less and opposite to $\vec{H}$.	Value $\vec{M}$ is low but in direction of $\vec{H}$.	$\vec{M}$ is very high and in direction of $\vec{H}$.
Examples	Bi, Cu, Ag, Pb, H_2O , Hg, H_2 , He, Ne, Au, Zn, Sb, NaCl, Diamond. (May be found in solid, liquid or gas).	Na, K, Mg, Mn, Sn, Pt, Al, O_2 (May be found in solid, liquid or gas.)	Fe, Co, Ni all their alloys, Fe_3O_4 , Gd, Alnico, etc. (Normally found only in solids) (crystalline solids)

MAGNETIC HYSTERESIS

Only Ferromagnetic materials show magnetic hysteresis, when Ferromagnetic material is placed in external magnetic field for magnetisation then B increases with H non-linearly along Oa . If H is again brought to zero then it decreases along path ab . Due to lagging behind of B with H this curve is known as hysteresis curve. [Lagging of B behind H is called hysteresis]



Cause of hysteresis : By removing external magnetising field ($H = 0$), the magnetic moment of some domains remains aligned in the applied direction of previous magnetising field which results into a residual magnetism.

- ◎ **Residual magnetism (ob) = B_r** $\equiv$ retentivity $\equiv$ remanence
Retentivity of a specimen is a measure of the magnetic field remaining in the ferromagnetic specimen when the magnetising field is removed.
- ◎ **Coercivity (oc) :** Coercivity is an measure of magnetising field required to destroy the residual magnetism of the ferromagnetic specimen.

Ferromagnetic materials

Soft magnetic materials

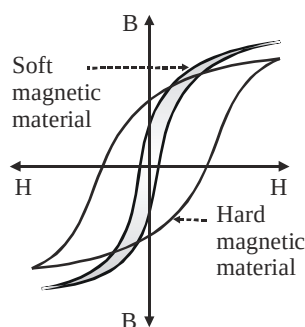
Low retentivity, low coercivity and small hysteresis loss.
suitable for making electromagnets, cores of transformers etc. Ex. Soft iron, (used in magnetic shielding)

Hard magnetic materials

High retentivity, high coercivity and large hysteresis loss
suitable for permanent magnet
Ex. Steel, Alnico

HYSTERESIS LOSS

- (i) The area of hysteresis loop for a ferromagnetic material is equal to the energy loss per cycle of magnetisation and demagnetisation per unit volume.



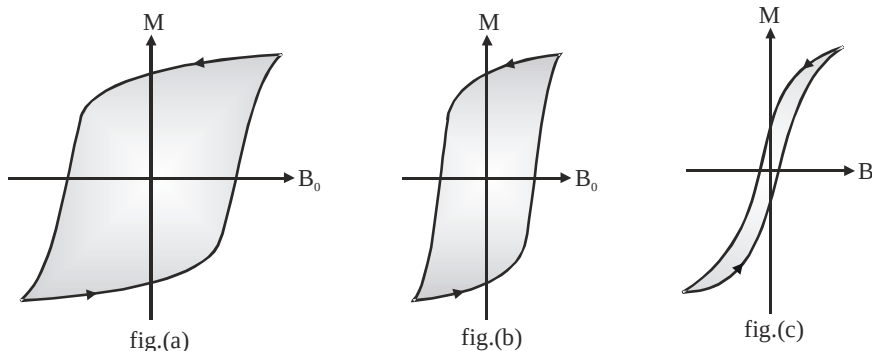
$$W_H = \oint B \cdot dH = \mu_0 \oint I \cdot dH$$

(ii) Its value is different for different materials.

(iii) The work done per cycle per unit volume of material is equal to the area of hysteresis loop.

$$\therefore \text{Total energy loss in material } W_H = V A n t \text{ joule} = \frac{V A n t}{J} \text{ calorie}$$

i.e $W_H = \text{volume of material} \times \text{area of hysteresis curve} \times \text{frequency} \times \text{time}.$



The materials of both (a) and (b) remain strongly magnetized when B_0 is reduced to zero. The material of (a) is also hard to demagnetize, it would be good for permanent magnets.

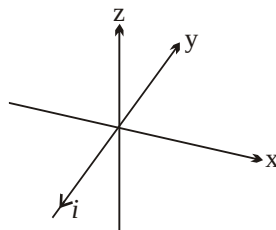
The material of (b) magnetizes and demagnetizes more easily, it could be used as a computer memory material.

The material of (c) would be useful for transformers and other alternating-current devices where zero hysteresis would be optimal.

EXERCISE (S-1)

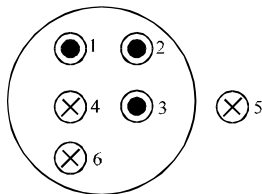
Biot savart law

1. Two long, straight wires, each carrying a current of 5 A, are placed along the X- and Y-axes respectively. The currents point along the positive directions of the axes. Find the magnetic field at the points (a) (1 m, 1 m), (b) (−1 m, 1 m), (c) (−1 m, −1 m) and (d) (1 m, −1 m).
MG0001
2. A circular loop of radius 4.0 cm is placed in a horizontal plane and carries an electric current of 5.0 A in the clockwise direction as seen from above. Find the magnetic field
(a) At a point 3.0 cm above the centre of the loop.
(b) At a point 3.0 cm below the centre of the loop.
MG0002
3. Two concentric circular coils X and Y of radii 16 cm and 10 cm, respectively, lie in the same vertical plane containing the north to south direction. Coil X has 20 turns and carries a current of 16A ; coil Y has 25 turns and carries a current of 18 A. The sense of the current in X is anticlockwise, and clockwise in Y, for an observer looking at the coils facing west. Give the magnitude and direction of the net magnetic field due to the coils at their centre.
(NCERT)
MG0003
4. A current element $\Delta \vec{\ell} = \Delta x \hat{i} - \Delta y \hat{j}$ carries 10 A current. It is placed at origin. Calculate magnetic field at point 'P' which is at position vector $\vec{r} = (\hat{i} + \hat{j})$ m with respect to origin. (where $\Delta x = \Delta y = 1\text{mm}$)
MG0004
5. A circular loop of radius r carries a current i . How should a long, straight wire carrying a current $4i$ be placed in the plane of the circle so that the magnetic field at the centre becomes zero ?
MG0005
6. A long straight wire carries a current of 10 A directed along the negative y-axis as shown in figure. A uniform magnetic field B_0 of magnitude 10^{-6} T is directed parallel to the x-axis. What is the resultant magnetic field at the following points?
(a) $x = 0, z = 2$ m; (b) $x = 2$ m, $z = 0$; (c) $x = 0, z = -0.5$ m

**MG0006**

Ampere's law

7. Six wires of current $I_1 = 1\text{A}$, $I_2 = 2\text{A}$, $I_3 = 3\text{A}$, $I_4 = 1\text{A}$, $I_5 = 5\text{A}$ and $I_6 = 4\text{A}$ cut the page perpendicularly at the points 1, 2, 3, 4, 5 and 6 respectively as shown in the figure. Find the value of the integral $\oint \vec{B} \cdot d\vec{\ell}$ around the closed path.



MG0007

Motion of charged particle

8. A charged particle (charge q , mass m) has velocity v_0 at origin in $+x$ direction. In space there is a uniform magnetic field B in $-z$ direction. Find the y coordinate of particle when it crosses y axis. **MG0008**
9. A beam of protons with a velocity $4 \times 10^5 \text{ m/s}$ enters a uniform magnetic field of 0.3 T at an angle of 60° to the magnetic field. Find the radius of the helical path taken by the proton beam. Also find the pitch of the helix (which is the distance travelled by a proton in the beam parallel to the magnetic field during one period of rotation). **MG0009**
10. An electron emitted by a heated cathode and accelerated through a potential difference of 2.0 kV , enters a region with uniform magnetic field of 0.15 T . Determine the trajectory of the electron if the field (a) is transverse to its initial velocity, (b) makes an angle of 30° with the initial velocity. **(NCERT)**

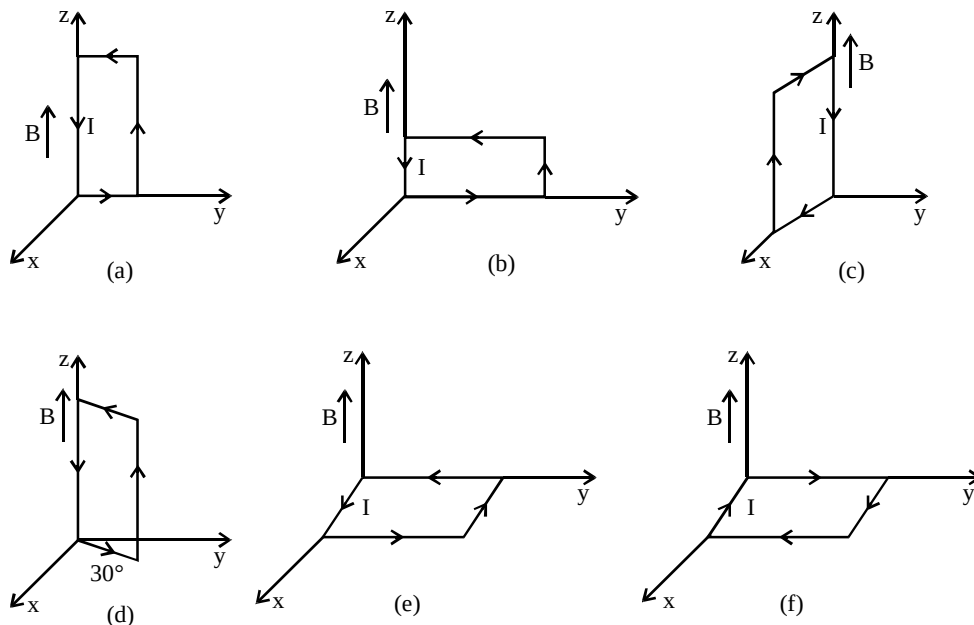
MG0010

Ampere force & torque

11. A straight horizontal conducting rod of length 0.45 m and mass 60g is suspended by two vertical wires at its ends. A current of 5.0 A is set up in the rod through the wires. **(NCERT)**
- (a) What magnetic field should be set up normal to the conductor in order that the tension in the wires is zero?
- (b) What will be the total tension in the wires if the direction of current is reversed keeping the magnetic field same as before? [Ignore the mass of the wires]. $g = 9.8 \text{ ms}^{-2}$. **MG0011**
12. (a) A circular coil of 30 turns and radius 8.0 cm carrying a current of 6.0 A is suspended vertically in a uniform horizontal magnetic field of magnitude 1.0 T . The field lines make an angle of 60° with the normal of the coil. Calculate the magnitude of the counter torque that must be applied to prevent the coil from turning. **(NCERT)**
- (b) Would your answer change, if the circular coil in (a) were replaced by a planar coil of some irregular shape that encloses the same area? (All other particulars are also unaltered.) **MG0012**

13. A uniform magnetic field of 3000 G is established along the positive z-direction. A rectangular loop of sides 10 cm and 5 cm carries a current of 12 A. What is the torque on the loop in the different cases shown in figure. What is the force on each case? Which case corresponds to stable equilibrium?

(NCERT)



MG0013

14. A circular coil of 20 turns and radius 10 cm is placed in a uniform magnetic field of 0.10 T normal to the plane of the coil. If the current in the coil is 5.0 A, what is the

(NCERT)

(a) total, torque on the coil,

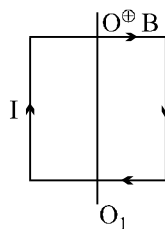
(b) total force on the coil,

(c) average force on each electron in the coil due to the magnetic field?

(The coil is made of copper wire of cross-sectional area 10^{-5} m^2 , and the free electron density in copper is given to be about 10^{29} m^{-3} .)

MG0014

15. A square current carrying loop made of thin wire and having a mass $m = 10\text{g}$ can rotate without friction with respect to the vertical axis OO_1 , passing through the centre of the loop at right angles to two opposite sides of the loop. The loop is placed in a homogeneous magnetic field with an induction $B = 10^{-1} \text{ T}$ directed at right angles to the plane of the drawing. A current $I = 2\text{A}$ is flowing in the loop. Find the period of small oscillations that the loop performs about its position of stable equilibrium.



MG0015

16. Two moving coil meters. M_1 and M_2 have the following particulars :

(NCERT)

$$R_1 = 10 \, \Omega, N_1 = 30,$$

$$A_1 = 3.6 \times 10^{-3} \, \text{m}^2, B_1 = 0.25 \, \text{T}$$

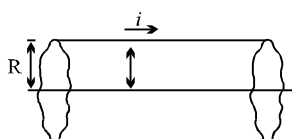
$$R_2 = 14 \, \Omega, N_2 = 42, A_2 = 1.8 \times 10^{-3} \, \text{m}^2, B_2 = 0.50 \, \text{T}$$

(The spring constants are identical for the two meters). Determine the ratio of (a) current sensitivity and (b) voltage sensitivity of M_2 and M_1 .

MG0016

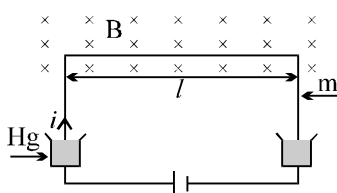
EXERCISE (S-2)

1. A cylindrical conductor of radius R carries a current along its length. The current density J , however, it is not uniform over the cross section of the conductor but is a function of the radius according to $J = br$, where b is a constant. Find an expression for the magnetic field B .
 (a) at $r_1 < R$ & (b) at distance $r_2 > R$, measured from the axis



MG0017

2. A U-shaped wire of mass m and length l is immersed with its two ends in mercury (see figure). The wire is in a homogeneous field of magnetic induction $\mathbf{B}$. If a charge, that is, a current pulse $q = \int i dt$, is sent through the wire, the wire will jump up. Calculate, the height h that the wire reaches, the size of the charge or current pulse, assuming that the time of the current pulse is very small in comparison with the time of flight. Make use of the fact that impulse of force equals $\int F dt$, which equals mv . Evaluate q for $B = 0.1 \text{ Wb/m}^2$, $m = 10 \text{ gm}$, $l = 20 \text{ cm}$ & $h = 3 \text{ meters}$. [$g = 10 \text{ m/s}^2$]



MG0018

3. A particle of mass $1 \times 10^{-26} \text{ kg}$ and charge $+1.6 \times 10^{-19} \text{ C}$ travelling with a velocity $1.28 \times 10^6 \text{ m/s}$ in the $+x$ direction enters a region in which a uniform electric field $\mathbf{E}$ and a uniform magnetic field of induction $\mathbf{B}$ are present such that $E_x = E_y = 0$, $E_z = -102.4 \text{ kV/m}$ and $B_x = B_z = 0$, $B_y = 8 \times 10^{-2} \text{ weber/m}^2$. The particle enters this region at the origin at time $t = 0$. Determine the location (x , y and z coordinates) of the particle at $t = 5 \times 10^{-6} \text{ s}$. If the electric field is switched off at this instant (with the magnetic field still present), what will be the position of the particle at $t = 7.45 \times 10^{-6} \text{ s}$?

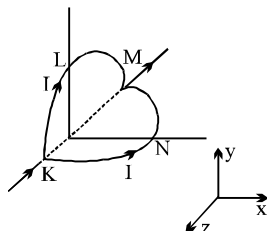
MG0019

4. Two coils each of 100 turns are held such that one lies in the vertical plane with their centres coinciding. The radius of the vertical coil is 20 cm and that of the horizontal coil is 30 cm. How would you neutralize the magnetic field of the earth at their common centre? What is the current to be passed through each coil? Horizontal component of earth's magnetic induction = 3.49×10^{-5} T and angle of dip = 30° .

MG0020

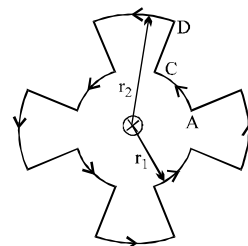
5. A circular loop of radius R is bent along a diameter and given a shape as shown in the figure. One of the semicircles (KNM) lies in the $x-z$ plane and the other one (KLM) in the $y-z$ plane with their centers at the origin. Current I is flowing through each of the semicircles as shown in figure.
- (i) A particle of charge q is released at the origin with a velocity $\mathbf{v} = -v_0 \hat{i}$. Find the instantaneous force $\mathbf{f}$ on the particle. Assume that space is gravity free.
- (ii) If an external uniform magnetic field $B \hat{j}$ is applied, determine the forces F_1 and F_2 on the semicircles KLM and KNM due to this field and the net force F on the loop.

[JEE 2000]



MG0021

6. A current of 10A flows around a closed path in a circuit which is in the horizontal plane as shown in the figure. The circuit consists of eight alternating arcs of radii $r_1 = 0.08$ m and $r_2 = 0.12$ m. Each arc subtends the same angle at the centre.
- (a) Find the magnetic field produced by this circuit at the centre.
- (b) An infinitely long straight wire carrying a current of 10A is passing through the centre of the above circuit vertically with the direction of the current being into the plane of the circuit. What is the force acting on the wire at the centre due to the current in the circuit? What is the force acting on the arc AC and the straight segment CD due to the current at the centre?



[JEE 2001]

MG0022

7. A particle of charge $+q$ and mass m moving under the influence of a uniform electric field $E \hat{i}$ and a magnetic field $B \hat{k}$ enters in I quadrant of a coordinate system at a point $(0, a)$ with initial velocity $v \hat{i}$ and leaves the quadrant at a point $(2a, 0)$ with velocity $-2v \hat{j}$. Find
- (a) Magnitude of electric field
- (b) Rate of work done by the electric field at point $(0, a)$
- (c) Rate of work done by both the fields at $(2a, 0)$.

MG0023

8. Electric charge q is uniformly distributed over a rod of length l . The rod is placed parallel to a long wire carrying a current i . The separation between the rod and the wire is a . Find the force needed to move the rod along its length with a uniform velocity v .

MG0024

9. There exists a uniform magnetic and electric field of magnitude 1 T and 1 V/m respectively along positive y-axis. A charged particle of mass 1 kg and charge 1 C moving with velocity 1 m/sec along x-axis is at origin at $t = 0$. If the coordinates of particle at time π seconds is given as (X, Y, Z) in meter, then find the value of $(XY + YZ + ZX) \times \frac{5}{\pi^2}$.

MG0025

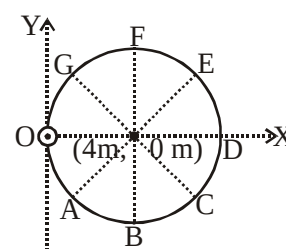
10. A proton beam passes without deviation through a region of space where there are uniform transverse mutually perpendicular electric and magnetic field with E and B . Then the beam strikes a grounded target. Find the force imparted by the beam on the target if the beam current is equal to I .

MG0026

11. An infinite uniform current carrying wire is kept along z-axis, carrying current I_0 in the direction of the positive z-axis. OABCDEFG represents a circle (where all the points are equally spaced), whose centre at point

$(4\text{m}, 0\text{ m})$ and radius 4 m as shown in the figure. If $\int_{DEF} \vec{B} \cdot d\vec{\ell} = \frac{\mu_0 I_0}{k}$ in S.I.

unit, then find the value of k .

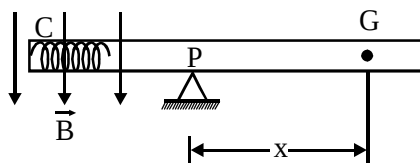


MG0027

12. A rod of length ℓ and total charge 'q' which is uniformly distributed is rotating with angular velocity ω about an axis passing through the centre of rod and perpendicular to rod. Find the magnitude of magnetic dipole moment (in Amp. m^2) of rod. (Take : $q = 4\text{C}$, $\omega = 6\text{ rad/s}$ and $\ell = 2\text{m}$.)

MG0028

13. A uniform wooden bar of mass $\frac{\pi}{100}\text{ kg}$ and radius of cross-section 10cm carries a light coil C of 100 turns. The bar is smoothly pivoted at P. If the coil carries a current 2A and subjected to an external magnetic field 10^{-2}T , the bar remains in equilibrium. Find the distance x (in cm) of the C.M. of the rod from the pivot.

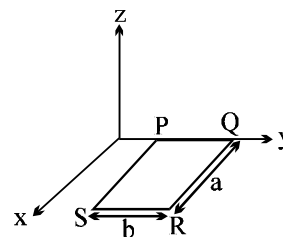


MG0029

14. 3 infinitely long thin wires each carrying current i in the same direction, are in the x - y plane of a gravity free space. The central wire is along the y -axis while the other two are along $x = \pm d$.
- Find the locus of the points for which the magnetic field B is zero.
 - If the central wire is displaced along the z -direction by a small amount & released, show that it will execute simple harmonic motion. If the linear density of the wires is λ , find the frequency of oscillation.

MG0030

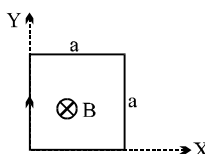
15. A rectangular loop PQRS made from a uniform wire has length a , width b and mass m . It is free to rotate about the arm PQ, which remains hinged along a horizontal line taken as the y -axis (see figure). Take the vertically upward direction as the z -axis. A uniform magnetic field $\vec{B} = (3\hat{i} + 4\hat{k})B_0$ exists in the region. The loop is held in the x - y plane and a current I is passed through it. The loop is now released and is found to stay in the horizontal position in equilibrium. [JEE 2002]



- What is the direction of the current I in PQ?
- Find the magnetic force on the arm RS.
- Find the expression for I in terms of B_0 , a , b and m .

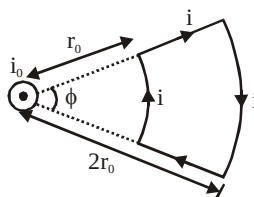
MG0031

16. A rectangular loop of wire is oriented with the left corner at the origin, one edge along X -axis and the other edge along Y -axis as shown in the figure. A magnetic field is into the page and has a magnitude that is given by $\beta = \alpha y$ where α is constant. Find the total magnetic force on the loop if it carries current i .



MG0032

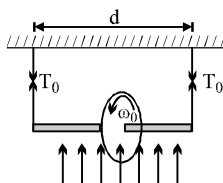
17. A closed loop carrying a current i is placed so that its plane is perpendicular to the long straight conductor which carries a current i_0 as shown in the figure. The torque acting on the current loop is $\alpha \mu_0$ N-m. Then find the value of α . (Given $i = 2$ A, $i_0 = 2$ A, $r_0 = 2\pi$ m, $\phi = 60^\circ$)



MG0033

18. A wheel of radius R having charge Q , uniformly distributed on the rim of the wheel is free to rotate about a light horizontal rod. The rod is suspended by light inextensible strings and a magnetic field B is applied as shown in the figure. The initial tensions in the strings are T_0 . If the breaking tension of the strings are $\frac{3T_0}{2}$, find the maximum angular velocity ω_0 with which the wheel can be rotate.

[JEE 2003]



MG0034

19. In a moving coil galvanometer, torque on the coil can be expressed as $\tau = ki$, where i is current through the wire and k is constant. The rectangular coil of the galvanometer having numbers of turns N , area A and moment of inertia I is placed in magnetic field B . Find
- k in terms of given parameters N , I , A and B .
 - the torsional constant of the spring, if a current i_0 produces a deflection of $\pi/2$ in the coil in reaching equilibrium position.
 - the maximum angle through which coil is deflected, if charge Q is passed through the coil almost instantaneously. (Ignore the damping in mechanical oscillations)

[JEE 2005]

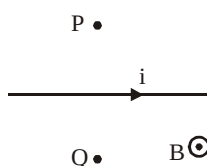
MG0035

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Biot savart law

1. A long, straight wire carrying a current of 1.0 A is placed horizontally in a uniform magnetic field $B = 1.0 \times 10^{-5}$ T pointing vertically upward (figure). The magnitude of the resultant magnetic field at the points P and Q, both situated at a distance of 2.0 cm from the wire in the same horizontal plane are respectively



- (A) zero, 20 μ T (B) 20 μ T, zero (C) zero, zero (D) 20 μ T, 20 μ T

MG0036

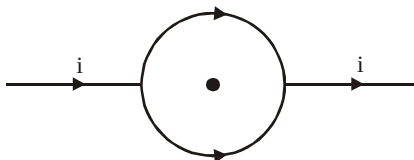
2. A horizontal overhead powerline is at a height of 4m from the ground and carries a current of 100 A from east to west. The magnetic field directly below it on the ground is ($\mu_0 = 4\pi \times 10^{-7} \text{TmA}^{-1}$)
- (A) 2.5×10^{-7} T southward (B) 5×10^{-6} T northward
(C) 5×10^{-6} T southward (D) 2.5×10^{-7} T northward

MG0037

3. Two parallel wires carry equal currents of 10 A along the same direction and are separated by a distance of 2.0 cm. Find the magnetic field at a point which is 2.0 cm away from each of these wires.
- (A) 3.4×10^{-4} T in a direction parallel to the plane of the wires and perpendicular to the wires
(B) 1.7×10^{-4} T in a direction parallel to the plane of the wires and parallel to the wires
(C) 1.7×10^{-4} T in a direction perpendicular to the plane of the wires and perpendicular to the wires
(D) 3.4×10^{-4} T in a direction perpendicular to the plane of the wires and parallel to the wires

MG0038

4. A conducting circular loop of radius a is connected to two long, straight wires. The straight wires carry a current i as shown in figure. Find the magnetic field B at the centre of the loop.



- (A) zero (B) $\frac{\mu_0 I}{2a}$ (C) $\frac{\mu_0 I}{a}$ (D) $\frac{\mu_0 I}{a} + \frac{\mu_0 I}{2\pi a}$

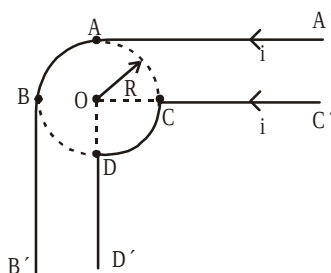
MG0039

5. A piece of wire carrying a current of 6.00 A is bent in the form of a circular arc of radius 10.0 cm, and it subtends an angle of 120° at the centre. Find the magnetic field B due to this piece of wire at the centre.

(A) zero (B) 1.26×10^{-5} T (C) 5×10^{-5} T (D) 7.2×10^{-5} T

MG0040

6. All straight wires are very long. Both AB and CD arc area of the same circle, both subtending right angles at the centre O. Then the magnetic field at O is



(A) $\frac{\mu_0 i}{4\pi R}$ (B) $\frac{\mu_0 i}{4\pi R} \sqrt{2}$ (C) $\frac{\mu_0 i}{2\pi R}$ (D) $\frac{\mu_0 i}{2\pi R} (\pi + 1)$

MG0041

7. Two identical long conducting wires AOB and COD are placed at right angles to each other. The wire AOB carries an electric current I_1 and COD carries a current I_2 . The magnetic field on a point lying at a distance d from O, in a direction perpendicular to the plane of the wires AOB and COD, will be given by-

(A) $\frac{\mu_0}{2\pi} \left(\frac{I_1 + I_2}{d} \right)^{1/2}$ (B) $\frac{\mu_0}{2\pi d} (I_1^2 + I_2^2)^{1/2}$ (C) $\frac{\mu_0}{2\pi d} (I_1 + I_2)$ (D) $\frac{\mu_0}{2\pi d} (I_1^2 + I_2^2)$

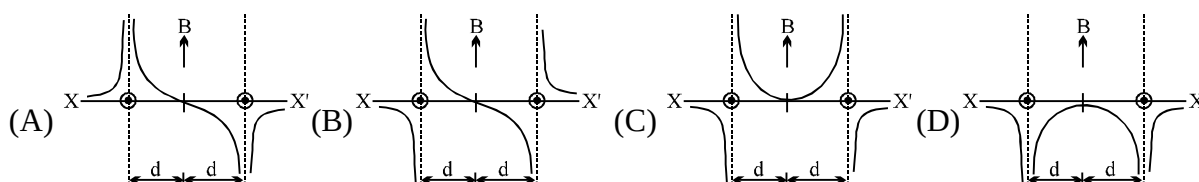
MG0042

8. The magnetic field due to a current carrying circular loop of radius 3 cm at a point on the axis at a distance of 4 cm from the centre is 54 μ T. What will be its value at the centre of the loop ?

(A) 250 μ T (B) 150 μ T (C) 125 μ T (D) 75 μ T

MG0043

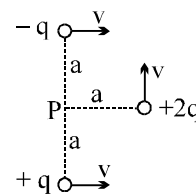
9. Two long parallel wires are at a distance 2d apart. They carry steady equal currents flowing out of the plane of the paper, as shown. The variation of the magnetic field B along the XX' is given by



MG0044

10. A charge $+2q$ moves vertically upwards with speed v , a second charge $-q$ moves horizontally to the right with the same speed v , and a third charge $+q$ moves horizontally to the right with the same speed v . The point P is located a perpendicular distance a away from each charge as shown in the figure. The magnetic field at point P is

- (A) Into the page with magnitude $\frac{\mu_0}{4\pi} \frac{2qv}{a^2}$
- (B) Into the page with magnitude $\frac{\mu_0}{4\pi} \frac{4qv}{a^2}$
- (C) Out of the page with magnitude $\frac{\mu_0}{4\pi} \frac{2qv}{a^2}$
- (D) Out the page with magnitude $\frac{\mu_0}{4\pi} \frac{4qv}{a^2}$



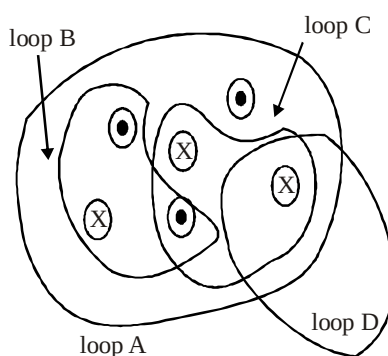
MG0045

11. Considering magnetic field along the axis of a circular loop of radius 1 meter, at what distance from the centre of the loop the magnetic field is $\frac{1}{2\sqrt{2}}$ times of its value at the centre ?
- (A) 1 m (B) 3 m (C) 5 m (D) 9 m

MG0046

Ampere's law

12. Consider six wires coming into or out of the page, all with the same current. Rank the line integral of the magnetic field (from most positive to most negative) taken counterclockwise around each loop shown.



- (A) $B > C > D > A$ (B) $B > C = D > A$ (C) $B > A > C = D$ (D) $C > B = D > A$

MG0047

13. **Statement-1** : Ampere law can be used to find magnetic field due to finite length of a straight current carrying wire.

Statement-2 : The magnetic field due to finite length of a straight current carrying wire is symmetric about the wire.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

MG0048

14. A long, cylindrical wire of radius b carries a current i distributed uniformly over its cross-section. Find the magnitude of the magnetic field at a point inside the wire at a distance a from the axis.

- (A) zero (B) $\frac{\mu_0 ib}{2\pi a^2}$ (C) $\frac{\mu_0 ia^2}{2\pi b^3}$ (D) $\frac{\mu_0 ia}{2\pi b^2}$

MG0049

15. A copper wire of diameter 1.6 mm carries a current of 20 A. Find the maximum magnitude of the magnetic field $\vec{B}$ due to this current.

- (A) 5.0 mT (B) 10 mT (C) 15 mT (D) 15.5 mT

MG0050

16. A closely wound solenoid 80 cm long has 5 layers of windings of 400 turns each. The diameter of the solenoid is 1.8 cm. If the current carried is 8.0 A, estimate the magnitude of B inside the solenoid near its centre.

(NCERT)

- (A) zero (B) $8\pi \times 10^{-3}$ T (C) $15\pi \times 10^{-3}$ T (D) $\pi \times 10^{-3}$ T

MG0051

17. A long solenoid has 200 turns per cm and carries a current i . The magnetic field at its centre is 6.28×10^{-2} weber/m². Another long solenoid has 100 turns per cm and it carries a current $i/3$. The value of the magnetic field at its centre is-

- (A) 1.05×10^{-2} weber/m² (B) 1.05×10^{-5} weber/m²
 (C) 1.05×10^{-3} weber/m² (D) 1.05×10^{-4} weber/m²

MG0052

18. A long straight wire of radius a carries a steady current i . The current is uniformly distributed across its cross-section. The ratio of the magnetic field at $\frac{a}{2}$ and $2a$ is-

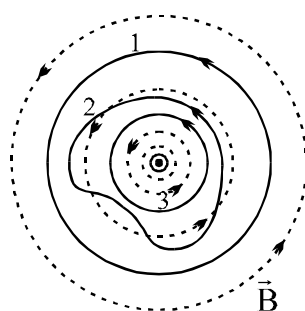
- (A) $\frac{1}{4}$ (B) 4 (C) 1 (D) $\frac{1}{2}$

MG0053

19. A current I flows along the length of an infinitely long, straight, thin walled pipe. Then-
- (A) the magnetic field is zero only on the axis of the pipe
 - (B) the magnetic field is different at different points inside the pipe
 - (C) the magnetic field at any point inside the pipe is zero
 - (D) the magnetic field at all points inside the pipe is the same, but not zero

MG0054

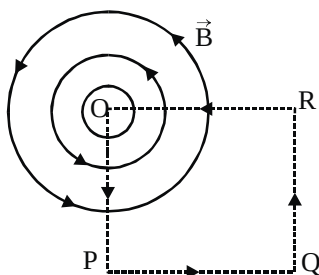
20. Consider the three closed loops drawn using solid line in the magnetic field (magnetic field lines are drawn using dotted line) of an infinite current-carrying wire normal to the plane of paper as shown. Rank the line integral of the magnetic field along each path in order of increasing magnitude



- (A) $1 > 2 > 3$ (B) $1 = 3 > 2$ (C) $1 = 2 = 3$ (D) $3 > 2 > 1$

MG0055

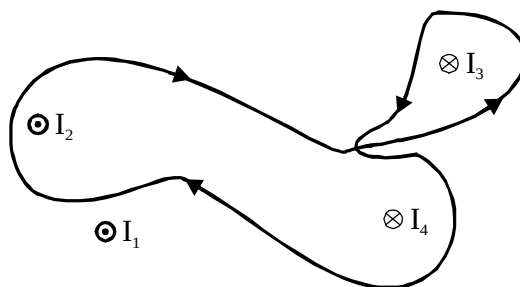
21. A current carrying wire (current = i) perpendicular to the plane of the paper produces a magnetic field, as shown in the figure. A square of side a is drawn with one of its vertices on the centre of the wire. The integral $\int \vec{B} \cdot d\vec{r}$ along $OPQRO$ has the value



- (A) $+\mu_0 i$ (B) $\frac{\mu_0 i}{8}$ (C) $\frac{\mu_0 i}{4}$ (D) $\frac{\mu_0 i}{2}$

MG0056

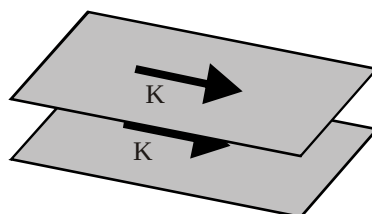
22. Four wires carrying current $I_1 = 2\text{A}$, $I_2 = 4\text{A}$, $I_3 = 6\text{A}$ and $I_4 = 8\text{A}$ respectively cut the page perpendicularly as shown in figure. The value of $\int \vec{B} \cdot d\vec{\ell}$ for the loop shown would be :-



- (A) $+2 \mu_0\text{-wb/m}$ (B) $-2 \mu_0\text{-wb/m}$ (C) $+10 \mu_0\text{-wb/m}$ (D) $-10 \mu_0\text{-wb/m}$

MG0057

23. What are the directions of the magnetic field between and outside a pair of two parallel large sheets carrying currents in the same directions, as illustrated in Figure (from the side shown)?



- (A) towards us between the plates and away from us above and below the plates.
 (B) toward us above the plates and away from us below plates and zero between plates.
 (C) towards us above and below the plates and zero between the plates
 (D) towards us between the plates and zero above and below the plates.

MG0058

Motion of charge particle

24. A charged particle enters a non-uniform uni-directional field such that initial velocity is parallel to magnetic field, then the radius of curvature of its path is (in standard notation):
 (A) mV/qB (B) 0 (C) ∞ (D) qB/mV

MG0059

25. A charge particle moves in a uniform magnetic field such that initial velocity is perpendicular to the magnetic field. No other force acts on the particle.
 (A) the motion is uniform rectilinear
 (B) the motion can be non uniform circular motion
 (C) the motion can be uniform circular motion
 (D) the motion must be uniform circular motion.

MG0060

26. A tightly-wound, long solenoid carries a current of 2.00A . An electron is found to execute a uniform circular motion inside the solenoid with a frequency of $1.00 \times 10^8\text{ rev/s}$. Find the number of turns per metre in the solenoid.
 (A) 500 Turns/m (B) 1020 Turns/m (C) 1232 Turns/m (D) 1420 Turns/m

MG0061

27. In a region, steady and uniform electric and magnetic fields are present. These two fields are parallel to each other. A charged particle is released from rest in this region. The path of the particle will be a-
 (A) helix (B) straight line (C) ellipse (D) circle

MG0062

28. An electron is moving along positive x-axis. A uniform electric field exists towards negative y-axis. What should be the direction of magnetic field of suitable magnitude so that net force of electron is zero
 (A) positive z- axis (B) negative z-axis (C) positive y-axis (D) negative y-axis

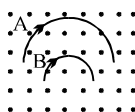
MG0063

29. An electron having kinetic energy T is moving in a circular orbit of radius R perpendicular to a uniform magnetic induction $\vec{B}$. If kinetic energy is doubled and magnetic induction tripled, the radius will become

- (A) $\frac{3R}{2}$ (B) $\sqrt{\frac{3}{2}} R$ (C) $\sqrt{\frac{2}{9}} R$ (D) $\sqrt{\frac{4}{3}} R$

MG0064

30. Two particles A and B of masses m_A and m_B respectively and having the same charge are moving in a plane. A uniform magnetic field exists perpendicular to this plane. The speeds of the particles are v_A and v_B respectively and the trajectories are as shown in the figure. Then [JEE, 2001 (Scr)]



- (A) $m_A v_A < m_B v_B$ (B) $m_A v_A > m_B v_B$
 (C) $m_A < m_B$ and $v_A < v_B$ (D) $m_A = m_B$ and $v_A = v_B$

MG0065

31. A charged particle moves in a magnetic field $\vec{B} = 10\hat{i}$ with initial velocity $\vec{u} = 5\hat{i} + 4\hat{j}$. The path of the particle will be
 (A) straight line (B) circle (C) helical (D) none

MG0066

32. An electron makes 3×10^5 revolutions per second in a circle of radius 0.5 angstrom. Find the magnetic field B at the centre of the circle.
 (A) 6×10^{-10} T (B) 12×10^{-10} T (C) 18×10^{-10} T (D) 24×10^{-10} T

MG0067

33. Electrons moving with different speeds enter a uniform magnetic field in a direction perpendicular to the field. They will move along circular paths.
 (A) of same radius
 (B) with larger radii for the faster electrons
 (C) with smaller radii for the faster electrons
 (D) either (B) or (C) depending on the magnitude of the magnetic field

MG0068

34. In the previous question, time periods of rotation will be :
 (A) same for all electrons
 (B) greater for the faster electrons
 (C) smaller for the faster electrons
 (D) either (B) or (C) depending on the magnitude of the magnetic field

MG0069

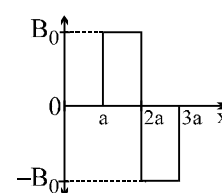
35. A particle of mass m and charge q moves with a constant velocity v along the positive x direction. It enters a region containing a uniform magnetic field B directed along the negative z direction, extending from $x = a$ to $x = b$. The minimum value of v required so that the particle can just enter the region $x > b$ is :-

JEE 2002 (screening)]

- (A) $q b B/m$ (B) $q(b - a) B/m$ (C) $q a B/m$ (D) $q(b + a) B/2m$

MG0070

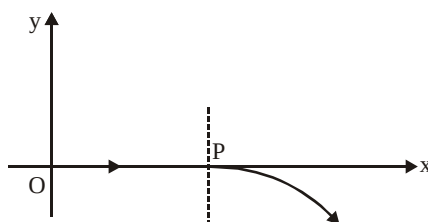
36. A magnetic field $\vec{B} = B_0 \hat{j}$ exists in the region $a < x < 2a$ and $\vec{B} = -B_0 \hat{j}$ in the region $2a < x < 3a$, where B_0 is a positive constant. A positive point charge moving with a velocity $\vec{v} = v_0 \hat{i}$, where v_0 is a positive constant, enters the magnetic field at $x = a$. The trajectory of the charge in this region can be like,



- (A) (B) (C) (D)

MG0071

37. For a positively charged particle moving in a $x - y$ plane initially along the x -axis, there is a sudden change in its path due to the presence of electric and/or magnetic field beyond P. The curved path is shown in the $x - y$ plane and is found to be non-circular. Which one of the following combinations is possible?
 [JEE 2004]



- (A) $\vec{E} = 0; \vec{B} = b\hat{j} + c\hat{k}$ (B) $\vec{E} = a\hat{i}; \vec{B} = c\hat{k} + a\hat{i}$
 (C) $\vec{E} = 0; \vec{B} = c\hat{j} + b\hat{k}$ (D) $\vec{E} = a\hat{i}; \vec{B} = c\hat{k} + b\hat{j}$

MG0072

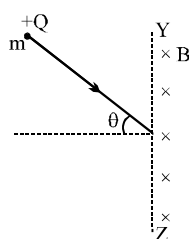
38. A particle having charge of 1 C, mass 1 kg and speed 1 m/s enters a uniform magnetic field, having magnetic induction of 1 T, at an angle $\theta = 30^\circ$ between velocity vector and magnetic induction. The pitch of its helical path is (in meters)

(A) $\frac{\sqrt{3}\pi}{2}$ (B) $\sqrt{3}\pi$ (C) $\frac{\pi}{2}$ (D) π

MG0073

39. A particle with charge $+Q$ and mass m enters a magnetic field of magnitude B , existing only to the right of the boundary YZ . The direction of the motion of the particle is perpendicular to the direction

of B . Let $T = 2\pi \frac{m}{QB}$. The time spent by the particle in the field will be



(A) $T\theta$ (B) $2T\theta$ (C) $T \left(\frac{\pi + 2\theta}{2\pi} \right)$ (D) $T \left(\frac{\pi - 2\theta}{2\pi} \right)$

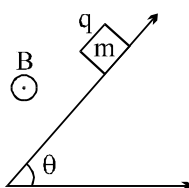
MG0074

40. In the previous question, if the particle has $-Q$ charge, the time spent by the particle in the field will be :-

(A) $T\theta$ (B) $2T\theta$ (C) $T \left(\frac{\pi + 2\theta}{2\pi} \right)$ (D) $T \left(\frac{\pi - 2\theta}{2\pi} \right)$

MG0075

41. A block of mass m & charge q is released on a long smooth inclined plane magnetic field B is constant, uniform, horizontal and parallel to surface as shown. Find the time from start when block loses contact with the surface.



(A) $\frac{m \cos \theta}{qB}$ (B) $\frac{m \operatorname{cosec} \theta}{qB}$ (C) $\frac{m \cot \theta}{qB}$ (D) none

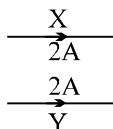
MG0076

42. In a cyclotron, a charged particle
 (A) undergoes acceleration all the time.
 (B) speeds up between the dees because of the magnetic field.
 (C) speeds up in a dee.
 (D) slows down within a dee and speeds up between dees.

MG0077

Ampere force & torque

43. In given figure, X and Y are two long straight parallel conductors each carrying a current of 2 A. The force on each conductor is F newtons. When the current in each is changed to 1 A and reversed in direction, the force on each is now



- (A) $F/4$ and unchanged in direction (B) $F/2$ and reversed in direction
(C) $F/2$ and unchanged in direction (D) $F/4$ and reversed in direction

MG0078

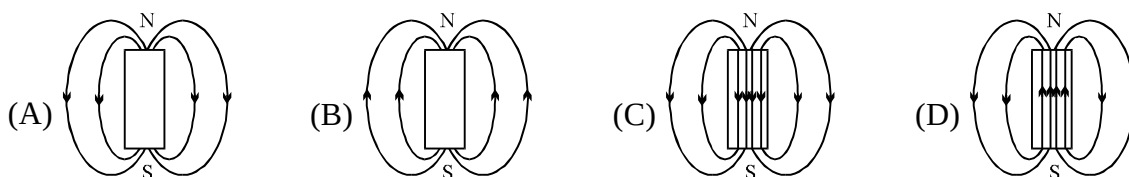
44. Two long and parallel straight wires A and B carrying currents of 8.0 A and 5.0 A in the same direction are separated by a distance of 4.0 cm. Estimate the force on a 10 cm section of wire A.

(NCERT)

- (A) 2×10^{-5} N ; attractive force normal to B towards A.
(B) 2×10^{-5} N ; attractive force normal to A towards B.
(C) 5×10^{-5} N ; attractive force normal to A towards B.
(D) 5×10^{-5} N ; attractive force normal to B towards A.

MG0079

45. The magnetic field lines due to a bar magnet are correctly shown in:- [JEE 2002 (screening)]



MG0080

46. A wire of mass 100 g carrying a current of 2A towards increasing x is in the form of $y = x^2$ ($-2\text{m} \leq x \leq +2\text{m}$). This wire is placed in a magnetic field $B = -0.02\hat{k}$ Tesla & gravity free region. The acceleration of the wire (in m/s^2) is :-

- (A) $-1.6\hat{j}$ (B) $-3.2\hat{j}$ (C) $1.6\hat{j}$ (D) $2.4\hat{j}$

MG0081

47. A circular loop of radius R carries a current I . Another circular loop of radius r ($r \ll R$) carries a current i and is placed at the centre of the larger loop. The planes of the two circles are at right angle to each other. Find the torque acting on the smaller loop.

- (A) zero (B) $\frac{\mu_0 \pi i I r^2}{4R}$ (C) $\frac{\mu_0 \pi i I r^2}{2R}$ (D) $\frac{\mu_0 \pi i I r^2}{R}$

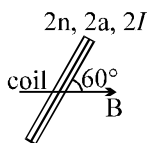
MG0082

48. A particle of charge q and mass m moves in a circular orbit of radius r with angular speed ω . The ratio of the magnitude of its magnetic moment to that of its angular momentum depends on.

- (A) ω and q (B) ω , q and m (C) q and m (D) ω and m

MG0083

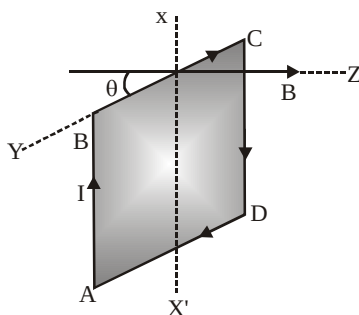
49. A rectangular coil PQ has $2n$ turns, an area $2a$ and carries a current $2I$, (refer figure). The plane of the coil is at 60° to a horizontal uniform magnetic field of flux density B . The torque on the coil due to magnetic force is :-



- (A) $BnaI \sin 60^\circ$ (B) $8BnaI \cos 60^\circ$ (C) $4naI B \sin 60^\circ$ (D) none

MG0084

50. The square loop ABCD, carrying a current I , is placed in a uniform magnetic field B , as shown. The loop can rotate about the axis XX' . The plane of the loop makes an angle θ ($\theta < 90^\circ$) with the direction of B . Through what angle will the loop rotate by itself before the torque on it becomes zero—

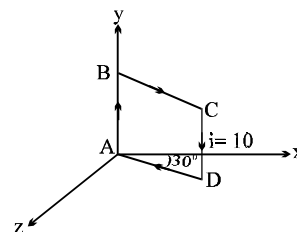


- (A) θ (B) $90^\circ - \theta$ (C) $90^\circ + \theta$ (D) $180^\circ - \theta$

MG0085

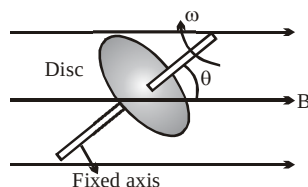
51. Figure shows a square current carrying loop ABCD of side 10 cm and current $i = 10\text{A}$. The magnetic moment $\vec{M}$ of the loop is

- (A) $(0.05) (\hat{i} - \sqrt{3}\hat{k}) \text{A} - \text{m}^2$
 (B) $(0.05) (\hat{j} + \hat{k}) \text{A} - \text{m}^2$
 (C) $(0.05) (\sqrt{3}\hat{i} + \hat{k}) \text{A} - \text{m}^2$
 (D) $(\hat{i} + \hat{k}) \text{A} - \text{m}^2$



MG0086

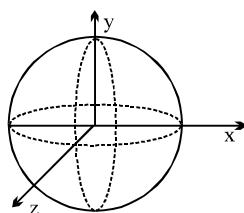
52. A disc of radius r and carrying positive charge q is rotating with an angular speed ω in a uniform magnetic field B about a fixed axis as shown in figure, such that angle made by axis of disc with magnetic field is θ . Torque applied by axis on the disc is



- (A) $\frac{q\omega r^2 B \sin \theta}{2}$, clockwise
 (B) $\frac{q\omega r^2 B \sin \theta}{4}$, anticlockwise
 (C) $\frac{q\omega r^2 B \sin \theta}{2}$, anticlockwise
 (D) $\frac{q\omega r^2 B \sin \theta}{4}$, clockwise

MG0087

53. Three rings, each having equal radius R , are placed mutually perpendicular to each other and each having its centre at the origin of co-ordinate system. If current I is flowing through each ring then the magnitude of the magnetic field at the common centre is



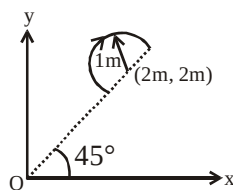
- (A) $\sqrt{3} \frac{\mu_0 I}{2R}$
 (B) zero
 (C) $(\sqrt{2} - 1) \frac{\mu_0 I}{2R}$
 (D) $(\sqrt{3} - \sqrt{2}) \frac{\mu_0 I}{2R}$

MG0088

54. Two mutually perpendicular long conducting wire carrying currents I_1 and I_2 lie in one plane. Locus of the point at which the magnetic induction is zero, is a
 (A) circle with centre as the point of intersection of the wire.
 (B) parabola with vertex as the point of intersection of the wire
 (C) straight line passing through the point of intersection of the wire
 (D) rectangular hyperbola

MG0089

55. A uniform magnetic field $\vec{B} = (3\hat{i} + 4\hat{j} + \hat{k})$ Tesla exist in a region of space. A semicircular wire of radius 1 m carrying a current of 1A having its centre at (2,2,0) m is placed on the X-Y plane as shown. The force on the semicircular wire will be



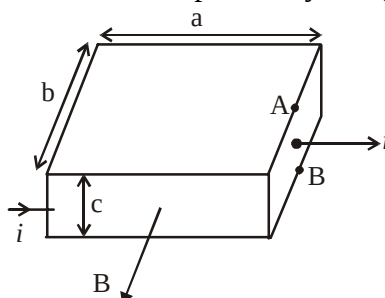
- (A) $\frac{1}{\sqrt{2}} (\hat{i} - \hat{j} + \hat{k})$ N
 (B) $\sqrt{2} (\hat{i} - \hat{j} + \hat{k})$ N
 (C) $\frac{1}{\sqrt{2}} (\hat{i} + \hat{j} - \hat{k})$ N
 (D) $\sqrt{2} (\hat{i} + \hat{j} - \hat{k})$ N

MG0090

56. A very long wire carrying current I is fixed along x -axis. Another parallel finite wire carrying a current in the opposite direction is kept at a distance d above the wire in xy plane. The second wire is free to move parallel to itself. The options available for its small displacements are in
 (i) +ve x direction (ii) +ve y direction (iii) +ve z direction
 Taking gravity in negative y direction, the nature of equilibrium of second wire is
 (A) stable for movement in x direction, unstable for movement in y direction, neutral for movement in z direction
 (B) stable for movement in y direction, unstable for movement in z direction, neutral for movement in x direction
 (C) stable for movement in z direction, unstable for movement in y direction, neutral for movement in x direction
 (D) stable for movement in y direction, unstable for movement in x direction, neutral for movement in z direction

MG0091

57. A current flows through a rectangular conductor in the presence of uniform magnetic field B pointing out of the page as shown. Then the potential difference $V_B - V_A$ is equal to
 (Assume charge carries in the conductor to be positively charged moving with speed v)



- (A) Bvc (B) $-Bvc$ (C) Bvb (D) $-Bvb$

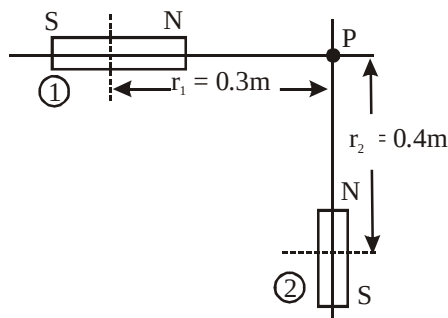
MG0092

58. A moving coil galvanometer has 150 equal divisions. Its current sensitivity is 10 divisions per milliampere and voltage sensitivity is 2 divisions per millivolt. In order that each division reads 1 V, the resistance in Ohm's needed to be connected in series with the coil will be- [AIEEE-2005]
 (A) 10^3 (B) 10^5 (C) 99995 (D) 9995

MG0093

SUPPLEMENT ON MAGNETIC PROPERTIES OF MATTER

59. Two short magnets of magnetic moment $2Am^2$ and $5Am^2$ respectively are placed along two lines drawn at right angle to each other on the sheet of paper as shown in the figure. What is the magnetic field at the point of intersection of their axis:



- (A) $2.15 \times 10^{-5} T$ (B) $215 \times 10^{-5} T$ (C) $2.15 \times 10^{-4} T$ (D) $21.5 \times 10^{-5} T$

MG0094

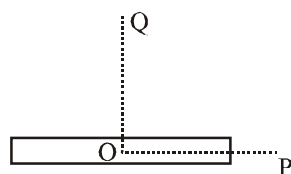
60. A steel wire of length L has a magnetic moment M . It is then bent into a semi-circular arc; the new magnetic moment will be :
 (A) M (B) $2M/\pi$ (C) M/L (D) $M \times L$

MG0095

61. Two bar magnets of the same mass, same length and breadth but having magnetic moments M and $2M$ are joined together north pole to north pole and suspended by a string. The time period of assembly in a magnetic field of strength H is 3 seconds. If now the polarity of one of the magnets is reversed and the combination is again made to oscillate in the same field, the time of oscillations is :
 (A) $\sqrt{3}$ sec (B) $3\sqrt{3}$ sec (C) 3 sec (D) 6 sec

MG0096

62. When magnetic field at P and Q is same then $OP/OQ = ?$



- (A) $\sqrt[3]{2}$ (B) $\frac{1}{\sqrt[3]{2}}$ (C) $2\sqrt{2}$ (D) $\frac{3}{2\sqrt{2}}$

MG0097

63. Similar pole each of pole strength m are placed at a distance of 1, 2, 4, 8,meters from the origin on the x-axis. Where do you place a similar pole n on the other side of the origin so that the origin becomes a neutral point :-
 (A) 0.5 m (B) 0.5774 m (C) 0.86 m (D) 1m

MG0098

64. Two identical poles each of pole strength 4×10^{-3} amp-m placed at the two corners of an equilateral triangle of side 20 cm. The magnetic field at the third corner is :-

- (A) $\sqrt{3} \times 10^{-8}$ N/amp-m (B) $\frac{\sqrt{3}}{4} \times 10^{-8}$ N/amp-m
 (C) zero (D) 10^{-8} N/amp-m

MG0099

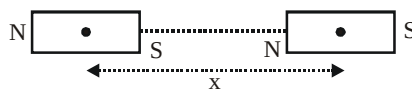
65. The magnetic field at a point on the axis of a dipole is 20 Wb/m² Northwards. What will be the magnetic induction at the point if the dipole be rotated anticlockwise by 90° :-
 (A) 10 Wb/m² Eastwards (B) 10 Wb/m² Southwards
 (C) 10 Wb/m² Northwards (D) 20 Wb/m² Southwards

MG0100

66. A bar magnet of magnetic moment $2A\text{-m}^2$ free to rotate about a vertical axis passing through its centre. The magnet is released from rest from east-west position. Then the kinetic energy of the magnet as it takes North-South position is (horizontal component of the earth's field is 25 μ T and angle of declination is zero :-
 (A) 25 μ J (B) 50 μ J (C) 100 μ J (D) 125 μ J

MG0101

67. The mid points of two small magnetic dipoles of length d in end-on positions, are separated by a distance x , ($x \gg d$). The force between them is proportional to x^{-n} where n is :-



- (A) 3 (B) 4 (C) 2 (D) 1

MG0102

68. A permanent magnet has shape of a thin disc with magnetic moment along its axis the radius of the disc is 1 cm. The magnetic field due to it can be assumed to be that due to molecular current I' flowing along the rim of the disc. If $\vec{B}$ at a distance of 10 cm from its centre on the axis is $30\mu\text{T}$, find I' (approximately) :-

- (A) 1 kA (B) 0.5 kA (C) 2.5 kA (D) 3 kA

MG0103

69. The angle of dip at a place is 30° and the intensity of the vertical component of the earth's magnetic field $V = 6 \times 10^{-5}\text{T}$. The total intensity of the earth's magnetic field at this place is :

- (A) $7 \times 10^{-5}\text{T}$ (B) $6 \times 10^{-5}\text{T}$ (C) $5 \times 10^{-5}\text{T}$ (D) $12 \times 10^{-5}\text{T}$

MG0104

70. The total intensity of the earth's magnetic field at the magnetic equator is 5 units. What is its value at a magnetic latitude of 37° degree?

- (A) $\sqrt{73}$ units (B) $\sqrt{52}$ units (C) 4 units (D) 3 units

MG0105

71. A magnet is suspended in the magnetic meridian with an untwisted wire. The upper end of wire is rotated through 180° to deflect the magnet by 30° from magnetic meridian. When the magnet is replaced by another magnet, the upper end of wire is rotated through 270° to deflect the magnet 30° from magnetic meridian. The ratio of magnetic moments of magnets is :-

- (A) 1 : 5 (B) 1 : 8 (C) 5 : 8 (D) 8 : 5

MG0106

72. The imagined magnet makes an angle of 11.3° with earth's axis. On the earth, there are two points where magnetic equator and geographic equator meet let the angle of dip and angle of declination at these point be θ & ϕ respectively :-

- (A) $\theta = \phi = 0^\circ$ (B) $\theta = 11.3^\circ, \phi = 0^\circ$
(C) $\theta = 0^\circ, \phi = 11.3^\circ$ (D) $\theta = \phi = 11.3^\circ$

MG0107

73. A specimen of iron of permeability 8×10^{-3} weber/ amp $\times$ metre is placed in a magnetic field of intensity 160 amp/metre. Then magnetic field in this iron is :

- (A) $20 \times 10^3\text{Wb/m}^2$ (B) 1.28Wb/m^2 (C) $5 \times 10^{-5}\text{Wb/m}^2$ (D) 0.8Wb/m^2

MG0108

74. Curie temperature is the temperature above which :
- (A) a paramagnetic material becomes diamagnetic
 - (B) a ferromagnetic material becomes diamagnetic
 - (C) a paramagnetic material becomes ferromagnetic
 - (D) a ferromagnetic material becomes paramagnetic

MG0109

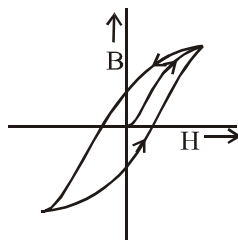
75. If a solution of ferromagnetic material is poured into a U-tube and one arm of this is placed between the poles of a strong magnet with the meniscus in line with the field, then the level of the solution will:
- (A) Rise
 - (B) Fall
 - (C) Rise till the liquid comes out
 - (D) Remain unchanged

MG0110

76. A sample of paramagnetic salt contains 2.0×10^{24} atomic dipoles each of dipole moment $1.5 \times 10^{-23} \text{ J T}^{-1}$. The sample is placed under a homogeneous magnetic field of 0.84 T, and cooled to a temperature of 4.2 K. The degree of magnetic saturation achieved is equal to 15%. What is the total dipole moment (approximate) of the sample for a magnetic field of 0.98 T and a temperature of 2.8 K? (Assume Curie's law).
- (A) 7.9 J T^{-1}
 - (B) 52.5 J T^{-1}
 - (C) 30 J T^{-1}
 - (D) 4.6 J T^{-1}

MG0111

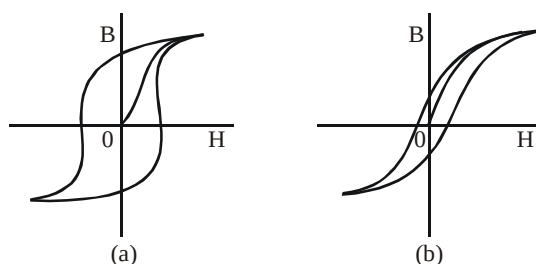
77. The B-H curve for a certain specimen is schematically shown by the given diagram. Which one of the following is the correct magnetic nature of the specimen?



- (A) Diamagnetic and not ferromagnetic or paramagnetic
- (B) Ferromagnetic and not diamagnetic or paramagnetic
- (C) Paramagnetic and not diamagnetic or ferromagnetic
- (D) Applicable to all the three types of magnetism mentioned above

MG0112

78. Two types of steel are characterized by the hysteresis loops shown in figure (a) and (b). The loops are obtained in the processes of magnetization and demagnetization of the steels. Which of the two types is best suited for using as the core of a transformer and which, for using as a permanent magnet ?



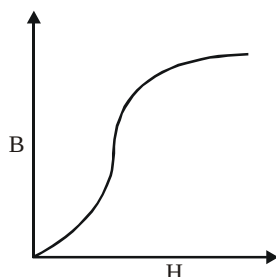
- (A) Both are best suited for transformer
 (B) Both are best suited for permanent magnet
 (C) A is best suited for permanent magnet & B is best suited for core of transformer
 (D) A is best suited for core of transformer & B is best suited for permanent magnet

MG0113

79. A material satisfied the relation $\mu_0(H + M) = 0$, where H and M are magnetic intensity and magnetisation respectively, then material is :-
 (A) nonmagnetic (B) paramagnetic (C) ferrormagnetic (D) diamagnetic

MG0114

80. The figure represent B-H curve for commercial iron. As H is increased, permeability :-



- (A) increases and becomes constant
 (B) increases, reaches a maximum and then decreases
 (C) decreases continuously till it becomes very small
 (D) decreases reaches a minimum and then increases

MG0115

81. **Statement-1** : The sensitivity of a moving coil galvanometer is increased by placing a suitable magnetic material as a core inside the coil.

and

Statement-2 : Soft iron has a high magnetic permeability and cannot be easily magnetized or demagnetized.

- (A) Statement-1 is True, Statement-2 is True ; statement-2 is a correct explanation for statement-1
 (B) Statement-1 is True, Statement-2 is True ; statement-2 is NOT a correct explanation for statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True

MG0116

82. Horizontal magnetic field of earth at Mumbai is 1.5×10^{-4} T in North direction. A small bar magnet of magnetic moment 10 Am^2 is kept on a horizontal table such that its North pole points due North. What is the magnetic field at 10 cm from centre of magnet at a point on its axis due North of it :-
 (A) 3.5×10^{-4} T (B) 5×10^{-5} T
 (C) 2.15×10^{-3} T (D) 1.15×10^{-3} T

MG0117

83. A compass needle kept on a horizontal table points in direction 30° west of north. A line is marked showing this direction and compass is now kept in vertical plane containing this line. The needle points in direction 30° below horizontal. If the plane of compass is still vertical but contains true north south direction at what angle to horizontal will needle point :-

(A) $\tan^{-1}\left(\frac{3}{2}\right)$ (B) $\tan^{-1}\left(\frac{2}{3}\right)$ (C) $\tan^{-1}(2)$ (D) $\tan^{-1}\left(\frac{1}{2}\right)$

MG0118

84. In a moving coil galvanometer the number of turns $N = 24$, area of the coil $A = 2 \times 10^{-3} \text{ m}^2$, and the magnetic field strength $B = 0.2 \text{ T}$. To increase its current sensitivity by 25% we :-
 (A) increase B to 0.30 T (B) decrease A to $1.5 \times 10^{-3} \text{ m}^2$
 (C) increase N to 30 (D) none of the above

MG0119

85. A thin rectangular magnet suspended freely has a period of oscillation equal to T. Now it is broken into two equal halves (each having half of the original length) and one piece is made to oscillate freely in the same field. If its period of oscillation is T' , the ratio T'/T is- [AIEEE - 2003]

(A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2}$ (C) 2 (D) $\frac{1}{4}$

MG0120

86. The magnetic lines of force inside a bar magnet: [AIEEE - 2003]
 (A) are from north-pole to south-pole of the magnet
 (B) do not exist
 (C) depend upon the area of cross-section of the bar magnet
 (D) are from south-pole to north-pole of the magnet

MG0121

87. The length of a magnet is large compared to its width and breadth. The time period of its oscillation in a vibration magnetometer is 2s. The magnet is cut along its length into three equal parts and three parts are then placed on each other with their like poles together. The time period of this combination will be- [AIEEE - 2004]
 (A) 2s (B) $\frac{2}{3}$ s (C) $2\sqrt{3}$ s (D) $2/\sqrt{3}$ s

MG0122

88. The materials suitable for making electromagnets should have- [AIEEE - 2004]
- (A) high retentivity and high coercivity
 - (B) low retentivity and low coercivity
 - (C) high retentivity and low coercivity
 - (D) low retentivity and high coercivity

MG0123

MULTIPLE CORRECT TYPE QUESTIONS

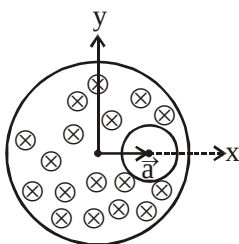
89. Consider three quantities $x = E/B$, $y = \sqrt{1/\mu_0\epsilon_0}$ and $z = \frac{l}{CR}$. Here, l is the length of a wire, C is a capacitance and R is a resistance. All other symbols have standard meanings.
- (A) x, y have the same dimensions
 - (B) y, z have the same dimensions
 - (C) z, x have the same dimensions
 - (D) none of the three pairs have the same dimensions.
90. The magnetic lines of force due to a straight current carrying wire will be:
- (A) circular for finite length of wire
 - (B) circular for semi-infinite wire
 - (C) circular for infinite wire
 - (D) Parabolic for finite wire
91. A current I flows along the length of an infinitely long, straight, solid pipe. Then correct statement(s)-
- (A) The magnetic field is zero only on the axis of the pipe
 - (B) The magnetic field is different at different points inside the pipe
 - (C) The magnetic field is maximum on surface
 - (D) The magnetic field at all points inside the pipe is the same, but not zero

MG0124

MG0125

MG0126

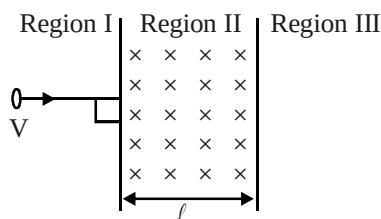
92. Figure shows cross-section view of a infinite cylindrical wire with a cylindrical cavity, current density is uniform $\vec{J} = -j_0 \hat{k}$ as shown in figure.



- (A) Field inside cavity is uniform
 (B) Field inside cavity is along $\vec{a}$
 (C) Field inside cavity is perpendicular to $\vec{a}$
 (D) If an electron is projected with velocity $v_0 \hat{j}$ it will move undeviated before colliding with cavity wall

MG0127

93. A particle of mass m and charge q , moving with velocity V enters Region II normal to the boundary as shown in the figure. Region II has a uniform magnetic field B perpendicular to the plane of the paper. The length of the Region II is ℓ . Choose the correct choice(s)



- (A) The particle enters Region III only if its velocity $V > \frac{q\ell B}{m}$
 (B) The particle enters Region III only if its velocity $V < \frac{q\ell B}{m}$
 (C) Path length of the particle in Region II is maximum when velocity $V = \frac{q\ell B}{m}$
 (D) Time spent in Region II is same for any velocity V as long as the particle returns to Region I

MG0128

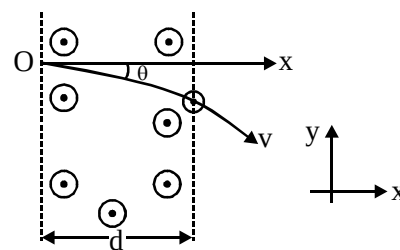
94. A particle of charge q is projected with a momentum $\vec{p} = p\hat{i}$ in the given region of magnetic field $\vec{B} = B\hat{k}$. It emerges from the magnetic field after deviating through an angle $\theta = 30^\circ$.

(A) The value of $\vec{p}$ is $2qBd$

(B) The value of $\vec{p}$ is qBd

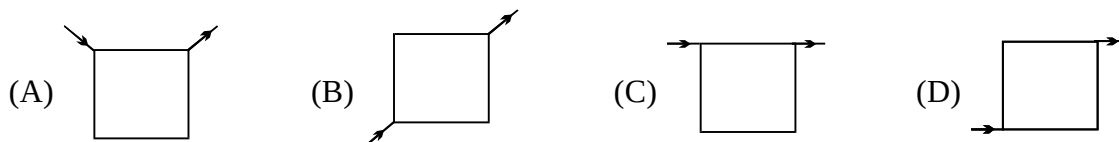
(C) Maximum change in momentum takes place for $d \geq \frac{p}{qB}$

(D) Maximum change in momentum takes place for $d \geq \frac{p}{2qB}$



MG0129

95. Current flows through uniform, square frames as shown. In which case is the magnetic field at the centre of the frame not zero. The wire current carrying shown outside coil and infinitely long.



MG0130

96. A long straight wire carries a current along the x -axis. Consider the points $A(0, 1, 0)$, $B(0, 1, 1)$, $C(1, 0, 1)$ and $D(1, 1, 1)$. Which of the following pairs of points will have magnetic fields of the same magnitude?

(A) A and B

(B) A and C

(C) B and C

(D) B and D

MG0131

97. Two identical charged particles enter a uniform magnetic field with same speed but at angles 30° and 60° with field. Let a , b and c be the ratio of their time periods, radii and pitches of the helical paths than

(A) $abc = 1$

(B) $abc > 1$

(C) $abc < 1$

(D) $a = bc$

MG0132

98. An electron is moving along the positive X -axis. You want to apply a magnetic field for a short time so that the electron may reverse its direction and move parallel to the negative X -axis. This can be done by applying the magnetic field along :-

(A) Y -axis

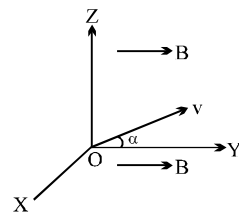
(B) Z -axis

(C) Y -axis only

(D) Z -axis only

MG0133

99. In a region of space, a uniform magnetic field B exists in the y -direction. A proton is fired from the origin, with its initial velocity v making a small angle α with the y -direction in the yz plane. In the subsequent motion of the proton,
- (A) its x -coordinate can never be positive
 (B) its x - and z -coordinates cannot both be zero at the same time
 (C) its z -coordinate can never be negative
 (D) its y -coordinate will be proportional to the square of its time elapsed

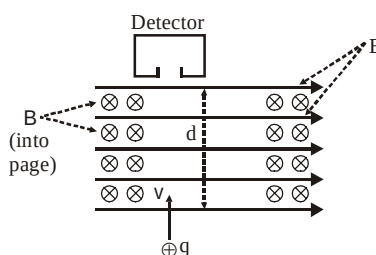


MG0134

COMPREHENSION TYPE QUESTIONS

Paragraph for Questions no. 100 to 103

A velocity filter uses the properties of electric and magnetic fields to select charged particles that are moving with a specific velocity. Charged particles with varying speeds are directed into the filter as shown in figure. The filter consists of an electric field E and a magnetic field B , each of constant magnitude, directed perpendicular to each other as shown. The particles that move straight through the filter with their direction unaltered by the fields have the specific filter speed, v_0 . Those with speeds to v_0 may experience sufficiently little deflection that they also enter the detector.



The charged particle will experience a force due to the electric field given by the relationship $\vec{F} = q\vec{E}$, where q is the charge of the particle and $\vec{E}$ is the electric field. The moving particle will also experience a force due to the magnetic field. This force acts to oppose the force due to the electric field. The strength of the force due to the magnetic field is given by the relationship $\vec{F} = q(\vec{v} \times \vec{B})$, where q is the charge of the particle, $\vec{v}$ is the speed of the particle, and $\vec{B}$ is the magnetic field strength. When the forces due to the two fields are equal and opposite, the net force on the particle will be zero, and the particle will pass through the filter with its path unaltered. The electric and magnetic field strengths can be adjusted to choose the specific velocity to be filtered. The effects of gravity can be neglected.

100. The electric and magnetic fields in the filter of figure are adjusted to detect particles with positive charge q of a certain speed, v_0 . Which of the following expressions is equal to this speed ?
- (A) $B/(q^2E)$ (B) $E/(q^2B)$ (C) B/E (D) E/B

MG0135

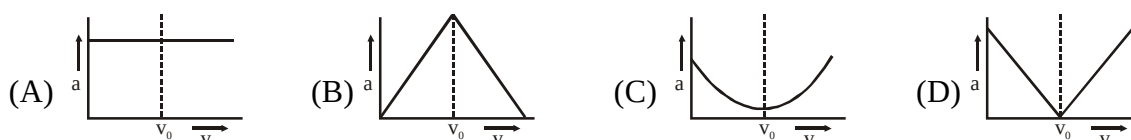
101. Which of the following is true about the velocity filter shown in figure ?
- It would not work with negatively charged particles
 - The wider the detector entrance, the more narrow the range of speed detected
 - The greater the distance d , the more narrow the range of speeds detected
 - The detector may not detect a charged particle with the desired filter speed if its charge is too high

MG0136

102. Which of the following statements is true regarding a charged particle that is moving through the filter at a speed that is less than the filter speed ?
- It experiences a greater force due to the magnetic field than due to the electric field
 - It experiences a greater force due to the electric field than due to the magnetic field
 - It experiences equal force due to both fields but greater acceleration due to the electric field
 - It experiences equal force due to both fields but greater acceleration due to the magnetic field

MG0137

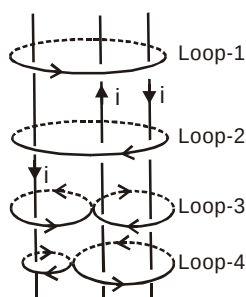
103. Particles of identical mass and charge are sent through the filter at varying speeds, and the magnitude of acceleration of each particle is recorded as it first begins to be deflected. If the filter is set to detect particles of speed v_0 , which one of the following is correct graph between acceleration and velocity of particle:



MG0138

MATRIX MATCH TYPE QUESTION

104. Three wires are carrying same constant current i in different directions. Four loops enclosing the wires in different manners are shown. The direction of $d\vec{\ell}$ is shown in figure.



Column-I

- Along closed loop-1
- Along closed loop-2
- Along closed loop-3
- Along closed loop-4

Column-II

- $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 i$
- $\oint \vec{B} \cdot d\vec{\ell} = -\mu_0 i$
- $\oint \vec{B} \cdot d\vec{\ell} = 0$
- Net work done by the magnetic force to move a unit charge along the loop is zero

MG0139

105. **Column-I** (Magnetic moment of)**Column-II**

- | | |
|--|-----------------------------|
| (A) a uniformly charged ring rotating uniformly about its axis | (p) $\frac{q\omega r^2}{5}$ |
| (B) a charged particle rotating uniformly about a point | (q) $\frac{q\omega r^2}{4}$ |
| (C) a uniformly charged disk rotating uniformly about its axis | (r) $\frac{q\omega r^2}{3}$ |
| (D) a uniformly charged spherical shell rotating uniformly about one of its diameter | (s) $\frac{q\omega r^2}{2}$ |
| (E) a uniformly charged sphere rotating | (t) $q\omega r^2$ |

MG0140

EXERCISE (O-2)

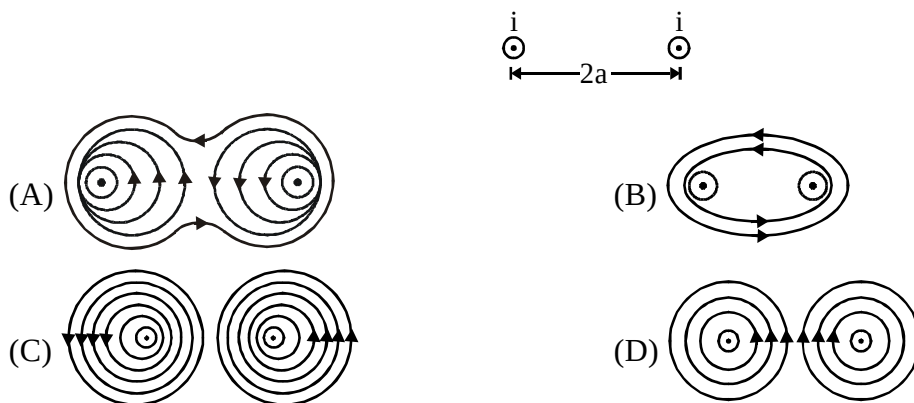
SINGLE CORRECT TYPE QUESTIONS

1. A ring like metallic conductor of resistance R and radius a , carries a constant current I . The ratio of the angular momentum L of the conduction electrons (about the axis of the ring) and the magnetic field B at the centre of the ring satisfy [where e and m represent the magnitudes of the electronic charge and mass]

(A) $\frac{B}{L} \propto \frac{e^2}{m}$ (B) $\frac{B}{L} \propto e.m$ (C) $\frac{B}{L} \propto \frac{e}{m}$ (D) $\frac{B}{L} \propto \frac{m}{e}$

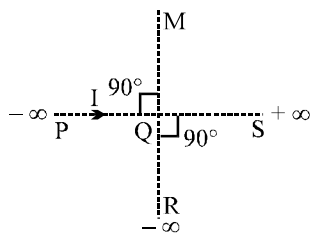
MG0141

2. Which of the following field patterns is correct for two long straight equal parallel current carrying wires ?



MG0142

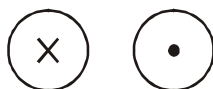
3. An infinitely long conductor PQR is bent to form a right angle as shown. A current I flows through PQR. The magnetic field due to this current at the point M is H_1 . Now, another infinitely long straight conductor QS is connected at Q so that the current in PQ remaining unchanged. The magnetic field at M is now H_2 . The ratio H_1/H_2 is given by :-



(A) $1/2$ (B) 1 (C) $2/3$ (D) 2

MG0143

4. Equal antiparallel currents are directed in two parallel wires so that one is out of the page and the other is into the page as shown. Compare the magnitude of the magnetic field B_2 at any arbitrary point equidistant from the wires to the magnitude of the field B_1 at that point from one wire alone :-



- (A) $B_2 > B_1$ for all equidistant points
 (B) $B_2 < B_1$ for all equidistant points
 (C) $B_2 > B_1$ for closer equidistant points only
 (D) $B_2 < B_1$ for closer equidistant points only

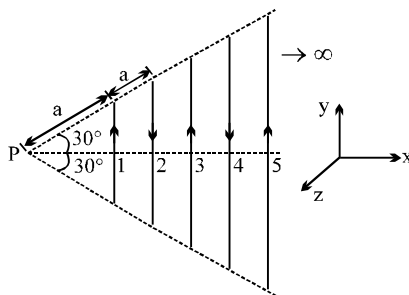
MG0144

5. A long straight wire along the z-axis carries a current I in the negative z direction. The magnetic vector field $\vec{B}$ at a point having coordinates (x, y) in the $z = 0$ plane is :- [JEE 2002 (screening)]

- (A) $\frac{\mu_0 I}{2\pi} \frac{(y\hat{i} - x\hat{j})}{(x^2 + y^2)}$ (B) $\frac{\mu_0 I}{2\pi} \frac{(x\hat{i} + y\hat{j})}{(x^2 + y^2)}$
 (C) $\frac{\mu_0 I}{2\pi} \frac{(x\hat{j} - y\hat{i})}{(x^2 + y^2)}$ (D) $\frac{\mu_0 I}{2\pi} \frac{(x\hat{i} - y\hat{j})}{(x^2 + y^2)}$

MG0145

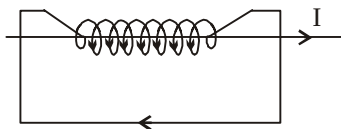
6. Infinite number of straight wires each carrying current I are equally placed as shown in the figure. Adjacent wires have current in opposite direction. Net magnetic field at point P is :-



- (A) $\frac{\mu_0 I}{4\pi} \frac{\ln 2}{\sqrt{3}a} \hat{k}$ (B) $\frac{\mu_0 I}{4\pi} \frac{\ln 4}{\sqrt{3}a} \hat{k}$ (C) $\frac{\mu_0 I}{4\pi} \frac{\ln 4}{\sqrt{3}a} (-\hat{k})$ (D) Zero

MG0146

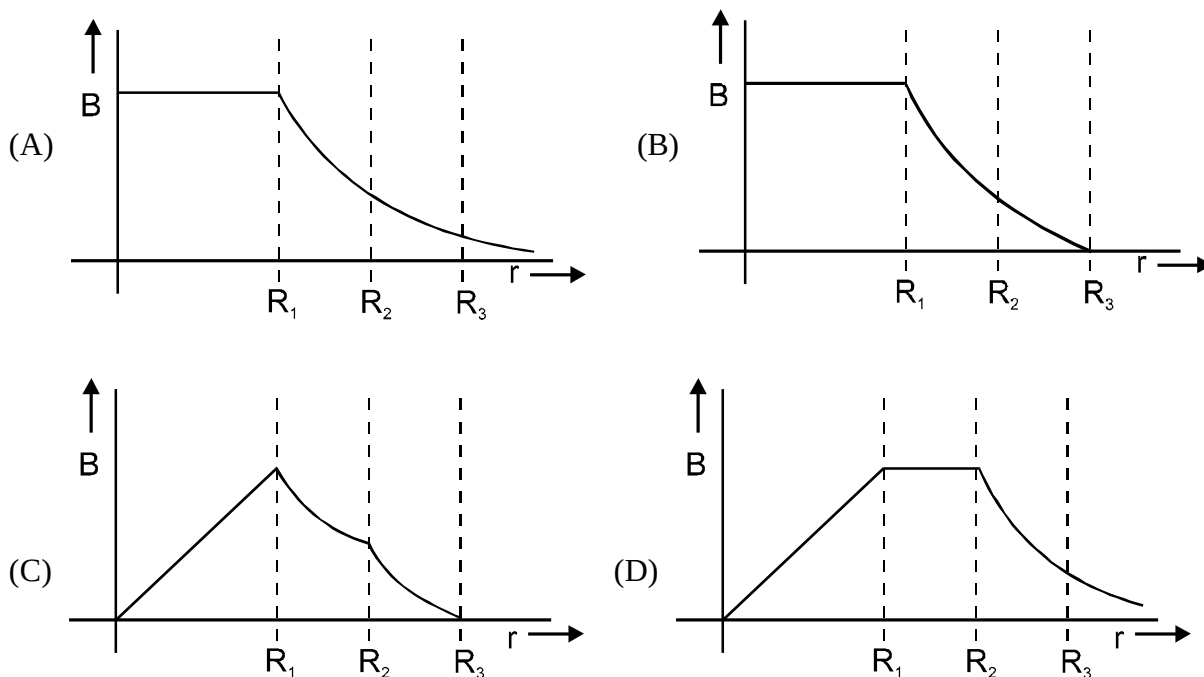
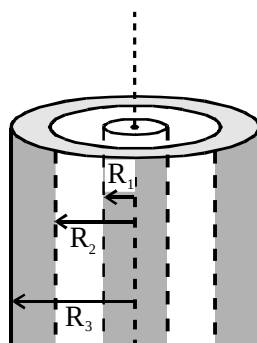
7. In the diagram shown, a wire carries current I . What is the value of the $\oint \vec{B} \cdot d\vec{s}$ (as in Ampere's law) on the helical loop shown in the figure? The integration is done in the sense shown. The loop has N turns and part of helical loop on which arrows are drawn is outside the plane of paper.



- (A) $-\mu_0(NI)$ (B) $\mu_0(I)$ (C) $\mu_0(NI)$ (D) Zero

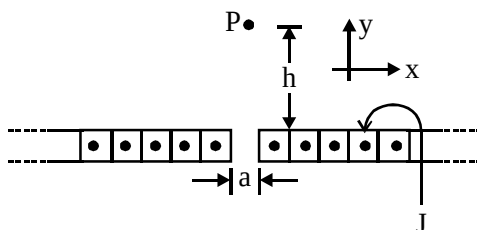
MG0147

8. A coaxial cable is made up of two conductors. The inner conductor is solid and is of radius R_1 & the outer conductor is hollow of inner radius R_2 and outer radius R_3 . The space between the conductors is filled with air. The inner and outer conductors are carrying currents of equal magnitudes and in opposite directions. Then the variation of magnetic field with distance from the axis is best plotted as:



MG0148

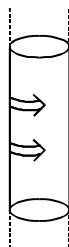
9. A small segment of length a is cut along z -axis from a infinite sheet having a surface current density J (current per unit width). The magnetic field at P is :-



- (A) $2\mu_0 J \left(1 - \frac{h}{a\pi}\right) \hat{i}$ (B) $\frac{\mu_0 J h}{2a\pi} \hat{i}$ (C) $\frac{\mu_0 J}{2} \left(\frac{a}{h\pi} - 1\right) \hat{i}$ (D) $-\frac{\mu_0 J}{2} \left(\frac{h}{a\pi} - 1\right) \hat{i}$

MG0149

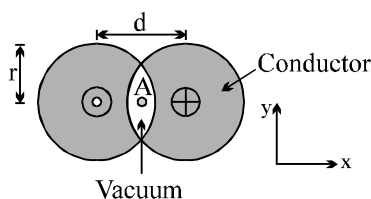
10. A hollow cylinder having infinite length and carrying uniform current per unit length λ along the circumference as shown. Magnetic field inside the cylinder is :-



- (A) $\frac{\mu_0 \lambda}{2}$ (B) $\mu_0 \lambda$ (C) $2\mu_0 \lambda$ (D) none

MG0150

11. Two long conductors are arranged as shown above to form overlapping cylinders, each of radius r , whose centers are separated by a distance d . Current of density J flows into the plane of the page along the shaded part of one conductor and an equal current flows out of the plane of the page along the shaded portion of the other, as shown. What are the magnitude and direction of the magnetic field at point A?



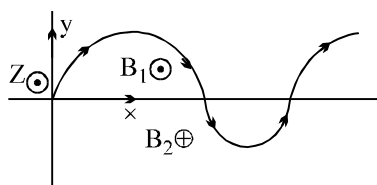
- (A) $(\mu_0/2\pi)\pi dJ$, in the $+y$ -direction (B) $(\mu_0/2\pi)d^2/r$, in the $+y$ -direction
(C) $(\mu_0/2\pi)4d^2J/r$, in the $-y$ -direction (D) $(\mu_0/2\pi)Jr^2/d$, in the $-y$ -direction

MG0151

12. An ionized gas contains both positive and negative ions. If it is subjected simultaneously to an electric field along the $+x$ direction and a magnetic field along the $+z$ direction, then [JEE 2000 (Scr)]
 (A) positive ions deflect towards $+y$ direction and negative ions towards $-y$ direction
 (B) all ions deflect towards $+y$ direction.
 (C) all ions deflect towards $-y$ direction
 (D) positive ions deflect towards $-y$ direction and negative ions towards $+y$ direction.

MG0152

13. At $t = 0$ a charge q is at the origin and moving in the y -direction with velocity $\vec{v} = v\hat{j}$. The charge moves in a magnetic field that is for $y > 0$ out of page and given by $B_1\hat{z}$ and for $y < 0$ into the page and given $-B_2\hat{z}$. The charge's subsequent trajectory is shown in the sketch. From this information, we can deduce that



- (A) $q > 0$ and $|B_1| < |B_2|$
 (B) $q < 0$ and $|B_1| < |B_2|$
 (C) $q > 0$ and $|B_1| > |B_2|$
 (D) $q < 0$ and $|B_1| > |B_2|$

MG0153

14. A particle of specific charge (charge/mass) α starts moving from the origin under the action of an electric field $\vec{E} = E_0\hat{i}$ and magnetic field $\vec{B} = B_0\hat{k}$. Its velocity at $(x_0, y_0, 0)$ is $(4\hat{i} - 3\hat{j})$. The value of x_0 is:

- (A) $\frac{13}{2} \frac{\alpha E_0}{B_0}$ (B) $\frac{16 \alpha B_0}{E_0}$ (C) $\frac{25}{2 \alpha E_0}$ (D) $\frac{5 \alpha}{2 B_0}$

MG0154

15. A particle of specific charge (q/m) is projected from the origin of coordinates with initial velocity $[u\hat{i} - v\hat{j}]$. Uniform electric and magnetic fields exist in the region along the $+y$ direction, of magnitude E and B . The particle will definitely return to the origin once if

- (A) $[vB/2\pi E]$ is an integer
 (B) $(u^2 + v^2)^{1/2} [B/\pi E]$ is an integer
 (C) $[vB/\pi E]$ is an integer
 (D) $[uB/\pi E]$ is an integer

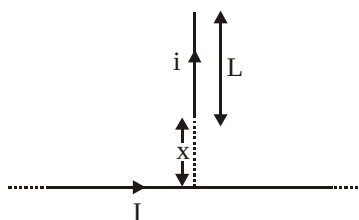
MG0155

16. An electron moving with a velocity $\vec{V}_1 = 2\hat{i}$ m/s at a point in a magnetic field experiences a force $\vec{F}_1 = -2\hat{j}$ N. If the electron is moving with a velocity $\vec{V}_2 = 2\hat{j}$ m/s at the same point, it experiences a force $\vec{F}_2 = +2\hat{i}$ N. The force the electron would experience if it were moving with a velocity $\vec{V}_3 = 2\hat{k}$ m/s at the same point is

- (A) zero
 (B) $2\hat{k}$ N
 (C) $-2\hat{k}$ N
 (D) information is insufficient

MG0156

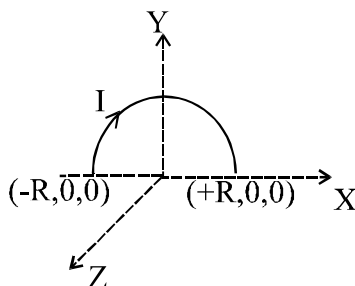
17. The magnetic force between wires as shown in figure is :-



- (A) $\frac{\mu_0 i I^2}{2\pi} \ln\left(\frac{x+\ell}{2x}\right)$ (B) $\frac{\mu_0 i I^2}{2\pi} \ln\left(\frac{2x+\ell}{2x}\right)$ (C) $\frac{\mu_0 i I}{2\pi} \ln\left(\frac{x+\ell}{x}\right)$ (D) None of these

MG0157

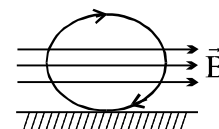
18. A semi circular current carrying wire having radius R is placed in x - y plane with its centre at origin 'O'. There is non-uniform magnetic field $\vec{B} = \frac{B_0 x}{2R} \hat{k}$ (here B_0 is +ve constant) is existing in the region. The magnetic force acting on semi circular wire will be along :-



- (A) $-x$ -axis (B) $+y$ -axis (C) $-y$ -axis (D) $+x$ -axis

MG0158

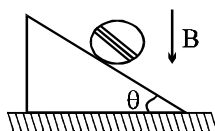
19. A conducting ring of mass 2 kg and radius 0.5 m is placed on a smooth horizontal plane. The ring carries a current $i = 4\text{ A}$. A horizontal magnetic field $B = 10\text{ T}$ is switched on at time $t = 0$ as shown in figure. The initial angular acceleration of the ring will be :-



- (A) $40\pi\text{ rad/s}^2$ (B) $20\pi\text{ rad/s}^2$ (C) $5\pi\text{ rad/s}^2$ (D) $15\pi\text{ rad/s}^2$

MG0159

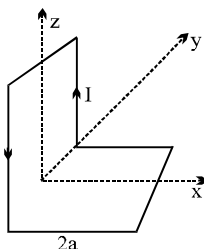
20. In the figure shown a coil of single turn is wound on a sphere of radius R and mass m . The plane of the coil is parallel to the plane and lies in the equatorial plane of the sphere. Current in the coil is i . The value of B if the sphere is in equilibrium is :-



- (A) $\frac{mg \cos \theta}{\pi i R}$ (B) $\frac{mg}{\pi i R}$ (C) $\frac{mg \tan \theta}{\pi i R}$ (D) $\frac{mg \sin \theta}{\pi i R}$

MG0160

21. A non-planar loop of conducting wire carrying a current I is placed as shown in the figure. Each of the straight sections of the loop is of length $2a$. The magnetic field due to this loop at the point $P(a, 0, a)$ points in the direction :-
[JEE, 2001 (Scr)]



- (A) $\frac{1}{\sqrt{2}}(-\hat{j} + \hat{k})$ (B) $\frac{1}{\sqrt{3}}(-\hat{j} + \hat{k} + \hat{i})$ (C) $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$ (D) $\frac{1}{\sqrt{2}}(\hat{i} + \hat{k})$

MG0161

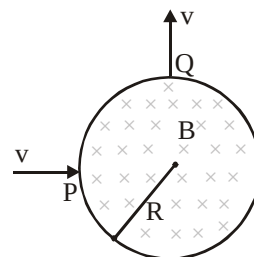
22. Two protons move parallel to each other, keeping distance r between them, both moving with same velocity $\vec{V}$. Then the ratio of the electric and magnetic force of interaction between them is :-

- (A) c^2/V^2 (B) $2c^2/V^2$ (C) $c^2/2V^2$ (D) None

MG0162

MULTIPLE CORRECT TYPE QUESTIONS

23. A particle of charge ' q ' and mass ' m ' enters normally (at point P) in a region of magnetic field with speed v . It comes out normally from Q after time T as shown in figure. The magnetic field B is present only in the region of radius R and is uniform. Initial and final velocities are along radial direction and they are perpendicular to each other. For this to happen, which of the following expression(s) is/are **CORRECT** :-



- (A) $B = \frac{mv}{qR}$ (B) $T = \frac{\pi R}{2v}$ (C) $T = \frac{\pi m}{2qB}$ (D) None of these

MG0163

24. A particle of mass m and charge q moving with velocity $\vec{v}$ enters a region of uniform magnetic field of induction $\vec{B}$. Then

- (A) its path in the region of the field is always circular
(B) its path in the region of the field is circular if $\vec{v} \cdot \vec{B} = 0$
(C) its path in the region of the field is a straight line if $\vec{v} \times \vec{B} = 0$
(D) distance travelled by the particle in time T does not depend on the angle between $\vec{v}$ and $\vec{B}$

MG0164

MATRIX MATCH TYPE QUESTION

25. A charged particle with some initial velocity is projected in a region where uniform electric and/or magnetic fields are present. In Column-I information about the existence of electric and/or magnetic field and direction of initial velocity of charged particle are given, while in column-II the possible paths of charged particle is mentioned. Match the entries of Column I with the entries of Column-II.

Column-I

Column-II

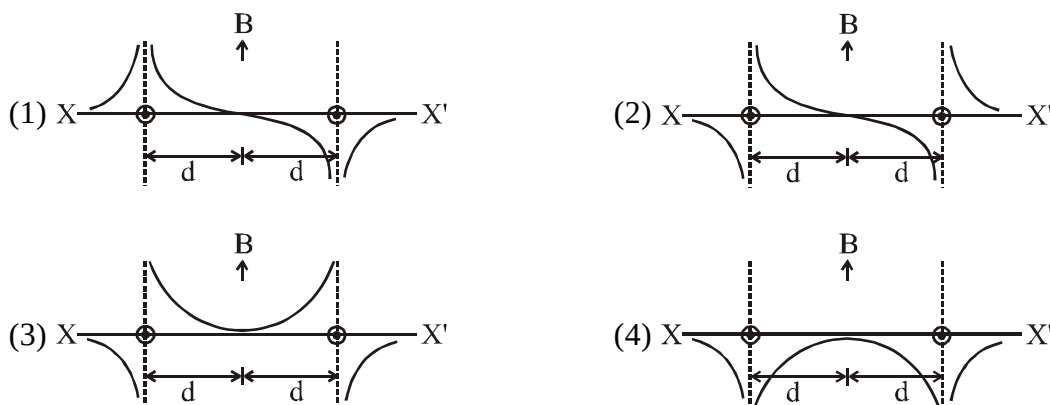
- | | |
|---|---|
| (A) $\vec{E} = 0, \vec{B} \neq 0$ and initial velocity is at an unknown angle with $\vec{B}$ | (P) Straight line |
| (B) $\vec{E} \neq 0, \vec{B} = 0$ and initial velocity is at an unknown angle with $\vec{E}$ | (Q) Parabola |
| (C) $\vec{E} \neq 0, \vec{B} \neq 0, \vec{E} \parallel \vec{B}$ and initial velocity is perpendicular to $\vec{E}$ | (R) Circular |
| (D) $\vec{E} \neq 0, \vec{B} \neq 0, \vec{E}$ is perpendicular to $\vec{B}$ and initial velocity is perpendicular to both $\vec{E}$ and $\vec{B}$ | (S) Helical path with nonuniform pitch
(T) Helical path with uniform pitch |

MG0165

EXERCISE-JM

1. Two long parallel wires are at a distance $2d$ apart. They carry steady equal currents flowing out of the plane of the paper as shown. The variation of the magnetic field B along the line XX' is given by:-

[AIEEE - 2010]



MG0166

2. A current I flows in an infinity long wire with cross section in the form of a semicircular ring of radius R . The magnitude of the magnetic induction along its axis is :-

[AIEEE - 2011]

- (1) $\frac{\mu_0 I}{2\pi R}$ (2) $\frac{\mu_0 I}{4\pi R}$ (3) $\frac{\mu_0 I}{\pi^2 R}$ (4) $\frac{\mu_0 I}{2\pi^2 R}$

MG0167

3. An electric charge $+q$ moves with velocity $\vec{V} = 3\hat{i} + 4\hat{j} + \hat{k}$, in an electromagnetic field given by $\vec{E} = 3\hat{i} + \hat{j} + 2\hat{k}$ $\vec{B} = \hat{i} + \hat{j} - 3\hat{k}$. The y -component of the force experienced by $+q$ is :

[AIEEE - 2010]

- (1) $2q$ (2) $11q$ (3) $5q$ (4) $3q$

MG0168

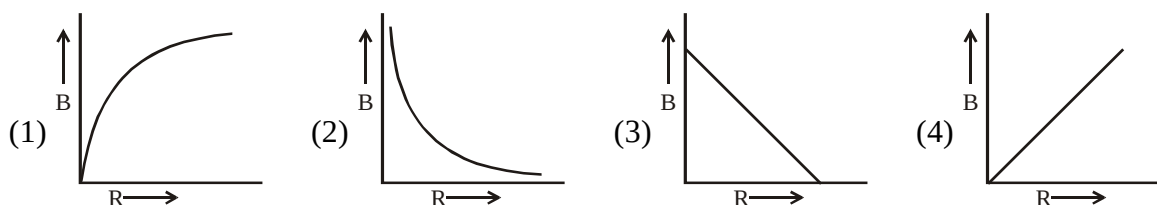
4. A thin circular disk of radius R is uniformly charged with density $\sigma > 0$ per unit area. The disk rotates about its axis with a uniform angular speed ω . The magnetic moment of the disk is :-

[AIEEE - 2011]

- (1) $2\pi R^4 \sigma \omega$ (2) $\pi R^4 \sigma \omega$ (3) $\frac{\pi R^4}{2} \sigma \omega$ (4) $\frac{\pi R^4}{4} \sigma \omega$

MG0169

5. A charge Q is uniformly distributed over the surface of non-conducting disc of radius R . The disc rotates about an axis perpendicular to its plane and passing through its centre with an angular velocity ω . As a result of this rotation a magnetic field of induction B is obtained at the centre of the disc. If we keep both the amount of charge placed on the disc and its angular velocity to be constant and vary the radius of the disc then the variation of the magnetic induction at the centre of the disc will be represented by the figure :- [AIEEE-2012]



MG0170

6. Two short bar magnets of length 1 cm each have magnetic moments 1.20 Am^2 and 1.00 Am^2 respectively. They are placed on a horizontal table parallel to each other with their N poles pointing towards the South. They have a common magnetic equator and are separated by a distance of 20.0 cm. The value of the resultant horizontal magnetic induction at the midpoint O of the line joining their centres is close to :- (Horizontal component of earth's magnetic induction is $3.6 \times 10^{-5} \text{ Wb/m}^2$)

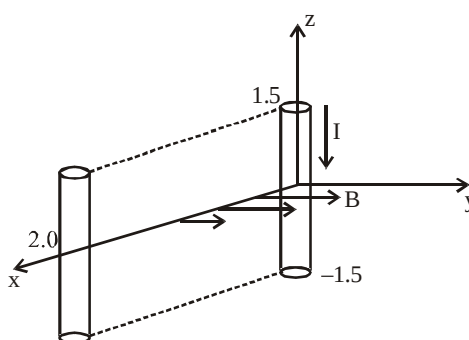
[JEE(Mains) - 2013]

- (1) $3.6 \times 10^{-5} \text{ Wb/m}^2$ (2) $2.56 \times 10^{-4} \text{ Wb/m}^2$
(3) $3.50 \times 10^{-4} \text{ Wb/m}^2$ (4) $5.80 \times 10^{-4} \text{ Wb/m}^2$

MG0171

7. A conductor lies along the z -axis at $-1.5 \leq z < 1.5 \text{ m}$ and carries a fixed current of 10.0 A in $-\hat{a}_z$ direction (see figure). For a field $\vec{B} = 3.0 \times 10^{-4} e^{-0.2x} \hat{a}_y \text{ T}$, find the power required to move the conductor at constant speed to $x = 2.0 \text{ m}$, $y = 0 \text{ m}$ in $5 \times 10^{-3} \text{ s}$. Assume parallel motion along the x -axis.

[JEE(Mains) - 2014]



- (1) 14.85 W (2) 29.7 W (3) 1.57 W (4) 2.97 W

MG0172

8. The coercivity of a small magnet where the ferromagnet gets demagnetized is $3 \times 10^3 \text{ A m}^{-1}$. The current required to be passed in a solenoid of length 10 cm and number of turns 100, so that the magnet gets demagnetized when inside the solenoid, is :

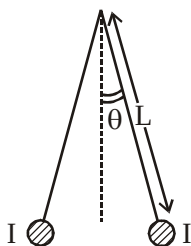
[JEE(Mains) - 2014]

- (1) 3 A (2) 6 A (3) 30 mA (4) 60 mA

MG0173

9. Two long current carrying thin wires, both with current I , are held by the insulating threads of length L and are in equilibrium as shown in the figure, with threads making an angle ' θ ' with the vertical. If wires have mass λ per unit length then the value of I is :- (g = gravitational acceleration)

[JEE(Mains) - 2015]

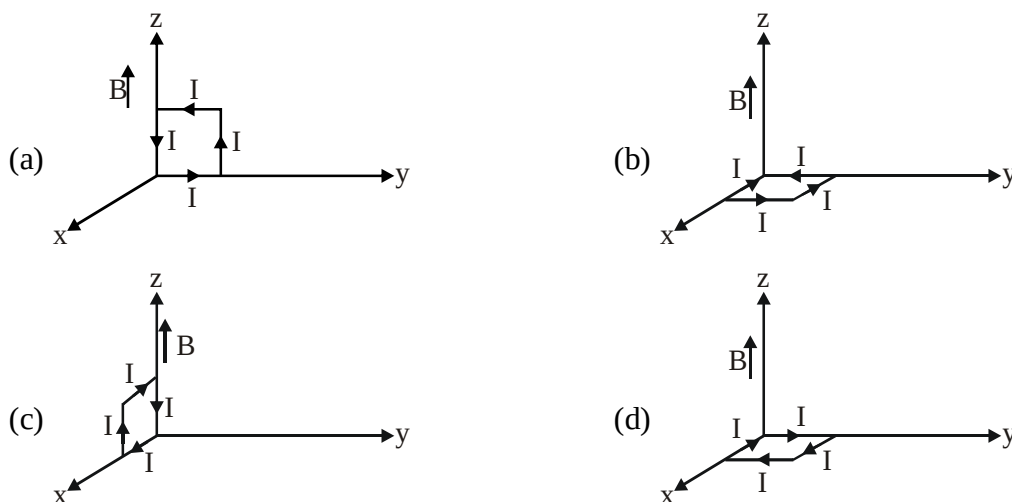


- (1) $2\sqrt{\frac{\pi g L}{\mu_0} \tan \theta}$ (2) $\sqrt{\frac{\pi \lambda g L}{\mu_0} \tan \theta}$ (3) $\sin \theta \sqrt{\frac{\pi \lambda g L}{\mu_0 \cos \theta}}$ (4) $2 \sin \theta \sqrt{\frac{\pi \lambda g L}{\mu_0 \cos \theta}}$

MG0174

10. A rectangular loop of sides 10 cm and 5 cm carrying a current I of 12 A is placed in different orientations as shown in the figures below :

[JEE(Mains) - 2015]



If there is a uniform magnetic field of 0.3 T in the positive z direction, in which orientations the loop would be in (i) stable equilibrium and (ii) unstable equilibrium ?

- (1) (b) and (d), respectively (2) (b) and (c), respectively
(3) (a) and (b), respectively (4) (a) and (c), respectively

MG0175

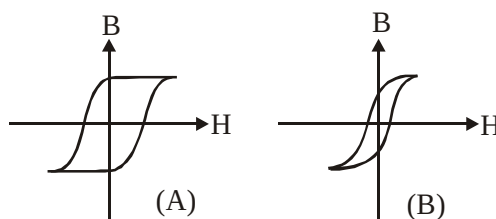
11. Two coaxial solenoids of different radii carry current I in the same direction. Let $\vec{F}_1$ be the magnetic force on the inner solenoid due to the outer one and $\vec{F}_2$ be the magnetic force on the outer solenoid due to the inner one. Then :

[JEE Main-2015]

- (1) $\vec{F}_1$ is radially inwards and $\vec{F}_2 = 0$
(2) $\vec{F}_1$ is radially outwards and $\vec{F}_2 = 0$
(3) $\vec{F}_1 = \vec{F}_2 = 0$
(4) $\vec{F}_1$ is radially inwards and $\vec{F}_2$ is radially outwards.

MG0176

12. Hysteresis loops for two magnetic materials A and B are given below :



These materials are used to make magnets for electric generators, transformer core and electromagnet core. Then it is proper to use ; **[JEE Main-2016]**

- (1) B for electromagnets and transformers.
- (2) A for electric generators and transformers.
- (3) A for electromagnets and B for electric transformers.
- (4) A for transformers and B for electric generators.

MG0177

13. Two identical wires A and B, each of length ' l ', carry the same current I . Wire A is bent into a circle of radius R and wire B is bent to form a square of side ' a '. If B_A and B_B are the values of magnetic field at the centres of the circle and square respectively, then the ratio $\frac{B_A}{B_B}$ is : **[JEE Main-2016]**

- (1) $\frac{\pi^2}{8\sqrt{2}}$
- (2) $\frac{\pi^2}{8}$
- (3) $\frac{\pi^2}{16\sqrt{2}}$
- (4) $\frac{\pi^2}{16}$

MG0178

14. A magnetic needle of magnetic moment $6.7 \times 10^{-2} \text{ Am}^2$ and moment of inertia $7.5 \times 10^{-6} \text{ kg m}^2$ is performing simple harmonic oscillations in a magnetic field of 0.01 T . Time taken for 10 complete oscillations is : **[JEE Main-2017]**
- (1) 6.98 s
 - (2) 8.76 s
 - (3) 6.65 s
 - (4) 8.89 s

MG0179

15. An electron, a proton and an alpha particle having the same kinetic energy are moving in circular orbits of radii r_e , r_p , r_α respectively in a uniform magnetic field B . The relation between r_e , r_p , r_α is:- **[JEE Main-2018]**
- (1) $r_e < r_p = r_\alpha$
 - (2) $r_e < r_p < r_\alpha$
 - (3) $r_e < r_\alpha < r_p$
 - (4) $r_e > r_p = r_\alpha$

MG0180

16. The dipole moment of a circular loop carrying a current I , is m and the magnetic field at the centre of the loop is B_1 . When the dipole moment is doubled by keeping the current constant, the magnetic field at the centre of the loop is B_2 . The ratio $\frac{B_1}{B_2}$ is : **[JEE Main-2018]**

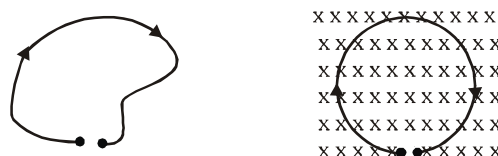
- (1) $\sqrt{3}$
- (2) $\sqrt{2}$
- (3) $\frac{1}{\sqrt{2}}$
- (4) 2

MG0181

EXERCISE-JA

1. A thin flexible wire of length L is connected to two adjacent fixed points and carries a current I in the clockwise direction, as shown in the figure. When the system is put in a uniform magnetic field of strength B going into the plane of the paper, the wire takes the shape of a circle. The tension in the wire is :

[2010, 3M]



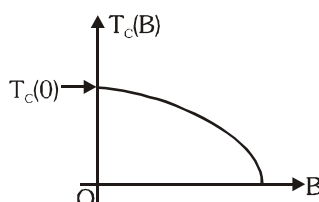
- (A) IBL (B) $\frac{IBL}{\pi}$ (C) $\frac{IBL}{2\pi}$ (D) $\frac{IBL}{4\pi}$

MG0182

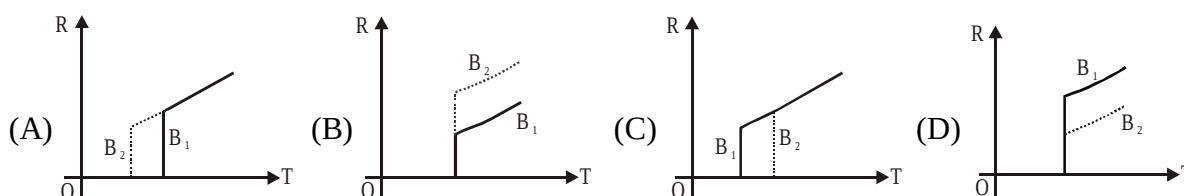
Paragraph for Question no. 2 and 3

Electrical resistance of certain materials, known as superconductors, changes abruptly from a nonzero value to zero as their temperature is lowered below a critical temperature $T_c(0)$. An interesting property of superconductors is that their critical temperature becomes smaller than $T_c(0)$ if they are placed in a magnetic field, i.e., the critical temperature $T_c(B)$ is a function of the magnetic field strength B . The dependence of $T_c(B)$ on B is shown in the figure.

[JEE2010]



2. In the graphs below, the resistance R of a superconductor is shown as a function of its temperature T for two different magnetic fields B_1 (solid line) and B_2 (dashed line). If B_2 is larger than B_1 , which of the following graphs shows the correct variation of R with T in these fields?



MG0183

3. A superconductor has $T_c(0) = 100$ K. When a magnetic field of 7.5 Tesla is applied, its T_c decreases to 75 K. For this material one can definitely say that when
- (A) $B = 5$ Tesla, $T_c(B) = 80$ K (B) $B = 5$ Tesla, $75 \text{ K} < T_c(B) < 100$ K
(C) $B = 10$ Tesla, $75 \text{ K} < T_c(B) < 100$ K (D) $B = 10$ Tesla, $T_c(B) = 70$ K

MG0184

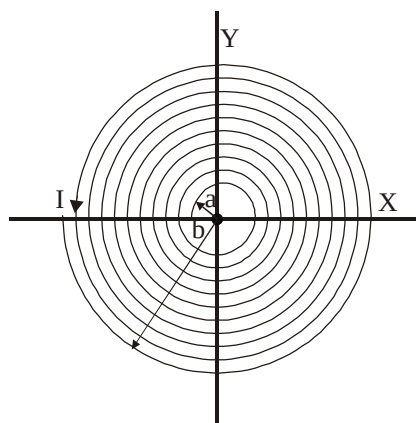
4. An electron and a proton are moving on straight parallel paths with same velocity. They enter a semi-infinite region of uniform magnetic field perpendicular to the velocity. Which of the following statement(s) is/are true? [IIT-JEE 2011]

- (A) they will never come out of the magnetic field region
(B) they will come out travelling along parallel paths
(C) they will come out of the same time
(D) they will come out at different times

MG0185

5. A long insulated copper wire is closely wound as a spiral of 'N' turns. The spiral has inner radius 'a' and outer radius 'b'. The spiral lies in the X-Y plane and a steady current 'I' flows through the wire. The Z-component of the magnetic field at the center of the spiral is [IIT-JEE 2011]

- (A) $\frac{\mu_0 NI}{2(b-a)} \ln\left(\frac{b}{a}\right)$
(B) $\frac{\mu_0 NI}{2(b-a)} \ln\left(\frac{b+a}{b-a}\right)$
(C) $\frac{\mu_0 NI}{2b} \ln\left(\frac{b}{a}\right)$
(D) $\frac{\mu_0 NI}{2b} \ln\left(\frac{b+a}{b-a}\right)$



MG0186

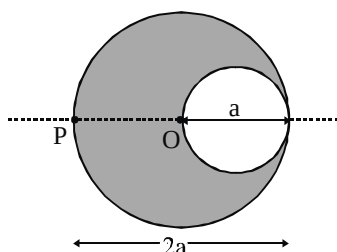
6. Consider the motion of a positive point charge in a region where there are simultaneous uniform electric and magnetic fields $\vec{E} = E_0 \hat{j}$ and $\vec{B} = B_0 \hat{j}$. At time $t = 0$, this charge has velocity $\vec{v}$ in the x-y plane, making an angle θ with the x-axis. Which of the following option (s) is (are) correct for time $t > 0$? [IIT-JEE 2012]

- (A) If $\theta = 0^\circ$, the charge moves in a circular path in the x-z plane.
(B) If $\theta = 0^\circ$, the charge undergoes helical motion with constant pitch along the y-axis.
(C) If $\theta = 10^\circ$, the charge undergoes helical motion with its pitch increasing with time, along the y-axis
(D) If $\theta = 90^\circ$, the charge undergoes linear but accelerated motion along the y-axis

MG0187

7. A cylindrical cavity of diameter a exists inside a cylinder of diameter $2a$ as shown in the figure. Both the cylinder and the cavity are infinitely long. A uniform current density J flows along the length. If the magnitude of the magnetic field at the point P is given by $\frac{N}{12} \mu_0 a J$, then the value of N is :

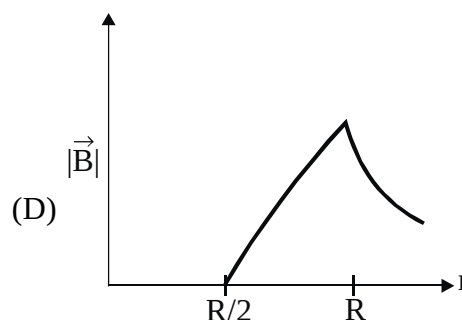
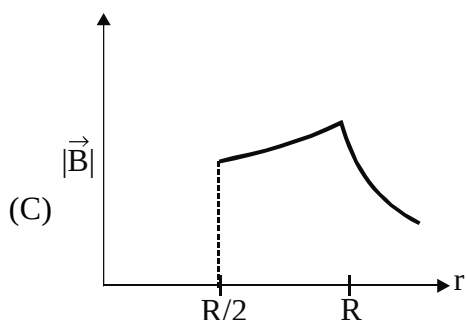
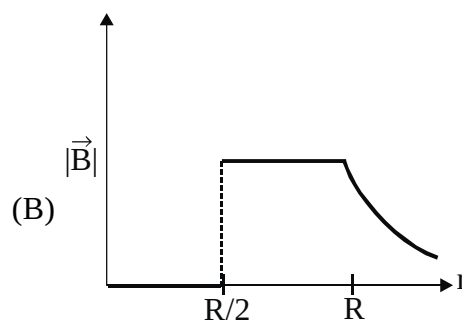
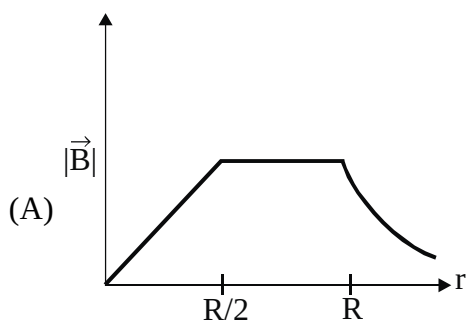
[IIT-JEE 2012]



MG0188

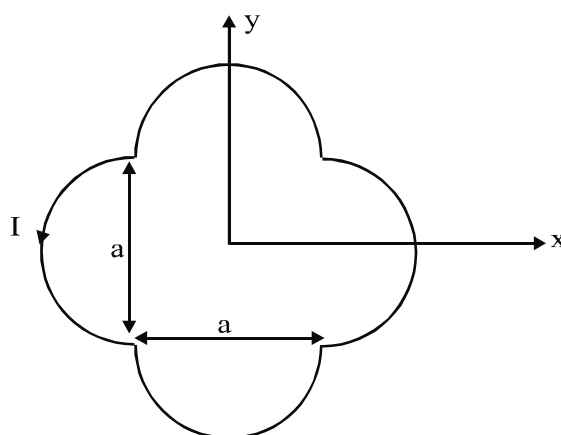
8. An infinitely long hollow conducting cylinder with inner radius $R/2$ and outer radius R carries a uniform current density along its length. The magnitude of the magnetic field, $|\vec{B}|$ as a function of the radial distance r from the axis is best represented by

[IIT-JEE 2012]



MG0189

9. A loop carrying current 'I' lies in the x-y plane as shown in the figure. The unit vector $\hat{k}$ is coming out of the plane of the paper. The magnetic moment of the current loop is [JEE 2012]



- (A) $a^2 I \hat{k}$ (B) $\left(\frac{\pi}{2} + 1\right) a^2 I \hat{k}$ (C) $-\left(\frac{\pi}{2} + 1\right) a^2 I \hat{k}$ (D) $(2\pi + 1) a^2 I \hat{k}$

MG0190

10. A particle of mass M and positive charge Q , moving with a constant velocity $\vec{u}_1 = 4\hat{i} \text{ ms}^{-1}$, enters a region of uniform static magnetic field normal to the x-y plane. The region of the magnetic field extends from $x = 0$ to $x = L$ from all values of y . After passing through this region, the particle emerges on the other side after 10 milliseconds with a velocity $\vec{u}_2 = 2(\sqrt{3}\hat{i} + \hat{j}) \text{ ms}^{-1}$. The correct statement(s) is (are) :- [IIT-JEE 2013]

- (A) The direction of the magnetic field is $-z$ direction.
 (B) The direction of the magnetic field is $+z$ direction.
 (C) The magnitude of the magnetic field $\frac{50\pi M}{3Q}$ units.
 (D) The magnitude of the magnetic field is $\frac{100\pi M}{3Q}$ units.

MG0191

11. A steady current I flows along an infinitely long hollow cylindrical conductor of radius R . This cylinder is placed coaxially inside an infinite solenoid of radius $2R$. The solenoid has n turns per unit length and carries a steady current I . Consider a point P at a distance r from the common axis. The correct statement(s) is (are) :- [JEE-Advanced-2013]

- (A) In the region $0 < r < R$, the magnetic field is non-zero
 (B) In the region $R < r < 2R$, the magnetic field is along the common axis
 (C) In the region $R < r < 2R$, the magnetic field is tangential to the circle of radius r , centred on the axis
 (D) In the region $r > 2R$, the magnetic field is non-zero

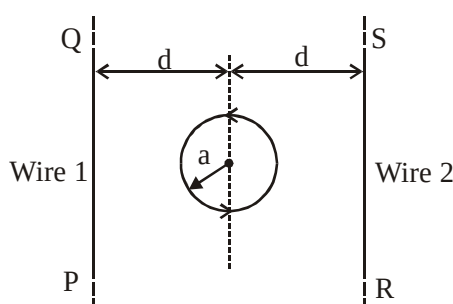
MG0192

12. Two parallel wires in the plane of the paper are distance X_0 apart. A point charge is moving with speed u between the wires in the same plane at a distance X_1 from one of the wires. When the wires carry current of magnitude I in the same direction, the radius of curvature of the path of the point charge is R_1 . In contrast, if the currents I in the two wires have directions opposite to each other, the

radius of curvature of the path is R_2 . If $\frac{X_0}{X_1} = 3$, and value of $\frac{R_1}{R_2}$ is:- [JEE-Advanced-2014]

Paragraph for Questions 13 & 14

The figure shows a circular loop of radius a with two long parallel wires (numbered 1 and 2) all in the plane of the paper. The distance of each wire from the centre of the loop is d . The loop and the wires are carrying the same current I . The current in the loop is in the counterclockwise direction if seen from above.



MG0193

13. When $d \approx a$ but wires are not touching the loop, it is found that the net magnetic field on the axis of the loop is zero at a height h above the loop. In that case
- current in wire 1 and wire 2 is the direction PQ and RS, respectively and $h \approx a$
 - current in wire 1 and wire 2 is the direction PQ and SR, respectively and $h \approx a$
 - current in wire 1 and wire 2 is the direction PQ and SR, respectively and $h \approx 1.2 a$
 - current in wire 1 and wire 2 is the direction PQ and RS, respectively and $h \approx 1.2 a$

MG0194

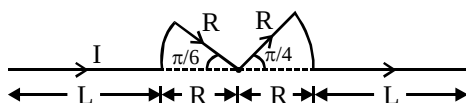
14. Consider $d \gg a$, and the loop is rotated about its diameter parallel to the wires by 30° from the position shown in the figure. If the currents in the wires are in the opposite directions, the torque on the loop at its new position will be (assume that the net field due to the wires is constant over the loop)

[JEE-Advanced-2014]

- (A) $\frac{\mu_0 I^2 a^2}{d}$ (B) $\frac{\mu_0 I^2 a^2}{2d}$ (C) $\frac{\sqrt{3} \mu_0 I^2 a^2}{d}$ (D) $\frac{\sqrt{3} \mu_0 I^2 a^2}{2d}$

MG0195

15. A conductor (shown in the figure) carrying constant current I is kept in the x - y plane in a uniform magnetic field $\vec{B}$. If F is the magnitude of the total magnetic force acting on the conductor, then the **CORRECT** statement(s) is (are) [JEE-Advanced-2015]



- (A) If $\vec{B}$ is along $\hat{z}$, $F \propto (L + R)$ (B) If $\vec{B}$ is along $\hat{x}$, $F = 0$
 (C) If $\vec{B}$ is along $\hat{y}$, $F \propto (L + R)$ (D) If $\vec{B}$ is along $\hat{z}$, $F = 0$

Answer Q.16, Q.17 and Q.18 by appropriately matching the information given in the three columns of the following table.

A charged particle (electron or proton) is introduced at the origin ($x = 0$, $y = 0$, $z = 0$) with a given initial velocity $\vec{v}$. A uniform electric field $\vec{E}$ and a uniform magnetic field $\vec{B}$ exist everywhere. The velocity $\vec{v}$, electric field $\vec{E}$ and magnetic field $\vec{B}$ are given in column 1, 2 and 3, respectively. The quantities E_0 , B_0 are positive in magnitude. [JEE-Advanced-2017]

Column-1	Column-2	Column-3
(I) Electron with $\vec{v} = 2 \frac{E_0}{B_0} \hat{x}$	(i) $\vec{E} = E_0 \hat{z}$	(P) $\vec{B} = -B_0 \hat{x}$
(II) Electron with $\vec{v} = \frac{E_0}{B_0} \hat{y}$	(ii) $\vec{E} = -E_0 \hat{y}$	(Q) $\vec{B} = B_0 \hat{x}$
(III) Proton with $\vec{v} = 0$	(iii) $\vec{E} = -E_0 \hat{x}$	(R) $\vec{B} = B_0 \hat{y}$
(IV) Proton with $\vec{v} = 2 \frac{E_0}{B_0} \hat{x}$	(iv) $\vec{E} = E_0 \hat{x}$	(S) $\vec{B} = B_0 \hat{z}$

MG0196

16. In which case will the particle move in a straight line with constant velocity ?
 (A) (II) (iii) (S) (B) (IV) (i) (S) (C) (III) (ii) (R) (D) (III) (iii) (P)

MG0197

17. In which case will the particle describe a helical path with axis along the positive z -direction ?
 (A) (II) (ii) (R) (B) (IV) (ii) (R) (C) (IV) (i) (S) (D) (III) (iii) (P)

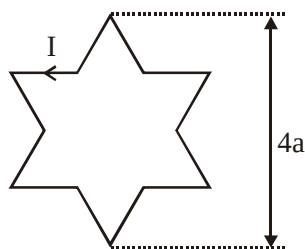
MG0198

18. In which case would the particle move in a straight line along the negative direction of y -axis (i.e., move along $-\hat{y}$) ?

- (A) (IV) (ii) (S) (B) (III) (ii) (P) (C) (II) (iii) (Q) (D) (III) (ii) (R)

MG0199

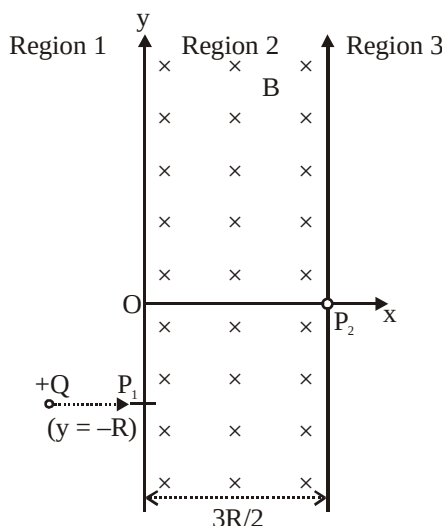
19. A symmetric star shaped conducting wire loop is carrying a steady state current I as shown in the figure. The distance between the diametrically opposite vertices of the star is $4a$. The magnitude of the magnetic field at the center of the loop is : [JEE-Advanced-2017]



- (A) $\frac{\mu_0 I}{4\pi a} 3[\sqrt{3}-1]$ (B) $\frac{\mu_0 I}{4\pi a} 6[\sqrt{3}-1]$ (C) $\frac{\mu_0 I}{4\pi a} 6[\sqrt{3}+1]$ (D) $\frac{\mu_0 I}{4\pi a} 3[2-\sqrt{3}]$

MG0200

20. A uniform magnetic field B exists in the region between $x = 0$ and $x = \frac{3R}{2}$ (region 2 in the figure) pointing normally into the plane of the paper. A particle with charge $+Q$ and momentum p directed along x -axis enters region 2 from region 1 at point $P_1(y = -R)$. Which of the following options(s) is/are correct ? [JEE-Advanced-2017]



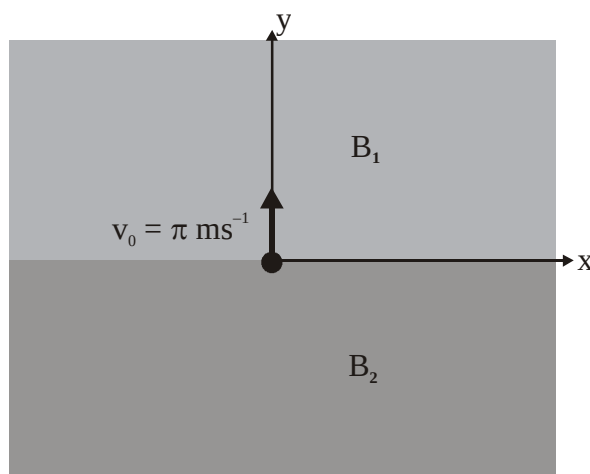
- (A) For $B = \frac{8}{13} \frac{p}{QR}$, the particle will enter region 3 through the point P_2 on x -axis
- (B) For $B > \frac{2}{3} \frac{p}{QR}$, the particle will re-enter region 1
- (C) For a fixed B , particle of same charge Q and same velocity v , the distance between the point P_1 and the point of re-entry into region 1 is inversely proportional to the mass of the particle.
- (D) When the particle re-enters region 1 through the longest possible path in region 2, the magnitude of the change in its linear momentum between point P_1 and the farthest point from y -axis is $\frac{p}{\sqrt{2}}$.

MG0201

21. Two infinitely long straight wires lie in the xy -plane along the lines $x = \pm R$. The wire located at $x = +R$ carries a constant current I_1 and the wire located at $x = -R$ carries a constant current I_2 . A circular loop of radius R is suspended with its centre at $(0, 0, \sqrt{3}R)$ and in a plane parallel to the xy -plane. This loop carries a constant current I in the clockwise direction as seen from above the loop. The current in the wire is taken to be positive if it is in the $+\hat{j}$ direction. Which of the following statements regarding the magnetic field $\vec{B}$ is (are) true ? **[JEE-Advanced-2018]**
- (A) If $I_1 = I_2$, then $\vec{B}$ cannot be equal to zero at the origin $(0, 0, 0)$
- (B) If $I_1 > 0$ and $I_2 < 0$, then $\vec{B}$ can be equal to zero at the origin $(0, 0, 0)$
- (C) If $I_1 < 0$ and $I_2 > 0$, then $\vec{B}$ can be equal to zero at the origin $(0, 0, 0)$
- (D) If $I_1 = I_2$, then the z -component of the magnetic field at the centre of the loop is $\left(-\frac{\mu_0 I}{2R}\right)$

MG0202

22. In the x - y -plane, the region $y > 0$ has a uniform magnetic field $B_1 \hat{k}$ and the region $y < 0$ has a another uniform magnetic field $B_2 \hat{k}$. A positively charged particle is projected from the origin along the positive y -axis with speed $v_0 = \pi \text{ ms}^{-1}$ at $t = 0$, as shown in the figure. Neglect gravity in this problem. Let $t = T$ be the time when the particle crosses the x -axis from below for the first time. If $B_2 = 4B_1$, the average speed of the particle, in ms^{-1} , along the x -axis in the time interval T is _____.

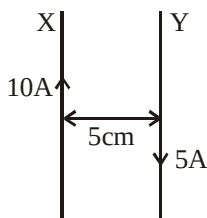
[JEE-Advanced-2018]

MG0203

MAGNETIC EFFECT OF CURRENT

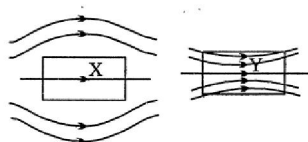
(CBSE Previous Year's Questions)

- Two long parallel straight wires X and Y separated by a distance of 5 cm in air carry currents of 10 A and 5 A respectively in opposite directions. Calculate the magnitude and direction of the force on a 20 cm length of the wire Y.



A circular coil of 100 turns, radius 10 cm carries a current of 5 A. It is suspended vertically in a uniform horizontal magnetic field of 0.5 T, the field lines making an angle of 60° with the normal to the plane of the coil. Calculate the torque that must be applied on it to prevent it from turning. [2; CBSE-2004]

- Using Biot-Savart law, deduce an expression for the magnetic field on the axis of a circular current carrying loop. Draw the magnetic field lines due to a circular current carrying loop. [3; CBSE-2004]
- A hydrogen ion of mass 'm' and charge 'q' travels with a speed 'v' in a circle of radius 'r' in a magnetic field of intensity 'B'. Write the equation in terms of these quantities only, relating the force on the ion to the required centripetal force. Hence derive an expression for its time period. [3; CBSE-2004]
- A uniform magnetic field gets modified as shown below, when two specimens X and Y are placed in it



(i) Identify the two specimens X and Y.

(ii) State the reason for the behaviour of the field lines in X and Y.

[3; CBSE-2004]

- Two wires of equal lengths are bent in the form of two loops. One of the loops is square shaped whereas the other loop is circular. These are suspended in a uniform magnetic field and the same current is passed through them. Which loop will experience greater torque? Give reasons.

[1; CBSE-2005]

- Write two characteristic properties to distinguish between diamagnetic and paramagnetic materials.

[2; CBSE-2005]

- Explain the principle and working of a cyclotron with the help of a Labeled diagram. A cyclotron's oscillator frequency is 10 MHz. What should be the operating magnetic field for accelerating protons? If the radius of its 'dees' is 60 cm, what is the kinetic energy of the proton beam produced by the accelerator? Express your answer in units of MeV.

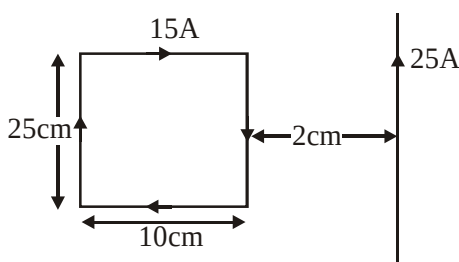
($e = 1.6 \times 10^{-19}$ C, $m_p = 1.67 \times 10^{-27}$ Kg, $1 \text{ MeV} = 1.602 \times 10^{-13}$ J)

[5; CBSE-2005]

8. Depict the magnetic field lines due to two straight, long, parallel conductors carrying currents I_1 and I_2 in the same direction. Hence deduce an expression for the force acting per unit length on one conductor due to the other. Is this force attractive or repulsive?

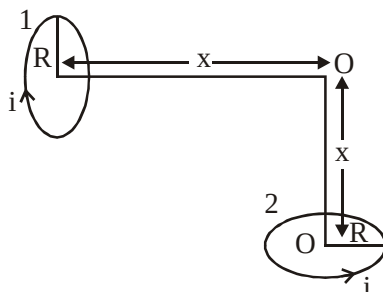
Figure shows a rectangular current-carrying loop placed 2 cm away from a long, straight, current-carrying conductor. What is the direction and magnitude of the net force acting on the loop?

[5; CBSE-2005]



9. Steel is preferred for making permanent magnets whereas soft iron is preferred for making electromagnets. Give one reason. [1; CBSE-2006]
10. Draw a neat and labelled diagram of a cyclotron. State the underlying principle and explain how a positively charged particle gets accelerated in this machine. Show mathematically that the cyclotron frequency does not depend upon the speed of the particle. [5; CBSE-2006]
11. State the Biot - Savart law for the magnetic field due to a current carrying element. Use this law to obtain a formula for magnetic field at the centre of a circular loop of radius R carrying a steady current I . Sketch the magnetic field lines for a current loop clearly indicating the direction of the field. [5; CBSE-2006]
12. An electron is moving along +ve x -axis in the presence of uniform magnetic field along +ve y -axis. What is the direction of the force acting on it? [1; CBSE-2007]
13. Explain, with the help of a labelled diagram, the principle and construction of a cyclotron. Deduce an expression for the cyclotron frequency and show that it does not depend on the speed of the charged particle. [3; CBSE-2007]
14. Distinguish the magnetic properties of dia, para- and ferro-magnetic substances in terms of (i) susceptibility, (ii) magnetic permeability and (iii) coercivity. Give one example of each of these materials. Draw the field lines due to an external magnetic field near a (i) diamagnetic, (ii) paramagnetic substance. [3; CBSE-2007]
15. What is the direction of the force acting on a charged particle q , moving with a velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$? [1; CBSE-2008]
16. Define magnetic susceptibility of a material Name two elements, one having positive susceptibility and the other having negative susceptibility. What does negative susceptibility signify? [2; CBSE-2008]

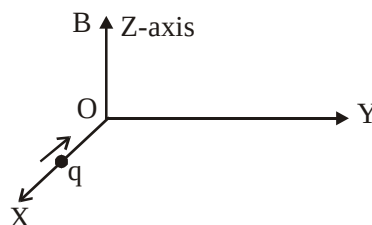
17. (a) Using Biot-Savart's law, derive an expression for the magnetic field at the centre of a circular coil of radius R , number of turns N , carrying current i .
 (b) Two small identical circular coils marked 1,2 carry equal currents and are placed with their geometric axes perpendicular to each other as shown in the figure. Derive an expression for the resultant magnetic field at O .



Draw a schematic diagram of a cyclotron. Explain its underlying principle and working, stating clearly the function of the electric and magnetic fields applied on a charged particle. Deduce an expression for the period of revolution and show that it does not depend on the speed of the charged particle.

[5; CBSE-2008]

18. Magnetic field lines can be entirely confined within the core of a toroid, but not within a straight solenoid. Why ?
 [1; CBSE-2009]
19. A charge ' q ' moving along the X -axis with a velocity v is subjected to a uniform magnetic field B acting along the Z -axis as it crosses the origin O .

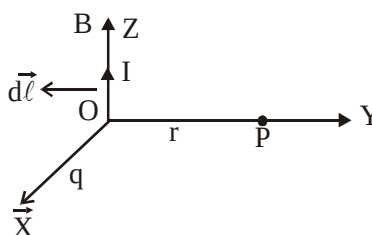


- (i) Trace its trajectory.
 (ii) Does the charge gain kinetic energy as it enters the magnetic field? Justify your answer.

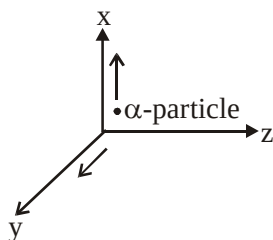
[2; CBSE-2009]

20. State Biot-Savart law.

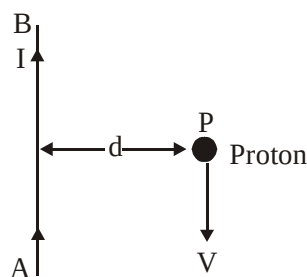
A current I flows in a conductor placed perpendicular to the plane of the paper. Indicate the direction of the magnetic field due to a small element $d\vec{\ell}$ at point P situated at a distance $\vec{r}$ from the element as shown in the figure.



21. Derive the expression for force per unit length between two long straight parallel current carrying conductors. Hence define one ampere. [3; CBSE-2009]
22. Explain the principle and working of a cyclotron with the help of a schematic diagram. Write the expression for cyclotron frequency. [3; CBSE-2009]
23. A beam of α particle projected along +x-axis, experiences a force due to a magnetic field along the +y – axis. What is the direction of the magnetic field? [1; CBSE-2010]



24. Deduce the expression for the magnetic dipole moment of an electron orbiting around the central nucleus. [2; CBSE-2010]
25. (a) With the help of a diagram, explain the principle and working of a moving coil galvanometer.
 (b) What is the importance of a radial magnetic field and how is it produced ?
 (c) Why is it that while using a moving coil galvanometer as a voltmeter a high resistance in series is required whereas in an ammeter a shunt is used ? [5; CBSE-2010]
26. (a) Derive an expression for the force between two long parallel current carrying conductors.
 (b) Use this expression to define S.I. unit of current.
 (c) A long straight wire AB carries a current I . A proton P travels with a speed v , parallel to the wire, at a distance d from it in a direction opposite to the current as shown in the figure. What is the force experienced by the proton and what is its direction ? [5; CBSE-2010]



27. Where on the surface of Earth is the angle of dip 90° ? [1; CBSE-2011]
28. Write the expression for Lorentz magnetic force on a particle of charge ' q ' moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$. Show that no work is done by this force on the charged particle. [2; CBSE-2011]
29. A steady current (I_1) flows through a long straight wire. Another wire carrying steady current (I_2) in the same direction is kept close and parallel to the first wire. Show with the help of a diagram how the magnetic field due to the current I_1 exerts a magnetic force on the second wire. Write the expression for this force. [2; CBSE-2011]

30. (a) State the principle of the working of moving coil galvanometer, giving its labelled diagram.
 (b) "Increasing the current sensitivity of a galvanometer may not necessarily increase its voltage sensitivity." Justify this statement.
 (c) Outline the necessary steps to convert a galvanometer of resistance R_g into an ammeter of a given range. [5; CBSE-2011]
31. (a) Using Ampere's circuital law, obtain the expression for the magnetic field due to a long solenoid at a point inside the solenoid on its axis.
 (b) In what respect is a toroid different from a solenoid? Draw and compare the pattern of the magnetic field lines in the two cases.
 (c) How is the magnetic field inside a given solenoid made strong? [5; CBSE-2011]
32. A circular coil of N turns and radius R carries a current I . It is unwound and rewound to make another coil of radius $R/2$, current I remaining the same. Calculate the ratio of the magnetic moments of the new coil and original coil. [2; CBSE-2012]
33. (a) Write the expression for the force, F , acting on a charged particle of charge ' q ', moving with a velocity $\vec{v}$ in the presence of both electric field $\vec{E}$ and magnetic field $\vec{B}$. Obtain the condition under which the particle moves undeflected through the fields.
 (b) A rectangular loop of size $l \times b$ carrying a steady current I is placed in uniform magnetic field $\vec{B}$. Prove that the torque $\vec{\tau}$ acting on the loop is given by $\vec{\tau} = \vec{m} \times \vec{B}$, where m is the magnetic moment of the loop.

OR

- (a) Explain, giving reason, the basic difference in converting a galvanometer into (i) a voltmeter and (ii) an ammeter
 (b) Two long straight parallel conductors carrying steady current I_1 and I_2 are separated by a distance ' d '. Explain briefly, with the help of a suitable diagram, how the magnetic field due to one conductor acts on the other. Hence deduce the expression for the force acting between the two conductors. Mention the nature of this force. [5; CBSE-2012]
34. A wire AB is carrying a steady current of 12 A and is lying on the table. Another wire CD carrying 5 A is held directly above AB at a height of 1 mm. Find the mass per unit length of the wire CD so that it remains suspended at its position when left free. Give the direction of the current flowing in CD with respect to that in AB. [Take the value of $g = 10 \text{ ms}^{-2}$] [CBSE-2013]
35. (a) Using Biot - Savart's law, derive the expression for the magnetic field in the vector form at a point on the axis of a circular current loop.
 (b) What does a toroid consist of? Find out the expression for the magnetic field inside a toroid for N turns of the coil having the average radius r and carrying a current I . Show that the magnetic field in the open space inside and exterior to the toroid is zero.

OR

- (a) Draw a schematic sketch of a cyclotron. Explain clearly the role of crossed electric and magnetic field in accelerating the charge. Hence derive the expression for the kinetic energy acquired by the particles.
 (b) An α -particle and a proton are released from the centre of the cyclotron and made to accelerate.
 (i) Can both be accelerated at the same cyclotron frequency? Give reason to justify your answer.
 (ii) When they are accelerated in turn, which of the two will have higher velocity at the exit slit of the dees? [CBSE-2013]

36. Using the concept of force between two infinitely long parallel current carrying conductors, define one ampere of current. [CBSE-2014]
37. Show diagrammatically the behaviour of magnetic field lines in the presence of (i) paramagnetic and (ii) diamagnetic substances. How does one explain this distinguishing feature? [CBSE-2014]
38. (a) Deduce an expression for the frequency of revolution of a charged particle in a magnetic field and show that it is independent of velocity or energy of the particle.
(b) Draw a schematic sketch of a cyclotron. Explain, giving the essential details of its construction, how it is used to accelerate the charged particles.

OR

- (a) Draw a labelled diagram of a moving coil galvanometer. Describe briefly its principle and working.
(b) Answer the following:
(i) Why is it necessary to introduce a cylindrical soft iron core inside the coil of a galvanometer?
(ii) Increasing the current sensitivity of a galvanometer may not necessarily increase its voltage sensitivity. Explain, giving reason. [CBSE-2014]
39. A cyclotron's oscillator frequency is 10 MHz. What should be the operating magnetic field for accelerating protons? If the radius of its 'dees' is 60 cm, calculate the kinetic energy (in MeV) of the proton beam produced by the accelerator. [3; CBSE-2015]
40. Deduce the expression for the torque $\vec{\tau}$ acting on a planar loop of area $\vec{A}$ and carrying current I placed in a uniform magnetic field $\vec{B}$. [3; CBSE-2015]
If the loop is free to rotate, what would be its orientation in stable equilibrium?
41. What can be the cause of helical motion of a charged particle? [1; CBSE-2016]
42. State Ampere's circuital law. Use this law to find magnetic field due to straight infinite current carrying wire. How are the magnetic field lines different from the electrostatic field lines? [3; CBSE-2016]

OR

State the principle of a cyclotron. Show that the time period of revolution of particles in a cyclotron is independent of their speeds. Why is this property necessary for the operation of a cyclotron?

43. Seema's uncle was advised by his doctor to have an MRI (Magnetic Resonance Imaging) scan of his brain. Her uncle felt it to be expensive and wanted to postpone it. [4; CBSE-2016]
When Seema learnt about this, she took the help of her family and also approached the doctor, who also offered a substantial discount. She then convinced her uncle to undergo the test to enable the doctor to know the condition of his brain. The information thus obtained greatly helped the doctor to treat him properly.
Based on the above paragraph, answer the following questions :
(a) What according to you are the values displayed by Seema, her family and the doctor?
(b) What could be the possible reason for MRI test to be so expensive?
(c) Assuming that MRI test was performed using a magnetic field of 0.1 T., find the minimum and maximum values of the force that the magnetic field could exert on a proton (charge = 1.6×10^{-19} C) moving with a speed of 10^4 m/s.

44. (a) When a bar magnet is pushed towards (or away) from the coil connected to a galvanometer, the pointer in the galvanometer deflects. Identify the phenomenon causing this deflection and write the factors on which the amount and direction of the deflection depends. State the laws describing this phenomenon. **[5; CBSE-2016]**
45. Find the condition under which the charged particles moving with different speeds in the presence of electric and magnetic field vectors can be used to select charged particles of a particular speed. **[2; CBSE-2017]**
46. Write two properties of a material suitable for making (a) a permanent magnet, and (b) an electromagnet. **[2; CBSE-2017]**
47. (a) State Biot – Savart law and express this law in the vector form. **[3; CBSE-2017]**
 (b) Two identical circular coils, P and Q each of radius R, carrying currents 1A and $\sqrt{3}$ A respectively, are placed concentrically and perpendicular to each other lying in the XY and YZ planes. Find the magnitude and direction of the net magnetic field at the centre of the coils.
48. A proton and an electron travelling along parallel paths enter a region of uniform magnetic field, acting perpendicular to their paths. Which of them will move in a circular path with higher frequency? **[3; CBSE-2018]**
49. A bar magnet of magnetic moment of 6J/T is aligned at 60° with a uniform external magnetic field of 0.44 T. Calculate (a) the work done in turning the magnet to align its magnetic moment (i) normal to the magnetic field, (ii) opposite to the magnetic field, and (b) the torque on the magnet in the final orientation in case (ii). **[3; CBSE-2018]**
50. (a) An iron ring of relative permeability μ_r has windings of insulated copper wire of n turns per metre. When the current in the windings is I, find the expression for the magnetic field in the ring.
 (b) The susceptibility of a magnetic material is 0.9853. Identify the type of magnetic material. Draw the modification of the field pattern on keeping a piece of this material in a uniform magnetic field. **[3; CBSE-2018]**

ANSWER KEY

EXERCISE (S-1)

1. **Ans.** (a) zero (b) $2\mu\text{T}$ along the Z-axis (c) zero and (d) $2\mu\text{T}$ along the negative Z-axis
2. **Ans.** $4.0 \times 10^{-5} \text{ T}$, downwards in both the cases
3. **Ans.** $5\pi \times 10^{-4} \text{ T} = 1.6 \times 10^{-3} \text{ T}$ towards west. 4. **Ans.** $7.07 \times 10^{-10} \hat{k} \text{ T}$
5. **Ans.** At a distance of $\frac{4r}{\pi}$ from the centre in such a way that the direction of the current in it is opposite to that in the nearest part of the circular wire.
6. **Ans.** (a) 0 (b) $1.41 \times 10^{-6} \text{ T}$, 45° in xz-plane, (c) $5 \times 10^{-6} \text{ T}$, +x-direction]
7. **Ans.** $\mu_0 \text{ weber.m}^{-1}$ 8. **Ans.** $\frac{2mv_0}{qB}$ 9. **Ans.** $1.2 \times 10^{-2} \text{ m}$, $4.37 \times 10^{-2} \text{ m}$
10. **Ans.** (a) Circular trajectory of radius 1.0 mm normal to B.
(b) Helical trajectory of radius 0.5 mm with velocity component $2.3 \times 10^7 \text{ ms}^{-1}$ along B.
11. **Ans.** (a) A horizontal magnetic field of magnitude 0.26 T normal to the conductor in such a direction that Fleming a left-hand rule gives a magnetic force upward.
(b) 1.176 N.
12. **Ans.** (a) 3.1 Nm, (b) No, the answer is unchanged because the formula $\tau = N I A \times B$ is true for a planar loop of any shape.
13. **Ans.** Use $\tau = IA \times B$ and $F = I l \times B$
(a) $1.8 \times 10^{-2} \text{ N m}$ along -y direction
(b) same as in (a)
(c) $1.8 \times 10^{-2} \text{ N m}$ along -x direction
(d) $1.8 \times 10^{-2} \text{ N m}$ at an angle of 240° with the +x direction
(e) zero
(f) zero
Force is zero in each case. Case (e) corresponds to stable, and case (f) corresponds to unstable equilibrium
14. **Ans.** (a) Zero, (b) zero, (c) force on each electron is $e v B = IB/(nA) = 5 \times 10^{-25} \text{ N}$. Note : Answer (c) denotes only the magnetic force.

15. **Ans.** $T_0 = 2\pi \sqrt{\frac{m}{6IB}} = 0.57 \text{ s}$

16. **Ans.** (a) 1.4, (b) 1

EXERCISE (S-2)

1. **Ans.** $B_1 = \frac{\mu_0 b r_1^2}{3}$, $B_2 = \frac{\mu_0 b R^3}{3r_2}$

2. **Ans.** $\sqrt{15} \text{ C}$

3. **Ans.** (6.4 m, 0, 0) (6.4m, 0, 2m)

4. **Ans.** $i_1 = 0.1110 \text{ A}$, $i_2 = 0.096 \text{ A}$

5. **Ans.** (i) $-\frac{\mu_0 I}{4R} q v_0 \hat{k}$ (ii) $F_1 = 2 I R B$ $F_2 = 2 I R B$, Net force = $F_1 + F_2 = 4 I R B \hat{i}$

6. Ans. (a) $6.6 \times 10^{-5} \text{T}$, (b) 0, 0, $2 \times 10^{-5} \ell \ln\left(\frac{3}{2}\right) \text{N}$ 7. Ans. (a) $\frac{3mv^2}{4qa}$, (b) $\frac{3mv^3}{4a}$, (c) zero

8. Ans. $\frac{\mu_0 i q v}{2\pi a}$ 9. Ans. 5 10. Ans. $\frac{m E I}{B e}$ 11. Ans. 8

12. Ans. 4 13. Ans. 020 14. Ans. $z = 0$, $x = \pm \frac{d}{\sqrt{3}}$, (ii) $\frac{I}{2\pi d} \sqrt{\frac{\mu_0}{\pi \lambda}}$

15. Ans. (a) current in loop PQRS is clockwise from P to QRS., (b) $\vec{F} = BI_0 b (3\hat{k} - 4\hat{i})$, (c) $I = \frac{mg}{6bB_0}$

16. Ans. $F = \alpha a^2 i \hat{j}$ 17. Ans. 4 18. Ans. $\omega = \frac{dT_0}{QR^2 B}$

19. Ans. (a) $k = NAB$, (b) $C = \frac{2i_0 NAB}{\pi}$, (c) $Q \times \sqrt{\frac{NAB\pi}{2li_0}}$

EXERCISE (O-1)

- | | | | | | |
|---|------------------|--------------|--|------------------|--------------|
| 1. Ans. (B) | 2. Ans. (C) | 3. Ans. (C) | 4. Ans. (A) | 5. Ans. (B) | 6. Ans. (C) |
| 7. Ans. (B) | 8. Ans. (A) | 9. Ans. (B) | 10. Ans. (D) | 11. Ans. (A) | 12. Ans. (C) |
| 13. Ans. (D) | 14. Ans. (D) | 15. Ans. (A) | 16. Ans. (B) | 17. Ans. (A) | 18. Ans. (C) |
| 19. Ans. (C) | 20. Ans. (C) | 21. Ans. (C) | 22. Ans. (B) | 23. Ans. (B) | 24. Ans. (C) |
| 25. Ans. (D) | 26. Ans. (D) | 27. Ans. (B) | 28. Ans. (B) | 29. Ans. (C) | 30. Ans. (B) |
| 31. Ans. (C) | 32. Ans. (A) | 33. Ans. (B) | 34. Ans. (A) | 35. Ans. (B) | 36. Ans. (A) |
| 37. Ans. (B) | 38. Ans. (B) | 39. Ans. (C) | 40. Ans. (D) | 41. Ans. (C) | 42. Ans. (A) |
| 43. Ans. (A) | 44. Ans. (B) | 45. Ans. (D) | 46. Ans. (C) | 47. Ans. (C) | 48. Ans. (C) |
| 49. Ans. (B) | 50. Ans. (C) | 51. Ans. (A) | 52. Ans. (D) | 53. Ans. (A) | 54. Ans. (C) |
| 55. Ans. (B) | 56. Ans. (B) | 57. Ans. (A) | 58. Ans. (D) | 59. Ans. (A) | 60. Ans. (B) |
| 61. Ans. (B) | 62. Ans. (A) | 63. Ans. (C) | 64. Ans. (A) | 65. Ans. (A) | 66. Ans. (B) |
| 67. Ans. (B) | 68. Ans. (B) | 69. Ans. (D) | 70. Ans. (B) | 71. Ans. (C) | 72. Ans. (C) |
| 73. Ans. (B) | 74. Ans. (D) | 75. Ans. (A) | 76. Ans. (A) | 77. Ans. (B) | 78. Ans. (C) |
| 79. Ans. (D) | 80. Ans. (B) | 81. Ans. (C) | 82. Ans. (C) | 83. Ans. (B) | 84. Ans. (C) |
| 85. Ans. (B) | 86. Ans. (D) | 87. Ans. (A) | 88. Ans. (B) | 89. Ans. (A,B,C) | |
| 90. Ans. (A,B,C) | 91. Ans. (A,B,C) | | 92. Ans. (A, C, D) | 93. Ans. (A,C,D) | |
| 94. Ans. (A, C) | 95. Ans. (C) | | 96. Ans. (B,D) | 97. Ans. (A,D) | |
| 98. Ans. (A,B) | 99. Ans. (A) | | 100. Ans. (D) | 101. Ans. (C) | |
| 102. Ans. (B) | 103. Ans. (D) | | 104. Ans. (A)-q,s; (B)-p,s; (C)-q,s; (D)-p,s | | |
| 105. Ans. (A)-s; (B)-s; (C)-q; (D)-r; (E)-p | | | | | |

EXERCISE (O-2)

- | | | | | | |
|------------------|---|--------------|--------------|------------------|--------------|
| 1. Ans. (C) | 2. Ans. (A) | 3. Ans. (C) | 4. Ans. (C) | 5. Ans. (A) | 6. Ans. (B) |
| 7. Ans. (C) | 8. Ans. (C) | 9. Ans. (C) | 10. Ans. (B) | 11. Ans. (A) | 12. Ans. (C) |
| 13. Ans. (A) | 14. Ans. (C) | 15. Ans. (C) | 16. Ans. (A) | 17. Ans. (C) | 18. Ans. (A) |
| 19. Ans. (A) | 20. Ans. (B) | 21. Ans. (D) | 22. Ans. (A) | 23. Ans. (A,B,C) | |
| 24. Ans. (B,C,D) | 25. Ans. (A) $\rightarrow$ (P,R, T) ; (B) $\rightarrow$ (P,Q) ; (C) $\rightarrow$ (S) ; (D) $\rightarrow$ (P) | | | | |

EXERCISE (JM)

- | | | | | | |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 1. Ans. (2) | 2. Ans. (3) | 3. Ans. (2) | 4. Ans. (4) | 5. Ans. (2) | 6. Ans. (2) |
| 7. Ans. (4) | 8. Ans. (1) | 9. Ans. (4) | 10. Ans. (1) | 11. Ans. (2) | 12. Ans. (1) |
| 13. Ans. (1) | 14. Ans. (3) | 15. Ans. (1) | 16. Ans. (2) | | |

EXERCISE (JA)

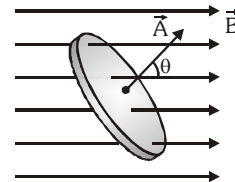
- | | | | | |
|------------------|-------------|-------------------------|----------------------|------------------|
| 1. Ans. (C) | 2. Ans. (A) | 3. Ans. (B) | 4. Ans. (BC,BD, BCD) | 5. Ans. (A) |
| 6. Ans. (C,D) | | 7. Ans. 5 | 8. Ans. (D) | 9. Ans. (B) |
| 11. Ans. (A, D) | | 12. Ans. 3 | 13. Ans. (C) | 14. Ans. (B) |
| 16. Ans. (A) | | 17. Ans. (C) | 18. Ans. (D) | 19. Ans. (B) |
| 21. Ans. (A,B,D) | | 22. Ans. 2 [1.99, 2.01] | | 20. Ans. (A) (B) |

ELECTROMAGNETIC INDUCTION & ALTERNATING CURRENT

KEY CONCEPT

MAGNETIC FLUX

The magnetic flux (ϕ) linked with a surface held in a magnetic field (B) is defined as the number of magnetic lines of force crossing that area (A). If θ is the angle between the direction of the field and normal to the area, (area vector) then $\phi = \vec{B} \cdot \vec{A} = BA \cos \theta$



FLUX LINKAGE

If a coil has more than one turn, then the flux through the whole coil is the sum of the fluxes through the individual turns. If the magnetic field is uniform, the flux through one turn is $\phi = BA \cos \theta$. If the coil has N turns, the total flux linkage $\phi = NBA \cos \theta$

- Magnetic lines of force are imaginary, magnetic flux is a real scalar physical quantity with dimensions

$$[\phi] = B \times \text{area} = \left[\frac{F}{IL} \right] [L^2] \quad \therefore B = \frac{F}{IL \sin \theta} \quad [\because F = BIL \sin \theta]$$

$$\therefore [\phi] = \left[\frac{MLT^{-2}}{AL} \right] [L^2] = [ML^2 T^{-2} A^{-1}]$$

SI UNIT of magnetic flux :

$\therefore [ML^2 T^{-2}]$ corresponds to energy

$$\frac{\text{joule}}{\text{ampere}} = \frac{\text{joule} \times \text{second}}{\text{coulomb}} = \text{weber (Wb)}$$

$$\text{or } T-m^2 \text{ (as tesla = Wb/m}^2\text{)} \quad \left[\text{ampere} = \frac{\text{coulomb}}{\text{second}} \right]$$

- For a given area flux will be maximum :

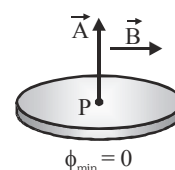
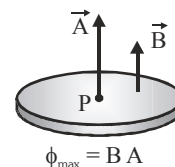
when magnetic field $\vec{B}$ is normal to the area

$$\theta = 0^\circ \Rightarrow \cos \theta = \text{maximum} = 1 \quad \phi_{\max} = BA$$

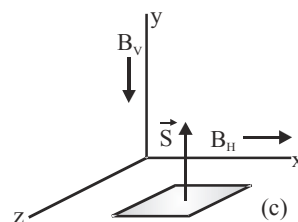
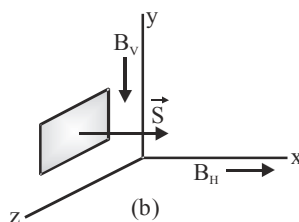
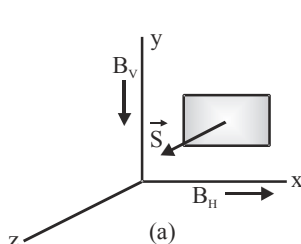
For a given area flux will be minimum :

when magnetic field $\vec{B}$ is parallel to the area

$$\theta = 90^\circ \Rightarrow \cos \theta = \text{minimum} = 0 \quad \phi_{\min} = 0$$



Ex. At a given plane, horizontal and vertical components of earth's magnetic field B_H and B_V are along x and y axes respectively as shown in figure. What is the total flux of earth's magnetic field associated with an area S , if the area S is in (a) x - y plane (b) y - z plane and (c) z - x plane ?



Sol. $\vec{B} = \hat{i} B_H - \hat{j} B_V = \text{constant}$, so $\phi = \vec{B} \cdot \vec{S}$ [$\vec{B} = \text{constant}$]

(a) For area in x-y plane $\vec{S} = S \hat{k}$, $\phi_{xy} = (\hat{i} B_H - \hat{j} B_V) \cdot (\hat{k} S) = 0$

(b) For area S in y-z plane $\vec{S} = S \hat{i}$, $\phi_{yz} = (\hat{i} B_H - \hat{j} B_V) \cdot (\hat{i} S) = B_H S$

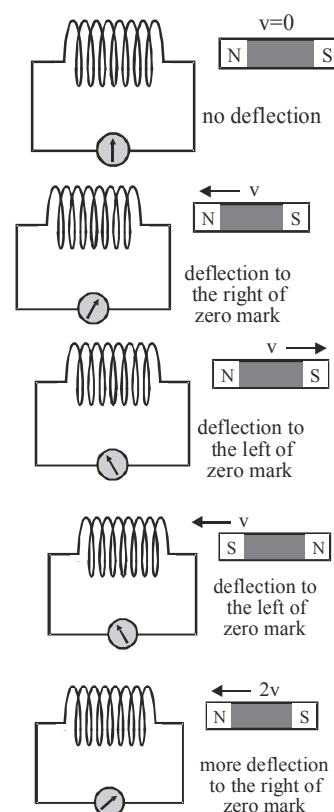
(c) For area S in z-x plane $\vec{S} = S \hat{j}$, $\phi_{zx} = (\hat{i} B_H - \hat{j} B_V) \cdot (\hat{j} S) = -B_V S$

Negative sign implies that flux is directed vertically downwards.

FARADAY'S EXPERIMENT

Faraday performed various experiments to discover and understand the phenomenon of electromagnetic induction. Some of them are :

- When the magnet is held stationary anywhere near or inside the coil, the galvanometer does not show any deflection.
- When the N-pole of a strong bar magnet is moved towards the coil, the galvanometer shows a deflection right to the zero mark.
- When the N-pole of a strong bar magnet is moved away from the coil, the galvanometer shows a deflection left to the zero mark.
- If the above experiments are repeated by bringing the S-pole of the magnet towards or away from the coil, the direction of current in the coil is opposite to that obtained in the case of N-pole.
- The deflection in galvanometer is more when the magnet moves faster and less when the magnet moves slow.



CONCLUSIONS

Whenever there is a relative motion between the source of magnetic field (magnet) and the coil, an emf is induced in the coil. When the magnet and coil move towards each other then the flux linked with the coil increases and emf is induced. When the magnet and coil move away from each other the magnetic flux linked with the coil decreases, again an emf is induced. This emf lasts so long the flux is changing.

Due to this emf an electric current start to flow and the galvanometer shows deflection.

The deflection in galvanometer last as long the relative motion between the magnet and coil continues.

Whenever relative motion between coil and magnet takes place an induced emf produced in coil. If coil is in closed circuit then current and charge is also induced in the circuit. This phenomenon is called electro magnetic induction.

FARADAY'S LAWS OF ELECTROMAGNETIC INDUCTION

Faraday's law of induction may be stated as follows:

The induced emf ε in a coil is proportional to the negative of the rate of change of magnetic flux:

$$\varepsilon = - \frac{d\Phi_B}{dt}$$

For a coil that consists of N loops, the total induced emf would be N times as large:

$$\varepsilon = -N \frac{d\Phi_B}{dt}$$

Thus, we see that an emf may be induced in the following ways:

- (i) by varying the magnitude of $\vec{B}$ with time (illustrated in Figure)

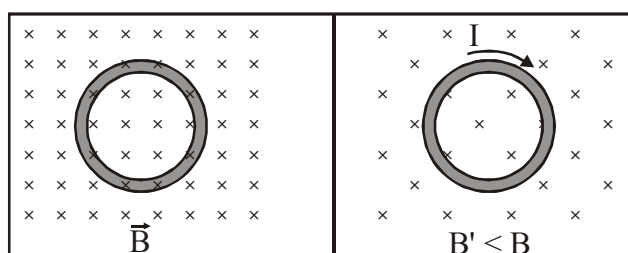


Figure : Inducing emf by varying the magnetic field strength

- (ii) by varying the magnitude of $\vec{A}$, i.e., the area enclosed by the loop with time (illustrated in Figure)

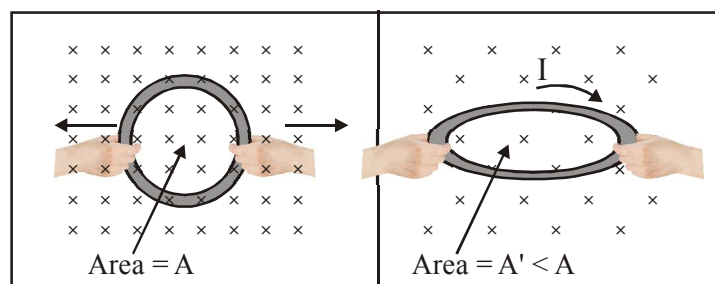


Figure : Inducing emf by changing the area of the loop

- (iii) varying the angle between $\vec{B}$ and the area vector $\vec{A}$ with time (illustrated in Figure)

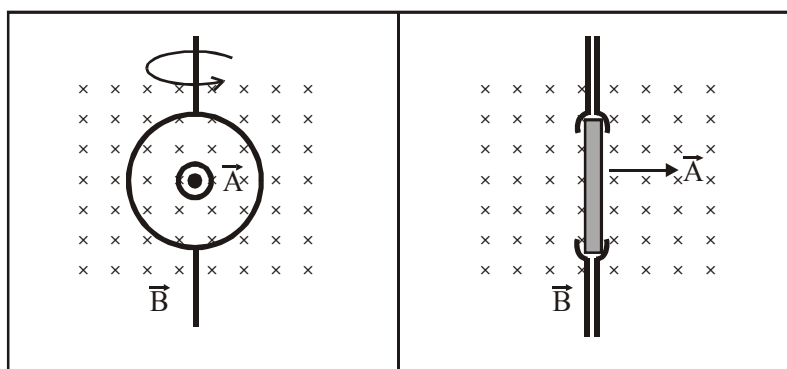


Figure : Inducing emf by varying the angle between $\vec{B}$ and $\vec{A}$

LENZ'S LAW

Lenz's Law :

The direction of the induced current is determined by Lenz's law:

The induced current produces magnetic fields which tend to oppose the change in magnetic flux that induces such currents.

To illustrate how Lenz's law works, let's consider a conducting loop placed in a magnetic field. We follow the procedure below:

1. Define a positive direction for the area vector $\vec{A}$.
2. Assuming that $\vec{B}$ is uniform, take the dot product of $\vec{B}$ and $\vec{A}$. This allows for the determination of the sign of the magnetic flux Φ_B .
3. Obtain the rate of flux change $d\Phi_B / dt$ by differentiation. There are three possibilities:

$$\frac{d\Phi_B}{dt} \begin{cases} > 0 \Rightarrow \text{induced emf } \varepsilon < 0 \\ < 0 \Rightarrow \text{induced emf } \varepsilon > 0 \\ = 0 \Rightarrow \text{induced emf } \varepsilon = 0 \end{cases}$$

4. Determine the direction of the induced current using the right-hand rule. With your thumb pointing in the direction of $\vec{A}$, curl the fingers around the closed loop. The induced current flows in the same direction as the way your fingers curl if $\varepsilon > 0$, and the opposite direction if $\varepsilon < 0$, as shown in Figure.

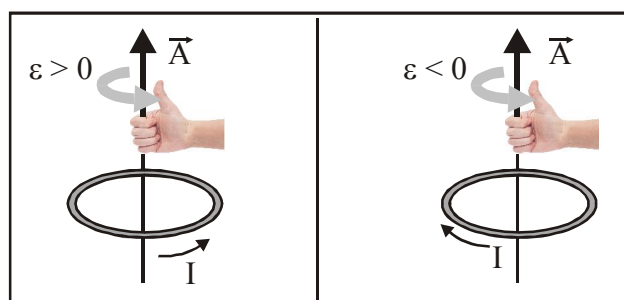


Figure : Determination of the direction of induced current by the right-hand rule

In Figure we illustrate the four possible scenarios of time-varying magnetic flux and show how Lenz's law is used to determine the direction of the induced current I .

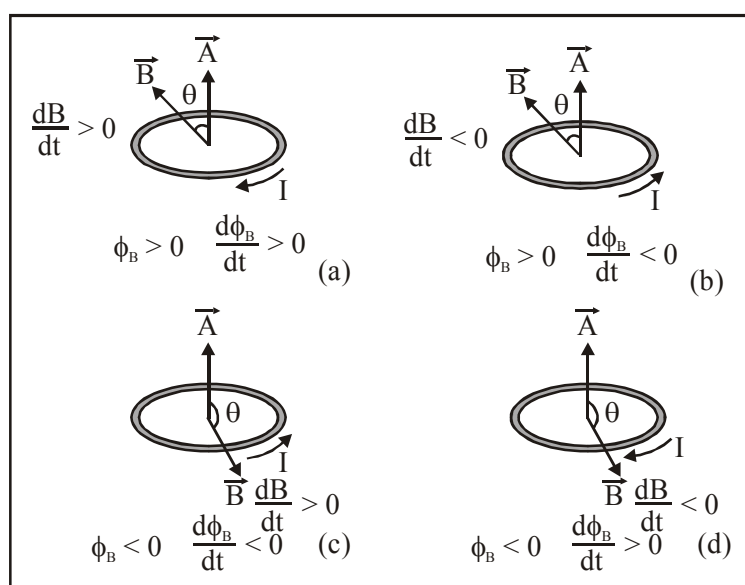


Figure : Direction of the induced current using Lenz's law

The above situations can be summarized with the following sign convention:

Φ_B	$d\Phi_B/dt$	ε	I
+	+	-	-
	-	+	+
-	+	-	-
	-	+	+

The positive and negative signs of I correspond to a counter clockwise and clockwise currents, respectively.

Ex. The radius of a coil decreases steadily at the rate of 10^{-2} m/s. A constant and uniform magnetic field of induction 10^{-3} Wb/m² acts perpendicular to the plane of the coil. What will be the radius of the coil when the induced e.m.f. in the $1\mu\text{V}$

Sol. Induced emf $e = \frac{d(BA)}{dt} = \frac{Bd(\pi r^2)}{dt} = 2B\pi r \frac{dr}{dt}$ radius of coil $r = \frac{e}{2B\pi \left(\frac{dr}{dt}\right)} = \frac{1 \times 10^{-6}}{2 \times 10^{-3} \times \pi \times 10^{-2}} = \frac{5}{\pi} \text{ cm}$

Ex. The ends of a search coil having 20 turns, area of cross-section 1 cm^2 and resistance 2 ohms are connected to a ballistic galvanometer of resistance 40 ohms. If the plane of search coil is inclined at 30° to the direction of a magnetic field of intensity 1.5 Wb/m^2 , coil is quickly pulled out of the field to a region of zero magnetic field, calculate the charge passed through the galvanometer.

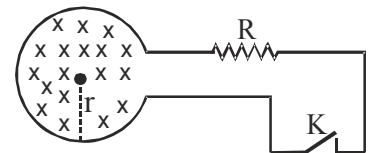
Sol. The total flux linked with the coil having turns N and area A is

$$\phi_1 = N(\vec{B} \cdot \vec{A}) = NBA \cos\theta = NBA \cos(90^\circ - 30^\circ) = \frac{NBA}{2}$$

when the coil is pulled out, the flux becomes zero, $\phi_2 = 0$ so change in flux is $\Delta\phi = \frac{NBA}{2}$

the charge flowed through the circuit is $q = \frac{\Delta\phi}{R} = \frac{NBA}{2R} = \frac{20 \times 1.5 \times 10^{-4}}{2 \times 42} = 0.357 \times 10^{-4} \text{ C}$

Ex. Shown in the figure is a circular loop of radius r and resistance R . A variable magnetic field of induction $B = B_0 e^{-t}$ is established inside the coil. If the key (K) is closed. Then calculate the electrical power developed right after closing the key.



Sol. Induced emf $e = \frac{d\phi}{dt} = \frac{d}{dt}(BA) = A \frac{dB}{dt} = \pi r^2 B_0 \frac{d}{dt}(e^{-t}) = -\pi r^2 B_0 e^{-t}$

At $t = 0$, $e_0 = B_0 e^{-0} \cdot \pi r^2 = B_0 \pi r^2$

The electric power developed in the resistor R just at the instant of closing the key is

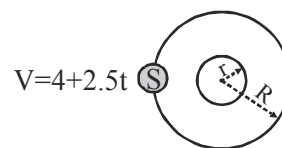
$$P = \frac{e_0^2}{R} = \frac{B_0^2 \pi^2 r^4}{R}$$

Ex. Two concentric coplanar circular loops made of wire, resistance per unit length $10^{-4} \Omega\text{m}^{-1}$, have diameters 0.2 m and 2 m . A time-varying potential difference $(4 + 2.5 t)$ volt is applied to the larger loop. Calculate the current in the smaller loop.

Sol. The magnetic field at the centre O due to the current in the larger loop is $B = \frac{\mu_0 I}{2R}$

If ρ is the resistance per unit length, then

$$I = \frac{\text{potential difference}}{\text{resistance}} = \frac{4 + 2.5t}{2\pi R \cdot \rho}$$



$$\therefore B = \frac{\mu_0}{2R} \cdot \frac{4 + 2.5t}{2\pi R \rho}$$

$\therefore r \ll R$, so the field B can be taken almost constant over the entire area of the smaller loop.

$$\therefore \text{the flux linked with the smaller loop is } \phi = B \times \pi r^2 = \frac{\mu_0}{2R} \cdot \frac{4 + 2.5t}{2\pi R \rho} \cdot \pi r^2$$

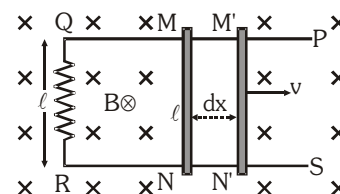
$$\text{Induced emf } e = \frac{d\phi}{dt} = \frac{\mu_0 r^2}{4R^2 \rho} \times 2.5$$

The corresponding current in the smaller loop is I' then

$$I' = \frac{e}{R} = \frac{\mu_0 r^2}{4R^2 \rho} \times 2.5 \times \frac{1}{2\pi r \rho} = \frac{2.5 \mu_0 r}{8\pi R^2 \rho^2} = \frac{2.5 \times 4\pi \times 10^{-7} \times 0.1}{8\pi \times (1)^2 \times (10^{-4})^2} = 1.25 \text{ A}$$

Induced emf by changing the area of the coil

A U shaped frame of wire, PQRS is placed in a uniform magnetic field B perpendicular to the plane and vertically inward. A wire MN of length ℓ is placed on this frame. The wire MN moves with a speed v in the direction shown. After time dt the wire reaches to the position $M'N'$ and distance covered = dx .



The change in area $\Delta A = \text{Length} \times \text{area} = \ell dx$

Change in the magnetic flux linked with the loop in the dt is $d\phi = B \times \Delta A = B \times \ell dx$

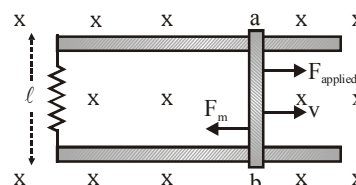
$$\text{induced emf } e = \frac{d\phi}{dt} = B \ell \frac{dx}{dt} = B \ell v \therefore \left[v = \frac{dx}{dt} \right]$$

If the resistance of circuit is R and the circuit is closed then the current through the circuit

$$I = \frac{e}{R} \Rightarrow I = \frac{Bv\ell}{R}$$

A magnetic force acts on the conductor in opposite direction of velocity

$$F_m = i \ell B = \frac{B^2 \ell^2 v}{R}$$



So, to move the conductor with a constant velocity v an

equal and opposite force F has to be applied in the conductor.

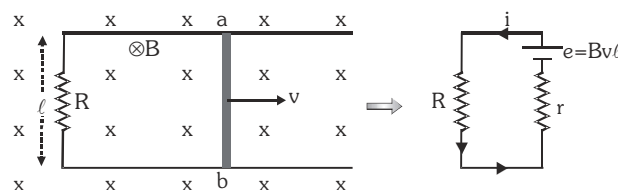
$$F = F_m = \frac{B^2 \ell^2 v}{R}$$

The rate at which work is done by the applied force is, $P_{\text{applied}} = Fv = \frac{B^2 \ell^2 v^2}{R}$

and the rate at which energy is dissipated in the circuit is, $P_{\text{dissipated}} = i^2 R = \left[\frac{Bv\ell}{R} \right]^2 R = \frac{B^2 \ell^2 v^2}{R}$

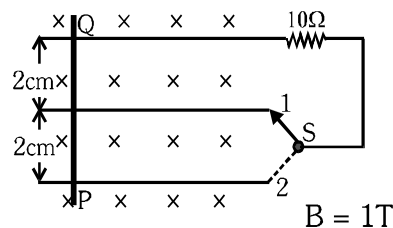
This is just equal to the rate at which work is done by the applied force.

- In the figure shown, we can replace the moving rod ab by a battery of emf $Bv\ell$ with the positive terminal at a and the negative terminal at b . The resistance r of the rod ab may be treated as the internal resistance of the battery.



Hence, the current in the circuit is $i = \frac{e}{R + r} = \frac{Bv\ell}{R + r}$

- Ex.** Wire PQ with negligible resistance slides on the three rails with 5 cm/sec . Calculate current in 10Ω resistance when switch S is connected to (a) position 1 (b) position 2



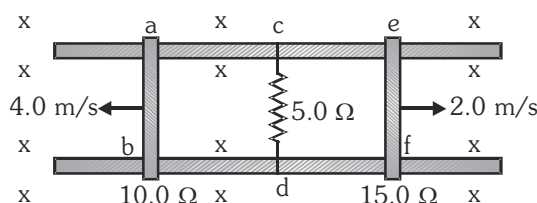
- Sol.** (a) For position 1

$$\text{Induced current } I = \frac{e}{R} = \frac{Bv\ell}{R} = \frac{1 \times 5 \times 10^{-2} \times 2 \times 10^{-2}}{10} = 0.1 \text{ mA}$$

- (b) For position 2

$$\text{Induced current } I = \frac{e}{R} = \frac{Bv(2\ell)}{R} = \frac{1 \times 5 \times 10^{-2} \times 4 \times 10^{-2}}{10} = 0.2 \text{ mA}$$

- Ex.** Two parallel rails with negligible resistance are 10.0 cm apart. They are connected by a 5.0Ω resistor. The circuit also contains two metal rods having resistance of 10.0Ω and 15.0Ω along the rails (fig). The rods are pulled away from the resistor at constant speeds 4.00 m/s and 2.00 m/s respectively. A uniform magnetic field of magnitude 0.01 T is applied perpendicular to the plane of the rails. Determine the current in the 5.0Ω resistor.



Sol. Two conductors are moving in uniform magnetic field, so motional emf will induced across them.

The rod ab will act as a source of emf $e_1 = Bv\ell = (0.01)(4.0)(0.1) = 4 \times 10^{-3} \text{ V}$

and internal resistance $r_1 = 10.0 \Omega$

Similarly, rod ef will also act as source of emf $e_2 = (0.01)(2.0)(0.1) = 2 \times 10^{-3} \text{ V}$

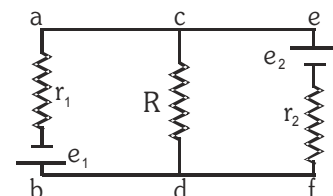
and internal resistance $r_2 = 15.0 \Omega$

From right hand rule : $V_b > V_a$ and $V_e > V_f$ Also $R = 5.0 \Omega$,

$$E_{eq} = \frac{e_1 r_2 - e_2 r_1}{r_1 + r_2} = \frac{6 \times 10^{-3} - 20 \times 10^{-3}}{15 + 10} = \frac{40}{25} \times 10^{-3} = 1.6 \times 10^{-3} \text{ volt}$$

$$r_{eq} = \frac{15 \times 10}{15 + 10} = 6 \Omega \quad \text{and}$$

$$I = \frac{E_{eq}}{r_{eq} + R} = \frac{1.6 \times 10^{-3}}{6 + 6} = \frac{1.6}{11} \times 10^{-3} = \frac{8}{55} \times 10^{-3} \text{ amp from d to c}$$



MOTIONAL EMF FROM LORENTZ FORCE

A conductor PQ is placed in a uniform magnetic field B , directed normal to the plane of paper outwards. PQ is moved with a velocity v , the free electrons of PQ also move with the same velocity. The

electrons experience a magnetic Lorentz force, $\vec{F}_m = (v \times \vec{B})$.

According to Fleming's left hand rule, this force acts in the direction PQ and hence the free electrons will move towards Q . A negative

charge accumulates at Q and a positive

charge at P . An electric field E is setup in the conductor from P to Q . Force exerted by electric field on the

free electrons is, $\vec{F}_e = e\vec{E}$

The accumulation of charge at the two ends continues till these two forces balance each other.

so $\vec{F}_m = -\vec{F}_e \Rightarrow e(\vec{v} \times \vec{B}) = -e\vec{E} \Rightarrow \vec{E} = -(\vec{v} \times \vec{B})$

The potential difference between the ends P and Q is $V = \vec{E} \cdot \vec{\ell} = (\vec{v} \times \vec{B}) \cdot \vec{\ell}$. It is the magnetic force on the moving free electrons that maintains the potential difference and produces the emf $\mathcal{E} = B\ell v$

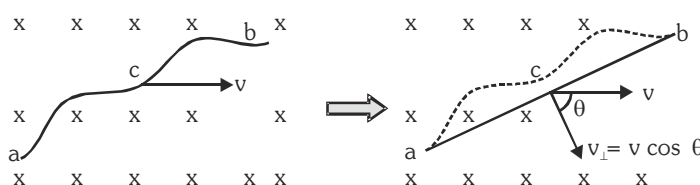
(for $\vec{B} \perp \vec{v} \perp \vec{\ell}$)

As this emf is produced due to the motion of a conductor, so it is called a motional emf.

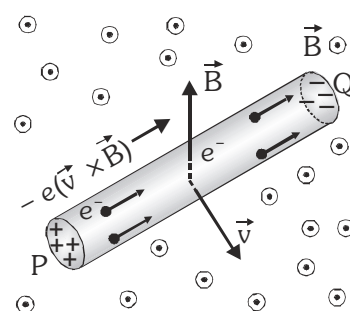
The concept of motional emf for a conductor can be generalized for any shape moving in any magnetic field uniform or not. For an element $d\vec{\ell}$ of conductor the contribution de to the emf is the

magnitude $d\ell$ multiplied by the component of $\vec{v} \times \vec{B}$ parallel to $d\vec{\ell}$, that is $de = (\vec{v} \times \vec{B}) \cdot d\vec{\ell}$

For any two points a and b the motional emf in the direction from b to a is,



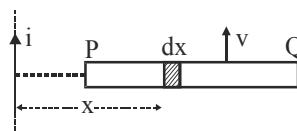
$$e = \int_b^a (\vec{v} \times \vec{B}) \cdot d\vec{\ell}$$



Motional emf in wire acb in a uniform magnetic field is the motional emf in an imaginary wire ab. Thus, $e_{acb} = e_{ab} = (\text{length of ab}) (v_{\perp}) (B)$, $v_{\perp}$ = the component of velocity perpendicular to both $\vec{B}$ and ab. From right hand rule : b is at higher potential and a at lower potential. Hence, $V_{ba} = V_b - V_a = (ab) (v \cos \theta) (B)$

Ex. A rod PQ of length L moves with a uniform velocity v parallel to a long straight wire carrying a current i, the end P remaining at a distance r from the wire. Calculate the emf induced across the rod. Take $v = 5.0$ m/s, $i = 100$ amp, $r = 1.0$ cm and $L = 19$ cm.

Sol. The rod PQ is moving in the magnetic field produced by the current-carrying long wire. The field is not uniform throughout the length of the rod (changing with distance). Let us consider a small element of length dx at distance x from wire. if magnetic field at the position of dx is B then emf induced



$$d\mathcal{E} = B v dx = \frac{\mu_0 i}{2\pi x} v dx$$

$\therefore$ emf $\mathcal{E}$ is induced in the entire length of the rod PQ is $\mathcal{E} = \int_P^Q d\mathcal{E} = \int_P^Q \frac{\mu_0 i}{2\pi x} v dx$

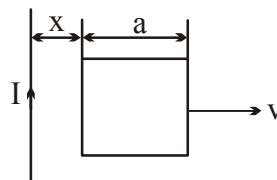
Now $x = r$ at P, and $x = r + L$ at Q. hence

$$\mathcal{E} = \frac{\mu_0 i v}{2\pi} \int_r^{r+L} \frac{dx}{x} = \frac{\mu_0 i v}{2\pi} [\log_e x]_r^{r+L} = \frac{\mu_0 i v}{2\pi} [\log_e (r + L) - \log_e r] = \frac{\mu_0 i v}{2\pi} \log \frac{r + L}{r}$$

Putting the given values :

$$\mathcal{E} = (2 \times 10^{-7}) (100) (5.0) \log_e \frac{1.0 + 19}{1.0} = 10^{-4} \log_e 20 \text{ Wb/s} = 3 \times 10^{-4} \text{ volt}$$

Ex. A square frame with side a and a long straight wire carrying a current I are located in the same plane as shown in Fig. The frame translates to the right with a constant velocity v. Find the emf induced in the frame as a function of distance x.



Ans. $\xi_i = \frac{\mu_0}{4\pi} \frac{2Ia^2v}{x(x+a)}$

Sol. Field, due to the current carrying wire, at a perpendicular distance x from it is given by,

$$B(x) = \frac{\mu_0 i}{2\pi x}$$

Motional emf is given by $\left| \int -(\vec{v} \times \vec{B}) \cdot d\vec{\ell} \right|$

There will be no induced emf in the segments (2) and (4) as, $\vec{v} \uparrow \uparrow d\vec{\ell}$ and magnitude of emf induced 1 and 3, will be

$$\xi_1 = v \left(\frac{\mu_0 i}{2\pi x} \right) a \text{ and } \xi_2 = v \left(\frac{\mu_0 i}{2\pi (a+x)} \right) a,$$

respectively, and their sense will be in the direction of $(\vec{v} \times \vec{B})$. So, emf induced in the network

$$\begin{aligned} &= \xi_1 - \xi_2 \text{ [as } \xi_1 > \xi_2] \\ &= \frac{av\mu_0 i}{2\pi} \left[\frac{1}{x} - \frac{1}{a+x} \right] = \frac{va^2\mu_0 i}{2\pi x(a+x)} \end{aligned}$$

Ex. A horizontal magnetic field B is produced across a narrow gap between square iron pole-pieces as shown. A closed square wire loop of side ℓ , mass m and resistance R is allowed to fall with the top of the loop in the

field. Show that the loop attains a terminal velocity given by $v = \frac{Rmg}{B^2 \ell^2}$

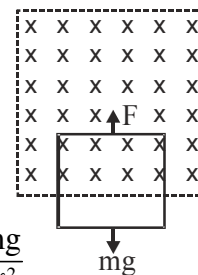
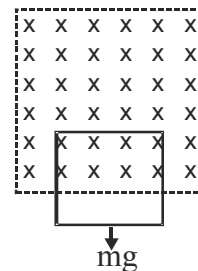
while it is between the poles of the magnet.

Sol. As the loop falls under gravity, the flux passing through it decreases and so an induced emf is set up in it. Then a force F which opposes its fall. When this force becomes equal to the gravity force mg , the loop attains a terminal velocity v .

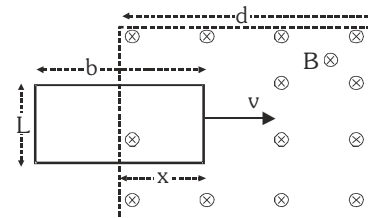
The induced emf $e = B v \ell$, and the induced current is $i = \frac{e}{R} = \frac{B v \ell}{R}$

The force experienced by the loop due to this current is $F = B \ell i = \frac{B^2 v \ell^2}{R}$

When v is the terminal (constant) velocity $F = mg$ or $\frac{B^2 v \ell^2}{R} = mg$ or $v = \frac{Rmg}{B^2 \ell^2}$



Ex. Figure shows a rectangular conducting loop of resistance R , width L , and length b being pulled at constant speed v through a region of width d in which a uniform magnetic field B is set up by an electromagnet. Let $L = 40 \text{ mm}$, $b = 10 \text{ cm}$, $d = 15 \text{ cm}$, $R = 1.6 \Omega$, $B = 2.0 \text{ T}$ and $v = 1.0 \text{ m/s}$



- Plot the flux ϕ through the loop as a function of the position x of the right side of the loop.
- Plot the induced emf as a function of the position of the loop.
- Plot the rate of production of thermal energy in the loop as a function of the position of the loop.

Sol. (i) When the loop is not in the field :

The flux linked with the loop $\phi = 0$

When the loop is entirely in the field :

Magnetic flux linked with the loop

$\phi = B L b$

$= 2 \times 40 \times 10^{-3} \times 10^{-1} = 8 \text{ mWb}$

When the loop is entering the field :

The flux linked with the loop $\phi = B L x$

When the loop is leaving the field :

The flux $\phi = B L [b - (x - d)]$

(ii) Induced emf is $e = -\frac{d\phi}{dt} = -\frac{d\phi}{dx} \frac{dx}{dt} = -\frac{d\phi}{dx} v$

$= -\text{slope of the curve of figure (i)} \times v$

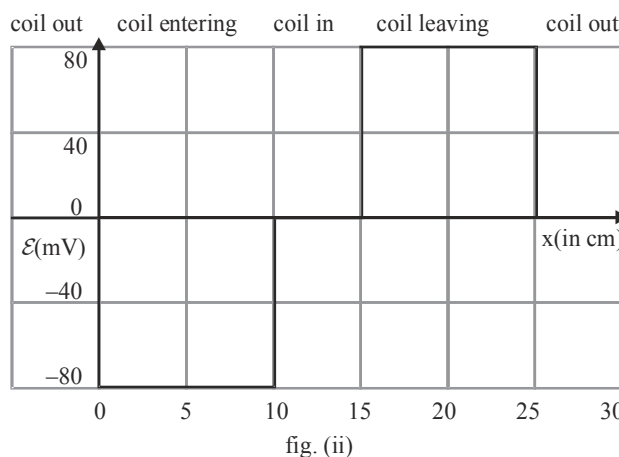
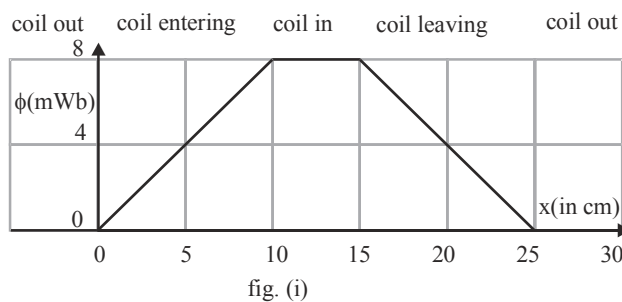
The emf for 0 to 10 cm :

$$e = -\frac{(8-0) \times 10^{-3}}{(10-0) \times 10^{-2}} \times 1 = -80 \text{ mV}$$

The emf for 10 to 15 cm : $e = 0 \times 1 = 0$

The emf for 15 to 25 cm :

$$e = -\frac{(0-8) \times 10^{-3}}{(25-15) \times 10^{-2}} \times 1 = +80 \text{ mV}$$

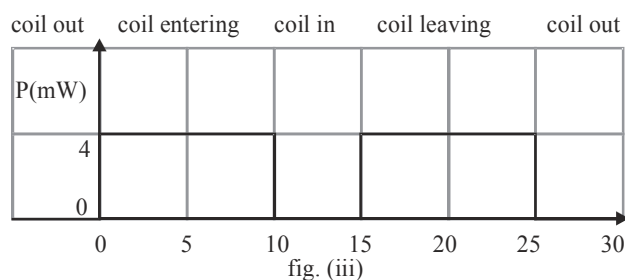


(iii) The rate of thermal energy production is $P = \frac{e^2}{R}$

$$\text{for 0 to 10 cm : } P = \frac{(80 \times 10^{-3})^2}{1.6} = 4 \text{ mW}$$

$$\text{for 10 to 15 cm : } P = 0$$

$$\text{for 15 to 25 cm : } P = \frac{(80 \times 10^{-3})^2}{1.6} = 4 \text{ mW}$$

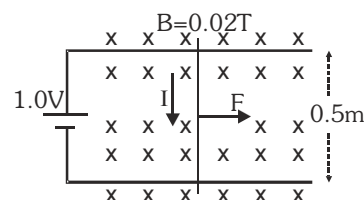


Ex. Two long parallel wires of zero resistance are connected to each other by a battery of 1.0 V. The separation between the wires is 0.5 m. A metallic bar, which is perpendicular to the wires and of resistance 10Ω , moves on these wires. When a magnetic field of 0.02 T is acting perpendicular to the plane containing the bar and the wires. Find the steady-state velocity of the bar. If the mass of the bar is 0.002 kg then find its velocity as a function of time.

Sol. The current in the 10Ω bar is $I = \frac{1.0 \text{ V}}{10 \Omega} = 0.1 \text{ A}$

The current carrying bar is placed in the magnetic field $\vec{B}$ (0.2 T) perpendicular to the plane of paper and directed downwards.

The magnetic force of the bar is $F = B I \ell = 0.02 \times 0.5 \times 0.1 = 1 \times 10^{-3} \text{ N}$



The moving bar cuts the lines of force of $\vec{B}$. If v be the instantaneous velocity of the bar, then the emf induced in the bar is $\mathcal{E} = B \ell v = 0.02 \times 0.5 \times v = 0.01 v$ volt. By Lenz's law, $\mathcal{E}$ will oppose the motion of the bar which will ultimately attain a steady velocity. In this state, the induced emf $\mathcal{E}$ will

be equal to the applied emf (1.0 volt). $\therefore 0.01 v = 1.0$ or $v = \frac{1.0}{0.01} = 100 \text{ ms}^{-1}$

Again, a magnetic force F acts on the bar. If m be the mass of the bar, the acceleration of the rod is

$$\frac{dv}{dt} = \frac{F}{m} \Rightarrow dv = \frac{F}{m} \cdot dt \quad \text{Integrating, } \int dv = \int \frac{F}{m} dt \Rightarrow v = \frac{F}{m} t + C \text{ (constant)}$$

If at $t = 0$, $v = 0$ then $C = 0$. $\therefore v = \frac{F}{m} t$ But $F = 1 \times 10^{-3} \text{ N}$, $m = 0.002 \text{ kg}$

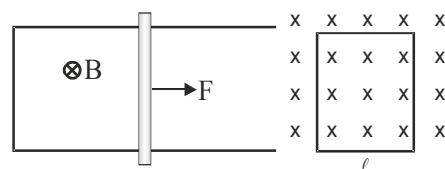
$$\therefore v = \frac{1 \times 10^{-3}}{0.002} t = 0.5 t$$

Ex. In figure, a rod closing the circuit moves along a U-shaped wire at a constant speed v under the action of the force F . The circuit is in a uniform magnetic field perpendicular to its plane. Calculate F if the rate generation of heat is P .

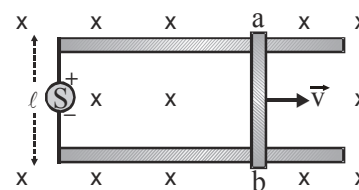
Sol. The emf induced across the ends of the rod, $\mathcal{E} = B \ell v$

Current in the circuit, $I = \frac{\mathcal{E}}{R} = \frac{B \ell v}{R}$ Magnetic force on the conductor, $F' = I \ell B$, towards left

$$\therefore \text{acceleration is zero } F' = F \Rightarrow B I \ell = F \text{ or } I = \frac{F}{B \ell} \therefore P = \mathcal{E} I = B \ell v \times \frac{F}{B \ell} = F v \therefore F = \frac{P}{v}$$



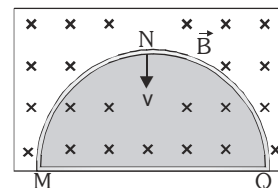
Ex. The diagram shows a wire ab of length ℓ and resistance R sliding on a smooth pair of rails with a velocity v towards right. A uniform magnetic field of induction B acts normal to the plane containing the rails and the wire inwards. S is a current source providing a constant I in the circuit. Determine the potential difference between a and b .



Sol. The wire ab which is moving with a velocity v is equivalent to an emf source of value $Bv\ell$ with its positive terminal towards a .

$$\therefore \text{Potential difference } V_a - V_b = Bv\ell - IR$$

Ex. A thin semicircular conducting ring of radius R is falling with its plane vertical in a horizontal magnetic induction $\vec{B}$ (fig.). At the position MNQ , the speed of the ring is v . What is the potential difference developed across the ring at the position MNQ ?



Sol. Let semiconductor ring falls through an infinitesimally small distance dx from its initial position MNQ to $M'Q'N'$ in time dt (fig). decrease in area of the ring inside the magnetic field,

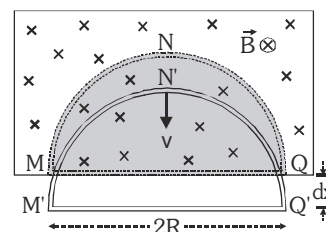
$$dA = -MQQ'M' = -M'Q' \times QQ' = -2R dx$$

$\therefore$ change in magnetic flux linked with the ring,

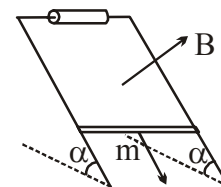
$$d\phi = B \times dA = B \times (-2R dx) = -2BR dx$$

$$\text{The potential difference developed across the ring, } e = -\frac{d\phi}{dt} = -\left[-2BR \frac{dx}{dt}\right] = 2BRv$$

the speed with which the ring is falling $v = \frac{dx}{dt}$



Ex. A copper connector of mass m slides down two smooth copper bars, set at an angle α to the horizontal, due to gravity (Fig.). At the top the bars are interconnected through a resistance R . The separation between the bars is equal to l . The system is located in a uniform magnetic field of induction B , perpendicular to the plane in which the connector slides. The resistances of the bars, the connector and the sliding contacts, as well as the self-inductance of the loop, are assumed to be negligible. Find the steady-state velocity of the connector.



Ans. $v = \frac{mgR \sin \alpha}{B^2 l^2}$

Sol. From Lenz's law, the current through the connector is directed from A to B . Here $\xi_{in} = vBl$ between A and B .

where v is the velocity of the rod at any moment.

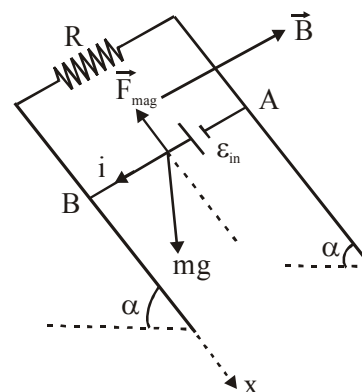
$$\text{For the rod, from } F_x = mw_x$$

$$\text{or, } mg \sin \alpha - i\ell B = mw$$

For steady state, acceleration of the rod must be equal to zero.

$$\text{Hence, } mg \sin \alpha = i\ell B \quad \dots\dots\dots (1)$$

$$\text{But, } i = \frac{\xi_{in}}{R} = \frac{vBl}{R}$$



from (1) and (2) $v = \frac{mg \sin \alpha R}{B^2 \ell^2}$

INDUCED E.M.F. DUE TO ROTATION OF A CONDUCTOR ROD IN A UNIFORM MAGNETIC FIELD

Let a conducting rod is rotating in a magnetic field around an axis passing through its one end, normal to its plane.

Consider an small element dx at a distance x from axis of rotation.

Suppose velocity of this small element = v

So, according to Lorentz's formula induced e.m.f. across this small element

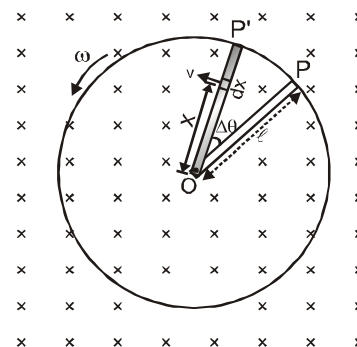
$$d\varepsilon = B v \cdot dx$$

$\because$ This small element dx is at distance x from O (axis of rotation)

$\therefore$ Linear velocity of this element dx is $v = \omega x$

substitute of value of v in eqⁿ (i) $d\varepsilon = B \omega x dx$

Every element of conducting rod is normal to magnetic field and moving in perpendicular direction to the field



So, net induced e.m.f. across conducting rod $\varepsilon = \int d\varepsilon = \int_0^l B \omega x dx = \omega B \left(\frac{x^2}{2} \right)_0^l$

or $\varepsilon = \frac{1}{2} B \omega \ell^2$ $\varepsilon = \frac{1}{2} B \times 2\pi f \times \ell^2$ [f = frequency of rotation]

$= B f (\pi \ell^2)$ area traversed by the rod $A = \pi \ell^2$ or $\varepsilon = BAf$

Ex. A wheel with 10 metallic spokes each 0.5 m long is rotated with angular speed of 120 revolutions per minute in a plane normal to the earth's magnetic field. If the earth's magnetic field at the given place is 0.4 gauss, find the e.m.f. induced between the axle and the rim of the wheel.

Sol. $\omega = 2\pi n = 2\pi \times \frac{120}{60} = 4\pi$, $B = 0.4 \text{ G} = 4 \times 10^{-5} \text{ T}$, length of each spoke = 0.5 m

induced emf $e = \frac{1}{2} B \omega \ell^2 = \frac{1}{2} \times 4 \times 10^{-5} \times 4\pi \times (0.5)^2 = 6.28 \times 10^{-5} \text{ volt}$

As all the ten spokes are connected with their one end at the axle and the other end at the rim, so they are connected in parallel and hence emf across each spoke is same. The number of spokes is immaterial.

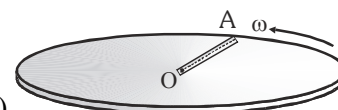
Ex. A horizontal copper disc of diameter 20 cm, makes 10 revolutions/sec about a vertical axis passing through its centre. A uniform magnetic field of 100 gauss acts perpendicular to the plane of the disc. Calculate the potential difference its centre and rim in volts.

Sol. $B = 100 \text{ gauss} = 100 \times 10^{-4} \text{ Wb/m}^2 = 10^{-2}$,
 $r = 10 \text{ cm} = 0.10 \text{ m}$, frequency of rotation = 10 rot/sec

The emf induced between centre and rim $\varepsilon = \frac{1}{2} B \omega \ell^2 = \frac{1}{2} B \omega r^2 (\because r = \ell)$

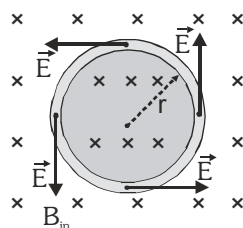
$\omega = 2\pi f = 2 \times 3.14 \times 10 = 62.8 \text{ s}^{-1}$

$\therefore \varepsilon = \frac{1}{2} \times 10 \times 62.8 \times (0.1)^2 = 3.14 \times 10^{-3} \text{ V} = 3.14 \text{ mV}$.



INDUCED ELECTRIC FIELD

When the magnetic field changes with time (let it increases with time) there is an induced electric field in the conductor caused by the changing magnetic flux.



Important properties of induced electric field :

- (i) It is non conservative in nature. The line integral of $\vec{E}$ around a closed path is not zero. When a charge q goes once around the loop, the total work done on it by the electric field is equal to q times the emf.

Hence
$$\oint \vec{E} \cdot d\vec{\ell} = e = -\frac{d\phi}{dt} \quad \dots(i)$$

This equation is valid only if the path around which we integrate is stationary.

- (ii) Due to symmetry, the electric field $\vec{E}$ has the same magnitude at every point on the circle and it is tangential at each point (figure).
 (iii) Being nonconservative field, so the concept of potential has no meaning for such a field.
 (iv) This field is different from the conservative electrostatic field produced by stationary charges.
 (v) The relation $\vec{F} = q\vec{E}$ is still valid for this field. (vi) This field can vary with time.

• For symmetrical situations $E\ell = \left| \frac{d\phi}{dt} \right| = A \left| \frac{dB}{dt} \right|$

ℓ = the length of closed loop in which electric field is to be calculated

A = the area in which magnetic field is changing.

Direction of induced electric field is the same as the direction of induced current.

Ex. The magnetic field at all points within the cylindrical region whose cross-section is indicated in the figure start increasing at a constant rate

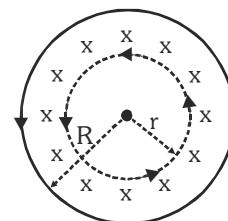
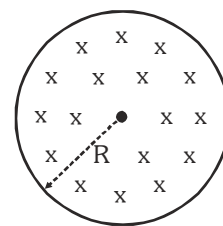
$\alpha \frac{\text{tesla}}{\text{second}}$. Find the magnitude of electric field as a function of r , the

distance from the geometric centre of the region.

Sol. For $r \leq R$:

$$\therefore E\ell = A \left| \frac{dB}{dt} \right|$$

$$\therefore E(2\pi r) = (\pi r^2) \alpha \Rightarrow E = \frac{r\alpha}{2} \Rightarrow E \propto r$$



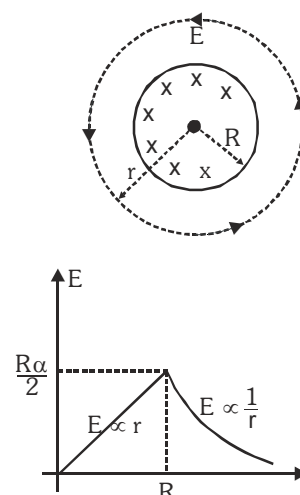
E-r graph is straight line passing through origin.

$$\text{At } r = R, \quad E = \frac{R\alpha}{2}$$

For $r \geq R$:

$$\therefore E \ell = A \left| \frac{dB}{dt} \right|$$

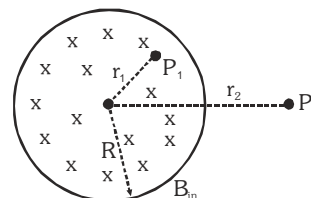
$$\therefore E (2\pi r) = (\pi R^2) \alpha \Rightarrow E = \frac{\alpha R^2}{2r} \Rightarrow E \propto \frac{1}{r}$$



Ex. For the situation described in figure the magnetic field changes with time according to,

$$B = (2.00 t^3 - 4.00 t^2 + 0.8) \text{ T and } r_2 = 2R = 5.0 \text{ cm}$$

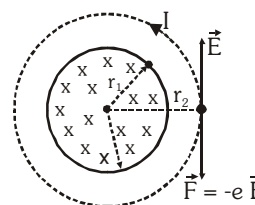
- Calculate the force on an electron located at P_2 at $t = 2.00 \text{ s}$
- What are the magnitude and direction of the electric field at P_1 when $t = 3.00 \text{ s}$ and $r_1 = 0.02 \text{ m}$.



Sol. $E \ell = A \left| \frac{dB}{dt} \right| \Rightarrow E = \frac{\pi R^2}{2\pi r_2} \frac{d}{dt} (2t^3 - 4t^2 + 0.8) = \frac{R^2}{2r_2} (6t^2 - 8t)$

- Force on electron at P_2 is $F = eE$

$$\begin{aligned} \therefore \text{at } t = 2 \text{ s } F &= \frac{1.6 \times 10^{-19} \times (2.5 \times 10^{-2})^2}{2 \times 5 \times 10^{-2}} \times [6(2)^2 - 8(2)] \\ &= \frac{1.6}{4} \times 2.5 \times 10^{-21} \times (24 - 16) = 8 \times 10^{-21} \text{ N at } t = 2 \text{ s,} \end{aligned}$$



$\frac{dB}{dt}$ is positive so it is increasing.

- direction of induced current and E are as shown in figure and hence force of electron having charge $-e$ is right perpendicular to r_2 downwards

$$(b) \text{ For } r_1 = 0.02 \text{ m and at } t = 3 \text{ s, } E = \frac{\pi r_1^2}{2\pi r_1} (6t^2 - 8t) = \frac{0.02}{2} \times [6(3)^2 - 8(3)]$$

$$= 0.3 \text{ V/m at } t = 3 \text{ sec, } \frac{dB}{dt}$$

is positive so B is increasing and hence direction of E is same as in case (a) and it is left perpendicular to r_1 upwards.

Generators :

One of the most important applications of Faraday's law of induction is to generators and motors. A generator converts mechanical energy into electric energy, while a motor converts electrical energy into mechanical energy.

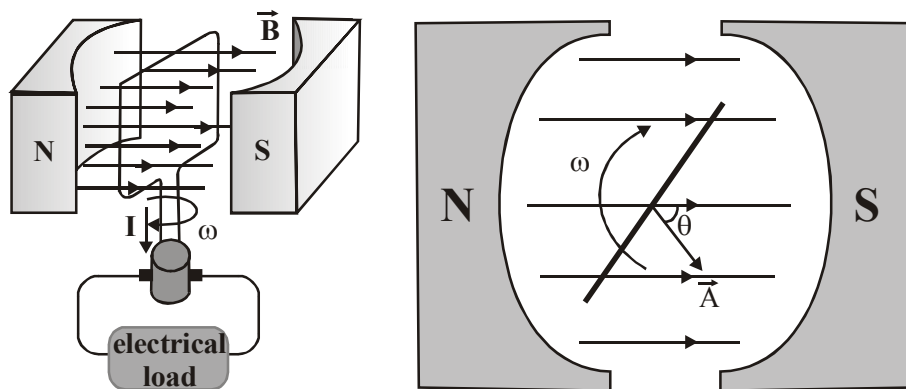


Figure : (a) A simple generator. (b) The rotating loop as seen from above.

Figure (a) is a simple illustration of a generator. It consists of an N -turn loop rotating in a magnetic field which is assumed to be uniform. The magnetic flux varies with time, thereby inducing an emf. From Figure (b), we see that the magnetic flux through the loop may be written as

$$\Phi_B = \vec{B} \cdot \vec{A} = BA \cos \theta = BA \cos \omega t$$

The rate of change of magnetic flux is

$$\frac{d\Phi_B}{dt} = -BA \omega \sin \omega t$$

Since there are N turns in the loop, the total induced emf across the two ends of the loop is

$$\varepsilon = -N \frac{d\Phi_B}{dt} = NBA \omega \sin \omega t$$

If we connect the generator to a circuit which has a resistance R , then the current generated in the

circuit is given by $I = \frac{|\varepsilon|}{R} = \frac{NBA\omega}{R} \sin \omega t$

The current is an alternating current which oscillates in sign and has an amplitude $I_0 = NBA\omega / R$. The power delivered to this circuit is

$$P = I |\varepsilon| = \frac{(NBA\omega)^2}{R} \sin^2 \omega t$$

On the other hand, the torque exerted on the loop is

$$\tau = \mu B \sin \theta = \mu B \sin \omega t$$

Thus, the mechanical power supplied to rotate the loop is

$$P_m = \tau \omega = \mu B \omega \sin \omega t$$

Since the dipole moment for the N -turn current loop is

$$\mu = NIA = \frac{N^2 A^2 B \omega}{R} \sin \omega t$$

the above expression becomes

$$P_m = \left(\frac{N^2 A^2 B \omega}{R} \sin \omega t \right) B \omega \sin \omega t = \frac{(NAB\omega)^2}{R} \sin^2 \omega t$$

As expected, the mechanical power put in is equal to the electric power output.

SELF INDUCTION

When the current through the coil changes, the magnetic flux linked with the coil also changes. Due to this change of flux a current induced in the coil itself according to lenz concept it opposes the change in magnetic flux. This phenomenon is called self induction and a factor by virtue of coil shows opposition for change in magnetic flux called coefficient of self inductance of coil.

Considering this coil circuit in two cases.

Case - I Current through the coil is constant

If $I \rightarrow B \rightarrow \phi$ (constant) $\Rightarrow$ No EMI

total flux of coil ($N\phi$) $\propto$ current through the coil

$$N\phi \propto I \Rightarrow N\phi = LI \quad L = \frac{N\phi}{I} = \frac{NBA}{I} = \frac{\phi_{\text{total}}}{I}$$

where L = coefficient of self inductance of coil

S I unit of L : $1 \frac{\text{Wb}}{\text{amp}} = 1 \text{ Henry} = 1 \frac{\text{N.m}}{\text{A}^2} = 1 \frac{\text{J}}{\text{A}^2}$ **Dimensions :** $[M^1 L^2 T^{-2} A^{-2}]$

Note : L is constant of coil it **does not depends on current** flow through the coil.

Case - II Current through the coil changes w.r.t. time

$$\text{If } \frac{dI}{dt} \rightarrow \frac{dB}{dt} \rightarrow \frac{d\phi}{dt} \Rightarrow \text{Static EMI} \Rightarrow N\phi = LI$$

$$-N \frac{d\phi}{dt} = -L \frac{dI}{dt}, (-N \frac{d\phi}{dt}) \text{ called total self induced emf of coil 'e}_s\text{'}$$

$$e_s = -L \frac{dI}{dt} \quad \text{S.I. unit of } L \rightarrow \frac{\text{V} \cdot \text{s}}{\text{A}}$$

SELF-INDUCTANCE OF A PLANE COIL

Total magnetic flux linked with N turns,

$$\phi = NBA = N \left(\frac{\mu_0 NI}{2r} \right) A = \frac{\mu_0 N^2 I}{2r} A = \frac{\mu_0 N^2 I}{2r} \times \pi r^2 = \frac{\mu_0 \pi N^2 r}{2} I \quad \text{But } \phi = LI \therefore L = \frac{\mu_0 \pi N^2 r}{2}$$

Ex. Self-Inductance of a Solenoid :

Compute the self-inductance of a solenoid with turns, length ℓ , and radius NR with a current I flowing through each turn, as shown in Figure.

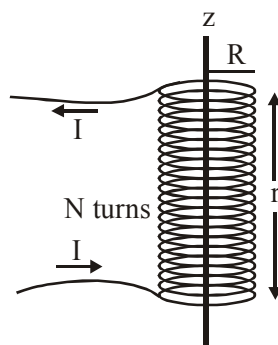


Figure : Solenoid

Solution :

Ignoring edge effects and applying Ampere's law, the magnetic field inside a solenoid is given by Eq. :

$$\vec{B} = \frac{\mu_0 N I}{\ell} \hat{k} = \mu_0 n I \hat{k}$$

where $n = N / \ell$ is the number of turns per unit length. The magnetic flux through each turn is

$$\Phi_B = BA = \mu_0 n I \cdot (\pi R^2) = \mu_0 n I \pi R^2$$

Thus, the self-inductance is
$$L = \frac{N \Phi_B}{I} = \mu_0 n^2 \pi R^2 \ell$$

We see that L depends only on the geometrical factors (n , R and ℓ) and is independent of the current I .

Ex. Self-Inductance of a Toroid :

Calculate the self-inductance of a toroid which consists of N turns and has a rectangular cross section, with inner radius a , outer radius b and height h , as shown in Figure (a).

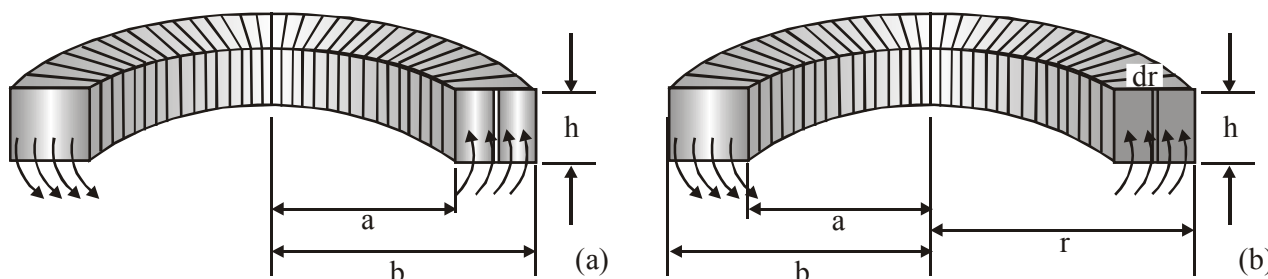


Figure : A toroid with N turns

Solution : According to Ampere's law discussed in section, the magnetic field is given by

$$\oint \vec{B} \cdot d\vec{s} = \oint B ds = B \oint ds = B (2\pi r) = \mu_0 N I$$

or
$$B = \frac{\mu_0 N I}{2\pi r}$$

The magnetic flux through one turn of the toroid may be obtained by integrating over the rectangular cross section, with $dA = h dr$ as the differential area element (figure-b)

$$\Phi_B = \iint \vec{B} \cdot d\vec{A} = \int_a^b \left(\frac{\mu_0 N I}{2\pi r} \right) h dr = \frac{\mu_0 N I h}{2\pi} \ln \left(\frac{b}{a} \right)$$

The total flux is $N \Phi_B$. Therefore, the self-inductance is

$$L = \frac{N \Phi_B}{I} = \frac{\mu_0 N^2 h}{2\pi} \ln \left(\frac{b}{a} \right)$$

Again, the self-inductance L depends only on the geometrical factors. Let's consider the situation where $a \gg b - a$. In this limit, the logarithmic term in the equation above may be expanded as

$$\ln \left(\frac{b}{a} \right) = \ln \left(1 + \frac{b-a}{a} \right) \approx \frac{b-a}{a}$$

and the self-inductance becomes

$$L \approx \frac{\mu_0 N^2 h}{2\pi} \cdot \frac{b-a}{a} = \frac{\mu_0 N^2 A}{2\pi a} = \frac{\mu_0 N^2 A}{\ell}$$

where $A = h(b-a)$ is the cross-sectional area, and $\ell = 2\pi a$. We see that the self inductance of the toroid in this limit has the same form as that of a solenoid.

MUTUAL INDUCTION

Whenever the current passing through primary coil or circuit change then magnetic flux neighbouring secondary coil or circuit will also change. Acc. to Lenz for opposition of flux change, so an emf induced in the neighbouring coil or circuit.

This phenomenon called as 'Mutual induction'. In case of mutual inductance for two coils situated close to each other, flux linked with the secondary due to current in primary.

Due to Air gap always $\phi_2 < \phi_1$ and $\phi_2 = B_1 A_2$ ($\theta = 0^\circ$).

Case - I When current through primary is constant

Total flux of secondary is directly proportional to current flow through the primary coil

$N_2 \phi_2 \propto I_1 \Rightarrow N_2 \phi_2 = MI_1$, $M = \frac{N_2 \phi_2}{I_1} = \frac{N_2 B_1 A_2}{I_1} = \frac{(\phi_1)_s}{I_p}$ where M : is coefficient of mutual induction.

Case - II When current through primary changes with respect to time

$$\text{If } \frac{dI_1}{dt} \rightarrow \frac{dB_1}{dt} \rightarrow \frac{d\phi_1}{dt} \rightarrow \frac{d\phi_2}{dt}$$

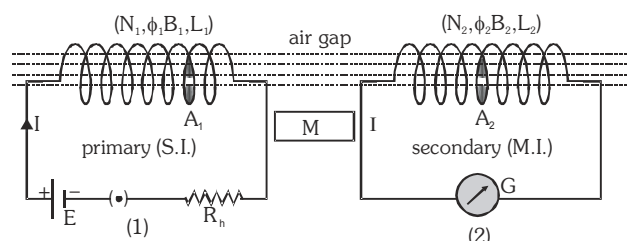
$$\Rightarrow \text{Static EMI } N_2 \phi_2 = MI_1 - N_2 \frac{d\phi_2}{dt} = -M \frac{dI_1}{dt}, \left[-N_2 \frac{d\phi_2}{dt} \right]$$

called total mutual induced emf of secondary coil e_m .

- The units and dimension of M are same as 'L'.
- The mutual inductance does not depends upon current through the primary and it is constant for circuit system.

'M' depends on :

- Number of turns (N_1, N_2).
- Area of cross section.
- Distance between two coils (As $d \downarrow = M \uparrow$).
- Coupling factor 'K'** between two coils.
- Coefficient of self inductance (L_1, L_2).
- Magnetic permeability of medium (μ_r).
- Orientation between two coils.



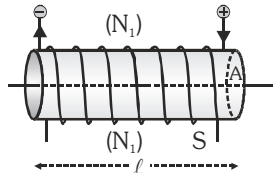
$$e_m = -M \left(\frac{dI_1}{dt} \right)$$

Secondary ← → Primary

DIFFERENT COEFFICIENT OF MUTUAL INDUCTANCE

- In terms of their number of turns
- In terms of their coefficient of self inductances
- In terms of their nos of turns (N_1, N_2)**

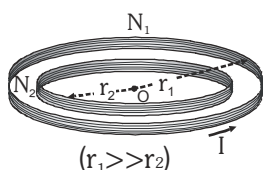
(a) Two co-axial solenoids :- ($M_{s_1s_2}$)



Coefficient of mutual inductance between two solenoids

$$M_{s_1s_2} = \frac{N_2 B_1 A}{I_1} = \frac{N_2}{I_1} \left[\frac{\mu_0 N_1 I_1}{\ell} \right] A \Rightarrow M_{s_1s_2} = \left[\frac{\mu_0 N_1 N_2 A}{\ell} \right]$$

(b) Two plane concentric coils ($M_{c_1c_2}$)



$$M_{c_1c_2} = \frac{N_2 B_1 A_2}{I_1} \text{ where } B_1 = \frac{\mu_0 N_1 I_1}{2r_1}, A_2 = \pi r_2^2$$

$$M_{c_1c_2} = \frac{N_2}{I_1} \left[\frac{\mu_0 N_1 I_1}{2r_1} \right] (\pi r_2^2) \Rightarrow M_{c_1c_2} = \frac{\mu_0 N_1 N_2 \pi r_2^2}{2r_1}$$

Two concentric loop :

$$M \propto \frac{r_2^2}{r_1} (r_1 \gg r_2)$$

Two concentric square loops :

$$M \propto \frac{b^2}{a}$$

A square and a circular loop

$$M \propto \frac{r^2}{a}$$

In terms of L_1 and L_2 : For two magnetically coupled coils :-

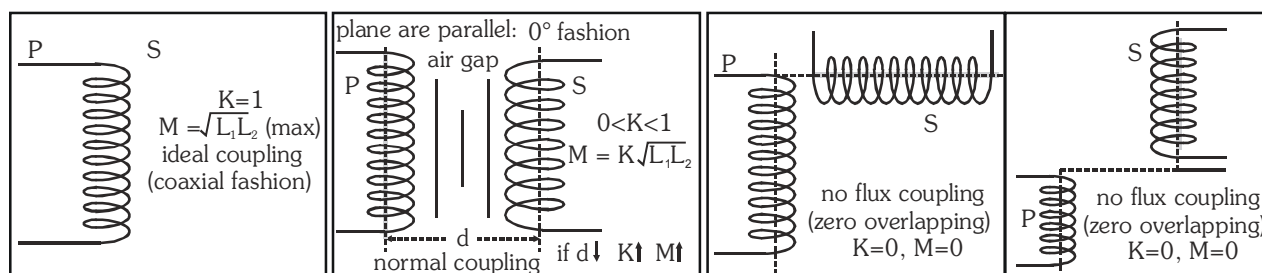
$$M = K\sqrt{L_1 L_2} \text{ here 'K' is coupling factor between two coils and its range } 0 \leq K \leq 1$$

• For ideal coupling $K_{\max} = 1 \Rightarrow M_{\max} = \sqrt{L_1 L_2}$ (where M is geometrical mean of L_1 and L_2)

• For real coupling ($0 < K < 1$) $M = K\sqrt{L_1 L_2}$

• Value of coupling factor 'K' decided from fashion of coupling.

• Different fashion of coupling



$$'K' \text{ also defined as } K = \frac{\phi_s}{\phi_p} = \frac{\text{mag. flux linked with secondary (s)}}{\text{mag. flux linked with primary (p)}}$$

INDUCTANCE IN SERIES AND PARALLEL

Two coil are connected in series : Coils are lying close together (M)

$$\text{If } M = 0, L = L_1 + L_2$$

$$\text{If } M \neq 0, L = L_1 + L_2 + 2M$$

(a) When current in both is in the same direction Then $L = (L_1 + M) + (L_2 + M)$

(b) When current flow in two coils are mutually in opposite directions.

$$L = L_1 + L_2 - 2M$$

Two coils are connected in parallel :

$$(a) \text{ If } M = 0 \text{ then } \frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2} \text{ or } L = \frac{L_1 L_2}{L_1 + L_2} \quad (b) \text{ If } M \neq 0 \text{ then } \frac{1}{L} = \frac{1}{(L_1 + M)} + \frac{1}{(L_2 + M)}$$

Ex. A coil is wound on an iron core and looped back on itself so that the core has two sets of closely wound wires in series carrying current in the opposite sense. What do you expect about its self-inductance ? Will it be larger or small ?

Sol. As the two sets of wire carry currents in opposite directions, their induced emf's also act in opposite directions. These induced emf's tend to cancel each other, making the self-inductance of the coil very small.

This situation is similar to two coils connected in series and producing fluxes in opposite directions. Therefore, their equivalent inductance must be $L_{eq} = L + L - 2M = L + L - 2L = 0$

Ex. A solenoid has 2000 turns wound over a length of 0.3 m. The area of cross-section is $1.2 \times 10^{-3} \text{ m}^2$. Around its central section a coil of 300 turns is closely wound. If an initial current of 2A is reversed in 0.25 s, find the emf induced in the coil.

$$\text{Sol. } M = \frac{\mu_0 N_1 N_2 A}{\ell} = \frac{4\pi \times 10^{-7} \times 2000 \times 300 \times 1.2 \times 10^{-3}}{0.3} = 3 \times 10^{-3} \text{ H}$$

$$\mathcal{E} = -M \frac{dI}{dt} = -3 \times 10^{-3} \left[\frac{-2 - 2}{0.25} \right] = 48 \times 10^{-3} \text{ V} = 48 \text{ mV}$$

ENERGY STORED IN INDUCTOR

The energy of a capacitor is stored in the electric field between its plates. Similarly, an inductor has the capability of storing energy in its magnetic field. An increasing current in an inductor causes an emf between its terminals.

$$\text{Power } P = \text{The work done per unit time} = \frac{dW}{dt} = -ei = -\left[L \frac{di}{dt} \right] i = -L i \frac{di}{dt}$$

here i = instantaneous current and L = inductance of the coil

$$\text{From } dW = -dU \text{ (energy stored) so } \frac{dW}{dt} = -\frac{dU}{dt} \quad \therefore \frac{dU}{dt} = Li \frac{di}{dt} \Rightarrow dU = Li di$$

The total energy U supplied while the current increases from zero to final value i is,

$$U = L \int_0^i i di = \frac{1}{2} L (i^2)_0^i \quad \therefore U = \frac{1}{2} L I^2$$

the energy stored in the magnetic field of an inductor when a current I is $= \frac{1}{2} L I^2$.

The source of this energy is the external source of emf that supplies the current.

- After the current has reached its final steady state value I , $\frac{di}{dt} = 0$ and no more energy is input to the inductor.
- When the current decreases from i to zero, the inductor acts as a source that supplies a total amount of energy $\frac{1}{2} L i^2$ to the external circuit. If we interrupt the circuit suddenly by opening a switch the current decreases very rapidly, the induced emf is very large and the energy may be dissipated in an arc the switch.

MAGNETIC ENERGY PER UNIT VOLUME OR ENERGY DENSITY

- The energy in an inductor is actually stored in the magnetic field within the coil. For a long solenoid its magnetic field can be assumed completely within the solenoid.

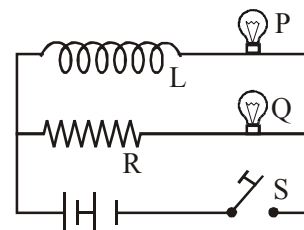
The energy U stored in the solenoid when a current I is,

$$U = \frac{1}{2} L I^2 = \frac{1}{2} (\mu_0 n^2 V) I^2 \quad (L = \mu_0 n^2 V) \quad (V = \text{Volume} = A\ell)$$

$$\text{The energy per unit volume } u = \frac{U}{V} = \frac{1}{2} \mu_0 n^2 I^2 = \frac{(\mu_0 n I)^2}{2 \mu_0} = \frac{B^2}{2 \mu_0} \quad (B = \mu_0 n I) \therefore u = \frac{1}{2} \frac{B^2}{\mu_0}$$

Ex. Figure shows an inductor L a resistor R connected in parallel to a battery through a switch. The resistance of R is same as that of the coil that makes L . Two identical bulb are put in each arm of the circuit.

- Which of two bulbs lights up earlier when S is closed?
- Will the bulbs be equally bright after some time?



- Sol.** (i) When switch is closed induced e.m.f. in inductor i.e. back e.m.f. delays the glowing of lamp P so lamp Q light up earlier.
- (ii) Yes. At steady state inductive effect becomes meaningless so both lamps become equally bright after some time.

Ex. A very small circular loop of area $5 \times 10^{-4} \text{ m}^2$, resistance 2 ohm and negligible inductance is initially coplanar and concentric with a much larger fixed circular loop of radius 0.1 m . A constant current of 1 ampere is passed in the bigger loop and the smaller loop is rotated with angular velocity $\omega \text{ rad/s}$ about a diameter. Calculate (a) the flux linked with the smaller loop (b) induced emf and induced current in the smaller loop as a function of time.

- Sol.** (a) The field at the centre of larger loop $B_1 = \frac{\mu_0 I}{2R} = \frac{2\pi \times 10^{-7}}{0.1} = 2\pi \times 10^{-6} \text{ Wb/m}^2$

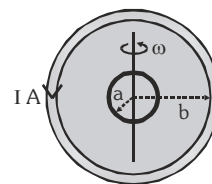
is initially along the normal to the area of smaller loop. Now as the smaller loop (and hence

normal to its plane) is rotating at angular velocity ω , with respect to $\vec{B}$ so the flux linked with the smaller loop at time t is, $\phi_2 = B_1 A_2 \cos \theta = (2\pi \times 10^{-6}) (5 \times 10^{-4}) \cos \omega t$

i.e., $\phi_2 = \pi \times 10^{-9} \cos \omega t \text{ Wb}$

- (b) The induced emf in the smaller loop

$$e_2 = -\frac{d\phi_2}{dt} = -\frac{d}{dt} (\pi \times 10^{-9} \cos \omega t) \\ = \pi \times 10^{-9} \omega \sin \omega t \text{ volt}$$

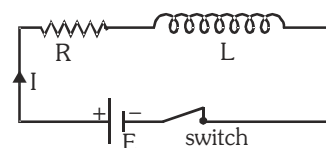


- (c) The induced current in the smaller loop is, $I_2 = \frac{e_2}{R} = \frac{1}{2} \pi \omega \times 10^{-9} \sin \omega t \text{ ampere}$

R-L DC CIRCUIT

Current Growth

- (i) **emf equation** $E = IR + L \frac{dI}{dt}$



(ii) **Current at any instant**

When key is closed the current in circuit increases exponentially with respect to time. The current in circuit

at any instant 't' given by $I = I_0 \left[1 - e^{-\frac{t}{\lambda}} \right]$

$t = 0$ (just after the closing of key) $\Rightarrow I = 0$

$t = \infty$ (some time after closing of key) $\Rightarrow I \rightarrow I_0$

(iii) Just after the closing of the key inductance behaves like open circuit and current in circuit is zero.

Open circuit, $t = 0$, $I = 0$ Inductor provide infinite resistance

(iv) Some time after closing of the key inductance behaves like simple connecting wire (short circuit) and current in circuit is constant.

Short circuit, $t \rightarrow \infty$, $I \rightarrow I_0$, Inductor provide zero resistance $I_0 = \frac{E}{R}$

(Final, steady, maximum or peak value of current) or ultimate current

Note : Peak value of current in circuit does not depends on self inductance of coil.

(v) **Time constant of circuit (λ)**

$\lambda = \frac{L}{R_{\text{sec}}}$ It is a time in which current increases up to 63% or 0.63 times of peak current value.

(vi) **Half life (T)**

It is a time in which current increases upto 50% or 0.50 times of peak current value.

$$I = I_0 (1 - e^{-t/\lambda}), t = T, I = \frac{I_0}{2} \Rightarrow \frac{I_0}{2} = I_0 (1 - e^{-T/\lambda}) \Rightarrow e^{-T/\lambda} = \frac{1}{2} \Rightarrow e^{T/\lambda} = 2$$

$$\frac{T}{\lambda} \log_e e = \log_e 2 \Rightarrow T = 0.693 \lambda \Rightarrow T = 0.693 \frac{L}{R_{\text{sec}}}$$

(vii) **Rate of growth of current at any instant :-**

$$\left[\frac{dI}{dt} \right] = \frac{E}{L} (e^{-t/\lambda}) \Rightarrow t = 0 \Rightarrow \left[\frac{dI}{dt} \right]_{\text{max}} = \frac{E}{L} \quad t = \infty \Rightarrow \left[\frac{dI}{dt} \right] \rightarrow 0$$

Note : Maximum or initial value of rate of growth of current does not depends upon resistance of coil.

Current Decay

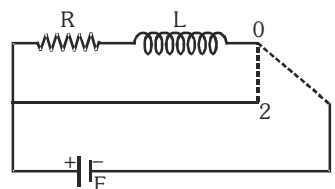
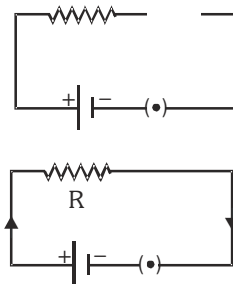
(i) **Emf equation** $IR + L \frac{dI}{dt} = 0$

(ii) **Current at any instant**

Once current acquires its final max steady value, if suddenly switch is put off then current start decreasing exponentially wrt to time. At switch put off condition $t = 0$, $I = I_0$, source emf E is cut off from circuit $I = I_0 (e^{-t/\lambda})$

Just after opening of key $t = 0 \Rightarrow I = I_0 = \frac{E}{R}$

Some time after opening of key $t \rightarrow \infty \Rightarrow I \rightarrow 0$



(iii) **Time constant (λ)**

It is a time in which current decreases up to 37% or 0.37 times of peak current value.

(iv) **Half life (T)**

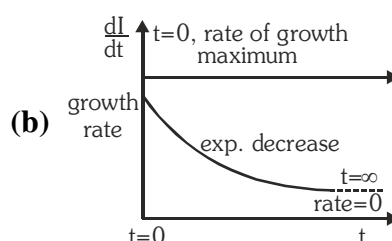
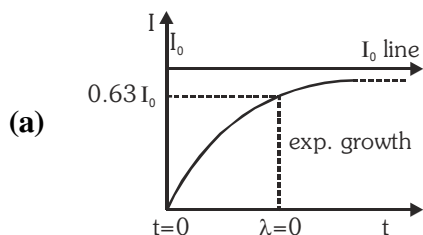
It is a time in which current decreases upto 50% or 0.50 times of peak current value.

(v) **Rate of decay of current at any instant**

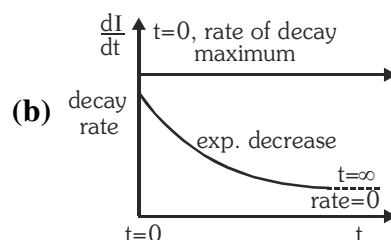
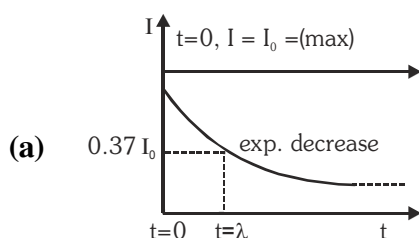
$$\left[-\frac{dI}{dt} \right] = \left[\frac{E}{L} \right] e^{-t/\lambda} \quad t=0, \left[-\frac{dI}{dt} \right]_{\max.} = \frac{E}{L} \quad t \rightarrow \infty \Rightarrow \left[-\frac{dI}{dt} \right] \rightarrow 0$$

Graph for R-L circuit :-

Current Growth :-



Current decay :-

**LC Oscillations :**

Consider an LC circuit in which a capacitor is connected to an inductor, as shown in Figure.

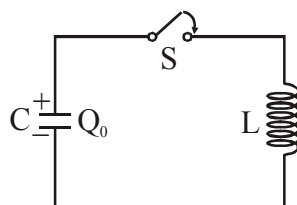


Figure LC Circuit

Suppose the capacitor initially has charge Q_0 . When the switch is closed, the capacitor begins to discharge and the electric energy is decreased. On the other hand, the current created from the discharging process generates magnetic energy which then gets stored in the inductor. In the absence of resistance, the total energy is transformed back and forth between the electric energy in the capacitor and the magnetic energy in the inductor. This phenomenon is called electromagnetic oscillation.

The total energy in the LC circuit at some instant after closing the switch is

$$U = U_C + U_L = \frac{1}{2} \frac{Q^2}{C} + \frac{1}{2} LI^2$$

The fact that U remains constant implies that

$$\frac{dU}{dt} = \frac{d}{dt} \left(\frac{1}{2} \frac{Q^2}{C} + \frac{1}{2} LI^2 \right) = \frac{Q}{C} \frac{dQ}{dt} + LI \frac{dI}{dt} = 0$$

$$\frac{Q}{C} + L \frac{d^2 Q}{dt^2} = 0$$

where $I = -dQ/dt$ (and $dI/dt = -d^2Q/dt^2$). Notice the sign convention we have adopted here. The negative sign implies that the current I is equal to the rate of decrease of charge in the capacitor plate immediately after the switch has been closed. The same equation can be obtained by applying the modified Kirchhoff's loop rule clockwise.

$$\frac{Q}{C} - L \frac{dI}{dt} = 0$$

followed by our definition of current.

The general solution to equation is $Q(t) = Q_0 \cos(\omega_0 t + \phi)$

where Q_0 is the amplitude of the charge and ϕ is the phase. The angular frequency ω_0 is given by

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

The corresponding current in the inductor is

$$I(t) = -\frac{dQ}{dt} = \omega_0 Q_0 \sin(\omega_0 t + \phi) = I_0 \sin(\omega_0 t + \phi)$$

where $I_0 = \omega_0 Q_0$. From the initial conditions $Q(t=0) = Q_0$ and $I(t=0) = 0$, the phase ϕ can be determined to $\phi = 0$. Thus, the solutions for the charge and the current in our LC circuit are

$$Q(t) = Q_0 \cos \omega_0 t$$

$$\text{and } I(t) = I_0 \sin \omega_0 t$$

The time dependence of $Q(t)$ and $I(t)$ are depicted in figure.

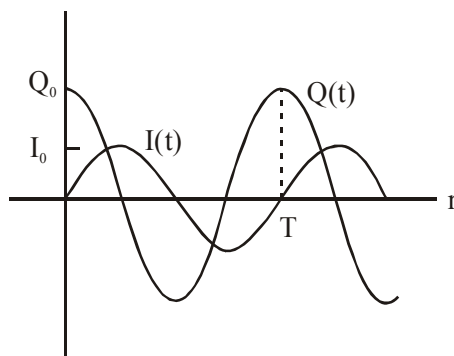


Figure : Charge and current in the LC circuit as a function of time

Using Eqs., we see that at any instant of time, the electric energy and the magnetic energies are given by

$$U_E = U_E = \frac{Q^2(t)}{2C} = \left(\frac{Q_0^2}{2C} \right) \cos^2 \omega_0 t$$

$$\text{and } U_B = \frac{1}{2} LI^2(t) = \frac{LI_0^2}{2} \sin^2 \omega_0 t = \frac{L(-\omega_0 Q_0)^2}{2} \sin^2 \omega_0 t = \left(\frac{Q_0^2}{2C} \right) \sin^2 \omega_0 t = \frac{Q_0^2}{2C}$$

The electric and magnetic energy oscillation is illustrated in figure.

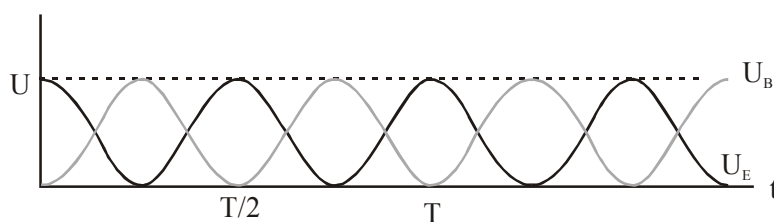


Figure : Electric and magnetic energy oscillations

The mechanical analog of the LC oscillations is the mass-spring system, shown in Figure.

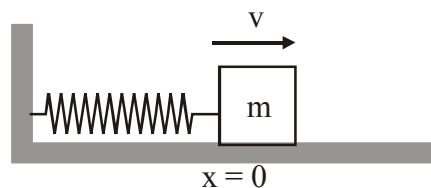


Figure : Mass-spring oscillations

If the mass is moving with a speed v and the spring having a spring constant k is displaced from its equilibrium by x , then the total energy of this mechanical system is

$$U = K + U_{sp} = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

where K and U_{sp} are the kinetic energy of the mass and the potential energy of the spring, respectively. In the absence of friction, U is conserved and we obtain

$$\frac{dU}{dt} = \frac{d}{dt} \left(\frac{1}{2}mv^2 + \frac{1}{2}kx^2 \right) = mv \frac{dv}{dt} + kx \frac{dx}{dt} = 0$$

Using $v = dx/dt$ and $dv/dt = d^2x/dt^2$, the above equation may be rewritten as

$$m \frac{d^2x}{dt^2} + kx = 0$$

The general solution for the displacement is

$$x(t) = x_0 \cos(\omega_0 t + \phi)$$

where $\omega_0 = \sqrt{\frac{k}{m}}$

is the angular frequency and x_0 is the amplitude of the oscillations. Thus, at any instant in time, the energy of the system may be written as

$$\begin{aligned} U &= \frac{1}{2} m x_0^2 \omega_0^2 \sin^2(\omega_0 t + \phi) + \frac{1}{2} k x_0^2 \cos^2(\omega_0 t + \phi) \\ &= \frac{1}{2} k x_0^2 [\sin^2(\omega_0 t + \phi) + \cos^2(\omega_0 t + \phi)] = \frac{1}{2} k x_0^2 \end{aligned}$$

In figure we illustrate the energy oscillations in the LC circuit and the mass spring system (harmonic oscillator).

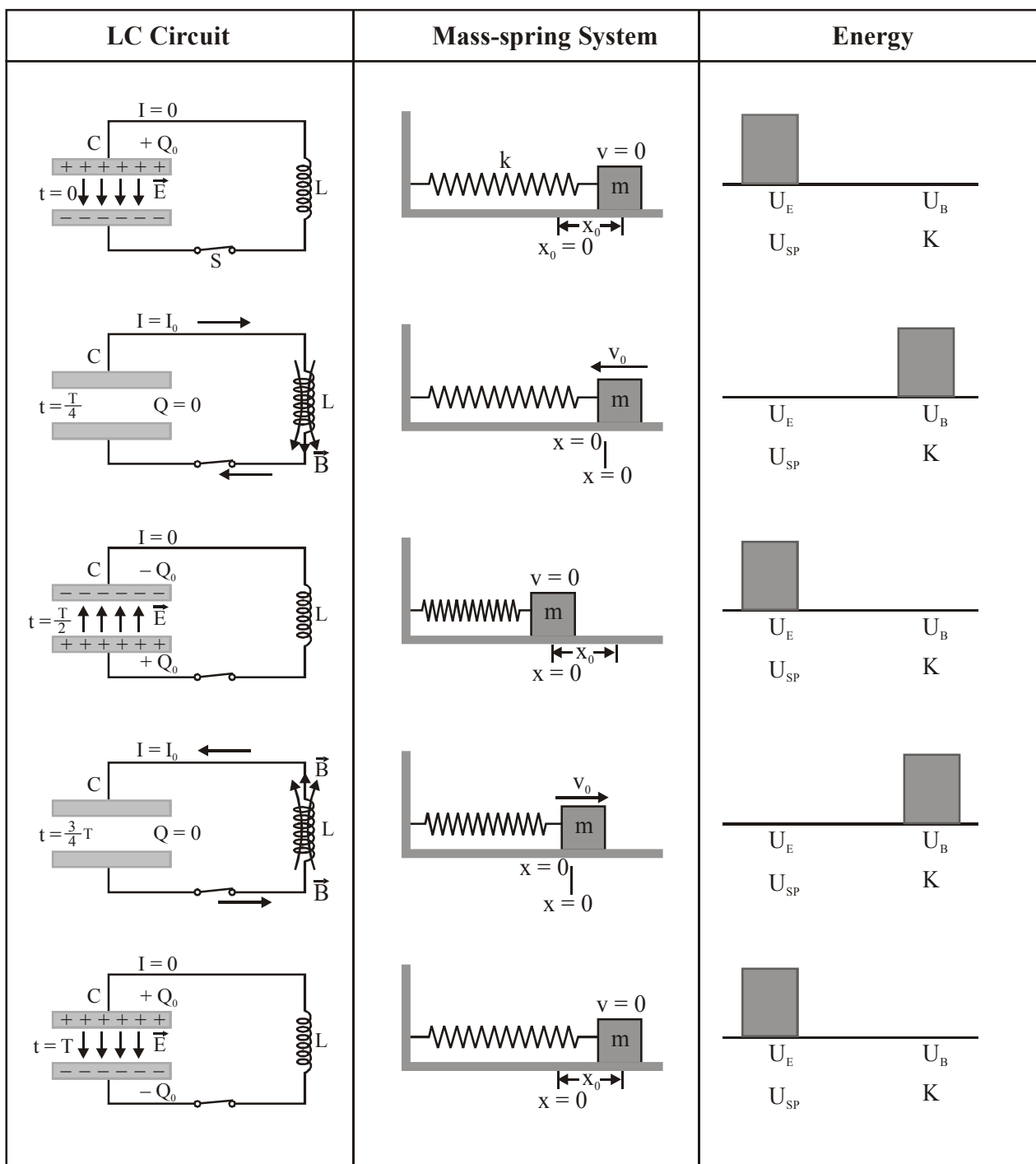


Figure : Energy oscillations in the LC Circuit and the mass-spring system

LC Circuit :

Ex. Consider the circuit shown in Figure. Suppose the switch which has been connected to point *a* for a long time is suddenly thrown to *b* at $t = 0$.

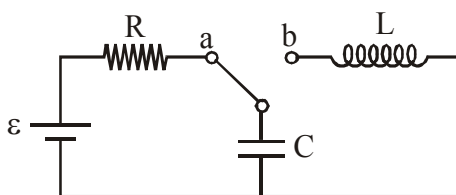


Figure : LC circuit

Find the following quantities :

- (a) the frequency of oscillation of the LC circuit.
- (b) the maximum charge that appears on the capacitor.
- (c) the maximum current in the inductor.
- (d) the total energy the circuit possesses at any time t .

Solution :

- (a) The (angular) frequency of oscillation of the LC circuit is given by $\omega = 2\pi f = 1/\sqrt{LC}$. Therefore, the frequency is :

$$f = \frac{1}{2\pi\sqrt{LC}}$$

- (b) The maximum charge stored in the capacitor before the switch is thrown to b is

$$Q = C\varepsilon$$

- (c) The energy stored in the capacitor before the switch is thrown is :

$$U_E = \frac{1}{2} C\varepsilon^2$$

On the other hand, the magnetic energy stored in the inductor is :

$$U_B = \frac{1}{2} LI^2$$

Thus, when the current is at its maximum, all the energy originally stored in the capacitor is now in the inductor :

$$\frac{1}{2} C\varepsilon^2 = \frac{1}{2} LI_0^2$$

This implies a maximum current

$$I_0 = \varepsilon \sqrt{\frac{C}{L}}$$

- (d) At any time, the total energy in the circuit would be equal to the initial energy that the capacitance stored, that is

$$U = U_E + U_B = \frac{1}{2} C\varepsilon^2$$

ALTERNATING CURRENT

ALTERNATING CURRENT AND VOLTAGE

Voltage or current is said to be alternating if it changes continuously in magnitude and periodically in direction with time. It can be represented by a sine curve or cosine curve

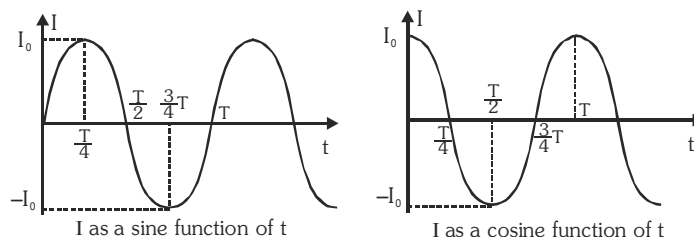
$$I = I_0 \sin \omega t \quad \text{or} \quad I = I_0 \cos \omega t$$

where I = Instantaneous value of current at time t ,

I_0 = Amplitude or peak value

$$\omega = \text{Angular frequency} \quad \omega = \frac{2\pi}{T} = 2\pi f$$

T = time period f = frequency



AMPLITUDE OF AC

The maximum value of current in either direction is called peak value or the amplitude of current. It is represented by I_0 . Peak to peak value = $2I_0$

PERIODIC TIME

The time taken by alternating current to complete one cycle of variation is called periodic time or time period of the current.

FREQUENCY

The number of cycles completed by an alternating current in one second is called the frequency of the current.

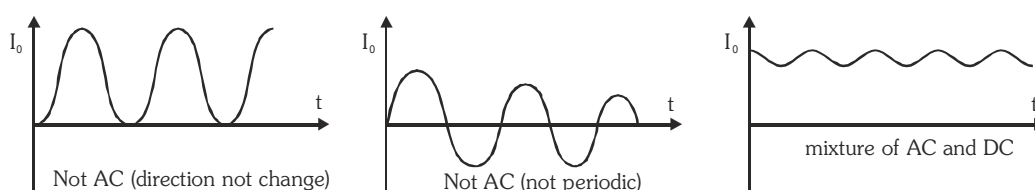
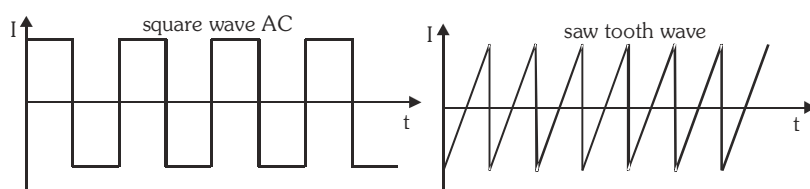
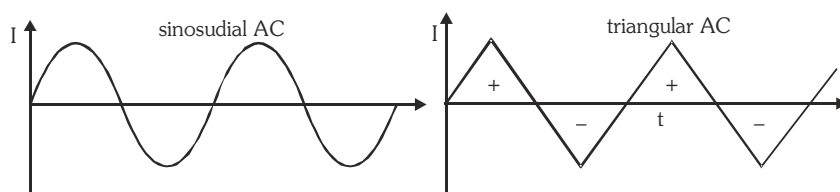
UNIT : cycle/s ; (Hz)

In India : $f = 50$ Hz, supply voltage = 220 volt

In USA : $f = 60$ Hz, supply voltage = 110 volt

CONDITION REQUIRED FOR CURRENT/ VOLTAGE TO BE ALTERNATING

- Amplitude is constant
 - Alternate half cycle is positive and half negative
- The alternating current continuously varies in magnitude and periodically reverses its direction.



AVERAGE VALUE OR MEAN VALUE

The mean value of A.C over any half cycle (either positive or negative) is that value of DC which would send same amount of charge through a circuit as is sent by the AC through same circuit in the same time.

$$\text{average value of current for half cycle } \langle I \rangle = \frac{\int_0^{T/2} I dt}{\int_0^{T/2} dt}$$

Average value of $I = I_0 \sin \omega t$ over the positive half cycle :

$$I_{av} = \frac{\int_0^{T/2} I_0 \sin \omega t dt}{\int_0^{T/2} dt} = \frac{2I_0}{\omega T} [-\cos \omega t]_0^{T/2} = \frac{2I_0}{\pi}$$

$$\begin{aligned} \langle \sin \theta \rangle &= \langle \sin 2\theta \rangle = 0 \\ \langle \cos \theta \rangle &= \langle \cos 2\theta \rangle = 0 \\ \langle \sin \theta \cos \theta \rangle &= 0 \\ \langle \sin^2 \theta \rangle &= \langle \cos^2 \theta \rangle = \frac{1}{2} \end{aligned}$$

- For symmetric AC, average value over full cycle = 0,
Average value of sinusoidal AC

Full cycle	(+ve) half cycle	(-ve) half cycle
0	$\frac{2I_0}{\pi}$	$-\frac{2I_0}{\pi}$

As the average value of AC over a complete cycle is zero, it is always defined over a half cycle which must be either positive or negative

MAXIMUM VALUE

- $I = a \sin \theta \Rightarrow I_{Max.} = a$
- $I = a + b \sin \theta \Rightarrow I_{Max.} = a + b$ (if a and $b > 0$)
- $I = a \sin \theta + b \cos \theta \Rightarrow I_{Max.} = \sqrt{a^2 + b^2}$
- $I = a \sin^2 \theta \Rightarrow I_{Max.} = a$ ($a > 0$)

ROOT MEAN SQUARE (rms) VALUE

It is value of DC which would produce same heat in given resistance in given time as is done by the alternating current when passed through the same resistance for the same time.

$$I_{rms} = \sqrt{\frac{\int_0^T I^2 dt}{\int_0^T dt}} \quad \text{rms value} = \text{Virtual value} = \text{Apparent value}$$

rms value of $I = I_0 \sin \omega t$:

$$I_{rms} = \sqrt{\frac{\int_0^T (I_0 \sin \omega t)^2 dt}{\int_0^T dt}} = \sqrt{\frac{I_0^2}{T} \int_0^T \sin^2 \omega t dt}$$

$$= I_0 \sqrt{\frac{1}{T} \int_0^T \left[\frac{1 - \cos 2\omega t}{2} \right] dt} = I_0 \sqrt{\frac{1}{T} \left[\frac{t}{2} - \frac{\sin 2\omega t}{2 \times 2\omega} \right]_0^T} = \frac{I_0}{\sqrt{2}}$$

- If nothing is mentioned then values printed in a.c circuit on electrical appliances, any given or unknown values, reading of AC meters are assumed to be RMS.

Current	Average	Peak	RMS	Angular frequency
$I_1 = I_0 \sin \omega t$	0	I_0	$\frac{I_0}{\sqrt{2}}$	ω
$I_2 = I_0 \sin \omega t \cos \omega t = \frac{I_0}{2} \sin 2\omega t$	0	$\frac{I_0}{2}$	$\frac{I_0}{2\sqrt{2}}$	2ω
$I_3 = I_0 \sin \omega t + I_0 \cos \omega t$	0	$\sqrt{2} I_0$	I_0	ω

- For above varieties of current $rms = \frac{\text{Peak value}}{\sqrt{2}}$

Ex. If $I = 2\sqrt{t}$ ampere then calculate average and rms values over $t = 2$ to 4 s

Sol. $\langle I \rangle = \frac{\int_2^4 2\sqrt{t} dt}{\int_2^4 dt} = \frac{4}{3} \frac{(t^{\frac{3}{2}})_2^4}{(t)_2^4} = \frac{2}{3} [8 - 2\sqrt{2}]$ and $I_{rms} = \sqrt{\frac{\int_2^4 (2\sqrt{t})^2 dt}{\int_2^4 dt}} = \sqrt{\frac{\int_2^4 4t dt}{2}} = \sqrt{2 \left[\frac{t^2}{2} \right]_2^4} = 2\sqrt{3} \text{ A}$

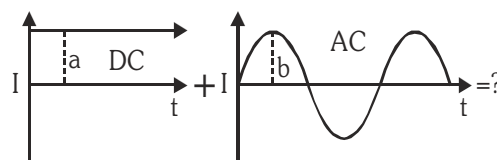
Ex. Find the time required for 50Hz alternating current to change its value from zero to rms value.

Sol. $\because I = I_0 \sin \omega t \therefore \frac{I_0}{\sqrt{2}} = I_0 \sin \omega t \Rightarrow \sin \omega t = \frac{1}{\sqrt{2}} \Rightarrow \omega t = \frac{\pi}{4}$
 $\Rightarrow \left(\frac{2\pi}{T} \right) t = \frac{\pi}{4} \Rightarrow t = \frac{T}{8} = \frac{1}{8 \times 50} = 2.5 \text{ ms}$

Ex. If $E = 20 \sin (100\pi t)$ volt then calculate value of E at $t = \frac{1}{600}$ s

Sol. At $t = \frac{1}{600}$ s $E = 20 \sin \left[100\pi \times \frac{1}{600} \right] = 20 \sin \left[\frac{\pi}{6} \right] = 20 \times \frac{1}{2} = 10 \text{ V}$

Ex. If a direct current of value a ampere is superimposed on an alternating current $I = b \sin \omega t$ flowing through a wire, what is the effective value of the resulting current in the circuit ?



Sol. As current at any instant in the circuit will be,

$$I = I_{DC} + I_{AC} = a + b \sin \omega t$$

$$\therefore I_{eff} = \sqrt{\frac{1}{T} \int_0^T I^2 dt} = \sqrt{\frac{1}{T} \int_0^T (a + b \sin \omega t)^2 dt} = \sqrt{\frac{1}{T} \int_0^T (a^2 + 2ab \sin \omega t + b^2 \sin^2 \omega t) dt}$$

but as $\frac{1}{T} \int_0^T \sin \omega t dt = 0$ and $\frac{1}{T} \int_0^T \sin^2 \omega t dt = \frac{1}{2}$ $\therefore I_{eff} = \sqrt{a^2 + \frac{1}{2} b^2}$

SOME IMPORTANT WAVE FORMS AND THEIR RMS AND AVERAGE VALUE

Nature of wave form	Wave-form	RMS Value	Average or mean Value
Sinusoidal		$\frac{I_0}{\sqrt{2}}$ $= 0.707 I_0$	$\frac{2I_0}{\pi}$ $= 0.637 I_0$
Half wave rectified		$\frac{I_0}{2}$ $= 0.5 I_0$	$\frac{I_0}{\pi}$ $= 0.318 I_0$
Full wave rectified		$\frac{I_0}{\sqrt{2}}$ $= 0.707 I_0$	$\frac{2I_0}{\pi}$ $0.637 I_0$

PHASE AND PHASE DIFFERENCE

(a) Phase

$$I = I_0 \sin (\omega t \pm \phi)$$

Initial phase = ϕ (it does not change with time)

Instantaneous phase = $\omega t \pm \phi$ (it changes with time)

- Phase decides both value and sign. © UNIT: radian

(b) Phase difference

Voltage $V = V_0 \sin (\omega t + \phi_1)$ Current $I = I_0 \sin (\omega t + \phi_2)$

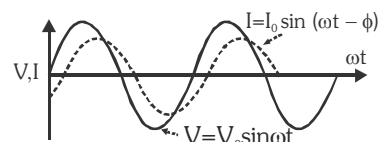
- Phase difference of I w.r.t. V $\phi = \phi_2 - \phi_1$
- Phase difference of V w.r.t. I $\phi = \phi_1 - \phi_2$

LAGGING AND LEADING CONCEPT

(a) V leads I or I lags V $\rightarrow$ It means, V reach maximum before I

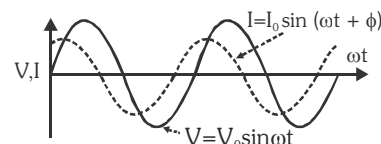
Let if $V = V_0 \sin \omega t$ then $I = I_0 \sin (\omega t - \phi)$

and if $V = V_0 \sin (\omega t + \phi)$ then $I = I_0 \sin \omega t$

(b) V lags I or I leads V $\rightarrow$ It means V reach maximum after I

Let if $V = V_0 \sin \omega t$ then $I = I_0 \sin (\omega t + \phi)$

and if $V = V_0 \sin (\omega t - \phi)$ then $I = I_0 \sin \omega t$

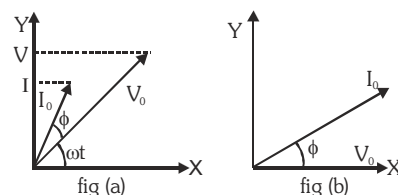


PHASOR AND PHASOR DIAGRAM

A diagram representing alternating current and voltage (of same frequency) as vectors (phasor) with the phase angle between them is called phasor diagram.

Let $V = V_0 \sin \omega t$ and $I = I_0 \sin (\omega t + \phi)$

In figure (a) two arrows represents phasors. The length of phasors represents the maximum value of quantity. The projection of a phasor on y-axis represents the instantaneous value of quantity



Ex. The Equation of current in AC circuit is $I = 4 \sin \left[100\pi t + \frac{\pi}{3} \right]$ A. Calculate.

- (i) RMS Value (ii) Peak Value (iii) Frequency (iv) Initial phase (v) Current at $t = 0$

Sol. (i) $I_{\text{rms}} = \frac{I_0}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$ A (ii) Peak value $I_0 = 4$ A

(iii) $\therefore \omega = 100\pi$ rad/s $\therefore$ frequency $f = \frac{100\pi}{2\pi} = 50$ Hz

(iv) Initial phase $= \frac{\pi}{3}$ (v) At $t = 0$, $I = 4 \sin \left[100\pi \times 0 + \frac{\pi}{3} \right] = 4 \times \frac{\sqrt{3}}{2} = 2\sqrt{3}$ A

Ex. If $I = I_0 \sin \omega t$, $E = E_0 \cos \left[\omega t + \frac{\pi}{3} \right]$. Calculate phase difference between E and I

Sol. $I = I_0 \sin \omega t$ and $E = E_0 \sin \left[\frac{\pi}{2} + \omega t + \frac{\pi}{3} \right]$ $\therefore$ phase difference $= \frac{\pi}{2} + \frac{\pi}{3} = \frac{5\pi}{6}$

Ex. If $E = 500 \sin (100\pi t)$ volt then calculate time taken to reach from zero to maximum.

Sol. $\therefore \omega = 100\pi \Rightarrow T = \frac{2\pi}{100\pi} = \frac{1}{50}$ s, time taken to reach from zero to maximum $= \frac{T}{4} = \frac{1}{200}$ s

Ex. Show that average heat produced during a cycle of AC is same as produced by DC with $I = I_{\text{rms}}$.

Sol. For AC, $I = I_0 \sin \omega t$, the instantaneous value of heat produced (per second) in a resistance R is,
 $H = I^2 R = I_0^2 \sin^2 \omega t \times R$ the average value of heat produced during a cycle is :

$$H_{\text{av}} = \frac{\int_0^T H dt}{\int_0^T dt} = \frac{\int_0^T (I_0^2 \sin^2 \omega t \times R) dt}{\int_0^T dt} = \frac{1}{2} I_0^2 R \quad \left[\because \int_0^T I_0^2 \sin^2 \omega t dt = \frac{1}{2} I_0^2 T \right]$$

$$\Rightarrow H_{\text{av}} = \left(\frac{I_0}{\sqrt{2}} \right)^2 R = I_{\text{rms}}^2 R \dots (i)$$

However, in case of DC, $H_{\text{DC}} = I^2 R \dots (ii)$ $\therefore I = I_{\text{rms}}$ so from equation (i) and (ii) $H_{\text{DC}} = H_{\text{av}}$
 AC produces same heating effects as DC of value $I = I_{\text{rms}}$. This is also why AC instruments which are based on heating effect of current give rms value.

DIFFERENT TYPES OF AC CIRCUITS

In order to study the behaviour of A.C. circuits we classify them into two categories :

- Simple circuits containing only one basic element i.e. resistor (R) or inductor (L) or capacitor (C) only.
- Complicated circuit containing any two of the three circuit elements R, L and C or all of three elements.

AC CIRCUIT CONTAINING PURE RESISTANCE

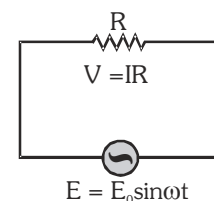
Let at any instant t the current in the circuit = I.

Potential difference across the resistance = I R.

with the help of kirchoff's circuital law $E - I R = 0$

$$\Rightarrow E_0 \sin \omega t = I R$$

$$\Rightarrow I = \frac{E_0}{R} \sin \omega t = I_0 \sin \omega t \quad (I_0 = \frac{E_0}{R} = \text{peak or maximum value of current})$$

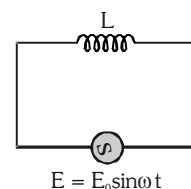
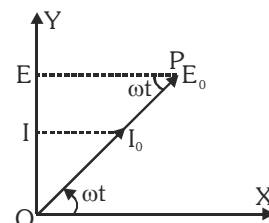
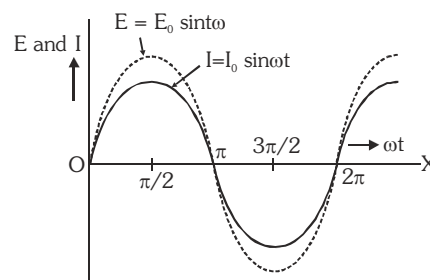


Alternating current developed in a pure resistance is also of sinusoidal nature. In an a.c. circuits containing pure resistance, the voltage and current are in the same phase. The vector or phasor diagram which represents the phase relationship between alternating current and alternating e.m.f. as shown in figure.

In the a.c. circuit having R only, as current and voltage are in the same phase, hence in fig. both phasors E_0 and I_0 are in the same direction, making an angle ωt with OX. Their projections on Y-axis represent the instantaneous values of alternating current and voltage.

i.e. $I = I_0 \sin \omega t$ and $E = E_0 \sin \omega t$.

Since $I_0 = \frac{E_0}{R}$, hence $\frac{I_0}{\sqrt{2}} = \frac{E_0}{R\sqrt{2}} \Rightarrow I_{rms} = \frac{E_{rms}}{R}$



When a.c. flows through the circuit, emf induced across inductance $= -L \frac{dI}{dt}$

Negative sign indicates that induced emf acts in opposite direction to that of applied emf.

Because there is no other circuit element present in the circuit other than inductance so with the help of

Kirchoff's circuital law $E + \left(-L \frac{dI}{dt}\right) = 0 \Rightarrow E = L \frac{dI}{dt}$ so we get

$$I = \frac{E_0}{\omega L} \sin \left(\omega t - \frac{\pi}{2} \right)$$

Maximum current $I_0 = \frac{E_0}{\omega L} \times 1 = \frac{E_0}{\omega L}$, Hence, $I = I_0 \sin \left(\omega t - \frac{\pi}{2} \right)$

In a pure inductive circuit current always lags behind the emf by $\frac{\pi}{2}$.

or alternating emf leads the a. c. by a phase angle of $\frac{\pi}{2}$.

Expression $I_0 = \frac{E_0}{\omega L}$ resembles the expression $\frac{E}{I} = R$.

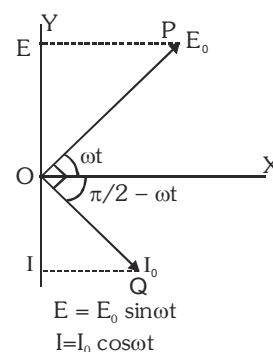
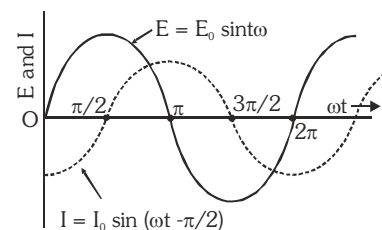
This non-resistive opposition to the flow of A.C. in a circuit is called the inductive reactance (X_L) of the circuit.

$X_L = \omega L = 2\pi f L$ where f = frequency of A.C.

Unit of X_L : ohm

$(\omega L) = \text{Unit of } L \times \text{Unit of } \omega = \text{henry} \times \text{sec}^{-1}$

$$= \frac{\text{Volt}}{\text{Ampere/sec}} \times \text{sec}^{-1} = \frac{\text{Volt}}{\text{Ampere}} = \text{ohm}$$

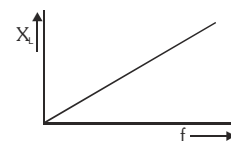


Inductive reactance $X_L \propto f$

Higher the frequency of A.C. higher is the inductive reactance offered by an inductor in an A.C. circuit.

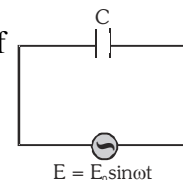
For d.c. circuit, $f = 0 \therefore X_L = \omega L = 2\pi f L = 0$

Hence, inductor offers no opposition to the flow of d.c. whereas a resistive path to a.c.



AC CIRCUIT CONTAINING PURE CAPACITANCE

A circuit containing an ideal capacitor of capacitance C connected with a source of alternating emf as shown in fig. The alternating e.m.f. in the circuit $E = E_0 \sin \omega t$. When alternating e.m.f. is applied across the capacitor a similarly varying alternating current flows in the circuit.



The two plates of the capacitor become alternately positively and negatively charged and the magnitude of the charge on the plates of the capacitor varies sinusoidally with time. Also the electric field between the plates of the capacitor varies sinusoidally with time. Let at any instant t charge on the capacitor = q

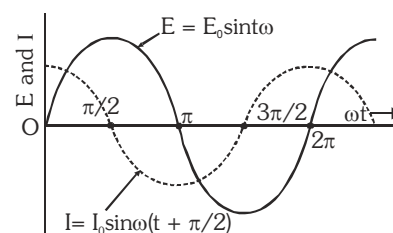
Instantaneous potential difference across the capacitor $E = \frac{q}{C}$

$$\Rightarrow q = C E \Rightarrow q = C E_0 \sin \omega t$$

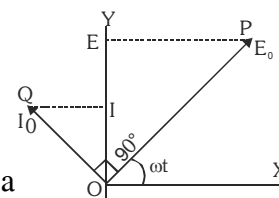
The instantaneous value of current

$$I = \frac{dq}{dt} = \frac{d}{dt}(C E_0 \sin \omega t) = C E_0 \omega \cos \omega t$$

$$\Rightarrow I = \frac{E_0}{(1/\omega C)} \sin\left(\omega t + \frac{\pi}{2}\right) = I_0 \sin\left(\omega t + \frac{\pi}{2}\right) \text{ where } I_0 = \omega C V_0$$



In a pure capacitive circuit, the current always leads the e.m.f. by a phase angle of $\pi/2$. The alternating emf lags behind the alternating current by a phase angle of $\pi/2$.



IMPORTANT POINTS

$\frac{E}{I}$ is the resistance R when both E and I are in phase, in present case they

differ in phase by $\frac{\pi}{2}$, hence $\frac{1}{\omega C}$ is not the resistance of the capacitor,

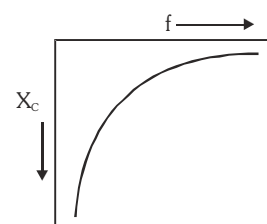
the capacitor offer opposition to the flow of A.C. This non-resistive opposition

to the flow of A.C. in a pure capacitive circuit is known as capacitive reactance X_C . $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$

Unit of X_C : ohm

Capacitive reactance X_C is inversely proportional to frequency of A.C. X_C decreases as the frequency increases.

This is because with an increase in frequency, the capacitor charges and discharges rapidly following the flow of current.



For d.c. circuit $f = 0$ $\therefore X_C = \frac{1}{2\pi fC} = \infty$ but has a very small value for a.c.

This shows that capacitor blocks the flow of d.c. but provides an easy path for a.c.

INDIVIDUAL COMPONENTS (R or L or C)

TERM	R	L	C
Circuit			
Supply Voltage	$V = V_0 \sin \omega t$	$V = V_0 \sin \omega t$	$V = V_0 \sin \omega t$
Current	$I = I_0 \sin \omega t$	$I = I_0 \sin (\omega t - \frac{\pi}{2})$	$I = I_0 \sin (\omega t + \frac{\pi}{2})$
Peak Current	$I_0 = \frac{V_0}{R}$	$I_0 = \frac{V_0}{\omega L}$	$I_0 = \frac{V_0}{1/\omega C} = V_0 \omega C$
Impedance (Ω)	$\frac{V_0}{I_0} = R$	$\frac{V_0}{I_0} = \omega L = X_L$	$\frac{V_0}{I_0} = \frac{1}{\omega C} = X_C$
$Z = \frac{V_0}{I_0} = \frac{V_{rms}}{I_{rms}}$	$R = \text{Resistance}$	$X_L = \text{Inductive reactance.}$	$X_C = \text{Capacitive reactance.}$
Phase difference	zero (in same phase)	$+\frac{\pi}{2}$ (V leads I)	$-\frac{\pi}{2}$ (V lags I)
Phasor diagram			
Variation of Z with f			
G, S_L, S_C (mho, seiman)	R does not depend on f $G = 1/R = \text{conductance.}$	$S_L = 1/X_L$ Inductive susceptance	$S_C = 1/X_C$ Capacitive susceptance
Behaviour of device in D.C. and A.C	Same in A C and D C	L passes DC easily (because $X_L = 0$) while gives a high impedance for the A.C. of high frequency ($X_L \propto f$)	C - blocks DC (because $X_C = \infty$) while provides an easy path for the A.C. of high frequency $\left[X_C \propto \frac{1}{f} \right]$
Ohm's law	$V_R = IR$	$V_L = IX_L$	$V_C = IX_C$

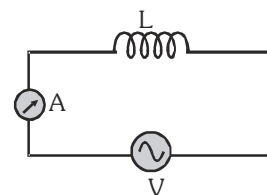
Ex. A capacitor of 50 pF is connected to an a.c. source of frequency 1kHz Calculate its reactance.

Sol. $X_C = \frac{1}{\omega C} = \frac{1}{2\pi \times 10^3 \times 50 \times 10^{-12}} = \frac{10^7}{\pi} \Omega$

Ex. In given circuit applied voltage $V = 50\sqrt{2} \sin 100\pi t$ volt and ammeter reading is 2A then calculate value of L

Sol. $V_{rms} = I_{rms} X_L$ $\therefore$ Reading of ammeter = I_{rms}

$$X_L = \frac{V_{rms}}{I_{rms}} = \frac{V_0}{\sqrt{2} I_{rms}} = \frac{50\sqrt{2}}{\sqrt{2} \times 2} = 25 \Omega \Rightarrow L = \frac{X_L}{\omega} = \frac{25}{100\pi} = \frac{1}{4\pi} \text{ H}$$



Ex. A 50 W, 100 V lamp is to be connected to an AC mains of 200 V, 50 Hz. What capacitance is essential to be put in series with the lamp ?

Sol. $\therefore$ resistance of the lamp $R = \frac{V_s^2}{W} = \frac{(100)^2}{50} = 200 \Omega$ and the maximum current $I = \frac{V}{R} = \frac{100}{200} = \frac{1}{2} \text{ A}$

$\therefore$ when the lamp is put in series with a capacitance and run at 200 V AC, from $V = IZ$

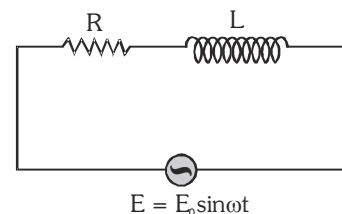
$$Z = \frac{V}{I} = \frac{200}{\frac{1}{2}} = 400 \Omega \quad \text{Now as in case of C-R circuit } Z = \sqrt{R^2 + \frac{1}{(\omega C)^2}},$$

$$\Rightarrow R^2 + \frac{1}{(\omega C)^2} = (400)^2 \quad \Rightarrow \frac{1}{(\omega C)^2} = 16 \times 10^4 - (200)^2 = 12 \times 10^4 \quad \Rightarrow \frac{1}{\omega C} = \sqrt{12} \times 10^2$$

$$\Rightarrow C = \frac{1}{100\pi \times \sqrt{12} \times 10^2} \text{ F} = \frac{100}{\pi \sqrt{12}} \mu\text{F} = 9.2 \mu\text{F}$$

RESISTANCE AND INDUCTANCE IN SERIES (R-L CIRCUIT)

A circuit containing a series combination of a resistance R and an inductance L, connected with a source of alternating e.m.f. E as shown in figure.



PHASOR DIAGRAM FOR L-R CIRCUIT

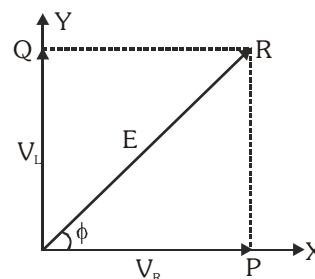
Let in a L-R series circuit, applied alternating emf is $E = E_0 \sin \omega t$. As R and L are joined in series, hence current flowing through both will be same at each instant. Let I be the current in the circuit at any instant and V_L and V_R the potential differences across L and R respectively at that instant.

Then $V_L = IX_L$ and $V_R = IR$

Now, V_R is in phase with the current while V_L leads the current by $\frac{\pi}{2}$.

So V_R and V_L are mutually perpendicular (Note : $E \neq V_R + V_L$)

The vector OP represents V_R (which is in phase with I), while OQ represents V_L (which leads I by 90°).



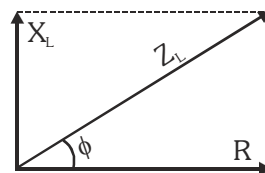
The resultant of V_R and V_L = the magnitude of vector OR $E = \sqrt{V_R^2 + V_L^2}$

Thus $E^2 = V_R^2 + V_L^2 = I^2 (R^2 + X_L^2) \Rightarrow I = \frac{E}{\sqrt{R^2 + X_L^2}}$

The phasor diagram shown in fig. also shows that in L-R circuit the applied emf E leads the current I or conversely the current I lags behind

the e.m.f. E by a phase angle ϕ $\tan \phi = \frac{V_L}{V_R} = \frac{IX_L}{IR} = \frac{X_L}{R} = \frac{\omega L}{R}$

$$\Rightarrow \phi = \tan^{-1} \left(\frac{\omega L}{R} \right)$$



Inductive Impedance Z_L :

In L-R circuit the maximum value of current $I_0 = \frac{E_0}{\sqrt{R^2 + \omega^2 L^2}}$ Here $\sqrt{R^2 + \omega^2 L^2}$ represents the

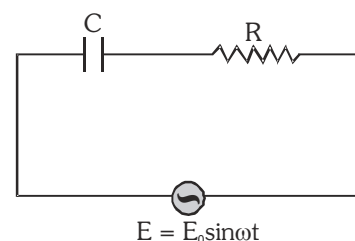
effective opposition offered by L-R circuit to the flow of a.c. through it. It is known as impedance

of L-R circuit and is represented by Z_L . $Z_L = \sqrt{R^2 + \omega^2 L^2} = \sqrt{R^2 + (2\pi fL)^2}$ The reciprocal of

impedance is called admittance $Y_L = \frac{1}{Z_L} = \frac{1}{\sqrt{R^2 + \omega^2 L^2}}$

RESISTANCE AND CAPACITOR IN SERIES (R-C CIRCUIT)

A circuit containing a series combination of a resistance R and a capacitor C , connected with a source of e.m.f. of peak value E_0 as shown in fig.



PHASOR DIAGRAM FOR R-C CIRCUIT

Current through both the resistance and capacitor will be same at every instant and the instantaneous potential differences across C and R are

$$V_C = I X_C \text{ and } V_R = I R$$

where X_C = capacitive reactance and I = instantaneous current.

Now, V_R is in phase with I , while V_C lags behind I by 90° .

The phasor diagram is shown in fig.

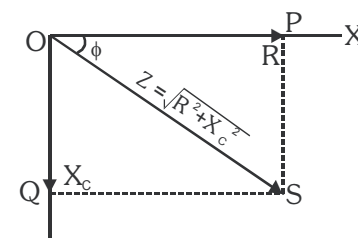
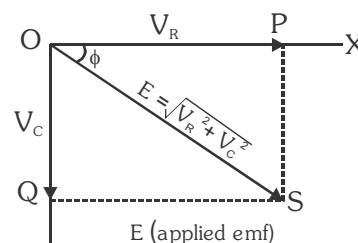
The vector OP represents V_R (which is in phase with I)

and the vector OQ represents V_C (which lags behind I by $\frac{\pi}{2}$).

The vector OS represents the resultant of V_R and V_C = the applied e.m.f. E .

$$\text{Hence } V_R^2 + V_C^2 = E^2 \Rightarrow E = \sqrt{V_R^2 + V_C^2}$$

$$\Rightarrow E^2 = I^2 (R^2 + X_C^2) \Rightarrow I = \frac{E}{\sqrt{R^2 + X_C^2}}$$



The term $\sqrt{R^2 + X_C^2}$ represents the effective resistance of the R-C circuit and called the capacitive

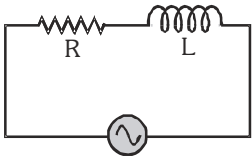
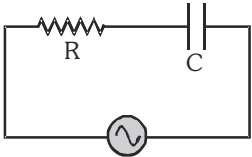
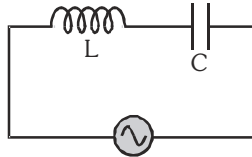
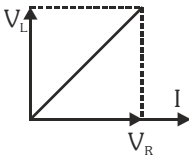
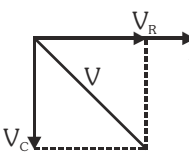
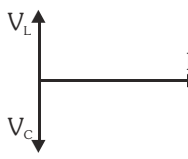
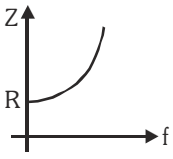
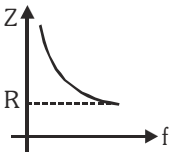
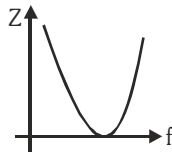
impedance Z_C of the circuit. Hence, in C-R circuit $Z_C = \sqrt{R^2 + X_C^2} = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}$

Capacitive Impedance Z_C :

In R-C circuit the term $\sqrt{R^2 + X_C^2}$ effective opposition offered by R-C circuit to the flow of a.c. through it. It is known as impedance of R-C circuit and is represented by Z_C . The phasor diagram also shows that in R-C circuit the applied e.m.f. lags behind the current I (or the current I leads the emf E) by a phase angle ϕ given by

$$\tan \phi = \frac{V_C}{V_R} = \frac{X_C}{R} = \frac{1/\omega C}{R} = \frac{1}{\omega CR}, \quad \tan \phi = \frac{X_C}{R} = \frac{1}{\omega CR} \Rightarrow \phi = \tan^{-1} \left(\frac{1}{\omega CR} \right)$$

COMBINATION OF COMPONENTS (R-L or R-C or L-C)

TERM	R-L	R-C	L-C
Circuit			
Phasor diagram	 $V^2 = V_R^2 + V_L^2$	 $V^2 = V_R^2 + V_C^2$	 $V = V_L - V_C \text{ (} V_L > V_C \text{)}$ $V = V_C - V_L \text{ (} V_C > V_L \text{)}$
Phase difference in between	V leads I ($\phi = 0$ to $\frac{\pi}{2}$) V and I	V lags I ($\phi = -\frac{\pi}{2}$ to 0)	V lags I ($\phi = -\frac{\pi}{2}$, if $X_C > X_L$) V leads I ($\phi = +\frac{\pi}{2}$, if $X_L > X_C$)
Impedance	$Z = \sqrt{R^2 + X_L^2}$	$Z = \sqrt{R^2 + (X_C)^2}$	$Z = X_L - X_C $
Variation of Z with f	as $f \uparrow, Z \uparrow$ 	as $f \uparrow, Z \downarrow$ 	as $f \uparrow, Z$ first $\downarrow$ then $\uparrow$ 
At very low f	$Z \simeq R \text{ (} X_L \rightarrow 0 \text{)}$	$Z \simeq X_C$	$Z \simeq X_C$
At very high f	$Z \simeq X_L$	$Z \simeq R \text{ (} X_C \rightarrow 0 \text{)}$	$Z \simeq X_L$

Ex. Calculate the impedance of the circuit shown in the figure.

Sol. $Z = \sqrt{R^2 + (X_c)^2} = \sqrt{(30)^2 + (40)^2} = \sqrt{2500} = 50 \Omega$

Ex. If $X_L = 50 \Omega$ and $X_C = 40 \Omega$ Calculate effective value of current in given circuit.

Sol. $Z = X_L - X_C = 10 \Omega$

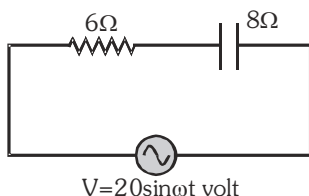
$$I_0 = \frac{V_0}{Z} = \frac{40}{10} = 4A \Rightarrow I_{\text{rms}} = \frac{4}{\sqrt{2}} = 2\sqrt{2} A$$

Ex. In given circuit calculate, voltage across inductor

Sol. $\because V^2 = V_R^2 + V_L^2 \quad \therefore V_L^2 = V^2 - V_R^2$

$$V_L \sqrt{V^2 - V_R^2} = \sqrt{(100)^2 - (60)^2} = \sqrt{6400} = 80 V$$

Ex. In given circuit find out (i) impedance of circuit (ii) current in circuit



Sol. (i) $Z = \sqrt{R^2 + X_C^2} = \sqrt{(6)^2 + (8)^2} = 10 \Omega$

(ii) $V = IZ \Rightarrow I = \frac{V_0}{Z} = \frac{20}{10} = 2A$ so $I_{\text{rms}} = \frac{2}{\sqrt{2}} = \sqrt{2} A$

Ex. When 10V, DC is applied across a coil current through it is 2.5 A, if 10V, 50 Hz A.C. is applied current reduces to 2 A. Calculate reactance of the coil.

Sol. For 10 V D.C. $\because V = IR \quad \therefore$ Resistance of coil $R = \frac{10}{2.5} = 4\Omega$ For 10 V A.C. $\leftarrow V = IZ$

$$\Rightarrow Z = \frac{V}{I} = \frac{20}{10} = 5\Omega$$

$$\because Z = \sqrt{R^2 + X_L^2} = 5 \Rightarrow R^2 + X_L^2 = 25 \Rightarrow X_L^2 = 5^2 - 4^2 \Rightarrow X_L = 3 \Omega$$

Ex. When an alternating voltage of 220V is applied across a device X, a current of 0.5 A flows through the circuit and is in phase with the applied voltage. When the same voltage is applied across another device Y, the same current again flows through the circuit but it leads the applied voltage by $\pi/2$ radians.

(a) Name the devices X and Y.

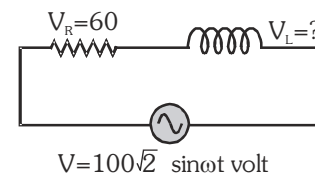
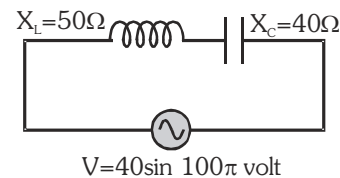
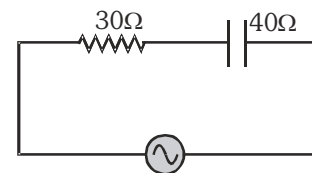
(b) Calculate the current flowing in the circuit when same voltage is applied across the series combination of X and Y.

Sol. (a) X is resistor and Y is a capacitor

(b) Since the current in the two devices is the same (0.5A at 220 volt)

When R and C are in series across the same voltage then

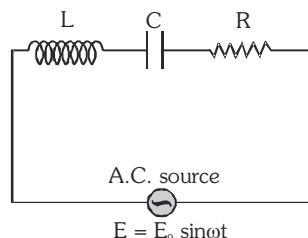
$$R = X_C = \frac{220}{0.5} = 440 \Omega \Rightarrow I_{\text{rms}} = \frac{V_{\text{rms}}}{\sqrt{R^2 + X_C^2}} = \frac{220}{\sqrt{(440)^2 + (440)^2}} = \frac{220}{440\sqrt{2}} = 0.35A$$



INDUCTANCE, CAPACITANCE AND RESISTANCE IN SERIES

(L-C-R SERIES CIRCUIT)

A circuit containing a series combination of an resistance R , a coil of inductance L and a capacitor of capacitance C , connected with a source of alternating e.m.f. of peak value of E_0 , as shown in fig.



PHASOR DIAGRAM FOR SERIES L-C-R CIRCUIT

Let in series LCR circuit applied alternating emf is $E = E_0 \sin \omega t$. As L, C and R are joined in series, therefore, current at any instant through the three elements has the same amplitude and phase.

However voltage across each element bears a different phase relationship with the current.

Let at any instant of time t the current in the circuit is I

Let at this time t the potential differences across L, C , and R

$$V_L = I X_L, V_C = I X_C \text{ and } V_R = I R$$

Now, V_R is in phase with current I but V_L leads I by 90°

While V_C lags behind I by 90° .

The vector OP represents V_R (which is in phase with I) the vector

OQ represent V_L (which leads I by 90°)

and the vector OS represents V_C (which lags behind I by 90°)

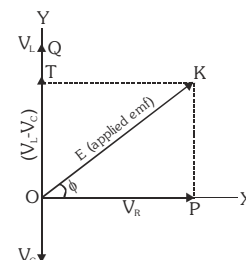
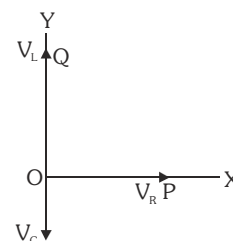
V_L and V_C are opposite to each other.

If $V_L > V_C$ (as shown in figure) the their resultant will be

$(V_L - V_C)$ which is represented by OT .

Finally, the vector OK represents the resultant of V_R and

$(V_L - V_C)$, that is, the resultant of all the three = applied e.m.f.

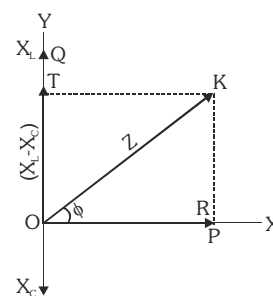


$$\text{Thus } E = \sqrt{V_R^2 + (V_L - V_C)^2} = I \sqrt{R^2 + (X_L - X_C)^2} \Rightarrow I = \frac{E}{\sqrt{R^2 + (X_L - X_C)^2}}$$

$$\text{Impedance } Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}$$

The phasor diagram also shown that in LCR circuit the applied e.m.f.

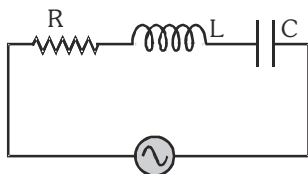
leads the current I by a phase angle ϕ $\tan \phi = \frac{X_L - X_C}{R}$



SERIES LCR AND PARALLEL LCR COMBINATION

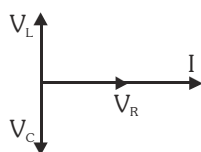
SERIES L-C-R CIRCUIT

1. Circuit diagram

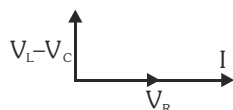


I same for R , L & C

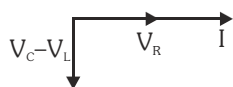
2. Phasor diagram



(i) If $V_L > V_C$ then



(ii) If $V_C > V_L$ then

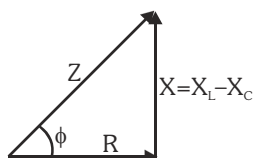


(iii) $V = \sqrt{V_R^2 + (V_L - V_C)^2}$

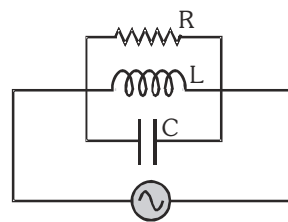
$$\text{Impedance } Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\tan \phi = \frac{X_L - X_C}{R} = \frac{V_L - V_C}{V_R}$$

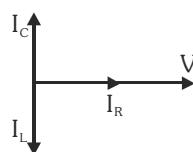
(iv) Impedance triangle



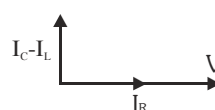
PARALLEL L-C-R CIRCUIT



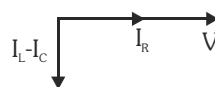
V same for R, L and C
 V same for R, L & C



(i) if $I_C > I_L$ then



(ii) if $I_L > I_C$ then

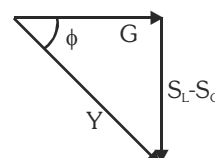


(iii) $I = \sqrt{I_R^2 + (I_L - I_C)^2}$

$$\text{Admittance } Y = \sqrt{G^2 + (S_L - S_C)^2}$$

$$\tan \phi = \frac{S_L - S_C}{G} = \frac{I_L - I_C}{I_R}$$

(iv) Admittance triangle



GOLDEN KEY POINTS

Series

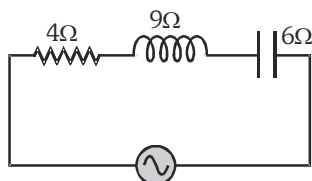
- (a) if $X_L > X_C$ then V leads I, ϕ (positive)
circuit nature inductive
- (b) if $X_C > X_L$ then V lags I, ϕ (negative)
circuit nature capacitive

Parallel

- (a) if $S_L > S_C$ ($X_L < X_C$) then V leads I, ϕ (positive)
circuit nature inductive
- (b) if $S_C > S_L$ ($X_C < X_L$) then V lags I, ϕ (negative)
circuit nature capacitive

- In A.C. circuit voltage for L or C may be greater than source voltage or current but it happens only when circuit contains L and C both and on R it never greater than source voltage or current.
- In parallel A.C. circuit phase difference between I_L and I_C is π

Ex. Find out the impedance of given circuit.



Sol. $Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{4^2 + (9 - 6)^2} = \sqrt{4^2 + 3^2} = \sqrt{25} = 5\Omega$

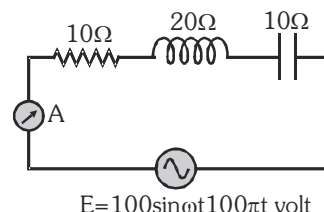
($\because X_L > X_C \therefore$ Inductive)

Ex. Find out reading of A C ammeter and also calculate the potential difference across, resistance and capacitor.

Sol. $Z = \sqrt{R^2 + (X_L - X_C)^2} = 10\sqrt{2}\Omega \Rightarrow I_0 = \frac{V_0}{Z} = \frac{100}{10\sqrt{2}} = \frac{10}{\sqrt{2}}\text{A}$

$\therefore$ ammeter reads RMS value, so its reading $= \frac{10}{\sqrt{2}} = 5\text{A}$

so $V_R = 5 \times 10 = 50\text{V}$ and $V_C = 5 \times 10 = 50\text{V}$



Ex. In LCR circuit with an AC source $R = 300\Omega$, $C = 20\mu\text{F}$, $L = 1.0\text{H}$, $E_{\text{rms}} = 50\text{V}$ and $f = 50/\pi\text{Hz}$. Find RMS current in the circuit.

Sol. $I_{\text{rms}} = \frac{E_{\text{rms}}}{Z} = \frac{E_{\text{rms}}}{\sqrt{R^2 + \left[\omega L - \frac{1}{\omega C}\right]^2}} = \frac{50}{\sqrt{300^2 + \left[2\pi \times \frac{50}{\pi} \times 1 - \frac{1}{20 \times 10^{-6} \times 2\pi \times \frac{50}{\pi}}\right]^2}}$

$\Rightarrow I_{\text{rms}} = \frac{50}{\sqrt{(300)^2 + \left[100 - \frac{10^3}{2}\right]^2}} = \frac{50}{100\sqrt{9+16}} = \frac{1}{10} = 0.1\text{A}$

RESONANCE

A circuit is said to be resonant when the natural frequency of circuit is equal to frequency of the applied voltage. For resonance both L and C must be present in circuit.

There are two types of resonance :

- (i) Series Resonance (ii) Parallel Resonance

SERIES RESONANCE

(a) At Resonance

$$(i) \quad X_L = X_C \quad (ii) \quad V_L = V_C \quad (iii) \quad \phi = 0 \text{ (V and I in same phase)}$$

$$(iv) \quad Z_{\min} = R \text{ (impedance minimum)} \quad (v) \quad I_{\max} = \frac{V}{R} \text{ (current maximum)}$$

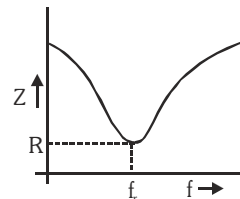
(b) Resonance frequency

$$\because X_L = X_C \Rightarrow \omega_r L = \frac{1}{\omega_r C} \Rightarrow \omega_r^2 = \frac{1}{LC} \Rightarrow \omega_r = \frac{1}{\sqrt{LC}} \Rightarrow f_r = \frac{1}{2\pi\sqrt{LC}}$$

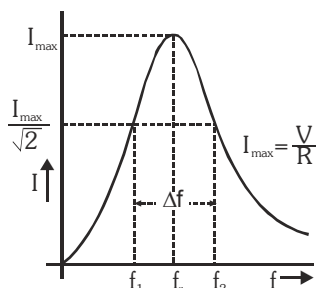
(c) Variation of Z with f

- (i) If $f < f_r$ then $X_L < X_C$ circuit nature capacitive, ϕ (negative)
 (ii) At $f = f_r$ then $X_L = X_C$ circuit nature, Resistive, $\phi = \text{zero}$
 (iii) If $f > f_r$ then $X_L > X_C$ circuit nature is inductive, ϕ (positive)

Variation of I with f as f increase, Z first decreases then increase



(d)



as f increase, I first increase then decreases

- At resonance impedance of the series resonant circuit is minimum so it is called 'acceptor circuit' as it most readily accepts that current out of many currents whose frequency is equal to its natural frequency. In radio or TV tuning we receive the desired station by making the frequency of the circuit equal to that of the desired station.

Half power frequencies

The frequencies at which, power become half of its maximum value called half power frequencies

$$\text{Band width} = \Delta f = f_2 - f_1$$

Quality factor Q : Q-factor of AC circuit basically gives an idea about stored energy & lost energy.

$$Q = 2\pi \frac{\text{maximum energy stored per cycle}}{\text{maximum energy loss per cycle}}$$

- (i) It represents the sharpness of resonance.
 (ii) It is unit less and dimension less quantity

$$(iii) \quad Q = \frac{(X_L)_r}{R} = \frac{(X_C)_r}{R} = \frac{2\pi f_r L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{f_r}{\Delta f} = \frac{f_r}{\text{band width}}$$

Magnification

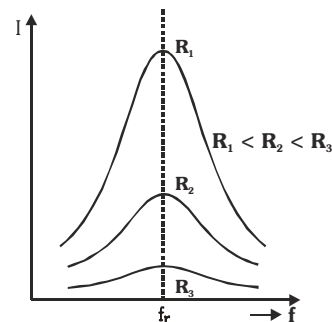
At resonance V_L or $V_C = QE$ (where E = supplied voltage)

So at resonance Magnification factor = Q-factor

Sharpness

Sharpness $\propto$ Quality factor $\propto$ Magnification factor

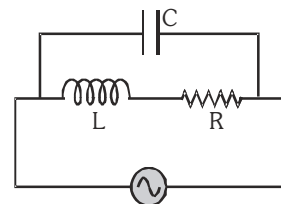
R decrease $\Rightarrow$ Q increases $\Rightarrow$ Sharpness increases



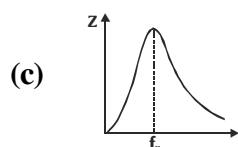
PARALLEL RESONANCE

(a) At resonance

- (i) $S_L = S_C$
- (ii) $I_L = I_C$
- (iii) $\phi = 0$
- (iv) $Z_{\max} = R$ (impedance maximum)
- (v) $I_{\min} = \frac{V}{R}$ (current minimum)



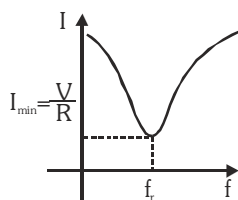
(b) Resonant frequency $f_r = \frac{1}{2\pi\sqrt{LC}}$



Variation of Z with f as f increases, Z first increases then decreases

- ⊙ If $f < f_r$ then $S_L > S_C$, ϕ (positive), circuit nature is inductive
- ⊙ If $f > f_r$ then $S_C > S_L$, ϕ (negative), circuit nature capacitive.

(d) Variation of I with f as f increases, I first decreases then increases

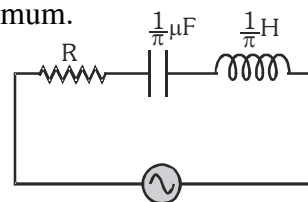


Note : For this circuit $f_r = \frac{1}{2\pi\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}} \Rightarrow Z_{\max} = \frac{L}{RC}$ For resonance $\frac{1}{LC} > \frac{R^2}{L^2}$

Ex. For what frequency the voltage across the resistance R will be maximum.

Sol. It happens at resonance

$$f = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{\frac{1}{\pi} \times 10^{-6} \times \frac{1}{\pi}}} = 500 \text{ Hz}$$

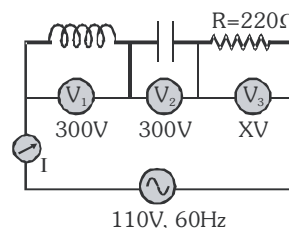


Ex. A capacitor, a resistor and a 40 mH inductor are connected in series to an AC source of frequency 60Hz, calculate the capacitance of the capacitor, if the current is in phase with the voltage. Also calculate the value of X and I.

Sol. At resonance

$$\omega L = \frac{1}{\omega C}, C = \frac{1}{\omega^2 L} = \frac{1}{4\pi^2 f^2 L} = \frac{1}{4\pi^2 \times (60)^2 \times 40 \times 10^{-3}} = 176\mu\text{F}$$

$$V = V_R \Rightarrow X = 110 \text{ V and } I = \frac{V}{R} = \frac{110}{220} = 0.5 \text{ A}$$



Ex. A coil, a capacitor and an A.C. source of rms voltage 24 V are connected in series, By varying the frequency of the source, a maximum rms current 6 A is observed, If this coil is connected to a battery of emf 12 V, and internal resistance 4Ω , then calculate the current through the coil.

Sol. At resonance current is maximum. $I = \frac{V}{R} \Rightarrow$ Resistance of coil $R = \frac{V}{I} = \frac{24}{6} = 4\Omega$

When coil is connected to battery, suppose I current flow through it then

$$I = \frac{E}{R+r} = \frac{12}{4+4} = 1.5 \text{ A}$$

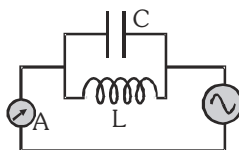
Ex. Radio receiver receives a message at 300m band, If the available inductance is 1 mH, then calculate required capacitance

Sol. Radio receives EM waves. (velocity of EM waves $c = 3 \times 10^8$ m/s)

$$\therefore c = f\lambda \Rightarrow f = \frac{3 \times 10^8}{300} = 10^6 \text{ Hz}$$

$$\text{Now } f = \frac{1}{2\pi\sqrt{LC}} = 1 \times 10^6 \Rightarrow C = \frac{1}{4\pi^2 L \times 10^{12}} = 25 \text{ pF}$$

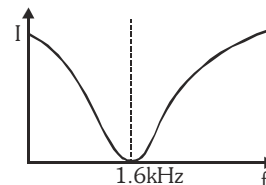
Ex. In a L-C circuit parallel combination of inductance of 0.01 H and a capacitor of $1 \mu\text{F}$ is connected to a variable frequency alternating current source as shown in figure. Draw a rough sketch of the current variation as the frequency is changed from 1kHz to 3kHz.



Sol. L and C are connected in parallel to the AC source,

$$\text{so resonance frequency } f = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.01 \times 10^{-6}}} = \frac{10^4}{2\pi} \approx 1.6 \text{ kHz}$$

In case of parallel resonance, current in L-C circuit at resonance is zero, so the I-f curve will be as shown in figure.



POWER IN AC CIRCUIT

The average power dissipation in LCR AC circuit

$$\text{Let } V = V_0 \sin \omega t \quad \text{and} \quad I = I_0 \sin (\omega t - \phi)$$

$$\text{Instantaneous power } P = (V_0 \sin \omega t)(I_0 \sin (\omega t - \phi)) = V_0 I_0 \sin \omega t (\sin \omega t \cos \phi - \sin \phi \cos \omega t)$$

$$\text{Average power } \langle P \rangle = \frac{1}{T} \int_0^T (V_0 I_0 \sin^2 \omega t \cos \phi - V_0 I_0 \sin \omega t \cos \omega t \sin \phi) dt$$

$$= V_0 I_0 \left[\frac{1}{T} \int_0^T \sin^2 \omega t \cos \phi dt - \frac{1}{T} \int_0^T \sin \omega t \cos \omega t \sin \phi dt \right] = V_0 I_0 \left[\frac{1}{2} \cos \phi - 0 \times \sin \phi \right]$$

$$\Rightarrow \langle P \rangle = \frac{V_0 I_0 \cos \phi}{2} = V_{\text{rms}} I_{\text{rms}} \cos \phi$$

Instantaneous power / Average power / actual power / Virtual power / apparent Peak power

power dissipated power/power loss Power/rms Power
 $P = VI$ $P = V_{rms} I_{rms} \cos \phi$ $P = V_{rms} I_{rms}$ $P = V_0 I_0$

- $I_{rms} \cos \phi$ is known as active part of current or wattfull current, workfull current. It is in phase with voltage.
- $I_{rms} \sin \phi$ is known as inactive part of current, wattless current, workless current. It is in quadrature (90°) with voltage.

Power factor :

Average power $\bar{P} = E_{rms} I_{rms} \cos \phi = \text{rms power} \times \cos \phi$

Power factor ($\cos \phi$) = $\frac{\text{Average power}}{\text{rms Power}}$ and $\cos \phi = \frac{R}{Z}$

Power factor : (i) is leading if I leads V (ii) is lagging if I lags V

GOLDEN KEY POINTS

$P_{av} \leq P_{rms}$				
Power factor varies from 0 to 1				
Pure/Ideal R	$\phi = 0$	V, I same Phase	Power factor = $\cos \phi = 1$ (maximum)	Average power $V_{rms} \cdot I_{rms}$
L	$+\frac{\pi}{2}$	V leads I	0	0
C	$-\frac{\pi}{2}$	V lags I	0	0
Choke coil	$+\frac{\pi}{2}$	V leads I	0	0

- At resonance power factor is maximum ($\phi = 0$ so $\cos \phi = 1$) and $P_{av} = V_{rms} I_{rms}$

Ex. A voltage of 10 V and frequency 10^3 Hz is applied to $\frac{1}{\pi}$ μ F capacitor in series with a resistor of 500Ω .

Find the power factor of the circuit and the power dissipated.

Sol. $\therefore X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 10^3 \times \frac{10^{-6}}{\pi}} = 500\Omega$ $\therefore Z = \sqrt{R^2 + X_C^2} = \sqrt{(500)^2 + (500)^2} = 500\sqrt{2} \Omega$

Power factor $\cos \phi = \frac{R}{Z} = \frac{500}{500\sqrt{2}} = \frac{1}{\sqrt{2}}$,

Power dissipated = $V_{rms} I_{rms} \cos \phi = \frac{V_{rms}^2}{Z} \cos \phi = \frac{(10)^2}{500\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{1}{10} \text{ W}$

Ex. If $V = 100 \sin 100 t$ volt and $I = 100 \sin (100 t + \frac{\pi}{3})$ mA for an A.C. circuit then find out

- (a) phase difference between V and I (b) total impedance, reactance, resistance
(c) power factor and power dissipated (d) components contains by circuits

Sol. (a) Phase difference $\phi = -\frac{\pi}{3}$ (I leads V)

(b) Total impedance $Z = \frac{V_0}{I_0} = \frac{100}{100 \times 10^{-3}} = 1 \text{ k}\Omega$

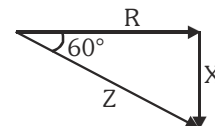
Now resistance $R = Z \cos 60^\circ = 1000 \times \frac{1}{2} = 500 \Omega$

reactance $X = Z \sin 60^\circ = 1000 \times \frac{\sqrt{3}}{2} = \frac{500}{\sqrt{3}} \Omega$

(c) $\phi = -60^\circ \Rightarrow$ Power factor $= \cos \phi = \cos (-60^\circ) = 0.5$ (leading)

Power dissipated $P = V_{\text{rms}} I_{\text{rms}} \cos \phi = \frac{100}{\sqrt{2}} \times \frac{0.1}{\sqrt{2}} \times \frac{1}{2} = 2.5 \text{ W}$

- (d) Circuit must contains R as $\phi \neq \frac{\pi}{2}$ and as ϕ is negative
so C must be their, (L may exist but $X_C > X_L$)



Ex. If power factor of a R-L series circuit is $\frac{1}{2}$ when applied voltage is $V = 100 \sin 100\pi t$ volt and resistance of circuit is 200Ω then calculate the inductance of the circuit.

Sol. $\cos \phi = \frac{R}{Z} \Rightarrow \frac{1}{2} = \frac{R}{Z} \Rightarrow Z = 2R \Rightarrow \sqrt{R^2 + X_L^2} = 2R \Rightarrow X_L = \sqrt{3} R$

$\omega L = \sqrt{3} R \Rightarrow L = \frac{\sqrt{3} R}{\omega} = \frac{\sqrt{3} \times 200}{100\pi} = \frac{2\sqrt{3}}{\pi} \text{ H}$

Ex. A circuit consisting of an inductance and a resistace joined to a 200 volt supply (A.C.). It draws a current of 10 ampere. If the power used in the circuit is 1500 watt. Calculate the wattless current.

Sol. Apparent power $= 200 \times 10 = 2000 \text{ W}$

$\therefore$ Power factor $\cos \phi = \frac{\text{True power}}{\text{Apparent power}} = \frac{1500}{2000} = \frac{3}{4}$

Wattless current $= I_{\text{rms}} \sin \phi = 10 \sqrt{1 - \left(\frac{3}{4}\right)^2} = \frac{10\sqrt{7}}{4} \text{ A}$

Ex. A coil has a power factor of 0.866 at 60 Hz. What will be power factor at 180 Hz.

Sol. Given that $\cos \phi = 0.866$, $\omega = 2\pi f = 2\pi \times 60 = 120\pi$ rad/s,

$$\omega' = 2\pi f' = 2\pi \times 180 = 360\pi \text{ rad/s}$$

$$\text{Now, } \cos \phi = R/Z \Rightarrow R = Z \cos \phi = 0.866 Z$$

$$\text{But } Z = \sqrt{R^2 + (\omega L)^2} \Rightarrow \omega L = \sqrt{Z^2 - R^2} = \sqrt{Z^2 - (0.866 Z)^2} = 0.5 Z$$

$$\therefore L = \frac{0.5Z}{\omega} = \frac{0.5Z}{120\pi}$$

When the frequency is changed to $\omega' = 2\pi \times 180 = 3 \times 120\pi = 300$ rad/s, then inductive reactance $\omega' L = 3 \omega L = 3 \times 0.5 Z = 1.5 Z$

$$\therefore \text{New impedance } Z' = \sqrt{[R + (\omega' L)^2]} = \sqrt{(0.866 Z)^2 + (1.5 Z)^2} = Z \sqrt{[(0.866)^2 + (1.5)^2]} = 1.732Z$$

$$\therefore \text{New power factor} = \frac{R}{Z'} = \frac{0.866 Z}{1.732 Z} = 0.5$$

CHOKE COIL

In a direct current circuit, current is reduced with the help of a resistance. Hence there is a loss of electrical energy $I^2 R$ per sec in the form of heat in the resistance. But in an AC circuit the current can be reduced by choke coil which involves very small amount of loss of energy. Choke coil is a copper coil wound over

a soft iron laminated core. This coil is put in series with the circuit in which current is to be reduced. It also known as ballast.

Circuit with a choke coil is a series L-R circuit. If resistance of choke coil = r (very small)

The current in the circuit $I = \frac{E}{Z}$ with $Z = \sqrt{(R+r)^2 + (\omega L)^2}$ So due to large inductance L of the coil, the current in the circuit is decreased appreciably. However, due to small resistance of the coil r ,

$$\text{The power loss in the choke } P_{av} = V_{rms} I_{rms} \cos \phi \rightarrow 0 \quad \because \cos \phi = \frac{r}{Z} = \frac{r}{\sqrt{r^2 + \omega^2 L^2}} \approx \frac{r}{\omega L} \rightarrow 0$$

Ex. A choke coil and a resistance are connected in series in an a.c circuit and a potential of 130 volt is applied to the circuit. If the potential across the resistance is 50 V. What would be the potential difference across the choke coil.

$$\text{Sol. } V = \sqrt{V_R^2 + V_L^2} \Rightarrow V_L = \sqrt{V^2 - V_R^2} = \sqrt{(130)^2 - (50)^2} = 120 \text{ V}$$

Ex. An electric lamp which runs at 80V DC consumes 10 A current. The lamp is connected to 100 V – 50 Hz ac source compute the inductance of the choke required.

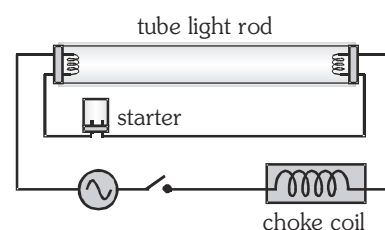
$$\text{Sol. Resistance of lamp } R = \frac{V}{I} = \frac{80}{10} = 8\Omega$$

Let Z be the impedance which would maintain a current of 10 A through the Lamp when it is run

$$\text{on 100 Volt a.c. then. } Z = \frac{V}{I} = \frac{100}{10} = 10 \Omega \text{ but } Z = \sqrt{R^2 + (\omega L)^2}$$

$$\Rightarrow (\omega L)^2 = Z^2 - R^2 = (10)^2 - (8)^2 = 36$$

$$\Rightarrow \omega L = 6 \Rightarrow L = \frac{6}{2\pi \times 50} = 0.02 \text{ H}$$



Ex. Calculate the resistance or inductance required to operate a lamp (60V, 10W) from a source of (100 V, 50 Hz)

Sol. (a) Maximum voltage across lamp = 60V

$$\therefore V_{\text{Lamp}} + V_R = 100 \quad \therefore V_R = 40\text{V}$$

$$\text{Now current through Lamp is} = \frac{\text{Wattage}}{\text{voltage}} = \frac{10}{60} = \frac{1}{6} \text{ A}$$

$$\text{But } V_R = IR \Rightarrow 40 = \frac{1}{6}(R) \Rightarrow R = 240 \Omega$$

(b) Now in this case $(V_{\text{Lamp}})^2 + (V_L)^2 = (V)^2$

$$(60)^2 + (V_L)^2 = (100)^2 \Rightarrow V_L = 80 \text{ V}$$

$$\text{Also } V_L = IX_L = \frac{1}{6}X_L \text{ so } X_L = 80 \times 6 = 480 \Omega = L(2\pi f) \Rightarrow L = 1.5 \text{ H}$$

A capacitor of suitable capacitance replace a choke coil in an AC circuit, the average power consumed in a capacitor is also zero. Hence, like a choke coil, a capacitor can reduce current in AC circuit without power dissipation.

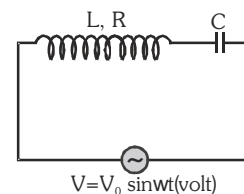
Cost of capacitor is much more than the cost of inductance of same reactance that's why choke coil is used.

Ex. A choke coil of resistance R and inductance L is connected in series with a capacitor C and complete combination is connected to a.c.

voltage, Circuit resonates when angular frequency of supply is $\omega = \omega_0$.

(a) Find out relation between ω_0 , L and C

(b) What is phase difference between V and I at resonance, is it changes when resistance of choke coil is zero.



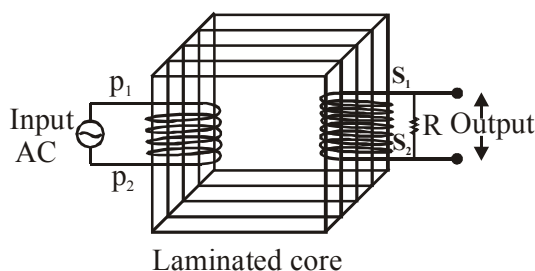
Sol. (a) At resonance condition $X_L = X_C \Rightarrow \omega_0 L = \frac{1}{\omega_0 C} \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$

(b) $\therefore \cos \phi = \frac{R}{Z} = \frac{R}{R} = 1 \quad \therefore \phi = 0^\circ$ No, It is always zero.

Transformers

One of the great advantages of ac over dc for electric-power distribution is that it is much easier to step voltage levels up and down with ac than with dc. The necessary conversion is accomplished by a static device called transformer using the principle of mutual induction.

The figure shows an idealised transformer which consists of two coils or windings, electrically insulated from each other but wound on the same core. The winding to which power is supplied is called primary, the winding from which power is delivered is called the secondary.



The ac source causes an alternating current in the primary which sets up an alternating flux in the core and this induces an emf in each winding of secondary in accordance with Faraday's law. For ideal transformer we assume that primary has negligible resistance and all the flux in core links both primary and secondary. The primary winding has N_1 turns and secondary has N_2 turns. When the magnetic flux changes because of changing currents in the two coils, the resulting induced emf are

$$E_1 = -N_1 \frac{d\phi_B}{dt} \text{ and } E_2 = -N_2 \frac{d\phi_B}{dt}$$

The flux per turn B is same in both primary and the secondary so that the emf per turn is same in each. The ratio of secondary emf E_2 to the primary emf E_1 is therefore equal at any instant to the ratio of secondary to primary turns.

$$\frac{E_2}{E_1} = \frac{N_2}{N_1}$$

If the windings have zero resistance, the induced emf 1 and 2 are equal to the terminal voltage across the primary and the secondary respectively, hence

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

If $N_2 > N_1$ then $V_2 > V_1$ and we have step up transformer, if $N_2 < N_1$ then $V_2 < V_1$ and we have a step down transformer.

If the transformer is assumed to be 100% efficient (no energy losses) the power input is equal to the power output i.e.

$$I_2 V_2 = I_1 V_1 \quad \therefore \quad \frac{I_2}{I_1} = \frac{N_1}{N_2}$$

All the currents and voltages derived above have same frequency as that of source. The equations obtained above apply to ideal transformers, although some energy is lost but well designed transformers have efficiency more than 95%, this is a good approximation. The causes of energy losses and their rectification is given below

Cause	Rectification
1. Due to poor design and air gaps in the core, all the flux due to primary does not pass through the secondary.	1. By winding the primary and secondary coil one over the other.
2. Resistance of windings causes $I^2 R$ loss.	2. In high current, low voltage, these are minimised by using thick wire.
3. The alternating magnetic flux induces eddy currents in the core and causes heating.	3. By using laminated core it can be reduced.
4. Alternating magnetising of core causes hysteresis loss.	4. It is kept minimum by using a magnetic material having low hysteresis loss. (e.g. soft iron)

Damped Harmonic Oscillator:

In the previous section we have considered several examples of oscillatory systems. In each case the amplitude of the oscillation does not depend on time and the system, once set to oscillate, will continue to do so forever. Physical systems are never as ideal as that. Even in the most controlled case there will be dissipative elements.

In each of these cases the amplitudes of the oscillation will gradually decrease and the motion will be 'damped'.

Most of the time we are interested in the motion of an object through air or other viscous fluids. The motion of the object is then subject to a resistive force called aerodynamic or viscous drag. Experimentally the resistive force is found to be proportional to the velocity of the body for low relative speed with respect to the medium. A laboratory model for a damped spring-mass system is shown in figure. Where a vertical spring has at its end a mass m which is connected to massless piston which moves through a liquid. For low speeds it is reasonable to write the frictional force as

$$F_f = -\alpha v \quad (\alpha \text{ in kg/s})$$

where α is a positive constant known as damping constant. The equation of motion of a harmonic oscillator becomes

$$m \frac{d^2x}{dt^2} = -kx - \alpha \frac{dx}{dt}$$

which can be rewritten as

$$\frac{d^2x}{dt^2} + 2\gamma \frac{dx}{dt} + \omega_0^2 x = 0$$

where $2\gamma = \alpha/m > 0$ and ω_0^2 , as before is equal to k/m . γ is known as damping coefficient

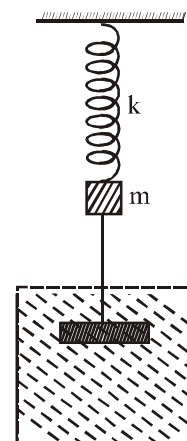
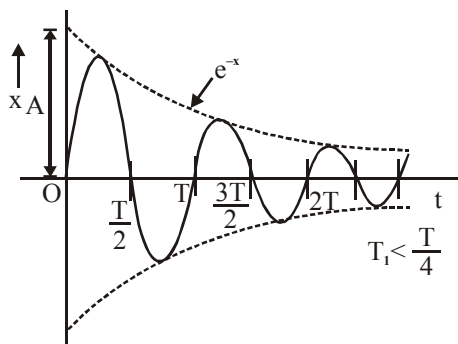
Damped Oscillations

If the dissipative forces are small, i.e. if $\gamma^2 < \omega_0^2$, the dominant force acting on the mass is the conservative restoring force. The system therefore shows oscillations, even though the amplitudes of oscillation continuously decrease with time.

we get $x = Ae^{-\gamma t} \sin(\Omega t + \phi)$

where the new angular frequency of oscillation Ω is given by $\Omega^2 = \omega_0^2 - \gamma^2$

i.e the frequency shift to a lower value due to damping. The amplitude of oscillation is no longer constant but decreases with time in an exponential fashion. The variation of x with time is shown in figure.



It may be seen that $\sin(\Omega t + \phi)$ is a periodic function with period $2\pi/\Omega$. when this is multiplied with $e^{-\gamma t}$ the zeroes of the product would still be at the same place where $\sin(\Omega t + \phi)$ becomes zero. The maxima however does not fall midway between the two minima because of the exponential $e^{-\gamma t}$ but occurs at a slightly earlier time.

Power Loss in a Damped Oscillation-Q-factor

Because of the work done against the nonconservative forces, the energy stored in an oscillator continuously decreases. The fraction of energy lost in a period with respect to its average value for that period is a measure of the quality of the oscillator. The less the value of this fraction is, the more is the ability of the oscillator to sustain periodic motion. Quantitatively, the quality of an oscillator is measured through a Q-factor defined as

$$Q = 2\pi \times \frac{\text{Average energy stored in one period}}{\text{average loss of energy in that period}}$$

If the damping is light i.e. $\gamma \ll \omega_0$, The instantaneous energy of the oscillator is

$$\begin{aligned} E(t) &= \frac{1}{2}mv^2 + \frac{1}{2}kx^2 \\ &= \frac{1}{2}mA^2\omega_0^2e^{-2\gamma t} \end{aligned}$$

where we have replaced k by $m\omega_0^2$. At $t = 0$ the energy of the oscillator is $E_0 = mA^2\omega_0^2/2$ so that

$$E(t) = E_0e^{-2\gamma t}$$

$$P(t) = \frac{dE}{dt} = -2\gamma E_0e^{-2\gamma t} = -2\gamma E(t)$$

Energy lost in one period can be approximately written as

$$\langle P(t) \rangle = \frac{2\pi}{\Omega} = 2\gamma \langle E(t) \rangle = \frac{2\pi}{\Omega}$$

where the angular bracket $\langle \dots \rangle$ denotes the time average value during a period at time t . The quality or Q-factor for an oscillator is

$$2\pi \frac{\langle E(t) \rangle}{2\gamma \langle E(t) \rangle} = \frac{\Omega}{2\gamma} \equiv \frac{\omega_0}{2\gamma}$$

It may be noted that Q is a dimensionless quantity, greater than unity by our assumption that $\gamma \ll \omega_0$.

Time to reach extreme:

The natural logarithm of two successive maxima is called logarithmic decrement and is denoted by λ . From Equation it follows that

$$\lambda = \frac{2\pi\gamma}{\Omega} = \frac{\pi}{Q}$$

Overdamped Motion

In equation if $\gamma > \omega_0$ we do not get oscillatory solution

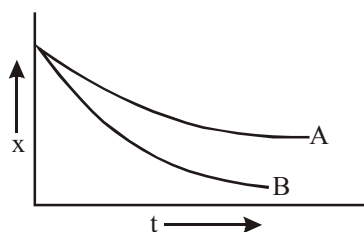
Critical Damping ($\gamma = \omega_0$)

The solution for the displacement is therefore

$$x = (A + Bt)e^{-\gamma t}$$

$$x' = -\frac{mg}{k}(1 + \gamma t)e^{-\gamma t}$$

It may be seen (hat once again there are no oscillation. As $t \rightarrow \infty$, $x \rightarrow 0$ so that the particle takes an infinite time before reaching the equilibrium position. The difference with the overdamped case is that the approach to the equilibrium is faster. The variation of displacement with time is shown in figure (curve B.)

**Forced Oscillations and Resonance**

The oscillations in a system can be indefinitely maintained by supplying energy continuously. In mechanical system this can be done by subjecting it to an external force which itself has harmonic time dependence. An interesting phenomenon occurs if the frequency of the external source is equal or nearly equal to the natural frequency of the system. The amplitude of oscillations is found to increase many folds in such cases. This is called resonance.

Forced Damped Oscillations

In the presence of resistive forces proportional to velocity, equation becomes

$$\frac{d^2x}{dt^2} + 2\gamma \frac{dx}{dt} + \omega_0^2 x = \frac{F_0}{m} \cos \omega t$$

Let us therefore try a solution of the type

$$x = A \cos \omega t + B \sin \omega t + Ce^{-\lambda t}$$

The steady - state displacement is then given by

$$x = C \cos (\omega t + \delta)$$

with

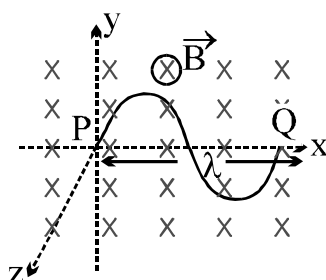
$$C = \frac{F_0}{m} \sqrt{\frac{1}{(\omega_0^2 - \omega^2)^2 + 4\omega^2\gamma^2}}$$

$$\text{and } \tan \delta = \frac{2\omega\gamma}{\omega^2 - \omega_0^2}$$

EXERCISE (S-1)

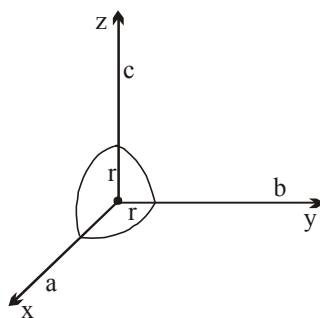
Faraday's law & Motional emf.

1. A wire forming one cycle of sine curve is moved in x-y plane with velocity $\vec{V} = V_x \hat{i} + V_y \hat{j}$. There exist a magnetic field $\vec{B} = -B_0 \hat{k}$. Find the motional emf develop across the ends PQ of wire.



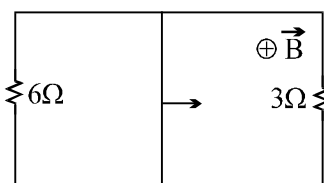
EM0001

2. A wire is bent into 3 circular segments of radius $r = 10$ cm as shown in figure. Each segment is a quadrant of a circle, ab lying in the xy plane, bc lying in the yz plane & ca lying in the zx plane.
(i) if a magnetic field B points in the positive x direction, what is the magnitude of the emf developed in the wire, when B increases at the rate of 3 mT/s ?
(ii) what is the direction of the current in the segment bc.



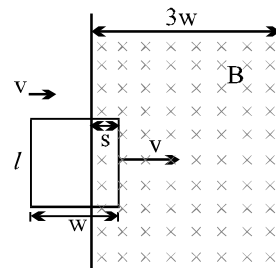
EM0002

3. A rectangular loop with a sliding connector of length $l = 1.0$ m is situated in a uniform magnetic field $B = 2$ T perpendicular to the plane of loop. Resistance of connector is $r = 2\Omega$. Two resistances of 6Ω and 3Ω are connected as shown in figure. Find the external force required to keep the connector moving with a constant velocity $v = 2$ m/s.

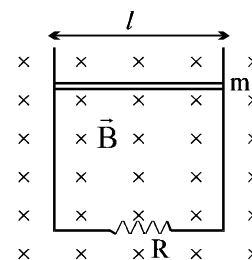


EM0003

4. A rectangular loop of dimensions l & w and resistance R moves with constant velocity V to the right as shown in the figure. It continues to move with same speed through a region containing a uniform magnetic field B directed into the plane of the paper & extending a distance $3w$. Sketch the flux, induced emf & external force acting on the loop as a function of the distance.

**EM0004**

5. A horizontal wire is free to slide on the vertical rails of a conducting frame as shown in figure. The wire has a mass m and length l and the resistance of the circuit is R . If a uniform magnetic field B is directed perpendicular to the frame, then find the terminal speed of the wire as it falls under the force of gravity.

**EM0005**

6. It is desired to measure the magnitude of field between the poles of a powerful loud speaker magnet. A small flat search coil of area 2 cm^2 with 25 closely wound turns, is positioned normal to the field direction, and then quickly snatched out of the field region. Equivalently, one can give it a quick 90° turn to bring its plane parallel to the field direction. The total charge flown in the coil (measured by a ballistic galvanometer connected to coil) is 7.5 mC . The combined resistance of the coil and the galvanometer is 0.50Ω . Estimate the field strength of magnet.

(NCERT)

EM0006

Induced electric field

7. There exists a uniform cylindrically symmetric magnetic field directed along the axis of a cylinder but varying with time as $B = kt$. If an electron is released from rest in this field at a distance of 'r' from the axis of cylinder, its acceleration, just after it is released would be (e and m are the electronic charge and mass respectively)

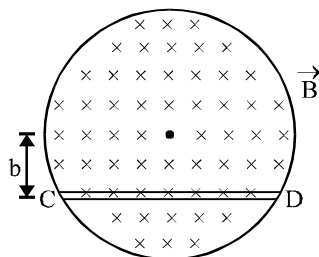
EM0007

8. An air-cored solenoid of length 30 cm. area of cross-section 25 cm^2 and number of turns 500, carries a current of 2.5 A. The current is suddenly switched off in a brief time of 10^{-3} s . How much is the average back emf induced across the ends of the open switch in the circuit? Ignore the variation in magnetic field near the ends of the solenoid. (NCERT)

(NCERT)

EM0008

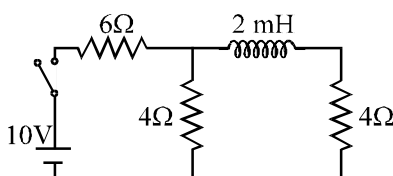
9. A uniform magnetic field $\vec{B}$ fills a cylindrical volumes of radius R . A metal rod CD of length l is placed inside the cylinder along a chord of the circular cross-section as shown in the figure. If the magnitude of magnetic field increases in the direction of field at a constant rate dB/dt , find the magnitude and direction of the EMF induced in the rod.



EM0009

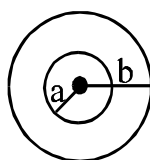
Inductance

10. In the given circuit, find the ratio of i_1 to i_2 where i_1 is the initial (at $t = 0$) current and i_2 is steady state (at $t = \infty$) current through the battery.



EM0010

11. Two concentric and coplanar circular coils have radii a and b ($b \gg a$) as shown in figure. Resistance of the inner coil is R . Current in the outer coil is increased from 0 to i , then find the total charge circulating the inner coil.

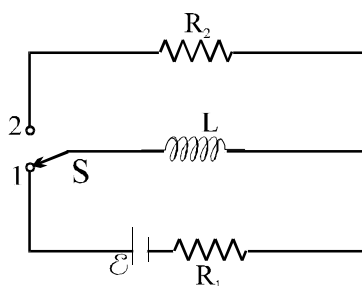


EM0011

12. Find the dimension of the quantity $\frac{L}{RCV}$, where symbols have usual meaning.

EM0012

13. In the circuit shown, initially the switch is in position 1 for a long time. Then the switch is shifted to position 2 for a long time. Find the total heat produced in R_2 .

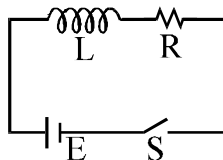


EM0013

14. An emf of 15 volt is applied in a circuit containing 5 H inductance and $10\ \Omega$ resistance. Find the ratio of the currents at time $t = \infty$ and $t = 1$ second.

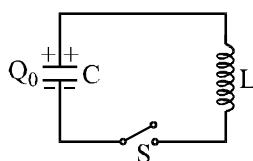
EM0014

15. In the circuit shown in figure switch S is closed at time $t = 0$. Find the charge which passes through the battery in one time constant.



EM0015

16. A capacitor C with a charge Q_0 is connected across an inductor through a switch S. If at $t = 0$, the switch is closed, then find the instantaneous charge q on the upper plate of capacitor.



EM0016

17. An inductor of inductance 2.0 mH, is connected across a charged capacitor of capacitance $5.0\ \mu\text{F}$ and the resulting LC circuit is set oscillating at its natural frequency. Let Q denote the instantaneous charge on the capacitor, and I the current in the circuit. It is found that the maximum value of Q is $200\ \mu\text{C}$.

- when $Q = 100\ \mu\text{C}$, what is the value of $|dI/dt|$?
- when $Q = 200\ \mu\text{C}$, what is the value of I ?
- Find the maximum value of I .
- when I is equal to one half its maximum value, what is the value of $|Q|$?

EM0017

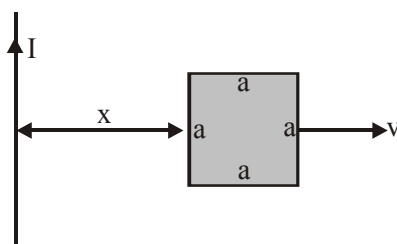
18. A pair of adjacent coils has a mutual inductance of 1.5 H. If the current in one coil changes from 0 to 20 A in 0.5 s, what is the change of flux linkage with the other coil?

(NCERT)

EM0018

19. (a) Obtain an expression for the mutual inductance between a long straight wire and a square loop of side a as shown in figure.
(b) Now assume that the straight wire carries a current of 50 A and the loop is moved to the right with a constant velocity, $v = 10\ \text{m/s}$. Calculate the induced emf in the loop at the instant when $x = 0.2\ \text{m}$. Take $a = 0.1\ \text{m}$ and assume that the loop has a large resistance.

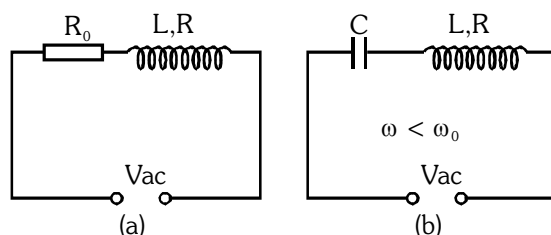
(NCERT)



EM0019

Alternating current

20. Draw the approximate voltage vector diagrams in the electric circuits shown in Fig. a, b. The external voltage V is assumed to be alternating harmonically with frequency ω .



EM0020

21. A current of 4 A flows in a coil when connected to a 12 V dc source. If the same coil is connected to a 12V, 50 rad/s ac source a current of 2.4 A flows in the circuit. Determine the inductance of the coil. Also find the power developed in the circuit if a 2500 μF capacitor is connected in series with the coil.

EM0021

22. An LCR series circuit with 100Ω resistance is connected to an ac source of 200 V and angular frequency 300 rad/s. When only the capacitance is removed, the current lags behind the voltage by 60° . When only the inductance is removed, the current leads the voltage by 60° . Calculate the current and the power dissipated in the LCR circuit.

EM0022

23. A series LCR circuit containing a resistance of 120Ω has angular resonance frequency $4 \times 10^5 \text{ rad s}^{-1}$. At resonance the voltages across resistance and inductance are 60 V and 40 V respectively. Find the values of L and C. At what frequency the current in the circuit lags the voltage by 45° ?

EM0023

24. Find the value of an inductance which should be connected in series with a capacitor of $5\mu\text{F}$, a resistance of 10Ω and an ac source of 50 Hz so that the power factor of the circuit is unity.

EM0024

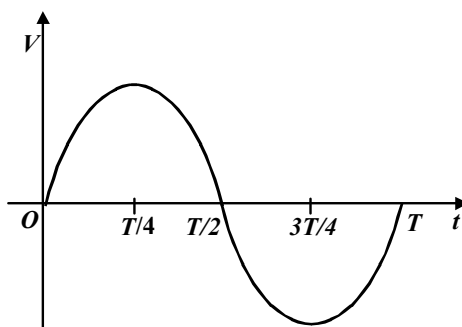
25. In an L-R series A.C circuit the potential difference across an inductance and resistance joined in series are respectively 12 V and 16V. Find the total potential difference across the circuit.

EM0025

26. In an LR series circuit, a sinusoidal voltage $V = V_0 \sin \omega t$ is applied. It is given that $L = 35 \text{ mH}$,

$R = 11 \Omega$, $V_{\text{rms}} = 220 \text{ V}$, $\frac{\omega}{2\pi} = 50 \text{ Hz}$ and $\pi = 22/7$. Find the amplitude of current in the steady state and

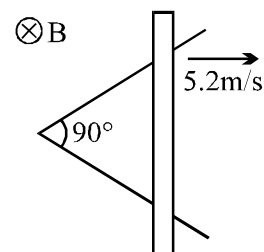
obtain the phase difference between the current and the voltage. Also plot the variation of current for one cycle on the given graph. [JEE 2004]



EM0026

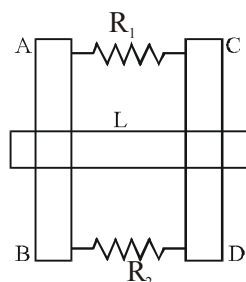
EXERCISE (S-2)

1. Two straight conducting rails form a right angle where their ends are joined. A conducting bar contact with the rails starts at vertex at the time $t = 0$ & moves symmetrically with a constant velocity of 5.2 m/s to the right as shown in figure. A 0.35 T magnetic field points out of the page. Calculate:
- The flux through the triangle by the rails & bar at $t = 3.0 \text{ s}$.
 - The emf around the triangle at that time.
 - In what manner does the emf around the triangle vary with time.



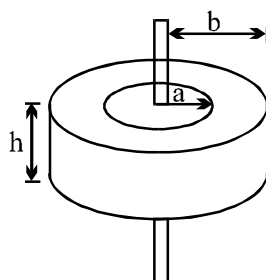
EM0027

2. Two parallel vertical metallic rails AB & CD are separated by 1 m . They are connected at the two ends by resistance R_1 & R_2 as shown in the figure. A horizontally metallic bar L of mass 0.2 kg slides without friction, vertically down the rails under the action of gravity. There is a uniform horizontal magnetic field of 0.6 T perpendicular to the plane of the rails, it is observed that when the terminal velocity is attained, the power dissipated in R_1 & R_2 are 0.76 W & 1.2 W respectively. Find the terminal velocity of bar L & value R_1 & R_2 .



EM0028

3. A long straight wire is arranged along the symmetry axis of a toroidal coil of rectangular cross-section, whose dimensions are given in the figure. The number of turns on the coil is N , and relative permeability of the surrounding medium is unity. Find the amplitude of the emf induced in this coil, if the current $i = i_m \cos \omega t$ flows along the straight wire.



EM0029

4. A metal rod of resistance 20Ω is fixed along a diameter of a conducting ring of radius 0.1 m and lies on x - y plane. There is a magnetic field $\vec{B} = (50\text{T})\hat{k}$. The ring rotates with an angular velocity $\omega = 20\text{ rad/sec}$ about its axis. An external resistance of 10Ω is connected across the centre of the ring and rim. Find the current through external resistance.

EM0030

5. A uniform but time varying magnetic field $B = Kt - C$; ($0 \leq t \leq C/K$), where K and C are constants and t is time, is applied perpendicular to the plane of the circular loop of radius ' a ' and resistance R . Find the total charge that will pass around the loop.

EM0031

6. A charged ring of mass $m = 50\text{ gm}$, charge 2 coulomb and radius $R = 2\text{ m}$ is placed on a smooth horizontal surface. A magnetic field varying with time at a rate of $(0.2\text{ t})\text{ Tesla/sec}$ is applied on to the ring in a direction normal to the surface of ring. Find the angular speed attained in a time $t_1 = 10\text{ sec}$. Assume that the magnetic field is cylindrically symmetric and covering the entire ring.

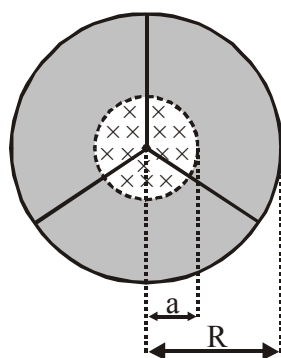
EM0032

7. A line charge λ per unit length is lodged uniformly onto the rim of a wheel of mass M and radius R . The wheel has light nonconducting spokes and is free to rotate without friction about its axis see figure. A uniform magnetic field extends over a circular region within the rim. It is given by,

$$\begin{aligned} \vec{B} &= -B_0\hat{k} & (r \leq a; a < R) \\ &= 0 & (\text{otherwise}) \end{aligned}$$

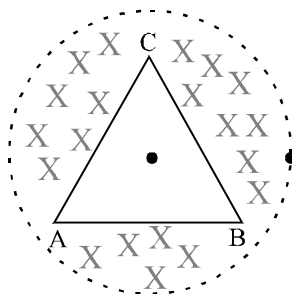
What is the angular velocity of the wheel after the field is suddenly switched off ?

(NCERT)



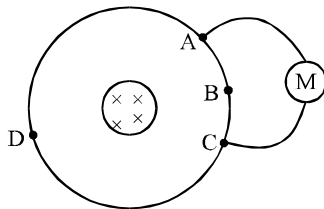
EM0033

8. A triangular wire frame (each side = 2 m) is placed in a region of time variant magnetic field having $dB/dt = \sqrt{3}\text{ T/s}$. The magnetic field is perpendicular to the plane of the triangle. The base of the triangle AB has a resistance 1Ω while the other two sides have resistance 2Ω each. The magnitude of potential difference between the points A and B will be



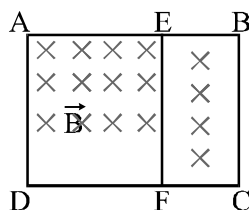
EM0034

9. A variable magnetic field creates a constant emf $\mathcal{E}$ in a conductor ABCDA. The resistances of portion ABC, CDA and AMC are R_1 , R_2 and R_3 respectively. What current will be shown by meter M? The magnetic field is concentrated near the axis of the circular conductor.



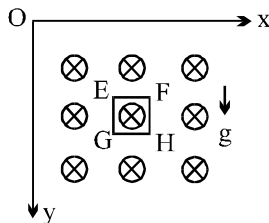
EM0035

10. A rectangular frame ABCD made of a uniform metal wire has a straight connection between E & F made of the same wire as shown in the figure. AEFD is a square of side 1 m & $EB = FC = 0.5$ m. The entire circuit is placed in a steadily increasing uniform magnetic field directed into the plane of the paper & normal to it. The rate of change of the magnetic field is 1 T/s , the resistance per unit length of the wire is $1 \Omega/\text{m}$. Find the current in segments AE, BE & EF.



EM0036

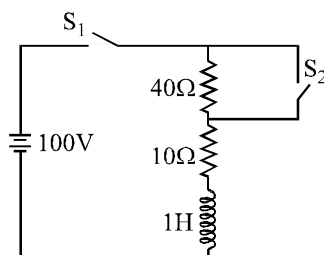
11. A magnetic field $\mathbf{B} = (B_0 y / a) \hat{k}$ is into the plane of paper in the $+z$ direction. B_0 and a are positive constants. A square loop EFGH of side a , mass m and resistance R , in x - y plane, starts falling under the influence of gravity. Note the directions of x and y axes in the figure. Find



- the induced current in the loop and indicate its direction,
- the total Lorentz force acting on the loop and indicate its direction,
- an expression for the speed of the loop, $v(t)$ and its terminal value.

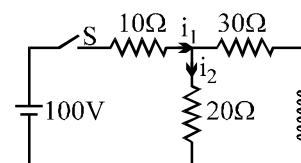
EM0037

12. In the circuit shown in the figure the switches S_1 and S_2 are closed at time $t = 0$. After time $t = (0.1) \ln 2$ sec, switch S_2 is opened. Find the current in the circuit at time $t = (0.2) \ln 2$ sec.



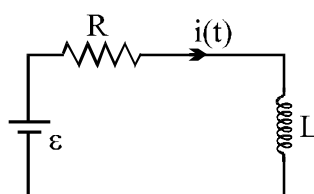
EM0038

13. Find the values of i_1 and i_2
- immediately after the switch S is closed.
 - long time later, with S closed.
 - immediately after S is open.
 - long time after S is opened.



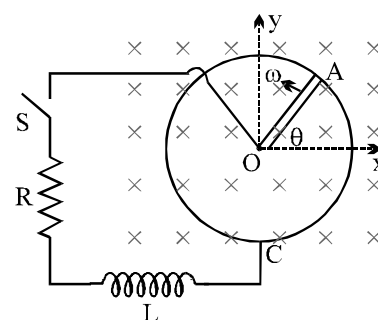
EM0039

14. Suppose the emf of the battery in the circuit shown varies with time t so the current is given by $i(t) = 3 + 5t$, where i is in amperes & t is in seconds. Take $R = 4\Omega$, $L = 6H$ & find an expression for the battery emf as function of time.



EM0040

15. A metal rod OA of mass m & length r is kept rotating with a constant angular speed ω in a vertical plane about a horizontal axis at the end O . The free end A is arranged to slide without friction along a fixed conducting circular ring in the same plane as that of rotation. A uniform & constant magnetic induction $\vec{B}$ is applied perpendicular & into the plane of rotation as shown in figure. An inductor L and an external resistance R are connected through a switch S between the point O & a point C on the ring to form an electrical circuit. Neglect the resistance of the ring and the rod. Initially, the switch is open.
- What is the induced emf across the terminals of the switch?
 - Obtain an expression for the current as a function of time after switch S is closed.
 - Obtain the time dependence of the torque required to maintain the constant angular speed, given that the rod OA was along the positive X -axis at $t = 0$.

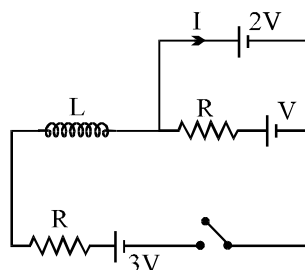


EM0041

16. A zero resistance coil of inductance L connects the upper ends of two vertical parallel long conductors. A horizontal sliding conductor, free to slide up and down, always maintaining contact with the vertical conductors, starts falling from rest at $t = 0$, due to its own weight mg . A uniform magnetic field of magnitude B exists in the region horizontally and perpendicular to the plane of the conductors. The distance between the vertical conductors is ' l '. After what time does the conductor come back to its starting position? Also find maximum speed achieved.

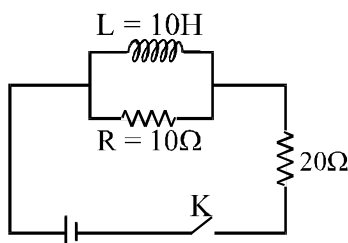
EM0042

17. In the LR circuit shown, what is the variation of the current I as a function of time? The switch is closed at time $t = 0$ sec.



EM0043

18. Two resistors of 10Ω and 20Ω and an ideal inductor of $10H$ are connected to a $2V$ battery as shown. The key K is shorted at time $t = 0$. Find the initial ($t = 0$) and final ($t \rightarrow \infty$) currents through battery.



EM0044

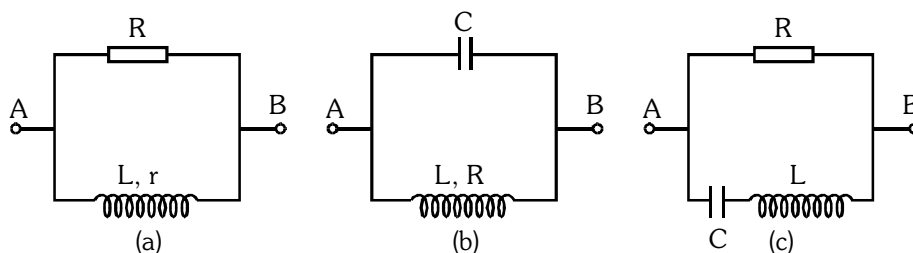
19. Two coils, 1 & 2, have a mutual inductance $= M$ and resistances R each. A current flows in coil 1, which varies with time as : $I_1 = kt^2$, where k is a constant and ' t ' is time. Find the total charge that has flown through coil 2, between $t = 0$ and $t = T$.

EM0045

20. A box P and a coil Q are connected in series with an ac source of variable frequency. The emf of source is $10 V$. Box P contains a capacitance of $1\mu F$ in series with a resistance of 32Ω while coil Q has a self-inductance $4.9 mH$ and a resistance of 68Ω series. The frequency is adjusted so that the maximum current flows in P and Q . Find the impedance of P and Q at this frequency. Also find the voltage across P and Q respectively.

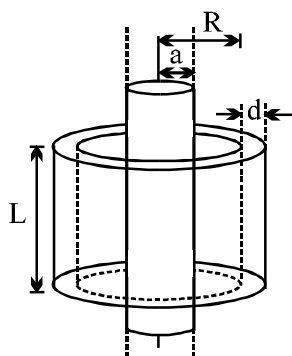
EM0046

21. Draw the approximate vector diagrams of currents in the circuits shown in Fig. The voltage applied across the points A and B is assumed to be sinusoidal; the parameters of each circuit are so chosen that the total current I_0 lags in phase behind the external voltage by an angle ϕ .



EM0047

22. A long solenoid of radius a and number of turns per unit length n is enclosed by cylindrical shell of radius R , thickness d ($d \ll R$) and length L . A variable current $i = i_0 \sin \omega t$ flows through the coil. If the resistivity of the material of cylindrical shell is ρ , find the induced current in the shell. [JEE 2005]



EM0048

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Faraday's law & motional emf.

1. A square of side 2 meters lies in the x - y plane in a region, where the magnetic field is given by $\vec{B} = B_0 (2\hat{i} + 3\hat{j} + 4\hat{k})T$, where B_0 is constant. The magnitude of flux passing through the square is:-

(A) $8 B_0 \text{ Wb}$. (B) $12 B_0 \text{ Wb}$. (C) $16 B_0 \text{ Wb}$. (D) $\sqrt{4 \times 29} B_0 \text{ Wb}$

EM0049

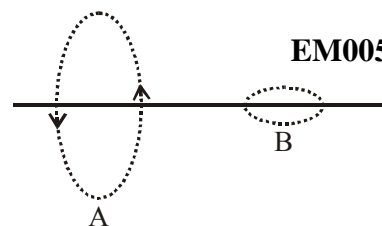
2. **Statement-1** : When a magnet is made to fall freely through a closed coil, its acceleration is always less than acceleration due to gravity.

and

Statement-2 : Current induced in the coil opposes the motion of the magnet, as per Lenz's law.

- (A) Statement-1 is true, Statement-2 is true ; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is true, Statement-2 is true, Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True

3. In the given figure the centre of a small conducting circular loop B lies on the axis of bigger circular loop A and their axis are mutually perpendicular. An anticlockwise (when viewed from the side of B) current in the loop A start increasing then :-

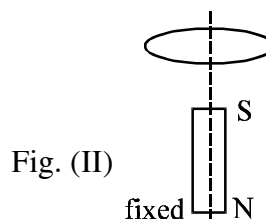
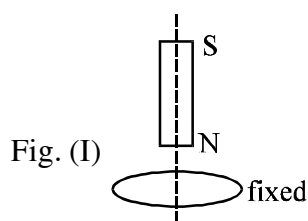


EM0050

- (A) current induced in the loop B is in clockwise direction (when viewed from above the B)
 (B) current induced in the loop B is in anti-clockwise direction (when viewed from above the B)
 (C) current must induced in the loop B but its direction can not be predicted
 (D) no current is induced in the loop B

EM0051

4. A vertical bar magnet is dropped from position on the axis of a fixed metallic coil as shown in figure-I. In figure-II the magnet is fixed and horizontal coil is dropped. The acceleration of the magnet and coil are a_1 and a_2 respectively then :-



- (A) $a_1 > g$, $a_2 > g$
 (C) $a_1 < g$, $a_2 < g$

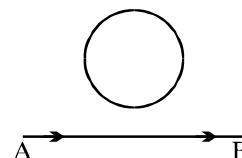
- (B) $a_1 > g$, $a_2 < g$
 (D) $a_1 < g$, $a_2 > g$

EM0052

5. Two identical coaxial circular loops carry a current i each circulating in the same direction. If the loops approach each other
- the current in each will decrease
 - the current in each will increase
 - the current in each will remain the same
 - the current in one will increase and in other will decrease

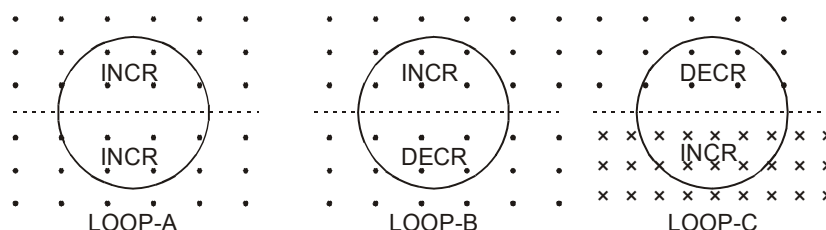
EM0053

6. In the arrangement shown in given figure current from A to B is increasing in magnitude. Induced current in the loop will
- have clockwise direction
 - have anticlockwise direction
 - be zero
 - oscillate between clockwise and anticlockwise



EM0054

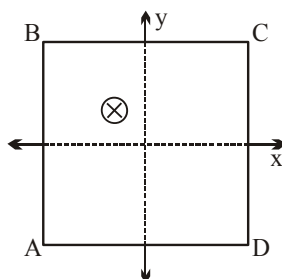
7. Three identical conducting circular loops are placed in uniform magnetic fields. Inside each loop, there are two magnetic field regions, separated by dashed line that coincides with a diameter, as shown. Magnetic fields may either be increasing (marked as INCR) or decreasing (marked as DECR) in magnitude at the same rates. If I_A , I_B and I_C are the magnitudes of the induced currents in the loops A, B and C respectively then choose the **CORRECT** relation :-



- (A) $I_A > I_B = I_C$ (B) $I_A = I_C > I_B$ (C) $I_A = I_B = I_C$ (D) $I_C > I_A > I_B$

EM0055

8. A square coil ABCD is placed in x-y plane with its centre at origin. A long straight wire, passing through origin, carries a current in negative z-direction. Current in this wire increases with time. The induced current in the coil is :



- (A) clockwise (B) anticlockwise (C) zero (D) alternating

EM0056

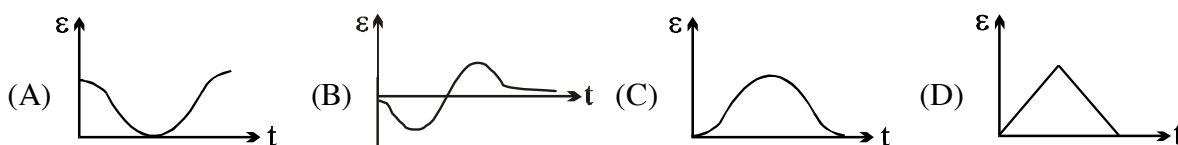
9. A short circuited coil is kept on the ground and a magnet is dropped on it as shown. The coil shows (when viewed from top)

- (A) anticlockwise current that increases in magnitude
 (B) anticlockwise current that remains constant
 (C) clockwise current that remains constant
 (D) clockwise current that increases in magnitude



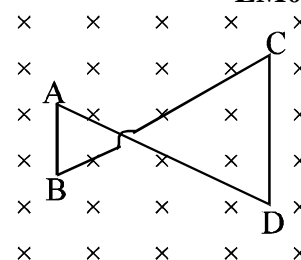
EM0057

10. The variation of induced emf (ϵ) with time (t) in a coil if a short bar magnet is moved along its axis with a constant velocity is best represented as [JEE 2004(Scr)]



11. A conducting wire frame is placed in a magnetic field which is directed into the paper. The magnetic field is increasing at a constant rate. The directions of induced currents in wires AB and CD are

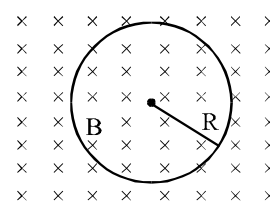
- (A) B to A and D to C
 (B) A to B and C to D
 (C) A to B and D to C
 (D) B to A and C to D



EM0058

12. A conducting loop of radius R is present in a uniform magnetic field B perpendicular to the plane of the ring. If radius R varies as a function of time ' t ', as $R = R_0 + t$. The e.m.f induced in the loop is

- (A) $2\pi(R_0 + t)B$ clockwise
 (B) $\pi(R_0 + t)B$ clockwise
 (C) $2\pi(R_0 + t)B$ anticlockwise
 (D) zero



EM0059

13. A thin wire of length $2m$ is perpendicular to the xy plane. It is moved with velocity $\vec{v} = (2\hat{i} + 3\hat{j} + \hat{k}) \text{ m/s}$ through a region of magnetic induction $\vec{B} = (\hat{i} + 2\hat{j}) \text{ Wb/m}^2$. Then potential difference induced between the ends of the wire :

- (A) 2 volts (B) 4 volts (C) 0 volts (D) none of these

EM0060

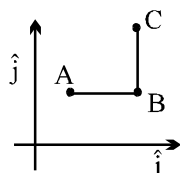
EM0061

14. A square loop of side a and resistance R is moved in the region of uniform magnetic field B (loop remaining completely inside field), with a velocity v through a distance x . The work done is :

(A) $\frac{B\ell^2 vx}{R}$ (B) $\frac{2B^2 \ell^2 vx}{R}$ (C) $\frac{4B^2 \ell^2 vx}{R}$ (D) 0

EM0062

15. There is a uniform magnetic field B normal to the xy plane. A conductor ABC has length $AB = l_1$, parallel to the x -axis, and length $BC = l_2$, parallel to the y -axis. ABC moves in the xy plane with velocity $v_x \hat{i} + v_y \hat{j}$. The potential difference between A and C is proportional to :-



(A) $v_x l_1 + v_y l_2$ (B) $v_x l_2 + v_y l_1$ (C) $v_x l_2 - v_y l_1$ (D) $v_x l_1 - v_y l_2$

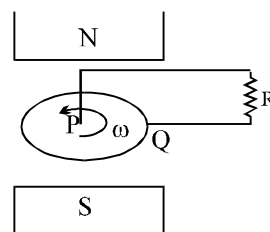
EM0063

16. A uniform but time variant magnetic field exists in a cylindrical region directed along the axis of cylinder of radius R . The graph of induced electric field at a given time v/s . r (r = distance from axis)



EM0064

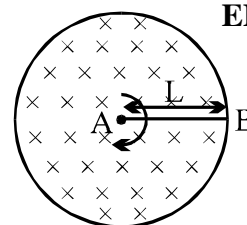
17. A metal disc rotates freely, between the poles of a magnet in the direction indicated. Brushes P and Q make contact with the edge of the disc and the metal axle. What current, if any, flows through R ?



- (A) a current from P to Q
 (B) a current from Q to P
 (C) no current, because the emf in the disc is opposed by the back emf
 (D) no current, because the emf induced in one side of the disc is opposed by the emf induced in the other side.
 (E) no current, because no radial emf is induced in the disc

EM0065

18. A copper rod AB of length L , pivoted at one end A , rotates at constant angular velocity ω , at right angles to a uniform magnetic field of induction B . The e.m.f developed between the mid point C of the rod and end B is



(A) $\frac{B\omega L^2}{4}$ (B) $\frac{B\omega L^2}{2}$ (C) $\frac{3B\omega L^2}{4}$ (D) $\frac{3B\omega L^2}{8}$

EM0066

- 19.** The e.m.f. induced in a coil of wire, which is rotating in a magnetic field, does not depend on
- (A) the angular speed of rotation (B) the area of the coil
(C) the number of turns on the coil (D) the resistance of the coil

EM0067

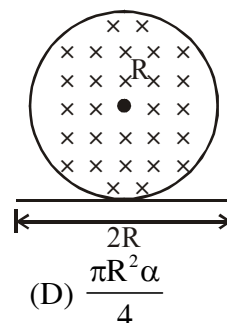
Induced electric field

- 20.** A ring of resistance 10Ω , radius 10cm and 100 turns is rotated at a rate 100 revolutions per second about its diameter is perpendicular to a uniform magnetic field of induction 10mT. The amplitude of the current in the loop will be nearly (Take : $\pi^2 = 10$)

- (A) 200A (B) 2A (C) 0.002A (D) none of these

EM0068

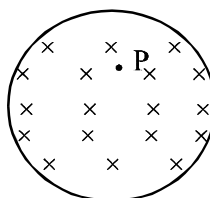
- 21.** A uniform but time varying magnetic field is present in a circular region of radius R . The magnetic field is perpendicular and into the plane of the loop and the magnitude of field is increasing at a constant rate α . There is a straight conducting rod of length $2R$ placed as shown in figure. The magnitude of induced emf across the rod is



- (A) $\pi R^2 \alpha$ (B) $\frac{\pi R^2 \alpha}{2}$ (C) $\frac{R^2 \alpha}{\sqrt{2}}$

EM0069

22. Figure shows a uniform magnetic field B confined to a cylindrical volume and is increasing at a constant rate. The instantaneous acceleration experienced by an electron placed at P is



- (A) zero (B) towards right (C) towards left (D) upwards

EM0070

- 23. Statement-1 :** For a charged particle moving from point P to point Q the net work done by an induced electric field on the particle is independent of the path connecting point P to point Q.

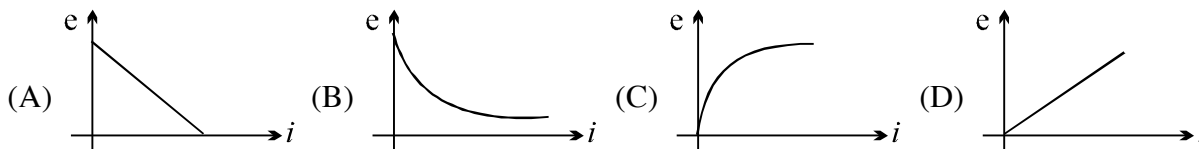
Statement-2 : The net work done by a conservative force on an object moving along closed loop is zero.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
- (C) Statement-1 is true, statement-2 is false.
- (D) Statement-1 is false, statement-2 is true.

EM0071

Inductance

24. In an L-R circuit connected to a battery of constant e.m.f. E switch S is closed at time $t = 0$. If e denotes the magnitude of induced e.m.f. across inductor and i the current in the circuit at any time t . Then which of the following graphs shows the variation of e with i ?



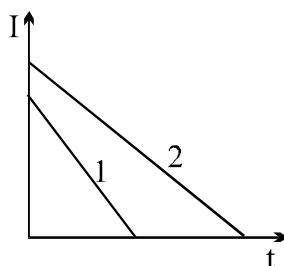
EM0072

25. A current of 2A is increasing at a rate of 4 A/s through a coil of inductance 2H . The energy stored in the inductor per unit time is :-

- (A) 2 J/s (B) 1 J/s (C) 16 J/s (D) 4 J/s

EM0073

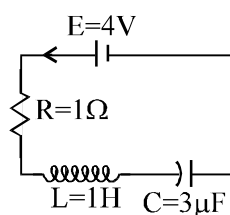
26. Two identical inductance carry currents that vary with time according to linear laws (as shown in figure). In which of two inductance is the self induction emf greater?



- (A) 1 (B) 2
(C) same (D) data are insufficient to decide

EM0074

27. The current in the given circuit is increasing with a rate $a = 4\text{ amp/s}$. The charge on the capacitor at an instant when the current in the circuit is 2 amp will be :



- (A) $4\mu\text{C}$ (B) $5\mu\text{C}$ (C) $6\mu\text{C}$ (D) none of these

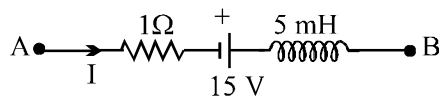
EM0075

28. A long solenoid of N turns has a self inductance L and area of cross section A . When a current i flows through the solenoid, the magnetic field inside it has magnitude B . The current i is equal to :

- (A) BAN/L (B) $BANL$ (C) BN/AL (D) B/ANL

EM0076

29. The network shown in the figure is part of a complete circuit. If at a certain instant, the current I is 5 A and it is decreasing at a rate of 10^3 As^{-1} then $V_B - V_A$ equals



- (A) 20 V (B) 15 V (C) 10 V (D) 5 V

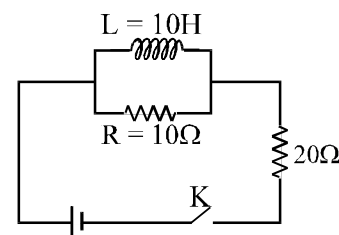
EM0077

30. In Problem 29, if I is reversed in direction, then $V_B - V_A$ equals

- (A) 5 V (B) 10 V (C) 15 V (D) 20 V

EM0078

31. Two resistors of 10Ω and 20Ω and an ideal inductor of 10H are connected to a 2V battery as shown. The key K is shorted at time $t = 0$. Find the initial ($t = 0$) and final ($t \rightarrow \infty$) currents through battery.



- (A) $\frac{1}{15} \text{ A}, \frac{1}{10} \text{ A}$ (B) $\frac{1}{10} \text{ A}, \frac{1}{15} \text{ A}$
(C) $\frac{2}{15} \text{ A}, \frac{1}{10} \text{ A}$ (D) $\frac{1}{15} \text{ A}, \frac{2}{25} \text{ A}$

EM0079

32. An inductor coil stores U energy when i current is passed through it and dissipates energy at the rate of P . The time constant of the circuit, when this coil is connected across a battery of zero internal resistance is :-

- (A) $\frac{4U}{P}$ (B) $\frac{U}{P}$ (C) $\frac{2U}{P}$ (D) $\frac{2P}{U}$

EM0080

33. A small coil of radius r is placed at the centre of a large coil of radius R , where $R \gg r$. The coils are coplanar. The coefficient of mutual inductance between the coils is :-

- (A) $\frac{\mu_0 \pi r^2}{2R}$ (B) $\frac{\mu_0 \pi r^2}{2R}$ (C) $\frac{\mu_0 \pi r^2}{2R^2}$ (D) $\frac{\mu_0 \pi r}{2R^2}$

EM0081

34. A long straight wire is placed along the axis of a circular ring of radius R . The mutual inductance of this system is :-

- (A) $\frac{\mu_0 R}{2}$ (B) $\frac{\mu_0 \pi R}{2}$ (C) $\frac{\mu_0}{2}$ (D) 0

EM0082

35. Two coils are placed close to each other. The mutual inductance of the pair of coils depends upon-

- (A) the rates at which currents are changing in the two coils
(B) relative position and orientation of the two coils
(C) the materials of the wires of the coils
(D) the currents in the two coils

[AIEEE - 2003]

EM0083

36. A small square loop of wire of side l is placed inside a large square loop of wire of side L ($L \gg l$). The loop are coplanar & their centres coincide. The mutual inductance of the system is proportional to :

(A) $\frac{l}{L}$ (B) $\frac{l^2}{L}$ (C) $\frac{L}{l}$ (D) $\frac{L^2}{l}$

EM0084

37. L , C and R represent physical quantities inductance, capacitance and resistance. The combination which has the dimensions of frequency is

(A) $\frac{1}{RC}$ and $\frac{R}{L}$ (B) $\frac{1}{\sqrt{RC}}$ and $\sqrt{\frac{R}{L}}$ (C) $\sqrt{LC}$ (D) $\frac{C}{L}$

EM0085

38. An ideal coil of 10 H is connected in series with a resistance of 5Ω and a battery of 5 V . 2 s after the connection is made, the current flowing (in ampere) in the circuit is :-

(A) $(1 - e)$ (B) e (C) e^{-1} (D) $(1 - e^{-1})$

EM0086

39. A coil of inductance 5 H is joined to a cell of emf 6 V through a resistance 10Ω at time $t = 0$. The emf across the coil at time $t = \ln \sqrt{2}\text{ s}$ is :

(A) 3 V (B) 1.5 V (C) 0.75 V (D) 4.5 V

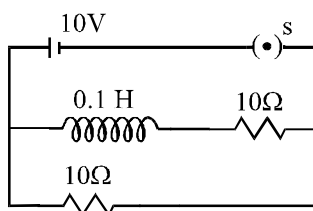
EM0087

40. For L-R circuit, the time constant is equal to

- (A) twice the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance
(B) ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance
(C) half the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance
(D) square of the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance

EM0088

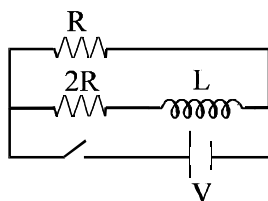
41. In the adjoining circuit, initially the switch S is open. The switch ' S ' is closed at $t = 0$. The difference between the maximum and minimum current that can flow in the circuit is



- (A) 2 Amp
(B) 3 Amp
(C) 1 Amp
(D) nothing can be concluded

EM0089

42. Find the ratio of time constant in build up and decay in the circuit as shown in figure :-



- (A) 1 : 1 (B) 3 : 2 (C) 2 : 3 (D) 1 : 3

EM0090

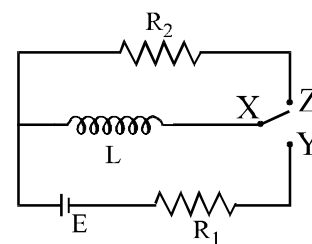
43. In the circuit shown, X is joined to Y for a long time, and then X is joined to Z. The total heat produced in R_2 is :

(A) $\frac{LE^2}{2R_1^2}$

(B) $\frac{LE^2}{2R_2^2}$

(C) $\frac{LE^2}{2R_1R_2}$

(D) $\frac{LE^2R_2}{2R_1^2}$



EM0091

44. In a L-R decay circuit, the initial current at $t = 0$ is I . The total charge that has flown through the resistor till the energy in the inductor has reduced to one-fourth its initial value, is

(A) LI/R

(B) $LI/2R$

(C) $LI\sqrt{2}/R$

(D) None

EM0092

45. The inductor in a L-C oscillation has a maximum potential difference of 16 V and maximum energy of 640 μJ . Find the value of capacitor in μF in L-C circuit.

(A) 5

(B) 4

(C) 3

(D) 2

EM0093

46. A condenser of capacity 6 μF is fully charged using a 6-volt battery. The battery is removed and a resistanceless 0.2 mH inductor is connected across the condenser. The current which is flowing through the inductor when one-third of the total energy is in the magnetic field of the inductor is :-

(A) 0.1 A

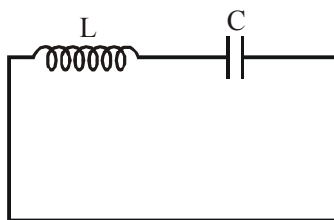
(B) 0.2 A

(C) 0.4 A

(D) 0.6 A

EM0094

47. In an LC circuit the capacitor has maximum charge q_0 . The value of $\left(\frac{dI}{dt}\right)_{\max}$ is :-



(A) $\frac{q_0}{LC}$

(B) $\frac{q_0}{\sqrt{LC}}$

(C) $\frac{q_0}{2LC}$

(D) $\frac{2q_0}{LC}$

EM0095

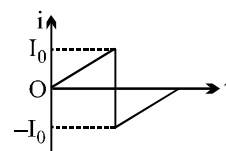
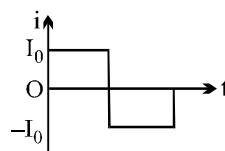
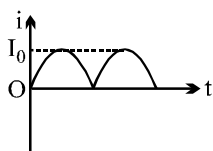
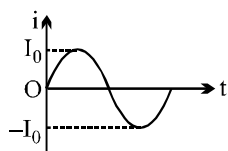
Alternating current

48. When 100 V DC is applied across a solenoid a current of 1 A flows in it. When 100 V AC is applied across the same coil, the current drops to 0.5 A. If the frequency of the AC source is 50 Hz, the impedance and inductance of the solenoid are:

(A) 100Ω , 0.93 H (B) 200Ω , 1.0 H (C) 10Ω , 0.86H (D) 200Ω , 0.55 H

EM0096

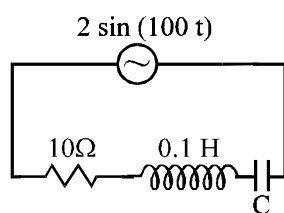
49. If I_1 , I_2 , I_3 and I_4 are the respective r.m.s. values of the time varying currents as shown in the four cases I, II, III and IV. Then identify the correct relations.



(A) $I_1 = I_2 = I_3 = I_4$ (B) $I_3 > I_1 = I_2 > I_4$ (C) $I_3 > I_4 > I_2 = I_1$ (D) $I_3 > I_2 > I_1 > I_4$

EM0097

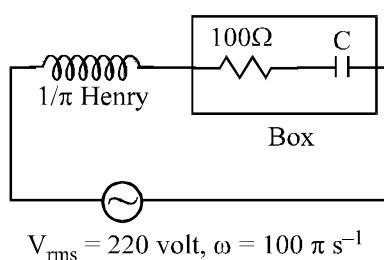
50. The power factor of the circuit is $1/\sqrt{2}$. The capacitance of the circuit is equal to



(A) $400\ \mu\text{F}$ (B) $300\ \mu\text{F}$ (C) $500\ \mu\text{F}$ (D) $200\ \mu\text{F}$

EM0098

51. In the circuit, as shown in the figure, if the value of R.M.S current is 2.2 ampere, the power factor of the box is



(A) $\frac{1}{\sqrt{2}}$ (B) 1 (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$

EM0099

52. The power in ac circuit is given by $P = E_{\text{rms}} I_{\text{rms}} \cos\phi$. The value of $\cos\phi$ in series LCR circuit at resonance is:

(A) zero (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{\sqrt{2}}$

EM0100

53. In ac circuit when ac ammeter is connected it reads i current. If a student uses dc ammeter in place of ac ammeter the reading in the dc ammeter will be:

(A) $\frac{i}{\sqrt{2}}$ (B) $\sqrt{2} i$ (C) $0.637 i$ (D) zero

EM0101

54. The phase difference between current and voltage in an AC circuit is $\pi/4$ radian. If the frequency of AC is 50 Hz, then the phase difference is equivalent to the time difference :

(A) 0.78 s (B) 15.7 ms (C) 0.25 s (D) 2.5 ms

EM0102

55. The effective value of current $i = 2 \sin 100 \pi t + 2 \sin(100 \pi t + 30^\circ)$ is :

(A) $\sqrt{2}$ Amp (B) $2\sqrt{2+\sqrt{3}}$ Amp (C) 4 Amp (D) None

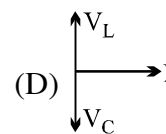
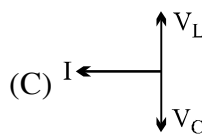
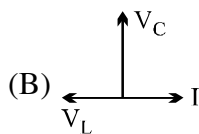
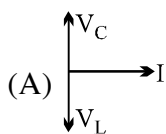
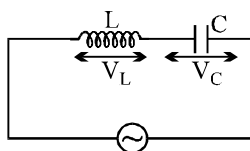
EM0103

56. In series LR circuit $X_L = 3R$. Now a capacitor with $X_C = R$ is added in series. Ratio of new to old power factor is

(A) 1 (B) 2 (C) $\frac{1}{\sqrt{2}}$ (D) $\sqrt{2}$

EM0104

57. The current I , potential difference V_L across the inductor and potential difference V_C across the capacitor in circuit as shown in the figure are best represented vectorially as



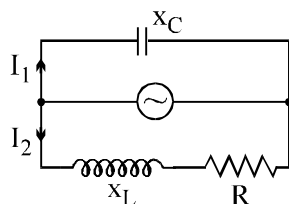
EM0105

58. In a series R-L-C circuit, the frequency of the source is half of the resonance frequency. The nature of the circuit will be

(A) capacitive (B) inductive (C) purely resistive (D) data insufficient

EM0106

59. In the shown AC circuit phase difference between currents I_1 and I_2 is



- (A) $\frac{\pi}{2} - \tan^{-1} \frac{X_L}{R}$ (B) $\tan^{-1} \frac{X_L - X_C}{R}$ (C) $\frac{\pi}{2} + \tan^{-1} \frac{X_L}{R}$ (D) $\tan^{-1} \frac{X_L - X_C}{R} + \frac{\pi}{2}$

EM0107

60. In a transformer, number of turns in the primary are 140 and that in the secondary are 280. If current in primary is 4A, then that in the secondary is

- (A) 4A (B) 2A (C) 6A (D) 10 A

EM0108

61. The primary of a 1 : 3 step - up transformer is connected to a source and the secondary is connected to a resistor R. The power dissipated by R in this situation is P. If R is connected directly to the source it will dissipate a power of :

- (A) P/9 (B) P/3 (C) P (D) 3P

EM0109

62. An ideal efficient transformer has a primary power input of 10kW. The secondary current when the transformer is on load is 25A. If the primary : secondary turns ratio is 8 : 1, then the potential difference applied to the primary coil is

- (A) $\frac{10^4 \times 8^2}{25}$ V (B) $\frac{10^4 \times 8}{25}$ V (C) $\frac{10^4}{25 \times 8}$ V (D) $\frac{10^4}{25 \times 8^2}$ V

EM0110

63. The core of any transformer is laminated so as to -

- (A) Make it light weight (B) Make it robust and strong
(C) Increase the secondary voltage (D) Reduce the energy loss due to eddy current

EM0111

64. If the difference between the equivalent inductance in the following figures is nL then find the value of n . Given coupling coefficient is $C = \sqrt{2}$ (Where coupling coefficient is defined as

$$C = \frac{\sqrt{L_1 L_2}}{M})$$

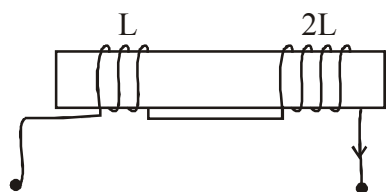


Figure (A)

(A) 2

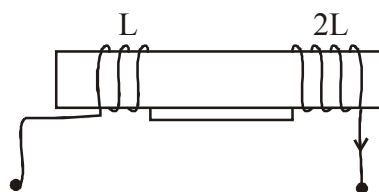


Figure (B)

(C) 4

(D) 5

EM0112

SUPPLEMENT FOR JEE-MAINS

65. Five particles undergo damped harmonic motion. Values for the spring constant k , the damping constant b , and the mass m are given below. Which leads to the smallest rate of loss of mechanical energy at the initial moment?

(A) $k = 100\text{N/m}$, $m = 50\text{g}$, $b = 8\text{g/s}$ (B) $k = 150\text{N/m}$, $m = 50\text{g}$, $b = 5\text{g/s}$
 (C) $k = 150\text{N/m}$, $m = 10\text{g}$, $b = 8\text{g/s}$ (D) $k = 200\text{N/m}$, $m = 8\text{g}$, $b = 6\text{g/s}$

EM0113

66. An RLC circuit has a capacitance of $12\mu\text{F}$, an inductance of 25mH , and a resistance of 60Ω . The current oscillates with an angular frequency of:

(A) $1.2 \times 10^3 \text{ rad/s}$ (B) $1.4 \times 10^3 \text{ rad/s}$ (C) $1.8 \times 10^3 \text{ rad/s}$ (D) $2.2 \times 10^3 \text{ rad/s}$
 (E) $2.6 \times 10^3 \text{ rad/s}$

EM0114

67. An RLC circuit has an inductance of 25mH and a capacitance of $5.0\mu\text{F}$. The charge on the capacitor does NOT oscillate but rather decays exponentially to zero. The resistance in the circuit must be:

(A) greater than or equal to $100\sqrt{2}\Omega$ (B) less than $100\sqrt{2}\Omega$ but greater than $50\sqrt{2}\Omega$
 (C) less than $50\sqrt{2}\Omega$ but greater than $25\sqrt{2}\Omega$ (D) less than $25\sqrt{2}\Omega$ but greater than 0

EM0115

68. Two underdamped oscillators are known to have the same natural frequency ω_0 . The mass and damping coefficient of the first oscillator are m_1 and b_1 , and the mass and damping coefficient of the second oscillator are m_2 and b_2 , respectively. A sinusoidal driving force of $F_{\text{ext}} = F_0 \cos \omega t$ is applied to each oscillator. Starting with ω far from ω_0 the driving force is tuned in order to observe resonant behavior. If $m_1 = 4m_2$ and $b_1 = 2b_2$, then which one of the following statements concerning the driven oscillations is correct?

(A) The resonant peak of the first driven oscillator is higher and narrower than that of the second oscillator.
 (B) The resonant peak of the first driven oscillator is higher and wider than that of the second oscillator.
 (C) The resonant peak of the first driven oscillator is lower and wider than that of the second oscillator.
 (D) The resonant peak of the first driven oscillator is lower and narrower than that of the second oscillator.

EM0116

69. A simple pendulum has a time period T if there is no air resistance. If a small air resistance is acting on the bob as it oscillates,

(A) The time period will be initially more than T and decreases with time.
 (B) The time period will be less than T initially and increases with time
 (C) The time period will be less than T and remains constant
 (D) The time period will be more than T and remains constant.

EM0117

70. A block is executing damped harmonic oscillation with time period T . Choose correct statement

(1) Time taken to go from extreme to mean position is $\frac{T}{4}$

(2) Time taken to go from one extreme to another is $\frac{T}{2}$

(3) Time taken to go from one extreme to another is less than $\frac{T}{2}$

(4) Time taken to go from one extreme to another is more than $\frac{T}{2}$

(A) 1,2 only

(B) 1,2,3 only

(C) 2 only

(D) 1,2,4 only

EM0118

71. The angular frequency of the damped oscillator is given by, $\omega = \sqrt{\frac{k}{m} - \frac{r^2}{4m^2}}$ where k is the spring

constant, m is the mass of the oscillator and r is damping constant. If the ratio $\frac{r^2}{mk}$ is 8%, the

change in time period compared to the undamped oscillator is approximately as follows:

(A) decreases by 8%

(B) decreases by 1%

(C) increases by 1%

(D) increases by 8%

EM0119

72. Two spheres of the same diameter but of different masses are suspended by strings of equal length. If the spheres are deflected from their positions of equilibrium, which of the two will have a greater oscillation period and which will have a greater logarithmic decrement if their oscillations occur in a real medium with viscosity?

(A) Heavier mass has larger time period & greater logarithmic decrement

(B) Lighter mass has larger time period & greater logarithmic decrement

(C) Lighter mass has larger time period but lesser logarithmic decrement

(D) Heavier mass has larger time period but lesser logarithmic decrement

EM0120

73. The amplitude of a simple pendulum, oscillating in air with a small spherical bob, decreases from 10 cm to 8 cm in 40 seconds. Assuming that Stokes law is valid, and ratio of the coefficient of viscosity of air to that of carbon dioxide is 1.3, the time in which amplitude of this pendulum will reduce from 10 cm to 5 cm in carbon dioxide will be close to ($\ln 5 = 1.601$, $\ln 2 = 0.693$).

(A) 231 s

(B) 208 s

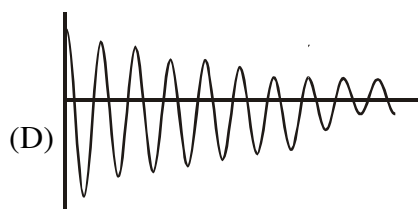
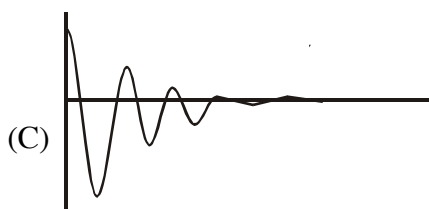
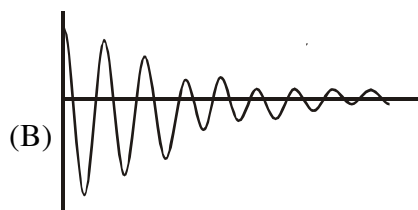
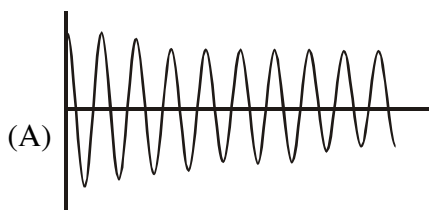
(C) 142 s

[JEE Main Online 2014]

(D) 161 s

EM0121

74. Which graph has the highest Q factor ?



EM0122

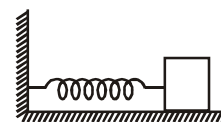
75. In the situation shown, the block can execute free oscillation (no damping) with angular frequency ω_1 . In presence of weak damping, it executes damped SHM with angular frequency ω_2 . When it is subjected to a sinusoidal force, it executes forced oscillation with maximum amplitude at angular frequency ω_3 (assume damping is present) :-

(A) $\omega_1 > \omega_2 > \omega_3$

(B) $\omega_1 > \omega_2 = \omega_3$

(C) $\omega_1 = \omega_2 = \omega_3$

(D) $\omega_1 > \omega_3 > \omega_2$



EM0123

76. In forced oscillation of a particle the amplitude is maximum for a frequency ω_1 of the force, while the energy is maximum for a frequency ω_2 of the force; then –

[AIEEE- 2004]

(A) $\omega_1 = \omega_2$

(B) $\omega_1 > \omega_2$

(C) $\omega_1 < \omega_2$ when damping is small and $\omega_1 > \omega_2$ when damping is large

(D) $\omega_1 < \omega_2$

EM0124

77. A pendulum with time period of 1s is losing energy due to damping. At certain time its energy is 45 J. If after completing 15 oscillations, its energy has become 15 J, its damping constant (in s^{-1}) is:-

[JEE Mains On line 2015]

(A) 2

(B) $\frac{1}{15} \ln 3$

(C) $\frac{1}{2}$

(D) $\frac{1}{30} \ln 3$

EM0125

MULTIPLE CORRECT TYPE QUESTIONS

78. The dimension of the ratio of magnetic flux and the resistance is equal to that of :

(A) induced emf

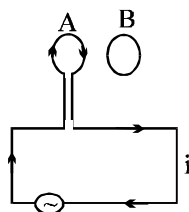
(B) charge

(C) inductance

(D) current

EM0126

79. Two circular coils A and B are facing each other as shown in figure. The current i through A can be altered



- (A) there will be repulsion between A and B if i is increased
 (B) there will be attraction between A and B if i is increased
 (C) there will be neither attraction nor repulsion when i is changed
 (D) attraction or repulsion between A and B depends on the direction of current. It does not depend whether the current is increased or decreased.

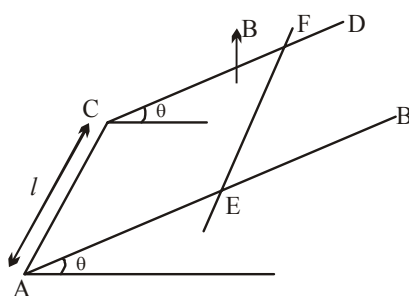
EM0127

80. A bar magnet is moved along the axis of copper ring placed far away from the magnet. Looking from the side of the magnet, an anticlockwise current is found to be induced in the ring. Which of the following may be true?

- (A) The south pole faces the ring and the magnet moves towards it.
 (B) The north pole faces the ring and the magnet moves towards it.
 (C) The south pole faces the ring and the magnet moves away from it.
 (D) The north pole faces the ring and the magnet moves away from it.

EM0128

81. AB and CD are smooth parallel rails, separated by a distance l , and inclined to the horizontal at an angle θ . A uniform magnetic field of magnitude B , directed vertically upwards, exists in the region. EF is a conductor of mass m , carrying a current i . For EF to be in equilibrium,



- (A) i must flow from E to F
 (B) $Bil = mg \tan \theta$
 (C) $Bil = mg \sin \theta$
 (D) $Bil = mg$

EM0129

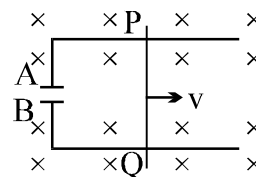
82. In the **previous question**, if B is normal to the plane of the rails

- (A) $Bil = mg \tan \theta$
 (B) $Bil = mg \sin \theta$
 (C) $Bil = mg \cos \theta$
 (D) equilibrium cannot be reached

EM0130

83. A conducting rod PQ of length $L = 1.0$ m is moving with a uniform speed $v = 20$ m/s in a uniform magnetic field $B = 4.0$ T directed into the paper. A capacitor of capacity $C = 10 \mu\text{F}$ is connected as shown in figure. Then

- (A) $q_A = +800 \mu\text{C}$ and $q_B = -800 \mu\text{C}$
 (B) $q_A = -800 \mu\text{C}$ and $q_B = +800 \mu\text{C}$
 (C) $q_A = 0 = q_B$
 (D) charge stored in the capacitor increases exponentially with time



EM0131

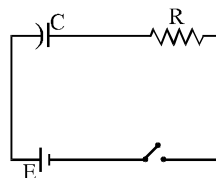
84. An LR circuit with a battery is connected at $t = 0$. Which of the following quantities is not zero just after the circuit

- (A) current in the circuit
 (B) magnetic field energy in the inductor
 (C) power delivered by the battery
 (D) emf induced in the inductor

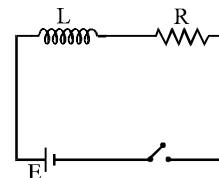
EM0132

85. The switches in figures (a) and (b) are closed at $t = 0$

- (A) The charge on C just after $t = 0$ is EC .
 (B) The charge on C long after $t = 0$ is EC .
 (C) The current in L just after $t = 0$ is E/R .
 (D) The current in L long after $t = 0$ is E/R .



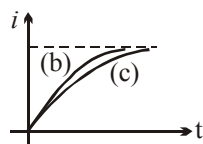
(a)



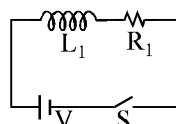
(b)

EM0133

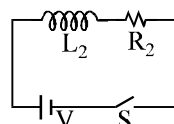
86. Current growth in two L-R circuits (b) and (c) as shown in figure (a). Let L_1 , L_2 , R_1 and R_2 be the corresponding values in two circuits. Then :-



(a)



(b)



(c)

- (A) $R_1 > R_2$

- (B) $R_1 = R_2$

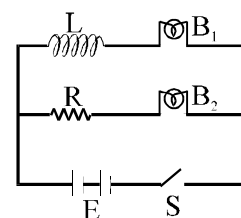
- (C) $L_1 > L_2$

- (D) $L_1 < L_2$

EM0134

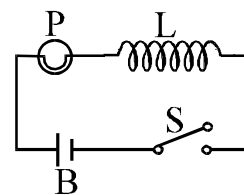
87. An inductor L , a resistance R and two identical bulbs B_1 and B_2 are connected to a battery through a switch S as shown in the figure. The resistance of coil having inductance L is also R . Which of the following statement gives the correct description of the happenings when the switch S is closed?

- (A) The bulb B_2 lights up earlier than B_1 and finally both the bulbs shine equally bright.
 (B) B_1 lights up earlier and finally both the bulbs acquire equal brightness.
 (C) B_2 lights up earlier and finally B_1 shines brighter than B_2 .
 (D) B_1 and B_2 light up together with equal brightness all the time.



EM0135

88. In figure, a lamp P is in series with an iron-core inductor L. When the switch S is closed, the brightness of the lamp rises relatively slowly to its full brightness than it would do without the inductor. This is due to



- (A) the low resistance of P
(B) the induced-emf in L
(C) the low resistance of L
(D) the high voltage of the battery B

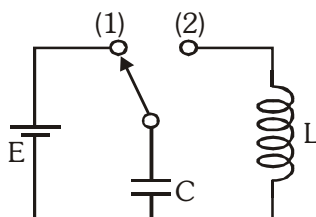
EM0136

89. Two different coils have self inductance 8mH and 2mH. The current in one coil is increased at a constant rate. The current in the second coil is also increased at the same constant rate. At a certain instant of time, the power given to the two coils is the same. At that time the current, the induced voltage and the energy stored in the first coil are I_1 , V_1 and W_1 respectively. Corresponding values for the second coil at the same instant are I_2 , V_2 and W_2 respectively. Then:

- (A) $\frac{I_1}{I_2} = \frac{1}{4}$ (B) $\frac{I_1}{I_2} = 4$ (C) $\frac{W_2}{W_1} = 4$ (D) $\frac{V_2}{V_1} = \frac{1}{4}$

EM0137

90. Initially key was placed on (1) till the capacitor got fully charged. Key is placed on (2) at $t = 0$. The time when the energy in both capacitor and inductor will be same-



- (A) $\frac{\pi\sqrt{LC}}{4}$ (B) $\frac{\pi\sqrt{LC}}{2}$ (C) $\frac{5\pi\sqrt{LC}}{4}$ (D) $\frac{5\pi\sqrt{LC}}{2}$

EM0138

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 91 to 93

The fact that a changing magnetic flux produces an electric field is basic to the operation of many high energy particle accelerators. Since the principle was first successfully applied to the acceleration of electrons (or β particles) in a device called the betatron, this method of acceleration is often given that name. The general idea involved is shown in figure.

An electromagnet is used to produce a changing flux through a circular loop defined by the doughnut-shaped vacuum chamber. We see that there will be an electric field E along the circular length of the doughnut, i.e. circling the magnet poles, given by

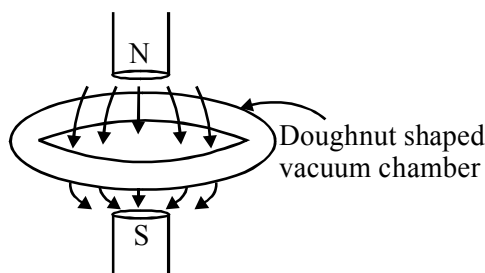
$$2\pi aE = \frac{d\phi}{dt}$$

Where a is the radius of the doughnut. Any charged particle inside the vacuum chamber will experience a force qE and will accelerate. Ordinarily, the charged particle would shoot out of the vacuum chamber and becomes lost.

However, if the magnetic field at the position of the doughnut is just proper to satisfy the relation
Centripetal force = magnetic force

$$\text{or } \frac{mv^2}{a} = qvB$$

then the charge will travel in a circle within the doughnut. By proper shaping of the magnet pole pieces, this relation can be satisfied. As a result, the charge will move at high speed along the loop within the doughnut. Each time it goes around the loop, it has, in effect, fallen through a potential difference equal to the induced, emf, namely $\varepsilon = \frac{d\phi}{dt}$. Its energy after n trips around the loop will be $q(n\varepsilon)$



91. Working of betatron is not based upon which of the following theories :-
 (A) Changing magnetic flux induces electric field
 (B) Charged particle at rest can be accelerated only by electric fields
 (C) magnetic fields can apply a force on moving charges which is perpendicular to both magnetic field and motion of the particle
 (D) Beta particles are emitted in radioactive decay process.

EM0139

92. Variable magnetic flux :-
 (A) Can change sinusoidally
 (B) Should increase all the time
 (C) Must becomes zero when induced field is maximum
 (D) None of these

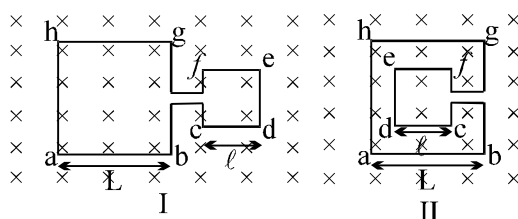
EM0139

93. Magnetic field which keeps the particles in circular path must :-
 (A) Remain a constant every where
 (B) Increase gradually which is proportional to K.E. of the particle
 (C) Increase gradually which is proportional to speed of the particle
 (D) None of these

EM0139

Paragraph for Question No. 94 to 97

The adjoining figure shows two different arrangements in which two square wire frames of same resistance are placed in a uniform constantly decreasing magnetic field B .



94. The value of magnetic flux in each case is given by
 (A) Case I: $\Phi = \pi(L^2 + \ell^2)B$; Case II: $\Phi = \pi(L^2 - \ell^2)B$
 (B) Case I: $\Phi = \pi(L^2 + \ell^2)B$; Case II: $\Phi = \pi(L^2 + \ell^2)B$
 (C) Case I: $\Phi = (L^2 + \ell^2)B$; Case II: $\Phi = (L^2 - \ell^2)B$
 (D) Case I: $\Phi = (L + \ell)^2B$; Case II: $\Phi = \pi(L - \ell)^2B$

EM0140

95. The direction of induced current in the case I is
 (A) from a to b and from c to d
 (B) from a to b and from f to e
 (C) from b to a and from d to c
 (D) from b to a and from e to f

EM0140

96. The direction of induced current in the case II is
 (A) from a to b and from c to d
 (B) from b to a and from f to e
 (C) from b to a and from c to d
 (D) from a to b and from d to c

EM0140

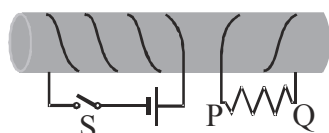
97. If I_1 and I_2 are the magnitudes of induced current in the cases I and II, respectively, then
 (A) $I_1 = I_2$ (B) $I_1 > I_2$ (C) $I_1 < I_2$ (D) nothing can be said

EM0140

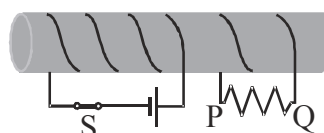
EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

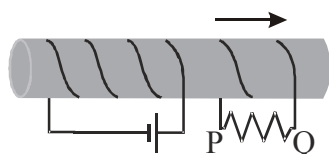
1. For each of the experiments (1, 2, 3, 4) shown in figure. Choose the **CORRECT** option(s) which shows the direction of current flow through the resistor PQ? Note that the wires are not always wrapped around the plastic tube in the same way.



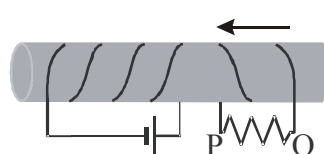
(1) S to be closed



(2) S to be opened



(3) Resistor coil PQ moves to right



(4) Resistor coil PQ moves to left

- | (1) | (2) | (3) | (4) |
|------------|--------|--------|--------|
| (A) P to Q | P to Q | P to Q | P to Q |
| (B) P to Q | Q to P | P to Q | Q to P |
| (C) Q to P | Q to P | Q to P | Q to P |
| (D) Q to P | Q to P | P to Q | P to Q |

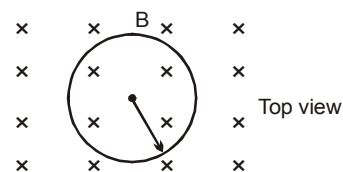
EM0141

2. An electron is moving in a circular orbit of radius R with an angular acceleration α . At the centre of the orbit is kept a conducting loop of radius r , ($r \ll R$). The e.m.f induced in the smaller loop due to the motion of the electron is :-

- | | |
|--|-------------------------------------|
| (A) zero, since charge on electron is constant | (B) $\frac{\mu_0 e r^2}{4R} \alpha$ |
| (C) $\frac{\mu_0 e r^2}{4\pi R} \alpha$ | (D) none of these |

EM0142

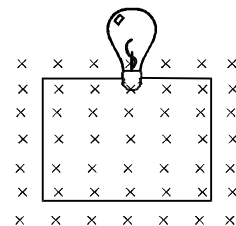
3. A non conducting ring (of mass m , radius r , having charge Q) is placed on a rough horizontal surface (in a region with transverse magnetic field). The field is increasing with time at the rate R and coefficient of friction between the surface and the ring is μ . For ring to remain in equilibrium μ should be greater than :-



- (A) $\frac{QrR}{mg}$ (B) $\frac{QrR}{2mg}$ (C) $\frac{QrR}{3mg}$ (D) $\frac{2QrR}{mg}$

EM0143

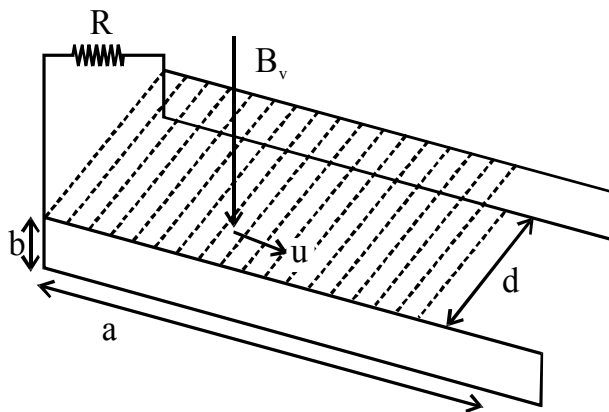
4. A square wire loop of 10.0 cm side lies at right angles to a uniform magnetic field of 20T. A 10 V light bulb is in a series with the loop as shown in the fig. The magnetic field is decreasing steadily to zero over a time interval Δt . The bulb will shine with full brightness if Δt is equal to :-



- (A) 20 ms (B) 0.02 ms (C) 2 ms (D) 0.2 ms

EM0144

5. The figure shows an apparatus suggested by Faraday to generate electric current from a flowing river. Two identical conducting plates of length a and width b are placed parallel facing one another on opposite sides of the river following with velocity u at a distance d apart. Now both the plates are connected by a load resistance R . Then the current through the load R is :- (Consider vertical component of the magnetic field produced by earth is B_v and the resistivity of river water is ρ .)



- (A) $\frac{B_v ub}{R}$ (B) $\frac{B_v ub}{R + \frac{\rho d}{ab}}$ (C) $\frac{B_v ud}{R + \frac{\rho d}{ab}}$ (D) None of the above

EM0145

6. The radius of a coil decreases steadily at the rate of 10^{-2} m/s. A constant and uniform magnetic field of induction 10^{-3} Wb/m² acts perpendicular to the plane of the coil. The radius of the coil when the induced e.m.f. in the coil is 1μ V, is :-

- (A) $\frac{2}{\pi}$ cm (B) $\frac{3}{\pi}$ cm (C) $\frac{4}{\pi}$ cm (D) $\frac{5}{\pi}$ cm

EM0146

7. A composite rod of length ℓ is one fourth insulator and remaining conductor is made to rotate freely with angular velocity ω , in a space free of any gravitational, electric & magnetic field. Then potential difference across the conducting region will be (rotation is about insulating end).

(A) $\frac{3m_e\omega^2\ell^2}{4e}$

(B) $\frac{1}{4} \frac{m_e\omega^2\ell^2}{e}$

(C) $\frac{1}{16} \frac{m_e\omega^2\ell^2}{e}$

(D) $\frac{15}{32} \frac{m_e\omega^2\ell^2}{e}$

EM0147

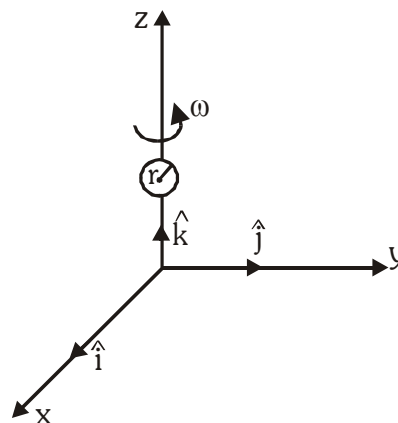
8. A circular loop wire of radius r rotates about the z -axis with angular velocity ω . The normal to the loop is always perpendicular to the z -axis. At time $t = 0$, the normal is parallel to the y -axis. An external magnetic field $\vec{B} = B_y\hat{j} + B_z\hat{k}$ is applied. The EMF $\varepsilon(t)$ induced in the loop is

(A) $\pi r^2 \omega B_y \sin \omega t$

(B) $\pi r^2 \omega B_z \cos \omega t$

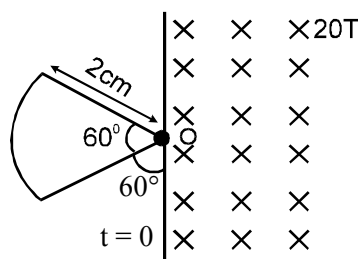
(C) $\pi r^2 \omega B_z \sin \omega t$

(D) $\pi r^2 \omega B_y \cos \omega t$



EM0148

9. A uniform magnetic field 20 T exists on right side of the boundary in a gravity free space as shown in figure. The given circular arc of radius 2 cm made of conducting wire of total resistance 4Ω is rotated around point O at a constant angular speed 2 rad per second. Power required to maintain the constant angular velocity between time interval $t = \frac{\pi}{6}$ s to $t = \frac{\pi}{3}$ s is :-



(A) $64 \mu W$

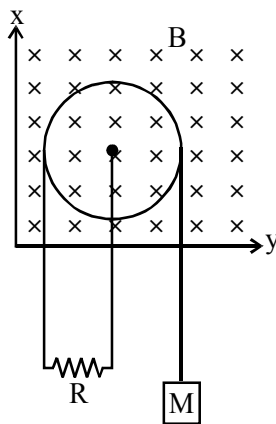
(B) $32 \mu W$

(C) $128 \mu W$

(D) $16 \mu W$

EM0149

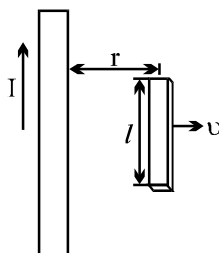
10. The block of mass (M) is connected by thread which is wound on a pulley, free to rotate about fixed horizontal axis as shown. A uniform magnetic field B exists in a horizontal plane. The disc is connected with the resistance R as shown. Calculate the terminal velocity of the block if it was released from rest. Treat pulley as uniform metallic disc of radius L .



- (A) $\frac{4mgR}{B^2L^2}$ (B) $\frac{3mgR}{4B^2L^2}$ (C) $\frac{2mgR}{B^2L^2}$ (D) $\frac{3mgR}{2B^2L^2}$

EM0150

11. A conducting rod moves with constant velocity v perpendicular to the long, straight wire carrying a current I as shown. Compute the emf generated between the ends of the rod.



- (A) $\frac{\mu_0 v I l}{\pi r}$ (B) $\frac{\mu_0 v I l}{2\pi r}$ (C) $\frac{2\mu_0 v I l}{\pi r}$ (D) $\frac{\mu_0 v I l}{4\pi r}$

EM0151

12. A metallic rod of length L and mass M is moving under the action of two unequal forces F_1 and F_2 (directed opposite to each other) acting at its ends along its length. Ignore gravity and any external magnetic field. If specific charge of electrons is (e/m) , then the potential difference between the ends of the rod in steady state must be

- (A) $|F_1 - F_2| mL/eM$ (B) $(F_1 + F_2) mL/eM$ (C) $[mL/eM] \ln [F_1/F_2]$ (D) None

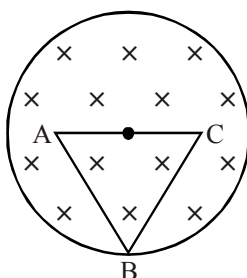
EM0152

13. Two parallel long straight conductors lie on a smooth surface. Two other parallel conductors rest on them at right angles so as to form a square of side a initially. A uniform magnetic field B exists at right angles to the plane containing the conductors. They all start moving out with a constant velocity v . If r is the resistance per unit length of the wire the current in the circuit will be

- (A) $\frac{Bv}{r}$ (B) $\frac{Br}{v}$ (C) Bvr (D) Bv

EM0153

14. An equilateral triangle ABC of side a is placed in the magnetic field with side AC and its centre coinciding with the centre of the magnetic field. The magnetic field varies with time as $B = ct$. The emf induced across side AB is :-



- (A) $\frac{\sqrt{3}}{4}a^2c$ (B) Zero (C) $\frac{\sqrt{3}}{8}a^2c$ (D) $\frac{(\sqrt{2}-1)}{2}a^2c$

EM0154

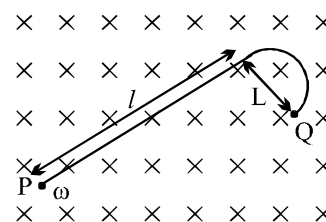
15. The magnetic field in a region is given by $\vec{B} = B_0 \left(1 + \frac{x}{a}\right) \hat{k}$. A square loop of edge-length d is placed with its edge along x & y axis. The loop is moved with constant velocity $\vec{V} = V_0 \hat{i}$. The emf induced in the loop is

- (A) $\frac{V_0 B_0 d^2}{a}$ (B) $\frac{V_0 B_0 d^2}{2a}$ (C) $\frac{V_0 B_0 a^2}{d}$ (D) None

EM0155

16. When a 'J' shaped conducting rod is rotating in its own plane with constant angular velocity ω , about one of its end P, in a uniform magnetic field $\vec{B}$ directed normally into the plane of paper then magnitude of emf induced across it will be

- (A) $B\omega\sqrt{L^2 + l^2}$
 (B) $\frac{1}{2}B\omega L^2$
 (C) $\frac{1}{2}B\omega(L^2 + l^2)$
 (D) $\frac{1}{2}B\omega l^2$



EM0156

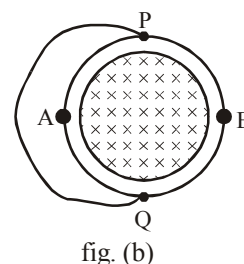
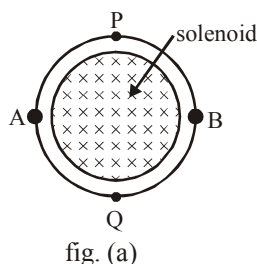
17. A rectangular coil of single turn, having area A , rotates in a uniform magnetic field B with an angular velocity ω about an axis perpendicular to the field. If initially the plane of coil is perpendicular to the field, then the average induced e.m.f. when it has rotated through 90° is :-

- (A) $\frac{\omega BA}{\pi}$ (B) $\frac{\omega BA}{2\pi}$ (C) $\frac{\omega BA}{4\pi}$ (D) $\frac{2\omega BA}{\pi}$

EM0157

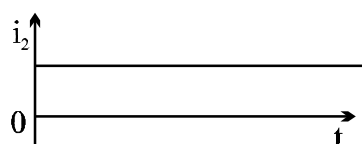
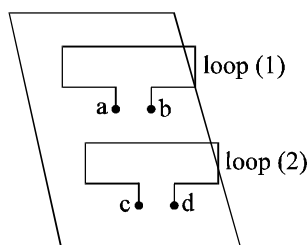
18. In figure (a) a solenoid produces a magnetic field whose strength increases into the plane of the page. An induced emf is established in a conduction loop surrounding the solenoid, and this emf lights bulbs A and B. In figure (b) point P and Q are shorted. After the short is inserted

- (A) Bulb A goes out, bulb B gets brighter
(B) Bulb B goes out, bulb A gets brighter
(C) Bulb A goes out, bulb B gets dimmer
(D) Bulb B goes out, bulb A gets dimmer

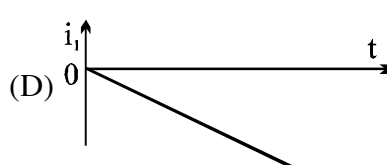
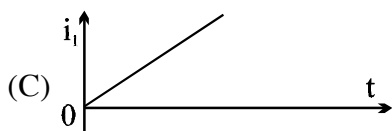
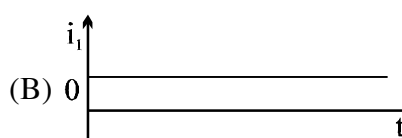
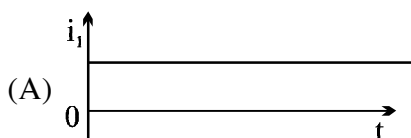


EM0158

19. An electric current i_1 can flow either direction through loop (1) and induced current i_2 in loop (2). Positive i_1 is when current is from 'a' to 'b' in loop (1) and positive i_2 is when the current is from 'c' to 'd' in loop (2). In an experiment, the graph of i_2 against time 't' is as shown below

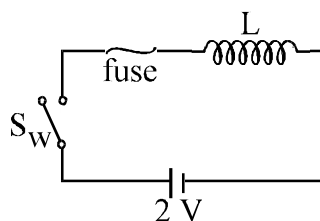


Which one(s) of the following graphs could have caused i_2 to behave as given above.



EM0159

20. In the circuit shown, the cell is ideal. The coil has an inductance of 4H and zero resistance. F is a fuse of zero resistance and will blow when the current through it reaches 5A. The switch is closed at $t = 0$. The fuse will blow :



- (A) just after $t=0$ (B) after 2s (C) after 5s (D) after 10s

EM0160

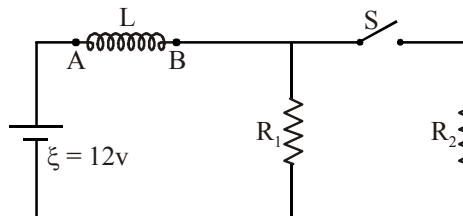
21. The circuit shown has been operating for a long time. The instant after the switch in the circuit labeled S is opened, what is the voltage across the inductor V_L and which labeled point (A or B) of the inductor is at a higher potential? Take $R_1 = 4.0 \Omega$, $R_2 = 8.0 \Omega$, and $L = 2.5 \text{ H}$.

(A) $V_L = 12 \text{ V}$; Point A is at the higher potential

(B) $V_L = 12 \text{ V}$; Point B is at the higher potential

(C) $V_L = 6 \text{ V}$; Point A is at the higher potential

(D) $V_L = 6 \text{ V}$; Point B is at the higher potential



EM0161

22. When a resistance R is connected in series with an element A, the electric current is found to be lagging behind the voltage by angle θ_1 . When the same resistance is connected in series with element B, current leads voltage by θ_2 . When R , A, B are connected in series, the current now leads voltage by θ . Assume same AC source is used in all cases, then

(A) $\theta = \theta_2 - \theta_1$

(B) $\tan \theta = \tan \theta_2 - \tan \theta_1$

(C) $\theta = \frac{\theta_1 + \theta_2}{2}$

(D) None of these

EM0162

23. An current is given by $I = I_0 + I_1 \sin \omega t$ then its rms value will be

(A) $\sqrt{I_0^2 + 0.5 I_1^2}$

(B) $\sqrt{I_0^2 + 0.5 I_0^2}$

(C) 0

(D) $I_0/\sqrt{2}$

EM0163

24. Power factor of an L-R series circuit is 0.6 and that of a C-R series circuit is 0.5. If the element (L, C, and R) of the two circuits are joined in series the power factor of this circuit is found to be 1. The ratio of the resistance in the L-R circuit to the resistance in the C-R circuit is

(A) 6/5

(B) 5/6

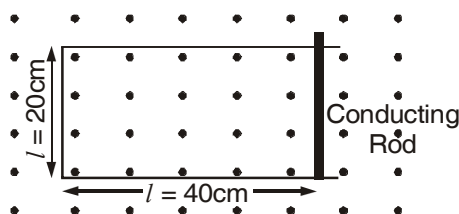
(C) $\frac{4}{3\sqrt{3}}$

(D) $\frac{3\sqrt{3}}{4}$

EM0164

MULTIPLE CORRECT TYPE QUESTIONS

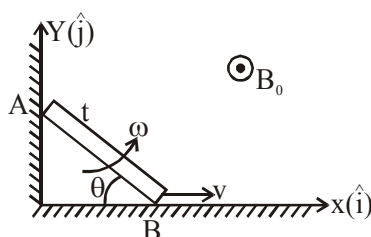
25. Figure shows a conducting rod of negligible resistance that can slide on smooth U-shaped rail made of wire of resistance $1 \Omega/\text{m}$. Position of the conducting rod at $t = 0$ is shown. A time dependent magnetic field $B = 2t$ Tesla is switched on at $t = 0$. After the magnetic field is switched on, the conducting rod is moved to the left perpendicular to the rails at constant speed 5 cm/s by some external agent.



- (A) The current in the loop at $t = 0$ due to induced emf is 0.16 A , clockwise
 (B) At $t = 2 \text{ s}$, induced emf has magnitude 0.08 V
 (C) The magnitude of the force required to move the conducting rod at constant speed 5 cm/s at $t = 2 \text{ s}$, is equal to 0.08 N
 (D) The magnitude of the force required to move the conducting rod at constant speed 5 cm/s at $t = 2 \text{ s}$, is equal to 0.16 N

EM0165

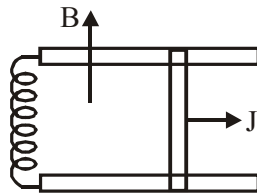
26. A thin conducting rod of length ℓ is moved such that its end B moves along the X-axis while end A moves along the Y-axis. A uniform magnetic field $B = B_0 \hat{k}$ exists in the region. At some instant, velocity of end B is v and the rod makes an angle of $\theta = 60^\circ$ with the X-axis as shown in the figure. Then, at this instant



- (A) angular speed of rod AB is $\omega = \frac{2v}{\sqrt{3}\ell}$
 (B) angular speed of rod AB is $\omega = \frac{\sqrt{3}v}{2\ell}$
 (C) e.m.f. induced in rod AB is $B\ell v\sqrt{3}$
 (D) e.m.f. induced in rod AB is $B\ell v/2\sqrt{3}$

EM0166

27. Two parallel resistanceless rails are connected by an inductor of inductance L at one end as shown in the figure. A magnetic field B exists in the space which is perpendicular to the plane of the rails. Now a conductor of length ℓ and mass m is placed transverse on the rails and given an impulse J towards the rightward direction. Then choose the **CORRECT** option (s).



- (A) Velocity of the conductor is half of the initial velocity after a displacement of the conductor

$$d = \sqrt{\frac{3J^2 L}{4B^2 \ell^2 m}}$$

- (B) Current flowing through the inductor at the instant when velocity of the conductor is half of the

$$\text{initial velocity is } i = \sqrt{\frac{3J^2}{4Lm}}$$

- (C) Velocity of the conductor is half of the initial velocity after a displacement of the conductor

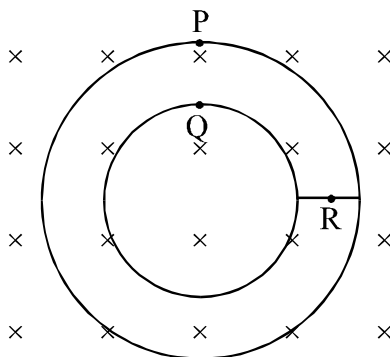
$$d = \sqrt{\frac{3J^2 L}{B^2 \ell^2 m}}$$

- (D) Current flowing through the inductor at the instant when velocity of the conductor is half of

$$\text{the initial velocity is } i = \sqrt{\frac{3J^2}{mL}}$$

EM0167

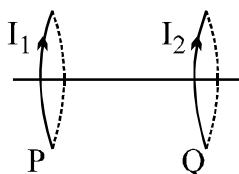
28. Figure shown plane figure made of a conductor located in a magnetic field along the inward normal to the plane of the figure. The magnetic field starts diminishing. Then the induced current



- (A) at point P is clockwise
(B) at point Q is anticlockwise
(C) at point Q is clockwise
(D) at point R is zero

EM0168

29. Two circular coils P & Q are fixed coaxially & carry currents I_1 and I_2 respectively

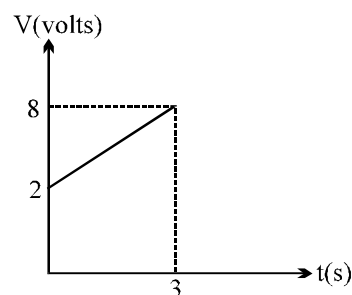


- (A) if $I_2 = 0$ & P moves towards Q, a current in the same direction as I_1 is induced in Q
 (B) if $I_1 = 0$ & Q moves towards P, a current in the opposite direction to that of I_2 is induced in P.
 (C) when $I_1 \neq 0$ and $I_2 \neq 0$ are in the same direction then the two coils tend to move apart.
 (D) when $I_1 \neq 0$ and $I_2 \neq 0$ are in opposite directions then the coils tends to move apart.

EM0169

30. A circuit element is placed in a closed box. At time $t = 0$, constant current generator supplying a current of 1 amp, is connected across the box. Potential difference across the box varies according to graph shown in figure. The element in the box is :

- (A) resistance of 2Ω
 (B) battery of emf 6V
 (C) inductance of 2H
 (D) capacitance of 0.5F



EM0170

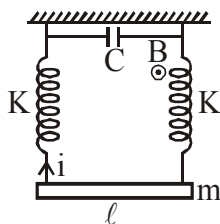
31. Two coils A and B have coefficient of mutual inductance $M = 2H$. The magnetic flux passing through coil A changes by 4 Weber in 10 seconds due to the change in current in B. Then
 (A) the change in current in B in this time interval is 0.5 A
 (B) the change in current in B in this time interval is 2A
 (C) the change in current in B in this time interval is 8A
 (D) a change in current of 1A in coil A will produce a change in flux passing through B by 4 Weber.

EM0171

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 32 and 33

In the figure shown a uniform conducting rod of mass m and length ℓ is suspended in vertical plane by two conducting springs of spring constant K . Upper end of spring are connected to each other by capacitor of capacitance C . A uniform horizontal magnetic field (B_0) perpendicular to plane of spring exists in space. Initially rod is in equilibrium but if centre of rod is pulled down and released, it performs SHM. Assume that the spring is small and neglect the magnetic force of interaction between circular section of springs & self inductance of rod.



32. Find time period of oscillation of rod :-

- (A) $2\pi\sqrt{\frac{m}{k}}$ (B) $2\pi\sqrt{\frac{B^2\ell^2C}{K}}$ (C) $\pi\sqrt{\frac{m+B^2\ell^2C}{K}}$ (D) $2\pi\sqrt{\frac{B^2\ell^2C+m}{2K}}$

EM0172

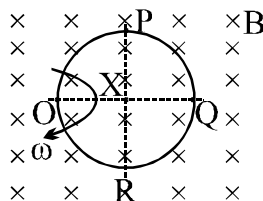
33. Choose correct options from following :-

- (A) Electrical energy stored in capacitor is maximum when rod is at its lower extreme position
 (B) Electrical energy stored in capacitor is maximum when rod is at its mean position
 (C) Current in rod is maximum at mean position of rod
 (D) If magnetic field is switched off then mean position of rod will change

EM0172

Paragraph for Question No. 34 to 36

A conducting ring of radius a is rotated about a point O on its periphery as shown in the figure in a plane perpendicular to uniform magnetic field B which exists everywhere. The rotational velocity is ω .



34. Choose the correct statement(s) related to the potential of the points P , Q and R

- (A) $V_P - V_O > 0$ and $V_R - V_O < 0$ (B) $V_P = V_R > V_O$
 (C) $V_O > V_P = V_Q$ (D) $V_Q - V_P = V_P - V_O$

EM0173

35. Choose the correct statement(s) related to the magnitude of potential differences

(A) $V_P - V_O = \frac{1}{2} B\omega a^2$

(B) $V_P - V_Q = \frac{1}{2} B\omega a^2$

(C) $V_Q - V_O = 2B\omega a^2$

(D) $V_P - V_R = 2B\omega a^2$

EM0173

36. Choose the correct statement(s) related to the induced current in the ring

(A) Current flows from Q $\longrightarrow$ P $\longrightarrow$ O $\longrightarrow$ R $\longrightarrow$ Q

(B) Current flows from Q $\longrightarrow$ R $\longrightarrow$ O $\longrightarrow$ P $\longrightarrow$ Q

(C) Current flows from Q $\longrightarrow$ P $\longrightarrow$ O and from Q $\longrightarrow$ R $\longrightarrow$ O

(D) No current flows

EM0173

Paragraph for question nos. 37 to 39

In a series L-R circuit, connected with a sinusoidal ac source, the maximum potential difference across L and R are respectively 3 volts and 4 volts.

37. At an instant the potential difference across resistor is 2 volts. The potential difference in volt, across the inductor at the same instant will be :

(A) $3 \cos 30^\circ$

(B) $3 \cos 60^\circ$

(C) $3 \cos 45^\circ$

(D) None of these

EM0174

38. At the same instant, the magnitude of the potential difference in volt, across the ac source may be

(A) $4 + 3\sqrt{3}$

(B) $\frac{4 + 3\sqrt{3}}{2}$

(C) $1 + \frac{\sqrt{3}}{2}$

(D) $2 + \frac{\sqrt{3}}{2}$

EM0174

39. If the current at this instant is decreasing the magnitude of potential difference at that instant across the ac source is

(A) Increasing

(B) Decreasing

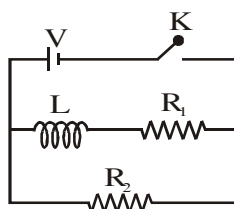
(C) Constant

(D) Cannot be said

EM0174

EXERCISE-JM

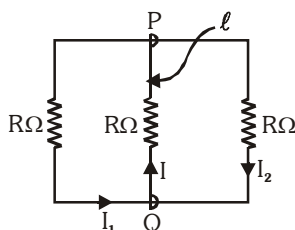
1. In the circuit show below, the key K is closed at $t = 0$. The current through the battery is : [AIEEE - 2010]



- (1) $\frac{V(R_1 + R_2)}{R_1 R_2}$ at $t = 0$ and $\frac{V}{R_2}$ at $t = \infty$ (2) $\frac{V R_1 R_2}{\sqrt{R_1^2 + R_2^2}}$ at $t = 0$ and $\frac{V}{R_2}$ at $t = \infty$
 (3) $\frac{V}{R_2}$ at $t = 0$ and $\frac{V(R_1 + R_2)}{R_1 R_2}$ at $t = \infty$ (4) $\frac{V}{R_2}$ at $t = 0$ and $\frac{V R_1 R_2}{\sqrt{R_1^2 + R_2^2}}$ at $t = \infty$

EM0175

2. A rectangular loop has a sliding connector PQ of length ℓ and resistance $R\Omega$ and it is moving with a speed v as shown. The set-up is placed in a uniform magnetic field going into the plane of the paper. The three currents I_1 , I_2 and I are :- [AIEEE - 2010]



- (1) $I_1 = I_2 = \frac{B\ell v}{6R}$, $I = \frac{B\ell v}{3R}$ (2) $I_1 = -I_2 = \frac{B\ell v}{R}$, $I = \frac{2B\ell v}{R}$
 (3) $I_1 = I_2 = \frac{B\ell v}{3R}$, $I = \frac{2B\ell v}{3R}$ (4) $I_1 = I_2 = I = \frac{B\ell v}{R}$

EM0176

3. In a series LCR circuit $R = 200\Omega$ and the voltage and the frequency of the main supply is 220 V and 50 Hz respectively. On taking out the capacitance from the circuit the current lags behind the voltage by 30° . On taking out the inductor from the circuit the current leads the voltage by 30° . The power dissipated in the LCR circuit is : [AIEEE - 2010]

- (1) 242 W (2) 305 W (3) 210 W (4) 0 W

EM0177

4. A boat is moving due east in a region where the earth's magnetic field is $5.0 \times 10^{-5} \text{ NA}^{-1} \text{ m}^{-1}$ due north and horizontal. The boat carries a vertical aerial 2m long. If the speed of the boat is 1.50 ms^{-1} , the magnitude of the induced emf in the wire of aerial is :- [AIEEE - 2011]
 (1) 0.50 mV (2) 0.15 mV (3) 1 mV (4) 0.75 mV

EM0178

5. A horizontal straight wire 20 m long extending from east to west is falling with a speed of 5.0 m/s, at right angles to the horizontal component of the earth's magnetic field $0.30 \times 10^{-4} \text{ Wb/m}^2$. The instantaneous value of the e.m.f. induced in the wire will be :- [AIEEE - 2011]
 (1) 6.0 mV (2) 3 mV (3) 4.5 mV (4) 1.5 mV

EM0179

6. A fully charged capacitor C with initial charge q_0 is connected to a coil of self inductance L at $t = 0$. The time at which the energy is stored equally between the electric and the magnetic fields is :- [AIEEE - 2011]

- (1) $2\pi\sqrt{LC}$ (2) $\sqrt{LC}$ (3) $\pi\sqrt{LC}$ (4) $\frac{\pi}{4}\sqrt{LC}$

EM0180

7. A coil is suspended in a uniform magnetic field, with the plane of the coil parallel to the magnetic lines of force. When a current is passed through the coil it starts oscillating; it is very difficult to stop. But if an aluminium plate is placed near to the coil, it stops. This is due to :- [AIEEE - 2012]
 (1) Electromagnetic induction in the aluminium plate giving rise to electromagnetic damping
 (2) Development of air current when the plate is placed
 (3) Induction of electrical charge on the plate
 (4) Shielding of magnetic lines of force as aluminium is a paramagnetic material

EM0181

8. If a simple pendulum has Significant amplitude (up to a factor of $1/e$ of original) only in the period between $t = 0$ s to $t = \tau$ s, then τ may be called the average life of the pendulum. When the spherical bob of the pendulum suffers a retardation (due to viscous drag) proportional to its velocity, with 'b' as the constant of proportionality, the average lifetime of the pendulum is (assuming damping is small) in seconds: [AIEEE-2012]

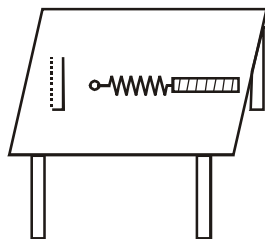
- (1) $\frac{1}{b}$ (2) $\frac{2}{b}$ (3) $\frac{0.693}{b}$ (4) b

EM0182

9. The amplitude of a damped oscillator decreases to 0.9 times its original magnitude in 5s. In another 10s it will decrease to a times its original magnitude, where a equals: [JEE Main-2013]
 (1) 0.81 (2) 0.729 (3) 0.6 (4) 0.7

EM0183

10. A metallic rod of length 'l' is tied to a string of length 2l and made to rotate with angular speed ω on a horizontal table with one end of the string fixed. If there is a vertical magnetic field 'B' in the region, the e.m.f. induced across the ends of the rod is : [JEE Main-2013]



- (1) $\frac{2B\omega l^2}{2}$ (2) $\frac{3B\omega l^2}{2}$ (3) $\frac{4B\omega l^2}{2}$ (4) $\frac{5B\omega l^2}{2}$

EM0184

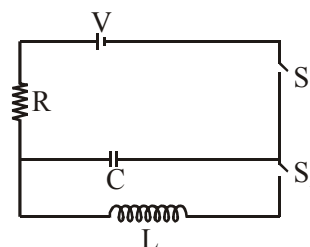
11. A circular loop of radius 0.3 cm lies parallel to a much bigger circular loop of radius 20 cm. The centre of the small loop is on the axis of the bigger loop. The distance between their centres is 15 cm. If a current of 2.0 A flows through the smaller loop, then the flux linked with bigger loop is :- [JEE Main-2013]

- (1) 9.1×10^{-11} weber (2) 6×10^{-11} weber (3) 3.3×10^{-11} weber (4) 6.6×10^{-9} weber

EM0185

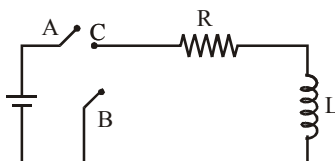
12. In an LCR circuit as shown below both switches are open initially. Now switch S_1 is closed, S_2 kept open, (q is charge on the capacitor and $\tau = RC$ is Capacitive time constant). Which of the following statement is correct? [JEE Main-2013]

- (1) Work done by the battery is half of the energy dissipated in the resistor
 (2) At $t = \tau$, $q = CV/2$
 (3) At $t = 2\tau$, $q = CV(1 - e^{-2})$
 (4) At $t = \frac{\tau}{2}$, $q = CV(1 - e^{-1})$



EM0186

13. In the circuit shown here, the point 'C' is kept connected to point 'A' till the current flowing through the circuit becomes constant. Afterward, suddenly, point 'C' is disconnected from point 'A' and connected to point 'B' at time $t = 0$. Ratio of the voltage across resistance and the inductor at $t = L/R$ will be equal to : [JEE Main-2014]

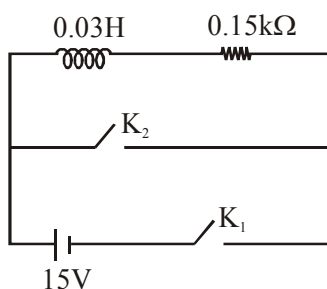


- (1) -1 (2) $\frac{1-e}{e}$ (3) $\frac{e}{1-e}$ (4) 1

EM0187

14. An inductor ($L = 0.03 \text{ H}$) and a resistor ($R = 0.15 \text{ k}\Omega$) are connected in series to a battery of 15V EMF in a circuit shown below. The key K_1 has been kept closed for a long time. Then at $t = 0$, K_1 is opened and key K_2 is closed simultaneously. At $t = 1\text{ms}$, the current in the circuit will be ($e^5 \approx 150$):-

[JEE Main-2015]

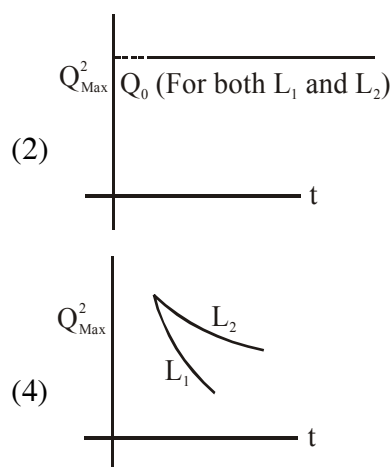
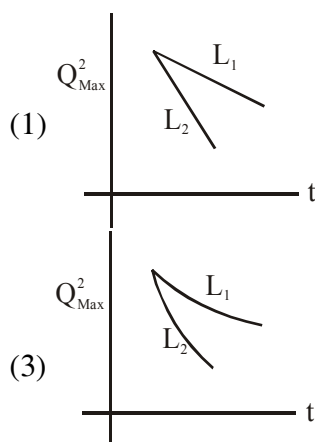
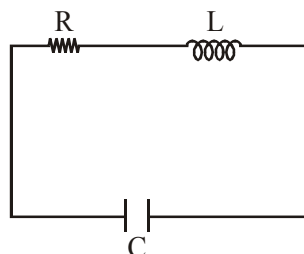


- (1) 6.7 mA (2) 0.67 mA (3) 100 mA (4) 67 mA

EM0188

15. An LCR circuit is equivalent to a damped pendulum. In an LCR circuit the capacitor is charged to Q_0 and then connected to the L and R as shown below. If a student plots graphs of the square of maximum charge (Q_{Max}^2) on the capacitor with time (t) for two different values L_1 and L_2 ($L_1 > L_2$) of L then which of the following represents this graph correctly? (plots are schematic and not drawn to scale)

[JEE Main-2015]

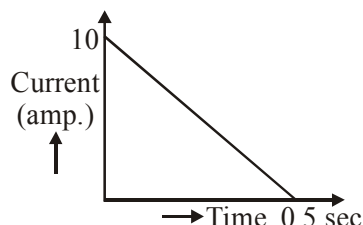


EM0189

16. An arc lamp requires a direct current of 10A at 80V to function. If it is connected to a 220V (rms), 50Hz AC supply, the series inductor needed for it to work is close to :- [JEE Main-2016]
 (1) 0.065 H (2) 80 H (3) 0.08 H (4) 0.044 H

EM0190

17. In a coil of resistance $100\ \Omega$, a current is induced by changing the magnetic flux through it as shown in the figure. The magnitude of change in flux through the coil is :- [JEE Main-2017]



- (1) 250 Wb (2) 275 Wb (3) 200 Wb (4) 225 Wb

EM0191

18. For an RLC circuit driven with voltage of amplitude v_m and frequency $\omega_0 = \frac{1}{\sqrt{LC}}$ the current exhibits resonance. The quality factor, Q is given by :- [JEE Main-2018]

- (1) $\frac{\omega_0 R}{L}$ (2) $\frac{R}{(\omega_0 C)}$ (3) $\frac{CR}{\omega_0}$ (4) $\frac{\omega_0 L}{R}$

EM0192

19. In an a. c. circuit, the instantaneous e.m.f. and current are given by

$$e = 100 \sin 30 t$$

$$i = 20 \sin \left(30t - \frac{\pi}{4} \right)$$

In one cycle of a.c., the average power consumed by the circuit and the wattless current are, respectively. [JEE Main-2018]

- (1) $\frac{1000}{\sqrt{2}}$, 10 (2) $\frac{50}{\sqrt{2}}$, 0 (3) 50, 0 (4) 50, 10

EM0193

EXERCISE - JA

1. An AC voltage source of variable angular frequency ω and fixed amplitude V_0 is connected in series with a capacitance C and an electric bulb of resistance R (inductance zero). When ω is increased

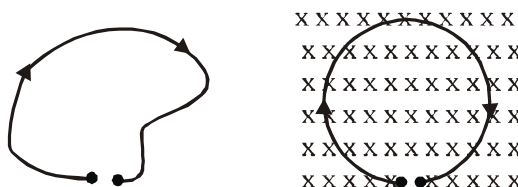
[JEE 2010]

- (A) the bulb glows dimmer
(B) the bulb glows brighter
(C) total impedance of the circuit is unchanged
(D) total impedance of the circuit increases

EM0194

2. A thin flexible wire of length L is connected to two adjacent fixed points and carries a current I in the clockwise direction, as shown in the figure. When the system is put in a uniform magnetic field of strength B going into the plane of the paper, the wire takes the shape of a circle. The tension in the wire is :-

[JEE 2010]



- (A) IBL
(B) $\frac{IBL}{\pi}$
(C) $\frac{IBL}{2\pi}$
(D) $\frac{IBL}{4\pi}$

EM0195

3. You are given many resistances, capacitors and inductors. These are connected to a variable DC voltage source (the first two circuits) or an AC voltage source of 50 Hz frequency (the next three circuits) in different ways as shown in **Column II**. When a current I (steady state for DC or rms for AC) flows through the circuit, the corresponding voltage V_1 and V_2 (indicated in circuits) are related as shown in **Column I**. Match the two [JEE 2010]

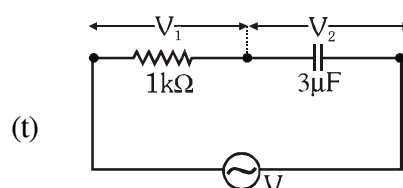
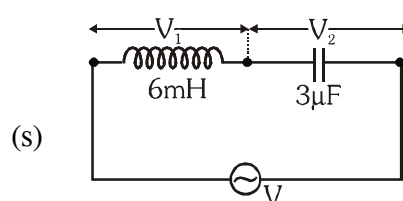
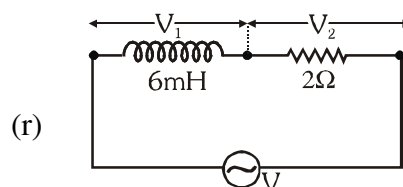
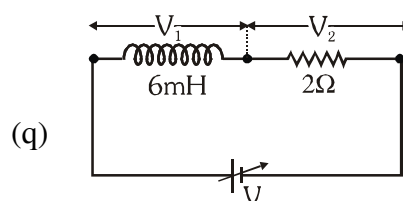
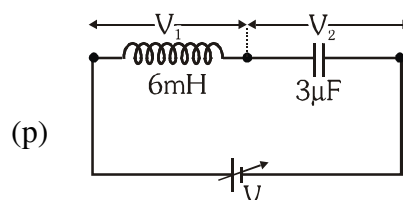
Column I

(A) $I \neq 0$, V_1 is proportional to I

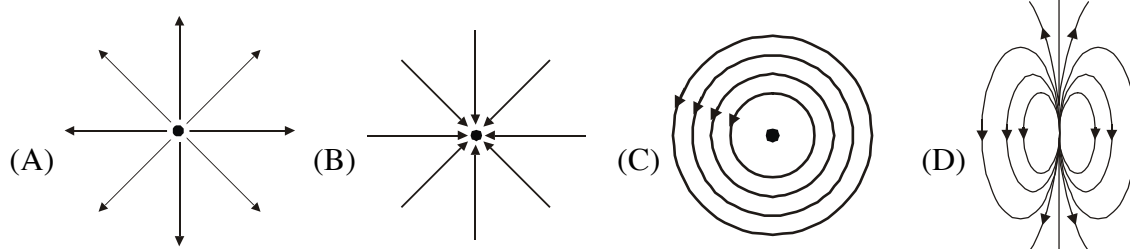
(B) $I \neq 0$, $V_2 > V_1$

(C) $V_1 = 0$, $V_2 = V$

(D) $I \neq 0$, V_2 is proportional to I

Column II

4. Which of the field patterns given below is valid for electric field as well as for magnetic field?



EM0196

[JEE 2011]

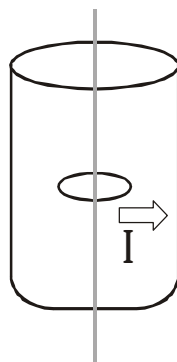
EM0197

5. A series R-C circuit is connected to AC voltage source. Consider two cases ; (A) when C is without a dielectric medium and (B) when C is filled with dielectric of constant 4. The current I_R through the resistor and voltage V_C across the capacitor are compared in the two cases. Which of the following is/are true? [JEE 2011]

(A) $I_R^A > I_R^B$ (B) $I_R^A < I_R^B$ (C) $V_C^A > V_C^B$ (D) $V_C^A < V_C^B$

EM0198

6. A long circular tube of length 10 m and radius 0.3 m carries a current I along its curved surface as shown. A wire-loop of resistance 0.005 ohm and of radius 0.1 m is placed inside the tube with its axis coinciding with the axis of the tube. The current varies as $I = I_0 \cos(300t)$ where I_0 is constant. If the magnetic moment of the loop is $N\mu_0 I_0 \sin(300t)$, then 'N' is [JEE 2011]



EM0199

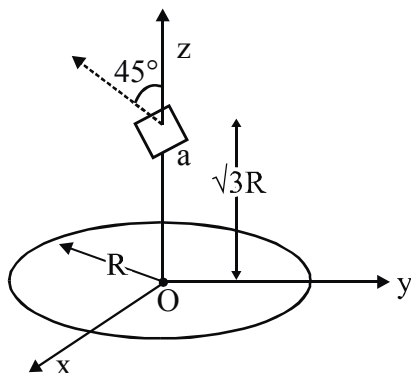
7. A series R-C combination is connected to an AC voltage of angular frequency $\omega = 500$ radian/s. If the impedance of the R-C circuit is $R\sqrt{1.25}$, the time constant (in millisecond) of the circuit is :-

[JEE 2011]

EM0200

8. A circular wire loop of radius R is placed in the x-y plane centred at the origin O. A square loop of side a ($a \ll R$) having two turns is placed with its centre at $z = \sqrt{3}R$ along the axis of the circular wire loop, as shown in figure. The plane of the square loop makes an angle of 45° with respect to the

z-axis. If the mutual inductance between the loops is given by $\frac{\mu_0 a^2}{2^{p/2} R}$, then the value of p is :- [JEE 2012]

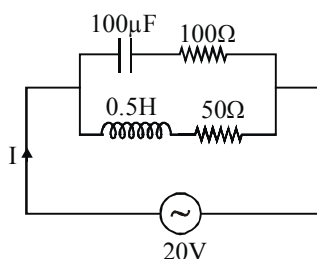


EM0201

9. A current carrying infinitely long wire is kept along the diameter of a circular wire loop, without touching it. The correct statement(s) is (are) [JEE 2012]
- (A) The emf induced in the loop is zero if the current is constant.
 (B) The emf induced in the loop is infinite if the current is constant.
 (C) The emf induced in the loop is zero if the current decreases at a steady rate.
 (D) The emf induced in the loop is finite if the current decreases at a steady rate.

EM0202

10. In the given circuit, the AC source has $\omega = 100 \text{ rad/s}$. Considering the inductor and capacitor to be ideal, the correct choice (s) is(are) [JEE 2012]



- (A) The current through the circuit, I is 0.3 A.
 (B) The current through the circuit, i is $0.3\sqrt{2} \text{ A}$.
 (C) The voltage across 100Ω resistor = $10\sqrt{2} \text{ V}$.
 (D) The voltage across 50Ω resistor = 10V.

EM0203

Paragraph for Questions 11 and 12

A point charge Q is moving in a circular orbit of radius R in the x - y plane with an angular velocity ω . This can be considered as equivalent to a loop carrying a steady current $\frac{Q\omega}{2\pi}$. A uniform magnetic field along the positive z -axis is now switched on, which increases at a constant rate from 0 to B in one second. Assume that the radius of the orbit remains constant. The application of the magnetic field induces an emf in the orbit. The induced emf is defined as the work done by an induced electric field in moving a unit positive charge around a closed loop. It is known that for an orbiting charge, the magnetic dipole moment is proportional to the angular momentum with a proportionality constant γ .

11. The change in the magnetic dipole moment associated with the orbit, at the end of the time interval of the magnetic field change is [JEE Advance-2013]

- (A) $-\gamma BQR^2$ (B) $-\gamma \frac{BQR^2}{2}$ (C) $\gamma \frac{BQR^2}{2}$ (D) γBQR^2

EM0204

12. The magnitude of the induced electric field in the orbit at any instant of time during the time interval of the magnetic field change is

- (A) $\frac{BR}{4}$ (B) $\frac{BR}{2}$ (C) BR (D) $2BR$

EM0204

Paragraph for Questions 13 and 14

A thermal power plant produces electric power of 600 kW and 4000 V, which is to be transported to a place 20 km away from the power plant for consumers' usage. It can be transported either directly with a cable of large current carrying capacity or by using a combination of step-up and step-down transformers at the two ends. The drawback of the direct transmission is the large energy dissipation. In the method using transformers, the dissipation is much smaller. In this method, a step-up transformer is used at the plant side so that the current is reduced to a smaller value. At the consumers' end, a step-down transformer is used to supply power to the consumers at the specified lower voltage. It is reasonable to assume that the power cable is purely resistive and the transformers are ideal with a power factor unity. All the currents and voltages mentioned are rms values. [JEE Advance-2013]

13. In the method using the transformers, assume that the ratio of the number of turns in the primary to that in the secondary in the step-up transformer is 1 : 10. If the power to the consumers has to be supplied at 200 V, the ratio of the number of turns in the primary to that in the secondary in the step-down transformer is

(A) 200 : 1 (B) 150 : 1 (C) 100 : 1 (D) 50 : 1

EM0205

14. If the direct transmission method with a cable of resistance $0.4 \Omega \text{ km}^{-1}$ is used, the power dissipation (in %) during transmission is

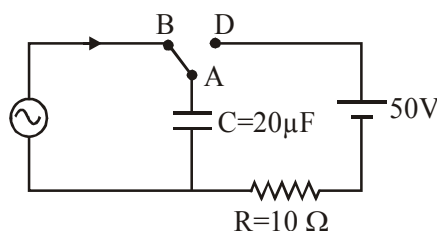
(A) 20 (B) 30 (C) 40 (D) 50

EM0205

15. At time $t = 0$, terminal A in the circuit shown in the figure is connected to B by a key and an alternating current $I(t) = I_0 \cos(\omega t)$, with $I_0 = 1 \text{ A}$ and $\omega = 500 \text{ rad s}^{-1}$ starts flowing in it with the initial direction

shown in the figure. At $t = \frac{7\pi}{6\omega}$, the key is switched from B to D. Now onwards only A and D are connected. A total charge Q flows from the battery to charge the capacitor fully. If $C = 20 \mu\text{F}$, $R = 10 \Omega$ and the battery is ideal with emf of 50 V, identify the correct statement (s).

[JEE Advance-2014]



(A) Magnitude of the maximum charge on the capacitor before $t = \frac{7\pi}{6\omega}$ is $1 \times 10^{-3} \text{ C}$.

(B) The current in the left part of the circuit just before $t = \frac{7\pi}{6\omega}$ is clockwise.

(C) Immediately after A is connected to D, the current in R is 10A

(D) $Q = 2 \times 10^{-3} \text{ C}$

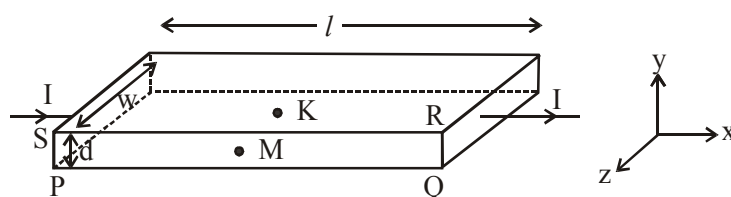
EM0206

Paragraph for Question No. 16 and 17

In a thin rectangular metallic strip a constant current I flows along the positive x -direction, as shown in the figure. The length, width and thickness of the strip are l , w and d , respectively.

A uniform magnetic field $\vec{B}$ is applied on the strip along the positive y -direction. Due to this, the charge carriers experience a net deflection along the z -direction. This results in accumulation of charge carriers on the surface PQRS and appearance of equal and opposite charges on the face opposite to PQRS. A potential difference along the z -direction is thus developed. Charge accumulation continues until the magnetic force is balanced by the electric force. The current is assumed to be uniformly distributed on the cross section of the strip and carried by electrons.

[JEE Advance-2015]



16. Consider two different metallic strips (1 and 2) of the same material. Their lengths are the same, widths are w_1 and w_2 and thicknesses are d_1 and d_2 , respectively. Two points K and M are symmetrically located on the opposite faces parallel to the x - y plane (see figure). V_1 and V_2 are the potential differences between K and M in strips 1 and 2, respectively. Then, for a given current I flowing through them in a given magnetic field strength B , the correct statement(s) is(are)

- (A) If $w_1 = w_2$ and $d_1 = 2d_2$, then $V_2 = 2V_1$
 (B) If $w_1 = w_2$ and $d_1 = 2d_2$, then $V_2 = V_1$
 (C) If $w_1 = 2w_2$ and $d_1 = d_2$, then $V_2 = 2V_1$
 (D) If $w_1 = 2w_2$ and $d_1 = d_2$, then $V_2 = V_1$

EM0207

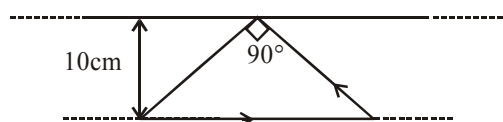
17. Consider two different metallic strips (1 and 2) of same dimensions (length l , width w and thickness d) with carrier densities n_1 and n_2 , respectively. Strip 1 is placed in magnetic field B_1 and strip 2 is placed in magnetic field B_2 , both along positive y -direction. Then V_1 and V_2 are the potential differences developed between K and M in strips 1 and 2, respectively. Assuming that the current I is the same for both the strips, the correct option(s) is(are)

- (A) If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = 2V_1$
 (B) If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = V_1$
 (C) If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = 0.5V_1$
 (D) If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = V_1$

EM0207

18. A conducting loop in the shape of right angled isosceles triangle of height 10 cm is kept such that the 90° vertex is very close to an infinitely long conducting wire (see the figure). The wire is electrically insulated from the loop. The hypotenuse of the triangle is parallel to the wire. The current in the triangular loop is in counterclockwise direction and increased at constant rate of 10 A s^{-1} . Which of the following statement(s) is(are) true?

[JEE Advance-2016]



- (A) The induced current in the wire is in opposite direction to the current along the hypotenuse.
 (B) There is a repulsive force between the wire and the loop
 (C) If the loop is rotated at a constant angular speed about the wire, an additional emf of $\left(\frac{\mu_0}{\pi}\right)$ volt is induced in the wire
 (D) The magnitude of induced emf in the wire is $\left(\frac{\mu_0}{\pi}\right)$ volt.

EM0208

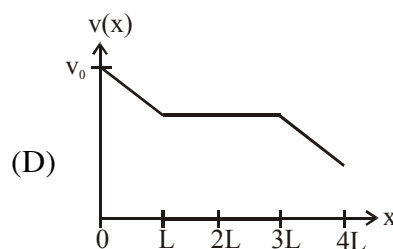
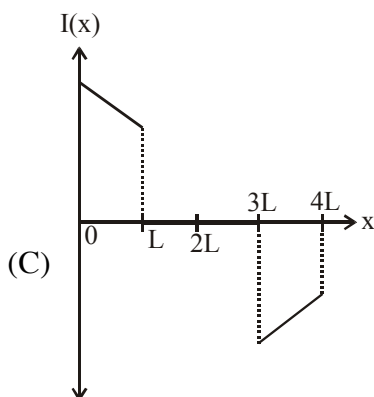
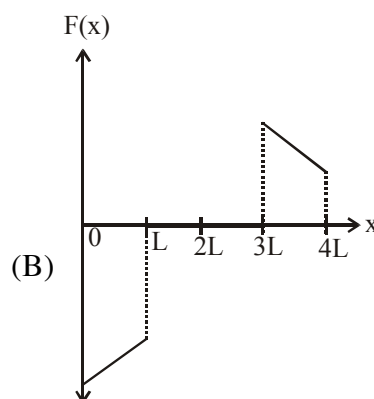
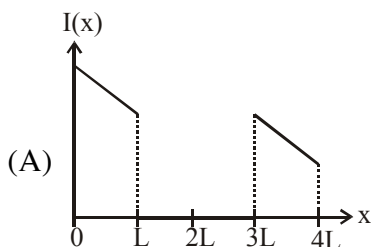
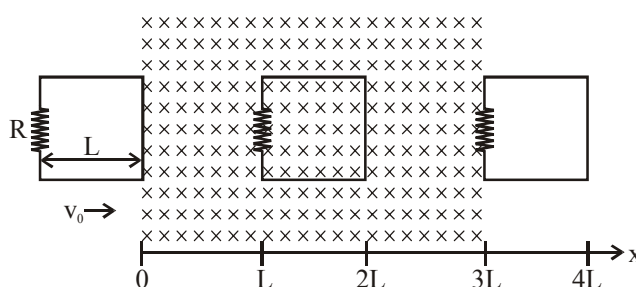
19. Two inductors L_1 (inductance 1 mH, internal resistance 3Ω) and L_2 (inductance 2mH, internal resistance 4Ω), and a resistor R (resistance 12Ω) are all connected in parallel across a 5V battery. The circuit is switched on at time $t = 0$. The ratio of the maximum to the minimum current ($I_{\max}/I_{\min}$) drawn from the battery is.

[JEE Advance-2016]

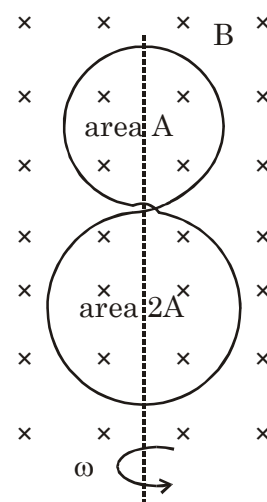
EM0209

20. A rigid wire loop of square shape having side of length L and resistance R is moving along the x -axis with a constant velocity v_0 in the plane of the paper. At $t = 0$, the right edge of the loop enters a region of length $3L$ where there is a uniform magnetic field B_0 into the plane of the paper, as shown in the figure. For sufficiently large v_0 , the loop eventually crosses the region. Let x be the location of the right edge of the loop. Let $v(x)$, $I(x)$ and $F(x)$ represent the velocity of the loop, current in the loop, and force on the loop, respectively, as a function of x . Counter-clockwise current is taken as positive. Which of the following schematic plot(s) is(are) correct? (Ignore gravity)

[JEE Advance-2016]



21. A circular insulated copper wire loop is twisted to form two loops of area A and $2A$ as shown in the figure. At the point of crossing the wires remain electrically insulated from each other. The entire loop lies in the plane (of the paper). A uniform magnetic field $\vec{B}$ points into the plane of the paper. At $t=0$, the loop starts rotating about the common diameter as axis with a constant angular velocity ω in the magnetic field. Which of the following options is/are correct? [JEE Advance-2017]

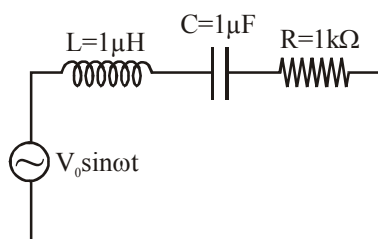


- (A) The rate of change of the flux is maximum when the plane of the loops is perpendicular to plane of the paper
(B) The net emf induced due to both the loops is proportional to $\cos \omega t$
(C) The emf induced in the loop is proportional to the sum of the areas of the two loops
(D) The amplitude of the maximum net emf induced due to both the loops is equal to the amplitude of maximum emf induced in the smaller loop alone

EM0211

22. In the circuit shown, $L = 1 \mu\text{H}$, $C = 1 \mu\text{F}$ and $R = 1 \text{ k}\Omega$. They are connected in series with an a.c. source $V = V_0 \sin \omega t$ as shown. Which of the following options is/are correct ?

[JEE Advance-2017]



- (A) The frequency at which the current will be in phase with the voltage is independent of R .
(B) At $\omega \sim 0$ the current flowing through the circuit becomes nearly zero
(C) At $\omega \gg 10^6 \text{ rad.s}^{-1}$, the circuit behaves like a capacitor.
(D) The current will be in phase with the voltage if $\omega = 10^4 \text{ rad.s}^{-1}$.

EM0212

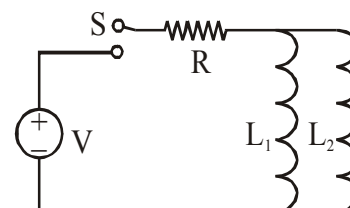
23. A source of constant voltage V is connected to a resistance R and two ideal inductors L_1 and L_2 through a switch S as shown. There is no mutual inductance between the two inductors. The switch S is initially open. At $t = 0$, the switch is closed and current begins to flow. Which of the following options is/are correct? [JEE Advance-2017]

- (A) The ratio of the currents through L_1 and L_2 is fixed at all times ($t > 0$)

- (B) After a long time, the current through L_1 will be $\frac{V}{R} \frac{L_2}{L_1 + L_2}$

- (C) After a long time, the current through L_2 will be $\frac{V}{R} \frac{L_1}{L_1 + L_2}$

- (D) At $t = 0$, the current through the resistance R is $\frac{V}{R}$



EM0213

24. The instantaneous voltages at three terminals marked X, Y and Z are given by

$$V_X = V_0 \sin \omega t$$

$$V_Y = V_0 \sin \left(\omega t + \frac{2\pi}{3} \right) \text{ and } V_Z = V_0 \sin \left(\omega t + \frac{4\pi}{3} \right)$$

An ideal voltmeter is configured to read rms value of the potential difference between its terminals. It is connected between points X and Y and then between Y and Z. The reading(s) of the voltmeter will be :-

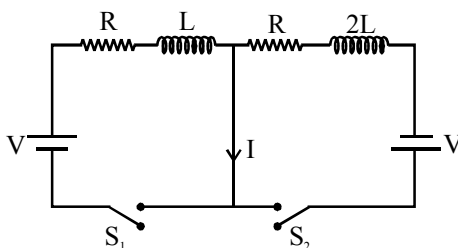
[JEE Advance-2017]

- (A) $V_{XY}^{\text{rms}} = V_0$ (B) $V_{YZ}^{\text{rms}} = V_0 \sqrt{\frac{1}{2}}$
 (C) Independent of the choice of the two terminals (D) $V_{XY}^{\text{rms}} = V_0 \sqrt{\frac{3}{2}}$

EM0214

25. In the figure below, the switches S_1 and S_2 are closed simultaneously at $t = 0$ and a current starts to flow in the circuit. Both the batteries have the same magnitude of the electromotive force (emf) and the polarities are as indicated in the figure. Ignore mutual inductance between the inductors. The current I in the middle wire reaches its maximum magnitude I_{max} at time $t = \tau$. Which of the following statement(s) is (are) true?

[JEE Advance-2018]



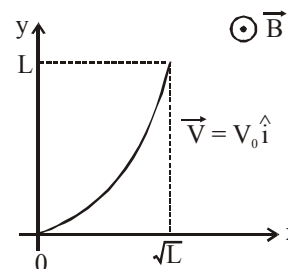
- (A) $I_{\text{max}} = \frac{V}{2R}$ (B) $I_{\text{max}} = \frac{V}{4R}$ (C) $\tau = \frac{L}{R} \ln 2$ (D) $\tau = \frac{2L}{R} \ln 2$

EM0215

26. A conducting wire of parabolic shape, initially $y = x^2$, is moving with velocity $\vec{V} = V_0 \hat{i}$ in a non-uniform magnetic field $\vec{B} = B_0 \left(1 + \left(\frac{y}{L} \right)^\beta \right) \hat{k}$, as shown in figure. If V_0 , B_0 , L and β are positive constants and $\Delta\phi$ is the potential difference developed between the ends of the wire, then the correct statement(s) is/are:

[JEE Advance-2019]

- (1) $|\Delta\phi|$ remains the same if the parabolic wire is replaced by a straight wire, $y = x$ initially, of length $\sqrt{2}L$
 (2) $|\Delta\phi|$ is proportional to the length of the wire projected on the y-axis.
 (3) $|\Delta\phi| = \frac{1}{2} B_0 V_0 L$ for $\beta = 0$
 (4) $|\Delta\phi| = \frac{4}{3} B_0 V_0 L$ for $\beta = 2$



EM0216

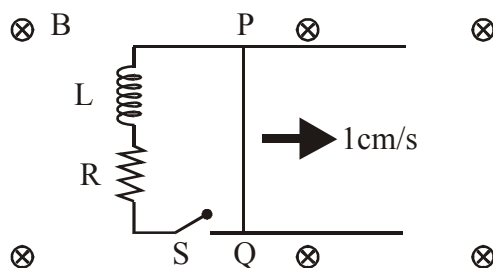
27. A 10 cm long perfectly conducting wire PQ is moving, with a velocity 1 cm/s on a pair of horizontal rails of zero resistance. One side of the rails is connected to an inductor $L = 1 \text{ mH}$ and a resistance $R = 1 \Omega$ as shown in figure. The horizontal rails, L and R lie in the same plane with a uniform magnetic field $B = 1 \text{ T}$ perpendicular to the plane. If the key S is closed at certain instant, the current in the circuit

1 millisecond is $x \times 10^{-3} \text{ A}$, where the value of x is _____.

[Assume the velocity of wire PQ remains constant (1 cm/s) after key S is closed.]

Given : $e^{-1} = 0.37$, where e is base of the natural logarithm]

[JEE Advance-2019]

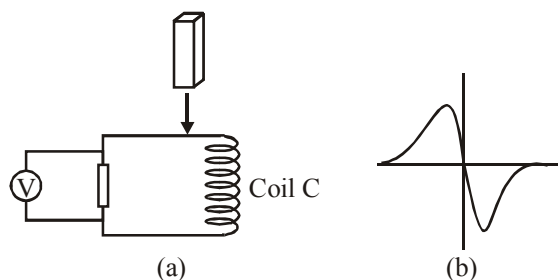


EM0217

ELECTROMAGNETIC INDUCTION & ALTERNATING CURRENT

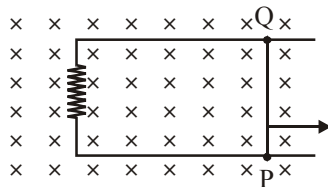
CBSE Previous Year's Questions

1. A solenoid with an iron core and a bulb are connected to a dc. source. How does the brightness of the bulb change, when the iron core is removed from the solenoid? [1; CBSE-2004]
2. Peak value of emf of an a.c. source is E_0 . What is its r.m.s. value? [1; CBSE-2004]
3. A bar magnet M is dropped so that it falls vertically through the coil C. The graph obtained for voltage produced across the coil vs time is shown in figure (b).
(i) Explain the shape of the graph.
(ii) Why is the negative peak longer than the positive peak? [2; CBSE-2004]



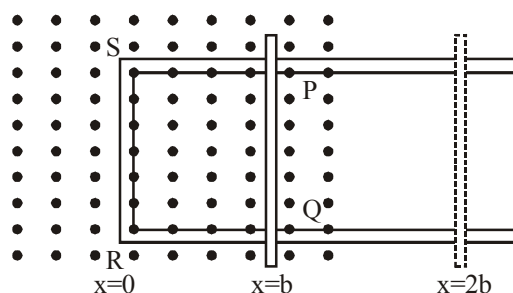
4. What is induced emf? Write Faraday's law of electromagnetic induction. Express it mathematically. A conducting rod of length ℓ with one end pivoted is rotated with a uniform angular speed in a vertical plane, normal to a uniform magnetic field 'B'. Deduce an expression for the emf induced in this rod. In India, domestic power supply is at 220 V, 50 Hz, while in USA it is 110 V, 50 Hz. Give one advantage and one disadvantage of 220 V supply over 110 V supply. [5; CBSE-2004]
5. A bulb and a capacitor are connected in series to an a.c. source of variable frequency. How will the brightness of the bulb change on increasing the frequency of the a.c. source? Give reason. [1; CBSE-2005]
6. A circular coil of radius 8 cm and 20 turns rotates about its vertical diameter with an angular speed of 50 s^{-1} in a uniform horizontal magnetic field of magnitude $3 \times 10^{-2} \text{ T}$. Find the maximum and average value of the emf induced in the coil. [2; CBSE-2005]
7. State the condition under which the phenomenon of resonance occurs in a series LCR circuit. Plot a graph showing variation of current with frequency of a.c. source in a series LCR circuit [2; CBSE-2005]
8. Define self-inductance and give its S. I. unit. Derive an expression for self-inductance of a long, air-cored solenoid of length ℓ , radius r , and having N number of turns. [3; CBSE-2005]
9. An alternating voltage of frequency f is applied across a series LCR circuit. Let f_r be the resonance frequency for the circuit. Will the current in the circuit lag, lead or remain in phase with the applied voltage when (i) $f > f_r$, (ii) $f < f_r$? Explain your answer in each case. [2; CBSE-2006]
10. When an inductor L and a resistor R in series are connected across a 12 V, 50 Hz supply, a current of 0.5 A flows in the circuit. The current differs in phase from applied voltage by $\pi/3$ radian. Calculate the value of R . [3; CBSE-2006]

11. A 0.5 m long metal rod PQ completes the circuit as shown in the figure. The area of the circuit is perpendicular to the magnetic field of flux density 0.15 T. If the resistance of the total circuit is 3Ω , calculate the force needed to move the rod in the direction as indicated with a constant speed of 2 ms^{-1} . [3 ; CBSE-2006]



12. What are eddy currents. How are these produced? in what sense are eddy currents considered undesirable in a transformer and how are these reduced in such a device? [3 ; CBSE-2006]
13. In a series LCR circuit, the voltages across an inductor, a capacitor and a resistor are 30 V, 30 V and 60 V respectively. What is the phase difference between the applied voltage and the current in the circuit ?
14. Calculate the current drawn by the primary of a transformer which steps down 200 V to 20 V to operate a device of resistance 20Ω . Assume the efficiency of the transformer to be 80%. [1 ; CBSE-2007]
15. An a.c. voltage of 100 V, 50 Hz is connected across a 20 ohm resistor and mH inductor in series. Calculate (i) impedance of the circuit, (ii) rms current in the circuit. [2 ; CBSE-2007]
16. Explain the term 'inductive reactance'. Show graphically the variation of inductive reactance with frequency of the applied alternating voltage. An a.c. voltage $E = E_0 \sin \omega t$ is applied across a pure inductor of inductance L. Show mathematically that the current flowing through it lags behind the applied voltage by a phase angle of $\pi/2$. [3 ; CBSE-2007]
17. Explain the term 'capacitive reactance'. Show graphically the variation of capacitive reactance with frequency of the applied alternating voltage. An a.c. voltage $E = E_0 \sin \omega t$ is applied across a pure capacitor of capacitance C. Show mathematically that the current flowing through it leads the applied voltage by a phase angle of $\pi/2$. [3 ; CBSE-2007]
18. Prove that an ideal capacitor, in an a. c. circuit does not dissipate power. [2 ; CBSE-2008]
19. Derive an expression for the impedance of a.c. circuit consisting of an inductor and a resistor. [2 ; CBSE-2008]
20. A metallic rod of length ℓ is rotated at a constant angular speed ω , normal to a uniform magnetic field B. Derive an expression for the current induced in the rod, if the resistance of the rod is R. [3 ; CBSE-2008]
21. An inductor 200mH, capacitor 500 μF , resistor 10Ω are connected in series with a 100 V, variable frequency ac. source. Calculate the [3 ; CBSE-2008]
- frequency at which the power factor of the circuit is unity
 - current amplitude at this frequency
 - Q-factor
22. (a) Define self inductance. Write its S.I. units.
(b) Derive an expression for self inductance of a long solenoid of length ℓ , cross-sectional area A having N number of turns. [3 ; CBSE-2009]

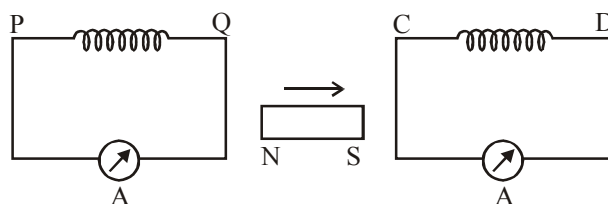
23. (a) Derive an expression for the average power consumed in a series LCR circuit connected to a.c. source, in which the phase difference between the voltage and the current in the circuit is ϕ .
 (b) Define the quality factor in an ac. circuit. Why should the quality factor have high value in receiving circuits? Name the factors on which it depends. [5; CBSE-2009]
24. (a) Derive the relationship between the peak, and the rms value of current in an ac. circuit, (b) Describe briefly, with the help of a labeled diagram, working of a step - up transformer. A step - up transformer converts a low voltage into high voltage. Does it not violate the principle of conservation of energy? Explain.
25. Define self-inductance of a coil. Write its S.I. units. [1; CBSE-2010]
26. Two identical loops, one of copper and the other of aluminium, are rotated with the same angular speed in the same magnetic field. Compare (i) the induced emf and (ii) the current produced in the two coils. Justify your answer. [2; CBSE-2010]
27. State Faraday's law of electromagnetic induction.
 Figure shows a rectangular conductor PQRS in which the conductor PQ is free to move in a uniform magnetic field B perpendicular to the plane of the paper. The field extends from $x = 0$ to $x = b$ and is zero for $x > b$. Assume that only the arm PQ possesses resistance r . When the arm PQ is pulled outward from $x=0$ with constant speed v , obtain the expressions for the flux and the induced emf. Sketch the variations of these quantities with distance $0 \leq X \leq 2b$. [5; CBSE-2010]



28. Draw a schematic diagram of a step-up transformer. Explain its working principle. Deduce the expression for the secondary to primary voltage in terms of the number of turns in the two coils. In an ideal transformer, how is this ratio related to the currents in the two coils? How is the transformer used in large scale transmission and distribution of electrical energy over long distances? [5; CBSE-2010]
29. What are eddy currents? Write any two applications of eddy currents. [2; CBSE-2011]
30. State the working of a.c. generator with the help of a labelled diagram. The coil of an a.c. generator having N turns, each of area A , is rotated with a constant angular velocity ω . Deduce the expression for the alternating e.m.f. generated in the coil. What is the source of energy generation in this device? [5; CBSE-2011]
31. Two bar magnets are quickly moved towards a metallic loop connected across a capacitor 'C' as shown in the figure. Predict the polarity of the capacitor. [1; CBSE-2011]



32. (a) Show that in an a.c. circuit containing a pure inductor, the voltage is ahead of current by $\pi/2$ in phase.
 (b) A horizontal straight wire of length L extending from east to west is falling with speed v at right angles to the horizontal component of Earth's magnetic field B .
 (i) Write the expression for the instantaneous value of the e.m.f. induced in the wire,
 (ii) What is the direction of the e.m.f. ?
 (iii) Which end of the wire is at the higher potential? **[5; CBSE-2011]**
33. A bar magnetic is moved in the direction indicated by the arrow between two coils PQ and CD. Predict the directions of induced current in each coil. **[1; CBSE-2012]**



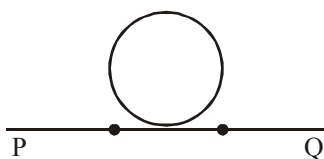
34. Mention the two characteristic properties of the material suitable for making core of a transformer. **[1; CBSE-2012]**
35. State the underlying principle of a transformer. How is the large scale transmission of electric energy over a long distances done with the use of transformers? **[2; CBSE-2012]**
36. A light bulb is rated 100 W for 220 V ac supply of 50 Hz. Calculate **[2; CBSE-2012]**
 (i) The resistance of the bulb (ii) The rms current through the bulb

OR

An alternative voltage given by $V = 140 \sin 314t$ is connected across a pure resistor of 50Ω . Find

- (i) the frequency of the source (ii) the rms current through the resistor **[2; CBSE-2012]**
37. A series LCR circuit is connected to an ac source. Using the phasor diagram, derive the expression for the source, explaining the nature of its variation. **[3; CBSE-2012]**
38. How does the mutual inductance of a pair of coils change when **[CBSE-2013]**
 (i) distance between the coils is increased and
 (ii) number of turns in the coils is increased?
39. The motion of copper plate is damped when it is allowed to oscillate between the two poles of a magnet. What is the cause of this damping? **[CBSE-2013]**
40. (a) For a given a.c, $i = i_m \sin \omega t$, show that the average power dissipated in a resistor R over a complete cycle is $\frac{1}{2} i_m^2 R$ **[CBSE-2013]**
 (b) A light bulb is rated at 100 W for a 220V a.c. supply. Calculate the resistance of the bulb.

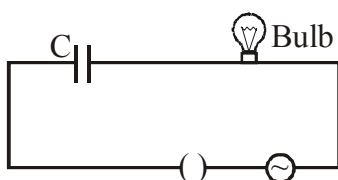
41. A conducting loop is held above a current carrying wire 'PQ' as shown in the figure. Depict the direction of the current induced in the loop when the current in the wire PQ is constantly increasing. [CBSE-2014]



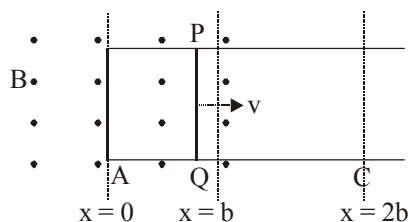
42. Why is the use of ac. voltage preferred over dc. voltage ? Give two reasons. [CBSE-2014]
43. A voltage $V = V_0 \sin \omega t$ is applied to a series LCR circuit. Derive the expression for the average power dissipated over a cycle. [CBSE-2014]
Under what condition is (i) no power dissipated even though the current flows through the circuit, (ii) maximum power dissipated in the circuit ? [CBSE-2014]
44. Define the term self-inductance of a solenoid. Obtain the expression for the magnetic energy stored in an inductor of self-inductance L to build up a current I through it. [CBSE-2014]
45. A planar loop of rectangular shape is moved within the region of a uniform magnetic field acting perpendicular to its plane. What is the direction and magnitude of the current induced in it ? [1; CBSE-2015]
46. Sunita and her friends visited an exhibition. The policeman asked them to pass through a metal detector. Sunita's friends were initially scared of it. Sunita, however, explained to them the purpose and working of the metal detector. [4; CBSE-2015]
Answer the following questions :
- (a) On what principle does a metal detector work ?
- (b) Why does the detector emit sound when a person carrying any metallic object walks through it ?
- (c) State any two qualities which Sunita displayed while explaining the purpose of walking through the detector. [4; CBSE-2015]
47. (a) State Faraday's law of electromagnetic induction. [5; CBSE-2015]
(b) Explain, with the help of a suitable example, how we can show that Lenz's law is a consequence of the principle of conservation of energy.
(c) Use the expression for Lorentz force acting on the charge carriers of a conductor to obtain the expression for the induced emf across the conductor of length l moving with velocity v through a magnetic field B acting perpendicular to its length. [5 ; CBSE-2015]

OR

- (a) Using phasor diagram, derive the expression for the current flowing in an ideal inductor connected to an a.c. source of voltage, $v = v_0 \sin \omega t$. Hence plot graphs showing variation of (i) applied voltage and (ii) the current as a function of ωt .
- (b) Derive an expression for the average power dissipated in a series LCR circuit.
48. (i) When an AC source is connected to an ideal capacitor, show that the average power supplied by the source over a complete cycle is zero. [3 ; CBSE-2016]
(ii) A bulb is connected in series with a variable capacitor and an A.C. source as shown. What happens to the brightness of the bulb when the key is plugged in and capacitance of the capacitor is gradually reduced ?



49. (b) Sketch the change in flux, emf and force when a conducting rod PQ of resistance R and length ℓ moves freely to and fro between A and C with speed v on a rectangular conductor placed in uniform magnetic field as shown in the figure. [5 ; CBSE-2016]

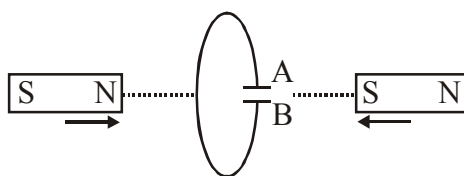


OR

In a series LCR circuit connected to an a.c. source of voltage $v = v_m \sin \omega t$, use phasor diagram to derive an expression for the current in the circuit.

Hence obtain the expression for the power dissipated in the circuit. Show that power dissipated at resonance is maximum.

50. Predict the polarity of the capacitor in the situation described below : [2; CBSE-2017]

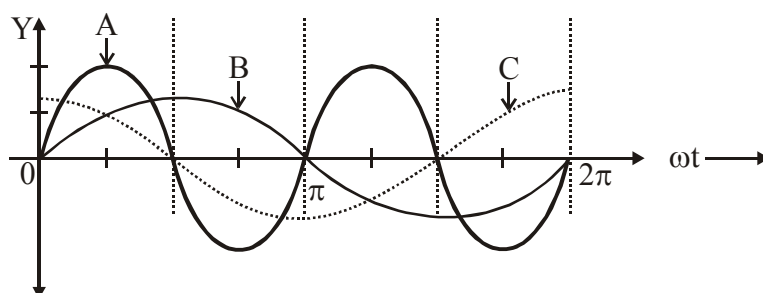


51. Define mutual inductance between a pair of coils. Derive an expression for the mutual inductance of two long coaxial solenoids of same length wound one over the other. [3; CBSE-2017]

OR

Define self-inductance of a coil. Obtain the expression for the energy stored in an inductor L connected across a source of emf.

52. A device 'X' is connected to an ac source $V = V_0 \sin \omega t$. The variation of voltage, current and power in one cycle is shown in the following graph : [5; CBSE-2017]



- Identify the device 'X'.
- Which of the curves A, B and C represent the voltage, current and the power consumed in the circuit? Justify your answer.
- How does its impedance vary with frequency of the ac source? Show graphically.
- Obtain an expression for the current in the circuit and its phase relation with ac voltage.

OR

- (a) Draw a labelled diagram of an ac generator. Obtain the expression for the emf induced in the rotating coil of N turns each of cross-sectional area A , in the presence of a magnetic field $\vec{B}$.
- (b) A horizontal conducting rod 10 m long extending from east to west is falling with a speed 5.0 ms^{-1} at right angles to the horizontal component of the Earth's magnetic field, $0.3 \times 10^{-4} \text{ Wb m}^{-2}$. Find the instantaneous value of the emf induced in the rod.
- 53.** The teachers of Geeta's school took the students on a study trip to a power generating station, located nearly 200 km away from the city. The teacher explained that electrical energy is transmitted over such a long distance to their city, in the form of alternating current (ac) raised to a high voltage. At the receiving end in the city, the voltage is reduced to operate the devices. As a result, the power loss is reduced. Geeta listened to the teacher and asked questions about how the ac is converted to a higher or lower voltage.
- (a) Name the device used to change the alternating voltage to a higher or lower value. State one cause for power dissipation in this device.
- (b) Explain with an example, how power loss is reduced if the energy is transmitted over long distances as an alternating current rather than a direct current.
- (c) Write two values each shown by the teachers and Geeta. **[4; CBSE-2018]**
- 54.** (a) State the principle of an ac generator and explain its working with the help of a labelled diagram. Obtain the expression for the emf induced in a coil having N turns each of cross-sectional area A , rotating with a constant angular speed ' ω ' in a magnetic field $\vec{B}$, directed perpendicular to the axis of rotation.
- (b) An aeroplane is flying horizontally from west to east with a velocity of 900 km/hour. Calculate the potential difference developed between the ends of its wings having a span of 20 m. The horizontal component of the Earth's magnetic field is $5 \times 10^{-4} \text{ T}$ and the angle of dip is 30° .

OR

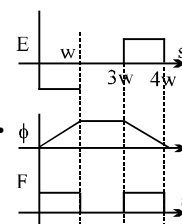
A device X is connected across an ac source of voltage $V = V_0 \sin \omega t$. the current through X is given

as $I = I_0 \sin \left(\omega t + \frac{\pi}{2} \right)$.

- (a) Identify the device X and write the expression for its reactance.
- (b) Draw graphs showing variation of voltage and current with time over one cycle of ac, for X.
- (c) How does the reactance of the device X vary with frequency of the ac? Show this variation graphically.
- (d) Draw the phasor diagram for the device X. **[5; CBSE-2018]**

ANSWER-KEY

EXERCISE (S-1)



1. Ans. $\lambda V_y B_0$ 2. Ans. (i) $2.4 \times 10^{-5} \text{ V}$ (ii) from c to b 3. Ans. 2 N

4. Ans.

5. Ans. $\frac{mgR}{B^2 \ell^2}$ 6. Ans. 0.75 T

7. Ans. $\frac{erk}{2m}$ directed along tangent to the circle of radius r, whose centre lies on the axis of cylinder.

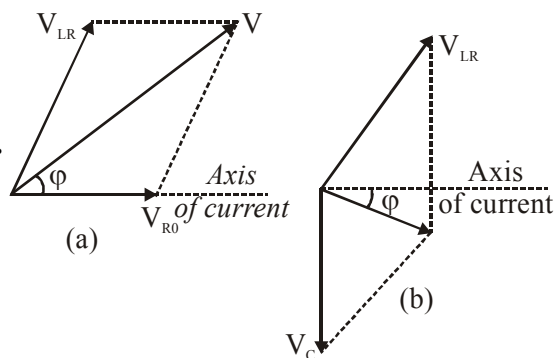
8. Ans. 6.5 V 9. Ans. $\frac{l}{2} \frac{dB}{dt} \sqrt{R^2 - \frac{l^2}{4}}$ 10. Ans. 0.8 11. Ans. $\frac{\mu_0 i a^2 \pi}{2Rb}$ 12. Ans. I^{-1}

13. Ans. $\frac{LE^2}{2R_1^2}$ 14. Ans. $\frac{e^2}{e^2 - 1}$ 15. Ans. $\frac{EL}{eR^2}$ 16. Ans. $q = Q_0 \sin\left(\sqrt{\frac{1}{LC}} t + \frac{\pi}{2}\right)$

17. Ans. (a) 10^4 A/s (b) 0 (c) 2A (d) $100\sqrt{3} \mu\text{C}$ 18. Ans. 30 Wb.

19. Ans. $\varepsilon = 1.7 \times 10^{-5} \text{ V}$

20. Ans.



21. Ans. 0.08 H, 17.28 W

22. Ans. 2A, 400W

23. Ans. 0.2 mH, $\frac{1}{32} \mu\text{F}$, $8 \times 10^5 \text{ rad/s}$

24. Ans. $\frac{20}{\pi^2} \cong 2\text{H}$

25. Ans. 20 V

26. Ans. 20 A, $\pi/4$,

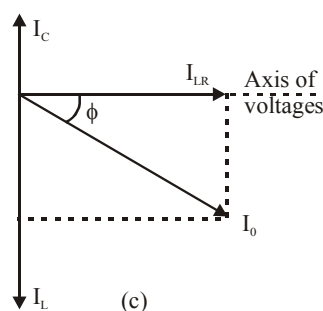
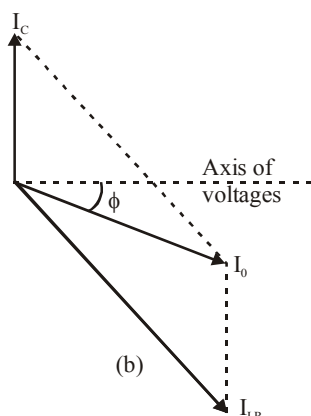
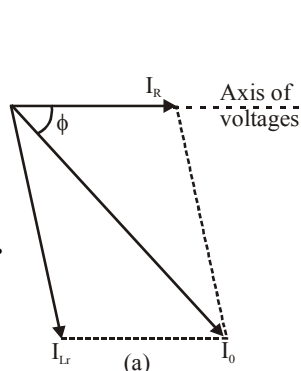
EXERCISE (S-2)

1. Ans. (i) 85.22 Tm^2 ; (ii) 56.8 V ; (iii) linearly2. Ans. $V = 1 \text{ ms}^{-1}$, $R_1 = 0.47 \Omega$, $R_2 = 0.30 \Omega$ 3. Ans. $\frac{\mu_0 h \omega i_m N}{2\pi} \ln \frac{b}{a}$ 4. Ans. $\frac{1}{3} A$ 5. Ans. $C \pi a^2 / R$ 6. Ans. 200 rad/sec 7. Ans. $-\frac{B \pi a^2 \lambda}{MR} \hat{k}$ 8. Ans. 0.4 V 9. Ans. $\frac{E R_1}{R_1 R_2 + R_2 R_3 + R_3 R_1}$ 10. Ans. $I_{EA} = \frac{7}{22} \text{ A}$; $I_{BE} = \frac{3}{11} \text{ A}$; $I_{FE} = \frac{1}{22} \text{ A}$ 11. Ans. (a) $i = \frac{B_0 a v}{R}$ in anticlockwise direction, $v =$ velocity at time t , (b) $F_{\text{net}} = B_0^2 a^2 v / R$,

$$(c) V = \frac{mgR}{B_0^2 a^2} \left(1 - e^{-\frac{B_0^2 a^2 t}{mR}} \right)$$

12. Ans. $67/32 \text{ A}$ 13. Ans. (i) $i_1 = i_2 = 10/3 \text{ A}$, (ii) $i_1 = 50/11 \text{ A}$; $i_2 = 30/11 \text{ A}$, (iii) $i_1 = 0$, $i_2 = 20/11 \text{ A}$, (iv) $i_1 = i_2 = 0$ 14. Ans. $42 + 20t \text{ volt}$ 15. Ans. (a) $E = \frac{1}{2} B \omega r^2$ (b) (i) $I = \frac{B \omega r^2 [1 - e^{-Rt/L}]}{2R}$, (ii) $\tau = \frac{mgr}{2} \cos \omega t + \frac{\omega B^2 r^4}{4R} (1 - e^{-Rt/L})$ 16. Ans. $2\pi \frac{\sqrt{mL}}{lB}$, $g \frac{\sqrt{mL}}{lB}$ 17. Ans. $-\frac{V}{R} e^{-\frac{Rt}{L}}$ 18. Ans. $\frac{1}{15} A, \frac{1}{10} A$ 19. Ans. $kMT^2/(R)$ 20. Ans. $77\Omega, 97.6\Omega, 7.7 \text{ V}, 9.76 \text{ V}$

21. Ans.

22. Ans. $I = \frac{(\mu_0 n i_0 \omega \cos \omega t) \pi a^2 (Ld)}{\rho 2\pi R}$

EXERCISE (O-1)

- | | | | | | |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 1. Ans. (C) | 2. Ans. (A) | 3. Ans. (D) | 4. Ans. (C) | 5. Ans. (A) | 6. Ans. (A) |
| 7. Ans. (B) | 8. Ans. (C) | 9. Ans. (A) | 10. Ans. (B) | 11. Ans. (A) | 12. Ans. (C) |
| 13. Ans. (A) | 14. Ans. (D) | 15. Ans. (C) | 16. Ans. (A) | 17. Ans. (A) | 18. Ans. (D) |
| 19. Ans. (D) | 20. Ans. (B) | 21. Ans. (D) | 22. Ans. (B) | 23. Ans. (D) | 24. Ans. (A) |
| 25. Ans. (C) | 26. Ans. (A) | 27. Ans. (C) | 28. Ans. (A) | 29. Ans. (B) | 30. Ans. (C) |
| 31. Ans. (A) | 32. Ans. (C) | 33. Ans. (B) | 34. Ans. (D) | 35. Ans. (B) | 36. Ans. (B) |
| 37. Ans. (A) | 38. Ans. (D) | 39. Ans. (A) | 40. Ans. (A) | 41. Ans. (C) | 42. Ans. (B) |
| 43. Ans. (A) | 44. Ans. (B) | 45. Ans. (A) | 46. Ans. (D) | 47. Ans. (A) | 48. Ans. (D) |
| 49. Ans. (B) | 50. Ans. (C) | 51. Ans. (A) | 52. Ans. (B) | 53. Ans. (D) | 54. Ans. (D) |
| 55. Ans. (D) | 56. Ans. (D) | 57. Ans. (D) | 58. Ans. (A) | 59. Ans. (C) | 60. Ans. (B) |
| 61. Ans. (A) | 62. Ans. (B) | 63. Ans. (D) | 64. Ans. (C) | | |

SUPPLEMENT FOR JEE-MAINS

- | | | | | | |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 65. Ans. (B) | 66. Ans. (B) | 67. Ans. (A) | 68. Ans. (D) | 69. Ans. (D) | 70. Ans. (C) |
| 71. Ans. (C) | 72. Ans. (B) | 73. Ans. (D) | 74. Ans. (A) | 75. Ans. (A) | 76. Ans. (A) |
| 77. Ans. (D) | | | | | |

MULTIPLE CORRECT TYPE QUESTIONS

- | | | | | | |
|--------------------|-----------------|-----------------|-----------------|--------------|--------------|
| 78. Ans. (B) | 79. Ans. (A) | 80. Ans. (B, C) | 81. Ans. (A, B) | 82. Ans. (B) | 83. Ans. (A) |
| 84. Ans. (D) | 85. Ans. (B, D) | 86. Ans. (B, D) | 87. Ans. (A) | 88. Ans. (B) | |
| 89. Ans. (A, C, D) | 90. Ans. (A, C) | | | | |

COMPREHENSION TYPE QUESTIONS

- | | | | | | |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 91. Ans. (D) | 92. Ans. (B) | 93. Ans. (C) | 94. Ans. (C) | 95. Ans. (C) | 96. Ans. (B) |
| 97. Ans. (B) | | | | | |

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

- | | | | | | |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 1. Ans. (A) | 2. Ans. (B) | 3. Ans. (B) | 4. Ans. (A) | 5. Ans. (C) | 6. Ans. (D) |
| 7. Ans. (D) | 8. Ans. (A) | 9. Ans. (D) | 10. Ans. (A) | 11. Ans. (B) | 12. Ans. (A) |
| 13. Ans. (A) | 14. Ans. (C) | 15. Ans. (A) | 16. Ans. (C) | 17. Ans. (D) | 18. Ans. (A) |
| 19. Ans. (D) | 20. Ans. (D) | 21. Ans. (D) | 22. Ans. (B) | 23. Ans. (A) | 24. Ans. (D) |

MULTIPLE CORRECT TYPE QUESTIONS

- | | | | | |
|--------------------|-----------------|-----------------|--------------------|-----------------|
| 25. Ans. (A, B, C) | 26. Ans. (A, D) | 27. Ans. (A, B) | 28. Ans. (A, C, D) | 29. Ans. (B, D) |
| 30. Ans. (D) | 31. Ans. (B) | | | |

COMPREHENSION TYPE QUESTIONS

- | | | | | | |
|--------------|--------------|-----------------|--------------|--------------|--------------|
| 32. Ans. (D) | 33. Ans. (B) | 34. Ans. (B, D) | 35. Ans. (C) | 36. Ans. (D) | 37. Ans. (A) |
| 38. Ans. (B) | 39. Ans. (A) | | | | |

EXERCISE-JM

- | | | | | | |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 1. Ans. (3) | 2. Ans. (3) | 3. Ans. (1) | 4. Ans. (2) | 5. Ans. (2) | 6. Ans. (4) |
| 7. Ans. (1) | 8. Ans. (2) | 9. Ans. (2) | 10. Ans. (4) | 11. Ans. (1) | 12. Ans. (3) |
| 13. Ans. (1) | 14. Ans. (2) | 15. Ans. (3) | 16. Ans. (1) | 17. Ans. (1) | 18. Ans. (4) |
| 19. Ans. (1) | | | | | |

EXERCISE-JA

- | | | | | | |
|-----------------------|------------------|--|----------------|----------------|----------------|
| 1. Ans. (B) | 2. Ans. (C) | 3. Ans. (A)-R,S,T; (B)-Q,R,S,T; (C)-P,Q; (D)-Q,R,S,T | | | |
| 4. Ans. (C) | 5. Ans. (B,C) | 6. Ans. 6 | 7. Ans. 4 | 8. Ans. 7 | 9. Ans. (A,C) |
| 10. Ans. (C) or (A,C) | | 11. Ans. (B) | 12. Ans. (B) | 13. Ans. (A) | 14. Ans. (B) |
| 15. Ans. (C,D) | 16. Ans. (A,D) | 17. Ans. (A,C) | 18. Ans. (B,D) | 19. Ans. 8 | 20. Ans. (C,D) |
| 21. Ans. (A,D) | 22. Ans. (A,B) | 23. Ans. (A,B,C) | | 24. Ans. (C,D) | |
| 25. Ans. (B,D) | 26. Ans. (1,2,4) | 27. Ans. (0.63) | | | |

S. No.	CHAPTER NAME	Pg.No.
1	ELASTICITY, THERMAL EXPANSION, CALORIMETRY & HEAT TRANSFER	01-74
2	KINETIC THEORY OF GASES & THERMODYNAMICS	75-148
3	FLUID MECHANICS	149-234



PHYSICS



Chapter 01 Elasticity, Thermal Expansion, Calorimetry & Heat Transfer

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Important Notes

[illegible]

ELASTICITY

A body is said to be rigid if the relative positions of its constituent particles remains unchanged when external deforming forces are applied to it. The nearest approach to a rigid body is diamond or carborundum.

Actually no body is perfectly rigid and every body can be deformed more or less by the application of suitable forces. All these deformed bodies however regain their original shape or size, when the deforming forces are removed.

The property of matter by virtue of which a body tends to regain its original shape and size after the removal of deforming forces is called elasticity.

SOME TERMS RELATED TO ELASTICITY :

- **Deforming Force**

External force which try to change in the length, volume or shape of the body is called deforming force.

- **Perfectly Elastic Body**

The body which perfectly regains its original form on removing the external deforming force, is defined as a perfectly elastic body. Ex. : quartz – Very nearly a perfect elastic body.

- **Plastic Body**

(a) The body which does not have the property of opposing the deforming force, is known as a plastic body.

(b) The bodies which remain in deformed state even after removed of the deforming force are defined as plastic bodies.

- **Internal restoring force**

When a external force acts at any substance then due to the intermolecular force there is a internal resistance produced into the substance called internal restoring force.

At equilibrium the numerical value of internal restoring force is equal to the external force.

STRESS

The internal restoring force acting per unit area of cross-section of the deformed body is called stress.

$$\text{Stress} = \frac{\text{Internal restoring force}}{\text{Area of cross section}} = \frac{F_{\text{internal}}}{A} = \frac{F_{\text{external}}}{A}$$

Stress depends on direction of force as well as direction of area of application so it is tensor.

SI UNIT : N-m^{-2}

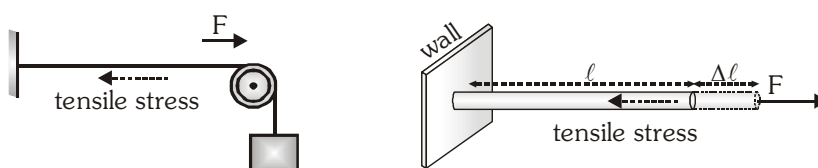
Dimensions : $\text{M}^1 \text{L}^{-1} \text{T}^{-2}$

There are three types of stress :-

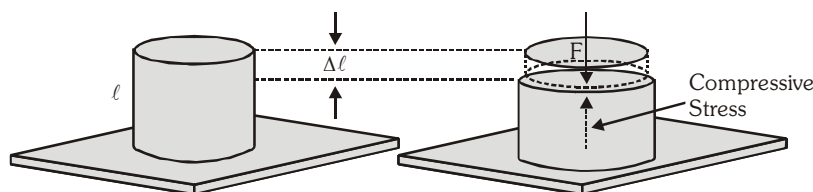
- **Longitudinal Stress**

When the stress is normal to the surface of body, then it is known as longitudinal stress. There are two types of longitudinal stress

(a) **Tensile Stress** : The longitudinal stress, produced due to increase in length of a body, is defined as tensile stress.



(b) Compressive Stress : The longitudinal stress, produced due to decrease in length of a body, is defined as compressive stress.

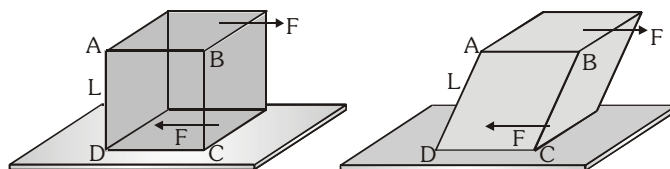


(c) Volume Stress

If equal normal forces are applied every one surface of a body, then it undergoes change in volume. The force opposing this change in volume per unit area is defined as volume stress.

(d) Tangential Stress or Shear Stress

When the stress is tangential or parallel to the surface of a body then it is known as shear stress. Due to this stress, the shape of the body changes or it gets twisted.



STRAIN

- The ratio of change of any dimension to its original dimension is called strain.

$$\text{Strain} = \frac{\text{change in size of the body}}{\text{original size of the body}}$$

- It is a unitless and dimensionless quantity.
- There are three types of strain : Type of strain depends upon the directions of applied force.

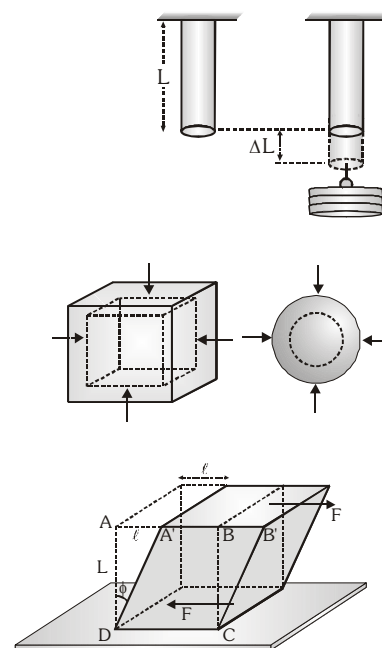
- Longitudinal strain** = $\frac{\text{change in length of the body}}{\text{initial length of the body}} = \frac{\Delta L}{L}$

- Volume strain** = $\frac{\text{change in volume of the body}}{\text{original volume of the body}} = \frac{\Delta V}{V}$

- Shear strain**

When a deforming force is applied to a body parallel to its surface then its shape (not size) changes. The strain produced in this way is known as shear strain. The strain produced due to change of shape of the body is known as shear strain.

$$\tan \phi = \frac{\ell}{L} \quad \text{or} \quad \phi = \frac{\ell}{L} = \frac{\text{displacement of upper face}}{\text{distance between two faces}}$$

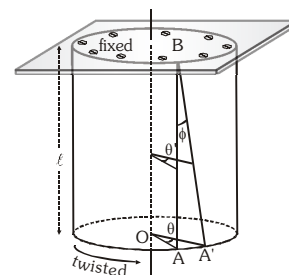


Relation Between angle of twist and Angle of shear

When a cylinder of length ' ℓ ' and radius ' r ' is fixed at one end and tangential force is applied at the other end, then the cylinder gets twisted. Figure shows the angle of shear $\angle ABA'$ and angle of twist $\angle AOA'$. Arc $AA' = r\theta$ and Arc $AA' = \ell\phi$ so $r\theta = \ell\phi$

$$\Rightarrow \phi = \frac{r\theta}{\ell}$$

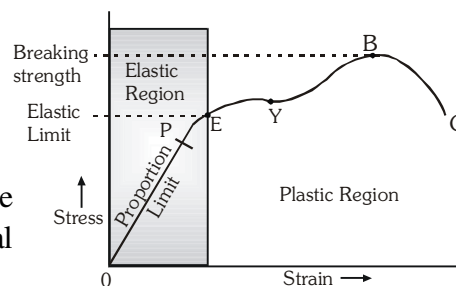
where θ = angle of twist, ϕ = angle of shear



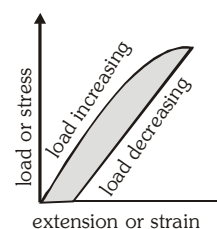
- When a material is under tensile stress restoring force are caused by intermolecular attraction while under compressive stress, the restoring force are due to intermolecular repulsion.
- If the deforming force is inclined to the surface at an angle θ such that $\theta \neq 0$ and $\theta \neq 90^\circ$ then both tangential and normal stress are developed.
- Linear strain in the direction of force is called **longitudinal strain** while in a direction perpendicular to force **lateral strain**.

STRESS – STRAIN GRAPH

- **Proportion Limit** : The limit in which Hooke's law is valid and stress is directly proportional to strain is called proportion limit. $\text{Stress} \propto \text{Strain}$
- **Elastic limit** : That maximum stress which on removing the deforming force makes the body to recover completely its original state.
- **Yield Point** : The point beyond elastic limit, at which the length of wire starts increasing without increasing stress, is defined as the yield point.
- **Breaking Point** : The position when the strain becomes so large that the wire breaks down at last, is called breaking point. At this position the stress acting in that wire is called breaking stress and strain is called breaking strain.



- **Elastic Hysteresis** : The strain persists even when the stress is removed. This lagging behind of strain is called elastic hysteresis. This is the reason why the values of strain for same stress are different while increasing the load and while decreasing the load.



- **Breaking Stress** : The stress required to cause actual fracture of a material is called the breaking stress. $\text{Breaking stress} = F/A$

- Breaking stress also measures the tensile strength.
- Metals with small plastic deformation are called brittle.
- Metals with large plastic deformation are called ductile.
- Elasticity restoring forces are strictly conservative only when the elastic hysteresis is zero. i.e. the loading and unloading stress – strain curves are identical.

Ex. Find out longitudinal stress and tangential stress on a fixed block.

Sol. Longitudinal or normal stress $\sigma_1 = \frac{100 \sin 30^\circ}{5 \times 2} = 5 \text{ N/m}^2$

$$\text{Tangential stress } \sigma_2 = \frac{100 \cos 30^\circ}{5 \times 2} = 5\sqrt{3} \text{ N/m}^2$$

Ex. The breaking stress of aluminium is $7.5 \times 10^8 \text{ dyne cm}^{-2}$. Find the greatest length of aluminium wire that can hang vertically without breaking. Density of aluminium is 2.7 g cm^{-3} .

Given : $g = 980 \text{ cm s}^{-2}$.

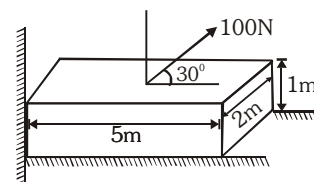
Sol. Let ℓ be the greatest length of the wire that can hang vertically without breaking.

Mass of wire $m = \text{cross-sectional area (A)} \times \text{length } (\ell) \times \text{density } (\rho)$, Weight of wire $= mg = A \ell \rho g$

This is equal to the maximum force that the wire can withstand.

$$\therefore \text{Breaking stress} = \frac{\ell A \rho g}{A} = \ell \rho g \Rightarrow 7.5 \times 10^8 = \ell \times 2.7 \times 980$$

$$\Rightarrow \ell = \frac{7.5 \times 10^8}{2.7 \times 980} \text{ cm} = 2.834 \times 10^5 \text{ cm} = 2.834 \text{ km}$$



HOOKE'S LAW

If the deformation is small, the stress in a body is proportional to the corresponding strain, this fact is

known as Hooke's Law. Within elastic limit : stress $\propto$ strain $\Rightarrow \frac{\text{stress}}{\text{strain}} = \text{constant}$

This constant is known as modulus of elasticity or coefficient of elasticity.

The modulus of elasticity depends only on the type of material used. It does not depend upon the value of stress and strain.

YOUNG'S MODULUS OF ELASTICITY 'Y'

- Within elastic limit the ratio of longitudinal stress and longitudinal strain is called Young's modulus of elasticity.

$$\bullet \quad Y = \frac{\text{longitudinal stress}}{\text{longitudinal strain}} = \frac{F/A}{\ell/L} = \frac{FL}{\ell A}$$

- Within elastic limit the force acting upon a unit area of a wire by which the length of a wire becomes double, is equivalent to the Young's modulus of elasticity of material of a wire.
- If L is the length of wire, r is radius and ℓ is the increase in length of the wire by suspending a weight

$$Mg \text{ at its one end then Young's modulus of elasticity of the material of wire } Y = \frac{(Mg/\pi r^2)}{(\ell/L)} = \frac{MgL}{\pi r^2 \ell}$$

- Unit of Y : N/m^2 • Dimensions of Y : $\text{M}^1\text{L}^{-1}\text{T}^{-2}$

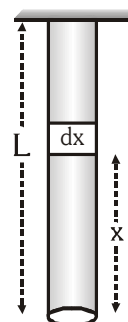
Increment of length due to own weight

Let a rope of mass M and length L is hanged vertically. As the tension of different point on the rope is different. Stress as well as strain will be different at different point.

(i) maximum stress at hanging point (ii) minimum stress at lower point

Consider a dx element of rope at x distance from lower end then tension $T = \left(\frac{M}{L}\right) x g$

$$\text{So stress} = \frac{T}{A} = \left(\frac{M}{L}\right) \frac{xg}{A}$$



Let increase in length of dx is dy then strain = $\frac{dy}{dx}$

So Young modulus of elasticity $Y = \frac{\text{stress}}{\text{strain}} = \frac{\frac{M}{L} \frac{gx}{A}}{\frac{dy}{dx}} \Rightarrow \left(\frac{M}{L}\right) \frac{gx}{A} dx = Y dy$

For full length of rope $\frac{Mg}{LA} \int_0^L x dx = Y \int_0^{\Delta \ell} dy \Rightarrow \frac{Mg}{LA} \frac{L^2}{2} = Y \Delta \ell \Rightarrow \Delta \ell = \frac{MgL}{2AY}$

[Since the stress is varying linearly we may apply average method to evaluate strain.]

BULK MODULUS OF ELASTICITY 'K' or 'B'

- Within elastic limit the ratio of the volume stress and the volume strain is called bulk modulus of elasticity.

$$K \text{ or } B = \frac{\text{volume stress}}{\text{volume strain}} = \frac{F/A}{-\frac{\Delta V}{V}} = \frac{\Delta P}{-\frac{\Delta V}{V}}$$

The minus sign indicates a decrease in volume with an increase in stress.

- Unit of K : $N m^{-2}$ or pascal
- Bulk modulus of an ideal gas is process dependence.

- For isothermal process $PV = \text{constant} \Rightarrow PdV + VdP = 0 \Rightarrow P = \frac{-dP}{dV/V}$ So bulk modulus = P

- For adiabatic process $PV^\gamma = \text{constant} \Rightarrow \gamma PV^{\gamma-1}dV + V^\gamma dP = 0$

$$\Rightarrow \gamma PdV + VdP = 0 \Rightarrow \gamma P = \frac{-dP}{dV/V} \text{ So bulk modulus} = \gamma P$$

- For any polytropic process $PV^n = \text{constant}$

$$\Rightarrow nPV^{n-1}dV + V^n dP = 0 \Rightarrow PdV + VdP = 0 \Rightarrow nP = \frac{-dP}{dV/V} \text{ So bulk modulus} = nP$$

- **Compressibility** : The reciprocal of bulk modulus of elasticity is defined as compressibility. $C = \frac{1}{K}$

MODULUS OF RIGIDITY ' η '

- Within elastic limit the ratio of shearing stress and shearing strain is called modulus of rigidity of a material.

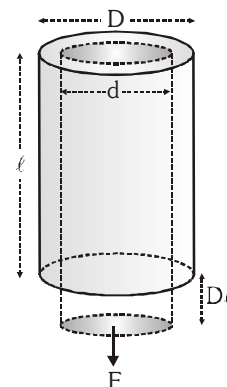
$$\eta = \frac{\text{shearing stress}}{\text{Shearing strain}} = \left(\frac{\frac{F_{\text{tan gential}}}{A}}{\phi} \right) = \frac{F_{\text{tan gential}}}{A\phi}$$

Note : Angle of shear ' ϕ ' always taking in radian

Poisson's Ratio (σ)

In elastic limit, the ratio of lateral strain and longitudinal strain is called Poisson's ratio.

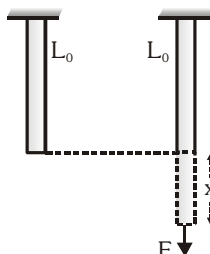
$$\sigma = \frac{\text{lateral strain}}{\text{longitudinal strain}} = \frac{\beta}{\alpha}; \beta = \frac{-\Delta D}{D} = \frac{d-D}{D} \text{ \& } \alpha = \frac{\Delta L}{L}$$



WORK DONE IN STRETCHING A WIRE

(Potential energy of a stretched wire)

When a wire is stretched, work is done against the interatomic forces, which is stored in the form of elastic potential energy.



For a wire of length L_0 stretched by a distance x , the restoring elastic force is :

$$F = \text{stress} \times \text{area} = Y \left[\frac{x}{L_0} \right] A$$

The work has to be done against the elastic restoring forces in stretching dx

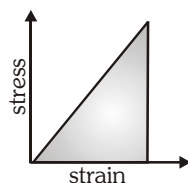
$$dW = Fdx = \frac{YA}{L_0} xdx$$

The total work done in stretching the wire from $x = 0$ to $x = \Delta \ell$ is, then

$$W = \int_0^{\Delta\ell} \frac{YA}{L_o} x dx = \frac{YA}{L_o} \left[\frac{x^2}{2} \right]_0^{\Delta\ell} = \frac{YA(\Delta\ell)^2}{2L_o}$$

$$W = \frac{1}{2} \times Y \times (\text{strain})^2 \times \text{original volume} = \frac{1}{2} (\text{stress}) (\text{strain}) (\text{volume})$$

- The value of K is maximum for solids and minimum for gases.



- For any ideal rigid body all three elastic modulus are infinite .
- η is the characteristic of solid material only as the fluids do not have fixed shape.
- Potential energy density = area under the stress–strain curve.
- Young's modulus = Slope of the stress–strain curve

Ex. A steel wire of 4.0 m in length is stretched through 2.00 mm. The cross-sectional area of the wire is 2.0 mm^2 . If Young's modulus of steel is $2.0 \times 10^{11} \text{ N/m}^2$ find (i) the energy density of wire (ii) the elastic potential energy stored in the wire.

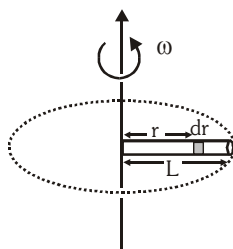
Sol. (i) The energy density of stretched wire = $\frac{1}{2} \times \text{stress} \times \text{strain} = \frac{1}{2} \times Y \times (\text{strain})^2$

$$= \frac{1}{2} \times 2.0 \times 10^{11} \times \left[\frac{2 \times 10^{-3}}{4} \right]^2 = 2.5 \times 10^4 \text{ J/m}^3$$

(ii) Elastic potential energy = energy density $\times$ volume = $2.5 \times 10^4 \times (2.0 \times 10^{-6}) \times 4.0 \text{ J}$
 $= 20 \times 10^{-2} = 0.20 \text{ J}$

Ex. A thin uniform metallic rod of length 0.5 m and radius 0.1 m rotates with an angular velocity 400 rad/s in a horizontal plane about a vertical axis passing through one of its ends. Calculate (a) tension in the rod and (b) the elongation of the rod. The density of material of the rod is 10^4 kg/m^3 and the Young's modulus is $2 \times 10^{11} \text{ N/m}^2$.

Sol. (a) Consider an element of length dr at a distance r from the axis of rotation as shown in figure. The centripetal force acting on this element will be $dT = dm r \omega^2 = (\rho A dr) r \omega^2$. As this force is provided by tension in the rod (due to elasticity), so the tension in the rod at a distance r from the axis of rotation will be due to the centripetal force due to all elements between $x = r$ to $x = L$



$$\text{i.e., } T = \int_r^L \rho A \omega^2 r dr = \frac{1}{2} \rho A \omega^2 [L^2 - r^2] \dots (i)$$

$$\text{So here } T = \frac{1}{2} \times 10^4 \times \pi \times 10^{-2} \times (400)^2 \left[\left(\frac{1}{2} \right)^2 - r^2 \right] = 8\pi \times 10^6 \left[\frac{1}{4} - r^2 \right] \text{ N}$$

(b) Now if dy is the elongation in the element of length dr at position r then strain

$$\frac{dy}{dr} = \frac{\text{stress}}{Y} = \frac{T}{AY} = \frac{1}{2} \frac{\rho \omega^2}{Y} [L^2 - r^2] dr$$

So the elongation of the whole rod

$$\Delta L = \frac{\rho \omega^2}{2Y} \int_0^L (L^2 - r^2) dr = \frac{1}{3} \frac{\rho \omega^2 L^3}{Y} = \frac{1}{3} \times \frac{10^4 \times (400)^2 (0.5)^3}{2 \times 10^{11}} = \frac{1}{3} \times 10^{-3} \text{ m}$$

Ex. Find the depth of lake at which density of water is 1% greater than at the surface.

Given compressibility $K = 50 \times 10^{-6} / \text{atm}$.

Sol. $B = \frac{\Delta P}{-\Delta V/V} \Rightarrow \frac{\Delta V}{V} = -\frac{\Delta P}{B}$

We know $P = P_{\text{atm}} + h\rho g$ and $m = \rho V = \text{constant}$

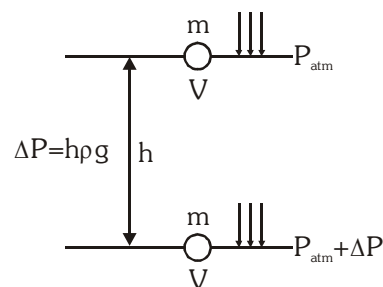
$$d\rho V + dV\rho = 0 \Rightarrow \frac{d\rho}{\rho} = -\frac{dV}{V} \text{ i.e. } \frac{\Delta\rho}{\rho} = \frac{\Delta P}{B}$$

$$\Rightarrow \frac{\Delta\rho}{\rho} = \frac{1}{100} = \frac{h\rho g}{B}$$

$$[\text{assuming } \rho = \text{constant}]; h\rho g = \frac{B}{100} = \frac{1}{100K}$$

$$\Rightarrow h\rho g = \frac{1 \times 1 \times 10^5}{100 \times 50 \times 10^{-6}}$$

$$h = \frac{10^5}{5000 \times 10^{-6} \times 1000 \times 10} = \frac{100 \times 10^3}{50} \text{ m} = 2 \text{ km}$$

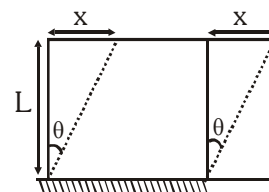


Ex. A rubber cube of side 5 cm has one side fixed while a tangential force equal to 1800 N is applied to opposite face. Find the shearing strain and the lateral displacement of the strained face. Modulus of rigidity for rubber is $2.4 \times 10^6 \text{ N/m}^2$.

Sol. Stress = $\frac{F}{A} = \eta \frac{x}{L}$

$$\text{Strain} = \theta = \frac{F}{A\eta} = \frac{1800}{25 \times 10^{-4} \times 2.4 \times 10^6} = \frac{180}{25 \times 24} = \frac{3}{10} = 0.3 \text{ radian}$$

$$\text{as } \frac{x}{L} = 0.3 \Rightarrow x = 0.3 \times 4 \times 10^{-2} = 1.5 \times 10^{-2} \text{ m} = 1.5 \text{ mm}$$



EFFECT OF TEMPERATURE ON ELASTICITY

When temperature is increased then due to weakness of inter molecular force the elastic properties in general decreases i.e. elastic constant decreases. Plasticity increases with temperature. For example, at ordinary room temperature, carbon is elastic but at high temperature, carbon becomes plastic. Lead is not much elastic at room temperature but when cooled in liquid nitrogen exhibit highly elastic behaviour. For a special kind of steel, elastic constants do not vary appreciably temperature. This steel is called 'INVAR steel'.

EFFECT OF IMPURITY ON ELASTICITY

Y is slightly increase by impurity. The inter molecular attraction force inside wire effectively increase by impurity due to this external force can be easily opposed.

TEMPERATURE & DIFFERENT TYPE OF TEMPERATURE SCALES

TEMPERATURE

- Temperature is a macroscopic physical quantity related to our sense of hot and cold.
- The natural flow of heat is from higher temperature to lower temperature, i.e. temperature determines the thermal state of a body whether it can give or receive heat.

DIFFERENT TYPES OF TEMPERATURE SCALES

- The Kelvin temperature scale is also known as thermodynamic scale. The SI unit of temperature is the kelvin and is defined as $(1/273.16)$ of the temperature of the triple point of water. The triple point of water is that point on a P–T diagram where the three phase of water, the solid, the liquid and the gas, can coexist in equilibrium.
- In addition to Kelvin temperature scale, there are other temperature scales also like Celsius, Fahrenheit, Reaumur, Rankine, etc. Temperature on one scale can be converted into other scale by using the following identity

$$\frac{\text{Reading on any scale} - \text{lower fixed point (LFP)}}{\text{Upper fixed point (UFP)} - \text{lower fixed point (LFP)}} = \text{constant for all scales}$$

Hence $\frac{C - 0^\circ}{100^\circ - 0^\circ} = \frac{F - 32^\circ}{212^\circ - 32^\circ} = \frac{K - 273.15}{373.15 - 273.15}$

- Different temperature scales :

Name of the scale	Symbol for each degree	Lower fixed point (LFP)	Upper fixed point (UFP)	Number of divisions on the scale
Celsius	°C	0°C	100°C	100
Fahrenheit	°F	32°F	212°F	180
Kelvin	K	273.15 K	373.15 K	100

Ex. Express a temperature of 60°F in degree celsius and in kelvin.

Sol. By using $\frac{C - 0^{\circ}}{100^{\circ} - 0^{\circ}} = \frac{F - 32^{\circ}}{212^{\circ} - 32^{\circ}} = \frac{K - 273.15}{373.15 - 273.15}$

$$\Rightarrow \frac{C - 0^\circ}{100^\circ - 0^\circ} = \frac{60^\circ - 32^\circ}{212^\circ - 32^\circ} = \frac{K - 273.15}{373.15 - 273.15} \Rightarrow C = 15.15^\circ \text{ C and } K = 288.7 \text{ K}$$

Ex. The temperature of an iron piece is heated from 30° to 90°C . What is the change in its temperature on the fahrenheit scale and on the kelvin scale?

Sol. $\Delta C = 90^\circ - 30^\circ = 60^\circ \text{C}$

Temperature difference on Fahrenheit Scale $\Delta F = \frac{9}{5} \Delta C = \frac{9}{5} (60^\circ\text{C}) = 108^\circ\text{F}$

Temperature difference on Kelvin Scale $\Delta K = \Delta C = 60K$

THERMAL EXPANSION

THERMAL EXPANSION

When matter is heated without any change in its state, it usually expands. According to atomic theory of matter, asymmetry in potential energy curve is responsible for thermal expansion. As with rise in temperature the amplitude of vibration increases and hence energy of atoms increases, hence the average distance between the atoms increases. So the matter as a whole expands.

- Thermal expansion is minimum in case of solids but maximum in case of gases because intermolecular force is maximum in solids but minimum in gases.
- Solids can expand in one dimension (Linear expansion), two dimension (Superficial expansion) and three dimension (Volume expansion) while liquids and gases usually suffer change in volume only.

Linear expansion :

$$\ell = \ell_0 (1 + \alpha \Delta\theta) \Rightarrow \Delta\ell = \ell_0 \alpha \Delta\theta$$

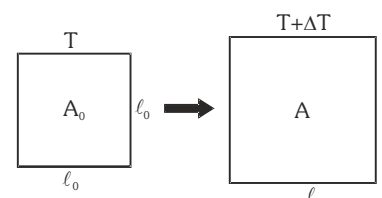


Superficial (areal) expansion :

$$A = A_0 (1 + \beta \Delta\theta)$$

Also $A_0 = \ell_0^2$ and $A = \ell^2$

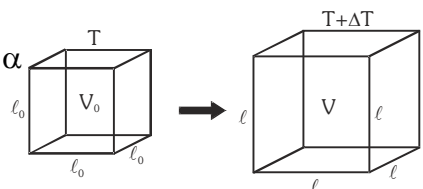
So $\ell^2 = \ell_0^2 (1 + \beta \Delta\theta) = [\ell_0 (1 + \alpha \Delta\theta)]^2 \Rightarrow \beta = 2\alpha$



Volume expansion :

$V = V_0 (1 + \gamma \Delta\theta)$ Also $V = \ell^3$ and $V_0 = \ell_0^3$ so $\gamma = 3\alpha$

$\Rightarrow 6\alpha = 3\beta = 2\gamma$ or $\alpha : \beta : \gamma = 1 : 2 : 3$

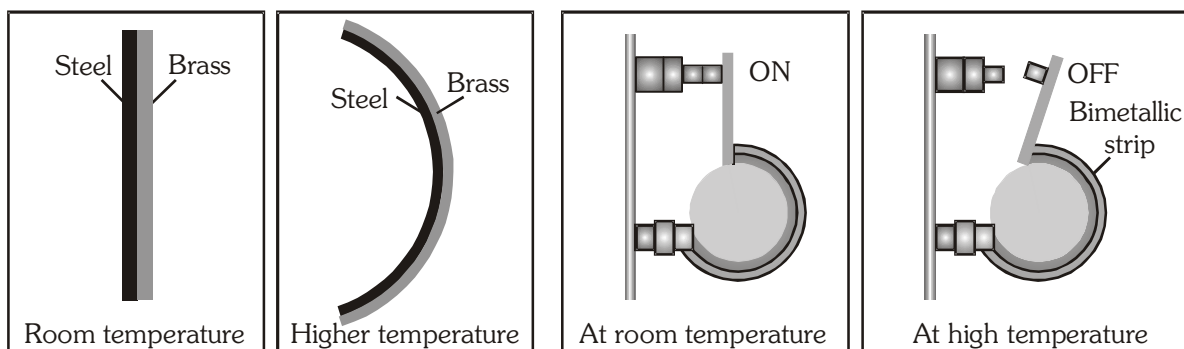


Contraction on heating :

Some rubber like substances contract on heating because transverse vibration of atoms of substance dominates over longitudinal vibration which is responsible for expansion.

Application of thermal Expansion in Solids

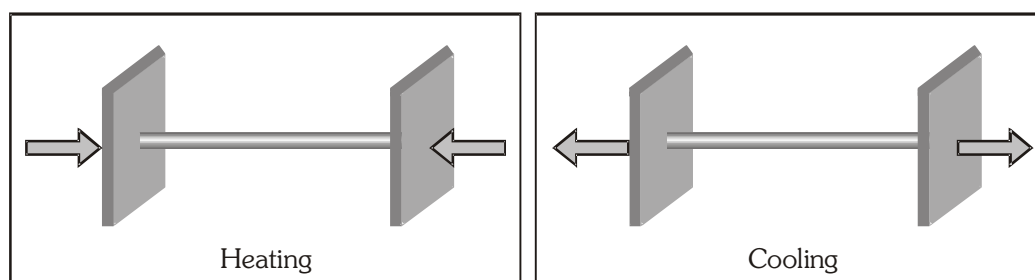
- (a) **Bi-metallic strip :** Two strips of equal length but of different materials (different coefficient of linear expansion) when joined together, it is called "Bi-metallic strip" and can be used in thermostat to break or make electrical contact. This strip has the characteristic property of bending on heating due to unequal linear expansion of the two metals. The strip will bend with metal of greater α on outer side.



- (b) **Effect of temperature on the time period of a simple pendulum :** A pendulum clock keeps proper time at temperature θ . If temperature is increased to θ' ($> \theta$) then due to linear expansion, length of pendulum increases and hence its time period will increase.

$$\text{Fractional change in time period } \frac{\Delta T}{T} = \frac{1}{2} \alpha \Delta \theta \quad (\because T \propto \sqrt{\ell} \therefore \frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta \ell}{\ell})$$

- Due to increment in its time period, a pendulum clock becomes slow in summer and will lose time. Loss of time in a time period $\Delta T = \frac{1}{2} \alpha \Delta \theta T$
 - The clock will lose time i.e. will become slow if $\theta' > \theta$ (in summer) and will gain time i.e. will become fast if $\theta' < \theta$ (in winter).
 - Since coefficient of linear expansion (α) is very small for invar, hence pendulums are made of invar to show the correct time in all seasons.
- (c) When a rod whose ends are rigidly fixed such as to prevent expansion or contraction, undergoes a change in temperature due to thermal expansion or contraction, a compressive or tensile stress is developed in it. Due to this thermal stress the rod will exert a large force on the supports. If the change in temperature of a rod of length L is $\Delta \theta$ then :-

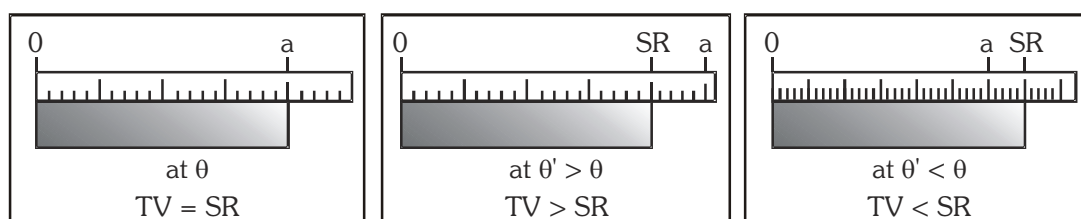


$$\text{Thermal strain} = \frac{\Delta L}{L} = \alpha \Delta \theta \quad \therefore \alpha = \frac{\Delta L}{L} \times \frac{1}{\Delta \theta} \quad \text{So thermal stress} = Y \alpha \Delta \theta$$

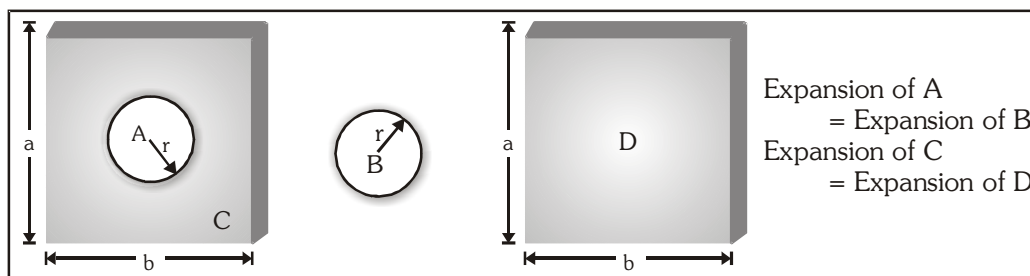
$$\therefore Y = \frac{\text{stress}}{\text{strain}}$$

$$\text{So force on the supports } F = Y A \alpha \Delta \theta$$

- (d) **Error in scale reading due to expansion or contraction:** If a scale gives correct reading at temperature θ . At temperature θ' ($> \theta$) due to linear expansion of scale, the scale will expand and scale reading will be lesser than true value so that,



- (e) **Expansion of cavity:** Thermal expansion of an isotropic object may be imagined as a photographic enlargement.



(f) **Some other application**

- When rails are laid down on the ground, space is left between the ends of two rails
- The transmission cable are not tightly fixed to the poles
- Test tubes, beakers and cubicles are made of pyrex-glass or silica because they have very low value of coefficient of linear expansion
- The iron rim to be put on a cart wheel is always of slightly smaller diameter than that of wheel
- A glass stopper jammed in the neck of a glass bottle can be taken out by warming the neck of the bottle.

Ex. A steel ruler exactly 20 cm long is graduated to give correct measurements at 20°C.

$$(\alpha_{\text{steel}} = 1.2 \times 10^{-5} \text{ } ^\circ\text{C}^{-1})$$

- Will it give readings that are too long or too short at lower temperatures?
- What will be the actual length of the ruler be when it is used in the desert at a temperature of 40°C?

Sol. (a) If the temperature decreases, the length of the ruler also decreases through thermal contraction. Below 20°C, each centimeter division is actually somewhat shorter than 1.0 cm, so the steel ruler gives readings that are too long.

- (b) At 40°C, the increases in length of the ruler is

$$\Delta \ell = \ell \alpha \Delta T = (20) (1.2 \times 10^{-5}) (40^\circ - 20^\circ) = 0.48 \times 10^{-2} \text{ cm}$$

$$\therefore \text{The actual length of the ruler is, } \ell' = \ell + \Delta \ell = 20.0048 \text{ cm}$$

Ex. A second's pendulum clock has a steel wire. The clock is calibrated at 20°C. How much time does the clock lose or gain in one week when the temperature is increased to 30°C? ($\alpha_{\text{steel}} = 1.2 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$)

Sol. The time period of second's pendulum is 2 second. As the temperature increases length time period increases. Clock becomes slow and it loses the time. The change in time period is

$$\Delta T = \frac{1}{2} T \alpha \Delta \theta = \left(\frac{1}{2} \right) (2) (1.2 \times 10^{-5}) (30^\circ - 20^\circ) = 1.2 \times 10^{-4} \text{ s}$$

$\therefore$ New Time period is

$$T' = T + \Delta T = (2 + 1.2 \times 10^{-4}) = 2.00012 \text{ s}$$

$\therefore$ Time lost in one week

$$\Delta t = \left(\frac{\Delta T}{T'} \right) t = \frac{(1.2 \times 10^{-4})}{(2.00012)} (7 \times 24 \times 3600) = 36.28 \text{ s}$$

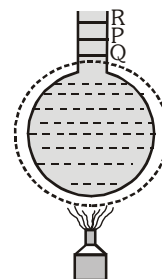
Thermal Expansion in Liquids

- Liquids do not have linear and superficial expansion but these only have volume expansion.
- Since liquids are always to be heated along with a vessel which contains them so initially on heating the system (liquid + vessel), the level of liquid in vessel falls (as vessel expands more since it absorbs heat and liquid expands less) but later on, it starts rising due to faster expansion of the liquid.

PQ → represents expansion of vessel

QR → represents the real expansion of liquid.

- The actual increase in the volume of the liquid
= The apparent increase in the volume of liquid + the increase in the volume of the vessel.
- Liquids have two coefficients of volume expansion.



(i) Co-efficient of apparent expansion (γ_a)

- It is due to apparent (that appears to be, but in not) increase in the volume of liquid if expansion of vessel containing the liquid is not taken into account.

$$\gamma_a = \frac{\text{Apparent expansion in volume}}{\text{Initial volume} \times \Delta\theta} = \frac{(\Delta V)}{V \times \Delta\theta}$$

(ii) Co-efficient of real expansion (γ_r)

- It is due to the actual increase in volume of liquid due to heating.

$$\gamma_r = \frac{\text{Real increase in volume}}{\text{Initial volume} \times \Delta\theta} = \frac{(\Delta V)}{V \times \Delta\theta}$$

- Also coefficient of expansion of flask $\gamma_{\text{vessel}} = \frac{(\Delta V)_{\text{vessel}}}{V \times \Delta\theta}$
- $\gamma_{\text{Real}} = \gamma_{\text{Apparent}} + \gamma_{\text{vessel}}$
- Change (apparent change) in volume in liquid relative to vessel is
 $\Delta V_{\text{app}} = V(\gamma_{\text{Real}} - \gamma_{\text{vessel}}) \Delta\theta = V(\gamma_r - 3\alpha)\Delta\theta$
 α = Coefficient of linear expansion of the vessel.

Ex. In figure shown, left arm of a U-tube is immersed in a hot water bath at temperature $t^\circ\text{C}$, and right arm is immersed in a bath of melting ice; the height of manometric liquid in respective columns is h_t and h_0 . Determine the coefficient of expansion of the liquid.

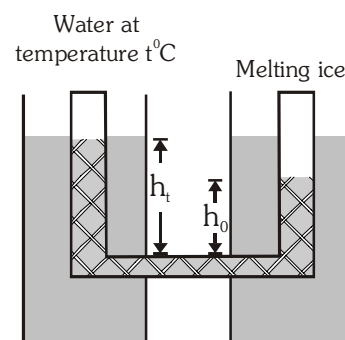
Sol. The liquid is in hydrostatic equilibrium $\Rightarrow \rho_t g h_t = \rho_0 g h_0$

Where, ρ_t is density of liquid in hot bath, ρ_0 is density of liquid in cold bath.

Volumes of a given mass M of liquid at temperatures t and 0°C

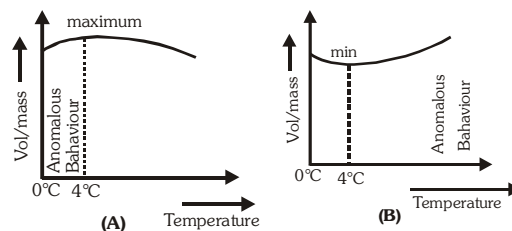
are related by $V_t = V_0(1 + \gamma t)$ Since $\rho_t V_t = \rho_0 V_0 \Rightarrow \rho_t = \frac{\rho_0 V_0}{V_t} = \frac{\rho_0}{(1 + \gamma t)}$

Since $h_t = \frac{\rho_0 h_0}{\rho_t} = h_0(1 + \gamma t)$ which on solving for γ , yields $\gamma = \frac{(h_t - h_0)}{h_0 t}$



Anomalous expansion of water

Generally matter expands on heating and contracts on cooling. In case of water, it expands on heating if its temperature is greater than 4°C . In the range 0°C to 4°C , water contracts on heating and expands on cooling, i.e. γ is negative. This behaviour of water in the range from 0°C to 4°C is called anomalous expansion.



This anomalous behaviour of water causes ice to form first at the surface of a lake in cold weather. As winter approaches, the water temperature increases initially at the surface. The water there sinks because of its increased density. Consequently, the surface reaches 0°C first and the lake becomes covered with ice. Aquatic life is able to survive the cold winter as the lake bottom remains unfrozen at a temperature of about 4°C . At 4°C , density of water is maximum while its specific volume is minimum.

Ex. The difference between lengths of a certain brass rod and of a steel rod is claimed to be constant at all temperatures. Is this possible ?

Sol. If L_B and L_S are the lengths of brass and steel rods respectively at a given temperature, then the lengths of the rods when temperature is changed by $\theta^{\circ}\text{C}$.

$$L'_B = L_B(1 + \alpha_B \Delta\theta) \text{ and } L'_S = L_S(1 + \alpha_S \Delta\theta) \quad \text{So that } L'_B = L'_S \Rightarrow (L_B - L_S) + (L_B \alpha_B - L_S \alpha_S) \Delta\theta$$

So $(L'_B - L'_S)$ will be equal to $(L_B - L_S)$ at all temperatures if, $L_B \alpha_B - L_S \alpha_S = 0$ [as $\Delta\theta \neq 0$] or $\frac{L_B}{L_S} = \frac{\alpha_S}{\alpha_B}$

i.e., the difference in the lengths of the two rods will be independent of temperature if the lengths are in the inverse ratio of their coefficients of linear expansion.

Ex. There are two spheres of same radius and material at same temperature but one being solid while the other hollow. Which sphere will expand more if

(a) they are heated to the same temperature, (b) same heat is given to them ?

Sol. (a) As thermal expansion of isotropic solids is similar to true photographic enlargement,

expansion of a cavity is same as if it had been a solid body of the same material

$$\text{i.e. } \Delta V = V\gamma \Delta\theta$$

As here V , γ and $\Delta\theta$ are same for both solid and hollow spheres treated (cavity) ;

so the expansion of both will be equal.

(b) If same heat is given to the two spheres due to lesser mass, rise in temperature of hollow sphere will be more [as $\Delta\theta = \frac{Q}{mc}$] and hence its expansion will be more [as $\Delta V = V\gamma\Delta\theta$].



CALORIMETRY

HEAT

When a hot body is put in contact with a cold one, the former gets colder and the latter warmer. From this observation it is natural to conclude that a certain quantity of heat has passed from the hot body to the cold one. Heat is a form of energy.

Heat is felt by its effects. Some of the effects of heat are :

- (a) Change in the degree of hotness (b) Expansion in length, surface area and volume
- (c) Change in state of a substance (d) Change in the resistance of a conductor
- (e) Thermo e.m.f. effect

SI UNIT : J (joule) Also measured in the unit calorie.

- **Calorie**

It is defined as the amount of heat required to raise the temperature of 1 g water by 1°C.

- **International calorie**

International calorie is the amount of heat required to raise the temperature of 1g water from 14.5 °C to 15.5 °C rise of temperature.

MECHANICAL EQUIVALENT OF HEAT

According to Joule, work may be converted into heat and vice-versa. The ratio of work done to heat

produced is always constant. $\frac{W}{H} = \text{constant (J)} \Rightarrow W = J H$

W must be in joule, irrespective of nature of energy or work and H must be in calorie.

J is called mechanical equivalent of heat. It is not a physical quantity but simply a conversion factor.

It converts unit of work into that of heat and vice-versa.

$J = 4.18 \text{ joule/cal}$ or $4.18 \times 10^3 \text{ joule per kilo-cal}$. For rough calculations we take $J = 4.2 \text{ joule/cal}$

SPECIFIC HEAT (s or c)

It is the amount of energy required to raise the temperature of unit mass of that substance by 1°C (or 1K) is called specific heat. It is represented by s or c.

If the temperature of a substance of mass m changes from T to T + dT when it exchanges an amount

of heat dQ with its surroundings then its specific heat is $c = \frac{1}{m} \frac{dQ}{dT}$

The specific heat depends on the pressure, volume and temperature of the substance.

For liquids and solids, specific heat measurements are most often made at a constant pressure as functions of temperature, because constant pressure is quite easy to produce experimentally.

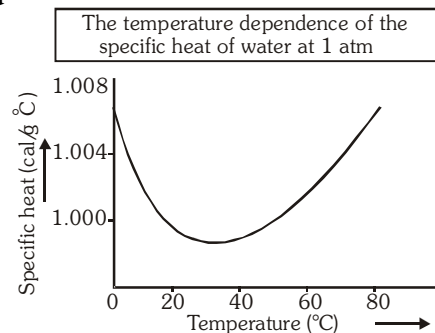
SI UNIT : joule/kg-K

CGS UNIT : cal/g -°C

Specific heat of water : $c_{\text{water}} = 1 \text{ cal/g-°C} = 1 \text{ cal/g-K} = 1 \text{ kcal/kg-K} = 4200 \text{ joule/kg-K}$

When a substance does not undergo a change of state (i.e., liquid remains liquid or solid remains solid), then the amount of heat required to raise the temperature of mass m of the substance by an amount $\Delta\theta$ is $Q = ms\Delta\theta$.

The temperature dependence of the specific heat of water at 1 atmospheric pressure is shown in figure. Its variation is less than 1% over the interval from 0 to 100°C. Such a small variation is typical for most solids and liquids, so their specific heats can generally be taken to be constant over fairly large temperature ranges.



- There are many processes possible to give heat to a gas.
A specific heat can be associated to each such process which depends on the nature of process.
- Value of specific heats can vary from zero (0) to infinity.
- Generally two types of specific heat are mentioned for a gas –
(a) Specific heat at constant volume (C_v) (b) Specific heat at constant pressure (C_p)
- These specific heats can be molar or gram.

MOLAR HEAT CAPACITY

The amount of energy needed to raise the temperature of one mole of a substance by 1°C (or 1 K) is called molar heat capacity. The molar heat capacity is the product of molecular weight and specific heat i.e.,

$$\text{Molar heat capacity } C = \text{Molecular weight (M)} \times \text{Specific heat (c)} \Rightarrow C = \frac{1}{\mu} \left(\frac{dQ}{dT} \right)$$

If the molecular mass of the substance is M and the mass of the substance is m then number of moles of the

$$\text{substance } \mu = \frac{m}{M} \Rightarrow C = \frac{M}{m} \left(\frac{dQ}{dT} \right) \quad \text{SI UNIT : J/mol-K}$$

THERMAL CAPACITY

The quantity of heat required to raise the temperature of the whole of that substance through 1°C is called thermal capacity. The thermal capacity of mass m of the whole of substance of specific heat s is ms

$$\text{Thermal capacity} = \text{mass} \times \text{specific heat}$$

Thermal capacity depends on property of material of the body and mass of the body.

SI UNIT : cal/°C or cal/K,

Dimensions : $ML^2 T^{-2} K^{-1}$

WATER EQUIVALENT OF A BODY

As the specific heat of water is unity so the thermal capacity of a body (ms) represents its water equivalent also.

- Mass of water having the same thermal capacity as the body is called the water equivalent of the body
- The water equivalent of a body is the amount of water that absorbs or gives out the same amount of heat as is done by the body when heated or cooled through 1°C.

$$\text{Water equivalent} = \text{mass of body} \times \text{specific heat of the material} \Rightarrow (w = ms).$$

LATENT HEAT OR HIDDEN HEAT

When state of a body changes, change of state takes place at constant temperature [melting point or boiling point] and heat released or absorbed is $Q = mL$ where L is latent heat. Heat is absorbed if solid converts into liquid (at melting point) or liquid converts into vapours (at boiling point) and heat is released if liquid converts into solid or vapours converts into liquid.

- **Latent heat of fusion**

It is the quantity of heat (in kilocalories) required to change its 1 kg mass from solid to liquid state at its melting point. Latent heat of fusion for ice : $80 \text{ kcal/kg} = 80 \text{ cal/g}$.

- **Latent heat of vaporization**

The quantity of heat required to change its 1 kg mass from liquid to vapour state at its boiling point.

Latent heat of vaporisation for water : $536 \text{ kcal/kg} = 536 \text{ cal/g}$

CHANGE OF STATE

- **Melting**

Conversion of solid into liquid state at constant temperature is known as melting.

- **Boiling**

Evaporation within the whole mass of the liquid is called boiling. Boiling takes place at a constant temperature known as boiling point. A liquid boils when the saturated vapour pressure on its surface is equal to atmospheric pressure. Boiling point reduces on decreasing pressure.

- **Evaporation**

Conversion of liquid into vapours at all temperatures is called evaporation. It is a surface phenomenon. Greater the temperature, faster is the evaporation. Smaller the boiling point of liquid, more rapid is the evaporation. Smaller the humidity, more is the evaporation. Evaporation increases on decreasing pressure that is why evaporation is faster in vacuum.

- **Heat of evaporation**

Heat required to change unit mass of liquid into vapour at a given temperature is called heat of evaporation at that temperature.

- **Sublimation**

Direct conversion of solid in to vapour state is called sublimation.

- **Heat of sublimation**

Heat required to change unit mass of solid directly into vapours at a given temperature is called heat of sublimation at that temperature.

- Camphor and ammonium chloride sublimes on heating in normal conditions.
 - A block of ice sublimes into vapours on the surface of moon because of very–very low pressure on its surface

- **Condensation**

The process of conversion from gaseous or vapour state to liquid state is known as condensation . These materials again get converted to vapour or gaseous state on heating.

PHASE OF A SUBSTANCE

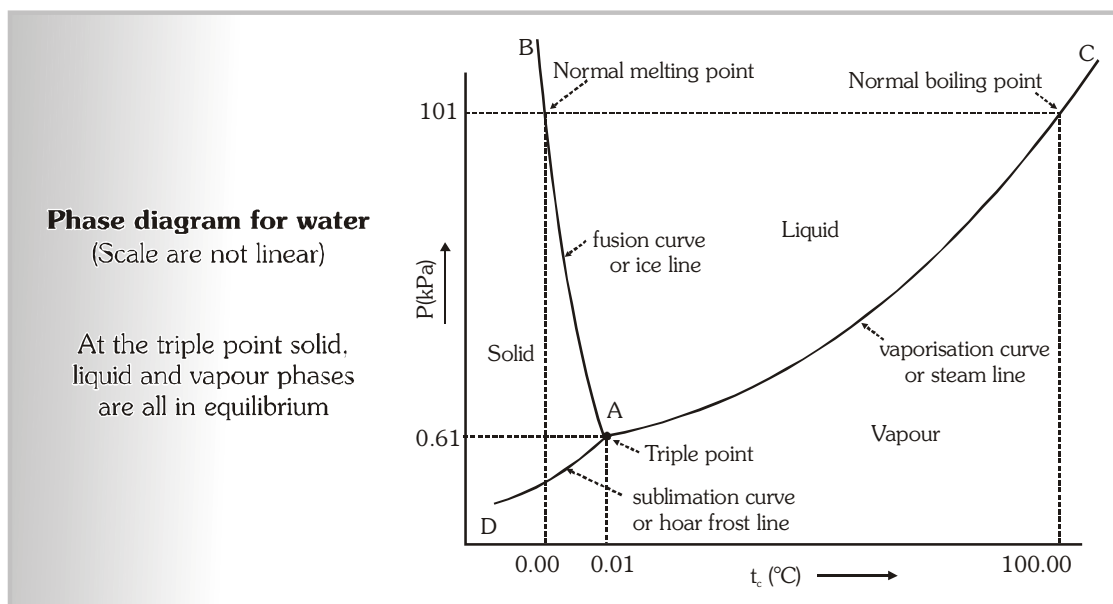
The phase of a substance is defined as its form which is homogeneous, physically distinct and mechanically separable from the other forms of that substance.

Phase diagram

- A phase diagram is a graph in which pressure (P) is represented along the y-axis and temperature (T) is represented along the x-axis.
- **Characteristics of Phase diagram**
 - Different phases of a substances can be shown on a phase diagram.
 - A region on the phase diagram represents a single phase of the substance, a curve represents equilibrium between two phases and a common point represents equilibrium between three phases.
 - A phase diagram helps to determine the condition under which the different phases are in equilibrium.
 - A phase diagram is useful for finding a convenient way in which a desired change of phase can be produced.

PHASE DIAGRAM FOR WATER

The phase diagram for water consists of three curves AB, AC and AD meeting each other at the point A, these curves divide the phase diagram into three regions.



Region to the left of the curve AB and above the curve AD represents the solid phase of water (ice). The region to the right of the curve AB and above the curve AC represents the liquid phase of water. The region below the curves AC and AD represents the gaseous phase of water (i.e. water vapour). A curve on the phase diagram represents the boundary between two phases of the substance.

Along any curve the two phases can coexist in equilibrium

- Along curve AB, ice and water can remain in equilibrium. This curve is called fusion curve or ice line. This curve shows that the melting point of ice decreases with increase in pressure.
- Along the curve AC, water and water vapour can remain in equilibrium. The curve is called vaporisation curve or steam line. *The curve shows that the boiling point of water increases with increase in pressure.*
- Along the curve AD, ice and water vapour can remain in equilibrium. This curve is called sublimation curve or hoar frost line.

TRIPLE POINT OF WATER

The three curves in the phase diagram of water meet at a single point A, which is called the triple point of water. The triple point of water represents the co-existence of all the three phases of water ice water and water vapour in equilibrium. The pressure corresponding to triple point of water is 6.03×10^{-3} atmosphere or 4.58 mm of Hg and temperature corresponding to it is 273.16K.

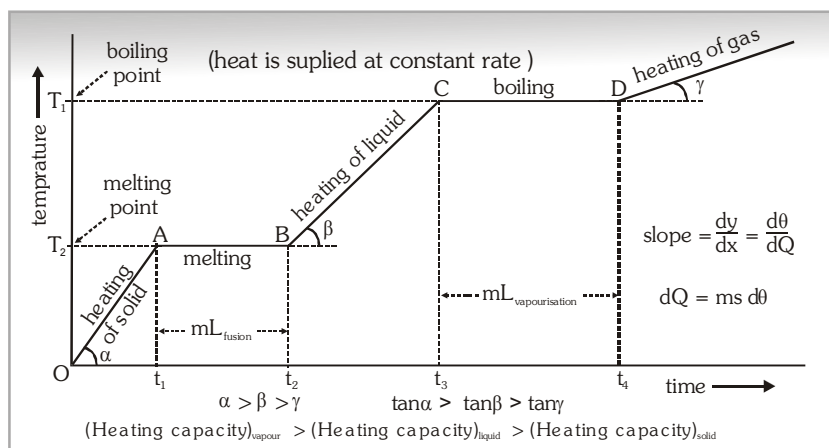
- **Significance of triple point of water**

Triple point of water represents a unique condition and it is used to define the absolute temperature. While making Kelvin's absolute scale, upper fixed point is 273.16 K and lower fixed point is 0 K.

One kelvin of temperature is fraction $\frac{1}{273.16}$ of the temperature of triple point of water.

HEATING CURVE

If to a given mass (m) of a solid, heat is supplied at constant rate and a graph is plotted between temperature and time as shown in figure is called heating curve.



- **In the region OA**

Rate of heat supply P is constant and temperature of solid is changing with time

So, $Q = mc_s \Delta T \Rightarrow P \Delta t = mc_s \Delta T$ [$\because Q = P \Delta t$] $\therefore \frac{\Delta T}{\Delta t} =$ The slope of temperature–time curve so

specific heat of solid $c_s \propto \frac{1}{\text{slope of line OA}}$ specific heat (or thermal capacity) is inversely proportional to the slope of temperature–time curve.

- **In the region AB**

Temperature is constant, so it represents change of state, i.e., melting of solid with melting point T_1 . At point A melting starts and at point B all solid is converted into liquid. So between A and B substance is partly solid and partly liquid. If L_F is the latent heat of fusion then

$$Q = mL_F \Rightarrow L_F = \frac{P(t_2 - t_1)}{m} \quad [\text{as } Q = P(t_2 - t_1) \Rightarrow L_F \propto \text{length of line AB}]$$

i.e., Latent heat of fusion is proportional to the length of line of zero slope.

$$[\text{In this region specific heat} \propto \frac{1}{\tan 0^\circ} = \infty]$$

- **In the region BC**

Temperature of liquid increases so specific heat (or thermal capacity) of liquid will be inversely proportional

$$\text{to the slope of line BC, } c_L \propto \frac{1}{\text{slope of line BC}}$$

- **In the region CD**

Temperature is constant, so it represents change of state, i.e., liquid is boiling with boiling point T_2 . At C all substance is in liquid state while at D is vapour state and between C and D partly liquid and partly gas. The length of line CD is proportional to latent heat of vaporisation, i.e., $L_v \propto \text{Length of line CD}$.

$$[\text{In this region specific heat} \propto \frac{1}{\tan 0^\circ} = \infty]$$

The line DE represents gaseous state of substance with its temperature increasing linearly with time. The reciprocal of slope of line will be proportional to specific heat or thermal capacity of substance in vapour state.

LAW OF MIXTURES

- When two bodies (one being solid and other liquid or both being liquid) at different temperatures are mixed, heat will be transferred from body at higher temperature to a body at lower temperature till both acquire same temperature. The body at higher temperature released heat while body at lower temperature absorbs it, so that Heat lost = Heat gained. Principle of calorimetry represents the law of conservation of heat energy.
- Temperature of mixture (T) is always $\geq$ lower temperature (T_L) and $\leq$ higher temperature (T_H),
 $T_L \leq T \leq T_H$

The temperature of mixture can never be lesser than lower temperature (as a body cannot be cooled below the temperature of cooling body) and greater than higher temperature (as a body cannot be heated above the temperature of heating body). Further more usually rise in temperature of one body is not equal to the fall temperature of the other body though heat gained by one body is equal to the heat lost by the other.

Ex. 5g ice at 0°C is mixed with 5g of steam at 100°C . What is the final temperature?

Sol. Heat required by ice to raise its temperature to 100°C ,

$$Q_1 = m_1 L_1 + m_1 c_1 \Delta\theta_1 = 5 \times 80 + 5 \times 1 \times 100 = 400 + 500 + 900 = 1800 \text{ cal}$$

$$\text{Heat given by steam when condensed } Q_2 = m_2 L_2 = 5 \times 536 = 2680 \text{ cal}$$

As $Q_2 > Q_1$. This means that whole steam is not even condensed.

Hence temperature of mixture will remain at 100°C .

Ex. A calorimeter of heat capacity 100 J/K is at room temperature of 30°C . 100 g of water at 40°C of specific heat 4200 J/kg-K is poured into the calorimeter. What is the temperature of water in calorimeter?

Sol. Let the temperature of water in calorimeter is t . Then heat lost by water = heat gained by calorimeter

$$(0.1) \times 4200 \times (40 - t) = 100 (t - 30) \Rightarrow 42 \times 40 - 42t = 10t - 300 \Rightarrow t = 38.07^\circ\text{C}$$

Ex. Find the quantity of heat required to convert 40 g of ice at -20°C into water at 20°C .

$$\text{Given } L_{\text{ice}} = 0.336 \times 10^6 \text{ J/kg.}$$

$$\text{Specific heat of ice} = 2100 \text{ J/kg-K, specific heat of water} = 4200 \text{ J/kg-K}$$

Sol. Heat required to raise the temperature of ice from -20°C to 0°C = $0.04 \times 2100 \times 20 = 1680 \text{ J}$

$$\text{Heat required to convert the ice into water at } 0^\circ\text{C} = mL = 0.04 \times 0.336 \times 10^6 = 13440 \text{ J}$$

$$\text{Heat required to heat water from } 0^\circ\text{C to } 20^\circ\text{C} = 0.04 \times 4200 \times 20 = 3360 \text{ J}$$

$$\text{Total heat required} = 1680 + 13440 + 3360 = 18480 \text{ J}$$

Ex. Steam at 100°C is passed into 1.1 kg of water contained in a calorimeter of water equivalent 0.02 kg at 15°C till the temperature of the calorimeter and its contents rises to 80°C . What is the mass of steam condensed? Latent heat of steam = 536 cal/g .

Sol. Heat required by (calorimeter + water)

$$Q = (m_1 c_1 + m_2 c_2) \Delta\theta = (0.02 + 1.1 \times 1) (80 - 15) = 72.8 \text{ kcal}$$

If m is mass of steam condensed, then heat given by steam

$$Q = mL + mc \Delta\theta = m \times 536 + m \times 1 \times (100 - 80) = 556 m \therefore 556 m = 72.8$$

$$\therefore \text{Mass of steam condensed } m = \frac{72.8}{556} = 0.130 \text{ kg}$$

MODE OF HEAT TRANSFER

Heat is a form of energy which transfers from a body at higher temperature to a body at lower temperature. The transfer of heat from one body to another may take place by any one of the following modes :

- **Conduction**

The process in which the material takes an active part by molecular action and energy is passed from one particle to another is called conduction. It is predominant in solids.

- **Convection**

The transfer of energy by actual motion of particle of medium from one place to another is called convection. It is predominant in fluids (liquids and gases).

- **Radiation**

Quickest way of transmission of heat is known as radiation. In this mode of energy transmission, heat is transferred from one place to another without effecting the intervening medium.

Conduction	Convection	Radiation
Heat Transfer due to Temperature difference	Heat transfer due to density difference	Heat transfer with out any medium
Due to free electron or vibration motion of molecules	Actual motion of particles	Electromagnetic radiation
Heat transfer in solid body (in mercury also)	Heat transfer in fluids (Liquid + gas)	All
Slow process	Slow process	Fast process (3×10^8 m/sec)
Irregular path	Irregular path	Straight line (like light)

THERMAL CONDUCTION

The process by which heat is transferred from hot part to cold part of a body through the transfer of energy from one particle to another particle of the body without the actual movement of the particles from their equilibrium positions is called conduction. The process of conduction only in solid body (except Hg) Heat transfer by conduction from one part of body to another continues till their temperatures become equal.

Steady state : When temperature of the each cross-section of the bar becomes constant through different for different cross-sections is called steady state.

- **Equation of thermal conduction**

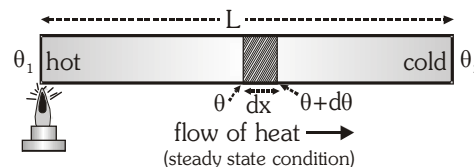
$$\text{Rate of heat flow } \frac{dQ}{dt} = -KA \frac{d\theta}{dx} \quad \frac{dQ}{dt} = \frac{KA}{L} (\theta_1 - \theta_2)$$

Cross section area = A ; Length = L

Thermal conductivity of material = K

- **Thermal (temperature) gradient**

The decrease in temperature with distance from hot end of the rod is known as temperature gradient or in the direction of heat energy flow, the rate of fall in temperature w.r.t. distance is called as temperature gradient. It is denoted by $-dT/dx$



- **Thermal conductivity (K) :** It's depends on nature of material.
 - **SI UNIT :** $\text{J s}^{-1} \text{m}^{-1} \text{K}^{-1}$ **Dimensions :** $\text{M}^1 \text{L}^1 \text{T}^{-3} \theta^{-1} \text{K}$

$\left[\begin{array}{l} \text{For Ag maximum is (410 W/mK)} \\ \text{For Freon minimum is 12 (0.008 W/mK)} \end{array} \right.$
 - For an ideal or perfect conductor of heat the value of $K = \infty$
 - For an ideal or perfect bad conductor or insulator the value of $K = 0$
 - For cooking the food, low specific heat and high conductivity utensils are most suitable.

APPLICATION OF THERMAL CONDUCTION

- In winter, the iron chairs appear to be colder than the wooden chairs.
- Cooking utensils are made of aluminium and brass whereas their handles are made of wood.
- Ice is covered in gunny bags to prevent melting of ice.
- We feel warm in woollen clothes and fur coat.
- Two thin blankets are warmer than a single blanket of double the thickness.
- Birds often swell their feathers in winter.
- A new quilt is warmer than old one.

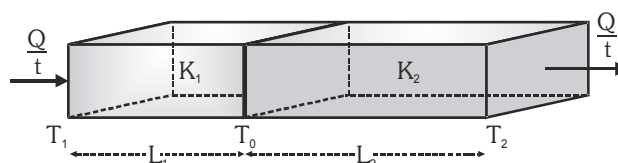
THERMAL RESISTANCE (R)

The thermal resistance of a body is a measure of its opposition of the flow of heat through it. $R = \frac{L}{KA}$

- **Heat flow through slabs in series**

$$R_{eq} = R_1 + R_2$$

$$\frac{L_1 + L_2}{K_{eq}A} = \frac{L_1}{K_1A} + \frac{L_2}{K_2A}$$



Equivalent thermal conductivity of the system is

$$K_{eq} = \frac{L_1 + L_2}{\frac{L_1}{K_1} + \frac{L_2}{K_2}} = \frac{\sum L_i}{\sum \frac{L_i}{K_i}}$$

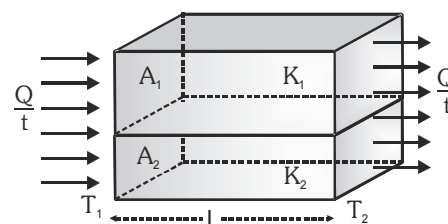
equivalent to

- **Heat flow through slabs in parallel**

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}, R = \frac{L}{KA}; \frac{K_{eq}}{L} (A_1 + A_2) = \frac{K_1 A_1}{L} + \frac{K_2 A_2}{L}$$

Equivalent thermal conductivity

$$K_{eq} = \frac{K_1 A_1 + K_2 A_2}{A_1 + A_2} = \frac{\sum K_i A_i}{\sum A_i}$$



equivalent to

GROWTH OF ICE ON LAKES

In winter atmospheric temperature falls below 0°C and water in the lake start freezing.

Let at time t thickness of ice on the surface of the lake = x and air temperature = $-\theta^\circ\text{C}$

The temperature of water in contact with the lower surface of ice = 0°C

Let area of the lake = A

Heat escaping through ice in time dt is $dQ = KA \frac{[0 - (-\theta)]}{x} dt$

Due to escape of this heat increasing extra thickness of ice $= dx$

Mass of this extra thickness of ice is $m = \rho V = \rho A \cdot dx$

$$dQ = mL = (\rho A \cdot dx) L$$

$$\therefore KA \frac{\theta}{x} dt = (\rho A \cdot dx) L \Rightarrow dt = \frac{\rho L}{K\theta} x dx$$

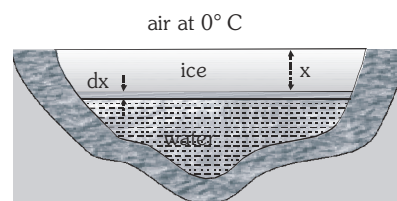
So time taken by ice to grow a thickness x is $t = \frac{\rho L}{K\theta} \int_0^x x dx = \frac{1}{2} \frac{\rho L}{K\theta} x^2$

So time taken by ice to grow from thickness x_1 to thickness x_2 is

$$t = t_2 - t_1 = \frac{1}{2} \frac{\rho L}{K\theta} (x_2^2 - x_1^2) \quad \text{and} \quad t \propto (x_2^2 - x_1^2)$$

Time taken to double and triple the thickness ratio $t_1 : t_2 : t_3 :: 1^2 : 2^2 : 3^2$

So $t_1 : t_2 : t_3 :: 1 : 4 : 9$



Ex. One end of a brass rod 2m long and having 1 cm radius is maintained at 250°C . When a steady state is reached, the rate of heat flow across any cross-section is 0.5 cal s^{-1} . What is the temperature of the other end $K = 0.26 \text{ cal s}^{-1} \text{ cm}^{-1} \text{ }^\circ\text{C}^{-1}$.

Sol. $\frac{Q}{t} = 0.5 \text{ cal s}^{-1}$; $r = 1 \text{ cm}$ $\therefore \text{Area } A = \pi r^2 = 3.142 \times 1 \text{ cm}^2 = 3.142 \text{ cm}^2$

$L = \text{Length of rod} = 2\text{m} = 200 \text{ cm}$, $T_1 = 250^\circ\text{C}$, $T_2 = ?$

$$\text{We know } \frac{Q}{t} = \frac{KA(T_1 - T_2)}{L} \text{ or } (T_1 - T_2) = \frac{Q}{t} \times \frac{\Delta x}{kA} = \frac{0.5 \times 200}{0.26 \text{ C}^{-1} \times 3.142} = 122.4^\circ\text{C}$$

$$\therefore T_2 = 250^\circ\text{C} - 122.4^\circ\text{C} = 127.6^\circ\text{C}$$

Ex. Steam at 373 K is passed through a tube of radius 10 cm and length 2 m. The thickness of the tube is 5 mm and thermal conductivity of the material is $390 \text{ W m}^{-1} \text{ K}^{-1}$, calculate the heat lost per second. The outside temp. is 0°C .

Sol. Using the relation $Q = \frac{KA(T_1 - T_2)t}{L}$

Here, heat is lost through the cylindrical surface of the tube.

$$A = 2\pi r (\text{radius of the tube}) (\text{length of the tube}) = 2\pi \times 0.1 \times 2 = 0.4 \pi \text{ m}^2$$

$$K = 390 \text{ W m}^{-1} \text{ K}^{-1}$$

$$T_1 = 373 \text{ K}, \quad T_2 = 0^\circ\text{C} = 273 \text{ K}, \quad L = 5 \text{ mm} = 0.005 \text{ m} \quad \text{and} \quad t = 1 \text{ s}$$

$$\therefore Q = \frac{390 \times 0.4 \pi \times (373 - 273) \times 1}{0.005} = \frac{390 \times 0.4 \pi \times 100}{0.005} = 98 \times 10^5 \text{ J}$$

Ex. The thermal conductivity of brick is $1.7 \text{ W m}^{-1} \text{ K}^{-1}$, and that of cement is $2.9 \text{ W m}^{-1} \text{ K}^{-1}$. What thickness of cement will have same insulation as the brick of thickness 20 cm.

Sol. Since $Q = \frac{KA(T_1 - T_2)t}{L}$. For same insulation by the brick and cement Q , A ($T_1 - T_2$) and t do not change. Hence, $\frac{K}{L}$ remain constant. If K_1 and K_2 be the thermal conductivities of brick and cement respectively and L_1 and L_2 be the required thickness then $\frac{K_1}{L_1} = \frac{K_2}{L_2}$ or $\frac{1.7}{20} = \frac{2.9}{L_2}$

$$\therefore L_2 = \frac{2.9}{1.7} \times 20 = 34.12 \text{ cm}$$

Ex. Two vessels of different material are identical in size and wall-thickness. They are filled with equal quantities of ice at 0°C . If the ice melts completely, in 10 and 25 minutes respectively then compare the coefficients of thermal conductivity of the materials of the vessels.

Sol. Let K_1 and K_2 be the coefficients of thermal conductivity of the materials, and t_1 and t_2 be the time in which ice melts in the two vessels. Since both the vessels are identical, so A and x in both the cases is same.

$$\text{Now, } Q = \frac{K_1 A (\theta_1 - \theta_2) t_1}{L} = \frac{K_2 A (\theta_1 - \theta_2) t_2}{L} \Rightarrow \frac{K_1}{K_2} = \frac{t_2}{t_1} = \frac{25 \text{ min}}{10 \text{ min}} = \frac{5}{2}$$

Ex. Three rods of material X and three rods of material Y are connected as shown in figure. All the rods are identical in length and cross-sectional area. If the end A is maintained at 60°C and the junction E at 10°C , calculate the temp. of the junctions B, C, D. The thermal conductivity of X is 0.92 CGS units and that of Y is 0.46 CGS units.

$$\text{Sol. } R_X \propto \frac{1}{K_X}, R_Y \propto \frac{1}{K_Y} \Rightarrow \frac{R_X}{R_Y} = \frac{K_Y}{K_X} = \frac{0.46}{0.92} = \frac{1}{2} \quad \text{Let } R_X = R \therefore R_Y = 2R$$

The total resistance $\Sigma R = R_Y + \text{effective resistance in the bridge}$

$$\Sigma R = 2R + \frac{2R \times 4R}{2R + 4R} = 2R + \frac{4}{3} R = \frac{10}{3} R \quad \& \quad \therefore \Delta\theta = \ell \times R$$

$$\text{Further } I_{BCE} (2R) = I_{BDE} (4R) \text{ and } I_{BCE} + I_{BDE} = I \Rightarrow I_{BCE} = \frac{2}{3} I \text{ and } I_{BDE} = \frac{1}{3} I$$

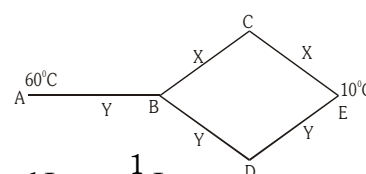
$$\text{For A and B } \theta_A - \theta_B = 60^\circ - \theta_B \Rightarrow 60 - \theta_B = 2R \times I \quad \dots(i)$$

$$\text{For B and C } \theta_B - \theta_C = \frac{2}{3} (I \times R) \quad \dots(ii) \quad \theta_C - \theta_E = \frac{2}{3} \times R \times I$$

$$\text{For A and E } \theta_A - \theta_E = 60 - 10 = 50 \Rightarrow \frac{10}{3} (R \times I) = 50 \quad \dots(iii) \therefore R \times I = 15$$

$$\therefore \theta_A - \theta_B - 2 \times 15 = 30, \theta_B = 60 - 30 = 30^\circ\text{C}, \theta_B - \theta_C = \left(\frac{2}{3}\right) \times 15 = 10$$

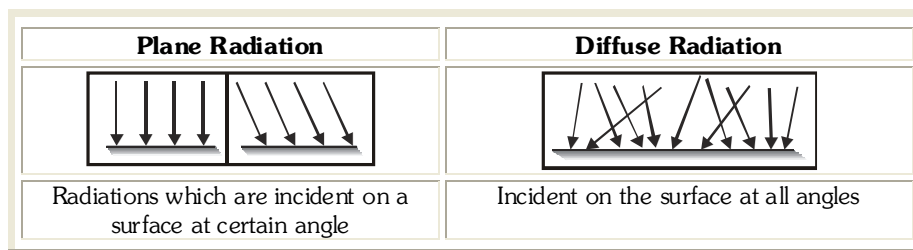
$$\therefore \theta_C = 30 - 10 = 20^\circ\text{C} \text{ Obviously, } \theta_C = \theta_D = 20^\circ\text{C}$$



Ex. Two plates of equal areas are placed in contact with each other. Their thickness are 2.0 cm and 5.0 cm respectively. The temperature of the external surface of the first plate is -20°C and that of the external surface of the second plate is 20°C . What will be the temperature of the contact surface if the plate (i) are of the same material, (ii) have thermal conductivities in the ratio 2 : 5.

All these and many other properties establish that heat radiation has nearly all the properties possessed by light and these are also electromagnetic waves with the only difference of wavelength or frequency. The wavelength of heat radiation is larger than that of visible light.

Types of thermal Radiation :– Two types of thermal radiation.



- When radiation passes through any medium then radiations slightly absorbed by medium according to its absorptive power so temperature of medium slightly increases.
- Heat radiation are always obtained in infra-red region of electromagnetic wave spectrum so they are called Infra red rays.

BASIC FUNDAMENTAL DEFINITIONS

Energy Density (u)

The radiation energy of whole wavelength (0 to ∞) present in unit volume at any point in space is defined as energy density. **SI UNIT** : J/m³

- Spectral energy density (u_λ)** : Energy density per unit spectral region. $u = \int_0^\infty u_\lambda d\lambda$

SI UNIT : J/m³ Å

- Absorptive power or absorptive coefficient 'a'** : The ratio of amount of radiation absorbed by a surface (Q_a) to the amount of radiation incident (Q) upon it is defined as the coefficient of absorption

$$a = \frac{Q_a}{Q} \text{ . It is unitless}$$

- Spectral absorptive power (a_λ)** $a_\lambda = \frac{Q a_\lambda}{Q_\lambda}$: Also called monochromatic absorptive coefficient

At a given wavelength $a = \int_0^\infty a_\lambda d\lambda$. For ideal black body a_λ and $a = 1$, a and a_λ are unitless

- Emissive power (e)** : The amount of heat radiation emitted by unit area of the surface in one second at a particular temperature. **SI UNIT** : J/m²s
- Spectral Emissive power (e_λ)** : The amount of heat radiation emitted by unit area of the body in one

second in unit spectral region at a given wavelength. Emissive power or total emissive power $e = \int_0^\infty e_\lambda d\lambda$

SI UNIT : W/m² Å

EMISSIONS (e)

- **Absolute emissivity or emissivity :** Radiation energy given out by a unit surface area of a body in unit time corresponding to unit temperature difference w.r.t. the surroundings is called Emissivity.

SI UNIT : W/m² °K

- **Relative emissivity (ϵ_r)** : $\epsilon_r = \frac{Q_{GB}}{Q_{IBB}} = \frac{e_{GB}}{E_{IBB}} = \frac{\text{emitted radiation by gray body}}{\text{emitted radiation by ideal black body}}$

GB = gray or general body, IBB = Ideal black body

- (i) No unit (ii) For ideal black body $e_r = 1$ (iii) range $0 < e_r < 1$

SPECTRAL, EMISSIVE, ABSORPTIVE AND TRANSMISSIVE POWER OF A GIVEN BODY SURFACE

Due to incident radiations on the surface of a body following phenomena occur by which the radiation is divided into three parts. (a) Reflection (b) Absorption (c) Transmission

- **From energy conservation**

$$Q = Q_r + Q_a + Q_t \Rightarrow \frac{Q_r}{Q} + \frac{Q_a}{Q} + \frac{Q_t}{Q} = 1 \Rightarrow r + a + t = 1$$

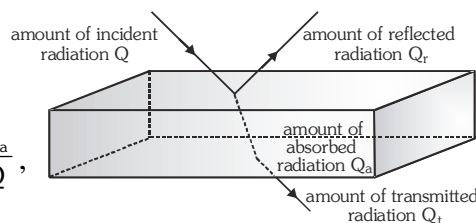
Reflective Coefficient $r = \frac{Q_r}{Q}$, Absorptive Coefficient $a = \frac{Q_a}{Q}$,

$$\text{Transmittive Coefficient } t = \frac{Q_t}{Q}$$

$$r = 1 \text{ and } a = 0, t = 0 \Rightarrow \text{Perfect reflector}$$

$a = 1$ and $r = 0, t = 0 \Rightarrow$ Ideal absorber (ideal black body)

$t = 1$ and $a = 0, \quad r = 0 \quad \Rightarrow \quad$ Perfect transmitter



$$\text{Reflection power (r)} = \left[\frac{Q_r}{Q} \times 100 \right] \%, \text{ Absorption power (a)} = \left[\frac{Q_a}{Q} \times 100 \right] \%$$

$$\text{Transmission power (t)} = \left[\frac{Q_t}{Q} \times 100 \right] \%$$

Ex. Total radiations incident on body = 400 J, 20% radiation reflected and 120 J absorbs. Then find out % of transmittive power

Sol. $Q = Q_t + Q_r + Q_a \Rightarrow 400 = 80 + 120 + Q_t \Rightarrow Q_t = 200$. So % of transmittive power is 50%

IDEAL BLACK BODY

- For a body surface which absorbs all incident thermal radiations at low temperature irrespective of their wave length and emitted out all these absorbed radiations at high temperature assumed to be an ideal black body surface.
- The identical parameters of an ideal black body is given by
 $a = a_{\lambda} = 1$ and $r = 0 = t, e_r = 1$
- The nature of emitted radiations from surface of ideal black body only depends on its temperature
- The radiations emitted from surface of ideal black body called as either full or white radiations.
- At low temperature surface of ideal black body is a perfect absorber and at a high temperature it proves to be a good emitter.
- An ideal black body need not be black colour (eg. Sun)



PREVOST'S THEORY OF HEAT ENERGY EXCHANGE

According to Prevost at every possible temperature (Not absolute temperature) there is a continuous heat energy exchange between a body and its surrounding and this exchange carry on for infinite time.

The relation between temperature difference of body with its surrounding decides whether the body experience cooling effect or heating effect.

When a cold body is placed in the hot surrounding : The body radiates less energy and absorbs more energy from the surrounding, therefore the temperature of body increases.

When a hot body placed in cooler surrounding : The body radiates more energy and absorb less energy from the surroundings. Therefore temperature of body decreases.

When the temperature of a body is equal to the temperature of the surrounding

The energy radiated per unit time by the body is equal to the energy absorbed per unit time by the body, therefore its temperature remains constant.

- At absolute zero temperature (0 kelvin) all atoms of a given substance remains in ground state, so, at this temperature emission of radiation from any substance is impossible, so Prevost's heat energy exchange theory does not applied at this temperature, so it is called **limited temperature** of prevosts theory.
- With the help of Prevost's theory rate of cooling of any body w.r.t. its surroundings can be worked out (applied to Stefan Boltzman law, Newton's law of cooling.)

KIRCHHOFF'S LAW

At a given temperature for all bodies the ratio of their spectral emissive power (e_λ) to spectral absorptive power (a_λ) is constant and this constant is equal to spectral emissive power (E_λ) of the ideal black body at same temperature

$$\frac{e_\lambda}{a_\lambda} = E_\lambda = \text{constant} \quad \left[\frac{e_\lambda}{a_\lambda} \right]_1 = \left[\frac{e_\lambda}{a_\lambda} \right]_2 = \text{constant} \quad e_\lambda \propto a_\lambda$$

Good absorbers are good emitters and bad absorbers are bad emitters

- For a constant temperature the spectral emissive power of an ideal black body is a constant parameter
- The practical confirmation of Kirchhoff's law carried out by Rishi apparatus and the main base of this apparatus is a Lesslie container.
- The main conclusion predicted from Kirchhof's law can be expressed as

Good absorber	$\rightleftharpoons$	Good emitter
Bad absorber	$\rightleftharpoons$	Bad emitter
(at Low temperature)		(at high temperature)

APPLICATIONS OF KIRCHOFF LAW

- In deserts days are hot and nights cold**

Sand is rough and black, so it is a good absorber and hence in deserts, days (When radiation from Sun is incident on sand) will be very hot. Now in accordance with Kirchhoff's Law, good absorber is a good emitter.

So nights (when sand emits radiation) will be cold.

STEFAN'S LAW

The amount of radiation emitted per second per unit area by a black body is directly proportional to the fourth power of its absolute temperature.

Amount of radiation emitted $E \propto T^4$ where T = temperature of ideal black body (in K)

$$E = \sigma T^4 \quad \text{This law is true for only ideal black body}$$

SI Unit : $E = \text{watt/m}^2$ $\sigma = \text{Stefen's constant} = 5.67 \times 10^{-8} \text{ watt /m}^2 \text{ K}^4$

Dimensions of σ : $M^1 L^0 T^{-3} \theta^{-4}$

Total radiation energy emitted out by surface of area A in time t : If T is constant.

$$\text{Ideal black body} \quad Q_{\text{IBB}} = \sigma A T^4 t \quad \text{and for any other body } Q_{\text{GB}} = e_r \sigma A T^4 t$$

Rate of emission of radiation

When Temperature of surrounding T_0 (Let $T_0 < T$)

$$\text{Rate of emission of radiation from ideal black body surface } E_1 = \sigma T^4$$

$$\text{Rate of absorption of radiation from surrounding } E_2 = \sigma T_0^4$$

$$\text{Net rate of loss of radiation from ideal black body surface is } E = E_1 - E_2 = \sigma T^4 - \sigma T_0^4 = \sigma (T^4 - T_0^4)$$

$$\text{Net loss of radiation energy from entire surface area in time } t \text{ is } Q_{\text{IBB}} = \sigma A (T^4 - T_0^4) t$$

$$\text{For any other body } Q_{\text{GB}} = e_r A \sigma (T^4 - T_0^4) t$$

If in time dt the net heat energy loss for ideal black body is dQ and because of this its temperature falls by $d\theta$

$$\text{Rate of loss of heat } R_H = \frac{dQ}{dt} = \sigma A (T^4 - T_0^4)$$

It is also equal to emitted power or radiation emitted per second

$$\text{Rate of fall in temperature (Rate of cooling) } R_F = \frac{d\theta}{dt} = \frac{\sigma A}{msJ} (T^4 - T_0^4) \left[\because \frac{dQ}{dt} = msJ \frac{d\theta}{dt} \right]$$

Note :

- (i) If all of T , T_0 , m , s , V , ρ , are same for different shape body then R_F and R_H will be maximum in the flat surface.
- (ii) If a solid and hollow sphere are taken with all the parameters same then hollow will cool down at fast rate.

- (iii) Rate of temperature fall, $R_F \propto \frac{1}{s} \propto \frac{d\theta}{dt}$ so $dt \propto s$. If condition in specific heat is $\Rightarrow s_1 > s_2 > s_3$

If all cooled same temperature i.e. temperature fall is also identical for all then required time $t \propto s \therefore t_1 > t_2 > t_3$

• **When a body cools by radiation the cooling depends on :**

- (i) Nature of radiating surface : greater the emissivity (e_r), faster will be the cooling.
- (ii) Area of radiating surface : greater the area of radiating surface, faster will be the cooling.
- (iii) Mass of radiating body : greater the mass of radiating body slower will be the cooling.
- (iv) Specific heat of radiating body : greater the specific heat of radiating body slower will be the cooling.
- (v) Temperature of radiating body : greater the temperature of radiating body faster will be the cooling.

Ex. The operating temperature of a tungsten filament in an incandescent lamp is 2000 K and its emissivity is 0.3. Find the surface area of the filament of a 25 watt lamp. Stefan's constant $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Sol. $\therefore$ Rate of emission = wattage of the lamp

$$\therefore W = Ae\sigma T^4 \Rightarrow A = \frac{W}{e\sigma T^4} = \frac{25}{0.3 \times 5.67 \times 10^{-8} \times (2000)^4} = 0.918 \text{ m}^2$$

NEWTON'S LAW OF COOLING

Rate of loss of heat $\left(\frac{dQ}{dt}\right)$ is directly proportional to excess of temperature of the body over that of

$$\text{surrounding. [(when } (\theta - \theta_0) \neq 35^\circ\text{C}] \quad \frac{dQ}{dt} \propto (\theta - \theta_0) \quad \Rightarrow \quad \frac{dQ}{dt} = ms \frac{d\theta}{dt}$$

θ = temperature of body [in $^\circ\text{C}$], θ_0 = temperature of surrounding, $\theta - \theta_0$ = excess of temperature ($\theta > \theta_0$)

If the temperature of body decrease $d\theta$ in time dt then rate of fall of temperature $-\frac{d\theta}{dt} \propto (\theta - \theta_0)$

Where negative sign indicates that the rate of cooling is decreasing with time.

Excess of temperature

If the temperature of body decreases from θ_1 to θ_2 and temperature of surroundings is θ_0 then average

$$\text{excess of temperature} = \left[\frac{\theta_1 + \theta_2}{2} - \theta_0\right] \Rightarrow \left[\frac{\theta_1 - \theta_2}{t}\right] = -K \left[\frac{\theta_1 + \theta_2}{2} - \theta_0\right]$$

Ex. If a liquid takes 30 seconds in cooling of 80°C to 70°C and 70 seconds in cooling 60°C to 50°C , then find the room temperature.

Sol. $\frac{\theta_1 - \theta_2}{t} = K \left[\frac{\theta_1 + \theta_2}{2} - \theta_0\right]$

In first case, $\frac{80 - 70}{30} = K \left[\frac{80 + 70}{2} - \theta_0\right] \quad \frac{1}{3} = K (75 - \theta_0) \dots (i)$

In second case, $\frac{60 - 50}{70} = K \left[\frac{60 + 50}{2} - \theta_0\right] \quad \frac{1}{7} = K (55 - \theta_0) \dots (ii)$

Equation (i) divide by equation (ii) $\frac{7}{3} = \frac{(75 - \theta_0)}{(55 - \theta_0)} \Rightarrow 385 - 7\theta_0 = 225 - 3\theta_0 \Rightarrow \theta_0 = \frac{160}{4} = 40^\circ\text{C}$

Limitations of Newton's Law

- Temperature difference should not exceed 35°C , $(\theta - \theta_0) \neq 35^\circ\text{C}$
- Loss of heat should only be by radiation.
- This law is an extended form of Stefan-Boltzman's law.

For Heating, Newton's law of heating $\frac{\theta_1 - \theta_2}{t} = +H \left[\theta_0 - \frac{\theta_1 + \theta_2}{2}\right]$ H heating constant.

Derivation of Newton's law from Steafen's Boltzman law

$$\frac{d\theta}{dt} = \frac{\sigma A}{msJ} (T^4 - T_0^4) \quad \left\{ \begin{array}{l} T - T_0 = \Delta T \\ T = T_0 + \Delta T \end{array} \right\}$$

$$\frac{d\theta}{dt} = \frac{\sigma A}{msJ} (T^4 - T_0^4) \quad T - T_0 = \Delta T, \Delta T \ll T_0$$

$$\frac{d\theta}{dt} = \frac{\sigma A}{msJ} [(T_0 + \Delta T)^4 - T_0^4] \quad \text{If } x \ll 1 \text{ then } (1 + x)^n = 1 + nx$$

$$\frac{d\theta}{dt} = \frac{\sigma A}{msJ} \left[T_0^4 \left(1 + \frac{\Delta T}{T_0} \right)^4 - T_0^4 \right] = \frac{\sigma A}{msJ} T_0^4 \left[\left(1 + \frac{\Delta T}{T_0} \right)^4 - 1 \right] = \frac{\sigma A}{msJ} T_0^4 \left[1 + 4 \frac{\Delta T}{T_0} - 1 \right]$$

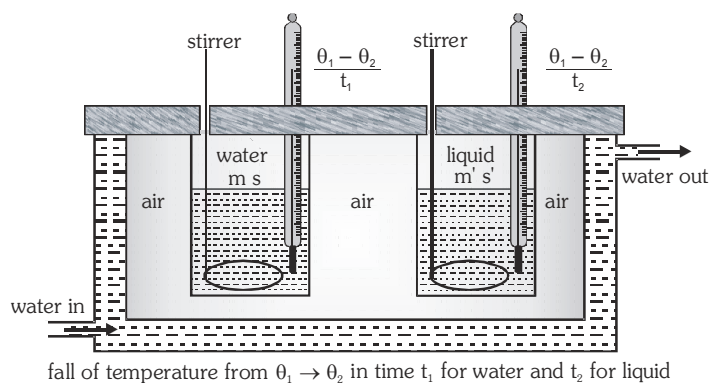
$$\frac{d\theta}{dt} = \left[4 \frac{\sigma A}{msJ} T_0^3 \right] \Delta T \Rightarrow \frac{d\theta}{dt} = K \Delta T \quad \text{constant } K = \frac{4\sigma A T_0^3}{msJ}$$

Newton's law of cooling $\frac{d\theta}{dt} \propto \Delta T$ (for small temperature difference)

APPLICATION OF NEWTON'S LAW OF COOLING

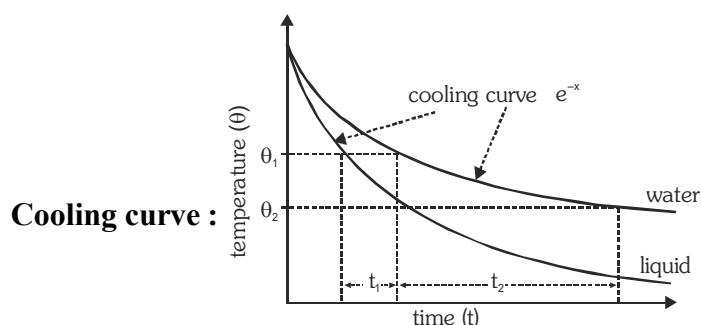
• To find out specific heat of a given liquid

If for the two given liquids their volume, radiating surface area, nature of surface, initial temperature are allowed to cool down in a common environments then rate of loss of heat of these liquids are equal.



$$\therefore \left[\frac{dQ}{dt} \right]_{\text{Water}} = \left[\frac{dQ}{dt} \right]_{\text{Liquid}} \quad \therefore (ms+w) \left[\frac{\theta_1 - \theta_2}{t_1} \right] = (m's'+w) \left[\frac{\theta_1 - \theta_2}{t_2} \right] \Rightarrow \frac{ms+w}{t_1} = \frac{m's'+w}{t_2}$$

where w = water equivalent of calorimeter.



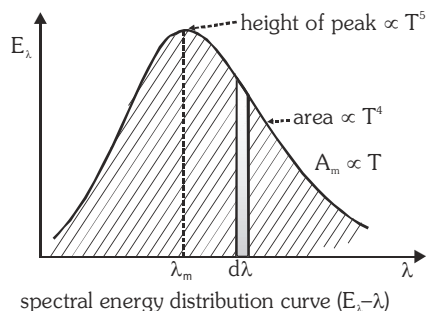
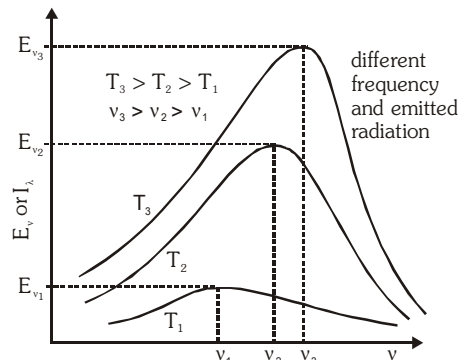
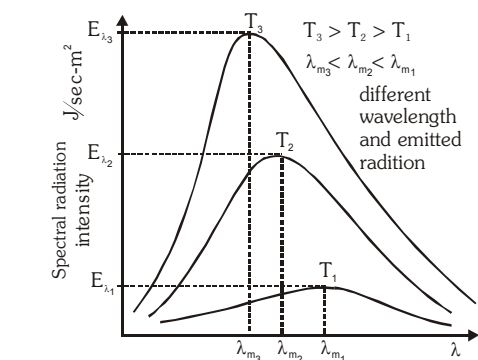
Ex. When a calorimeter contains 40g of water at 50°C, then the temperature falls to 45°C in 10 minutes. The same calorimeter contains 100g of water at 50°C, it takes 20 minutes for the temperature to become 45°C. Find the water equivalent of the calorimeter.

Sol. $\frac{m_1 s_1 + W}{t_1} = \frac{m_2 s_2 + W}{t_2}$ where W is the water equivalent

$$\Rightarrow \frac{40 \times 1 + W}{10} = \frac{100 \times 1 + W}{20} \Rightarrow 80 + 2W = 100 + W \Rightarrow W = 20 \text{ g}$$

SPECTRAL ENERGY DISTRIBUTION CURVE OF BLACK BODY RADIATIONS

Practically given by : Lumers and Pringshem Mathematically given by : Plank



$$(i) \quad \lambda_m \propto \frac{1}{T}$$

$$(ii) \quad E_{\lambda_m} \propto T^5$$

$$(iii) \quad \text{Area} \int_0^\infty E_\lambda d\lambda = E = \sigma T^4 \quad \frac{A_1}{A_2} = \left[\frac{T_1}{T_2} \right]^4$$

- Spectral energy distribution curves are continuous. At any temperature in between possible wavelength ($0 - \infty$) radiation emitted but for different wavelength quantity of radiations are different.
- As the wave length increases, the amount of radiation emitted first increase, becomes maximum and then decreases.
- At a particular temperature the area enclosed between the spectral energy curve shows the spectral

emissive power of the body. $\text{Area} = \int_0^\infty E_\lambda d\lambda = E = \sigma T^4$

WEIN'S DISPLACEMENT LAW

The wavelength corresponding to maximum emission of radiation decrease with increasing temperature $\left[\lambda_m \propto \frac{1}{T} \right]$. This is known as Wein's displacement law. $\lambda_m T = b$ where b Wein's constant $= 2.89 \times 10^{-3} \text{ mK}$.

Dimensions of $\mathbf{b}$: $\text{M}^0 \text{L}^1 \text{T}^0 \theta^1$

Relation between frequency and temperature $v_m = \frac{c}{h}T$

Ex. The temperature of furnace is 2000°C , in its spectrum the maximum intensity is obtained at about $4000\text{\AA}$. If the maximum intensity is at $2000\text{\AA}$ calculate the temperature of the furnace in $^{\circ}\text{C}$.

Sol. by using $\lambda_m T = b$, $4000 (2000+273) = 2000(T) \Rightarrow T = 4546\text{K}$

The temperature of furnace = $4546 - 273 = 4273^{\circ}\text{C}$

SOLAR CONSTANT 'S'

The Sun emits radiant energy continuously in space of which an in significant part reaches the Earth. The solar radiant energy received per unit area per unit time by a black surface held at right angles to the Sun's rays and placed at the mean distance of the Earth (in the absence of atmosphere) is called solar constant.

The solar constant S is taken to be 1340 watts/m^2 or $1.937 \text{ Cal/cm}^2\text{--minute}$

• Temperature of the Sun

Let R be the radius of the Sun and ' d ' be the radius of Earth's orbit around the Sun. Let E be the energy emitted by the Sun per second per unit area. The total energy emitted by the Sun in one second = $E \cdot A = E \times 4\pi R^2$. (This energy is falling on a sphere of radius equal to the radius of the Earth's orbit around the Sun i.e., on a sphere of surface area $4\pi d^2$)

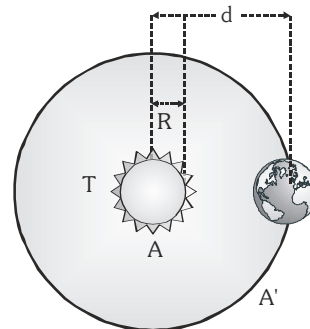
So, The energy falling per unit area of Earth = $\frac{4\pi R^2 \times E}{4\pi d^2} = \frac{E R^2}{d^2}$

$R = 7 \times 10^8 \text{ m}$, $d = 1.5 \times 10^{11} \text{ m}$, $s = 5.7 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Solar constant $S = \frac{E R^2}{d^2}$

By Stefan's Law $E = \sigma T^4$

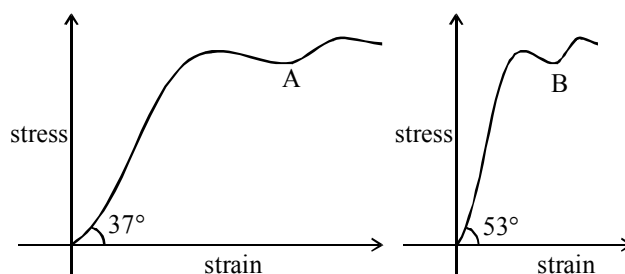
$$S = \frac{\sigma T^4 R^2}{d^2} \Rightarrow T = \left[\frac{S \times d^2}{\sigma \times R^2} \right]^{\frac{1}{4}} = \left[\frac{1340 \times (1.5 \times 10^{11})^2}{5.7 \times 10^{-8} \times (7 \times 10^8)^2} \right]^{\frac{1}{4}} = 5732 \text{ K}$$



EXERCISE (S-1)

Elasticity

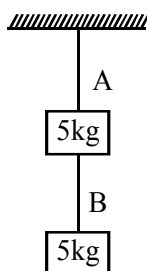
1. A steel wire of length 4.5 m and a copper wire of length 3.5 m are stretched same amount under a given load. If ratio of youngs modulli of steel to that of copper is $\frac{12}{7}$, then what is the ratio of cross sectional area of steel wire to copper wire ?
EH0001
2. Diagram shows stress-strain graph for two material A & B. The graphs are drawn to scale.



The ratio of young modulli of material A to material B is.

EH0002

3. Two identical wires A & B of same material are loaded as shown in figure. If the elongation in wire B is 1.5 mm, what is the elongation in A. (Mass of A & B can be neglected)

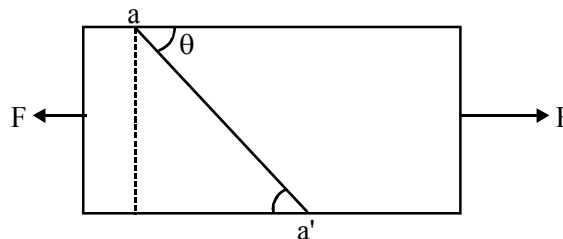


EH0003

4. A wire of length L and radius r is clamped rigidly at one end. When the other end of the wire is pulled by a force f , its length increases by l . Another wire of the same material of length $2L$ and radius $2r$, is pulled by a force $2f$. Find the increase in length of this wire.

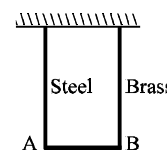
EH0004

5. Consider a long steel bar under a tensile stress due to forces $\vec{F}$ acting at the edges along the length of the bar (Fig.). Consider a plane making an angle θ with the length. What are the tensile and shearing stresses on this plane?
- (a) For what angle is the tensile stress a maximum?
- (b) For what angle is the shearing stress a maximum?



EH0005

6. A light rigid bar AB is suspended horizontally from two vertical wires, one of steel and one of brass, as shown in figure. Each wire is 2.00 m long. The diameter of the steel wire is 0.60 mm and the length of the bar AB is 0.20 m. When a mass of 10 kg is suspended from the centre of AB bar remains horizontal.
- (i) What is the tension in each wire?
- (ii) Calculate the extension of the steel wire and the energy stored in it.
- (iii) Calculate the diameter of the brass wire.
- (iv) If the brass wire were replaced by another brass wire of diameter 1 mm, where should the mass be suspended so that AB would remain horizontal? The Young modulus for steel = 2.0×10^{11} Pa, the Young modulus for brass = 1.0×10^{11} Pa.



EH0006

Calorimetry

7. One day in the morning, Ramesh filled up $1/3$ bucket of hot water from geyser, to take bath. Remaining $2/3$ was to be filled by cold water (at room temperature) to bring mixture to a comfortable temperature. Suddenly Ramesh had to attend to something which would take some times, say 5-10 minutes before he could take bath. Now he had two options: (i) fill the remaining bucket completely by cold water and then attend to the work, (ii) first attend to the work and fill the remaining bucket just before taking bath. Which option do you think would have kept water warmer? Explain.
8. An aluminium container of mass 100 gm contains 200 gm of ice at -20°C . Heat is added to the system at the rate of 100 cal/s. Find the temperature of the system after 4 minutes (specific heat of ice = 0.5 and $L = 80$ cal/gm, specific heat of Al = 0.2 cal/gm/ $^\circ\text{C}$)
9. A hot liquid contained in a container of negligible heat capacity loses temperature at rate 3 K/min, just before it begins to solidify. The temperature remains constant for 30 min. Find the ratio of specific heat capacity of liquid to specific latent heat of fusion is in K^{-1} (given that rate of losing heat is constant).
10. Two 50 gm ice cubes are dropped into 250 gm of water into a glass. If the water was initially at a temperature of 25°C and the temperature of ice -15°C . Find the final temperature of water. (specific heat of ice = 0.5 cal/gm/ $^\circ\text{C}$ and $L = 80$ cal/gm). Find final amount of water and ice.

EH0007

EH0008

EH0009

EH0010

11. A flow calorimeter is used to measure the specific heat of a liquid. Heat is added at a known rate to a stream of the liquid as it passes through the calorimeter at a known rate. Then a measurement of the resulting temperature difference between the inflow and the outflow points of the liquid stream enables us to compute the specific heat of the liquid. A liquid of density 0.2 g/cm^3 flows through a calorimeter at the rate of $10 \text{ cm}^3/\text{s}$. Heat is added by means of a 250-W electric heating coil, and a temperature difference of 25°C is established in steady-state conditions between the inflow and the outflow points. Find the specific heat of the liquid.

EH0011

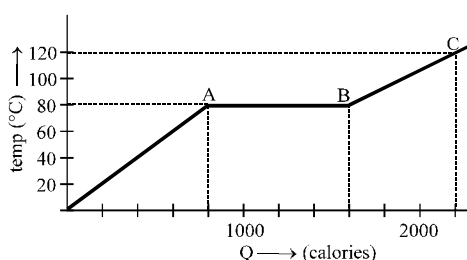
12. Two identical calorimeter A and B contain equal quantity of water at 20°C . A 5 gm piece of metal X of specific heat $0.2 \text{ cal g}^{-1} (\text{C}^\circ)^{-1}$ is dropped into A and a 5 gm piece of metal Y into B. The equilibrium temperature in A is 22°C and in B 23°C . The initial temperature of both the metals is 40°C . Find the specific heat of metal Y in $\text{cal g}^{-1} (\text{C}^\circ)^{-1}$.

EH0012

13. The temperature of 100 gm of water is to be raised from 24°C to 90°C by adding steam to it. Calculate the mass of the steam required for this purpose.

EH0013

14. A substance is in the solid form at 0°C . The amount of heat added to this substance and its temperature are plotted in the following graph. If the relative specific heat capacity of the solid substance is 0.5 , find from the graph



- the mass of the substance ;
- the specific latent heat of the melting process, and
- the specific heat of the substance in the liquid state.

EH0014

Thermal expansion

15. If two rods of length L and $2L$ having coefficients of linear expansion α and 2α respectively are connected so that total length becomes $3L$, determine the average coefficient of linear expansion of the composite rod.

EH0015

16. A clock pendulum made of invar has a period of 0.5 sec at 20°C . If the clock is used in a climate where average temperature is 30°C , approximately. How much fast or slow will the clock run in 10^6 sec . ($\alpha_{\text{invar}} = 1 \times 10^{-6}/^\circ\text{C}$)

EH0016

17. An iron bar (Young's modulus $= 10^{11} \text{ N/m}^2$, $\alpha = 10^{-6}/^\circ\text{C}$) 1 m long and 10^{-3} m^2 in area is heated from 0°C to 100°C without being allowed to bend or expand. Find the compressive force developed inside the bar.

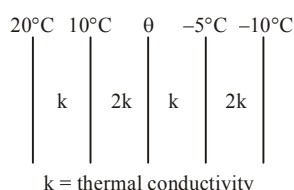
EH0017

Conduction

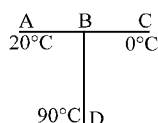
18. A thin walled metal tank of surface area 5m^2 is filled with water and contains an immersion heater dissipating 1 kW . The tank is covered with 4 cm thick layer of insulation whose thermal conductivity is 0.2 W/m/K . The outer face of the insulation is 25°C . Find the temperature of the tank in the steady state

EH0018

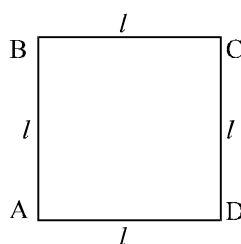
19. The figure shows the face and interface temperature of a composite slab containing of four layers of two materials having identical thickness. Under steady state condition, find the value of temperature θ .

**EH0019**

20. Three conducting rods of same material and cross-section are shown in figure. Temperature of A, D and C are maintained at 20°C , 90°C and 0°C . Find the ratio of length BD and BC if there is no heat flow in AB.

**EH0020**

21. In the square frame of side l of metallic rods, the corners A and C are maintained at T_1 and T_2 respectively. The rate of heat flow from A to C is ω . If A and D are instead maintained T_1 & T_2 respectively. Find the total rate of heat flow.

**EH0021**

22. One end of copper rod of uniform cross-section and of length 1.5 meters is in contact with melting ice and the other end with boiling water. At what point along its length should a temperature of 200°C be maintained, so that in steady state, the mass of ice melting is equal to that of steam produced in the same interval of time? Assume that the whole system is insulated from the surroundings.

EH0022

Radiation

23. Two spheres of same radius R have their densities in the ratio $8 : 1$ and the ratio of their specific heats are $1 : 4$. If by radiation their rates of fall of temperature are same, then find the ratio of their rates of losing heat.

EH0023

24. A solid receives heat by radiation over its surface at the rate of 4 kW . The heat convection rate from the surface of solid to the surrounding is 5.2 kW , and heat is generated at a rate of 1.7 kW over the volume of the solid. The rate of change of the average temperature of the solid is 0.5°Cs^{-1} . Find the heat capacity of the solid.

EH0024

25. A solid copper cube and sphere, both of same mass & emissivity are heated to same initial temperature and kept under identical conditions. What is the ratio of their initial rate of fall of temperature?

EH0025

26. A vessel containing 100 gm water at 0°C is suspended in the middle of a room. In 15 minutes the temperature of the water rises by 2°C . When an equal amount of ice is placed in the vessel, it melts in 10 hours. Calculate the specific heat of fusion of ice.

EH0026

27. The maximum in the energy distribution spectrum of the sun is at $4753 \text{ \AA}$ and its temperature is 6050K . What will be the temperature of the star whose energy distribution shows a maximum at $9506 \text{ \AA}$.

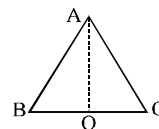
EH0027

28. A pan filled with hot food cools from 50.1°C to 49.9°C in 5 sec . How long will it take to cool from 40.1°C to 39.9°C if room temperature is 30°C ?

EH0028

EXERCISE (S-2)

1. Three aluminium rods of equal length form an equilateral triangle ABC. Taking O (mid point of rod BC) as the origin. Find the increase in Y-coordinate of center of mass per unit change in temperature of the system. Assume the length of the each rod is 2m, and $\alpha_{al} = 4\sqrt{3} \times 10^{-6} / ^\circ\text{C}$



EH0029

2. A metal rod A of 25cm lengths expands by 0.050 cm, when its temperature is raised from 0°C to 100°C . Another rod B of a different metal of length 40cm expands by 0.040 cm for the same rise in temperature. A third rod C of 50cm length is made up of pieces of rods A and B placed end to end expands by 0.03 cm on heating from 0°C to 50°C . Find the lengths of each portion of the composite rod.

EH0030

3. A wire of cross-sectional area $4 \times 10^{-4} \text{ m}^2$, modulus of elasticity $2 \times 10^{11} \text{ N/m}^2$ and length 1 m is stretched between two vertical rigid poles. A mass of 1 kg is suspended at its middle. Calculate the angle it makes with the horizontal.

EH0031

4. A copper calorimeter of mass 100 gm contains 200 gm of a mixture of ice and water. Steam at 100°C under normal pressure is passed into the calorimeter and the temperature of the mixture is allowed to rise to 50°C . If the mass of the calorimeter and its contents is now 330 gm, what was the ratio of ice and water in the beginning? Neglect heat losses.

Given : Specific heat capacity of copper = $0.42 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$,

Specific heat capacity of water = $4.2 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$,

Specific heat of fusion of ice = $3.36 \times 10^5 \text{ J kg}^{-1}$

Latent heat of condensation of steam = $22.5 \times 10^5 \text{ J kg}^{-1}$

EH0032

5. An isosceles triangle is formed with a rod of length l_1 and coefficient of linear expansion α_1 for the base and two thin rods each of length l_2 and coefficient of linear expansion α_2 for the two pieces, if the distance between the apex and the midpoint of the base remain unchanged as the temperatures

varied show that $\frac{l_1}{l_2} = 2\sqrt{\frac{\alpha_2}{\alpha_1}}$.

EH0033

6. A steel drill making 180 rpm is used to drill a hole in a block of steel. The mass of the steel block and the drill is 180 gm. If the entire mechanical work is used up in producing heat and the rate of raise in temperature of the block and the drill is 0.5°C/s . Find (a) the rate of working of the drill in watts, and (b) the torque required to drive the drill. Specific heat of steel = 0.1 and $J = 4.2 \text{ J/cal}$. Use : $P = \tau \omega$

EH0034

7. Ice at -20°C is filled upto height $h = 10\text{ cm}$ in a uniform cylindrical vessel. Water at temperature $\theta^{\circ}\text{C}$ is filled in another identical vessel upto the same height $h = 10\text{ cm}$. Now, water from second vessel is poured into first vessel and it is found that level of upper surface falls through $\Delta h = 0.5\text{ cm}$ when thermal equilibrium is reached. Neglecting thermal capacity of vessels, change in density of water due to change in temperature and loss of heat due to radiation, calculate initial temperature θ of water. Given,
- Density of water : $\rho_w = 1\text{ gm cm}^{-3}$ Density of ice : $\rho_i = 0.9\text{ gm/cm}^3$
 Specific heat of water : $s_w = 1\text{ cal/gm }^{\circ}\text{C}$ Specific heat of ice : $s_i = 0.5\text{ cal/gm}^{\circ}\text{C}$
 Specific latent heat of ice : $L = 80\text{ cal/gm}$

EH0035

8. A highly conducting solid cylinder of radius a and length l is surrounded by a co-axial layer of a material having thermal conductivity K and negligible heat capacity. Temperature of surrounding space (out side the layer) is T_0 , which is higher than temperature of the cylinder. If heat capacity per unit volume of cylinder material is s and outer radius of the layer is b , calculate time required to increase temperature of the cylinder from T_1 to T_2 . Assume end faces to be thermally insulated.

EH0036

9. A lagged stick of cross section area 1 cm^2 and length 1 m is initially at a temperature of 0°C . It is then kept between 2 reservoirs of temperature 100°C and 0°C . Specific heat capacity is $10\text{ J/kg}^{\circ}\text{C}$ and linear mass density is 2 kg/m . Find



- (a) temperature gradient along the rod in steady state.
 (b) total heat absorbed by the rod to reach steady state.

EH0037

10. A cylindrical block of length 0.4 m and an area of cross-section 0.04 m^2 is placed coaxially on a thin metal disc of mass 0.4 kg and of the same cross-section. The upper face of the cylinder is maintained at a constant temperature of 400 K and the initial temperature of the disc is 300 K . If the thermal conductivity of the material of the cylinder is 10 watt/m-K and the specific heat of the material of the disc is 600 J/kg-K , how long will it take for the temperature of the disc to increase to 350 K ? Assume, for purposes of calculation, the thermal conductivity of the disc to be very high and the system to be thermally insulated except for the upper face of the cylinder.

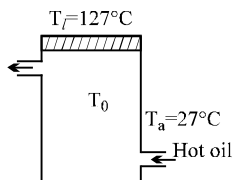
EH0038

11. A liquid takes 5 minutes to cool from 80°C to 50°C . How much time will it take to cool from 60°C to 30°C ? The temperature of surrounding is 20°C . Use exact method.

EH0039

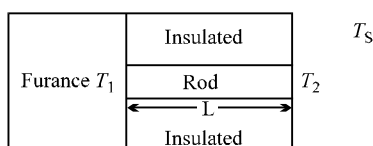
12. Hot oil is circulated through an insulated container with a wooden lid at the top whose conductivity $K = 0.149 \text{ J/(m}^\circ\text{C}\cdot\text{sec)}$, thickness $t = 5 \text{ mm}$, emissivity $= 0.6$. Temperature of the top of the lid in steady state is at $T_l = 127^\circ$. If the ambient temperature $T_a = 27^\circ\text{C}$. Calculate [JEE' 2003]
- (a) rate of heat loss per unit area due to radiation from the lid.

(b) temperature of the oil. (Given $\sigma = \frac{17}{3} \times 10^{-8}$)



EH0040

13. One end of a rod of length L and cross-sectional area A is kept in a furnace of temperature T_1 . The other end of the rod is kept at a temperature T_2 . The thermal conductivity of the material of the rod is K and emissivity of the rod is e . It is given that $T_2 = T_s + \Delta T$ where $\Delta T \ll T_s$, T_s being the temperature of the surroundings. If $\Delta T \propto (T_1 - T_s)$, find the proportionality constant. Consider that heat is lost only by radiation at the end where the temperature of the rod is T_2 . [JEE 2004]



EH0041

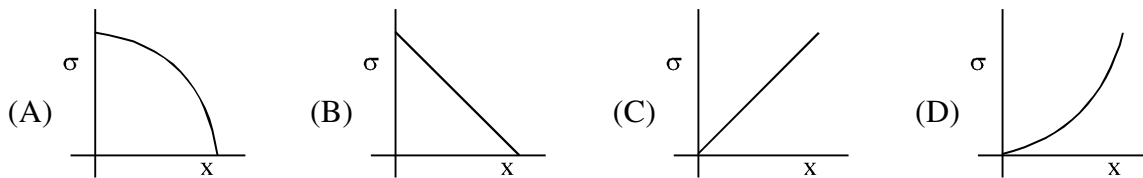
EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Elasticity

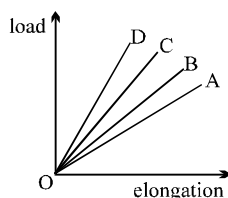
- The maximum load a wire can withstand without breaking, when its length is reduced to half of its original length, will
(A) be double. (B) be half. (C) be four times. (D) remain same.
EH0042
- The temperature of a wire is doubled. The Young's modulus of elasticity
(A) will also double. (B) will become four times.
(C) will remain same. (D) will decrease.
EH0043
- A spring is stretched by applying a load to its free end. The strain produced in the spring is
(A) volumetric. (B) shear.
(C) longitudinal and shear. (D) longitudinal.
EH0044
- Overall changes in volume and radii of a uniform cylindrical steel wire are 0.2% and 0.002% respectively when subjected to some suitable force. Longitudinal tensile stress acting on the wire is :-
($Y = 2.0 \times 10^{11} \text{ Nm}^{-2}$)
(A) $3.2 \times 10^9 \text{ Nm}^{-2}$ (B) $3.2 \times 10^7 \text{ Nm}^{-2}$ (C) $3.6 \times 10^9 \text{ Nm}^{-2}$ (D) $3.9 \times 10^8 \text{ Nm}^{-2}$
EH0045
- A solid sphere of radius R made of material of bulk modulus K is surrounded by a liquid in a cylindrical container. A massless piston of area A floats on the surface of the liquid. When a mass m is placed on the piston to compress the liquid, the fractional change in the radius of the sphere $\delta R/R$ is
(A) mg/AK (B) $mg/3AK$ (C) mg/A (D) $mg/3AR$
EH0046
- A wire suspended vertically from one of its ends is stretched by attaching a weight of 200 N to the lower ends. The weight stretches the wire by 1mm. Then the elastic energy stored in the wire is-
[AIEEE - 2003]
(A) 0.2 J (B) 10 J (C) 20 J (D) 0.1 J
EH0047

7. A uniform rod rotating in gravity free region with certain constant angular velocity. The variation of tensile stress with distance x from axis of rotation is best represented by which of the following graphs.



EH0048

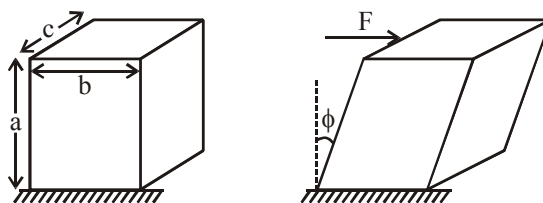
8. The load versus strain graph for four wires of the same material is shown in the figure. The thickest wire is represented by the line :-



- (A) OB (B) OA (C) OD (D) OC

EH0049

9. A cuboidal block of sides a , b & c is fixed on ground. The top is pushed by a horizontal force F as shown. The angle ϕ by which the block deforms is : (η is modulus of rigidity)



- (A) $\frac{F}{ab\eta}$ (B) $\frac{F}{ac\eta}$ (C) $\frac{F}{bc\eta}$ (D) $\frac{F}{\sqrt{b^2 + c^2} \eta}$

EH0050

Calorimetry

10. Heat is associated with

- (A) kinetic energy of random motion of molecules.
 (B) kinetic energy of orderly motion of molecules.
 (C) total kinetic energy of random and orderly motion of molecules.
 (D) kinetic energy of random motion in some cases and kinetic energy of orderly motion in other.

EH0051

11. Equal amount of heat energy are transferred into equal mass of ethyl alcohol and water sample. The rise in temperature of water sample is 25°C . The temperature rise of ethyl alcohol will be.
(Specific heat of ethyl alcohol is one half of the specific heat of water).
(A) 12.5°C (B) 25°C
(C) 50°C (D) It depends on the rate of energy transfer.

EH0052

12. A block of mass 2.5 kg is heated to temperature of 500°C and placed on a large ice block. What is the maximum amount of ice that can melt (approx.). Specific heat for the body = $0.1\text{ Cal/gm}^{\circ}\text{C}$.
(A) 1 kg (B) 1.5 kg (C) 2 kg (D) 2.5 kg

EH0053

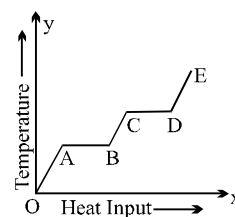
13. 10 gm of ice at 0°C is kept in a calorimeter of water equivalent 10 gm . How much heat should be supplied to the apparatus to evaporate the water thus formed? (Neglect loss of heat)
(A) 6200 cal (B) 7200 cal (C) 13600 cal (D) 8200 cal

EH0054

14. A continuous flow water heater (geyser) has an electrical power rating = 2 kW and efficiency of conversion of electrical power into heat = 80% . If water is flowing through the device at the rate of 100 cc/sec , and the inlet temperature is 10°C , the outlet temperature will be
(A) 12.2°C (B) 13.8°C (C) 20°C (D) 16.5°C

EH0055

15. A solid material is supplied with heat at a constant rate. The temperature of material is changing with heat input as shown in the figure. What does slope DE represents?
(A) latent heat of liquid
(B) latent heat of vapour
(C) heat capacity of vapour
(D) inverse of heat capacity of vapour



EH0056

16. A block of ice with mass m falls into a lake. After impact, a mass of ice $m/5$ melts. Both the block of ice and the lake have a temperature of 0°C . If L represents the heat of fusion, the minimum distance the ice fell before striking the surface is

- (A) $\frac{L}{5g}$ (B) $\frac{5L}{g}$ (C) $\frac{gL}{5m}$ (D) $\frac{mL}{5g}$

EH0057

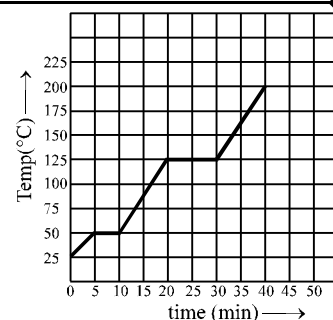
17. The specific heat of a metal at low temperatures varies according to $S = aT^3$ where a is a constant and T is the absolute temperature. The heat energy needed to raise unit mass of the metal from $T = 1\text{ K}$ to $T = 2\text{ K}$ is :-

- (A) $3a$ (B) $\frac{15a}{4}$ (C) $\frac{2a}{3}$ (D) $\frac{12a}{5}$

EH0058

18. The graph shown in the figure represent change in the temperature of 5 kg of a substance as it absorbs heat at a constant rate of 42 kJ min^{-1} . The latent heat of vapourization of the substance is :

(A) 630 kJ kg^{-1}
 (B) 126 kJ kg^{-1}
 (C) 84 kJ kg^{-1}
 (D) 12.6 kJ kg^{-1}



EH0059

19. The density of a material A is 1500 kg/m^3 and that of another material B is 2000 kg/m^3 . It is found that the heat capacity of 8 volumes of A is equal to heat capacity of 12 volumes of B. The ratio of specific heats of A and B will be

(A) 1 : 2 (B) 3 : 1 (C) 3 : 2 (D) 2 : 1

EH0060

20. 2 kg ice at -20°C is mixed with 5 kg water at 20°C . Then final amount of water in the mixture would be; Given specific heat of ice = $0.5 \text{ cal/g}^\circ\text{C}$, specific heat of water = $1 \text{ cal/g}^\circ\text{C}$, Latent heat of fusion of ice = 80 cal/g .

[JEE' (Scr) 2003]

(A) 6 kg (B) 5 kg (C) 4 kg (D) 2 kg

EH0061

21. Some steam at 100°C is passed into 1.1 kg of water contained in a calorimeter of water equivalent 0.02 kg at 15°C so that the temperature of the calorimeter and its contents rises to 80°C . What is the mass of steam condensing. (in kg)

(A) 0.130 (B) 0.065 (C) 0.260 (D) 0.135

EH0062

22. Find the amount of heat supplied to decrease the volume of an ice water mixture by 1 cm^3 without any change in temperature. ($\rho_{\text{ice}} = 0.9 \rho_{\text{water}}$, $L_{\text{ice}} = 80 \text{ cal/gm}$).

(A) 360 cal (B) 500 cal (C) 720 cal (D) none of these

EH0063

Thermal expansion

23. The radius of a metal sphere at room temperature T is R , and the coefficient of linear expansion of the metal is α . The sphere is heated a little by a temperature ΔT so that its new temperature is $T + \Delta T$. The increase in the volume of the sphere is approximately

(A) $2\pi R \propto \Delta T$ (B) $\pi R^2 \propto \Delta T$ (C) $4\pi R^3 \propto \Delta T / 3$ (D) $4\pi R^3 \propto \Delta T$

EH0064

24. A hole is made in a metal plate, when the temperature of metal is raised then the diameter of the hole will :-

(A) Decrease
 (B) Increase
 (C) Remain same
 (D) Answer depends upon the initial temperature of the metal

EH0065

25. A rod of length 2m rests on smooth horizontal floor. If the rod is heated from 0°C to 20°C . Find the longitudinal strain developed? ($\alpha = 5 \times 10^{-5}/^\circ\text{C}$)

(A) 10^{-3} (B) 2×10^{-3} (C) Zero (D) None

EH0066

26. A steel tape gives correct measurement at 20°C . A piece of wood is being measured with the steel tape at 0°C . The reading is 25 cm on the tape, the real length of the given piece of wood must be:
 (A) 25 cm (B) <25 cm (C) >25 cm (D) can not say

EH0067

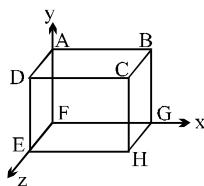
27. The bulk modulus of copper is 1.4×10^{11} Pa and the coefficient of linear expansion is $1.7 \times 10^{-5} (\text{C}^{\circ})^{-1}$. What hydrostatic pressure is necessary to prevent a copper block from expanding when its temperature is increased from 20°C to 30°C ?
 (A) 6.0×10^5 Pa (B) 7.1×10^7 Pa (C) 5.2×10^6 Pa (D) 40 atm

EH0068

28. A thin copper wire of length L increase in length by 1% when heated from temperature T_1 to T_2 . What is the percentage change in area when a thin copper plate having dimensions $2L \times L$ is heated from T_1 to T_2 ?
 (A) 1% (B) 2% (C) 3% (D) 4%

EH0069

29. A cuboid ABCDEFGH is anisotropic with $\alpha_x = 1 \times 10^{-5} /^{\circ}\text{C}$, $\alpha_y = 2 \times 10^{-5} /^{\circ}\text{C}$, $\alpha_z = 3 \times 10^{-5} /^{\circ}\text{C}$. Coefficient of superficial expansion of faces can be



- (A) $\beta_{ABCD} = 5 \times 10^{-5} /^{\circ}\text{C}$ (B) $\beta_{BCGH} = 4 \times 10^{-5} /^{\circ}\text{C}$
 (C) $\beta_{CDEH} = 3 \times 10^{-5} /^{\circ}\text{C}$ (D) $\beta_{EFGH} = 2 \times 10^{-5} /^{\circ}\text{C}$

EH0070

30. The coefficient of apparent expansion of a liquid in a copper vessel is C and in a silver vessel is S . The coefficient of volume expansion of copper is γ_c . What is the coefficient of linear expansion of silver?
 (A) $\frac{(C + \gamma_c + S)}{3}$ (B) $\frac{(C - \gamma_c + S)}{3}$ (C) $\frac{(C + \gamma_c - S)}{3}$ (D) $\frac{(C - \gamma_c - S)}{3}$

EH0071

31. Two rods one of aluminium of length l_1 having coefficient of linear expansion α_a , and other steel of length l_2 having coefficient of linear expansion α_s are joined end to end. The expansion in both the

rods is same on variation of temperature. Then the value of $\frac{l_1}{l_1 + l_2}$ is

[JEE' (Scr) 2003]

- (A) $\frac{\alpha_s}{\alpha_a + \alpha_s}$ (B) $\frac{\alpha_s}{\alpha_a - \alpha_s}$ (C) $\frac{\alpha_a + \alpha_s}{\alpha_s}$ (D) None of these

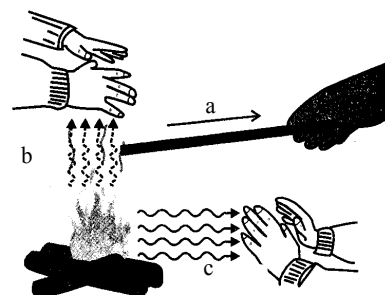
EH0072

32. An open vessel is filled completely with oil which has same coefficient of volume expansion as that of the vessel. On heating both oil and vessel,
- (A) the vessel can contain more volume and more mass of oil
 (B) the vessel can contain same volume and same mass of oil
 (C) the vessel can contain same volume but more mass of oil
 (D) the vessel can contain more volume but same mass of oil

EH0073

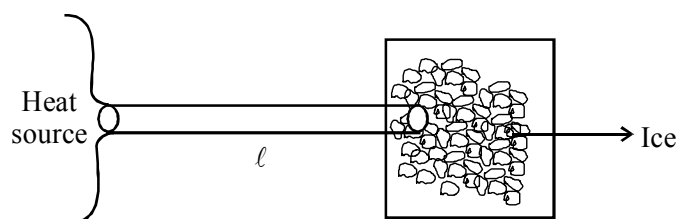
Conduction

33. Diagram shows a heat source 'S' and three position of heat recover (hand). The main mode of heat transfer is given as 'a', 'b' & 'c'. Choose the correct matching :-
- (A) a – conduction ; b – convection ; c – radiation
 (B) b – conduction ; a – convection ; c – radiation
 (C) a – conduction ; c – convection ; b – radiation
 (D) c – conduction ; b – radiation ; a – convection

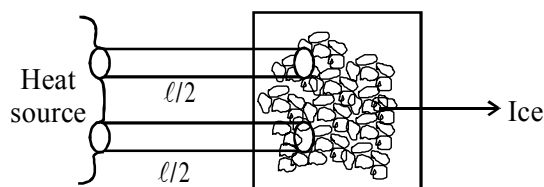


EH0074

34. A rod of length ' ℓ ' and cross-section 'A' is used to melt a piece of ice as shown.



Now if the rod broken into two equal parts and is arranged as shown.

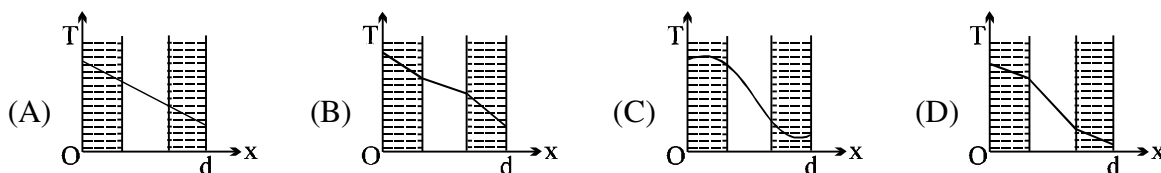


Time taken to melt ice in second use becomes.

- (A) Half (B) One-fourth (C) Twice (D) Four times
35. One end of a 2.35m long and 2.0cm radius aluminium rod ($K = 235 \text{ W.m}^{-1}\text{K}^{-1}$) is held at 20°C . The other end of the rod is in contact with a block of ice at its melting point. The rate in kg.s^{-1} at which ice melts is [Take latent heat of fusion for ice as $\frac{10}{3} \times 10^5 \text{ J.kg}^{-1}$]
- (A) $48\pi \times 10^{-6}$ (B) $24\pi \times 10^{-6}$ (C) $2.4\pi \times 10^{-6}$ (D) $4.8\pi \times 10^{-6}$

EH0076

36. The wall with a cavity consists of two layers of brick separated by a layer of air. All three layers have the same thickness and the thermal conductivity of the brick is much greater than that of air. The left layer is at a higher temperature than the right layer and steady state condition exists. Which of the following graphs predicts correctly the variation of temperature T with distance d inside the cavity?



EH0077

37. A wall has two layer A and B each made of different material, both the layers have the same thickness. The thermal conductivity of the material A is twice that of B. Under thermal equilibrium the temperature difference across the wall B is 36°C . The temperature difference across the wall A is
- (A) 6°C (B) 12°C (C) 18°C (D) 72°C

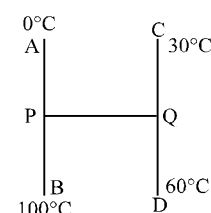
EH0078

38. Two identical conducting rods are first connected independently to two vessels, one containing water at 100°C and the other containing ice at 0°C . In the second case, the rods are joined end to end and connected to the same vessels. Let q_1 and q_2 g/s be the rate of melting of ice in the two cases respectively. The ratio q_2/q_1 is
- [JEE' 2004 (Scr.)]

- (A) $1/2$ (B) $2/1$ (C) $4/1$ (D) $1/4$

EH0079

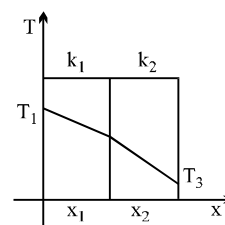
39. Three identical rods AB, CD and PQ are joined as shown. P and Q are mid points of AB and CD respectively. Ends A, B, C and D are maintained at 0°C , 100°C , 30°C and 60°C respectively. The direction of heat flow in PQ is
- (A) from P to Q (B) from Q to P
(C) heat does not flow in PQ (D) data not sufficient



EH0080

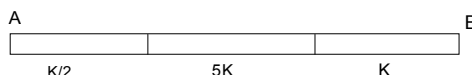
40. The temperature drop through each layer of a two layer furnace wall is shown in figure. Assume that the external temperature T_1 and T_3 are maintained constant and $T_1 > T_3$. If the thickness of the layers x_1 and x_2 are the same, which of the following statements are correct.

- (A) $k_1 > k_2$
(B) $k_1 < k_2$
(C) $k_1 = k_2$ but heat flow through material (1) is larger than through (2)
(D) $k_1 = k_2$ but heat flow through material (1) is less than that through (2)



EH0081

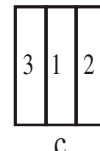
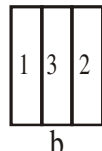
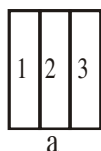
41. A composite rod made of three rods of equal length and cross-section as shown in the fig. The thermal conductivities of the materials of the rods are $K/2$, $5K$ and K respectively. The end A and end B are at constant temperatures. All heat entering the end A goes out of the end B, there being no loss of heat from the sides of the bar. The effective thermal conductivity of the bar is



- (A) $15K/16$ (B) $6K/13$ (C) $5K/16$ (D) $2K/13$.

EH0082

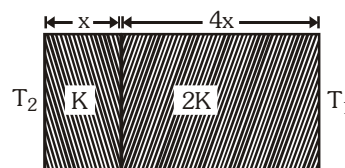
42. Figure shows three different arrangements of materials 1, 2 and 3 to form a wall. Thermal conductivities are $k_1 > k_2 > k_3$. The left side of the wall is 20°C higher than the right side. Temperature difference ΔT across the material 1 has following relation in three cases :



- (A) $\Delta T_a > \Delta T_b > \Delta T_c$ (B) $\Delta T_a = \Delta T_b = \Delta T_c$ (C) $\Delta T_a = \Delta T_b > \Delta T_c$ (D) $\Delta T_a = \Delta T_b < \Delta T_c$

EH0083

43. The temperature of the two outer surfaces of a composite slab, consisting of two materials having coefficients of thermal conductivity K and $2K$ and thickness x and $4x$, respectively are T_2 and T_1 ($T_2 > T_1$). The rate of



heat transfer through the slab, in a steady state is $\left(\frac{A(T_2 - T_1)K}{x}\right)f$, with f

equals to-

[AIEEE - 2004]

- (A) 1 (B) $1/2$ (C) $2/3$ (D) $1/3$

EH0084

Radiation

44. A black metal foil is warmed by radiation from a small sphere at temperature ' T ' and at a distance ' d '. It is found that the power received by the foil is P . If both the temperature and distance are doubled, the power received by the foil will be :

- (A) $16P$ (B) $4P$ (C) $2P$ (D) P

EH0085

45. The rate of emission of radiation of a black body at 273°C is E , then the rate of emission of radiation of this body at 0°C will be :-

- (A) $\frac{E}{16}$ (B) $\frac{E}{4}$ (C) $\frac{E}{8}$ (D) 0

EH0086

46. The power radiated by a black body is P and it radiates maximum energy around the wavelength λ_0 . If the temperature of the black body is now changed so that it radiates maximum energy around wavelength $3/4\lambda_0$, the power radiated by it will increase by a factor of :-
 (A) $4/3$ (B) $16/9$ (C) $64/27$ (D) $256/81$

EH0087

47. Spheres P and Q are uniformly constructed from the same material which is a good conductor of heat and the radius of Q is thrice the radius of P. The rate of fall of temperature of P is x times that of Q when both are at the same surface temperature. The value of x is :
 (A) $1/4$ (B) $1/3$ (C) 3 (D) 4

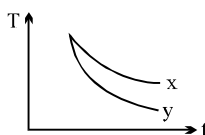
EH0088

48. Two spheres of the same material have radii 1 m and 4 m and temperatures 4000 K and 2000 K respectively. The ratio of the energy radiated per second by the first sphere to that by the second is-
 (A) $1 : 1$ (B) $16 : 1$ (C) $4 : 1$ (D) $1 : 9$

EH0089

49. If emissivity of bodies X and Y are e_x and e_y and absorptive power are A_x and A_y then

[JEE' (Scr) 2003]



- (A) $e_y > e_x ; A_y > A_x$ (B) $e_y < e_x ; A_y < A_x$ (C) $e_y > e_x ; A_y < A_x$ (D) $e_y = e_x ; A_y = A_x$

EH0090

50. Three discs A, B, and C having radii 2 m, 4 m and 6 m respectively are coated with carbon black on their outer surfaces. The wavelengths corresponding to maximum intensity are 300 nm, 400 nm and 500 nm respectively. The power radiated by them are Q_A , Q_B and Q_C respectively.

[JEE' 2004 (Scr.)]

- (A) Q_A is maximum (B) Q_B is maximum (C) Q_C is maximum (D) $Q_A = Q_B = Q_C$

EH0091

51. A black body calorimeter filled with hot water cools from 60°C to 50°C in 4 min and 40°C to 30°C in 8 min. The approximate temperature of surrounding is :
 (A) 10°C (B) 15°C (C) 20°C (D) 25°C

EH0092

52. A system S receives heat continuously from an electrical heater of power 10W. The temperature of S becomes constant at 50°C when the surrounding temperature is 20°C . After the heater is switched off, S cools from 35.1°C to 34.9°C in 1 minute. The heat capacity of S is
 (A) $100\text{J}/^\circ\text{C}$ (B) $300\text{J}/^\circ\text{C}$ (C) $750\text{J}/^\circ\text{C}$ (D) $1500\text{J}/^\circ\text{C}$

EH0093

53. If the temperature of the sun were to increase from T to $2T$ and its radius from R to $2R$, then the ratio of the radiant energy received on earth to what it was previously, will be-
 (A) 4 (B) 16 (C) 32 (D) 64

[AIEEE - 2004]

EH0094

MULTIPLE CORRECT TYPE QUESTIONS

Elasticity

54. A wire is suspended from the ceiling and stretched under the action of a weight F suspended from its other end. The force exerted by the ceiling on it is equal and opposite to the weight.
- (A) Tensile stress at any cross section A of the wire is F/A .
 (B) Tensile stress at any cross section is zero.
 (C) Tensile stress at any cross section A of the wire is $2F/A$.
 (D) Tension at any cross section A of the wire is F .

EH0095

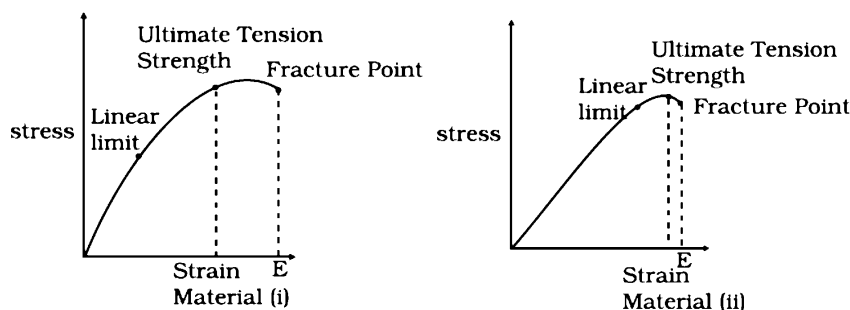
55. A copper and a steel wire of the same diameter are connected end to end. A deforming force F is applied to this composite wire which causes a total elongation of 1cm. The two wires will have
- (A) the same stress. (B) different stress. (C) the same strain. (D) different strain.

EH0096

56. A body of mass M is attached to the lower end of a metal wire, whose upper end is fixed. The elongation of the wire is l .
- (A) Loss in gravitational potential energy of M is Mgl
 (B) The elastic potential energy stored in the wire is Mgl
 (C) The elastic potential energy stored in the wire is $1/2 Mgl$
 (D) Heat produced is $1/2 Mgl$.

EH0097

57. The stress-strain graphs for two materials are shown in figure (assume same scale).



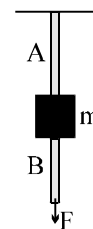
- (A) Material (ii) is more elastic than material (i) and hence material (ii) is more brittle.
 (B) Material (i) and (ii) have the same elasticity and the same brittleness.
 (C) Material (ii) is elastic over a larger region of strain as compared to (i).
 (D) Material (ii) is more brittle than material (i).

EH0098

58. A composite rod consists of a steel rod of length 25 cm and area $2A$ and a copper rod of length 50cm and area A . The composite rod is subjected to an axial load F . If the Young's modulus of steel and copper are in the ratio 2 : 1.
- (A) the extension produced in copper rod will be more .
 (B) the extension in copper and steel parts will be in the ratio 2 : 1.
 (C) the stress applied to the copper rod will be more.
 (D) no extension will be produced in the steel rod.

EH0099

59. The wires A and B shown in the figure are made of the same material and have radii r_A and r_B respectively. The block between them has a mass m . When the force F is $mg/3$, one of the wires breaks.



- (A) A breaks if $r_A = r_B$
 (B) A breaks if $r_A < 2r_B$
 (C) Either A or B may break if $r_A = 2r_B$
 (D) The lengths of A and B must be known to predict which wire will break

EH0100

Calorimetry

60. Mark the **CORRECT** options:

- (A) A system X is in thermal equilibrium with Y but not with Z . System Y and Z may be in thermal equilibrium with each other.
 (B) A system X is in thermal equilibrium with Y but not with Z . Systems Y and Z are not in thermal equilibrium with each other.
 (C) A system X is neither in thermal equilibrium with Y nor with Z . The systems Y and Z must be in thermal equilibrium with each other.
 (D) A system X is neither in thermal equilibrium with Y nor with Z . The system Y and Z may be in thermal equilibrium with each other.

EH0101

61. 50 gm ice at -10°C is mixed with 20 gm steam at 100°C . When the mixture finally reaches its steady state inside a calorimeter of water equivalent 1.5 gm then : [Assume calorimeter was initially at 0°C , Take latent heat of vaporization of water = 540 cal/gm, Latent heat of fusion of water = 80 cal/gm, specific heat capacity of water = 1 cal/gm $^\circ\text{C}$, specific heat capacity of ice = 0.5 cal/gm $^\circ\text{C}$]
 (A) Mass of water remaining is : 67.4 gm (B) Mass of steam remaining is : 2.6 gm
 (C) Mass of water remaining is : 67.87 gm (D) Mass of steam remaining is : 2.13 gm

EH0102

Thermal expansion

62. When the temperature of a copper coin is raised by 80°C , its diameter increases by 0.2%.
 (A) Percentage rise in the area of a face is 0.4 %
 (B) Percentage rise in the thickness is 0.4 %
 (C) Percentage rise in the volume is 0.6 %
 (D) Coefficient of linear expansion of copper is $0.25 \times 10^{-4} \text{ }^\circ\text{C}^{-1}$.

EH0103

Radiation

63. Two metallic sphere A and B are made of same material and have got identical surface finish. The mass of sphere A is four times that of B. Both the spheres are heated to the same temperature and placed in a room having lower temperature but thermally insulated from each other.
 (A) The ratio of heat loss of A to that of B is $2^{4/3}$.
 (B) The ratio of heat loss of A to that of B is $2^{2/3}$.
 (C) The ratio of the initial rate of cooling of A to that of B is $2^{-2/3}$.
 (D) The ratio of the initial rate of cooling of A to that of B is $2^{-4/3}$.

EH0104

64. Two bodies A and B have thermal emissivities of 0.01 and 0.81 respectively. The outer surface areas of the two bodies are the same. The two bodies radiate energy at the same rate. The wavelength λ_B , corresponding to the maximum spectral radiance in the radiation from B, is shifted from the wavelength corresponding to the maximum spectral radiance in the radiation from A by $1.00 \mu\text{m}$. If the temperature of A is 5802 K ,
- (A) the temperature of B is 1934 K (B) $\lambda_B = 1.5 \mu\text{m}$
 (C) the temperature of B is 11604 K (D) the temperature of B is 2901 K

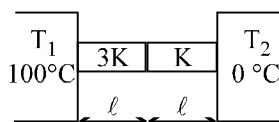
EH0105

COMPREHENSION TYPE QUESTIONS

Conduction

Paragraph for Question No. 65 to 67

Two rods A and B of same cross-sectional area A and length l connected in series between a source ($T_1 = 100^\circ\text{C}$) and a sink ($T_2 = 0^\circ\text{C}$) as shown in figure. The rod is laterally insulated



65. The ratio of the thermal resistance of the rods is

(A) $\frac{R_A}{R_B} = \frac{1}{3}$ (B) $\frac{R_A}{R_B} = 3$ (C) $\frac{R_A}{R_B} = \frac{3}{4}$ (D) $\frac{4}{3}$

EH0106

66. If T_A and T_B are the temperature drops across the rod A and B, then

(A) $\frac{T_A}{T_B} = \frac{3}{1}$ (B) $\frac{T_A}{T_B} = \frac{1}{3}$ (C) $\frac{T_A}{T_B} = \frac{3}{4}$ (D) $\frac{T_A}{T_B} = \frac{4}{3}$

EH0106

67. If G_A and G_B are the temperature gradients across the rod A and B, then

(A) $\frac{G_A}{G_B} = \frac{3}{1}$ (B) $\frac{G_A}{G_B} = \frac{1}{3}$ (C) $\frac{G_A}{G_B} = \frac{3}{4}$ (D) $\frac{G_A}{G_B} = \frac{4}{3}$

EH0106

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

1. A cylindrical wire of radius 1 mm, length 1 m, Young's modulus $= 2 \times 10^{11} \text{ N/m}^2$, poisson's ratio $\mu = \pi/10$ is stretched by a force of 100 N. Its radius will become
(A) 0.99998 mm (B) 0.99999 mm (C) 0.99997 mm (D) 0.99995 mm

EH0107

2. A thermally insulated vessel contains some water at 0°C . The vessel is connected to a vacuum pump to pump out water vapour. This results in some water getting frozen. It is given Latent heat of vaporization of water at $0^\circ\text{C} = 21 \times 10^5 \text{ J/kg}$ and latent heat of freezing of water $= 3.36 \times 10^5 \text{ J/kg}$. The maximum percentage amount of water that will be solidified in this manner will be :-
(A) 86.2% (B) 33.6% (C) 21% (D) 24.36%

EH0108

3. Ice at 0°C is added to 200 g of water initially at 70°C in a vacuum flask. When 50 g of ice has been added and has all melted the temperature of the flask and contents is 40°C . When a further 80g of ice has been added and has all melted, the temperature of the whole is 10°C . Calculate the specific latent heat of fusion of ice. [Take $S_w = 1 \text{ cal/gm } ^\circ\text{C}$.]
(A) $3.8 \times 10^5 \text{ J/kg}$ (B) $1.2 \times 10^5 \text{ J/kg}$ (C) $2.4 \times 10^5 \text{ J/kg}$ (D) $3.0 \times 10^5 \text{ J/kg}$

EH0109

4. The coefficient of linear expansion of copper is $17 \times 10^{-6} (^\circ\text{C})^{-1}$. A copper statue is 93 m tall on the summer morning of temperature 25°C . What is maximum order of increase in magnitude of the height in statue (maximum temperature of day is 45°C)
(A) 0.1 mm (B) 1 mm (C) 10 mm (D) 100 mm

EH0110

5. The coefficients of thermal expansion of steel and a metal X are respectively 12×10^{-6} and $2 \times 10^{-6} \text{ per } ^\circ\text{C}$. At 40°C , the side of a cube of metal X was measured using a steel vernier callipers. The reading was 100 mm. Assuming that the calibration of the vernier was done at 0°C , then the actual length of the side of the cube at 0°C will be
(A) $> 100 \text{ mm}$ (B) $< 100 \text{ mm}$
(C) $= 100 \text{ mm}$ (D) data insufficient to conclude

EH0111

6. The volume of the bulb of a mercury thermometer at 0°C is V_0 and cross section of the capillary is A_0 . The coefficient of linear expansion of glass is α_g per $^\circ\text{C}$ and the cubical expansion of mercury γ_m per $^\circ\text{C}$. If the mercury just fills the bulb at 0°C , what is the length of mercury column in capillary at $T^\circ\text{C}$.

$$(A) \frac{V_0 T (\gamma_m + 3\alpha_g)}{A_0 (1 + 2\alpha_g T)} \quad (B) \frac{V_0 T (\gamma_m - 3\alpha_g)}{A_0 (1 + 2\alpha_g T)} \quad (C) \frac{V_0 T (\gamma_m + 2\alpha_g)}{A_0 (1 + 3\alpha_g T)} \quad (D) \frac{V_0 T (\gamma_m - 2\alpha_g)}{A_0 (1 + 3\alpha_g T)}$$

EH0112

7. A rod of length 2m at 0°C and having expansion coefficient $\alpha = (3x + 2) \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ where x is the distance (in cm) from one end of rod. The length of rod at 20°C is :
 (A) 2.124 m (B) 3.24 m (C) 2.0120 m (D) 3.124 m

EH0113

8. A liquid is given some heat.

Statement A : Some liquid evaporates.

Statement B : The liquid starts boiling.

(A) A implies B and B implies A

(B) B implies A but, A does not imply B

(C) A implies B but B does not imply A

(D) Neither A implies B nor B implies A

EH0114

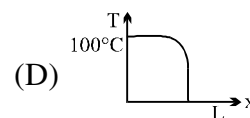
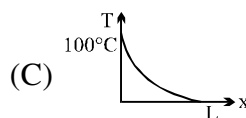
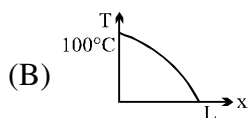
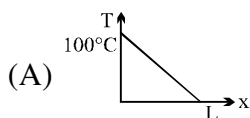
9. A long solid cylinder is radiating power. It is remoulded into a number of smaller cylinders, each of which has the same length as original cylinder. Each small cylinder has the same temperature as the original cylinder. The total radiant power emitted by the pieces is twice that emitted by the original cylinder. How many smaller cylinders are there ? Neglect the energy emitted by the flat faces of cylinder.
 (A) 3 (B) 4 (C) 5 (D) 6

EH0115

10. Four rods of same material with different radii r and length ℓ are used to connect two reservoirs of heat at different temperatures. Which one will conduct most heat ?
 (A) $r = 2\text{cm}$, $\ell = 0.5\text{m}$ (B) $r = 2\text{cm}$, $\ell = 2\text{m}$
 (C) $r = 0.5\text{cm}$, $\ell = 0.5\text{m}$ (D) $r = 1\text{cm}$, $\ell = 1\text{m}$

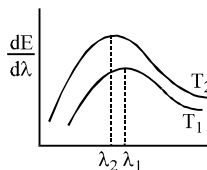
EH0116

11. A rod of length L and uniform cross-sectional area has varying thermal conductivity which changes linearly from $2K$ at end A to K at the other end B. The ends A and B of the rod are maintained at constant temperature 100°C and 0°C , respectively. At steady state, the graph of temperature : $T = T(x)$ where x = distance from end A will be



EH0117

12. The spectral emissive power E_λ for a body at temperature T_1 is plotted against the wavelength and area under the curve is found to be A . At a different temperature T_2 the area is found to be $9A$. Then $\lambda_1/\lambda_2 =$



(A) 3

(B) $1/3$

(C) $1/\sqrt{3}$

(D) $\sqrt{3}$

EH0118

13. 'Gulab Jamuns' (assumed to be spherical) are to be heated in an oven. They are available in two sizes, one twice bigger (in radius) than the other. Pizzas (assumed to be discs) are also to be heated in oven. They are also in two sizes, one twice big (in radius) than the other. All four are put together to be heated to oven temperature. Choose the correct option from the following:
- (A) Both size gulab jamuns will get heated in the same time.
 (B) Smaller gulab jamuns are heated before bigger ones.
 (C) Smaller pizzas are heated before bigger ones.
 (D) Bigger pizzas are heated before smaller ones.

EH0119

14. An experiment is performed to measure the specific heat of copper. A lump of copper is heated in an oven, then dropped into a beaker of water. To calculate the specific heat of copper, the experimenter must know or measure the value of all of the quantities below EXCEPT the
- (A) heat capacity of water and beaker
 (B) original temperature of the copper and the water
 (C) final (equilibrium) temperature of the copper and the water
 (D) time taken to achieve equilibrium, after the copper is dropped into the water

EH0120

15. One end of a conducting rod is maintained at temperature 50°C and at the other end, ice is melting at 0°C . The rate of melting of ice is doubled if:
- (A) the temperature is made 200°C and the area of cross-section of the rod is doubled
 (B) the temperature is made 100°C and length of rod is made four times
 (C) area of cross-section of rod is halved and length is doubled
 (D) the temperature is made 100°C and the area of cross-section of rod and length both are doubled.

EH0121

16. A black body is at a temperature of 2880 K . The energy of radiation emitted by this object with wavelength between 499 nm and 500 nm is U_1 , between 999 nm and 1000 nm is U_2 and between 1499 nm and 1500 nm is U_3 . The Wien constant $b = 2.88 \times 10^6\text{ nm K}$. Then
- (A) $U_1 = 0$ (b) $U_3 = 0$ (C) $U_1 > U_2$ (D) $U_2 > U_1$

EH0122

MULTIPLE CORRECT TYPE QUESTIONS

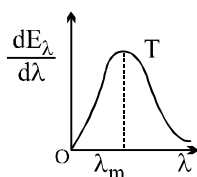
17. A bimetallic strip is formed out of two identical strips one of copper and the other of brass. The coefficient of linear expansion of the two metals are α_C and α_B . On heating, the temperature of the strip goes up by ΔT and the strip bends to form an arc of radius of curvature R . Then R is
- (A) proportional to ΔT (B) inversely proportional to ΔT
 (C) proportional to $|\alpha_B - \alpha_C|$ (D) inversely proportional to $|\alpha_B - \alpha_C|$

EH0123

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 18 and 19

The figure shows a radiant energy spectrum graph for a black body at a temperature T .



18. Choose the **CORRECT** statement(s)
- (A) The radiant energy is not equally distributed among all the possible wavelengths
 - (B) For a particular wavelength the spectral intensity is maximum
 - (C) The area under the curve is equal to the total rate at which heat is radiated by the body at that temperature
 - (D) None of these
- EH0124**
19. If the temperature of the body is raised to a higher temperature T' , then choose the correct statement(s)
- (A) The intensity of radiation for every wavelength increases
 - (B) The maximum intensity occurs at a shorter wavelength
 - (C) The area under the graph increases
 - (D) The area under the graph is proportional to the fourth power of temperature

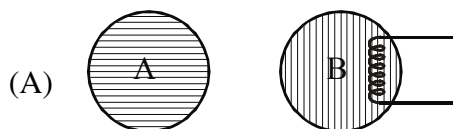
EH0124

MATRIX MATCH TYPE QUESTION

20. A & B are two black bodies of radii r_A and r_B respectively, placed in surrounding of temperature T_0 . At steady state the temperature of A & B is T_A & T_B respectively.

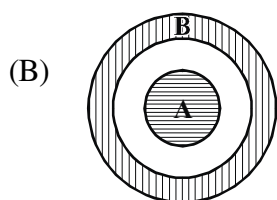
Column I

Column II



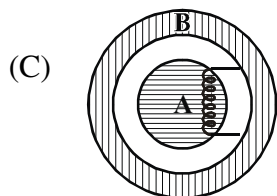
- A & B are solid sphere
- $r_A = r_B$
- Body 'B' is being heated by a heater of constant power 'P'

(P) $T_A = T_B$



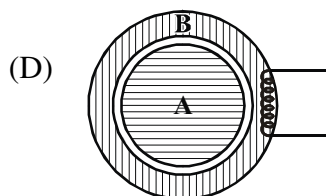
- B is thin spherical shell
- A is a solid sphere
- $r_A < r_B$

(Q) $T_A < T_B$



- B is thin spherical shell
- A is a solid sphere
- $r_A < r_B$
- Body A is being heated by a heater of constant power 'P'

(R) Heat received by A is more than heat radiated by it at steady state.



- B is thin spherical shell
- A is a solid sphere
- $r_A \approx r_B$
- Body B is being heated by a heater of constant power 'P'

(S) Radiation spectrum of A & B is distinguishable

(T) Steady state can't be achieved

21. A sample 'A' of liquid water and a sample B of ice of equal mass are kept in 2 nearby containers so that they can exchange heat with each other but are thermally insulated from the surroundings. The graphs in column-II show the sketch of temperature T of samples versus time t . Match with appropriate description in column-I.

Column I

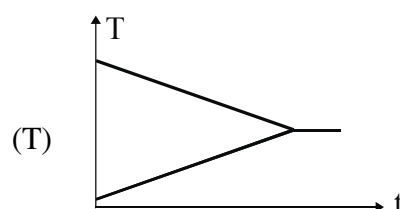
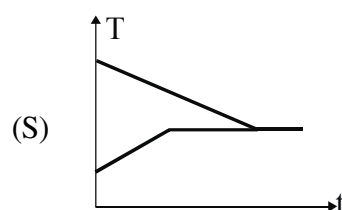
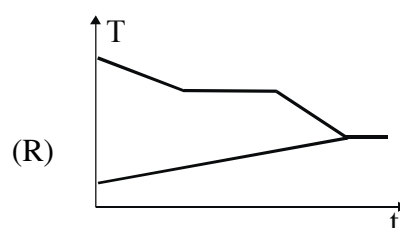
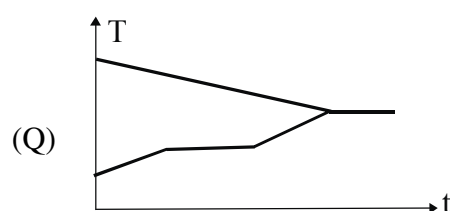
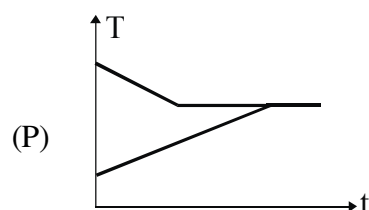
- (A) Equilibrium temperature is above melting point of ice.

- (B) At least some of water freezes.

- (C) At least some of ice melts.

- (D) Equilibrium temperature is below freezing point of water

Column II



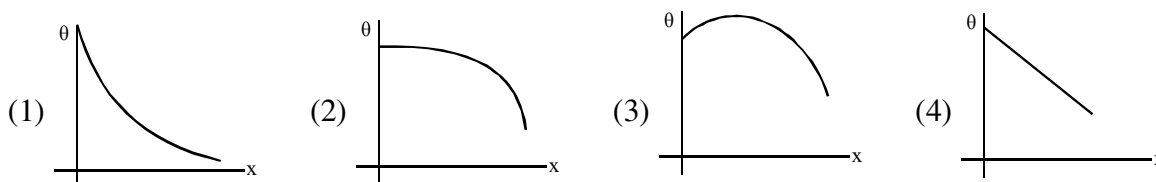
EH0126

EXERCISE (JM)

1. Two wires are made of the same material and have the same volume. However wire 1 has cross-sectional area A and wire 2 has cross-sectional area $3A$. If the length of wire 1 increases by Δx on applying force F , how much force is needed to stretch wire 2 by the same amount ? [AIEEE-2009]
- (1) $6F$ (2) $9F$ (3) F (4) $4F$

EH0127

2. A long metallic bar is carrying heat from one of its ends to the other end under steady-state. The variation of temperature θ along the length x of the bar from its hot end is best described by which of the following figures ? [AIEEE - 2009]

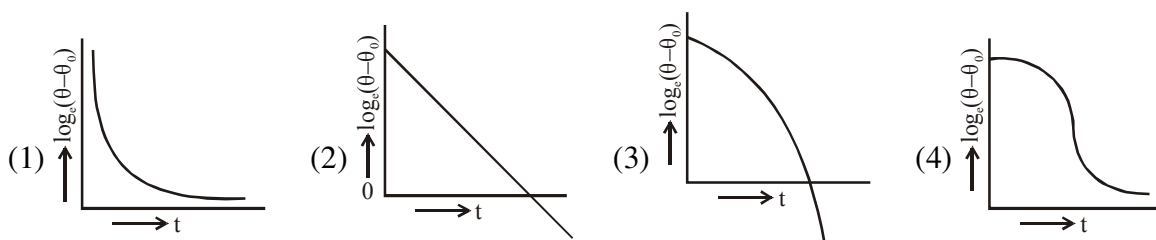


EH0128

3. 100 g of water is heated from 30°C to 50°C . Ignoring the slight expansion of the water, the change in its internal energy is (specific heat of water is 4184 J/kg/K) :- [AIEEE - 2011]
- (1) 84 kJ (2) 2.1 kJ (3) 4.2 kJ (4) 8.4 kJ

EH0129

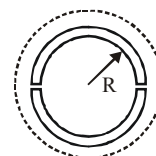
4. A liquid in a beaker has temperature $\theta(t)$ at time t and θ_0 is temperature of surroundings, then according to Newton's law of cooling the correct graph between $\log_e (\theta - \theta_0)$ and t is :- [AIEEE 2012]



EH0130

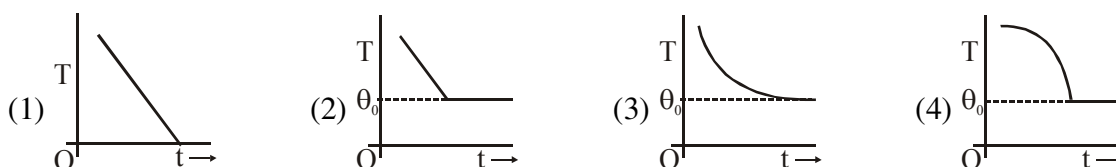
5. A wooden wheel of radius R is made of two semicircular parts (see figure). The two parts are held together by a ring made of a metal strip of cross sectional area S and Length L . L is slightly less than $2\pi R$. To fit the ring on the wheel, it is heated so that its temperature rises by ΔT and it just steps over the wheel. As it cools down to surrounding temperature, it presses the semicircular parts together. If the coefficient of linear expansion of the metal is α , and its Young's modulus is Y , the force that one part of the wheel applies on the other part is : [AIEEE 2012]

- (1) $2SY\alpha\Delta T$
 (2) $2\pi SY\alpha\Delta T$
 (3) $SY\alpha\Delta T$
 (4) $\pi SY\alpha\Delta T$



EH0131

6. If a piece of metal is heated to temperature θ and then allowed to cool in a room which is at temperature θ_0 the graph between the temperature T of the metal and time t will be closed to: [JEE-Main- 2013]



EH0132

7. The pressure that has to be applied to the ends of a steel wire of length 10 cm to keep its length constant when its temperature is raised by 100°C is :

(For steel Young's modulus is $2 \times 10^{11} \text{ N m}^{-2}$ and coefficient of thermal expansion is $1.1 \times 10^{-5} \text{ K}^{-1}$)

[JEE-Main-2014]

- (1) $2.2 \times 10^7 \text{ Pa}$ (2) $2.2 \times 10^6 \text{ Pa}$ (3) $2.2 \times 10^8 \text{ Pa}$ (4) $2.2 \times 10^9 \text{ Pa}$

EH0133

8. Three rods of Copper, Brass and Steel are welded together to form a Y-shaped structure. Area of cross-section of each rod = 4 cm^2 . End of copper rod is maintained at 100°C where as ends of brass and steel are kept at 0°C . Lengths of the copper, brass and steel rods are 46, 13 and 12 cms respectively. The rods are thermally insulated from surroundings except at ends. Thermal conductivities of copper, brass and steel are 0.92, 0.26 and 0.12 CGS units respectively. Rate of heat flow through copper rod is :

[JEE-Main-2014]

- (1) 4.8 cal/s (2) 6.0 cal/s (3) 1.2 cal/s (4) 2.4 cal/s

EH0134

9. A pendulum made of a uniform wire of cross sectional area A has time period T . When an additional mass M is added to its bob, the time period changes to T_M . If the Young's modulus of the material of

the wire is Y then $\frac{1}{Y}$ is equal to :- (g = gravitational acceleration)

[JEE-Main-2015]

- (1) $\left[1 - \left(\frac{T_M}{T}\right)^2\right] \frac{A}{Mg}$ (2) $\left[1 - \left(\frac{T}{T_M}\right)^2\right] \frac{A}{Mg}$ (3) $\left[\left(\frac{T_M}{T}\right)^2 - 1\right] \frac{A}{Mg}$ (4) $\left[\left(\frac{T_M}{T}\right)^2 - 1\right] \frac{Mg}{A}$

EH0135

10. A pendulum clock loses 12s a day if the temperature is 40°C and gains 4s a day if the temperature is 20°C . The temperature at which the clock will show correct time, and the coefficient of linear expansion (α) of the metal of the pendulum shaft are respectively :- [JEE-Main-2016]

- (1) 55°C ; $\alpha = 1.85 \times 10^{-2} / ^{\circ}\text{C}$ (2) 25°C ; $\alpha = 1.85 \times 10^{-5} / ^{\circ}\text{C}$
(3) 60°C ; $\alpha = 1.85 \times 10^{-4} / ^{\circ}\text{C}$ (4) 30°C ; $\alpha = 1.85 \times 10^{-3} / ^{\circ}\text{C}$

EH0136

11. A copper ball of mass 100 gm is at a temperature T . It is dropped in a copper calorimeter of mass 100gm, filled with 170 gm of water at room temperature. Subsequently, the temperature of the system is found to be 75°C . T is given by : (Given : room temperature = 30°C , specific heat of copper = $0.1 \text{ cal/gm}^{\circ}\text{C}$) [JEE-Main-2017]

- (1) 1250°C (2) 825°C (3) 800°C (4) 885°C

EH0137

12. A man grows into a giant such that his linear dimensions increase by a factor of 9. Assuming that his density remains same, the stress in the leg will change by a factor of : [JEE-Main-2017]

- (1) 81 (2) $\frac{1}{81}$ (3) 9 (4) $\frac{1}{9}$

EH0138

13. An external pressure P is applied on a cube at 0°C so that it is equally compressed from all sides. K is the bulk modulus of the material of the cube and α is its coefficient of linear expansion. Suppose we want to bring the cube to its original size by heating. The temperature should be raised by :

[JEE-Main-2017]

- (1) $\frac{3\alpha}{PK}$ (2) $3PK\alpha$ (3) $\frac{P}{3\alpha K}$ (4) $\frac{P}{\alpha K}$

EH0139

14. A solid sphere of radius r made of a soft material of bulk modulus K is surrounded by a liquid in a cylindrical container. A massless piston of area a floats on the surface of the liquid, covering entire cross section of cylindrical container. When a mass m is placed on the surface of the piston to compress

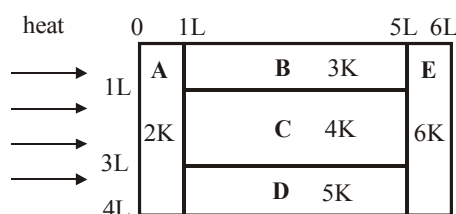
the liquid, the fractional decrement in the radius of the sphere, $\left(\frac{dr}{r}\right)$, is : [JEE-Main-2018]

- (1) $\frac{Ka}{3mg}$ (2) $\frac{mg}{3Ka}$ (3) $\frac{mg}{Ka}$ (4) $\frac{Ka}{mg}$

EH0140

EXERCISE (JA)

1. A metal rod AB of length $10x$ has its one end A in ice at 0°C and the other end B in water at 100°C . If a point P on the rod is maintained at 400°C , then it is found that equal amounts of water and ice evaporate and melt per unit time. The latent heat of evaporation of water is 540 cal/g and latent heat of melting of ice is 80 cal/g . If the point P is at a distance of λx from the ice end A, find the value of λ . [Neglect any heat loss to the surrounding]
[JEE 2009]
EH0141
2. Two spherical bodies A (radius 6 cm) and B (radius 18 cm) are at temperatures T_1 and T_2 , respectively. The maximum intensity in the emission spectrum of A is at 500 nm and in that of B is at 1500 nm . Considering them to be black bodies, what will be the ratio of the rate of total energy radiated by A to that of B?
[JEE 2010]
EH0142
3. A piece of ice (heat capacity $= 2100\text{ J kg}^{-1} ^\circ\text{C}^{-1}$ and latent heat $= 3.36 \times 10^5\text{ J kg}^{-1}$) of mass m grams is at -5°C at atmospheric pressure. It is given 420 J of heat so that the ice starts melting. Finally when the ice–water mixture is in equilibrium, it is found that 1 gm of ice has melted. Assuming there is no other heat exchange in the process, the value of m is
[JEE 2010]
EH0143
4. A 0.1 kg mass is suspended from a wire of negligible mass. The length of the wire is 1 m and its cross-sectional area is $4.9 \times 10^{-7}\text{ m}^2$. If the mass is pulled a little in the vertically downward direction and released, it performs simple harmonic motion of angular frequency 140 rad s^{-1} . If the Young's modulus of the material of the wire is $n \times 10^9\text{ Nm}^{-2}$, the value of n is
[IIT-JEE 2010]
EH0144
5. A composite block is made of slabs A, B, C, D and E of different thermal conductivities (given in terms of a constant K) and sizes (given in terms of length, L) as shown in the figure. All slabs are of same width. Heat ' Q ' flows only from left to right through the blocks. Then in steady state
[JEE 2011]



- (A) heat flow through A and E slabs are same
- (B) heat flow through slab E is maximum
- (C) temperature difference across slab E is smallest
- (D) heat flow through C = heat flow through B + Heat flow through D

EH0145

6. Steel wire of length 'L' at 40°C is suspended from the ceiling and then a mass 'm' is hung from its free end. The wire is cooled down from 40° to 30° C to regain its original length 'L'. The coefficient of linear thermal expansion of the steel is $10^{-5}/^{\circ}\text{C}$, Young's modulus of steel is 10^{11} N/m^2 and radius of the wire is 1 mm. Assume that $L \gg$ diameter of the wire. Then the value of 'm' in kg is nearly

[JEE 2011]

EH0146

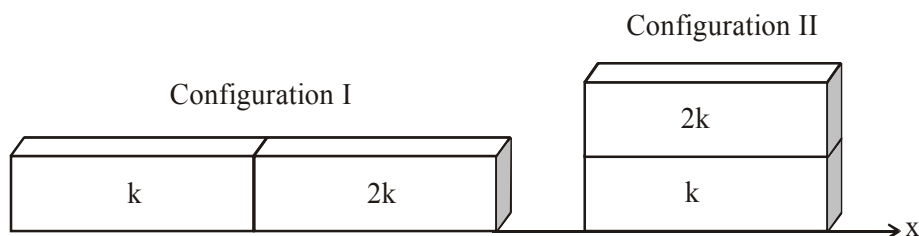
7. Three very large plates of same area are kept parallel and close to each other. They are considered as ideal black surfaces and have very high thermal conductivity. The first and third plates are maintained at temperatures 2T and 3T respectively. The temperature of the middle (i.e. second) plate under steady state condition is

[JEE 2012]

(A) $\left(\frac{65}{2}\right)^{1/4} T$ (B) $\left(\frac{97}{4}\right)^{1/4} T$ (C) $\left(\frac{97}{2}\right)^{1/4} T$ (D) $(97)^{1/4} T$

EH0147

8. Two rectangular blocks, having identical dimensions, can be arranged either in configuration I or in configuration II as shown in the figure. One of the blocks has thermal conductivity k and the other 2k. The temperature difference between the ends along the x-axis is the same in both the configurations. It takes 9s to transport a certain amount of heat from the hot end to the cold end in the configuration I. The time to transport the same amount of heat in the configuration II is :- [JEE-Advance-2013]



- (A) 2.0 s (B) 3.0 s (C) 4.5 s (D) 6.0 s

EH0148

9. One end of a horizontal thick copper wire of length 2L and radius 2R is welded to an end of another horizontal thin copper wire of length L and radius R. When the arrangement is stretched by applying forces at two ends, the ratio of the elongation in the thin wire to that in the thick wire is :-

[JEE-Advance-2013]

- (A) 0.25 (B) 0.50 (C) 2.00 (D) 4.00

EH0149

10. Parallel rays of light of intensity $I = 912 \text{ Wm}^{-2}$ are incident on a spherical black body kept in surroundings of temperature 300 K . Take Stefan-Boltzmann constant $\sigma = 5.7 \times 10^{-8} \text{ Wm}^{-2} \text{ K}^{-4}$ and assume that the energy exchange with the surroundings is only through radiation. The final steady state temperature of the black body is close to :-

[JEE-Advance-2014]

- (A) 330 K (B) 660 K (C) 990 K (D) 1550 K

EH0150

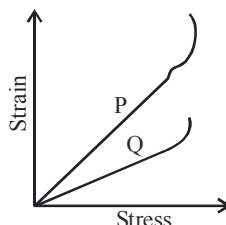
11. Two spherical stars A and B emit blackbody radiation. The radius of A is 400 times that of B and A emits 10^4 times the power emitted from B. The ratio $\left(\frac{\lambda_A}{\lambda_B}\right)$ of their wavelengths λ_A and λ_B at which the peaks occur in their respective radiation curves is.

[JEE-Advance-2015]

EH0151

12. In plotting stress versus strain curves for two materials P and Q, a student by mistake puts strain on the y-axis and stress on the x-axis as shown in the figure. Then the correct statement(s) is (are):-

[JEE-Advance-2015]

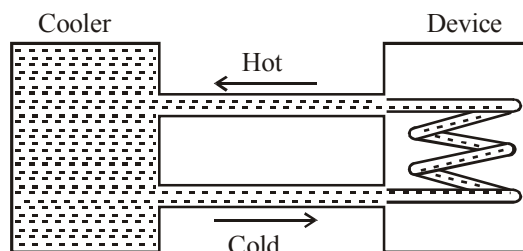


- (A) P has more tensile strength than Q (B) P is more ductile than Q
(C) P is more brittle than Q (D) The Young's modulus of P is more than that of Q

EH0152

13. A water cooler of storage capacity 120 litres can cool water at constant rate of P watts. In a closed circulation system (as shown schematically in the figure), the water from the cooler is used to cool an external device that generates constantly 3 kW of heat (thermal load). The temperature of water fed into the device cannot exceed 30°C and the entire stored 120 litres of water is initially cooled to 10°C . The entire system is thermally insulated. The minimum value of P (in watts) for which the device can be operated for 3 hours is :

[JEE-Advance-2016]



(Specific heat of water is $4.2 \text{ kJ kg}^{-1} \text{ K}^{-1}$ and the density of water is 1000 kg m^{-3})

- (A) 1600 (B) 2067 (C) 2533 (D) 3933

EH0153

14. A metal is heated in a furnace where a sensor is kept above the metal surface to read the power radiated (P) by the metal. The sensor has a scale that displays $\log_2(P/P_0)$, where P_0 is a constant. When the metal surface is at a temperature of 487°C , the sensor shows a value 1. Assume that the emissivity of the metallic surface remains constant. What is the value displayed by the sensor when the temperature of the metal surface is raised to 2767°C ?

[JEE-Advance-2016]

EH0154

15. The ends Q and R of two thin wires, PQ and RS, are soldered (joined) together. Initially each of the wires has a length of 1 m at 10°C . Now the end P is maintained at 10°C , while the end S is heated and maintained at 400°C . The system is thermally insulated from its surroundings. If the thermal conductivity of wire PQ is twice that of the wire RS and the coefficient of linear thermal expansion of PQ is $1.2 \times 10^{-5} \text{ K}^{-1}$, the change in length of the wire PQ is

[JEE-Advance-2016]

- (A) 0.78 mm (B) 0.90 mm (C) 1.56 mm (D) 2.34 mm

EH0155

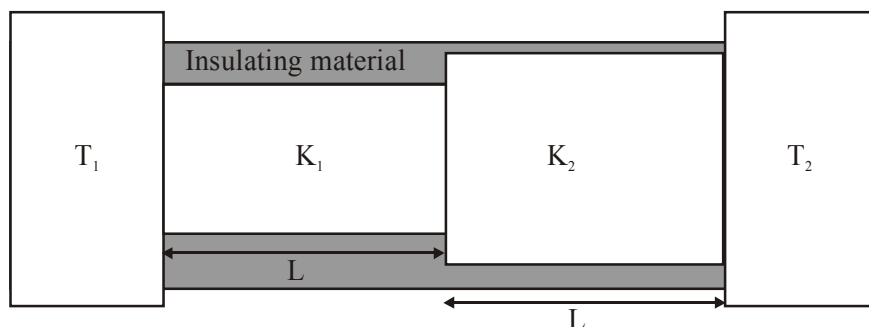
16. A human body has a surface area of approximately 1 m^2 . The normal body temperature is 10 K above the surrounding room temperature T_0 . Take the room temperature to be $T_0 = 300 \text{ K}$. For $T_0 = 300 \text{ K}$, the value of $\sigma T_0^4 = 460 \text{ W m}^{-2}$ (where σ is the Stefan-Boltzmann constant). Which of the following options is/are correct ?

[JEE-Advance-2017]

- (A) The amount of energy radiated by the body in 1 second is close to 60 Joules
(B) If the surrounding temperature reduces by a small amount $\Delta T_0 \ll T_0$, then to maintain the same body temperature the same (living) human being needs to radiate $\Delta W = 4\sigma T_0^3 \Delta T_0$ more energy per unit time
(C) Reducing the exposed surface area of the body (e.g. by curling up) allows humans to maintain the same body temperature while reducing the energy lost by radiation
(D) If the body temperature rises significantly then the peak in the spectrum of electromagnetic radiation emitted by the body would shift to longer wavelengths

EH0156

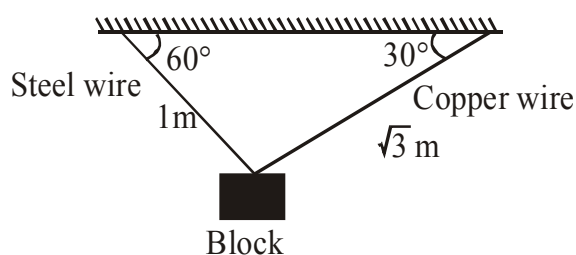
17. Two conducting cylinders of equal length but different radii are connected in series between two heat baths kept at temperatures $T_1 = 300\text{ K}$ and $T_2 = 100\text{ K}$, as shown in the figure. The radius of the bigger cylinder is twice that of the smaller one and the thermal conductivities of the materials of the smaller and the larger cylinders are K_1 and K_2 respectively. If the temperature at the junction of the two cylinders in the steady state is 200 K , then $K_1/K_2 = \underline{\hspace{2cm}}$. [JEE-Advance-2018]



EH0157

18. A block of weight 100 N is suspended by copper and steel wires of same cross sectional area 0.5 cm^2 and, length $\sqrt{3}\text{ m}$ and 1 m , respectively. Their other ends are fixed on a ceiling as shown in figure. The angles subtended by copper and steel wires with ceiling are 30° and 60° , respectively. If elongation in copper wire is $(\Delta\ell_c)$ and elongation in steel wire is $(\Delta\ell_s)$, then the ratio $\frac{\Delta\ell_c}{\Delta\ell_s}$ is _____. [Young's modulus for copper and steel are $1 \times 10^{11}\text{ N/m}^2$ and $2 \times 10^{11}\text{ N/m}^2$ respectively]

[JEE-Advance-2019]



EH0158

19. A liquid at 30°C is poured very slowly into a Calorimeter that is at temperature of 110°C . The boiling temperature of the liquid is 80°C . It is found that the first 5 gm of the liquid completely evaporates. After pouring another 80 gm of the liquid the equilibrium temperature is found to be 50°C . The ratio of the Latent heat of the liquid to its specific heat will be _____ $^{\circ}\text{C}$.

[Neglect the heat exchange with surrounding]

[JEE-Advance-2019]

EH0159

ANSWER KEY

EXERCISE (S-1)

1. Ans. 0.75 2. Ans. $\frac{9}{16}$ 3. Ans. 3mm 4. Ans. ℓ'
5. Ans. (a) $\theta = \frac{\pi}{2}$ (b) $\theta = \pi/4$.
6. Ans. (i) 50 N, (ii) 0.045 J, 1.8×10^{-3} m (iii) 8.4×10^{-4} m, (iv) $x = 0.12$ m 7. Ans. The first one
8. Ans. 25.5°C 9. Ans. 1/90 10. Ans. 0°C , 125/4 g ice, 1275/4 g water
11. Ans. 5000 J/ $^\circ\text{C}$ kg 12. Ans. 27/85 13. Ans. 12 gm
14. Ans. (i) 0.02kg, (ii) 40,000 cal kg^{-1} , (iii) 750 cal $\text{kg}^{-1}\text{K}^{-1}$ 15. Ans. $5\alpha/3$
16. Ans. 5 sec slow 17. Ans. 10, 000 N 18. Ans. 65°C 19. Ans. 5°C
20. Ans. 7/2 21. Ans. $(4/3)\omega$ 22. Ans. 10.34 cm 23. Ans. 2:1
24. Ans. 1000 J ($^\circ\text{C}$) $^{-1}$ 25. Ans. $\left(\frac{6}{\pi}\right)^{1/3}$ 26. Ans. 80 kcal/kg 27. Ans. 3025 K
28. Ans. 10 sec

EXERCISE (S-2)

1. Ans. 4×10^{-6} m/ $^\circ\text{C}$ 2. Ans. 10cm, 40cm 3. Ans. 1/200 rad 4. Ans. 1 : 1.26
6. Ans. (a) 37.8 J/s (Watts), (b) 2.005 N-m 7. Ans. 45°C
8. Ans. $\frac{a^2 s}{2K} \log_e \left(\frac{b}{a}\right) \log_e \left(\frac{T_0 - T_1}{T_0 - T_2}\right)$ 9. Ans. (a) -100°C (b) 1000 J
10. Ans. 166.3 sec 11. Ans. 10 min. 12. Ans. (a) 595 watt/ m^2 ; (b) $T_0 \approx 420$ K
13. Ans. $\frac{K}{4e\sigma LT_s^3 + K}$

EXERCISE (O-1)

1. Ans. (D) 2. Ans. (D) 3. Ans. (C) 4. Ans. (D) 5. Ans. (B) 6. Ans. (D)
7. Ans. (A) 8. Ans. (C) 9. Ans. (C) 10. Ans. (A) 11. Ans. (C) 12. Ans. (B)
13. Ans. (D) 14. Ans. (B) 15. Ans. (D) 16. Ans. (A) 17. Ans. (B) 18. Ans. (C)
19. Ans. (D) 20. Ans. (A) 21. Ans. (A) 22. Ans. (C) 23. Ans. (D) 24. Ans. (B)
25. Ans. (C) 26. Ans. (B) 27. Ans. (B) 28. Ans. (B) 29. Ans. (C) 30. Ans. (C)
31. Ans. (A) 32. Ans. (D) 33. Ans. (A) 34. Ans. (B) 35. Ans. (C) 36. Ans. (D)
37. Ans. (C) 38. Ans. (D) 39. Ans. (A) 40. Ans. (A) 41. Ans. (A) 42. Ans. (B)
43. Ans. (D) 44. Ans. (B) 45. Ans. (A) 46. Ans. (D) 47. Ans. (C) 48. Ans. (A)
49. Ans. (A) 50. Ans. (B) 51. Ans. (B) 52. Ans. (D) 53. Ans. (D) 54. Ans. (A,D)
55. Ans. (A,D) 56. Ans. (A,C,D) 57. Ans. (C,D) 58. Ans. (A,C)
59. Ans. (A,B,C) 60. Ans. (B,D) 61. Ans. (A,B) 62. Ans. (A,C,D)
63. Ans. (A,C) 64. Ans. (A,B) 65. Ans. (A) 66. Ans. (B)
67. Ans. (B)

EXERCISE (O-2)

- | | | | | | |
|---|--------------|---------------------------------------|--------------|----------------|----------------|
| 1. Ans. (D) | 2. Ans. (A) | 3. Ans. (A) | 4. Ans. (C) | 5. Ans. (A) | 6. Ans. (B) |
| 7. Ans. (C) | 8. Ans. (B) | 9. Ans. (B) | 10. Ans. (A) | 11. Ans. (B) | 12. Ans. (D) |
| 13. Ans. (B) | 14. Ans. (D) | 15. Ans. (D) | 16. Ans. (D) | 17. Ans. (B,D) | 18. Ans. (A,B) |
| 19. Ans. (A,B,C,D) | | 20. Ans. A - Q,S; B - P; C - S; D - P | | | |
| 21. Ans. A - Q; B - P,R; C - Q,S; D - R | | | | | |

EXERCISE (JM)

- | | | | | | |
|--------------|--------------|-------------|--------------|--------------|--------------|
| 1. Ans. (2) | 2. Ans. (4) | 3. Ans. (4) | 4. Ans. (2) | 5. Ans. (1) | 6. Ans. (3) |
| 7. Ans. (3) | 8. Ans. (1) | 9. Ans. (3) | 10. Ans. (2) | 11. Ans. (4) | 12. Ans. (3) |
| 13. Ans. (3) | 14. Ans. (2) | | | | |

EXERCISE (JA)

- | | | | | | |
|-------------------------|--------------|-----------------|-------------------|-----------------------|------------|
| 1. Ans. 9 | 2. Ans. 9 | 3. Ans. 8 | 4. Ans. 4 | 5. Ans. (ACD or ABCD) | |
| 6. Ans. 3 | 7. Ans. (C) | 8. Ans. (A) | 9. Ans. (C) | 10. Ans. (A) | 11. Ans. 2 |
| 12. Ans. (A,B) | 13. Ans. (B) | 14. Ans. 9 | 15. Ans. (A) | 16. Ans. (C) | |
| 17. Ans. 4 [3.99, 4.01] | | 18. Ans. (2.00) | 19. Ans. (270.00) | | |

Important Notes



Chapter Contents

02

Kinetic Theory of Gases & Thermodynamics

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Important Notes

[illegible]

KINETIC THEORY OF GASES

The properties of the gases are entirely different from those of solid and liquid. In case of gases, thermal expansion is very large as compared to solids and liquids. To state the conditions of a gas, its volume, pressure and temperature must be specified.

Intermolecular force Solid > liquid > real gas > ideal gas (zero)

Potential energy Solid < liquid < real gas < ideal gas (zero)

Internal energy, internal kinetic energy, internal potential energy

At a given temperature for solid, liquid and gas:

- (i) Internal kinetic energy : Same for all
- (ii) Internal potential Energy : Maximum for ideal gas (PE = 0) and Minimum for solids (PE = -ve)
- (iii) Internal Energy : Maximum for Ideal gas and Minimum for solid

At a given temperature for rared and compressed gas :

- (i) Internal kinetic energy → Same
- (ii) Internal potential energy → $(PE)_{\text{Rared}} > (PE)_{\text{compressed}}$
- (iii) Internal Energy → $(U)_{\text{Rared}} > (U)_{\text{compressed}}$

	N.T.P.	S.T.P.
Temperature	0° C = 273.15 K	0.01° C = 273.16K
Pressure	1 atm = 1.01325 × 10 ⁵ N/m ² = 1.01325 × 10 ⁵ pascal	1 atm
Volume	22.4 litre	22.4 litre

IDEAL GAS CONCEPT

- A gas which follows all gas laws and gas equation at every possible temperature and pressure is known as ideal or perfect gas.
- Volume of gas molecules is negligible as compared to volume of container so volume of gas = volume of container (Except 0K)
- No intermolecular force act between gas molecules.
- Potential energy of ideal gas is zero so internal energy of ideal gas is perfectly translational K.E. of gas. It is directly proportional to absolute temperature.

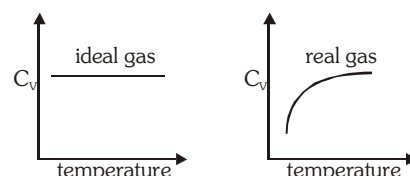
So, internal energy depends only and only on its temperature.

$$E_{\text{trans}} \propto T$$

$$\text{For a substance } U = U_{\text{KE}} + U_{\text{PE}}$$

U_{KE} : depends only on T, U_{PE} : depends upon intermolecular forces (Always negative)

- Specific heat of ideal gas is constant quantity and it does not change with temperature
- All real gases behaves as ideal gas at high temperature and low pressure.
- Volume expansion coefficient (α) and pressure expansion coefficient (β) is same for a ideal gas and value of each is $\frac{1}{273}$ per °C $\alpha = \beta = \frac{1}{273}$ per °C
- Gas molecule have point mass and negligible volume and velocity is very high (10⁷ cm/s). That's why there is no effect of gravity on them.



EQUATION OF STATE FOR IDEAL GAS

$$PV = \mu RT \quad \text{where } \mu = \text{number of moles of gas} \Rightarrow PV = \frac{M}{M_w} RT = \left[\frac{mN}{mN_0} \right] RT = \left[\frac{R}{N_0} \right] N T = NkT$$

Ex. By increasing temperature of gas by 5°C its pressure increases by 0.5% from its initial value at constant volume then what is initial temperature of gas ?

Sol. $\therefore$ At constant volume $T \propto P \therefore \frac{\Delta T}{T} \times 100 = \frac{\Delta P}{P} \times 100 = 0.5 \Rightarrow T = \frac{5 \times 100}{0.5} = 1000\text{K}$

Ex. Calculate the value of universal gas constant at STP.

Sol. Universal gas constant is given by $R = \frac{PV}{T}$

One mole of all gases at S.T.P. occupy volume $V = 22.4 \text{ litre} = 22.4 \times 10^{-3} \text{ m}^3$

$$P = 760 \text{ mm of Hg} = 760 \times 10^{-3} \times 13.6 \times 10^3 \times 9.80 \text{ N m}^{-2} \quad T = 273 \text{ K}$$

$$\therefore R = \frac{760 \times 10^{-3} \times 13.6 \times 10^3 \times 9.80 \times 22.4 \times 10^{-3}}{273} = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$$

Ex. A closed container of volume 0.02 m^3 contains a mixture of neon and argon gases at a temperature of 27°C and pressure of $1 \times 10^5 \text{ Nm}^2$. The total mass of the mixture is 28 g . If the gram molecular weights of neon and argon are 20 and 40 respectively, find the masses of the individual gases in the container, assuming them to be ideal. Given : $R = 8.314 \text{ J/mol/K}$.

Sol. Let m gram be the mass of neon. Then, the mass of argon is $(28 - m)\text{g}$.

$$\text{Total number of moles of the mixture, } \mu = \frac{m}{20} + \frac{28 - m}{40} = \frac{28 + m}{40} \quad \dots(i)$$

$$\text{Now, } \mu = \frac{PV}{RT} = \frac{1 \times 10^5 \times 0.02}{8.314 \times 300} = 0.8 \quad \dots(ii)$$

$$\text{By (i) and (ii), } \frac{28 + m}{40} = 0.8 \Rightarrow 28 + m = 32 \Rightarrow m = 4 \text{ gram or mass of argon} = (28 - 4)\text{g} = 24 \text{ g}$$

Ex. Calculate the temperature of the Sun if density is 1.4 g cm^{-3} , pressure is 1.4×10^9 atmosphere and average molecular weight of gases in the Sun is 2 g/mole . [Given $R = 8.4 \text{ J mol}^{-1} \text{ K}^{-1}$]

Sol. $PV = \mu RT \Rightarrow T = \frac{PV}{\mu R} \dots(i) \quad \text{But } \mu = \frac{M}{M_w} \text{ and } \rho = \frac{M}{V} \therefore \mu = \frac{\rho V}{M_w}$

$$\text{From equation (i) } T = \frac{PVM_w}{\rho VR} = \frac{PM_w}{\rho R} = \frac{1.4 \times 10^9 \times 1.01 \times 10^5 \times 2 \times 10^{-3}}{1.4 \times 1000 \times 8.4} = 2.4 \times 10^7 \text{ K}$$

Ex. At the top of a mountain a thermometer reads 7°C and barometer reads 70 cm of Hg. At the bottom of the mountain they read 27°C and 76 cm of Hg respectively. Compare the density of the air at the top with that at the bottom.

Sol. By gas equation $PV = \frac{M}{M_w}RT \Rightarrow \frac{P}{\rho T} = \frac{R}{M_w} \left[\because \mu = \frac{M}{M_w} \text{ and } \frac{M}{V} = \rho \right]$

Now as M_w and R are same for top and bottom $\left[\frac{P}{\rho T} \right]_T = \left[\frac{P}{\rho T} \right]_B$ So $\frac{\rho_T}{\rho_B} = \frac{P_T}{P_B} \times \frac{T_B}{T_T} = \frac{70}{76} \times \frac{300}{280} = \frac{75}{76} = 0.9868$

Ex. During an experiment an ideal gas is found to obey an additional law $VP^2 = \text{constant}$. The gas is initially at temperature T and volume V . What will be the temperature of the gas when it expands to a volume $2V$.

Sol. By gas equation $PV = \mu RT$ and $VP^2 = \text{constant}$ on eliminating P

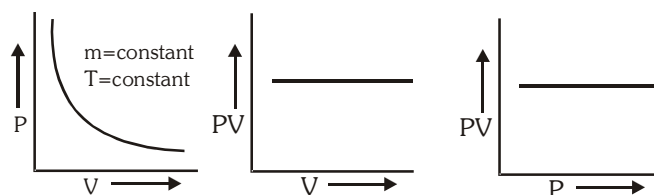
$$\left[\frac{A}{\sqrt{V}} \right] V = \mu RT \Rightarrow \sqrt{V} = \frac{\mu R}{A} T \therefore \frac{\sqrt{V_1}}{\sqrt{V_2}} = \left[\frac{T_1}{T_2} \right] \Rightarrow \frac{\sqrt{V}}{\sqrt{2V}} = \frac{T}{T'} \Rightarrow T' = (\sqrt{2}) T$$

GAS LAWS

Boyle's Law

According to it for a given mass of an ideal gas at constant temperature, the volume of a gas is inversely proportional to its pressure, i.e., $V \propto \frac{1}{P}$ if m and $T =$

Constant



Ex. A sample of oxygen with volume of 500 cc at a pressure of 2 atm is compressed to a volume of 400 cc. What pressure is needed to do this if the temperature is kept constant ?

Sol. Temperature is constant, so $P_1 V_1 = P_2 V_2 \therefore P_2 = P_1 \frac{V_1}{V_2} = 2 \left[\frac{500}{400} \right] = 2.5 \text{ atm}$

Ex. An air bubble doubles in radius on rising from bottom of a lake to its surface. If the atmosphere pressure is equal to that due to a column of 10 m of water, then what will be the depth of the lake. (Assuming that surface tension is negligible) ?

Sol. Given that constant temperature, we use $P_1 V_1 = P_2 V_2$

$$P_2 = (10 \text{ dg (for water column)}) \quad P_1 = (10+h) \text{ dg (where } h = \text{depth of lake)}$$

$$V_1 = \frac{4\pi}{3} r^3, \quad V_2 = \frac{4\pi}{3} (2r)^3 = 8 \left(\frac{4\pi}{3} r^3 \right) = 8V_1 \text{ Thus for } P_2 V_2 = P_1 V_1,$$

$$\text{We have } 10 \text{ dg } (8V_1) = (10+h) \text{ dg } V_1 \Rightarrow 80 = 10+h \Rightarrow h = 70 \text{ m}$$

Ex. A vessel of volume $8.0 \times 10^{-3} \text{ m}^3$ contains an ideal gas at 300 K and 200 k Pa. The gas is allowed to leak till the Pressure falls to 125 kPa. Calculate the amount of the gas leaked assuming that the temperature remains constant.

Sol. As the gas leaks out, the volume and the temperature of the remaining gas do not change. The number of moles of the gas in the vessel is given by $n = \frac{PV}{RT}$.

The number of moles in the vessel before the leakage is $n_1 = \frac{P_1 V}{RT}$ and that after the leakage is $n_2 = \frac{P_2 V}{RT}$.

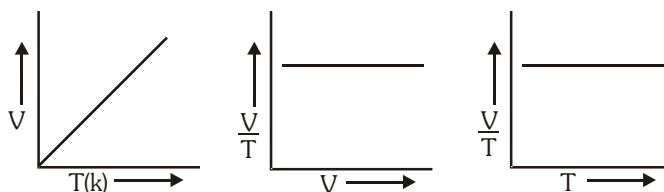
The amount leaked is $n_1 - n_2 = \frac{(P_1 - P_2)V}{RT} = \frac{(200 - 125) \times 10^3 \times 8.0 \times 10^{-3}}{8.3 \times 300} = 0.24 \text{ mole}$

• Charles's Law

According to it for a given mass of an ideal gas at constant pressure, volume of a gas is directly proportional to its absolute temperature,

i.e.

$V \propto T$ if m and $P = \text{Constant}$



Ex. 1500 ml of a gas at a room temperature of 23°C is inhaled by a person whose body temperature is 37°C , if the pressure and mass stay constant, what will be the volume of the gas in the lungs of the person?

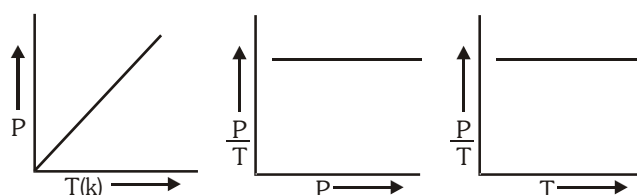
Sol. $T_1 = 273 + 37 = 310 \text{ K}$; $T_2 = 273 + 23 = 296 \text{ K}$. Pressure and amount of the gas are kept constant,

$$\text{So } \frac{V_1}{T_1} = \frac{V_2}{T_2} \therefore V_2 = V_1 \times \frac{T_2}{T_1} = 1500 \times \frac{293}{310} = 1417.74 \text{ ml}$$

Gay-Lussac's Law

According to it, for a given mass of an ideal gas at constant volume, pressure of a gas is directly proportional to its absolute temperature,

i.e., $P \propto T$ if m and $V = \text{constant}$



Ex. A sample of O_2 is at a pressure of 1 atm when the volume is 100 ml and its temperature is 27°C . What will be the temperature of the gas if the pressure becomes 2 atm and volume remains 100 ml.

Sol. $T_1 = 273 + 27 = 300 \text{ K}$

$$\text{For constant volume } \frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow T_2 = T_1 \times \frac{P_2}{P_1} = 300 \times \frac{2}{1} = 600 \text{ K} = 600 - 273 = 327^\circ\text{C}$$

Avogadro's Law

According to it, at same temperature and pressure of equal volumes of all gases contain equal number of molecules, i.e., $N_1 = N_2$ if P, V and T are same.

The kinetic theory of gases

- Rudolph Clausius (1822–88) and James Clark Maxwell (1831–75) developed the kinetic theory of gases in order to explain gas laws in terms of the motion of the gas molecules. The theory is based on following assumptions as regards to the motion of molecules and the nature of the gases.

Basic postulates of Kinetic theory of gases

- Every gas consists of extremely small particles known as molecules. The molecules of a given gas are all identical but are different than those of another gas.
- The molecules of a gas are identical, spherical, rigid and perfectly elastic point masses.
- The size is negligible in comparison to inter molecular distance (10^{-9} m)

Assumptions regarding motion :

- Molecules of a gas keep on moving randomly in all possible directions with all possible velocities.
- The speed of gas molecules lies between zero and infinity (very high speed).
- The number of molecules moving with most probable speeds is maximum.

Assumptions regarding collision:

- The gas molecules keep on colliding among themselves as well as with the walls of the containing vessel. These collisions are perfectly elastic. (i.e., the total energy before collision = total energy after the collisions.)

Assumptions regarding force:

- No attractive or repulsive force acts between gas molecules.
- Gravitational attraction among the molecules is ineffective due to extremely small masses and very high speed of molecules.

Assumptions regarding pressure:

- Molecules constantly collide with the walls of the container due to which their momentum changes. This change in momentum is transferred to the walls of the container. Consequently, pressure is exerted by gas molecules on the walls of the container.

Assumptions regarding density:

- The density of gas is constant at all points of the container.

PROPERTIES/ASSUMPTIONS OF IDEAL GAS

- The molecules of a gas are in a state of continuous random motion. They move with all possible velocities in all possible directions. They obey Newton's law of motion.
- Mean momentum = 0; Mean velocity = 0. $\langle \vec{v} \rangle = 0$; $\langle v^2 \rangle \neq 0$ (Non zero); $\langle v^3 \rangle = \langle v^5 \rangle = 0$
- The average distance travelled by a molecule between two successive collisions is called the mean free path (λ_m) of the molecule.
- The time during which a collision takes place is negligible as compared to the time taken by the molecule to cover the mean free path so the ratio of time of collision to free time of motion is $10^{-8} : 1$.
- When a gas is taken into a vessel it is uniformly distributed in the entire volume of the vessel such that its density, molecular density, motion of molecules etc. are all identical for all directions, therefore root mean velocity

$$\overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2} \rightarrow \text{equal Pressure exerted by the gas in all directions } P_x = P_y = P_z = P \rightarrow \text{equal}$$

- All these assumptions can be justified, if the number of gas molecules is taken to be very large

EXPRESSION FOR PRESSURE OF AN IDEAL GAS

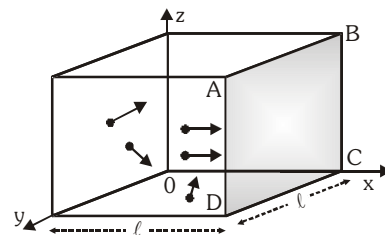
Consider an ideal gas enclosed in a cubical vessel of length ℓ . Suppose there are 'N' molecules in a gas which are moving with velocities $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_N$.

If we consider any single molecule than its instantaneous velocity

$\vec{v}$ can be expressed as $\vec{v} = v_x \hat{i} + v_y \hat{j} + v_z \hat{k}$

Due to random motion of the molecule $v_x = v_y = v_z$

$$|\vec{v}| = v_x \sqrt{3} = v_y \sqrt{3} = v_z \sqrt{3} = \sqrt{v_x^2 + v_y^2 + v_z^2}$$



Suppose a molecule of mass m is moving with a velocity v_x towards the face ABCD. It strikes the face of the cubical vessel and returns back to strike the opposite face.

Change in momentum of the molecule per collision $\Delta p = -mv_x - mv_x = -2mv_x$

Momentum transferred to the wall of the vessel per molecule per collision $\Delta p = 2mv_x$

The distance travelled by the molecule in going to face ABCD and coming back is 2ℓ .

So, the time between two successive collision is $\Delta t = \frac{2\ell}{v_x}$

Number of collision per sec per molecule is $f_c = \frac{v_x}{2\ell} = \frac{\text{molecule velocity}}{\text{mean free path}}$, $f_c = \frac{v_{rms}}{\lambda_m}$ or $f_c = \frac{v_m}{\lambda_m}$

Hence momentum transferred in the wall per second by the molecule is = force on the wall

$$\text{force } F = (2mv_x) \frac{v_x}{2\ell} = \frac{mv_x^2}{\ell} = \frac{mv^2}{3\ell}$$

Pressure exerted by gas molecule $P = \frac{F}{A} = \frac{1}{3} \frac{mv^2}{\ell \times A} \Rightarrow P = \frac{1}{3} \frac{mv^2}{V} [\because A \times \ell = V]$

Pressure exerted by gas $P = \sum \frac{1}{3} \frac{mv^2}{V} = \sum \frac{1}{3} \frac{mv^2}{V} \times \frac{N}{N} = \frac{1}{3} \frac{mN}{V} \sum \frac{v^2}{N} = \frac{1}{3} \frac{mN}{V} v_{rms}^2$

$$v_{rms}^2 = \frac{3PV}{M} = \frac{3\mu RT}{\mu M_w} \Rightarrow v_{rms} = \sqrt{\frac{3RT}{M_w}}, P = \frac{1}{3} \frac{M}{V} v_{rms}^2 = \frac{1}{3} \rho v_{rms}^2$$

• Average number of molecules for each wall = $\frac{N}{6}$. No. of molecules along each axis = $\frac{N}{3}$ ($N_x = N_y = N_z$)

• $\vec{v}_x^2 = \vec{v}_y^2 = \vec{v}_z^2 = \frac{\vec{v}_{rms}^2}{3}$ Root mean square velocity along any axis for gas molecule is $(v_{rms})_x = (v_{rms})_y$

$$= (v_{rms})_z = \frac{v_{rms}}{\sqrt{3}}$$

All gas laws and gas equation can be obtained by expression of pressure of gas (except Joule's law)

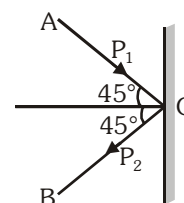
Ex. The mass of a hydrogen molecule is 3.32×10^{-27} kg. If 10^{23} molecules are colliding per second on a stationary wall of area 2 cm^2 at an angle of 45° to the normal to the wall and reflected elastically with a speed 10^3 m/s . Find the pressure exerted on the wall will be (in N/m^2)

Sol. As the impact is elastic $\therefore |\vec{p}_1| = |\vec{p}_2| = p = mv = 3.32 \times 10^{-24} \text{ kg m/s}$

The change in momentum along the normal $\Delta p = |\vec{p}_2 - \vec{p}_1| = 2p \cos 45^\circ = \sqrt{2}p$

If f is the collision frequency then force applied on the wall $F = \frac{\Delta p}{\Delta t} = \Delta p \times f = \sqrt{2}pf$

$$\therefore \text{Pressure } P = \frac{F}{A} = \frac{\sqrt{2}pf}{A} = \frac{\sqrt{2} \times 3.32 \times 10^{-24} \times 10^{23}}{2 \times 10^{-4}} = 2.347 \times 10^3 \text{ N/m}^2$$

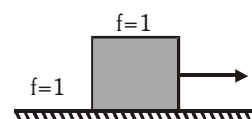


DEGREE OF FREEDOM (f)

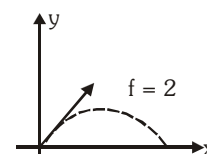
- The number of independent ways in which a molecule or an atom can exhibit motion or have energy is called its degrees of freedom.

For example

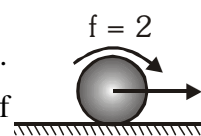
- (a) Block has one degree of freedom, because it is confined to move in a straight line and has only one translational degree of freedom.



- (b) The projectile has two degrees of freedom because it is confined to move in a plane and so it has two translational degrees of freedom.



- (c) The sphere has two degrees of freedom: one rotational and another translational. Similarly, a particle free to move in space will have three translational degrees of freedom.



Note : In pure rolling sphere has one degree of freedom as $KE = \frac{1}{2}mv^2 (1 + \frac{K^2}{R^2}) = \frac{7}{10}mv^2$

- The degrees of freedom are of three types :
 - Translational Degree of freedom :** Maximum three degrees of freedom are there corresponding to translational motion.
 - Rotational Degree of freedom :** The number of degrees of freedom in this case depends on the structure of the molecule.
 - Vibrational Degree of freedom :** It is exhibited at high temperatures.

Degree of freedom for different gases according to atomicity of gas at low temperature

Atomicity of gas	Translational	Rotational	Total	
Monoatomic Ex. Ar, Ne, Ideal gas etc	3	0	3	
Diatomic Ex. O ₂ , Cl ₂ , N ₂ etc.	3	2	5	

At high temperatures a diatomic molecule has 7 degrees of freedom. (3 translational, 2 rotational and 2 vibrational)

Ex. Calculate the total number of degrees of freedom possessed by the molecules in one cm^3 of H_2 gas at NTP.

Sol. 22400 cm^3 of every gas contains 6.02×10^{23} molecules.

$$\therefore \text{Number of molecules in } 1 \text{ cm}^3 \text{ of } \text{H}_2 \text{ gas} = \frac{6.02 \times 10^{23}}{22400} = 0.26875 \times 10^{20}$$

Number of degrees of freedom of a H_2 gas molecule = 5

$$\therefore \text{Total number of degrees of freedom of } 0.26875 \times 10^{20} \times 5 = 1.34375 \times 10^{20}.$$

MAXWELL'S LAW OF EQUIPARTITION OF ENERGY

The total kinetic energy of a gas molecules is equally distributed among its all degree of freedom and the energy associated with each degree of freedom at absolute temperature T is $\frac{1}{2}kT$

For one molecule of gas

$$\text{Energy related with each degree of freedom} = \frac{1}{2}kT$$

$$\text{Energy related with all degree of freedom} = \frac{f}{2}kT \quad \because \bar{v}_x^2 = \bar{v}_y^2 = \bar{v}_z^2 = \frac{\bar{v}_{\text{rms}}^2}{3} \Rightarrow \frac{1}{2}m\bar{v}_{\text{rms}}^2 = \frac{3}{2}kT$$

$$\text{So energy related with one degree of freedom} = \frac{1}{2}m \frac{\bar{v}_{\text{rms}}^2}{3} = \frac{3}{2} \frac{kT}{3} = \frac{1}{2}kT$$

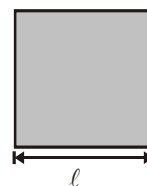
Ex. A cubical box of side 1 meter contains helium gas (atomic weight 4) at a pressure of 100 N/m^2 . During an observation time of 1 second, an atom travelling with the root-mean-square speed parallel to one of the edges of the cube, was found to make 500 hits with a particular wall, without any collision with other atoms. Take $R = \frac{25}{3} \text{ J/mol-K}$ and $k = 1.38 \times 10^{-23} \text{ J/K}$.

- Evaluate the temperature of the gas.
- Evaluate the average kinetic energy per atom.
- Evaluate the total mass of helium gas in the box.

Sol. Volume of the box = 1 m^3 , Pressure of the gas = 100 N/m^2 . Let T be the temperature of the gas

(a) Time between two consecutive collisions with one wall = $\frac{1}{500} \text{ sec}$

This time should be equal to $\frac{2\ell}{v_{\text{rms}}}$, where ℓ is the side of the cube.



$$2\ell v_{\text{rms}} = \frac{1}{500} \Rightarrow v_{\text{rms}} = 1000 \text{ m/s} \therefore \sqrt{\frac{3RT}{M}} = 1000 \Rightarrow T = \frac{(1000)^2 M}{3R} = \frac{(10)^6 (3 \times 10^{-3})}{3 \left(\frac{25}{3}\right)} = 160 \text{ K}$$

(b) Average kinetic energy per atom = $\frac{3}{2}kT = \frac{3}{2} [(1.38 \times 10^{-23}) \times 160] \text{ J} = 3.312 \times 10^{-21} \text{ J}$

(c) From $PV = nRT = \frac{m}{M}RT$, Mass of helium gas in the box $m = \frac{PVM}{RT}$

$$\text{Substituting the values, } m = \frac{(100)(1)(4 \times 10^{-3})}{\left(\frac{25}{3}\right)(160)} = 3.0 \times 10^{-4} \text{ kg}$$

DIFFERENT K.E. OF GAS (INTERNAL ENERGY)

- Translatory kinetic energy (E_T)** $E_T = \frac{1}{2} M v_{rms}^2 = \frac{3}{2} PV$

Kinetic energy of volume V is $= \frac{1}{2} M v_{rms}^2$ Note : Total internal energy of ideal gas is kinetic

- Energy per unit volume or energy density (E_v)**

$$E_v = \frac{\text{Total energy}}{\text{Volume}} = \frac{E}{V} ; E_v = \frac{1}{2} \left[\frac{M}{V} \right] v_{rms}^2 = \frac{1}{2} \rho v_{rms}^2 \because P = \frac{2}{3} \left[\frac{1}{2} \rho v_{rms}^2 \right] \therefore E_v = \frac{3}{2} P$$

- Molar K.E. or Mean Molar K.E. (E)**

$$E = \frac{1}{2} M_w v_{rms}^2 \text{ for } N_0 \text{ molecules or } M_w \text{ (gram)} \quad E = \frac{3}{2} RT = \frac{3}{2} N_0 kT$$

- Molecular kinetic energy or mean molecular K.E. ($\bar{E}$)**

$$E = \frac{1}{2} M_w v_{rms}^2, \bar{E} = \frac{E}{N_0} = \frac{3}{2} \frac{RT}{N_0} \Rightarrow \bar{E} = \frac{3}{2} kT$$

GOLDEN KEY POINT

- Except 0 K, at any temperature T , $E > E_m > \bar{E}$
- At a common temperature, for all ideal gas

E and $\bar{E}$ are same while E_m is different and depends upon nature of gas (M_w or m)

- For thermal equilibrium of gases, temperature of each gas is same and this temperature called as temperature of mixture (T_m) which can be find out on basis of conservation of energy (All gases are

of same atomicity). $T_m = \frac{\sum NT}{\sum N} = \frac{N_1 T_1 + N_2 T_2 + \dots + N_n T_n}{N_1 + N_2 + \dots + N_n}$

- 1 mole gas :** Mean kinetic energy $= \frac{3}{2} RT$; Total kinetic energy $= \frac{f}{2} RT$

1 molecule of gases : Mean kinetic energy $= \frac{3}{2} kT$; Total kinetic energy $= \frac{f}{2} kT$

$f \rightarrow$ Degree of freedom

Ex. Two ideal gases at temperature T_1 and T_2 are mixed. There is no loss of energy. If the masses of molecules of the two gases are m_1 and m_2 and number of their molecules are n_1 and n_2 respectively. Find the temperature of the mixture.

Sol. Total energy of molecules of first gas $= \frac{3}{2} n_1 kT_1$, Total energy of molecules of second gas $= \frac{3}{2} n_2 kT_2$

Let temperature of mixture be T then total energy of molecules of mixture $= \frac{3}{2} k(n_1 + n_2)T$

$$\therefore \frac{3}{2} (n_1 + n_2) kT = \frac{3}{2} k(n_1 T_1 + n_2 T_2) \Rightarrow T = \frac{n_1 T_1 + n_2 T_2}{(n_1 + n_2)}$$

Ex. The first excited state of hydrogen atom is 10.2 eV above its ground state. What temperature is needed to excite hydrogen atoms to first excited level.

Sol. K.E. of the hydrogen atom $\frac{3}{2} kT = 10.2 \text{ eV} = 10.2 \times (1.6 \times 10^{-19}) \text{ J}$

$$\Rightarrow T = \frac{2}{3} \times \frac{10.2 \times 1.6 \times 10^{-19}}{1.38 \times 10^{-23}} = 7.88 \times 10^4 \text{ K}$$

DIFFERENT SPEEDS OF GAS MOLECULES

• Average velocity

Because molecules are in random motion in all possible direction in all possible velocity. Therefore,

the average velocity of the gas in molecules in container is zero. $\langle \vec{v} \rangle = \frac{\vec{v}_1 + \vec{v}_2 + \dots + \vec{v}_N}{N} = 0$

rms speed of molecules $v_{\text{rms}} = \sqrt{\frac{3P}{\rho}} = \sqrt{\frac{3RT}{M_w}} = \sqrt{\frac{3kT}{m}} = 1.73 \sqrt{\frac{kT}{m}}$

Mean speed of molecules : By maxwell's velocity distribution law v_M or $\langle |\vec{v}| \rangle = v_{\text{mean}}$

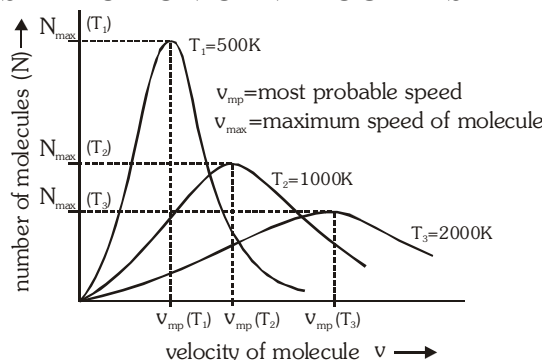
$$\langle |\vec{v}| \rangle = v_{\text{mean}} = \frac{|\vec{v}_1| + |\vec{v}_2| + \dots + |\vec{v}_N|}{N} = \sqrt{\frac{8P}{\pi\rho}} = \sqrt{\frac{8RT}{\pi M_w}} = \sqrt{\frac{8kT}{\pi m}} = 1.59 \sqrt{\frac{kT}{m}}$$

Most probable speed of molecules (v_{mp})

At a given temperature, the speed to which maximum number of molecules belongs is called as most

probable speed (v_{mp}) $v_{\text{mp}} = \sqrt{\frac{2P}{\rho}} = \sqrt{\frac{2RT}{M_w}} = \sqrt{\frac{2kT}{m}} = 1.41 \sqrt{\frac{kT}{m}}$

MAXWELL'S LAW OF DISTRIBUTION OF VELOCITIES



GOLDEN KEY POINT

- At any given temperature graph drawn in between molecular velocity and number of molecules is known as velocity distribution curve.
- The velocities of molecules of a gas are in between zero and infinity ($0 - \infty$)
- With the increase in the temperature, the most probable velocity and maximum molecule velocity both increases.
- The number of molecules within certain velocity range is constant although the velocity of molecule changes continuously at particular temperature.
- The area enclosed between the ($N - v$) curve and the velocity axis presents the total number of molecules.

On the basis of velocity distribution Maxwell established gives the law of equipartition of energy for gases of any temperature.

Ex. The velocities of ten particles in ms^{-1} are 0, 2, 3, 4, 4, 4, 5, 5, 6, 9. Calculate

(i) average speed and (ii) rms speed (iii) most probable speed.

Sol. (i) average speed, $v_{\text{av}} = \frac{0+2+3+4+4+4+5+5+6+9}{10} = \frac{42}{10} = 4.2 \text{ ms}^{-1}$

(ii) rms speed, $v_{\text{rms}} = \left[\frac{(0)^2 + (2)^2 + (3)^2 + (4)^2 + (4)^2 + (4)^2 + (5)^2 + (5)^2 + (6)^2 + (9)^2}{10} \right]^{1/2} = \left[\frac{228}{10} \right]^{1/2} = 4.77 \text{ ms}^{-1}$

(iii) most probable speed $v_{\text{mp}} = 4 \text{ m/s}$

Ex. At what temperature, will the root mean square velocity of hydrogen be double of its value at S.T.P., pressure remaining constant ?

Sol. Let v_1 be the r.m.s. velocity at S.T.P. and v_2 be the r.m.s. velocity at unknown temperature T_2 .

$$\therefore \frac{v_1^2}{v_2^2} = \frac{T_1}{T_2} \quad \text{or} \quad T_2 = T_1 \left[\frac{v_2}{v_1} \right]^2 = 273 \times (2)^2 = 273 \times 4 = 1092 \text{ K} = (1092 - 273) = 819^\circ\text{C}$$

Ex. Calculate rms velocity of oxygen molecule at 27°C

Sol. Temperature, $T = 27^\circ\text{C} \Rightarrow 273 + 27 = 300 \text{ K}$,

Molecular weight of oxygen $= 32 \times 10^{-3} \text{ kg}$ and $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$

$$\text{rms velocity is } v_{\text{rms}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3 \times 8.31 \times 300}{32 \times 10^{-3}}} = 483.5 \text{ ms}^{-1}$$

Ex. Calculate the kinetic energy of a gram molecule of argon at 127°C .

Sol. Temperature, $T = 127^\circ\text{C} = 273 + 127 = 400 \text{ K}$, $R = 8.31 \text{ J/mol K}$

$$\text{K.E. per gram molecule of argon} = \frac{3}{2} R T = \frac{3}{2} \times 8.31 \times 400 = 4986 \text{ J}$$

THERMODYNAMICS

Branch of physics which deals with the inter-conversion between heat energy and any other form of energy is known as thermodynamics. In this branch of physics we deal with the processes involving heat, work and internal energy. In this branch of science the conversion of heat into mechanical work and vice versa is studied.

- **Thermodynamical System**

The system which can be represented in of pressure (P), volume (V) and temperature (T), is known thermodynamic system. A specified portion of matter consisting of one or more substances on which the effects of variables such as temperature, volume and pressure are to be studied, is called a system. e.g. A gas enclosed in a cylinder fitted with a piston is a system.

- **Surroundings**

Anything outside the system, which exchanges energy with the system and which tends to change the properties of the system is called its surroundings.

- **Heterogeneous System**

A system which is not uniform throughout is said to be heterogeneous. e.g. A system consisting of two or more immiscible liquids.

- **Homogeneous System**

A system is said to be homogeneous if it is completely uniform throughout. e.g. Pure solid or liquid.

- **Isolated System**

A system in which there can be no exchange of matter and energy with the surroundings is said to be an isolated system.

- **Universe**

The system and its surroundings are together known as the universe.

- **Thermodynamic variables of the system**

(i) Composition (μ) (ii) Temperature (T) (iii) Volume (V) (iv) Pressure (P)

- **Thermodynamic state**

The state of a system can be described completely by composition, temperature, volume and pressure. If a system is homogeneous and has definite mass and composition, then the state of the system can be described by the remaining three variables namely temperature, pressure and volume. These variables are interrelated by equation $PV = \mu RT$. The thermodynamic state of the system is its condition as identified by two independent thermodynamic variables (P, V or P, T or V, T).

- **Zeroth law of thermodynamics**

If objects A and B are separately in thermal equilibrium with a third object C (say thermometer), then objects A and B are in thermal equilibrium with each other. Zeroth law of thermodynamics introduce the concept of temperature. Two objects (or systems) are said to be in thermal equilibrium if their temperatures are the same.

In measuring the temperature of a body, it is important that the thermometer be in the thermal equilibrium with the body whose temperature is to be measured.

- **Thermal equilibrium**

Thermal equilibrium is a situation in which two objects in thermal contact cease to exchange energy by the process of heat. Heat is the transfer of energy from one object to another object as a result of a difference in temperature between them.

• **Internal Energy**

Internal energy of a system is the energy possessed by the system due to molecular motion and molecular configuration. The energy due to molecular motion is called internal kinetic energy (U_k) and that due to molecular configuration is called internal potential energy (U_p). $dU = dU_k + dU_p$

If there no intermolecular forces, then $dU_p = 0$ and $dU = dU_k = m c_v dT$

c_v = Specific heat at constant volume and dT = Infinitesimal change in temperature

m = Mass of system

M = Molecular weight

Molar heat capacity $C_v = Mc_v$

For μ -moles of ideal gas $dU = \mu C_v dT = \frac{m}{M} C_v dT$

Internal energy in the absence of inter-molecular forces is simply the function of temperature and state only, it is independent of path followed. $\Delta U = U_f - U_i$

U_i = Internal energies in initial state and U_f = Internal energies in final state

• **Thermodynamic Processes**

In the thermodynamic process pressure, volume, temperature and entropy of the system change with time.

Thermodynamic process is said to take place if change occurs in the state of a thermodynamic system.

• **Sign convention used for the study of thermodynamic processes**

Heat gained by a system	Positive
Heat lost by a system	Negative
The work done by a system	Positive
Work done on the system	Negative
Increase in the internal energy of system	Positive
Decrease in the internal energy of system	Negative

• **Indicator Diagram or P-V Diagram**

In the equation of state of a gas $PV = \mu RT$

Two thermodynamic variables are sufficient to describe the behavior of a thermodynamic system.

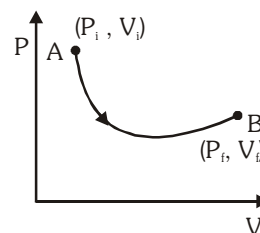
If any two of the three variables P , V and T are known then the third can be calculated.

P - V diagram is a graph between the volume V and the pressure P of the system.

The volume is plotted against X-axis while the pressure is plotted against Y-axis.

The point A represents the initial stage of the system. Initial pressure of the system is P_i and initial volume of the system V_i .

The point B represents the final state of the system. P_f and V_f are the final pressure and final volume respectively of the system. The points between A and B represent the intermediate states of the system. With the help of the indicator diagram we calculate the amount of work done by the gas or on the gas during expansion or compression.



WORK DONE BY THERMODYNAMIC SYSTEM

One of the simple example of a thermodynamic system is a gas in a cylinder with a movable piston.

- If the gas expands against the piston**

Gas exerts a force on the piston and displace it through a distance and does work on the piston.

- If the piston compresses the gas**

When piston moved inward, work is done on the gas.

- The work associated with volume changes**

If pressure of gas on the piston = P .

Then the force on the piston due to gas is $F = PA$

When the piston is pushed outward an infinitesimal distance dx , the work done by the gas is $dW = F \times dx = PA \, dx$

The change in volume of the gas is $dV = A \, dx$, $\therefore dW = P \, dV$

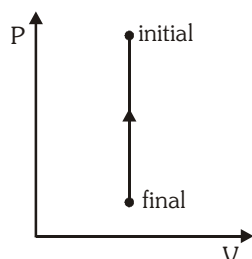
For a finite change in volume from V_i to V_f , this equation is then integrated between V_i to V_f to find the

$$\text{net work done } W = \int_{V_i}^{V_f} dW = \int_{V_i}^{V_f} P \, dV$$

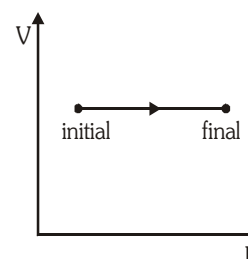
Hence the work done by a gas is equal to the area under P - V graph.

Following different cases are possible.

(i) Volume is constant

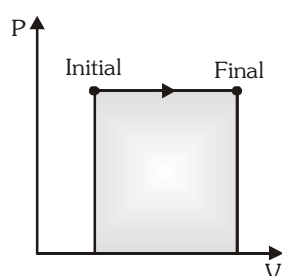


or

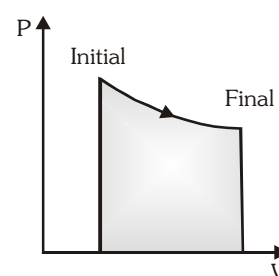


$$V = \text{constant and } W_{AB} = 0$$

(ii) Volume is increasing



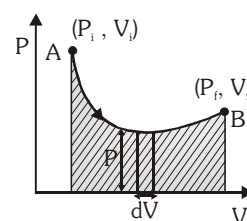
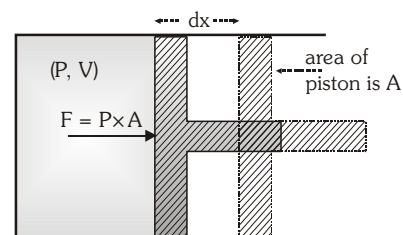
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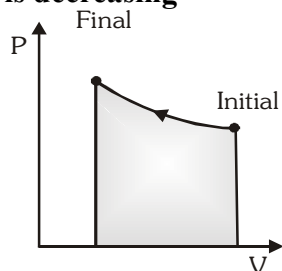
V is increasing

$$W_{AB} > 0$$

$$W_{AB} = \text{Shaded area}$$



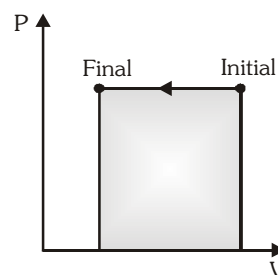
(iii) Volume is decreasing



V is decreasing

$$W_{AB} < 0$$

or



$$W_{AB} = - \text{Shaded area}$$

FIRST LAW OF THERMODYNAMICS

Let a gas in a cylinder with a moveable piston changes from an initial equilibrium state to a final equilibrium state.

System change its state through path 'a' :

The heat absorbed by the system in this process = δQ_a

The work done by the system = δW_a

• **Again for path 'b' :**

Heat absorbed by the system = δQ_b , Work done by the system = δW_b .

It is experimental fact that the $\delta Q_a - \delta W_a = \delta Q_b - \delta W_b$

Both δQ and δW depend on the thermodynamic path taken between two equilibrium states, but difference ($\delta Q - \delta W$) does not depends on path in between two definite states of the system.

So, there is a function (internal energy) of the thermodynamic coordinates (P, V and T) whose final value (U_f) minus its initial value (U_i) equals the change $\delta Q - \delta W$ in the process.

• **$dU = \delta Q - \delta W$.** This is the first law of thermodynamics.

Heat supplied to the system and work done by the system are path dependent so they are denoted by δQ and δW respectively. Change in internal energy $\Delta U = U_f - U_i$ does not depends on path it depends only on initial and final positions of the system. So, it is denoted by dU (or ΔU)

• **First Law of Thermodynamics**

If some quantity of heat is supplied to a system capable of doing external work, then the quantity of heat absorbed by the system is equal to the sum of the increase in the internal energy of the system and the external work done by the system. $\delta Q = dU + \delta W$ or $Q = W + \Delta U$

* This law is applicable to every process in nature

* The first law of thermodynamics introduces the concept of internal energy.

* The first law of thermodynamics is based on the law of conservation of energy.

* δQ , dU and δW must be expressed in the same units (either in units of work or in units of heat).

* This law is applicable to all the three phases of matter, i.e., solid, liquid and gas.

* dU is a characteristic of the state of a system, it may be any type of internal energy—translational kinetic energy, rotational kinetic energy, binding energy etc.

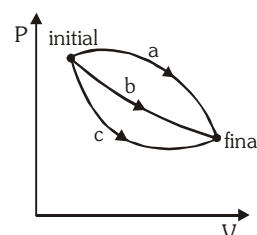
• **Limitations of first law of thermodynamics :**

It does not explain the direction of heat flow and it does not explain how much amount of heat given will be converted into work.

• **Significance of the first law of thermodynamics :**

The first law of thermodynamics tells us that it is impossible to get work from any machine without giving it an equivalent amount of energy.

Ex. The pressure in monoatomic gas increases linearly from $4 \times 10^5 \text{ Nm}^{-2}$ to $8 \times 10^5 \text{ Nm}^{-2}$ when its volume increases from 0.2 m^3 to 0.5 m^3 . Calculate.



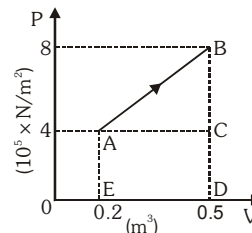
- (i) Work done by the gas, (ii) Increase in the internal energy,
 (iii) Amount of heat supplied, (iv) Molar heat capacity of the gas $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$

Sol. $P_1 = 4 \times 10^5 \text{ Nm}^{-2}$ $P_2 = 8 \times 10^5 \text{ Nm}^{-2}$, $V_1 = 0.2 \text{ m}^3$, $V_2 = 0.5 \text{ m}^3$

- (i) Work done by the gas = Area under P-V graph (Area ABCDEA)

$$= \frac{1}{2} (AE + BD) \times AC = \frac{1}{2} (4 \times 10^5 + 8 \times 10^5) \times (0.5 - 0.2)$$

$$= \frac{1}{2} \times 12 \times 10^5 \times 0.3 = 1.8 \times 10^5 \text{ J}$$



- (ii) Increase in internal energy $\Delta U = C_v (T_2 - T_1) = \frac{C_v}{R} R(T_2 - T_1) = \frac{C_v}{R} (P_2 V_2 - P_1 V_1)$
 For monoatomic gas

$$C_v = \frac{3}{2} R \therefore \Delta U = \frac{3}{2} [(8 \times 10^5 \times 0.5) - (4 \times 10^5 \times 0.2)] = \frac{3}{2} [4 \times 10^5 - 0.8 \times 10^5] = 4.8 \times 10^5 \text{ J}$$

- (iii) $Q = \Delta U + W = 4.8 \times 10^5 + 1.8 \times 10^5 = 6.6 \times 10^5 \text{ J}$

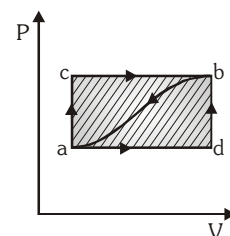
(iv) $C = \frac{Q}{\eta \Delta T} = \frac{QR}{\eta R \Delta T} = \frac{QR}{\eta (P_2 V_2 - P_1 V_1)} = \frac{6.6 \times 10^5 \times 8.31}{1 \times 3.2 \times 10^5} = 17.14 \text{ J/mole K}$

Ex. When a system is taken from state a to state b, in figure along the path $a \rightarrow c \rightarrow b$, 60 J of heat flow into the system, and 30 J of work is done :

- (i) How much heat flows into the system along the path $a \rightarrow d \rightarrow b$ if the work is 10 J.

- (ii) When the system is returned from b to a along the curved path, the work done by the system is -20 J. Does the system absorb or liberate heat, and how much?

- (iii) If, $U_a = 0$ and $U_d = 22 \text{ J}$, find the heat absorbed in the process $a \rightarrow d$ and $d \rightarrow b$.



Sol. For the path a, c, b, $\Delta U = Q - W = 60 - 30 = 30 \text{ J}$ or $U_b - U_a = 30 \text{ J}$

- (i) Along the path a, d, b, $Q = \Delta U + W = 30 + 10 = 40 \text{ J}$

- (ii) Along the curved path b, a, $Q = (U_a - U_b) + W = (-30) + (-20) = -50 \text{ J}$, heat flows out the system

- (iii) $Q_{ad} = 32 \text{ J}$; $Q_{db} = 8 \text{ J}$

ISOMETRIC OR ISOCHORIC PROCESS

- Isochoric process is a thermodynamic process that takes place at constant volume of the system, but pressure and temperature varies for change in state of the system.

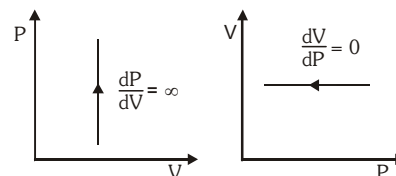
Equation of state $P = \text{constant} \times T$ (P and T are variable, V is constant)

Work done In this process volume remains constant $\Delta V = 0$ or $dV = 0 \Rightarrow W = \int_{V_i}^{V_f} P dV = 0$

Form of first Law $Q = \Delta U$

It means whole of the heat supplied is utilized for change in internal energy of the system. $Q = \Delta U = \mu C_v \Delta T$

Slope of the P-V curve $\frac{dP}{dV} = \infty$



Specific heat at constant volume (C_v)

The quantity of heat required to raise the temperature of 1 gram mole gas through 1°C at constant volume is equal to the specific heat at constant volume.

- A gas enclosed in a cylinder having rigid walls and a fixed piston. When heat is added to the gas, there would be no change in the volume of the gas.

- When a substance melts, the change in volume is negligibly small. So, this may be regarded as a nearly isochoric process.
- Heating process in pressure cooker is an example of isometric process.

Ex. An ideal gas has a specific heat at constant pressure $C_p = \frac{5R}{2}$. The gas is kept in a closed vessel of volume 0.0083 m^3 at a temperature of 300K and a pressure of $1.6 \times 10^6 \text{ Nm}^{-2}$. An amount of $2.49 \times 10^4 \text{ J}$ of heat energy is supplied to the gas. Calculate the final temperature and pressure of the gas.

Sol. $C_v = C_p - R = \frac{5R}{2} - R = \frac{3R}{2}$, $\Delta V = 0$, $T_1 = 300 \text{ K}$, $V = 0.0083 \text{ m}^3$, $P_1 = 1.6 \times 10^6 \text{ Nm}^{-2}$

From first law of thermodynamics $Q = \Delta U + P\Delta V \Rightarrow \Delta U = Q = 2.49 \times 10^4 \text{ J}$

From gas equation $n = \frac{PV}{RT} = \frac{1.6 \times 10^6 \times 0.0083}{8.3 \times 300} = \frac{16}{3}$

$$\therefore \Delta U = nC_v\Delta T \Rightarrow \Delta T = \frac{\Delta U}{nC_v} = \frac{2.49 \times 10^4 \times 6}{3 \times 8.3 \times 16} = 375 \text{ K}$$

Final temperature = $300 + 375 = 675\text{K}$

According to pressure law $P \propto T \Rightarrow \frac{P_2}{P_1} = \frac{T_2}{T_1} \Rightarrow P_2 = \frac{T_2}{T_1} \times P_1 = \frac{1.6 \times 10^6 \times 675}{300} = 3.6 \times 10^6 \text{ Nm}^{-2}$

Ex. 5 moles of oxygen is heated at constant volume from 10°C to 20°C . What will be change in the internal energy of the gas? The gram molecular specific heat of oxygen at constant pressure.

$C_p = 8 \text{ cal/mole}$ and $R = 8.36 \text{ joule/mole } ^\circ\text{C}$.

Sol. $\therefore C_v = C_p - R = 8 - 2 = 6 \text{ cal/mole } ^\circ\text{C}$

$\therefore$ Heat absorbed by 5 moles of oxygen at constant volume

$$Q = nC_v\Delta T = 5 \times 6 (20 - 10) = 30 \times 10 = 300 \text{ cal}$$

At constant volume $\Delta V = 0$. $\therefore \Delta W = 0$

$\therefore$ From first law of thermodynamics $Q = \Delta U + W \Rightarrow 300 = \Delta U + 0 \Rightarrow \Delta U = 300 \text{ cal}$

ISOBARIC PROCESS

Isobaric process is a thermodynamic process that takes place at constant pressure, but volume and temperature varies for change in state of the system.

- Equation of state** $V = \text{constant} \times T$ or $V \propto T$
- Work done** In this process pressure remains constant $\Delta P = 0$

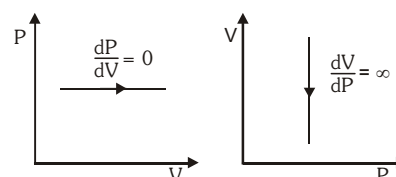
$$\text{Work done } W = \int_{V_i}^{V_f} P dV = P(V_f - V_i)$$

- Form of first Law** $Q = \Delta U + P(V_f - V_i)$
 $\mu C_p dT = \mu C_v dT + P(V_f - V_i)$

It is clear that heat supplied to the system is utilized for :

- Increasing internal energy and
- Work done against the surrounding atmosphere.

- Slope of the PV curve :** $\left(\frac{dP}{dV}\right)_{\text{isobaric}} = 0$



• **Specific heat at constant pressure (C_p)**

The quantity of heat required to raise the temperature of 1 gram mole gas through 1°C at constant pressure is equal to the specific heat. Heating of water at atmospheric pressure. • Melting of solids and boiling of liquids at atmospheric pressure.

Ex. At normal pressure and 0°C temperature the volume of 1 kg of ice is reduced by 91 cm^3 on melting. Latent heat of melting of ice is $3.4 \times 10^5\text{ J/kg}$. Calculate the change in the internal energy when 2kg of ice melts at normal pressure and 0°C . ($P=1.01 \times 10^5\text{ Nm}^{-2}$)

Sol. Heat energy absorbed by 2 kg of ice for melting $Q = mL = 2 \times 3.4 \times 10^5 = 6.8 \times 10^5\text{ J}$

Change in volume of 2 kg of ice $= 2 \times 91 = 182\text{ cm}^3 = 182 \times 10^{-6}\text{ m}^3$

$$\therefore W = P\Delta V = 1.01 \times 10^5 \times (-182 \times 10^{-6}) = -18.4\text{ J}$$

Since, work is done on ice so work W is taken -ve. Now from first law of thermodynamics

$$Q = \Delta U + W \Rightarrow \Delta U = Q - W = 6.8 \times 10^5 - (-18.4) = (6.8 \times 10^5 + 18.4)\text{ J}$$

Ex. What amount of heat must be supplied to $2.0 \times 10^{-2}\text{ kg}$ of nitrogen (at room temperature) to raise the temperature by 45°C at constant pressure. Molecular mass of $\text{N}_2 = 28$, $R = 8.3\text{ J mol}^{-1}\text{ K}^{-1}$.

Sol. Here $m = 2 \times 10^{-2}\text{ kg}$, $\Rightarrow n = \frac{m}{M} = \frac{2 \times 10^{-2}}{28 \times 10^{-3}} = \frac{5}{7}$ & $C_p = \frac{7}{2}R$

$$\therefore Q = nC_p\Delta T = \frac{5}{7} \times \frac{7}{2} \times 8.3 \times 45 = 933.75\text{ J}$$

ISOTHERMAL PROCESS

In this process pressure and volume of system change but temperature remains constant.

In an isothermal process, the exchange of heat between the system and the surroundings is allowed.

Isothermal process is carried out by either supplying heat to the substance or by extracting heat from it.

A process has to be extremely slow to be isothermal.

• **Equation of state**

$$P V = \text{constant } (\mu RT) \quad (T \text{ is constant})$$

• **Work Done**

Consider μ moles of an ideal gas, enclosed in a cylinder, at absolute temperature T , fitted with a frictionless piston. Suppose that gas undergoes an isothermal expansion from the initial state (P_1, V_1) to the final state (P_2, V_2).

$$\therefore \text{Work done : } W = \int_{V_1}^{V_2} \frac{\mu RT}{V} dV = \mu RT \int_{V_1}^{V_2} \frac{dV}{V} = \mu RT [\log_e V]_{V_1}^{V_2}$$

$$= \mu RT [\log_e V_2 - \log_e V_1] = \mu RT \log_e \left[\frac{V_2}{V_1} \right]$$

$$\Rightarrow W = 2.303 \mu RT \log_{10} \left[\frac{P_1}{P_2} \right] \quad [\because P_1 V_1 = P_2 V_2]$$

Form of First Law

There is no change in temperature and internal energy of the system depends on temperature only

$$\text{So} \quad \Delta U = 0, Q = 2.303 \mu RT \log_{10} \left[\frac{V_2}{V_1} \right]$$

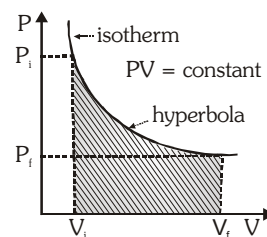
It is clear that Whole of the heat energy supplied to the system is utilized by the system in doing external work. There is no change in the internal energy of the system.

Slope of the isothermal curve

For an isothermal process, $PV = \text{constant}$

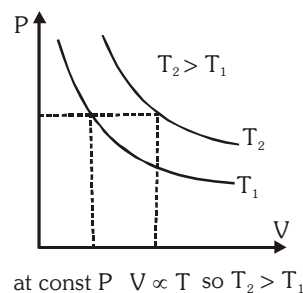
$$\text{Differentiating, } PdV + VdP = 0 \quad \Rightarrow \quad VdP = -PdV \Rightarrow \frac{dP}{dV} = -\frac{P}{V}$$

$$\text{Slope of isothermal curve, } \left[\frac{dP}{dV} \right]_{\text{isothermal}} = -\frac{P}{V}$$



For a given system :

- The product of the pressure and volume of a given mass of a perfect gas remains constant in an isothermal process.
- Boyle's law is obeyed in an isothermal process.
- A graph between pressure and volume of a given mass of a gas at constant temperature is known as isotherm or isothermal of the gas.
- Two isotherms for a given gas at two different temperatures T_1 and T_2 are shown in figure.
- The curves drawn for the same gas at different temperatures are mutually parallel and do not cut each other.
- If two isotherms intersect each other at a single point we get same value of P and V at intersection point.
- $PV = \mu RT_1$ for temperature T_1 and $PV = \mu RT_2$ for temperature T_2 .



It means $T_1 = T_2$ which is not possible.

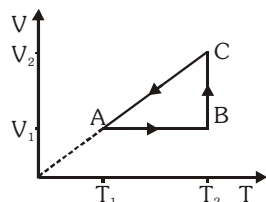
- An ideal gas enclosed in a conducting cylinder fitted with a conducting piston. Let the gas be allowed to expand very-very slowly.
- This shall cause a very slow cooling of the gas, but heat will be conducted into the cylinder from the surrounding. Hence the temperature of the gas remains constant. If the gas is compressed very-very slowly, heat will be produced, but this heat will be conducted to the surroundings and the temperature of the gas shall remain constant.
- The temperature of a substance remains constant during melting. So, the melting process is an isothermal process.
- Boiling is an isothermal process, when a liquid boils, its temperature remains constant.
- If sudden changes are executed in a vessel of infinite conductivity then they will be isothermal.

Ex. Two moles of a gas at 127°C expand isothermally until its volume is doubled. Calculate the amount of work done.

Sol. $n = 2, \quad T = 127 + 273 = 400\text{K}, \quad \frac{V_2}{V_1} = 2$

From formula $W = 2.3026 nRT \log_{10} \frac{V_2}{V_1} = 2.3026 \times 2 \times 8.3 \times 400 \times \log_{10} 2$
 $= 2.3026 \times 2 \times 8.3 \times 400 \times 0.3010 \approx 4.6 \times 10^3 \text{ J}$

Ex. Figure shows a process ABCA performed on an ideal gas.



Find the net heat given to the system during the process.

Sol. Since the process is cyclic, hence the change in internal energy is zero.

The heat given to the system is then equal to the work done by it.

The work done in part AB is $W_1 = 0$ (the volume remains constant). The part BC represents an isothermal process so that the work done by the gas during this part is $W_2 = nRT_2 \ln \frac{V_2}{V_1}$

During the part CA $V \propto T$ So, V/T is constant and hence, $P = \frac{nRT}{V}$ is constant

The work done by the gas during the part CA is $W_3 = P(V_1 - V_2) = nRT_1 - nRT_2 = -nR(T_2 - T_1)$.

The net work done by the gas in the process ABCA is $W = W_1 + W_2 + W_3 = nR \left[T_2 \ln \frac{V_2}{V_1} - (T_2 - T_1) \right]$

The same amount of heat is given to the gas.

ADIABATIC PROCESS

It is that thermodynamic process in which pressure, volume and temperature of the system change but there is no exchange of heat between the system and the surroundings.

A sudden and quick process will be adiabatic since there is no sufficient time available for exchange of heat so process adiabatic.

Equation of state : $PV = \mu RT$

Equation for adiabatic process $PV^\gamma = \text{constant}$

Work done

Let initial state of system is (P_1, V_1, T_1) and after adiabatic change final state of system is (P_2, V_2, T_2) then we can write $P_1 V_1^\gamma = P_2 V_2^\gamma = K$ (here K is const.)

So $W = \int_{V_1}^{V_2} P dV = K \int_{V_1}^{V_2} V^{-\gamma} dV = K \left[\frac{V^{-\gamma+1}}{-\gamma+1} \right]_{V_1}^{V_2} = \frac{K}{(-\gamma+1)} [V_2^{-\gamma+1} - V_1^{-\gamma+1}]$ ($\because K = P_1 V_1^\gamma = P_2 V_2^\gamma$)

$$\Rightarrow W = \frac{1}{(\gamma-1)} [P_1 V_1^\gamma V_1^{-\gamma} \cdot V_1 - P_2 V_2^\gamma V_2^{-\gamma} \cdot V_2] = \frac{1}{(\gamma-1)} [P_1 V_1 - P_2 V_2]$$

$$\Rightarrow W = \frac{\mu R}{(\gamma-1)} (T_1 - T_2) \quad (\because PV = \mu RT)$$

Form of first law : $dU = -\delta W$

It means the work done by an ideal gas during adiabatic expansion (or compression) is on the cost of change in internal energy proportional to the fall (or rise) in the temperature of the gas.

If the gas expands adiabatically, work is done by the gas. So, W_{adia} is positive.

The gas cools during adiabatic expansion and $T_1 > T_2$.

If the gas is compressed adiabatically, work is done on the gas. So, W_{adia} is negative.

The gas heats up during adiabatic compression and $T_1 < T_2$.

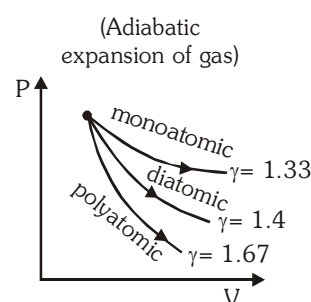
Slope of the adiabatic curve

For an adiabatic process, $PV^\gamma = \text{constant}$

Differentiating, $P^\gamma V^{\gamma-1} dV + V^\gamma dP = 0$

$$\Rightarrow V^\gamma dP = -\gamma PV^{\gamma-1} dV \Rightarrow \frac{dP}{dV} = -\frac{\gamma PV^{\gamma-1}}{V^\gamma} = -\gamma \frac{P}{V} = \gamma \left(-\frac{P}{V} \right)$$

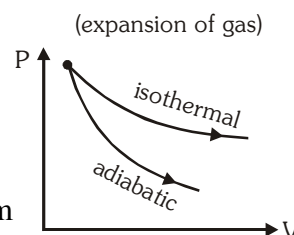
Slope of adiabatic curve, $\left[\frac{dP}{dV} \right]_{\text{adiabatic}} = -\frac{\gamma P}{V}$



Slope of adiabatic is greater than the slope of isotherm

$$\left[\frac{dP}{dV} \right]_{\text{adia}} = \gamma \left[-\frac{P}{V} \right] = \gamma \left[\frac{dP}{dV} \right]_{\text{iso}} \Rightarrow \frac{\text{slope of adiabatic changes}}{\text{slope of isothermal changes}} = \gamma$$

Since γ is always greater than one so an adiabatic is steeper than an isotherm



Examples of adiabatic process

- A gas enclosed in a thermally insulated cylinder fitted with a non-conducting piston. If the gas is compressed suddenly by moving the piston downwards, some heat is produced. This heat cannot escape the cylinder. Consequently, there will be an increase in the temperature of the gas.
- If a gas is suddenly expanded by moving the piston outwards, there will be a decrease in the temperature of the gas.
- Bursting of a cycle tube.
- Propagation of sound waves in a gas.
- In diesel engines burning of diesel without spark plug is done due to adiabatic compression of diesel vapour and air mixture

Ex. Why it is cooler at the top of a mountain than at sea level?

Sol. Pressure decreases with height. Therefore if hot air rises, it suffers adiabatic expansion.

From first law of thermodynamics $\Delta Q = \Delta U + \Delta W \Rightarrow \Delta U = -\Delta W$ [$\because \Delta Q = 0$]

This causes a decrease in internal energy and hence a fall of temperature.

Ex. 2m^3 volume of a gas at a pressure of $4 \times 10^5 \text{ Nm}^{-2}$ is compressed adiabatically so that its volume becomes 0.5m^3 . Find the new pressure. Compare this with the pressure that would result if the compression was isothermal. Calculate work done in each process. $\gamma = 1.4$

Sol. $V_1 = 2\text{m}^3$, $P_1 = 4 \times 10^5 \text{ Nm}^{-2}$, $V_2 = 0.5\text{m}^3$

In adiabatic process $P_1 V_1^\gamma = P_2 V_2^\gamma \Rightarrow P_2 = 4 \times 10^5 \left[\frac{2}{0.5} \right]^{1.4} = 4 \times 10^5 (4)^{1.4} = 2.8 \times 10^6 \text{ Nm}^{-2}$

In isothermal process $P_1 V_1 = P_2 V_2 \Rightarrow P_2 = \frac{P_1 V_1}{V_2} = \frac{4 \times 10^5 \times 2}{0.5} = 1.6 \times 10^6 \text{ Nm}^{-2}$.

Now work done in adiabatic process $W = \frac{P_2 V_2 - P_1 V_1}{\gamma - 1} = \frac{(2.8 \times 10^6 \times 0.5) - (4 \times 10^5 \times 2)}{1.4 - 1} = 1.48 \times 10^6 \text{ J}$.

Work done in isothermal process $W = 2.3026 RT \log \frac{V_2}{V_1} = 2.3026 P_1 V_1 \log \frac{V_2}{V_1}$
 $= 2.3026 \times 4 \times 10^5 \times 2 \times \log \left[\frac{0.5}{2.0} \right] = 2.3026 \times 4 \times 10^5 \times 2 \log \left(\frac{1}{4} \right) = -1.1 \times 10^6 \text{ J}$

Ex. Two samples of a gas initially at same temperature and pressure are compressed from a volume V to $\frac{V}{2}$. One sample is compressed isothermally and the other adiabatically. In which sample is the pressure greater?

Sol. Let initial volume, $V_1 = V$ and pressure, $P_1 = P$, final volume, $V_2 = \frac{V}{2}$ and final pressure, $P_2 = ?$

For isothermal compression $P_2 V_2 = P_1 V_1$ or $P_2 = \frac{P_1 V_1}{V_2} = \frac{PV}{\frac{V}{2}} = 2P$

For adiabatic compression $P_2' = P_1 \left[\frac{V_1}{V_2} \right]^\gamma \Rightarrow P_2' = P \left[\frac{V}{V/2} \right]^\gamma = 2^\gamma P$

$\Rightarrow P_2' = 2^\gamma P$ $\gamma > 1 \therefore 2^\gamma > 2$ and $P_2' > P_2$

Pressure during adiabatic compression is greater than the pressure during isothermal compression.

Ex. Calculate the work done when 1 mole of a perfect gas is compressed adiabatically. The initial pressure and volume of the gas are 10^5 N/m^2 and 6 litre respectively. The final volume of the gas is 2 litres.

Molar specific heat of the gas at constant volume is $\frac{3R}{2}$ $[(3)^{5/3} = 6.19]$

Sol. For an adiabatic change $PV^\gamma = \text{constant}$ $P_1 V_1^\gamma = P_2 V_2^\gamma$

As molar specific heat of gas at constant volume $C_v = \frac{3}{2}R$

$C_p = C_v + R = \frac{3}{2}R + R = \frac{5}{2}R \Rightarrow \gamma = \frac{C_p}{C_v} = \frac{\frac{5}{2}R}{\frac{3}{2}R} = \frac{5}{3}$

$$\therefore P_2 = \left[\frac{V_1}{V_2} \right]^\gamma P_1 = \left[\frac{6}{2} \right]^{5/3} \times 10^5 = (3)^{5/3} \times 10^5 = 6.19 \times 10^5 \text{ N/m}^2$$

$$\begin{aligned} \text{Work done } W &= \frac{1}{1-\gamma} [P_2 V_2 - P_1 V_1] = \frac{1}{1-(5/3)} [6.19 \times 10^5 \times 2 \times 10^{-3} - 10^5 \times 6 \times 10^{-3}] \\ &= - \left[\frac{2 \times 10^2 \times 3}{2} (6.19 - 3) \right] = -3 \times 10^2 \times 3.19 = -957 \text{ joules} \end{aligned}$$

– ive sign shows external work done on the gas

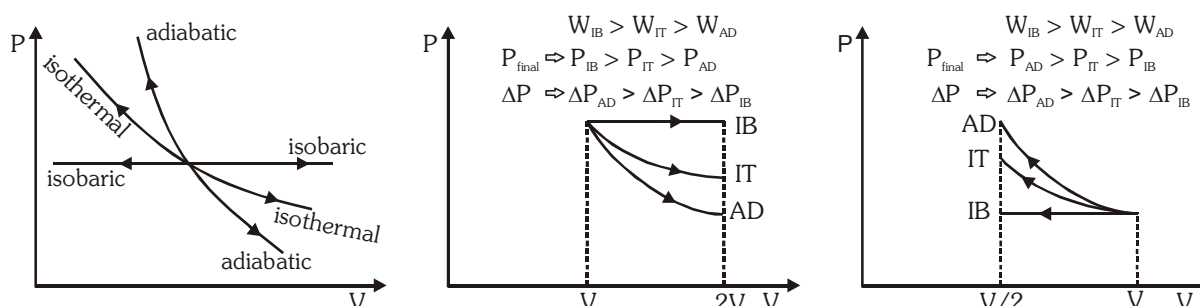
Ex. A motor tyre pumped to a pressure of 3 atm. It suddenly bursts. Calculate the fall in temperature due to adiabatic expansion. The temperature of air before expansion is 27°C. Given $\gamma=1.4$.

Sol. We know that $T_2^\gamma P_2^{1-\gamma} = T_1^\gamma P_1^{1-\gamma} \Rightarrow \left[\frac{T_2}{T_1} \right]^\gamma = \left[\frac{P_1}{P_2} \right]^{1-\gamma} \Rightarrow \left[\frac{T_2}{300} \right]^{1.4} = \left[\frac{3}{1} \right]^{1-1.4}$

$$\Rightarrow \left[\frac{T_2}{300} \right]^{1.4} = \left[\frac{1}{3} \right]^{0.4} \Rightarrow T_2 = 219.2 \text{ K} \Rightarrow T_1 - T_2 = (300 - 219.2) \text{ K} = 80.8 \text{ K}$$

GOLDEN KEY POINT

- When a gas expands its volume increases, then final pressure is less for adiabatic expansion. But, when a gas compresses its volume decreases, then the final pressure is more in case of adiabatic compression.



First Law of Thermodynamics Applied to Different Processes

Process	Q	ΔU	W
Cyclic	W	0	Area of the closed curve
Isochoric	ΔU	$\mu C_V \Delta T$ (μ mole of gas)	0
Isothermal	W	0	$\mu RT \log_e \left[\frac{V_f}{V_i} \right] = \mu RT \log_e \left[\frac{P_i}{P_f} \right]$
Adiabatic	0	$-W$	$\frac{\mu R(T_f - T_i)}{1 - \gamma}$
Isobaric	$\mu C_p \Delta T$	$\mu C_V \Delta T$	$P(V_f - V_i) = \mu R(T_f - T_i)$

Ex. Plot $P - V$, $V - T$ graph corresponding to the $P - T$ graph for an ideal gas shown in figure. Explain your answers.

Sol. For process AB $T = \text{constant}$ so $P \propto \frac{1}{V}$
 For process BC $P = \text{constant}$ so $V \propto T$
 For process CD $T = \text{constant}$ so $V \propto \frac{1}{P}$
 For process DA $P = \text{constant}$ so $V \propto T$

FREE EXPANSION

Take a thermally insulated bottle with ideal gas at some temperature T_1 and, by means of a pipe with a stopcock, connect this to another insulated bottle which is evacuated. If we suddenly open the stopcock, the gas will rush from the first bottle into the second until the pressures are equalized.

Experimentally, we find that this process of free expansion does not change the temperature of the gas – when the gas attains equilibrium and stops flowing, the final temperature of both bottles are equal to the initial temperature T_1 .

This process is called a free expansion.

The change in the internal energy of the gas can be calculated by applying the first law of thermodynamics to the free-expansion process.

The process is adiabatic because of the insulation, So $Q = 0$.

No part of the surroundings moves so the system does no work on its surroundings.

- **For ideal gas** $(\delta W)_{\text{ext.}} = \text{Work done against external atmosphere}$
 $= P dV = 0$ (because $P = 0$)
 $(\delta W)_{\text{int.}} = \text{Work done against internal molecular forces} = 0$
 $\delta Q = dU + \delta W \Rightarrow 0 = dU + 0$

The internal energy does not change $dU = 0$ So U and T are constant.

The initial and final states of this gas have the same internal energy.

Which implies that the internal energy of an ideal gas does not depend on the volume at all.

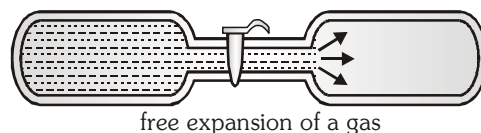
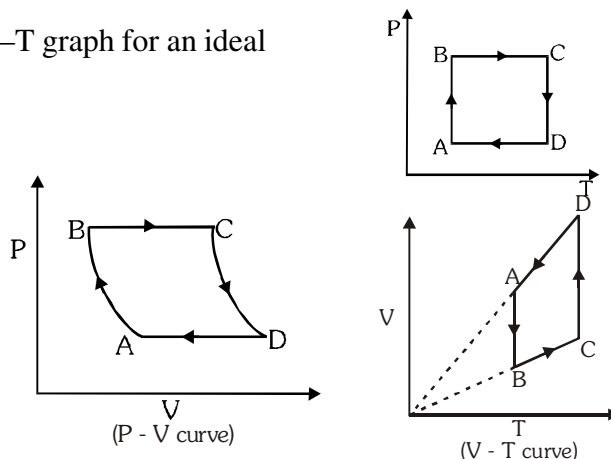
The free-expansion process has led us to the following conclusion :

The internal energy $U(T)$ of an ideal gas depends only on the temperature.

- **For real gas**

In free expansion of real gases, measurements show that the temperature changes slightly in a free expansion. Which implies that the internal energy of a real gas depends on the volume also.

$$\begin{aligned} \delta Q &= 0 & (\delta W)_{\text{ext.}} &= 0 & (\because P = 0) \\ (\delta W)_{\text{int.}} &\neq 0 & & & (\text{Intermolecular forces are present in real gases}) \\ \delta Q &= dU + \delta W & \Rightarrow 0 &= dU + (\delta W)_{\text{int.}} & \Rightarrow dU = -(\delta W)_{\text{int.}} \\ & & \Rightarrow U &\text{decreases. So } T \text{ decreases.} \end{aligned}$$



RELATION BETWEEN DEGREE OF FREEDOM AND SPECIFIC HEAT OF GAS

Energy related with each degree of freedom = $\frac{1}{2}kT$, Energy related with all degree of freedom = $\frac{f}{2}kT$

Internal energy of one mole of ideal gas (total K.E.) $U = \frac{f}{2}RT$ for Isometric process (volume constant)

$$\delta W = 0$$

By first law of thermodynamics $\delta Q = \delta W + dU \Rightarrow C_v dT = dU \Rightarrow C_v = \frac{dU}{dT}$

$$C_v = \frac{dU}{dT} = \frac{f}{2}R = \frac{R}{\gamma - 1} \quad . C_p = C_v + R = \left[\frac{f}{2} + 1 \right] R = \frac{\gamma R}{\gamma - 1} \quad \text{and} \quad \gamma = \frac{C_p}{C_v} = 1 + \frac{2}{f}$$

$$C_v = \frac{R}{\gamma - 1} \quad , \quad C_p = \frac{\gamma R}{\gamma - 1} \quad \text{and} \quad \gamma = 1 + \frac{2}{f}$$

General expression for C (C_p or C_v) in the process $PV^x = \text{constant}$ $C = \frac{R}{\gamma - 1} + \frac{R}{1 - x}$

For isobaric process $P = \text{constant}$ so $x = 0$ $\therefore C = C_p = \frac{R}{\gamma - 1} + R = C_v + R$

For isothermal process, $PV = \text{constant}$ so $x = 1$ $\therefore C = \infty$

For adiabatic process $PV^\gamma = \text{constant}$ so $x = \gamma$ $\therefore C = 0$

Values of f , U , C_v , C_p and γ for different gases are shown in table below.

Atomicity of gas	f	C_v	C_p	γ
Monoatomic	3	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{3} = 1.67$
Diatomic	5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5} = 1.4$
Triatomic and Triatomic linear	7	$\frac{7}{2}R$	$\frac{9}{2}R$	$\frac{9}{7} = 1.28$
Poly atomic Triangular Non-linear	6	$\frac{6}{2}R = 3R$	$\frac{8}{2}R = 4R$	$\frac{4}{3} = 1.33$

- $1 < \gamma < 2$
- If atomicity of gases is same U , C_p , C_v and γ is same for gas mixture.
- If in a gas mixture gases are of different atomicity, then for gas mixture γ changes according to following condition. Diatomic $\gamma_1 \leq \text{mixture} \leq \gamma_2$ mono atomic where $\gamma_1 < \gamma_2$
- If 'f' is the degree of freedom per molecule for a gas, then

$$\text{Total energy of each molecule} = \frac{fkT}{2}$$

$$\text{Total energy per mole of gas} = N_0 \frac{f}{2} kT = \frac{f}{2} RT$$

- According to kinetic theory of gases, the molecule are not interacting with each other. So potential energy is zero and internal energy of gas molecules is only their kinetic energy.
- For ' μ ' mole of a gas : Internal energy at temperature T is $U = \frac{\mu f R T}{2} = \mu C_V T$
- Change in internal energy is given by $dU = \frac{\mu f R}{2} (dT) = \mu C_V dT$
This change is process independent.

C_p is greater than C_v

If a gas is heated at constant volume, the gas does no work against external pressure. In this case, the whole of the heat energy supplied to the gas is spend in raising the temperature of the gas.

If a gas is heated at constant pressure, its volume increases. In this case, heat energy is required for the following two purpose :

- To increase the volume of the gas against external pressure.
- To increase the temperature of 1 mole of gas through 1 K.

Thus, more heat energy is required to raise the temperature of 1 mole of gas through 1 K when it is heated at constant pressure than when it is heated at constant volume. $\therefore C_p > C_v$

The difference between C_p and C_v is equal to thermal equivalent of the work done by the gas in expanding against external pressure.

Mayer's formula : $C_p - C_v = R$

$\therefore$ At constant pressure $dQ = \mu C_p dT$, $dU = \mu C_v dT$ & $dW = PdV = \mu R dT$

Now from first law of thermodynamics $dQ = dW + dU$

$$\Rightarrow \mu C_p dT = \mu R dT + \mu C_v dT \Rightarrow C_p = R + C_v \Rightarrow C_p - C_v = R$$

Ex. Calculate the difference between two specific heats of 1 g of helium gas at NTP. Molecular weight of helium = 4 and $J = 4.186 \times 10^7$ erg cal⁻¹.

Sol. Gas constant for 1 g of helium, $r = \frac{R}{M_w} = \frac{PV}{T \times M_w} = \frac{76 \times 13.6 \times 981 \times 22400}{273 \times 4} = 2.08 \times 10^7$ erg g⁻¹ K⁻¹

$$C_p - C_v = \frac{r}{J} = \frac{2.08 \times 10^7}{4.186 \times 10^7} = 0.5 \text{ cal g}^{-1} \text{ K}^{-1}$$

Ex. Calculate the molar specific heat at constant volume. Given : specific heat of hydrogen at constant pressure is 6.85 cal mol⁻¹ K⁻¹ and density of hydrogen = 0.0899 g cm⁻³. One mole of gas = 2.016 g, $J = 4.2 \times 10^7$ erg cal⁻¹ and 1 atmosphere = 10^6 dyne cm⁻².

Sol. Since the density of hydrogen is 0.0899 g cm⁻³ therefore volume occupied by 0.0899 g of hydrogen

at NTP is 1000 cm³. So, volume of 1 mole (2.016 g) of gas, $V = \frac{1000}{0.0899} \times 2.016 \text{ cm}^3$

$$C_p - C_v = \frac{R}{J} = \frac{PV}{T \times J} = \frac{10^6 \times 1000 \times 2.016}{0.0899 \times 273 \times 4.2 \times 10^7} = 1.96 \text{ cal mol}^{-1} \text{ K}^{-1}$$

$$\therefore C_v = C_p - 1.96 = (6.85 - 1.96) = 4.89 \text{ cal mol}^{-1} \text{ K}^{-1}$$

Ex. The specific heat of argon at constant volume is 0.075 kcal/kg K. Calculate its atomic weight, [$R = 2 \text{ cal/mol K}$]

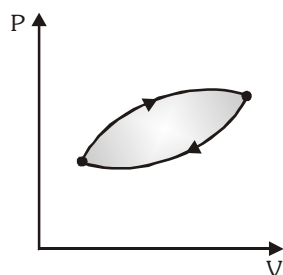
Sol. As argon is monoatomic, its molar specific heat at constant volume will be

$$C_V = \frac{3}{2}R = \frac{3}{2} \times 2 = 3 \text{ cal/mol K}, C_V = M_w c_v \text{ and } c_v = 0.075 \text{ cal/g K}$$

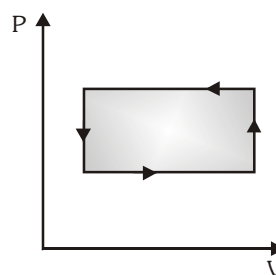
$$\text{So } 3 = M_w \times 0.075 \Rightarrow M_w = \frac{3}{0.075} = 40 \text{ gram/mole}$$

Cyclic process

Cyclic process is that thermodynamic process in which the system returns to its initial stage after undergoing a series of changes. A typical cyclic process is represented on PV diagram as shown in figure.

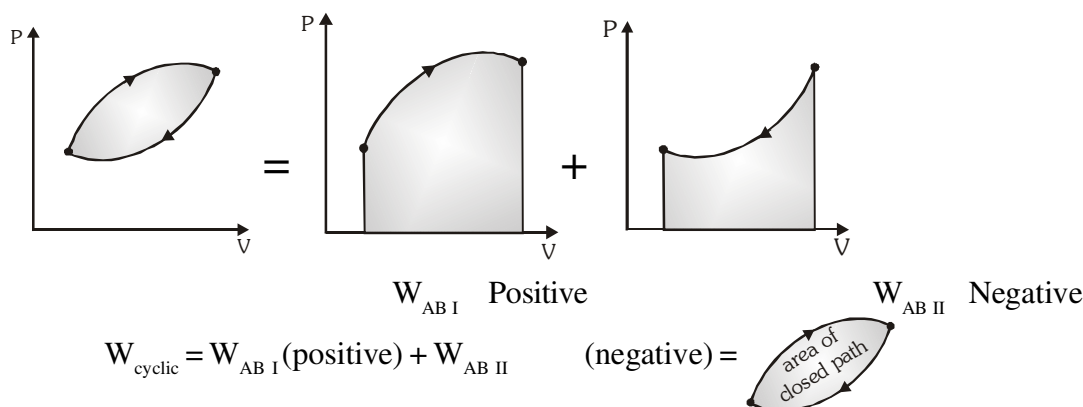


$$W_{\text{clockwise cycle}} = + \text{ Shaded area}$$



$$W_{\text{anticlockwise cycle}} = - \text{ Shaded area}$$

WORK DONE IN CLOCKWISE CYCLE

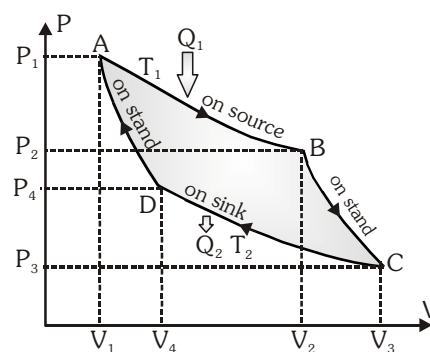


Similarly work done in anticlockwise cyclic process is negative.

8. CARNOT CYCLE

Carnot devised an ideal engine which is based on a reversible cycle of four operations in succession :

- (i) isothermal expansion, $A \rightarrow B$
- (ii) adiabatic expansion, $B \rightarrow C$
- (iii) isothermal compression $C \rightarrow D$
- (iv) adiabatic compression. $D \rightarrow A$



Main parts of Carnot's engine are**Source of heat**

It is a hot body of very large heat capacity kept at a constant high temperature T_1 .

Mechanical arrangements and working substance

It is a cylinder whose walls are perfectly non-conducting and its base is perfectly conducting fitted with non-conducting piston. This piston move without any friction. Ideal gas enclosed in cylinder as a working substance.

Heat sink

It is a cold body at low temperature T_2 . It is a body of large heat capacity.

Stand

It is of two types : conducting and nonconducting.

Working

Through a set of reversible processes the working substance is taken back to initial condition to get maximum work from this type of ideal engine.

Processes of Carnot's cycle can be denoted by an indicator diagram.

Isothermal expansion A $\rightarrow$ B

Initially the cylinder is taken to be in thermal equilibrium with the high temperature T_1 , this is initial state of working substance denoted by point A (P_1, V_1, T_1).

After that the piston is allowed to move outward slowly. With the movement of the piston the process is very slow so that it is isothermal.

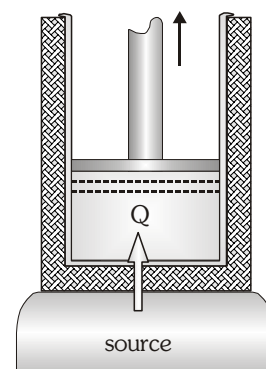
Heat from reservoir flows through the base of cylinder into the gas so temperature of the gas remains T_1 .

Gas expand and receive heat Q_1 from source and gets state B(P_2, V_2, T_1)

This heat input Q_1 to the gas from path A to B is utilized for doing work W_1 .

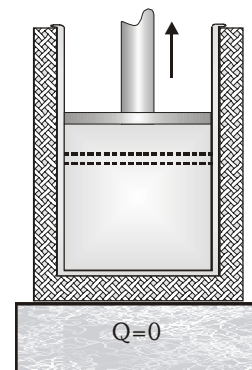
By path A to B the heat input to the gas = the work done against the external pressure.

$$W_1 = Q_1 = \int_{V_1}^{V_2} PdV = \int_{V_1}^{V_2} \frac{\mu RT_1}{V} dV = \mu RT_1 \ln \frac{V_2}{V_1}$$



Adiabatic expansion B → C

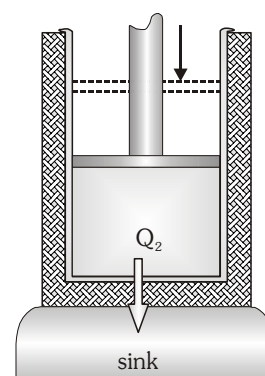
Now cylinder is put in contact with a non-conducting stand and piston is allowed to move outward, because no heat can enter in or leave out so the expansion of gas is adiabatic. The temperature falls to T_2 K and gas describes the adiabatic process from B to **point C** (P_3, V_3, T_2) during this expansion more work is done (W_2) at the expense of the internal energy.



Work done in adiabatic path BC is $W_2 = \frac{\mu R}{\gamma - 1} (T_1 - T_2)$

Isothermal compression C → D

Now the gas cylinder is placed in contact with sink at temperature T_2 . The piston is moved slowly inward so that heat produced during compression passes to the sink. The gas is isothermally compressed from C to **point D**. (P_4, V_4, T_2) The heat rejected Q_2 to the cold reservoir (sink) at T_2 occurs over this path.

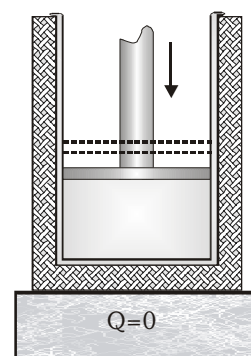


Amount of work done on gas W_3 = amount of heat rejected to the sink

$$Q_2 = W_3 = \mu RT_2 \ln \left(\frac{V_4}{V_3} \right) \Rightarrow Q_2 = \mu RT_2 \ln \left(\frac{V_4}{V_3} \right)$$

Adiabatic compression D → A

The cylinder is removed from the sink and is put in contact with insulating stand now piston moves inward. Heat is not allowed to go out and it increases the internal energy of the system. Now work is done on the gas during adiabatic compression from state D to initial point A (P_1, V_1, T_1). No heat exchanges occur over the adiabatic path.



Work done on the system $W_4 = \frac{\mu R}{\gamma - 1} (T_2 - T_1)$

This cycle of operations is called a Carnot cycle.

In first two steps work is done by engine W_1 and W_2 are positive

In last two steps work is done on gas W_3 and W_4 are negative

The work done in complete cycle W = the area of the closed part of the P-V cycle.

$$W = W_1 + W_2 + W_3 + W_4$$

$$\therefore W = \mu RT_1 \ln \frac{V_2}{V_1} + \frac{\mu R}{\gamma - 1} (T_1 - T_2) + \mu RT_2 \ln \frac{V_4}{V_3} + \frac{\mu R}{\gamma - 1} (T_2 - T_1) = \mu RT_1 \ln \frac{V_2}{V_1} + \mu RT_2 \ln \frac{V_4}{V_3}$$

$$\text{Efficiency of Carnot Engine, } \eta = \frac{W}{Q_1} = \frac{\mu RT_1 \ln \frac{V_2}{V_1} + \mu RT_2 \ln \frac{V_4}{V_3}}{\mu RT_1 \ln \frac{V_2}{V_1}}$$

B to C and D to A are adiabatic paths

$$\text{so } T_1 V_2^{(\gamma-1)} = T_2 V_3^{(\gamma-1)} \text{ and } T_1 V_1^{(\gamma-1)} = T_2 V_4^{(\gamma-1)} \Rightarrow \frac{V_2}{V_1} = \frac{V_3}{V_4}$$

$$\eta = \frac{T_1 - T_2}{T_1} = \frac{Q_1 - Q_2}{Q_1} \Rightarrow \eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1} \quad \frac{Q_1}{T_1} = \frac{Q_2}{T_2}$$

$$\eta = \frac{T_1 - T_2}{T_1} \times 100\% \Rightarrow \eta = \frac{Q_1 - Q_2}{Q_1} \times 100\%$$

The efficiency for the Carnot engine is the best that can be obtained for any heat engine.

The efficiency of a Carnot engine is never 100% because it is 100% only if temperature of sink $T_2 = 0\text{K}$ and $T_1 = \infty$ which is impossible.

CARNOT THEOREM

No irreversible engine (I) can have efficiency greater than Carnot reversible engine (R) working

$$\text{between same hot and cold reservoirs. } \eta_R > \eta_I \Rightarrow 1 - \frac{T_2}{T_1} > 1 - \frac{Q_2}{Q_1}$$

9. SECOND LAW OF THERMODYNAMICS

The first law of thermodynamics is a generalization of the law of conservation of energy to include heat energy. It tells us that heat and mechanical work are mutually interconvertible.

Second law of thermodynamics tells us in what conditions heat can be converted into useful work.

The following three conditions must be fulfilled to utilize heat for useful work :

- (i) A device called engine with a working substance is essential.
- (ii) The engine must work in a reversible cyclic process.
- (iii) The engine must operate between two temperatures. It will absorb heat from a hot body (called source), and converts a part of it into useful work and reject the rest to a cold body (called sink).

There are two conventional statements of second law :

Kelvin-Planck Statement

It is impossible for an engine working between a cyclic process to extract heat from a reservoir and convert completely into work. In other words, 100% conversion of heat into work is impossible.

Clausius Statement

It is impossible for a self-acting machine, unaided by any external agency to transfer heat from a cold to hot reservoir. In other words heat can not in itself flow from a colder to a hotter body.

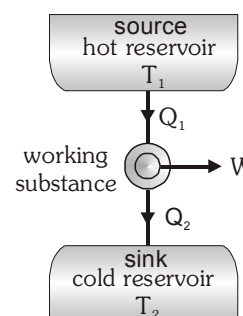
10. HEAT ENGINE

Heat engine is a device which converts heat into work.

Three parts of a heat engine:

- Source of high temperature reservoir at temperature T_1
- Sink or low temperature reservoir at temperature T_2
- Working substance.

In a cycle of heat engine the working substance extracts heat Q_1 from source, does some work W and rejects remaining heat Q_2 to the sink.



Efficiency of heat engine $\eta = \frac{\text{work done (W)}}{\text{heat taken from source (Q}_1\text{)}}$

$$\eta = \frac{T_1 - T_2}{T_1} = \frac{Q_1 - Q_2}{Q_1}$$

Ex. A carnot engine working between 400 K and 800 K has a work output of 1200 J per cycle. What is the amount of heat energy supplied to the engine from source per cycle?

Sol. $W = 1200 \text{ J}$, $T_1 = 800 \text{ K}$, $T_2 = 400 \text{ K}$

$$\therefore \eta = 1 - \frac{T_2}{T_1} = \frac{W}{Q_1} \Rightarrow 1 - \frac{400}{800} = \frac{1200}{Q_1} \Rightarrow 0.5 = \frac{1200}{Q_1}$$

Heat energy supplied by source $Q_1 = \frac{1200}{0.5} = 2400 \text{ joule per cycle}$

Ex. The temperatures T_1 and T_2 of the two heat reservoirs in an ideal carnot engine are 1500°C and 500°C respectively. Which of the following : increasing T_1 by 100°C or decreasing T_2 by 100°C would result in a greater improvement in the efficiency of the engine?

Sol. $T_1 = 1500^\circ\text{C} = 1500 + 273 = 1773 \text{ K}$ and $T_2 = 500^\circ\text{C} = 500 + 273 = 773 \text{ K}$.

The efficiency of a carnot's engine $\eta = 1 - \frac{T_2}{T_1}$

When the temperature of the source is increased by 100°C , keeping T_2 unchanged, the new temperature of the source is $T'_1 = 1500 + 100 = 1600^\circ\text{C} = 1873 \text{ K}$. The efficiency becomes

$$\eta' = 1 - \frac{T_2}{T'_1} = 1 - \frac{773}{1873} = 0.59$$

On the other hand, if the temperature of the sink is decreased by 100°C , keeping T_1 unchanged, the new temperature of the sink is $T'_2 = 500 - 100 = 400^\circ\text{C} = 673 \text{ K}$. The efficiency now becomes

$$\eta'' = 1 - \frac{T'_2}{T_1} = 1 - \frac{673}{1773} = 0.62$$

Since η'' is greater than η' , decreasing the temperature of the sink by 100°C results in a greater efficiency than increasing the temperature of the source by 100°C .

Ex. A heat engine operates between a cold reservoir at temperature $T_2 = 300$ K and a hot reservoir at temperature T_1 . It takes 200 J of heat from the hot reservoir and delivers 120 J of heat to the cold reservoir in a cycle. What could be the minimum temperature of hot reservoir?

Sol. Work done by the engine in a cycle is $W = 200 - 120 = 80$ J. $\eta = \frac{W}{Q} = \frac{80}{200} = 0.4$

From Carnot's Theorem $0.4 \leq 1 - \frac{T_2}{T_1} = 1 - \frac{300}{T_1}$ or $\frac{300}{T_1} \leq 0.6$ or $T_1 \geq \frac{300}{0.6}$ or $T_1 \geq 500$

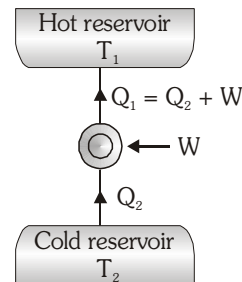
11. REFRIGERATOR

It is inverse of heat engine. It extracts heat (Q_2) from a cold reservoir, External work W is done on it and rejects heat (Q_1) to hot reservoir. The coefficient of performance of a refrigerator.

$$\beta = \frac{\text{heat extracted from cold reservoir}}{\text{work done on refrigerator}} = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2} = \frac{1}{\frac{Q_1}{Q_2} - 1}$$

For Carnot reversible refrigerator $\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$

$$\therefore \beta = \frac{Q_2}{W} = \frac{1}{\left[\frac{Q_1}{Q_2} - 1\right]} = \frac{1}{\left[\frac{T_1}{T_2} - 1\right]} \Rightarrow \beta = \frac{T_2}{T_1 - T_2}$$



Ex. A Carnot engine works as a refrigerator between 250 K and 300 K. If it receives 750 cal of heat from the reservoir at the lower temperature. Calculate the amount of heat rejected at the higher temperature.

Sol. $T_1 = 300$ K $T_2 = 250$ K $Q_2 = 750$ $Q_1 = ?$

$$\frac{Q_1}{Q_2} = \frac{T_1}{T_2} \quad Q_1 = \frac{300}{250} \times 750 = 900 \text{ cal}$$

Ex. The temperature inside & outside of refrigerator are 260 K and 315 K respectively. Assuming that the refrigerator cycle is reversible, calculate the heat delivered to surroundings for every joule of work done.

Sol. $T_2 = 260$ K, $T_1 = 315$ K, $W = 1$ joule

Coefficient of performance of Carnot refrigerator $\beta = \frac{Q_2}{W} = \frac{T_2}{T_1 - T_2}$

$$\therefore \frac{Q_2}{1} = \frac{260}{315 - 260} = \frac{260}{55} \Rightarrow Q_2 = \frac{260}{55} = 4.7 \text{ J}$$

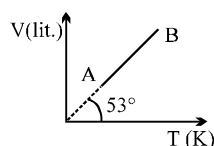
Ex. A refrigerator takes heat from water at 0°C and transfer it to room at 27°C . If 100 kg of water is converted in ice at 0°C then calculate the workdone. (Latent heat of ice is 3.4×10^5 J/kg)

Sol. Coefficient of performance (COP) $= \frac{T_2}{T_1 - T_2} = \frac{273}{300 - 273} = \frac{273}{27}$

$$W = \frac{Q_2}{\text{COP}} = \frac{mL}{\text{COP}} = \frac{100 \times 3.4 \times 10^5}{273/27} = \frac{100 \times 3.4 \times 10^5 \times 27}{273} = 3.36 \times 10^6 \text{ J}$$

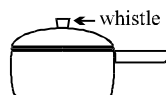
EXERCISE (S-1)

1. V-T curve for 2 moles of a gas is straight line as shown in the graph here. Find the pressure of gas at A.



KT0001

2. An empty pressure cooker of volume 10 litres contains air at atmospheric pressure 10^5 Pa and temperature of 27°C . It contains a whistle which has area of 0.1 cm^2 and weight of 100 g. What should be the temperature of air inside so that the whistle is just lifted up?



KT0002

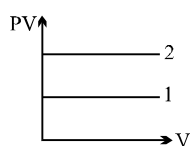
3. Examine the following plots and predict whether in

(i) $P_1 < P_2$ and $T_1 > T_2$, in

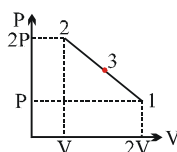
(ii) $T_1 = T_2 < T_3$, in

(iii) $V_1 > V_2$, in

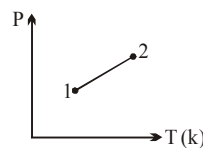
(iv) $P_1 > P_2$ or otherwise.



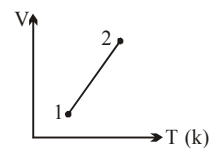
(i)



(ii)



(iii)



(iv)

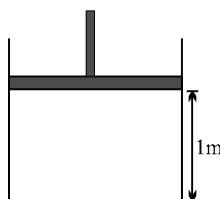
KT0003

4. Find the molecular mass of a gas if the specific heats of the gas are $C_p = 0.2\text{ cal/gm}^\circ\text{C}$ and $C_v = 0.15\text{ cal/gm}^\circ\text{C}$. [Take $R = 2\text{ cal/mole}^\circ\text{C}$]

KT0004

5. The piston cylinder arrangement shown contains a diatomic gas at temperature 300 K. The cross-sectional area of the cylinder is 1 m^2 . Initially the height of the piston above the base of the cylinder is 1 m. The temperature is now raised to 400 K at constant pressure. Find the new height of the piston above the base of the cylinder. If the piston is now brought back to its original height without any heat loss, find the new equilibrium temperature of the gas. You can leave the answer in fraction.

[JEE' 2004]



KT0005

6. A mixture of 4 gm helium and 28 gm of nitrogen is enclosed in a vessel of constant volume 300 K. Find the quantity of heat absorbed by the mixture to doubled the root mean velocity of its molecules. (R = Universal gas constant)

KT0006

7. An insulated container containing monoatomic gas of molar mass m is moving with a velocity v_0 . If the container is suddenly stopped, find the change in temperature. [JEE 2003]

KT0007

8. If heat is added at constant volume, 6300 J of heat are required to raise the temperature of an ideal gas by 150 K. If instead, heat is added at constant pressure, 8800 joules are required for the same temperature change. When the temperature of the gas changes by 300 K, determine the change in the internal energy of the gas.

KT0008

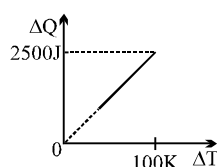
9. Ideal diatomic gas is taken through a process $\Delta Q = 2\Delta U$. Find the molar heat capacity for the process (where ΔQ is the heat supplied and ΔU is change in internal energy)

KT0009

10. A cylinder with a movable piston contains 3 moles of hydrogen at standard temperature and pressure. The walls of the cylinder are made of a heat insulator, and the piston is insulated by having a pile of sand on it. By what factor does the pressure of the gas increase if the gas is compressed to half its original volume ?

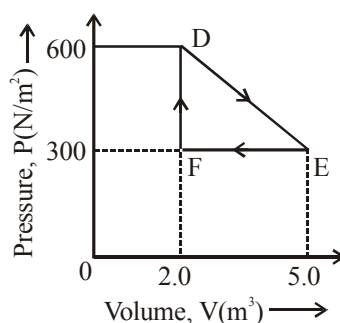
KT0010

11. One mole of a gas mixture is heated under constant pressure, and heat required ΔQ is plotted against temperature difference acquired. Find the value of γ for mixture.



KT0011

12. A thermodynamic system is taken from an original state to an intermediate state by the linear process shown in figure. Its volume is then reduced to the original value from E to F by an isobaric process. Calculate the total work done by the gas from D to E to F

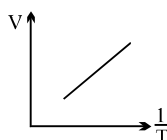


KT0012

13. In changing the state of a gas adiabatically from an equilibrium state A to another equilibrium state B, an amount of work equal to 22.3 J is done on the system. If the gas is taken from state A to B via a process in which the net heat absorbed by the system is 9.35 cal, how much is the net work done by the system in the latter case ? (Take 1 cal = 4.19 J)

KT0013

14. One mole of an ideal monoatomic gas undergoes a process as shown in the figure. Find the molar specific heat of the gas in the process.

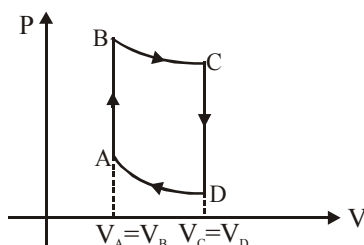


KT0014

15. Two cylinders A and B of equal capacity are connected to each other via a stopcock. A contains a gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stopcock is suddenly opened. Answer the following :
- What is the final pressure of the gas in A and B ?
 - What is the change in internal energy of the gas ?
 - What is the change in the temperature of the gas ?
 - Do the intermediate states of the system (before settling to the final equilibrium state) lie on its P-V-T surface ?

KT0015

16. A cycle followed by an engine (made of one mole of perfect gas in a cylinder with a piston) is shown in figure.



A to B : volume constant

B to C : adiabatic

C to D : volume constant

D to A : adiabatic

$$V_C = V_D = 2V_A = 2V_B$$

- In which part of the cycle heat is supplied to the engine from outside?
- In which part of the cycle heat is being given to the surrounding by the engine?
- What is the work done by the engine in one cycle? Write your answer in term of P_A , P_B , V_A
- What is the efficiency of the engine?

$$[\gamma = \frac{5}{3} \text{ for the gas}], (C_V = \frac{3}{2}R \text{ for one mole})$$

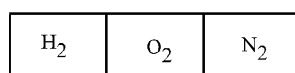
KT0016

EXERCISE (S-2)

1. A barometer is faulty. When the true barometer reading are 73 and 75 cm of Hg, the faulty barometer reads 69 cm and 70 cm respectively.
- What is the total length of the barometer tube?
 - What is the true reading when the faulty barometer reads 69.5 cm ?
 - What is the faulty barometer reading when the true barometer reads 74 cm?

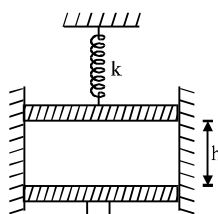
KT0017

2. A vessel of volume $V = 30\text{ l}$ is separated into three equal parts by stationary semipermeable thin membranes as shown in the Figure. The left, middle and right parts are filled with $m_{\text{H}_2} = 30\text{ g}$ of hydrogen, $m_{\text{O}_2} = 160\text{ g}$ of oxygen, and $m_{\text{N}_2} = 70\text{ g}$ of nitrogen respectively. The left partition lets through only hydrogen, while the right partition lets through hydrogen and nitrogen. What will be the pressure in each part of the vessel after the equilibrium has been set in if the vessel is kept at a constant temperature $T = 300\text{ K}$?



KT0018

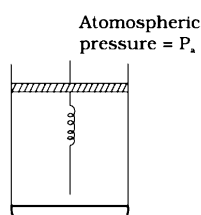
3. An ideal gas at NTP is enclosed in a adiabatic vertical cylinder having area of cross section $A = 27\text{ cm}^2$, between two light movable pistons as shown in the figure. Spring with force constant $k = 3700\text{ N/m}$ is in a relaxed state initially. Now the lower piston is moved upwards a height $h/2$, h being the initial length of gas column. It is observed that the upper piston moves up by a distance $h/16$. Find h taking γ for the gas to be 1.5. Also find the final temperature of the gas.



KT0019

4. Consider one mole of perfect gas in a cylinder of unit cross section with a piston attached (Fig.). A spring (spring constant k) is attached (unstretched length L) to the piston and to the bottom of the cylinder. Initially the spring is unstretched and the gas is in equilibrium. A certain amount of heat Q is supplied to the gas causing an increase of volume from V_0 to V_1 .

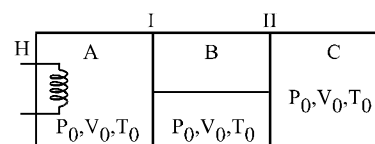
- What is the initial pressure of the system?
- What is the final pressure of the system?
- Using the first law of thermodynamics, write down a relation between Q , P_a , V , V_0 and k .



KT0020

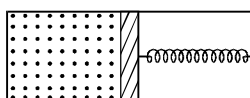
5. The figure shows an insulated cylinder divided into three parts A, B and C. Pistons I and II are connected by a rigid rod and can move without friction inside the cylinder. Piston I is perfectly conducting while piston II is perfectly insulating. The initial state of the gas ($\gamma = 1.5$) present in each compartment A, B and C is as shown. Now, compartment A is slowly given heat through a heater H such that the final volume of C becomes $\frac{4V_0}{9}$. Assume the gas to be ideal and find.

- Final pressures in each compartment A, B and C
- Final temperatures in each compartment A, B and C
- Heat supplied by the heater
- Work done by gas in A and B.
- Heat flowing across piston I.



KT0021

6. A thermally insulated vessel is divided into two equal parts by a heat-insulating piston which can move in the vessel without the friction. The left part of the vessel contains one mole of an ideal monatomic gas, & the right part is empty. The piston is connected to the right wall of the vessel through a spring whose length in free state is equal to the length of the vessel as shown in the figure. Determine the heat capacity C of the system, neglecting the heat capacities of the vessel, piston and spring.



KT0022

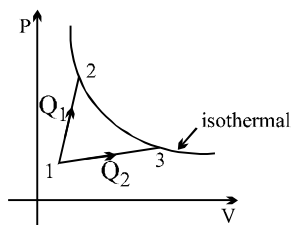
7. A weightless piston divides a thermally insulated cylinder into two parts of volumes V and $3V$. 2 moles of an ideal gas at pressure $P = 2$ atmosphere are confined to the part with volume $V = 1$ litre. The remainder of the cylinder is evacuated. The piston is now released and the gas expands to fill the entire space of the cylinder. The piston is then pressed back to the initial position. Find the increase of internal energy in the process and final temperature of the gas. The ratio of the specific heat of the gas $\gamma = 1.5$.

KT0023

8. Two moles of an ideal monoatomic gas are confined within a cylinder by a massless & frictionless spring loaded piston of cross-sectional area $4 \times 10^{-3} \text{ m}^2$. The spring is, initially in its relaxed state. Now the gas is heated by an electric heater, placed inside the cylinder, for some time. During this time, the gas expands and does 50 J of work in moving the piston through a distance 0.10 m. The temperature of the gas increases by 50 K. Calculate the spring constant & the heat supplied by the heater. [$P_0 = 10^5 \text{ Pa}$]

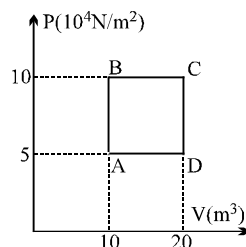
KT0024

9. A gas takes part in two processes in which it is heated from the same initial state 1 to the same final temperature. The processes are shown on the P-V diagram by the straight line 1-2 and 1-3. 2 and 3 are the points on the same isothermal curve. Q_1 and Q_2 are the heat transfer along the two processes. Then in which case will the heat transfer be more.



KT0025

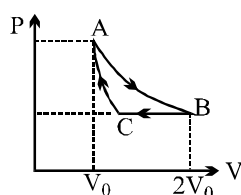
10. A sample of 2 kg of monoatomic Helium (assumed ideal) is taken through the process ABC and another sample of 2 kg of the same gas is taken through the process ADC as in figure. Given, molecular mass of Helium = 4



- What is the temperature of Helium in each of the states A, B, C & D ?
- Is there any way of telling afterwards which sample of Helium went through the process ABC and which went through the process ADC? Write Yes or No.
- How much is the heat involved in each of the processes ABC & ADC.

KT0026

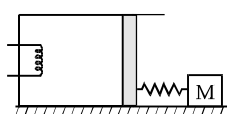
11. In a cyclic process ABCA consisting of isothermal expansion AB, isobaric compression BC and adiabatic compression CA, find the efficiency of cycle (Given : $T_A = T_B = 400$ K, $\gamma = 1.5$)



KT0027

12. An adiabatic cylinder has 8 gram of helium. A light smooth adiabatic piston is connected to a light spring of force constant 300 N/m. The other end of the spring is connected with a block of mass 1 kg kept on a rough horizontal surface of coefficient of friction $\mu = 0.3$. Area of cross-section of cylinder is $A = 25 \text{ cm}^2$. Initially the spring is in a relaxed position and the temperature of the gas is 400 K. The gas is heated slowly for some time by means of an electric heater so as to bring the block M on the verge of motion. Take $P_{\text{atm}} = 10^5 \text{ N/m}^2$. Find

- the work done by the gas
- the final temperature
- heat supplied by the heater



KT0028

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

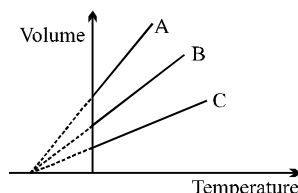
KTG

- Find the approx. number of molecules contained in a vessel of volume 7 litres at 0°C at 1.3×10^5 pascal
(A) 2.4×10^{23} (B) 3×10^{23} (C) 6×10^{23} (D) 4.8×10^{23} **KT0029**
- An ideal gas mixture filled inside a balloon expands according to the relation $PV^{2/3} = \text{constant}$. The temperature inside the balloon is
(A) increasing (B) decreasing (C) constant (D) can't be said **KT0030**
- At a temperature T K, the pressure of 4.0g argon in a bulb is p . The bulb is put in a bath having temperature higher by 50K than the first one. 0.8g of argon gas had to be removed to maintained original pressure. The temperature T is equal to
(A) 510 K (B) 200 K (C) 100 K (D) 73 K **KT0031**
- 28 gm of N_2 gas is contained in a flask at a pressure of 10 atm and at a temperature of 57° . It is found that due to leakage in the flask, the pressure is reduced to half and the temperature reduced to 27°C . The quantity of N_2 gas that leaked out is :-
(A) 11/20 gm (B) 20/11 gm (C) 5/63 gm (D) 63/5 gm **KT0032**
- A container X has volume double that of container Y and both are connected by a thin tube. Both contains same ideal gas. The temperature of X is 200K and that of Y is 400K. If mass of gas in X is m then in Y it will be:
(A) $m/8$ (B) $m/6$ (C) $m/4$ (D) $m/2$ **KT0033**
- When 2 gms of a gas are introduced into an evacuated flask kept at 25°C the pressure is found to be one atmosphere. If 3 gms of another gas added to the same flask the pressure becomes 1.5 atmospheres. The ratio of the molecular weights of these gases will be :-
(A) 1 : 3 (B) 3 : 1 (C) 2 : 3 (D) 3 : 2 **KT0034**
- A rigid tank contains 35 kg of nitrogen at 6 atm. Sufficient quantity of oxygen is supplied to increase the pressure to 9 atm, while the temperature remains constant. Amount of oxygen supplied to the tank is :
(A) 5 kg (B) 10 kg (C) 20 kg (D) 40 kg **KT0035**
- During an experiment an ideal gas obeys an addition equation of state $P^2V = \text{constant}$. The initial temperature and pressure of gas are T and V respectively. When it expands to volume $2V$, then its temperature will be:
(A) T (B) $\sqrt{2} T$ (C) $2 T$ (D) $2\sqrt{2} T$ **KT0036**

9. A vessel contains 1 mole of O_2 gas (molar mass 32) at a temperature T . The pressure of the gas is P . An identical vessel containing one mole of He gas (molar mass 4) at a temperature $2T$ has a pressure of
 (A) $P/8$ (B) P (C) $2P$ (D) $8P$

KT0037

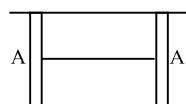
10. The expansion of an ideal gas of mass m at a constant pressure P is given by the straight line B. Then the expansion of the same ideal gas of mass $2m$ at a pressure $2P$ is given by the straight line



- (A) C (B) A (C) B (D) none

KT0038

11. A cylindrical tube of cross-sectional area A has two air tight frictionless pistons at its two ends. The pistons are tied with a straight piece of metallic wire. The tube contains a gas at atmospheric pressure P_0 and temperature T_0 . If temperature of the gas is doubled then the tension in the wire is



- (A) $4 P_0 A$ (B) $P_0 A/2$ (C) $P_0 A$ (D) $2 P_0 A$

KT0039

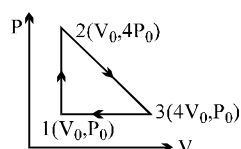
12. One mole of an ideal diatomic gas is taken through the cycle as shown in the figure.

$1 \rightarrow 2$: isochoric process

$2 \rightarrow 3$: straight line on P-V diagram

$3 \rightarrow 1$: isobaric process

The average molecular speed of the gas in the states 1, 2 and 3 are in the ratio



- (A) $1 : 2 : 2$ (B) $1 : \sqrt{2} : \sqrt{2}$ (C) $1 : 1 : 1$ (D) $1 : 2 : 4$

KT0040

13. One mole of an ideal gas at STP is heated in an insulated closed container until the average speed of its molecules is doubled. Its pressure would therefore increase by factor.

- (A) 1.5 (B) $\sqrt{2}$ (C) 2 (D) 4

KT0041

14. Three particles have speeds of $2u$, $10u$ and $11u$. Which of the following statements is correct?

- (A) The r.m.s. speed exceeds the mean speed by about u .
 (B) The mean speed exceeds the r.m.s. speed by about u .
 (C) The r.m.s. speed equals the mean speed.
 (D) The r.m.s. speed exceeds the mean speed by more than $2u$.

KT0042

15. One mole of an ideal gas is contained with in a cylinder by a frictionless piston and is initially at temperature T . The pressure of the gas is kept constant while it is heated and its volume doubles. If R is molar gas constant, the work done by the gas in increasing its volume is :-

(A) $RT \ln 2$ (B) $1/2 RT$ (C) RT (D) $3/2 RT$

KT0043

16. A gas mixture consists of 2 moles of oxygen and 4 moles of argon at temperature T . Neglecting all vibrational modes, the total internal energy of the system is :-

(A) $4 RT$ (B) $15 RT$ (C) $9 RT$ (D) $11 RT$

KT0044

First Law of Thermodynamics :

17. In thermodynamic process pressure of a fixed mass of gas is changed in such a manner that the gas releases 30 joule of heat and 18 joule of work was done on the gas. If the initial internal energy of the gas was 60 joule, then, the final internal energy will be :

(A) 32 joule (B) 48 joule (C) 72 joule (D) 96 joule

KT0045

18. Two monoatomic ideal gas at temperature T_1 and T_2 are mixed. There is no loss of energy. If the masses of molecules of the two gases are m_1 and m_2 and number of their molecules are n_1 and n_2 respectively. The temperature of the mixture will be :

(A) $\frac{T_1 + T_2}{n_1 + n_2}$ (B) $\frac{T_1}{n_1} + \frac{T_2}{n_2}$ (C) $\frac{n_2 T_1 + n_1 T_2}{n_1 + n_2}$ (D) $\frac{n_1 T_1 + n_2 T_2}{n_1 + n_2}$

KT0046

19. An ideal gas expands isothermally from a volume V_1 to V_2 and then compressed to original volume V_1 adiabatically. Initial pressure is P_1 and final pressure is P_3 . The total work done is W . Then

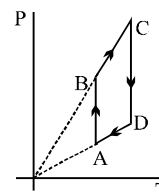
[JEE' 2004 (Scr)]

(A) $P_3 > P_1, W > 0$ (B) $P_3 < P_1, W < 0$
(C) $P_3 > P_1, W < 0$ (D) $P_3 = P_1, W = 0$

KT0047

20. Pressure versus temperature graph of an ideal gas is shown in figure

(A) During the process AB work done by the gas is positive
(B) During the process CD work done by the gas is negative
(C) During the process BC internal energy of the gas is increasing
(D) None



KT0048

21. A polyatomic gas with six degrees of freedom does 25J of work when it is expanded at constant pressure. The heat given to the gas is :-

(A) 100J (B) 150J (C) 200J (D) 250J

KT0049

22. A reversible adiabatic path on a P-V diagram for an ideal gas passes through state A where $P = 0.7 \times 10^5 \text{ N/m}^2$ and $V = 0.0049 \text{ m}^3$. The ratio of specific heat of the gas is 1.4. The slope of path at A is :

(A) $2.0 \times 10^7 \text{ Nm}^{-5}$ (B) $1.0 \times 10^7 \text{ Nm}^{-5}$ (C) $-2.0 \times 10^7 \text{ Nm}^{-5}$ (D) $-1.0 \times 10^7 \text{ Nm}^{-5}$

KT0050

23. The adiabatic Bulk modulus of a diatomic gas at atmospheric pressure is

(A) 0 Nm^{-2} (B) 1 Nm^{-2} (C) $1.4 \times 10^4 \text{ Nm}^{-2}$ (D) $1.4 \times 10^5 \text{ Nm}^{-2}$

KT0051

24. A given quantity of an ideal gas is at pressure P and absolute temperature T. The isothermal bulk modulus of the gas is :

(A) $2P/3$ (B) P (C) $3P/2$ (D) $2P$

KT0052

25. One mole of an ideal gas at temperature T_1 expands according to the law $\frac{P}{V^2} = a$ (constant). The work done by the gas till temperature of gas becomes T_2 is :

(A) $\frac{1}{2} R(T_2 - T_1)$ (B) $\frac{1}{3} R(T_2 - T_1)$ (C) $\frac{1}{4} R(T_2 - T_1)$ (D) $\frac{1}{5} R(T_2 - T_1)$

KT0053

26. The first law of thermodynamics can be written as $\Delta U = \Delta Q + \Delta W$ for an ideal gas. Which of the following statements is correct?

(A) ΔU is always zero when no heat enters or leaves the gas
 (B) ΔW is the work done by the gas in this written law.
 (C) ΔU is zero when heat is supplied and the temperature stays constant
 (D) $\Delta Q = -\Delta W$ when the temperature increases very slowly.

KT0054

27. 2 moles of a monoatomic gas are expanded to double its initial volume, through a process $P/V = \text{constant}$. If its initial temperature is 300 K, then which of the following is not true.

(A) $\Delta T = 900 \text{ K}$ (B) $\Delta Q = 3200 \text{ R}$ (C) $\Delta Q = 3600 \text{ R}$ (D) $W = 900 \text{ R}$

KT0055

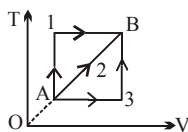
28. A student records ΔQ , ΔU & ΔW for a thermodynamic cycle $A \rightarrow B \rightarrow C \rightarrow A$. Certain entries are missing. Find correct entry in following options.

	AB	BC	CA
ΔW	40J		30J
ΔU		50J	
ΔQ	150J	10J	

(A) $W_{BC} = -70 \text{ J}$ (B) $\Delta Q_{CA} = 130 \text{ J}$ (C) $\Delta U_{AB} = 190 \text{ J}$ (D) $\Delta U_{CA} = -160 \text{ J}$

KT0056

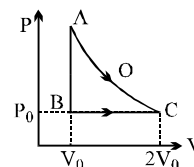
29. A given mass of a gas expands from a state A to the state B by three paths 1, 2 and 3 as shown in T-V indicator diagram. If W_1 , W_2 and W_3 respectively be the work done by the gas along the three paths, then



- (A) $W_1 > W_2 > W_3$ (B) $W_1 < W_2 < W_3$ (C) $W_1 = W_2 = W_3$ (D) $W_1 < W_2$, $W_1 > W_3$

KT0057

30. An ideal gas is taken from point A to point C on P-V diagram through two process AOC and ABC as shown in the figure. Process AOC is isothermal



- (A) Process AOC requires more heat than process ABC.
(B) Process ABC requires more heat than process AOC.
(C) Both process AOC & ABC require same amount of heat.
(D) Data is insufficient for comparison of heat requirement for the two processes.

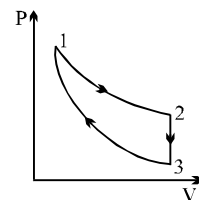
KT0058

31. Monoatomic, diatomic and triatomic gases whose initial volume and pressure are same, are compressed till their volume becomes half the initial volume.

- (A) If the compression is adiabatic then monoatomic gas will have maximum final pressure.
(B) If the compression is adiabatic then triatomic gas will have maximum final pressure.
(C) If the compression is adiabatic then their final pressure will be same.
(D) If the compression is isothermal then their final pressure will be different.

KT0059

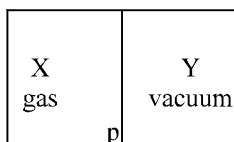
32. Three processes form a thermodynamic cycle as shown on P-V diagram for an ideal gas. Process $1 \rightarrow 2$ takes place at constant temperature (300K). Process $2 \rightarrow 3$ takes place at constant volume. During this process 40J of heat leaves the system. Process $3 \rightarrow 1$ is adiabatic and temperature T_3 is 275K. Work done by the gas during the process $3 \rightarrow 1$ is :-



- (A) - 40J (B) - 20J (C) +40J (D) +20J

KT0060

33. A closed container is fully insulated from outside. One half of it is filled with an ideal gas X separated by a plate P from the other half Y which contains a vacuum as shown in figure. When P is removed, X moves into Y. Which of the following statements is correct?



- (A) No work is done by X (B) X decreases in temperature
(C) X increases in internal energy (D) X doubles in pressure

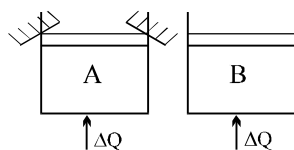
KT0061

34. Water is heated in an open pan where the air pressure is 10^5 Pa. The water remains a liquid, which expands by a small amount as it is heated. Determine the ratio of the heat absorbed by the water to the work done by water. (γ for water = $10^{-3}/^\circ\text{C}$, $S = 1 \text{ cal/gm}^\circ\text{C}$)

(A) 4.2×10^3 (B) 4.2×10^5 (C) 4.2×10^2 (D) 4.2×10^4

KT0062

35. Two identical vessels A & B contain equal amount of ideal monoatomic gas. The piston of A is fixed but that of B is free. Same amount of heat is absorbed by A & B. If B's internal energy increases by 100 J the change in internal energy of A is :-



(A) 100 J (B) $\frac{500}{3}$ J (C) 250 J (D) none of these

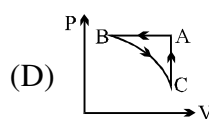
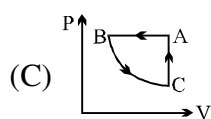
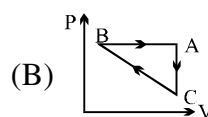
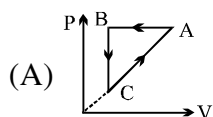
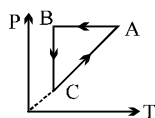
KT0063

36. An ideal gas expands from volume V_1 to V_2 . This may be achieved by either of the three processes: isobaric, isothermal and adiabatic. Let ΔU be the change in internal energy of the gas, Q be the quantity of heat added to the system and W be the work done by the gas. Identify which of the following statements is false for ΔU ?

- (A) ΔU is least under adiabatic process.
 (B) ΔU is greatest under adiabatic process.
 (C) ΔU is greatest under the isobaric process.
 (D) ΔU in isothermal process lies in-between the values obtained under isobaric and adiabatic processes.

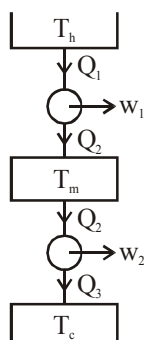
KT0064

37. A cyclic process ABCA is shown in PT diagram. When presented on PV, it would



KT0065

38. Suppose that two heat engines are connected in series, such that the heat exhaust of the first engine is used as the heat input of the second engine as shown in figure. The efficiencies of the engines are η_1 and η_2 , respectively. The net efficiency of the combination is given by



- (A) $\eta_{\text{net}} = \eta_2 + (1 - \eta_1)\eta_2$ (B) $\eta_{\text{net}} = \frac{\eta_1}{(1 - \eta_1)\eta_2}$
 (C) $\eta_{\text{net}} = \eta_1 + (1 - \eta_1)\eta_2$ (D) $\eta_{\text{net}} = \frac{1 - \eta_1}{(1 - \eta_2)\eta_2}$

KT0066

39. During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The ratio C_P/C_V for the gas is- [AIEEE-2003]
 (A) 4/3 (B) 2 (C) 5/3 (D) 3/2

KT0067

40. "Heat cannot be itself flow from a body at lower temperature to a body at higher temperature" is a statement or consequence of- [AIEEE - 2003]
 (A) second law of thermodynamics (B) conservation of momentum
 (C) conservation of mass (D) first law of thermodynamics

KT0068

41. A carnot engine takes 3×10^6 cal of heat from a reservoir at 627°C and gives it to a sink at 27°C . The work done by the engine is- [AIEEE - 2003]
 (A) 4.2×10^6 J (B) 8.4×10^6 J (C) 16.8×10^6 J (D) zero

KT0069

MULTIPLE CORRECT TYPE QUESTIONS

42. Two gases have the same initial pressure, volume and temperature. They expand to the same final volume, one adiabatically and the other isothermally
 (A) The final temperature is greater for the isothermal process
 (B) The final pressure is greater for the isothermal process
 (C) The work done by the gas is greater for the isothermal process
 (D) All the above options are incorrect
43. On the P-T graph of an ideal gas,
 (A) adiabatic process will be a straight line
 (B) isochoric process will be a straight line passing through the origin
 (C) adiabatic curve will have a positive slope
 (D) the slope of adiabatic curve will decrease with increase in T

KT0070

KT0071

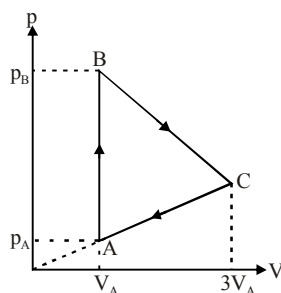
44. During the melting of a slab of ice at 273 K at atmospheric pressure
 (A) Positive work is done by the ice-water system on the atmosphere
 (B) Positive work is done on the ice-water system by the atmosphere
 (C) The internal energy of the ice-water increases
 (D) The internal energy of ice-water system decreases

KT0072

COMPREHENSION TYPE QUESTIONS

Paragraph for question nos. 45 and 46

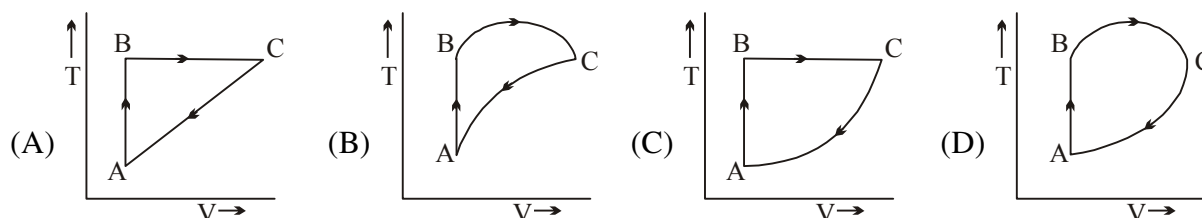
A sample of ideal gas is taken through the cyclic process shown in the figure. The temperature of the gas at state A is $T_A = 200$ K. At states B and C the temperature of the gas is the same.



45. Net work done by the gas in the process is
 (A) $2P_A V_A$ (B) $4P_A V_A$ (C) $6P_A V_A$ (D) $8P_A V_A$

KT0073

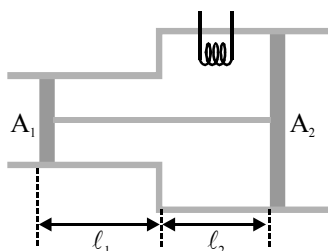
46. Which of the following graphs best represent the cyclic process in T-V diagram.



KT0073

Paragraph for Question Nos. 47 to 49

The walls of the two connecting cylinders shown in the figure are adiabatic (thermally insulating). The cross-sectional areas of the parts are A_1 and A_2 . There is a well-fitting but freely moving, thermally insulating piston in each cylinder, at a distance $\ell_1 = \ell_2 = \ell$ from the point where the cross-sectional area changes. The pistons are connected to each other by a thin and rigid rod. The enclosed volume contains air. The temperature and air pressure are T_0 and p_0 both inside and outside. The heater filament inside is operated for time t at a power of P .



47. Mark the **CORRECT** statement :-

- (A) Process performed is adiabatic (B) Piston shifts towards left
(C) Process performed is polytropic (D) Piston shifts towards right

KT0074

48. What is change in temperature (Q is heat energy supplied, v_0 is initial volume) :-

- (A) $\Delta T = \frac{2}{5} \frac{QT_0}{p_0 V_0}$ (B) $\Delta T = \frac{2}{7} \frac{QT_0}{p_0 V_0}$
(C) $\Delta T = \frac{5}{7} \frac{QT_0}{p_0 V_0}$ (D) $\Delta T = \frac{QT_0}{p_0 V_0}$

KT0074

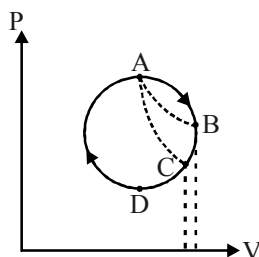
49. What is displacement (x) of piston till equilibrium is reached :-

- (A) $x = \frac{2Pt}{(A_2 - A_1)7p_0}$ (B) $x = \frac{2Pt}{(A_2 - A_1)5p_0}$
(C) $x = \frac{2Pt}{(A_2 - A_1)3p_0}$ (D) $x = \frac{2Pt}{(A_2 - A_1)9p_0}$

KT0074

MATRIX MATCH TYPE QUESTIONS

50. For an ideal gas a process PV diagram is a circle. An adiabat from A passes through C. An isotherm from A passes through B. We take a part of the circular cyclic process. Comment on the sign of the quantity of column-I.



Column-I

- (A) Heat given to the gas in going from A to C along circle
(B) Heat given to the gas in going from B to C along circle
(C) Heat given to the gas in going from C to D along circle

Column-II

- (P) Positive
(Q) Negative
(R) Zero
(S) can't be said

KT0075

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

1. An open and wide glass tube is immersed vertically in mercury in such a way that length 0.05 m extends above mercury level. The open end of the tube is closed and the tube is raised further by 0.43m. The length of air column above mercury level in the tube will be : Take $P_{\text{atm}} = 76$ cm of mercury
 (A) 0.215 m (B) 0.2 m (C) 0.1 m (D) 0.4 m

KT0076

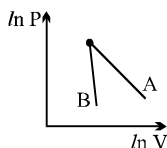
2. A barometer tube, containing mercury, is lowered in a vessel containing mercury until only 50 cm of the tube is above the level of mercury in the vessel. If the atmospheric pressure is 75 cm of mercury, what is the pressure at the top of the tube ?
 (A) 33.3 kPa (B) 66.7 kPa (C) 3.33 MPa (D) 6.67 MPa

KT0077

3. At temperature T , N molecules of gas A each having mass m and at the same temperature $2N$ molecules of gas B each having mass $2m$ are filled in a container. The mean square velocity of molecules of gas B is v^2 and mean square of x component of velocity of molecules of gas A is w^2 . The ratio of w^2/v^2 is :
 (A) 1 (B) 2 (C) 1/3 (D) 2/3

KT0078

4. The figure, shows the graph of logarithmic reading of pressure and volume for two ideal gases A and B undergoing adiabatic process. From figure it can be concluded that



- (A) gas B is diatomic (B) gas A and B both are diatomic
 (C) gas A is monoatomic (D) gas B is monoatomic & gas A is diatomic

KT0079

5. When unit mass of water boils to become steam at 100°C , it absorbs Q amount of heat. The densities of water and steam at 100°C are ρ_1 and ρ_2 respectively and the atmospheric pressure is p_0 . The increase in internal energy of the water is

- (A) Q (B) $Q + p_0 \left(\frac{1}{\rho_1} - \frac{1}{\rho_2} \right)$ (C) $Q + p_0 \left(\frac{1}{\rho_2} - \frac{1}{\rho_1} \right)$ (D) $Q - p_0 \left(\frac{1}{\rho_1} + \frac{1}{\rho_2} \right)$

KT0080

6. A cylinder made of perfectly non conducting material closed at both ends is divided into two equal parts by a heat proof piston. Both parts of the cylinder contain the same masses of a gas at a temperature $t_0 = 27^\circ\text{C}$ and pressure $P_0 = 1$ atm. Now if the gas in one of the parts is slowly heated to $t = 57^\circ\text{C}$ while the temperature of first part is maintained at t_0 the distance moved by the piston from the middle of the cylinder will be (length of the cylinder = 84 cm)

- (A) 3 cm (B) 5 cm (C) 2 cm (D) 1 cm

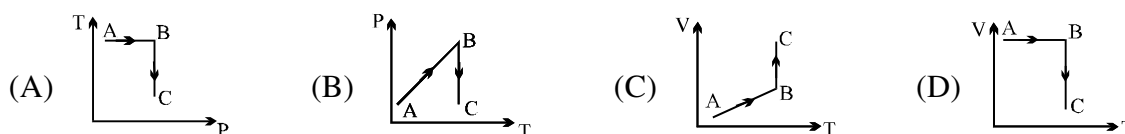
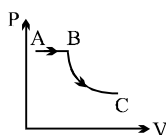
KT0081

7. 1 kg of a gas does 20 kJ of work and receives 16 kJ of heat when it is expanded between two states. A second kind of expansion can be found between the initial and final state which requires a heat input of 9 kJ. The work done by the gas in the second expansion is :

(A) 32 kJ (B) 5 kJ (C) -4 kJ (D) 13 kJ

KT0082

8. A process is shown in the diagram. Which of the following curves may represent the same process?



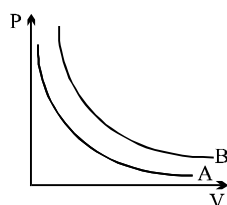
KT0083

9. An ideal gas expands in such a way that $PV^2 = \text{constant}$ throughout the process.

(A) The graph of the process of T-V diagram is a parabola.
 (B) The graph of the process of T-V diagram is a straight line.
 (C) Such an expansion is possible only with heating.
 (D) Such an expansion is possible only with cooling.

KT0084

10. Figure shows the pressure P versus volume V graphs for two different gas sample at a given temperature. M_A and M_B are masses of two samples, n_A and n_B are numbers of moles. Which of the following **must be incorrect**.



(A) $M_A > M_B$ (B) $M_A < M_B$ (C) $n_A > n_B$ (D) $n_A < n_B$

KT0085

11. A vertical cylinder with heat-conducting walls is closed at the bottom and is fitted with a smooth light piston. It contains one mole of an ideal gas. The temperature of the gas is always equal to the surrounding's temperature, T_0 . The piston is moved up slowly to increase the volume of the gas to η times. Which of the following is incorrect?

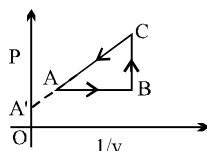
(A) Work done by the gas is $RT_0 \ln \eta$.
 (B) Work done against the atmosphere is $RT_0(\eta - 1)$.
 (C) There is no change in the internal energy of the gas.
 (D) The final pressure of the gas is $\frac{1}{(\eta - 1)}$ times its initial pressure.

KT0086

12. A gas is enclosed in a vessel at a constant temperature at a pressure of 5 atmosphere and volume 4 litre. Due to a leakage in the vessel, after some time, the pressure is reduced to 4 atmosphere. As a result, the
- volume of the gas decreased by 20%
 - average K.E. of gas molecule decreases by 20%
 - 20% of the gas escaped due to the leakage
 - 25% of the gas escaped due to the leakage

KT0087

13. An enclosed ideal gas is taken through a cycle as shown in the figure. Then

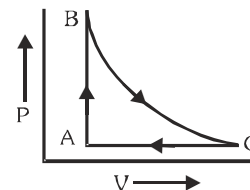


- Along AB, temperature decreases while along BC temperature increases
- Along AB, temperature increases while along BC the temperature decreases.
- Along CA work is done by the gas and the internal energy remains constant.
- Along CA work is done on the gas and internal energy of the gas increases.

KT0088

14. One mole of monoatomic ideal gas undergoes a cyclic process ABCA as shown in figure. Process BC is adiabatic. The temperatures at A, B and C are 300, 600 and 450K respectively. Choose the correct statement(s).

- In process CA change in internal energy is $225R$.
- In process AB change in internal energy is $-150R$.
- In process BC change in internal energy is $-225R$.
- Change in internal energy during the whole cyclic process is $+150R$.



KT0089

MULTIPLE CORRECT TYPE QUESTIONS

15. Let v_{av} , v_{rms} and v_p respectively denote mean speed, root mean square speed and the most probable speed of the molecule in an ideal monoatomic gas at absolute temperature T . The mass of a molecule is m then :

- no molecule can have speed greater than $\sqrt{2} v_{rms}$
- no molecule can have speed less than $v_p / \sqrt{2}$
- $v_p < v_{av} < v_{rms}$
- the average kinetic energy of a molecule is $\frac{3}{4} m v_p^2$

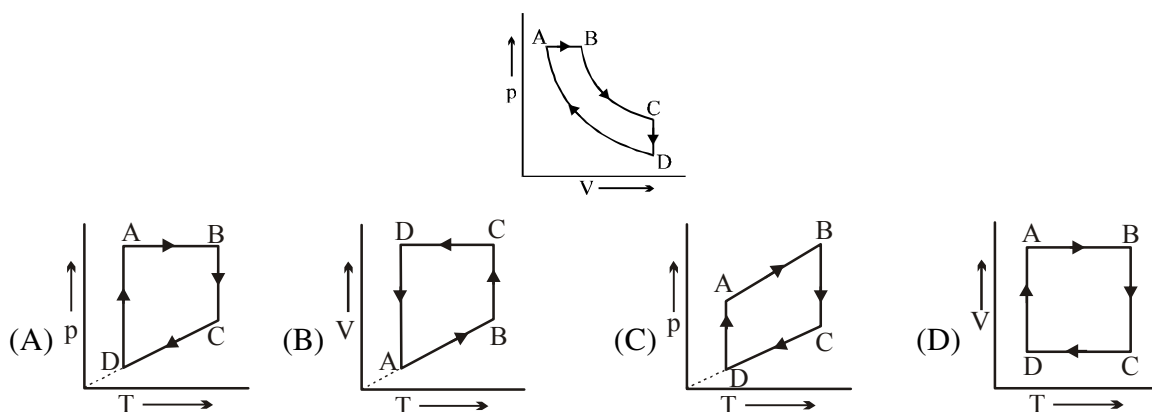
KT0090

16. A gas expands such that its initial and final temperature are equal. Also, the process followed by the gas traces a straight line on the P-V diagram :

- The temperature of the gas remains constant throughout.
- The temperature of the gas first increases and then decreases
- The temperature of the gas first decreases and then increases
- The straight line has a negative slope.

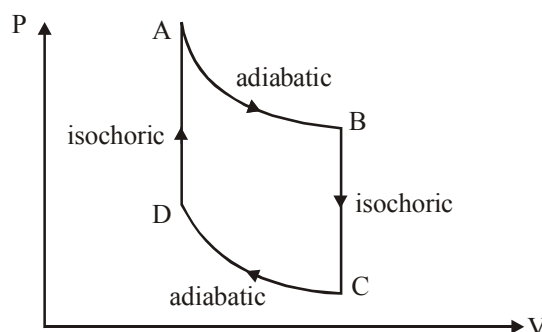
KT0091

17. A cyclic process ABCD is shown in the p - V diagram. Which of the following curves represents the same process if BC & DA are isothermal processes



KT0092

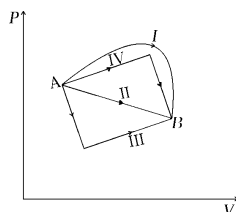
18. Two moles of helium gas is taken through the cycle ABCDA as shown in the figure. If $T_A = 1000$ K, $2P_A = 3P_B = 6P_C$.



- (A) work done by the gas in the process A to B is 3741 J.
 (B) heat lost by the gas in the process B to C is 10600 J.
 (C) temperature T_D is 2000 K.
 (D) none of these

KT0093

19. Figure shows the P - V diagram of an ideal gas undergoing a change of state from A to B. Four different parts I, II, III and IV as shown in the figure may lead to the same change of state.



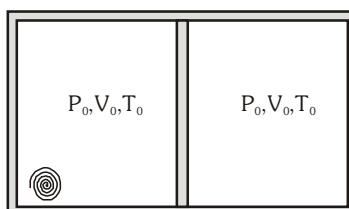
- (A) Change in internal energy is same in IV and III cases, but not in I and II.
 (B) Change in internal energy is same in all the four cases.
 (C) Work done is maximum in case I
 (D) Work done is minimum in case II.

KT0094

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 20 to 22

One mole of a monoatomic ideal gas occupies two chambers of a cylinder partitioned by means of a movable piston. The walls of the cylinder as well as the piston are thermal insulators. Initially equal amounts of gas fill both the chambers at (P_0, V_0, T_0) . A coil is burnt in the left chamber which absorbs heat and expands, pushing the piston to the right. The gas on the right chamber is compressed until to pressure becomes $32 P_0$.



20. The final volume of left chamber is :-

(A) $\frac{V_0}{8}$

(B) $\frac{15}{8} V_0$

(C) $\frac{7}{8} V_0$

(D) $\frac{9}{8} V_0$

KT0095

21. The work done on the gas in the right chamber is :-

(A) $\frac{9}{2} P_0 V_0$

(B) $-\frac{9}{2} P_0 V_0$

(C) $\frac{13}{2} P_0 V_0$

(D) $\frac{17}{2} P_0 V_0$

KT0095

22. The change in internal energy of the gas in the left chamber is :-

(A) $\frac{186}{4} RT_0$

(B) $\frac{177}{4} RT_0$

(C) $\frac{59}{2} RT_0$

(D) $\frac{131}{4} RT_0$

KT0095

MATRIX MATCH TYPE QUESTION

23.

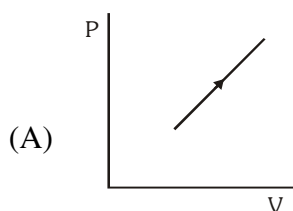
Column I

(Pressure volume graph)

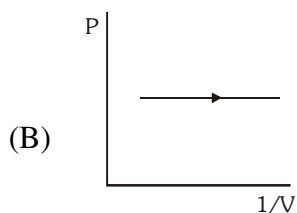
Column II

(W is work done by gas,

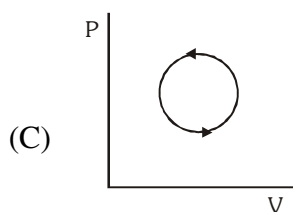
Q heat supply to gas)



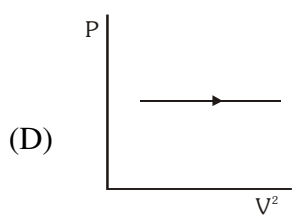
(P) $W > 0$



(Q) $W < 0$



(R) $Q > 0$



(S) $Q < 0$

KT0096

SUPPLEMENT FOR JEE-MAINS

- Two engines A and B have their sources at 400 K and 350 K and sinks at 350 K and 300 K respectively. Which engine is more efficient and by how much ?
- A carnot engine working between 400 K and 800 K has a work output of 1200 J per cycle. What is the amount of heat energy supplied to the engine from source per cycle?
- A carnot engine works as a refrigerator between 250 K and 300 K. If it receives 750 cal of heat from the reservoir at the lower temperature. Calculate the amount of heat rejected at the higher temperature.
- The temperature insides & outside of refrigerator are 260 K and 315 K respectively. Assuming that the refrigerator cycle is reversible, calculate the heat delivered to surroundings for every joule of work done.
- A refrigerator takes heat from water at 0°C and transfer it to room at 27°C. If 100 kg of water is converted in ice at 0°C then calculate the workdone. (Latent heat of ice is 3.4×10^5 J/kg)
- The coefficient of performance of a refrigerator working between 10° C and 20° C is :—
(A) 1 (B) 28.3 (C) 29.3 (D) 3.53
- A refrigerator freezes 5 kg of water at 0°C into ice at 0°C in a time interval of 20 minutes. Assume that room temperature is 20°C. Calculate the minimum power needed to accomplish it ?
(A) 24.4 Watt (B) 0.1025 Watt (C) 0.0244 Watt (D) 102.5 Watt
- A carnot engine whose source temperature is at 400 K takes 100 kcal of heat at this temperature in each cycle and gives 70 kcal to the sink. Calculate (i) the temperature of the sink and (ii) the efficiency of the engine.
(A) 280 K, 30 % (B) 270 K, 30% (C) 280 K, 10 % (D) None of these
- An ionization gauge installed in the artificial satellite showed that 1 cm³ of the atmosphere contained about a thousand million particles of gas at a height of 300 km from the Earth's surface. Find the mean free path of the gas particles at this height. Take the diameter of the particles equal to 2×10^{-10} m.
(A) $\ell = 5.6$ km (B) $\ell = 2.6$ km (C) $\ell = 8.4$ km (D) $\ell = 1.1$ km
- What maximum number of molecules of a gas should be contained in 1 cm³ of a spherical vessels with a diameter of 15 cm so that the molecules do not collide with each other? The diameter of a gas molecule is 3×10^{-8} cm.
(A) $1.7 \times 10^{18}/\text{CC}$ (B) $1.7 \times 10^{16}/\text{CC}$ (C) $2.5 \times 10^{19}/\text{CC}$ (D) $1.1 \times 10^{14}/\text{CC}$
- An ideal refrigerator operates according to the reverse Carnot cycle and transmits heat from a cold source with water at a temperature of 0°C to a boiler with water at a temperature of 100 °C. What amount of water must be frozen in the cooler to convert 1 kg of water into vapour in the boiler?
(A) 4.94 kg (B) 3.24 kg (C) 5.63 kg (D) 2.12 kg
- Oxygen (10 grammes) is heated from $t_1 = 50^\circ\text{C}$ to $t_2 = 150^\circ\text{C}$. Find the change in entropy if the oxygen is heated: (1) isochorically, (2) isobarically.
(A) $\Delta S_V = 1.76$ J/K; $\Delta S_P = 2.46$ J/K (B) $\Delta S_V = \Delta S_P = 1.76$ J/K
(C) $\Delta S_V = \Delta S_P = 2.46$ J/K (D) $\Delta S_V = 2.46$ J/K; $\Delta S_P = 1.76$ J/K
- An ideal gas expands into a vacuum in a rigid vessel. As a result there is :
(A) a change in entropy (B) an increase of pressure
(C) a change in temperature (D) a decrease of internal energy
(E) a change in phase